Pierre Deligne

Collected Papers

English edition

A source-aligned Codex edition

Editorial note

This volume is intended for continuous mathematical reading. Each work is presented as a self-contained English text, retaining the source's mathematical notation, numbered structure, diagrams, and bibliography as closely as the edition permits. The French and English editions are independent books rather than an interleaved bilingual layout. Texts are prepared by source-aligned transcription and translation against identified printed witnesses. Detailed source identities, correction history, and archival evidence accompany the public record instead of interrupting the papers themselves.

Coverage note. This sparse edition contains D001–D012, D014, D015, D021, D025–D029, and D036 in numerical order. D013, D016–D020, D022–D024, and D030–D035 are not included or claimed complete in this volume; later complete works remain eligible for inclusion despite those gaps.

Contents

CONGRUENCES FOR THE NUMBER OF SUBGROUPS OF ORDER p^k IN A FINITE GROUP

P. DELIGNE

The method used by Serre [1, p. 147] to prove the Sylow theorem gives results more general than that theorem. We spell them out here.

We use the following notation: p denotes a prime number; for every integer n, $v(n)$ is the exponent of the largest power of p dividing n; finally, G is a finite group of order g.

For completeness, recall the following lemma (Lazard [2, p. 71]).

Lemma 1.

a) *If $n = \sum c_i p^i$, with $0 \leq c_i < p$ the base-p expansion of n, then*

$$v(n!) = \frac{1}{p-1}\left(n - \sum c_i\right).$$

b)

$$v\binom{r}{s} \geq v(r) - v(s).$$

For a), one has[1]

$$v(p^k!) = \sum_{r=1}^{k} \left\lfloor \frac{p^k}{p^r} \right\rfloor = \sum_{r=1}^{k} p^{k-r} = \sum_{i=0}^{k-1} p^i = \frac{p^k - 1}{p-1},$$

which proves the assertion in this case. It is clear that, if $0 \leq c < p$, then $v((cp^k)!) = c\, v(p^k!)$, and that, if $m < p^{v(n)}$, then

$$v((n+m)!) = v(n!) + v(m!).$$

These additivity properties are the same as those of the right-hand side, whence the formula.

Part b) follows easily by returning to the definitions and taking account of the algorithm which computes the digits c_i of a sum from the digits of its summands. Equality holds when r is a power of p.

Proposition 1.

$$\binom{p^k n}{p^k m} \equiv \binom{n}{m} \pmod{p^{v(n)+2}}$$

if $p \neq 2$, or if n is odd. The congruence is always true modulo $p^{v(n)+1}$.

Proof. Let X and Y be two indeterminates. One has

$$(X+Y)^{p^k} = X^{p^k} + Y^{p^k} + pQ,$$

where Q is a polynomial of degree $< p^k$ in each variable. The binomial formula then gives

$$(X+Y)^{p^k n} = (X^{p^k} + Y^{p^k})^n + np(X^{p^k} + Y^{p^k})^{n-1} Q \tag{1}$$

[1] The source's intermediate sum is negative; we replace it by the exact Legendre calculation (source defect R1).

modulo the largest power of p which divides all the numbers $p^i\binom{n}{i}$, $i \geq 2$. These have valuation at least
$$i + v(n) - v(i) \geq v(n) + 2$$
if $p \neq 2$. If $p = 2$, one always has $i > v(i)$ (Cantor), which gives the exponent $v(n) + 1$.

The binomial coefficient $\binom{p^k n}{p^k m}$ is the coefficient of $X^{p^k m} Y^{p^k(n-m)}$.[2] The second term on the right-hand side of (1) contains no term of this type, since none of its exponents is divisible by p^k. Thus the coefficient of this monomial on the right-hand side of (1) is $\binom{n}{m}$. This proves the assertion when $p \neq 2$. If $p = 2$, the preceding already proves the congruence modulo $2^{v(n)+1}$. In addition, $Q \equiv X^{2^{k-1}} Y^{2^{k-1}} \pmod 2$, so $Q^2 \equiv X^{2^k} Y^{2^k} \pmod 4$, and the next term in the binomial expansion gives
$$\binom{2^k n}{2^k m} \equiv \binom{n}{m} + 2n(n-1)\binom{n-2}{m-1} \pmod{2^{v(n)+2}},$$
which gives the asserted improvement when n is odd. $\square$

Theorem.
Let S be a Sylow p-subgroup of G, let $s = v(g)$, and let d_k be the number of subgroups of G of order p^{s-k}, for $0 \leq k \leq s$. Then:

(i) *$d_k \equiv 1 \pmod p$;*

(ii) *if S is cyclic or abelian of type $(p, \dots, p)$, then d_k is congruent modulo p^{k+1} to the number of subgroups of S of index p^k;*

(iii) *if S is not cyclic, then*
$$d_1 \equiv 1 + p \pmod{p^2}.$$

Let G act by left translations on the set of its subsets with p^k elements. If such a subset P of G has stabilizer H, then P is a union of right cosets of H, and hence H is a group of order p^l with $l \leq k$; one has $l = k$ if and only if P is a translate, on the right or on the left, of a subgroup of order p^k. The number of elements in the orbit of P is divisible by p^{s-l}. Therefore $\binom{g}{p^k}$ is congruent modulo p^{s-k+1} to the number of subsets of G which are translates of a subgroup of order p^k. By Proposition 1, it is also congruent to gp^{-k}. There are $d_{s-k}\, g\, p^{-k}$ such subsets. Thus
$$gp^{-k} \equiv d_{s-k}\, g\, p^{-k} \pmod{p^{s-k+1}},$$
and the first assertion follows.

We now consider case (ii), with S abelian of type $(p, \dots, p)$. Let $G_{i,j}$ be the number of subgroups of order p^i in a subgroup of order p^j; this is the number of points of a Grassmannian. For $i > j$, $G_{i,j} = 0$; for $i \leq j$,
$$G_{i,j} = \frac{(p^j - 1)\cdots(p^j - p^{i-1})}{(p^i - 1)\cdots(p^i - p^{i-1})} = \widetilde{G}_i \widetilde{G}_{j-i} \widetilde{G}_j^{-1},$$
where
$$\widetilde{G}_i = [(1 - p^i)\cdots(1 - p)]^{-1} \quad (i \geq 0), \qquad \widetilde{G}_i = 0 \quad (i < 0).$$
Here $\widetilde{G}_i = (-1)^i G_i$, where the source defines $G_i = [(p^i - 1)\cdots(p - 1)]^{-1}$. This sign normalization leaves $G_{i,j}$ unchanged.[3]

[2] The printed source instead has $X^{p^k m} Y^{p^k n}$. For $m > 0$ that monomial has total degree $p^k(m+n) > p^k n$, so its coefficient in $(X+Y)^{p^k n}$ is zero. The displayed $Y^{p^k(n-m)}$ is the source repair recorded as R2 in the editorial register.

[3] The source later claims $G_k^{-1} \equiv G_{k+1}^{-1} \pmod{p^{k+1}}$, which is false for its printed definition. The displayed normalization gives $\widetilde{G}_{k+1}^{-1} = \widetilde{G}_k^{-1}(1 - p^{k+1})$ and repairs the argument exactly (source defect A1).

We know that p^i divides the number u_i of subsets of G with p^s elements whose stabilizer has order p^{s-i}. Put $g = p^s h$. Counting in two ways the pairs formed by a subgroup of order p^{s-i} and a subset with p^s elements fixed by it gives, for $0 \le i, j \le s$,

$$\begin{cases} d_i \dbinom{p^i h}{p^i} = \sum_j G_{s-i,s-j} u_j & (2) \\ \qquad p^i \mid u_i & (3) \\ \qquad d_s = 1 & (4) \end{cases}$$

Now work in the localization of $\mathbf{Z}$ at p, or in $\mathbf{Z}_p$. Let e_i be the quotient of the left-hand side of (2) by $\binom{p^s h}{p^s}\widetilde{G}_i$. By Proposition 1, and since $\widetilde{G}_i$ is prime to p for $i \ge 0$, there are numbers v_i, $0 \le i \le s$, such that

$$\begin{cases} d_i \equiv \widetilde{G}_i e_i \pmod{p^{i+1}} & (5) \\ e_i = \sum_j \widetilde{G}_{i-j} v_j & (2') \\ p^i \mid v_i & (3') \\ e_s = \widetilde{G}_s^{-1} & (4') \end{cases}$$

Analogous identities hold in the group S. To show that they determine d_i modulo p^{i+1}, it is enough to check that the identities, for $0 \le i, j \le s$,

$$\begin{cases} e_i = \sum_j \widetilde{G}_{i-j} v_j & (2'') \\ p^i \mid v_i & (3'') \\ e_s = 0 & (4'') \end{cases}$$

imply $p^{i+1} \mid e_i$.

From $(2'')$ one obtains

$$e_i - e_{i+1} = \sum_j (\widetilde{G}_{i-j} - \widetilde{G}_{i+1-j}) v_j.$$

Since $\widetilde{G}_k^{-1} \equiv \widetilde{G}_{k+1}^{-1} \pmod{p^{k+1}}$, one also has

$$p^{i+1-j} \mid \widetilde{G}_{i-j} - \widetilde{G}_{i+1-j}, \qquad p^{i+1} \mid e_i - e_{i+1}.$$

The conclusion follows from $(4'')$.

The same technique, in a simpler form, applies when S is cyclic.

We turn to case (iii). The subgroups of S of index p are all normal. Their intersection S' is the Frattini subgroup of S. For every subgroup T of S, if $TS' = S$, then $T = S$. In particular, if S is not cyclic, then S/S' is not cyclic either, and the subgroups of S of index p are identified with the hyperplanes of a vector space over $\mathbf{Z}/p$ of dimension > 1; their number is congruent to $1 + p \pmod{p^2}$.

The preceding equations leave the following system.[4]

$$\begin{cases} d_0 h = u_0, \\ d_1 \binom{ph}{p} = (1+p) u_0 + u_1 \pmod{p^2}, \\ \binom{p^s h}{p^s} = u_0 + u_1 \pmod{p^2}. \end{cases}$$

[4] In its third line the source prints the lower argument p; the next source line itself uses p^s. We restore p^s here (source defect R3). The final subscript after u is damaged in the scan; the counting identities force the intended right-hand side $u_0 + u_1$.

By Proposition 1, $\binom{p^s h}{p^s} \equiv \binom{ph}{p} \equiv h \pmod{p^2}$. Hence

$$d_1 h \equiv (1+p)d_0 h + (h - d_0 h) \pmod{p^2}, \qquad \text{and}$$
$$d_1 \equiv 1 + pd_0 \pmod{p^2}.$$

The assertion follows because $d_0 \equiv 1 \pmod{p}$.

References

[1] Serre, J.-P., *Corps locaux. Act. sc. et ind.* Hermann, 1962.
[2] Lazard, M., Groupes analytiques p-adiques. *Publ. Math. I.H.E.S.*, no. 26, 1965.

Appendix to *Residues and Duality* (Lect. Notes in Math. 20, Springer-Verlag 1966, pp. 404–421)

APPENDIX: COHOMOLOGY WITH PROPER SUPPORT AND CONSTRUCTION OF THE FUNCTOR $f^!$.

by P. DELIGNE[1]

Verdier has shown that, in the topological case, the formalism of Poincare duality reduces to local problems upstairs (see [1]). To transpose his construction to the schematic setting, one needs a theory of cohomology "with proper support" for coherent sheaves.

Unless explicitly stated otherwise, all preschemes considered are noetherian and all presheaves are quasi-coherent.

No. 1. Formalities of pro-objects.

Proposition 1. *Let C be a U-category*[2] *in which finite inductive limits exist. Let h be a functor $C^\circ \to$ (ens). The following conditions are equivalent:*

(i) *h is an inductive limit, over a small*[3] *filtered ordered set, of representable functors.*
(ii) *h is an inductive limit, over a small filtered category, of representable functors.*
(iii) *h transforms finite projective limits into finite inductive limits, and there is a small set of objects such that every element of an $h(X)$ factors through one of them.*

The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are trivial, and (iii) $\Rightarrow$ (ii) is standard: h is the limit of the h_F over the category of pairs (F, α), where $\alpha \in h(F)$ and F lies in the subcategory of C stable under finite inductive limits generated by the given small set. It remains to prove that (ii) $\Rightarrow$ (i). If $\mathcal{L}$ and $\mathcal{M}$ are two filtered categories, a functor $F : \mathcal{L} \to \mathcal{M}$ is called cofinal if $\varinjlim_{\mathcal{L}} h_{F(L)}$ is the final functor. For every $G : \mathcal{M} \to \mathcal{N}$, one then has $\varinjlim_{\mathcal{M}} G = \varinjlim_{\mathcal{L}} GF$. Thus it remains to prove:

Lemma 1. *For every small filtered category $\mathcal{L}$, there exists a cofinal functor from a small filtered ordered set to $\mathcal{L}$.*

[1] This is an expanded version of a letter from P. Deligne to R. Hartshorne (letter of March 3, 1966). The footnotes were added by the copyist.
[2] U denotes a fixed universe throughout.
[3] "small" means "$\in U$".

The first projection $\mathcal{L} \times \mathbb{N} \to \mathcal{L}$ is a cofinal functor, where $\mathbb{N}$ has its natural order. This permits one to reduce to the case where $\operatorname{Ob} \mathcal{L}$, preordered by the relation $\operatorname{Hom}(X, Y) \neq \varnothing$, has no greatest element. Order by inclusion the set E of finite subcategories of $\mathcal{L}$ having a single final object, where finite means having a finite set of arrows. Define a functor from E to $\mathcal{L}$ by sending each of these categories to its final object. Under the hypotheses made, E is filtered and this functor is cofinal.

The functors satisfying the equivalent conditions of Proposition 1 are the Ind-objects of C. They form a full subcategory $\operatorname{Ind} C$ of $\underline{\operatorname{Hom}}(C^\circ, (\mathrm{Ens}))$, stable under inductive limits. The assignment $C \mapsto \operatorname{Ind} C$ is a functor $(\mathrm{cat}) \to (\mathrm{cat})$. If filtered inductive limits exist in C, then for every $h \in \operatorname{Ind} C$ the functor $X \mapsto \operatorname{Hom}(h, h_X)$ is corepresentable; hence there is a functor, denoted $\varinjlim$, from $\operatorname{Ind} C$ to C. In particular, one always has a functor $\operatorname{Ind} \operatorname{Ind} C \to \operatorname{Ind} C$, and every functor $C \to \operatorname{Ind} D$ extends to a functor $\operatorname{Ind} C \to \operatorname{Ind} D$. In $\operatorname{Ind} C$, filtered inductive limits are exact and will be denoted "$\varinjlim$"; in particular, identifying C with a full subcategory of $\operatorname{Ind} C$, if $(X_i)_{i \in I}$ is a filtered inductive system in C, then

$$\varinjlim_{i \in I} X_i = \varinjlim_{i \in I} \text{“} \varinjlim \text{”} X_i.$$

A functor on $\operatorname{Ind} C$ is representable if and only if it transforms finite projective limits into finite inductive limits, filtered inductive limits into filtered inductive limits, and its restriction to C satisfies the smallness condition of Proposition 1(iii). Indeed, Proposition 1 then shows that its restriction to C is an Ind-object, to which the functor is everywhere equal because of the condition on limits.

The preceding statements, except Proposition 1(iii) and the preceding assertion, remain true if one uses everywhere limits of sequences, or countable limits, which is the same thing.

By duality one defines the category $\operatorname{pro} C$ of pro-objects, namely the full subcategory of $\underline{\operatorname{Hom}}(C, (\mathrm{Ens}))^\circ$.

Proposition 2. *Let X be a quasi-compact, quasi-separated prescheme, not necessarily noetherian. The category of quasi-coherent sheaves on X is equivalent to the category of Ind-objects of the category of quasi-coherent sheaves of finite presentation on X.*

The map is "$\varinjlim$" $\mathcal{F}_i \mapsto \varinjlim \mathcal{F}_i$. If $\mathcal{F}$ is of finite presentation, then for every filtered inductive system $(\mathcal{G}_i)_{i \in I}$,

$$\operatorname{Hom}_X(\mathcal{F}, \varinjlim \mathcal{G}_i) = \varinjlim \operatorname{Hom}_X(\mathcal{F}, \mathcal{G}_i).$$

By a finite gluing argument, which commutes with $\varinjlim$, one reduces to the affine case. Thus the preceding functor is fully faithful. Its image is stable under filtered inductive limits and finite sums. It remains only to prove that for every $x \in X$, every $\mathcal{F}$ on X, and every $s \in \mathcal{F}_x$, there exists a sheaf $\mathcal{G}$ of finite presentation and a map from $\mathcal{G}$ to $\mathcal{F}$ whose image contains s.

The sheaf $\mathcal{F}$ will be a limit of images of sheaves of finite presentation, and each of these will be a quotient of a sheaf of finite presentation by a limit of sub-sheaves of finite type.

On a quasi-compact neighbourhood U of x, let $f : \mathcal{C} \to \mathcal{F}$ be a map with $\mathcal{C}$ of finite presentation and with s in its image. It is enough to extend $\mathcal{C}$ and f to all of X; proceeding step by step, it is enough to know how to extend them to $U \cup V$ when V is affine. This amounts to carrying out an extension from $U \cap V$ to V.

Lemma 2. *Let $\mathcal{C}$ be a sheaf of finite presentation on a quasi-compact open U of an affine scheme X, let $\mathcal{F}$ be a sheaf on X, and let f be a map from $\mathcal{C}$ to $\mathcal{F}|U$. It is possible to extend $\mathcal{C}$ and f to $\overline{\mathcal{C}}$ and $\overline{f}$, defined on X, with $\overline{\mathcal{C}}$ of finite presentation.*

Let ι be the inclusion of U in X, and let $\mathcal{C}_1$ be the fiber product of $\iota_*\mathcal{C}$ and $\mathcal{F}$ over $\iota_*\iota^*\mathcal{F}$. Then $\mathcal{C}_1$ extends $\mathcal{C}$ and maps to $\mathcal{F}$. If one writes $\mathcal{C}_1$ as a limit of its sub-sheaves of finite type, then, since U is quasi-compact and $\mathcal{C}$ is of finite presentation, one of these already extends $\mathcal{C}$. Writing the latter as a quotient of $\mathcal{O}^n$ by a limit of sub-sheaves of finite type of $\mathcal{O}^n$, one sees in the same way that $\mathcal{C}_1$ may be replaced by a sheaf of finite presentation.

Corollary 1. *A contravariant additive functor $F : (q\ coh) \to Ab$ is representable if and only if it is left exact and transforms filtered inductive limits into filtered inductive limits.*

Corollary 2. *Every sheaf of finite presentation defined on a quasi-compact open U of X extends to a sheaf of finite presentation on X.*

Let $\mathcal{A}$ be an abelian category. We shall have to work essentially in $\operatorname{pro} D^b\mathcal{A}$, the full subcategory of $\operatorname{pro} D(\mathcal{A})$ consisting of the "$\varprojlim$" K_i for which the cohomology of K_i is uniformly bounded as i varies. The functor $\mathbb{H}^p$ extends to a functor $\operatorname{pro} D(\mathcal{A}) \to \operatorname{pro} \mathcal{A}$.

Proposition 3. *The functors $\mathbb{H}^p$ form a conservative[4] family of functors on $\operatorname{pro} D^b\mathcal{A}$.*

[4] That is, if u is an arrow such that all $\mathbb{H}^p(u)$ are invertible, then u is invertible.

Let $f : K \to L$ be an arrow of pro $D^b\mathcal{A}$ such that the maps $\mathbb{H}^p(K) \to \mathbb{H}^p(L)$ are isomorphisms in pro $\mathcal{A}$. We must check that, for every M in $D^b(\mathcal{A})$,

$$\operatorname{Hom}(K, M) \longleftarrow \operatorname{Hom}(L, M)$$

is an isomorphism. Write $K = \text{“}\varprojlim\text{''} K_\alpha$. There is a convergent spectral sequence

$$\operatorname{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \operatorname{Ext}^{p+q}(K_\alpha, M),$$

whence, on passing to the limit, another convergent spectral sequence

$$\varinjlim \operatorname{Ext}^p(\mathbb{H}^q K_\alpha, M) \Longrightarrow \varinjlim \operatorname{Ext}^{p+q}(K_\alpha, M),$$

since q is uniformly bounded. Using the fact that every functor, here Ext, passes to pro-objects, and using the functor $\varinjlim : \text{Ind Ab} \to \text{Ab}$, we may rewrite this as

$$\varinjlim \operatorname{Ext}^p(\mathbb{H}^q K, M) \Longrightarrow \varinjlim \operatorname{Ext}^{p+q}(K, M) = \operatorname{Hom}(K[-p-q], M).$$

This spectral sequence remains valid for $M \in D(\mathcal{A})$. The proposition follows immediately from the existence of these spectral sequences.

No. 2. Fundamental lemmas.

Recall that all preschemes considered are noetherian.

Proposition 4 (Duality theorem for an open immersion)**.** *Let U be an open subset of X, let j be the inclusion, let $\mathcal{J}$ be an ideal defining the closed complement, let $\mathcal{F}$ be a coherent sheaf on U, and let $\mathcal{G}$ be quasi-coherent on X. Let $\overline{\mathcal{F}}$ be a coherent extension of $\mathcal{F}$. Then*

$$\operatorname{Hom}_U(\mathcal{F}, j^*\mathcal{G}) = \operatorname{Hom}_X(\text{“}\varprojlim\text{''} \mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}). \tag{5}$$

The map $\varinjlim \operatorname{Hom}_X(\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}) \to \operatorname{Hom}_U(\mathcal{F}, \mathcal{G})$ is injective: if the image under f of $\mathcal{J}^n\overline{\mathcal{F}}$ in $\mathcal{G}$ has its support in the complement of U, then it is annihilated by a power of $\mathcal{J}$, say $\mathcal{J}^k$, and the image of $\mathcal{J}^{n+k}\overline{\mathcal{F}}$ is zero.

Now assume that X is affine, and let $f \in \operatorname{Hom}_U(\mathcal{F}, \mathcal{G})$. For k sufficiently large, every section s of $\mathcal{J}^k\overline{\mathcal{F}}$ over X has an image over U which extends to a section of $\mathcal{G}$ over X. Replacing $\overline{\mathcal{F}}$ by $\mathcal{J}^k\overline{\mathcal{F}}$, we may

[5] This formula is of course well known; cf., for example, P. Gabriel's thesis.

suppose that we have a diagram

$$\begin{tikzcd} 0 \arrow[r] & R \arrow[r] \arrow[d, dotted] & \mathcal{O}^n \arrow[r] \arrow[d, equal] & \overline{\mathcal{F}} \arrow[r] \arrow[d, dotted] & 0 \\ 0 \arrow[r] & R_1 \arrow[r] & \mathcal{O}^n \arrow[r] & \mathcal{G} & \end{tikzcd}$$

in which the dotted arrows are defined on U.

For ℓ sufficiently large, by Krull's theorem and by the fact that $R/(R \cap R_1)$ is concentrated outside U, one has $R \cap \mathcal{J}^\ell \cdot \mathcal{O}^n \subset R_1$, which permits f to be extended to a map $\overline{f} : \mathcal{J}^\ell\overline{\mathcal{F}} \to \mathcal{G}$.

In the general case, let U_i be a finite affine covering of X, and let U_{ijk} be finite affine coverings of $U_{ij} = U_i \cap U_j$. Since $\varinjlim$ commutes with finite products, one obtains

$$\begin{tikzcd} 0 \arrow[r] & \mathrm{Hom}_U(\mathcal{F}, j^*\mathcal{G}) \arrow[r] & \prod_i \mathrm{Hom}_{U_i\cap U}(\mathcal{F}, j^*\mathcal{G}) \arrow[r, shift left] \arrow[r, shift right] & \prod_{ijk} \mathrm{Hom}_{U_{ijk}\cap U}(\mathcal{F}, j^*\mathcal{G}) \\ 0 \arrow[r] & \varinjlim \mathrm{Hom}_X(\mathcal{J}^n\overline{\mathcal{F}}, j^*\mathcal{G}) \arrow[r] \arrow[u] & \prod_i \varinjlim \mathrm{Hom}_{U_i}(\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}) \arrow[r, shift left] \arrow[r, shift right] \arrow[u] & \prod_{ijk} \varinjlim \mathrm{Hom}_{U_{ijk}}(\mathcal{J}^n\overline{\mathcal{F}}, \mathcal{G}) \arrow[u] \end{tikzcd}$$

which completes the proof.

Proposition 4 gives an explicit formula for the functor (coh on U) $\to$ pro (coh on X) left adjoint to j^*. We shall denote this functor by $j_!$ (extension by zero)[6]; it follows from Krull's theorem that it is exact. This also follows from the fact that, since X is noetherian, an injective object of the category of quasi-coherent sheaves on X, restricted to U, remains injective as a quasi-coherent sheaf on U.

Proposition 5 (Independence of the compactification)**.**

$$\begin{tikzcd} U \arrow[r, hook] \arrow[d, equal] & X \arrow[d, "f"] \\ U \arrow[r, hook] & Y \end{tikzcd}$$

Let $f : X \to Y$ be a proper morphism inducing an isomorphism between the open subset U of Y and $f^{-1}(U)$. Let $\mathcal{J}$ be a sheaf of ideals defining $Y \setminus U$, and put $\mathcal{I} = f^*\mathcal{J}$. If $\mathcal{F}$ is coherent on X, then

(i) *for $k > 0$, the projective system $\mathbb{R}^k f_*\mathcal{I}^n\mathcal{F}$ is essentially zero, that is, it defines the zero pro-object;*

(ii) *for $k = 0$, for all sufficiently large n, $f_*\mathcal{I}^{n+1}\mathcal{F} = \mathcal{J} \cdot f_*\mathcal{I}^n\mathcal{F}$.*

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[6] This terminology and notation plainly conflict with the generally accepted ones, for example in Godement's book and in SGA 1962. The reader may prefer to read $\mathfrak{J}_!$ instead of $j_!$.

One has

$$\bigoplus_{n=0}^{\infty} \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \quad \text{is a finite type } \bigoplus_{n=0}^{\infty} \mathcal{J}^n\text{-module (EGA III).}$$

Thus, for n sufficiently large, the map $\mathcal{J} \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \to \mathbb{R}^k f_* \mathcal{I}^{n+1} \mathcal{F}$ is surjective; (ii) follows. If $k > 0$, $\mathbb{R}^k f_*$ is zero on U, so there exists N such that $\mathcal{J}^N \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$ is zero for every n. The diagram

$$\begin{tikzcd}
\mathcal{J}^p \otimes \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \arrow[r] \arrow[d] & \mathbb{R}^k f_* \mathcal{I}^{n+p} \mathcal{F} \arrow[r] \arrow[d] & 0 \quad (\text{if } n \geq n_0) \\
\mathcal{J}^p \cdot \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} \arrow[r] & \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F} &
\end{tikzcd}$$

proves that for N sufficiently large and $k > 0$ the map

$$\mathbb{R}^k f_* \mathcal{I}^{n+N} \mathcal{F} \to \mathbb{R}^k f_* \mathcal{I}^n \mathcal{F}$$

is zero, which is (i).

No. 3. Definition of $\mathbb{R}f_!$.

A morphism f, respectively a composable pair of morphisms, respectively a triple, will be called compactifiable if one can respectively find commutative diagrams

$$\begin{tikzcd}
X \arrow[r, hook] \arrow[d, "f"'] & \overline{X} \arrow[dl] \\
S &
\end{tikzcd}
\qquad
\begin{tikzcd}
X \arrow[r, hook] \arrow[d, "g"'] & \overline{X} \arrow[r, hook] & \overline{\overline{X}} \arrow[dl] \\
Y \arrow[r, hook] \arrow[d, "f"'] & \overline{Y} \arrow[dl] & \\
S & &
\end{tikzcd}
\qquad
\begin{tikzcd}
X \arrow[r, hook] \arrow[d, "h"'] & \overline{X} \arrow[r, hook] \arrow[dl] & \overline{\overline{X}} \arrow[r, hook] \arrow[dl] & \overline{\overline{\overline{X}}} \arrow[dl] \\
Y \arrow[r, hook] \arrow[d, "g"'] & \overline{Y} \arrow[r, hook] \arrow[dl] & \overline{\overline{Y}} \arrow[dl] & \\
Z \arrow[r, hook] \arrow[d, "f"'] & \overline{Z} \arrow[dl] & & \\
S & & &
\end{tikzcd}$$

in which the horizontal arrows are open immersions and the oblique arrows are proper. Every compactifiable morphism is separated and of finite type, and Nagata asserts in [2] that his proof gives the converse for integral noetherian schemes, but the hypotheses he imposes on the base are not clear.[7]

We propose, for every compactifiable f, to define

$$\mathbb{R}f_! : \operatorname{pro} D^b_{\mathrm{coh}}(X) \to \operatorname{pro} D^b_{\mathrm{coh}}(S).$$

Recall that the category $D^b_{\mathrm{coh}}(S)$, the bounded derived category of coherent sheaves on S, is a full subcategory of the derived category of

[7]Mumford is said to have checked, via Nagata's proof, that every separated morphism of finite type of noetherian schemes is compactifiable.

all sheaves, not necessarily quasi-coherent, on S; its essential image consists of complexes with bounded coherent cohomology.

If f is proper, one has $\mathbb{R}f_* : D^b_{\mathrm{coh}}(X) \to D^b_{\mathrm{coh}}(S)$ of finite dimension, and it induces $\mathbb{R}f_! : \operatorname{pro} D^b_{\mathrm{coh}}(X) \to \operatorname{pro} D^b_{\mathrm{coh}}(S)$. If f is an open immersion, $f_! : (\text{coh on } X) \to \operatorname{pro}(\text{coh on } S)$ induces

$$D^b(\text{coh on } X) \to D^b \operatorname{pro}(\text{coh on } S),$$

which maps to $\operatorname{pro} D^b_{\mathrm{coh}}(S)$, and this map extends to

$$\mathbb{R}f_! : \operatorname{pro} D^b_{\mathrm{coh}}(X) \to \operatorname{pro} D^b_{\mathrm{coh}}(S).$$

If a coherent complex K on X is extended to $\overline{K}$ on S, its image is "$\varprojlim$" $\mathcal{J}^n\overline{K}$, where $\mathcal{J}$ defines $S \setminus X$.

For a composite $f = gh$, with g proper and h an open immersion, set $\mathbb{R}f_! = \mathbb{R}g_! \, \mathbb{R}h_!$. One must verify:

Independence of the compactification.

Consider a commutative diagram

$$\begin{tikzcd}
 & X' \arrow[dd, "g"] \\
X \arrow[ur, "j'"] \arrow[dr, "j''"'] \arrow[dd, "f"'] & \\
 & X'' \arrow[dl, "\bar f"] \\
S &
\end{tikzcd}$$

where j' and j'' are open immersions, and $\bar f$ and g are proper maps. We have $j'X = g^{-1}j''X \cap j'X$. We must prove that

$$\mathbb{R}\bar f_* \, \mathbb{R}j''_! = \mathbb{R}(\bar f g)_* \, \mathbb{R}j'_!,$$

or equivalently, since $\mathbb{R}(\bar f g)_* = \mathbb{R}\bar f_* \, \mathbb{R}g_*$, that

$$\mathbb{R}j''_! = \mathbb{R}g_* \, \mathbb{R}j'_!.$$

One reduces to the case where X is dense in X', so as to be in the situation of Proposition 5. Let K be a bounded coherent complex on X, extended to $\overline{K}$ on X'. Then $g_*\overline{K}$ extends K on X'', and one has a map

$$g_*\mathcal{J}'^n\overline{K} \to \mathbb{R}g_*\mathcal{J}'^n\overline{K}, \tag{1}$$

where $\mathcal{J}'$ denotes a sheaf of ideals defining $X' \setminus X$. Proposition 5(ii) shows that $j''_!K =$ "$\varprojlim$" $g_*\mathcal{J}'^n\overline{K}$. Passing to the "limit", one obtains

$$\text{“}\varprojlim\text{”}\, \mathbb{R}^p g_*\mathcal{J}'^n\overline{K}^q \Longrightarrow \text{“}\varprojlim\text{”}\, \mathbb{R}^{p+q} g_*\mathcal{J}'^n\overline{K}.$$

Proposition 5(i) shows that this spectral sequence degenerates, and Proposition 3 implies that the arrow deduced from (1) by passing to the “limit” is an isomorphism. This gives the desired formula.

It remains to verify the following.

(a) For every diagram of the form

$$
\begin{tikzcd}
& X' \arrow[d, "h"] \\
X \arrow[ur, "j'"] \arrow[r, "j''"'] \arrow[dr, "j'''"'] \arrow[dd, "f"'] & X'' \arrow[d, "g"] \\
& \bar{X} \arrow[dl, "\bar{f}"] \\
S &
\end{tikzcd}
$$

one has a compatibility

$$
\begin{aligned}
\mathbb{R}\bar{f}_* \, \mathbb{R}j'''_! &= \mathbb{R}\bar{f}_* \, \mathbb{R}g_* \, \mathbb{R}j''_! \\
&= \mathbb{R}\bar{f}_* \, \mathbb{R}g_* \, \mathbb{R}h_* \, \mathbb{R}j'_! \\
&= \mathbb{R}\bar{f}_* \, \mathbb{R}(gh)_* \, \mathbb{R}j'_! \\
&= \mathbb{R}f_* \, \mathbb{R}j'''_!.
\end{aligned}
$$

Since two compactifications can always be dominated by a third, for example $X' \times_S X''$, this is enough to prove independence of the choices.

(b) For every compactifiable pair, an identification

$$\mathbb{R}(fg)_! = \mathbb{R}f_! \, \mathbb{R}g_!.$$

(c) For every compactifiable triple, a compatibility

$$
\begin{array}{ccc}
\mathbb{R}(fgh)_! & = & \mathbb{R}(fg)_! \, \mathbb{R}h_! \\
\| & & \| \\
\mathbb{R}f_! \, \mathbb{R}(gh)_! & = & \mathbb{R}f_! \, \mathbb{R}g_! \, \mathbb{R}h_!
\end{array}
$$

One would then have proved:

Theorem 1. *For $f : X \to Y$ compactifiable, $K \in pro\, D^b_{coh}(X)$ and $L \in pro\, D^b_{coh}(Y)$, put*

$$\mathrm{Hom}_f(K, L) = \mathrm{Hom}_Y(\mathbb{R}f_! K, L).$$

Modulo compactifiability questions, this makes the categories

$$pro\, D^b_{coh}(X)$$

into a cofibrant category over noetherian preschemes, with the arrows compactifiable.

No. 4. Definition of $\mathbb{R}f^!$.

It follows from a general theorem of Verdier [1] that the functor $\mathbb{R}f_* : D(X) \to D(S)$ has a right adjoint $D^+(S) \to D^+(X)$. It deserves a name, say $\mathbb{R}f^!$, only when f is proper, and its definition is not directly suited for computation.

First, one must make explicit a procedure for computing $\mathbb{R}f_*$ at the level of complexes, of the form

$$\mathbb{R}f_* K = f_* C^*(K),$$

where C^* is a finite acyclic resolution depending functorially and exactly on the object to which it is applied, and compatible with filtered inductive limits. Then, for every injective I on S, the functor $\mathrm{Hom}(f_* C^p \mathcal{F}, I)$ is exact in the quasi-coherent sheaf $\mathcal{F}$ on X and transforms $\varinjlim$ into $\varprojlim$; hence it is representable by $f_p^! I$, an injective quasi-coherent sheaf. The $f_p^! I$ form a complex, which defines $\mathbb{R}f^! : D^+S \to D^+X$, right adjoint to $\mathbb{R}f_* : D(X) \to D(S)$.

Here is how to define such a resolution.

1) <u>If S is separated</u>: take a finite covering of X by opens affine over S, for instance affine opens, and put $\mathbb{R}f_* K = f_* \check{C}(\mathcal{U}, K)$, the alternating Cech complex.

2) <u>Otherwise</u>: the canonical flasque resolutions have two defects here.

(a) $f_* C^* \mathcal{F}$ is no longer quasi-coherent. This causes no problem as long as, as here, quasi-coherent injectives are injective as sheaves.

(b) C^* does not commute with filtered inductive limits. This defect is easy to correct: write a quasi-coherent sheaf as $\mathcal{F} = \varinjlim \mathcal{F}_i$, with the $\mathcal{F}_i$ of finite presentation, and put $C'^* \mathcal{F} = \varinjlim C^* \mathcal{F}_i$. This does not depend on the choices.

For an open immersion, take $\mathbb{R}f^! K = f^* K$.

For a composite morphism $f = gh$, with g proper and h an open immersion, take $\mathbb{R}f^! = \mathbb{R}g^! \mathbb{R}h^!$. Combining Proposition 4 with the preceding construction gives:

Theorem 2 (Duality)**.** *Let $f : X \to S$ be compactifiable, $f = gh$. The functors*

$$\mathbb{R}f_! : \mathit{pro}\, D^b_{coh}(X) \to \mathit{pro}\, D^b_{coh}(S) \quad \text{and} \quad \mathbb{R}f^! : D^+(S) \to D^+(X)$$

are adjoint to one another.

One should verify the independence of the compactification and a formalism of compatibilities and identifications analogous to the case of $\mathbb{R}f_!$. Particular cases of the formula sheet follow from the adjunction formula, by uniqueness of a genuine adjoint.

In the general case one may use the following proposition.

Proposition 6. *Let $f : X \to S$ be compactifiable and let $L \in D^+_{\mathrm{qc}}(S)$. The fiber at $x \in X$ of $\mathcal{H}^p \mathbb{R} f^! L$ is given by*

$$(\mathcal{H}^p \mathbb{R} f^! L)_x = \varinjlim_{x \in U} \operatorname{Hom}^p_S(\mathbb{R} f_!(U, \mathcal{O}_U), L).$$

One may assume that X is proper and that L is given as a complex of injective quasi-coherent sheaves. Let U be an affine open of X, and let $\mathcal{J}$ be an ideal defining $X \setminus U$. Then

$$\begin{aligned}(\mathcal{H}^p \mathbb{R} f^! L)(U) &= \mathbb{H}^p(\mathbb{R} f^! L(U)) = \mathbb{H}^p \operatorname{Hom}_U(\mathcal{O}, \mathbb{R} f^! L) \\ &= \mathbb{H}^p \varinjlim \operatorname{Hom}_X(\mathcal{J}^n, \mathbb{R} f^! L) = \varinjlim \operatorname{Hom}^p_S(\mathbb{R} f_* \mathcal{J}^n, L) \\ &= \operatorname{Hom}^p_S(\mathbb{R} f_!(U, \mathcal{O}_U), L),\end{aligned}$$

and the proposition follows.

We shall assume for $\mathbb{R} f^!$ a variance formalism analogous to that of Theorem 1 for $\mathbb{R} f_!$.

If one wants to make the definition of $\mathbb{R} f^!$ more explicit in the proper case, one can observe that if a quasi-coherent sheaf $\mathcal{F}$ represents a functor F, then for every open U and every $\mathcal{J}$ defining $X \setminus U$, one has

$$\mathcal{F}(U) = \varinjlim \operatorname{Hom}(\mathcal{J}^n, \mathcal{F}) = \varinjlim F(\mathcal{J}^n).$$

No. 5. Computation of $\mathbb{R} f^!$.

For a finite morphism, $\mathbb{R} f^!$ is what one wants; to see this, it is enough to take the identity as resolving functor in the computation of $\mathbb{R} f_*$. For projective space, the uniqueness of an adjoint leaves no choice; one uses the fact that the adjunction formula is available for $\mathbb{R} f_! = \mathbb{R} f_* : D(X) \to D(S)$. We shall return to the arrow. For an open immersion, $\mathbb{R} f^! = \mathbb{R} f^*$.

Thus for every quasi-projective open U of X over S, one can compute the restriction of $\mathbb{R} f^!$ to U. In particular, if one can check locally that $\mathbb{R}^q f^! K = 0$ for $q \neq p$, then one knows $\mathbb{R} f^! K$, since an object of the derived category with only one non-zero cohomology sheaf is determined by that sheaf. Recall an important case where this condition is satisfied, with $K = \mathcal{O}$.

Proposition 7. *Let $f : X \to Y$ be compactifiable. Suppose that locally on X and on Y one can find a diagram*

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "g"'] & & Y \arrow[dl, "h"] \\ & S & \end{tikzcd}$$

where g and h are locally complete intersections of relative dimensions d' and d'', and let $d = d' - d''$.

Then $\mathbb{R}f^!\mathcal{O}$ is reduced to a single cohomology sheaf, situated in degree $-d$, and that sheaf is invertible.

Indeed, $\mathbb{R}g^!\mathcal{O} = \mathbb{R}f^!\,\mathbb{R}h^!\mathcal{O}$. We know, by local calculations in projective space, that $\mathbb{R}g^!\mathcal{O}$ and $\mathbb{R}h^!\mathcal{O}$ are of the indicated type, situated in degrees $-d'$ and $-d''$. Since everything is local, one may suppose that $\mathbb{R}^{-d''}h^!\mathcal{O}$ is isomorphic to $\mathcal{O}$, and the assertion follows.

To pass from this to more general complexes, one would first have to prove formulas of the form

$$\mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L) = \mathbb{R}f^!(K) \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L.$$

Only the definition of the arrow is problematic; its being an isomorphism is a local question. Of course boundedness restrictions are needed. I have checked none of the following in detail.

First method for defining the arrow.

Assume that K and L lie in $D^b_{\mathrm{coh}}(S)$, not merely in the quasi-coherent setting, and also that

(a) $\mathbb{R}f^!K$ is bounded, it being automatically coherent, which is a local problem;

(b) $K \overset{\mathbb{L}}{\otimes} L$ is bounded;

(c) $\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L$ is bounded.

To define

$$\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L \to \mathbb{R}f^!(K \overset{\mathbb{L}}{\otimes} L),$$

it is enough by duality to define

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L) \to K \overset{\mathbb{L}}{\otimes} L.$$

By Kunneth,

$$\mathbb{R}f_!(\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} \mathbb{L}f^*L) = \mathbb{R}f_!\,\mathbb{R}f^!K \overset{\mathbb{L}}{\otimes} L,$$

and the required arrow is induced by the adjunction arrow $\mathbb{R}f_!\,\mathbb{R}f^!K \to K$.

Second method.

Assume that f is compactifiable, flat, and locally a relative complete intersection. Then $\mathbb{R}f^!\mathcal{O} = \omega[d]$, where ω is invertible. Let $\mathbb{R}f_!$ denote a procedure for computing $\mathbb{R}f_!$ at the level of complexes, namely such a procedure for a compactification $\bar{f} : \bar{X} \to S$ of f, assumed to be of the type described in No. 4 and such that $C^p\mathcal{F}$ is zero for $p > d$, which can be arranged by truncation. Then there is an arrow

$$\mathbb{R}^d \bar{f}_* \mathcal{F}[-d] \to \mathbb{R}\bar{f}_*\mathcal{F} \quad \text{for every } \mathcal{F} \text{ on } \bar{X},$$

since the last cohomology sheaf maps to its complex. From this one obtains morphisms of complexes

$$\mathrm{Hom}_S(\mathbb{R}\bar{f}_*\mathcal{F}, I) \to \mathrm{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}[-d], I) \quad \text{for every complex } I \text{ on } S.$$

Let I be injective. The functor $\mathbb{R}^d\bar{f}_*\mathcal{F}[-d]$ is right exact in $\mathcal{F}$, so the functor in $\mathcal{F}$ on the right of the arrow is representable. Define I_1 by

$$\mathrm{Hom}_S(\mathbb{R}^d\bar{f}_*\mathcal{F}, I) = \mathrm{Hom}_{\bar{X}}(\mathcal{F}, I_1) \quad \text{and} \quad \mathbb{R}f_1^! I = I_1[d].$$

The preceding construction defines

$$\mathbb{R}f^! I \to \mathbb{R}f_1^! I,$$

and, for a complex of injectives, $\mathbb{R}f_1^!$ is defined term by term.

It remains to evaluate $\mathbb{R}f_1^! I$ on affine opens U of X. It is given by $\varinjlim \mathrm{Hom}(\mathbb{R}^d f_! \mathcal{J}^n, I)$. Since this problem is local upstairs, it is easy to show by projective methods that the dual of the ind-object

$$\text{“}\varprojlim\text{”}\, \mathbb{R}^d f_! \mathcal{J}^n = \mathbb{R}^d f_!(U, \mathcal{O})$$

is $f_*(U, \omega)$, represented as the limit of its submodules of sections having a pole of order at least n outside U, with $n \to \infty$.

This gives

$$\mathbb{R}f_1^! I = \omega \otimes f^* I[d],$$

and the general formula

$$\mathbb{R}f^! K = f^* K \otimes \omega[d].$$

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LEFSCHETZ THEOREM
AND DEGENERATION CRITERIA
FOR SPECTRAL SEQUENCES

by P. DELIGNE[1]

1. General theorems

Let $\mathcal{A}$ be an abelian category and let $D^b(\mathcal{A})$ be the full subcategory of its derived category formed by the bounded complexes. Recall that if T is a cohomological functor (see [8] or [9], p. 10) from $D^b(\mathcal{A})$ to an abelian category $\mathcal{B}$, Verdier [8] has defined a spectral sequence, functorial in $X \in D^b(\mathcal{A})$:

$$E_2^{pq} = T(H^q(X)[p]) \Longrightarrow T(X[p+q]). \tag{1.1}$$

Proposition (1.2). Let $X \in D^b(\mathcal{A})$. The following conditions are equivalent:

(i) for every T as above, the spectral sequence (1.1) degenerates;

(ii) the object X is isomorphic, in $D^b(\mathcal{A})$, to $\sum_i H^i(X)[-i]$.

(i) ⇒ (ii). Denote by T_i the cohomological functor in K defined by

$$T_i(K) = \mathrm{Hom}_{D^b(\mathcal{A})}(H^i(X), K).$$

For each of the T_i $(i \in \mathbb{Z})$, the spectral sequence (1.1) is written

$$E_2^{pq} = \mathrm{Ext}^p(H^i(X), H^q(K)) \Longrightarrow \mathrm{Hom}_{D^b(\mathcal{A})}(H^i(X), K[p+q]), \tag{1.3}$$

and the homomorphism from $\mathrm{Hom}(H^i(X), K[i])$ to

$$E_2^{0i} = \mathrm{Hom}(H^i(X), H^i(K))$$

which follows from it is none other than the one coming from the functoriality of H^0.

If the spectral sequences (1.3) degenerate, the maps

$$\mathrm{Hom}(H^i(X)[-i], K) \longrightarrow \mathrm{Hom}(H^i(X), H^i(K))$$

are surjective. By hypothesis, this is so for $K = X$; hence there exist morphisms a_i from $H^i(X)[-i]$ to X, inducing the identity on the i-th cohomology object. Moreover, since $H^i(X) = 0$ except for finitely many values of i, the a_i vanish except for finitely many i, which allows one to define

$$a = \sum_i a_i : \sum_i H^i(X)[-i] \longrightarrow X.$$

By construction, a induces the identity on cohomology, hence is an isomorphism.

(ii) ⇒ (i). The spectral sequence (1.1) is an additive functor in X, and degenerates when X has only one nonzero cohomology object; hence it also degenerates if X is a sum of complexes having only one nonzero cohomology object.

Remark (1.4). It follows from the preceding proof that it is enough to verify condition (i) for cohomological functors of the form $\mathrm{Hom}_{D^b(\mathcal{A})}(A, *)$, with $A \in \mathrm{Ob}(\mathcal{A})$. By duality, it would also be enough to consider only the functors $\mathrm{Hom}_{D^b(\mathcal{A})}(*, A)$, since condition (ii) is self-dual.

[1]Aspirant at the F.N.R.S.

Theorem (1.5). Let $X \in D^b(\mathcal{A})$, let n be an integer, and let u be an endomorphism of degree 2 of X,

$$u : X \longrightarrow X[2],$$

whose iterates induce isomorphisms

$$u^i : H^{n-i}(X) \xrightarrow{\sim} H^{n+i}(X) \qquad (i \geq 0).$$

Under these conditions, X is isomorphic to $\sum_i H^i(X)[-i]$ in the category $D^b(\mathcal{A})$.

We verify condition (i) of (1.2). Let T be a cohomological functor from $D^b(\mathcal{A})$ to $\mathcal{B}$, and put $T^p(K) = T(K[p])$.

The primitive part of $H^{n-i}(X)$, denoted ${}_0H^{n-i}(X)$, is the kernel of the homomorphism

$$u^{i+1} : H^{n-i}(X) \longrightarrow H^{n+i+2}(X).$$

One checks that, for $i \geq 0$, the maps

$$\begin{aligned} \bigoplus_{k\geq 0} u^k : \bigoplus_{k\geq 0} {}_0H^{n-i-2k}(X) &\xrightarrow{\sim} H^{n-i}(X), \\ \bigoplus_{k\geq 0} u^{k+i} : \bigoplus_{k\geq 0} {}_0H^{n-i-2k}(X) &\xrightarrow{\sim} H^{n+i}(X) \end{aligned} \tag{1.6}$$

are isomorphisms.

Since the spectral sequence (1.1) is functorial in X and compatible with translations of degrees, the endomorphism u of X defines an endomorphism of degree 2 of the spectral sequence,

$$u : E_r^{pq} \longrightarrow E_r^{p,q+2},$$

commuting with the differentials d_r $(r \geq 2)$.

When $r = 2$, the endomorphism u of the spectral sequence

$$u : T^p(H^q(X)) \longrightarrow T^p(H^{q+2}(X))$$

is obtained from the morphism $u : H^q(X) \to H^{q+2}(X)$ by functoriality of T^p.

Let $i \geq 0$. Denote by ${}_0E_r^{p,n-i}$ the kernel of the homomorphism

$$u^{i+1} : E_r^{p,n-i} \longrightarrow E_r^{p,n+i+2}$$

(the primitive part of the spectral sequence). For $r = 2$, the decompositions (1.6) induce, by functoriality of each T^p, decompositions

$$\begin{aligned} \bigoplus_{k\geq 0} u^k : \bigoplus_{k\geq 0} {}_0E_2^{p,n-i-2k} &\xrightarrow{\sim} E_2^{p,n-i}, \\ \bigoplus_{k\geq 0} u^{k+i} : \bigoplus_{k\geq 0} {}_0E_2^{p,n-i-2k} &\xrightarrow{\sim} E_2^{p,n+i}. \end{aligned} \tag{1.7}$$

We prove by induction on r that the differentials d_r $(r \geq 2)$ are zero. The induction hypothesis implies that $E_2 = E_r$, so that the decomposition (1.7) applies to the term E_r of the spectral sequence. Since u commutes with the d_r, it will be enough to prove that d_r vanishes on the primitive part of E_r.

The following diagram is commutative:

$$\begin{tikzcd} {}_0E_r^{p,n-i} \arrow[r, "d_r"] \arrow[d, "u^{i+1}"] & E_r^{p+r,n-i-r+1} \arrow[d, "u^{i+1}"] \\ E_r^{p,n+i+2} \arrow[r, "d_r"] & E_r^{p+r,n+i-r+3}. \end{tikzcd}$$

The left vertical arrow is zero by definition of primitivity. The right vertical arrow is a monomorphism, because the arrow

$$u^{i+r-1} : E_r^{p+r,n-(i+r-1)} \longrightarrow E_r^{p+r,n+(i+r-1)}$$

is an isomorphism[2] and $r - 1 \geq 1$. It follows that d_r is zero.

[2]The printed source gives $E_r^{p,n+(i+r-1)}$ as the target; the index $p + r$, required by the bidegree of d_r, is restored here.

Remark (1.8). One should take care that in general one cannot choose an isomorphism between X and $\sum_i H^i(X)[-i]$ which transforms u into the sum of the maps $H^i(u) : H^i(X) \to H^{i+2}(X)$. One easily obtains a counterexample by taking $H^i(X) = 0$ for $i \neq n$.

One can define an isomorphism, "more beautiful than the others", between X and $\sum_i H^i(X)[-i]$, but it will not be used here.

Remark (1.9). Let $X \in D^b(\mathcal{A})$, let n and s be integers, and let u be an endomorphism of degree s (respectively $2s$) of X. Suppose that $H^i(X) = 0$ when i is not of the form ks $(0 \leq k \leq n)$ (respectively $0 \leq k \leq 2n$), and that u^{n-2i} (respectively u^i) induces an isomorphism between $H^{is}(X)$ and $H^{(n-i)s}(X)$ (respectively between $H^{(n-i)s}(X)$ and $H^{(n+i)s}(X)$). Under these hypotheses, the proof of (1.5) still shows that X is isomorphic to $\sum_i H^i(X)[-i]$.

Remark (1.10). Let $(X_i)_{i\in\mathbb{Z}}$ be a family of objects of $D^b(\mathcal{A})$ and let u be a family of degree-2 homomorphisms $u_i : X_i \to X_{i+1}[2]$. If, for all i and j, u_i induces an isomorphism between $H^{n-j}(X_i)$ and $H^{n+j}(X_{i+j})$, it is still true that each X_i is isomorphic to the sum of its cohomology objects. To see this, it is enough to apply the preceding arguments formally to

$$u : \sum_i X_i \longrightarrow \sum_i X_i,$$

even if this sum does not exist in $D^b(\mathcal{A})$.

The same remark applies to the variant (1.9).

Theorem (1.11). Let $X \in D^b(\mathcal{A})$ and suppose that X admits degree-0 endomorphisms $\pi_i : X \to X$ such that $H^j(\pi_i) = \delta_{ij}$. Then X is isomorphic to $\sum_i H^i(X)[-i]$ in the category $D^b(\mathcal{A})$.

We again apply criterion (1.2) (i). The diagram

$$\begin{tikzcd} E_r^{p,q} \arrow[r, "d_r"] \arrow[d, "\pi_q"] & E_r^{p+r,q-r+1} \arrow[d, "\pi_q"] \\ E_r^{p,q} \arrow[r, "d_r"] & E_r^{p+r,q-r+1} \end{tikzcd}$$

is commutative; the first column is the identity and the second is zero, so that $d_r = 0$.

Corollary (1.12). Let $X \in D^b(\mathcal{A})$ be the sum of a finite family of objects $X_k \in D^b(\mathcal{A})$. If

$$X \simeq \sum_i H^i(X)[-i],$$

then, for every k,

$$X_k \simeq \sum_i H^i(X_k)[-i].$$

Let j_k be the canonical injection of X_k into X, let pr_k be the canonical projection of X onto X_k, and let π_i be an endomorphism of X such that $H^j(\pi_i) = \delta_{ij}$. For every k, the $\mathrm{pr}_k\, \pi_i j_k$ satisfy the hypothesis of (1.11).

Remark (1.13). One checks that, in order that the isomorphism between X and $\sum_i H^i(X)[-i]$ can be chosen so that the π_i are identified with the pr_i, it is necessary and sufficient that the π_i be pairwise orthogonal idempotents. There is then a unique isomorphism with this property and inducing the identity on cohomology.

2. Applications

Let $f : X \to Y$ be a continuous map between topological spaces, or even a morphism of topoi, let A be a sheaf of rings on X, and let $\mathcal{F}$ be a sheaf of A-modules. If $u \in H^2(X, A)$, then u defines a degree-2 endomorphism in $D^b(X)$, again denoted

$$u : \mathcal{F} \longrightarrow \mathcal{F}[2].$$

By functoriality, u defines a degree-2 endomorphism

$$Rf_*(u) : Rf_*\mathcal{F} \longrightarrow Rf_*\mathcal{F}[2].$$

We shall say that $(X, f, u, \mathcal{F}, n)$ satisfies the Lefschetz condition, or that $\mathcal{F}$ satisfies the Lefschetz condition relative to u, if, for every $i \geq 0$, the maps obtained by iterating i times the arrow $Rf_*(u)$,

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F} \longrightarrow R^{n+i}f_*\mathcal{F} \qquad (i \geq 0),$$

are isomorphisms. Theorem (1.5) gives here:

Proposition (2.1). If $\mathcal{F}$ satisfies the Lefschetz condition relative to u, then, in $D^b(Y)$,

$$Rf_*\mathcal{F} \simeq \sum_i R^i f_*\mathcal{F}[-i]$$

and in particular the Leray spectral sequence

$$H^p(Y, R^q f_*\mathcal{F}) \Longrightarrow H^{p+q}(X, \mathcal{F})$$

degenerates.

If one had wished, one could have supposed that $\mathcal{F}$ was an (A, f^*B)-bimodule, where B is a sheaf of rings on Y, and stated (2.1) in $D^b(Y, B)$.

(2.2) The case of schemes, endowed with the étale topology, does not enter into the framework of (2.1). In what follows we shall suppose that the theory of $\mathbb{Z}_\ell$-sheaves and $\mathbb{Q}_\ell$-sheaves (not necessarily constructible) has been made, and that the derived category of the category of these "sheaves" is reasonable. As far as the degeneration statements for spectral sequences are concerned, the reader who has no confidence in Jouanolou's future work may verify, by returning to the proof of (1.5), that one can dispense with it. Let ℓ be a prime number. Throughout what follows, assume that ℓ is invertible on all schemes under consideration.

Recall that on every scheme X the $\mathbb{Z}_\ell$-sheaf $\mathbb{Z}_\ell(1)$ is defined by the formula

$$\mathbb{Z}_\ell(1) = \varprojlim_n \mu_{\ell^n}.$$

This sheaf is an invertible $\mathbb{Z}_\ell$-module, which permits one to define, for every $\mathbb{Z}_\ell$-sheaf or $\mathbb{Q}_\ell$-sheaf $\mathcal{F}$, the "sheaf" $\mathcal{F}(i)$ by the formula

$$\mathcal{F}(i) = \mathcal{F} \otimes_{\mathbb{Z}_\ell} \mathbb{Z}_\ell(1)^{\otimes i} \qquad (i \in \mathbb{Z}).$$

For every morphism $f : X \to Y$ of schemes and every pair of $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaves $\mathcal{F}$ and $\mathcal{G}$ on X and Y respectively, one has

$$Rf_*(\mathcal{F})(i) = Rf_*(\mathcal{F}(i)) \qquad \text{and} \qquad f^*(\mathcal{G}(i)) = f^*(\mathcal{G})(i). \tag{2.3}$$

If $\mathcal{L}$ is an invertible sheaf on X, its first ℓ-adic Chern class lies in $H^2(X, \mathbb{Z}_\ell(1))$.

If $f : X \to Y$ is a morphism of schemes, if $\mathcal{F}$ is a $\mathbb{Z}_\ell$-sheaf, respectively a $\mathbb{Q}_\ell$-sheaf, on X, and if $u \in H^2(X, \mathbb{Z}_\ell(1))$, respectively $u \in H^2(X, \mathbb{Q}_\ell(1))$, then $\mathcal{F}$ will be said to satisfy the Lefschetz condition relative to u if, for every $i \geq 0$, the maps obtained by iterating i times the arrow $Rf_*(u)$,

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F} \longrightarrow R^{n+i}f_*\mathcal{F}(i) \qquad (i \geq 0),$$

are isomorphisms. By (2.3), this implies that all the arrows

$$Rf_*(u)^i : R^{n-i}f_*\mathcal{F}(k) \longrightarrow R^{n+i}f_*\mathcal{F}(k+i)$$

are isomorphisms. Applying (1.10), one obtains:

Proposition (2.4). With the preceding notation, if $\mathcal{F}$ satisfies the Lefschetz condition relative to u, then, in $D^b(Y, \mathbb{Z}_\ell)$, respectively $D^b(Y, \mathbb{Q}_\ell)$, one has

$$Rf_*\mathcal{F} \simeq \sum_i R^i f_*\mathcal{F}[-i] \tag{2.5}$$

and in particular the Leray spectral sequence

$$H^p(Y, R^q f_*\mathcal{F}) \Longrightarrow H^{p+q}(X, \mathcal{F})$$

degenerates.

If g is a morphism of schemes from Y to S, (2.4) also implies the degeneration of the Leray spectral sequence

$$R^p g_* R^q f_*\mathcal{F} \Longrightarrow R^{p+q}(gf)_*\mathcal{F}.$$

(2.6.1) It remains to review a few interesting cases in which the Lefschetz condition is satisfied; the justifying details are given in (2.7). Let $f : X \to Y$, u, $\mathcal{F}$, and n be given. In the topological case, it is known from Godement [2], II, (4.7.1), that if f is proper and Y locally paracompact, the formation of higher direct images commutes with passage to fibers, so that the Lefschetz condition is checked fiber by fiber. In the case of schemes, the same reduction applies, the reference to Godement being replaced by SGA 4, XII, (5.1). Still in the case of schemes, if f is proper and smooth and $\mathcal{F}$ is twisted constant, that is, smooth in a newer terminology, then the $R^i f_*\mathcal{F}$ are still twisted constant (cf. SGA 4, XVI, (2.2)); hence, if Y is connected, it is even enough to verify the Lefschetz condition on one geometric fiber.

In all the examples given, f will be a proper smooth continuous map, respectively a proper smooth morphism of schemes, of relative topological dimension $2n$, respectively algebraic dimension n. A continuous map is called smooth if locally on the source it is isomorphic to the projection from $\mathrm{R}^n \times Y$ to Y.

(2.6.2) If f is a proper smooth morphism of Kähler varieties, the constant sheaf $\mathbb{C}$ satisfies the Lefschetz condition relative to the fundamental 2-class of X.

(2.6.3) If X is a complex relative scheme (see [3]), projective and smooth over the locally paracompact topological space Y, and endowed with a relatively ample invertible sheaf $\mathcal{O}(1)$, the constant sheaf $\mathbb{C}$ satisfies the Lefschetz condition relative to the first Chern class of $\mathcal{O}(1)$.

(2.6.4) Suppose that f is a projective smooth morphism of schemes, with Y connected, and that a relatively ample invertible sheaf $\mathcal{O}(1)$ on X is given. If one can lift to characteristic 0 a geometric fiber of f, and its polarization, then the constant $\mathbb{Q}_\ell$-adic sheaf $\mathbb{Q}_\ell$ satisfies the Lefschetz condition relative to the ℓ-adic first Chern class of $\mathcal{O}(1)$.

It is known that one can lift to characteristic 0 in the preceding sense in the following cases:

- X is a Severi–Brauer variety over Y;
- X is a nonsingular complete intersection of hypersurfaces in a Severi–Brauer variety over Y, for the polarization induced by the canonical polarization of the Severi–Brauer variety;
- X is a polarized abelian scheme over Y (Mumford, to appear).

It is conjectured that (2.6.4) remains true when no lifting to characteristic 0 is known; this is proved only for surfaces.

(2.6.5) If f is a projective smooth morphism of relative dimension 2 and if X is endowed with a relatively ample invertible sheaf $\mathcal{O}(1)$, then the constant sheaf $\mathbb{Q}_\ell$ satisfies the Lefschetz condition relative to the first Chern class of $\mathcal{O}(1)$.

(2.7) In cases (2.6.2) and (2.6.3), the given cohomology 2-class on X induces on the fibers of f the 2-class defined by the Kähler structure of the fibers. Thus, by (2.6.1), to prove (2.6.2) and (2.6.3) it suffices to prove (2.6.2) in the special case where Y is a point. The assertion then follows from Hodge theory and the Hodge–Lepage decomposition (Weil [10], IV, no. 6, corollary to theorem 5). The proof given in loc. cit. still applies if $\mathcal{F}$ is a locally constant sheaf of finite-dimensional $\mathbb{C}$-vector spaces, endowed with a locally constant positive definite Hermitian structure.

By (2.6.1), to verify (2.6.4) it suffices to treat the case where Y is the spectrum of a field of characteristic 0, and the Lefschetz principle allows one to suppose $Y = \mathrm{Spec}(\mathbb{C})$.

It is then known (cf. SGA 4, XI, (4.4)) that, for every twisted constant $\mathbb{Q}_\ell$-adic sheaf $\mathcal{F}$ on X, if $\mathcal{F}^{an}$ denotes the locally constant sheaf of $\mathbb{Q}_\ell$-vector spaces on the topological space X^{an} deduced from $\mathcal{F}$, then the algebraic and transcendental cohomology $\mathbb{Q}_\ell$-vector spaces of $\mathcal{F}$ and $\mathcal{F}^{an}$ are isomorphic:

$$H^i(X, \mathcal{F}) \xrightarrow{\sim} H^i(X^{an}, \mathcal{F}^{an}). \tag{2.7.2}$$

Choosing an embedding of $\mathbb{Q}_\ell$ into $\mathbb{C}$, one obtains isomorphisms

$$H^i(X, \mathcal{F}) \otimes_{\mathbb{Q}_\ell} \mathbb{C} \xrightarrow{\sim} H^i(X^{an}, \mathcal{F}^{an}) \otimes_{\mathbb{Q}_\ell} \mathbb{C}. \tag{2.7.3}$$

Over $\mathbb{C}$, the exponential map defines the so-called canonical isomorphism between $\mathbb{Z}/\ell^n$ and μ_{ℓ^n}, hence between $\mathbb{Z}_\ell$ and $\mathbb{Z}_\ell(1)$. The image, under the composite isomorphism

$$H^2(X, \mathbb{Z}_\ell(1)) \longrightarrow H^2(X, \mathbb{Z}_\ell) \longrightarrow H^2(X^{an}, \mathbb{Z}) \otimes \mathbb{Z}_\ell,$$

of the ℓ-adic first Chern class of $\mathcal{O}(1)$ coincides with the transcendental first Chern class of $\mathcal{O}(1)$. The isomorphism (2.7.3) therefore reduces the case of (2.6.4) to which we had come to (2.6.3).

For the proof of (2.6.5), see Kleiman [6], pp. 28 ff. The essential point is the following theorem (Weil [11], p. 127).

(2.8) Let S be a smooth projective surface over a field k, and let D be a smooth curve on S such that $\mathcal{O}(D)$ is ample. Then the composite map

$$\mathrm{Pic}^0(S) \longrightarrow \mathrm{Pic}^0(D) = \mathrm{Alb}(D) \longrightarrow \mathrm{Alb}(S)$$

is an isogeny.[3]

Remark (2.9). The degeneration theorem for the spectral sequence (2.2) deduced from (2.6.4) was conjectured by Grothendieck, from considerations of "weights", when Y is projective and smooth over an algebraically closed field.

Remark (2.10). Serre pointed out to me that the degeneration theorems for spectral sequences deduced from (2.6.2) and (2.6.3) can also be proved by an easy extension of the method used by Blanchard ([1], II, 1), at least if Poincaré duality is available on the base. Conversely, the method followed here completes Theorems II (1.1) and II (1.2) of loc. cit.; the proof of the implication (ii) ⇒ (i) is essentially Blanchard's, who had to suppose the higher direct-image sheaves constant.

Theorem (2.11). Let $f : X \to Y$ be a proper smooth continuous map (2.6.1) of topological spaces of relative dimension $2n$, with Y locally paracompact, let k be a commutative field of characteristic zero, and let $u \in H^0(Y, R^2 f_* k)$. Suppose that:

a) the sheaf $R^1 f_* k$ is constant, i.e. simple;

b) the iterated cup products
$$u^i \wedge : R^{n-i} f_* k \longrightarrow R^{n+i} f_* k$$
are isomorphisms.

Then the following conditions are equivalent:

(i) the class u is induced by an element of $H^2(X, k)$;

(ii) the transgression
$$d_2 : H^0(Y, R^2 f_* k) \longrightarrow H^2(Y, f_* k)$$
is zero on u;

[3] The printed source gives $\mathrm{Pic}^0(X)$ and $\mathrm{Alb}(X)$ here, although the surface introduced in the statement is S; S is restored.

(iii) in $\mathrm{D}^b(Y,k)$, one has

$$Rf_*k \simeq \bigoplus_i R^i f_*k[-i].$$

The implication (i) $\Rightarrow$ (iii) is contained in (2.1). Assertion (iii) implies the degeneration of the whole Leray spectral sequence

$$H^p(Y, R^q f_*k) \Longrightarrow H^{p+q}(X,k), \tag{2.12}$$

and a fortiori (ii). It remains to prove that (ii) implies (i).

Let $x \in Y$ and $X_x = f^{-1}(x)$. The induced class $u_x^n \in H^{2n}(X_x,k)$ is nonzero on every connected component of X_x, which is therefore orientable.[4] Let Tr_u denote the composite map

$$H^{2n}(X_x,k) \xrightarrow{(u_x^n\wedge)^{-1}} H^0(X_x,k) = H_0(X_x,k) \xrightarrow{\mathrm{Tr}} k.$$

Poincaré duality implies that the bilinear form

$$H^{n-i}(X_x,k) \times H^{n+i}(X_x,k) \longrightarrow k, \qquad (x,y) \longmapsto \mathrm{Tr}_u(xy)$$

is a perfect duality. This construction globalizes and gives

$$\mathrm{Tr}_u : R^{2n} f_*k \longrightarrow k.$$

First prove that in (2.12), $d_2 u \in H^2(Y, R^1 f_*k)$ is zero. If $x \in H^0(Y, R^1 f_*k)$, then for degree reasons $xu^n = 0$, hence

$$d_2(xu^n) = d_2x\, u^n - nxu^{n-1} d_2 u = 0.$$

By b), the map

$$u^{n-1}\wedge : H^0(Y, R^1 f_*k) \longrightarrow H^0(Y, R^{2n-1} f_*k)$$

is an isomorphism. Thus, if (ii) holds, for every $y \in H^0(Y, R^{2n-1} f_*k)$ one has

$$y\, d_2 u = 0,$$

and in particular

$$\mathrm{Tr}_u(y\, d_2 u) = 0. \tag{2.13}$$

Let V be a k-vector space such that $R^1 f_*k$ is isomorphic to the constant sheaf V. Then $d_2 u \in H^2(Y,V)$ and, by (2.13), for every linear form w on V, the image of $d_2 u$ under w in $H^2(Y,k)$ is zero. Thus $d_2 u = 0$.

If $d_2 u = 0$, the proof of (1.5) shows that all differentials d_2 of (2.12) are zero, so that $E_2 = E_3$. For degree reasons, $u^{n+1} = 0$, hence

$$d_3 u^{n+1} = (n+1) u^n d_3 u = 0.$$

Since $d_3 u \in H^3(Y, R^0 f_*k)$, b) implies that $d_3 u = 0$. For degree reasons $d_r u = 0$ for $r > 3$, and u therefore comes from an element of $H^2(X,k)$.

(2.14) In the scheme-theoretic setting, the preceding result remains valid, but it is usable only in the "geometric" case, more precisely when $\mathbb{Q}_\ell$ is isomorphic to $\mathbb{Q}_\ell(1)$ on Y.

(2.15) Corollary (1.12) allows some of the preceding statements to be extended to certain nonconstant sheaves.

Proposition (2.16). Let g and f be two composable continuous maps, respectively two morphisms of schemes,

$$X \xrightarrow{g} Y \xrightarrow{f} Z,$$

and let $\mathcal{F}$ be a sheaf, respectively a $\mathbb{Z}_\ell$-sheaf or $\mathbb{Q}_\ell$-sheaf, on X. Suppose that

a) $Rg_*\mathcal{F} \in \mathrm{D}^b(Y)$ and $Rg_*\mathcal{F} \simeq \bigoplus_i R^i g_*\mathcal{F}[-i]$;

b) $R(fg)_*\mathcal{F} \in \mathrm{D}^b(Z)$ and $R(fg)_*\mathcal{F} \simeq \bigoplus_i R^i (fg)_*\mathcal{F}[-i]$.

[4] After introducing x, the printed source switches to X_s and u_s^n; the subscript is regularized to x here.

Then, for every k, one has
$$Rf_*(R^k g_*\mathcal{F}) \simeq \bigoplus_i R^i f_*(R^k g_*\mathcal{F})[-i].$$
By b), the complex
$$R(fg)_*\mathcal{F} = Rf_* Rg_*\mathcal{F} \simeq \bigoplus_k Rf_*(R^k g_*\mathcal{F}[-k])$$
is a sum of its cohomology objects; by (1.12), the complexes $Rf_* R^k g_*\mathcal{F}$, which up to a shift are direct factors thanks to a), have the same property.

Proposition (2.17). Let $\mathcal{E}$ be a locally free vector bundle on a scheme S, let $\mathcal{F}$ be a twisted constant ℓ-adic sheaf on $\mathbb{P}(\mathcal{E})$, and let f be the projection from $\mathbb{P}(\mathcal{E})$ to S. In the derived category, one has
$$Rf_*\mathcal{F} \simeq \bigoplus_i R^i f_*\mathcal{F}[-i].$$
Projective space is simply connected and has torsion-free cohomology (SGA 1, XI, 1.1 and SGA 4, XVI, 2.2). Hence
$$\mathcal{F} = f^* f_*\mathcal{F}, \qquad Rf_*\mathcal{F} = Rf_*\mathbb{Z}_\ell \overset{\mathrm{L}}{\otimes} f_*\mathcal{F},$$
which reduces the proof to the case $\mathcal{F} = \mathbb{Z}_\ell$. Let u be the Chern class of the invertible sheaf $\mathcal{O}(1)$ on $\mathbb{P}(\mathcal{E})$. Then $\mathbb{Z}_\ell$ is known to satisfy the Lefschetz condition relative to u, and (2.2) concludes the proof.

(2.18) From (2.17), with the help of (2.16), one obtains the same result for flag bundles of all kinds defined by $\mathcal{E}$. In the topological setting, the preceding applies directly to vector bundles over $\mathbb{C}$, and extends to vector bundles over $\mathbb{H}$ by (1.9). For real vector bundles, one must restrict to the case where $\mathcal{F}$ comes from a sheaf of 2-torsion on the base.

Remark (2.19). Let X be an abelian scheme of relative dimension g over a base Y. For every integer n, multiplication by n, denoted $[n]$, defines an endomorphism
$$[n]^* : Rf_*\mathbb{Q}_\ell \longrightarrow Rf_*\mathbb{Q}_\ell.$$
Taking rational linear combinations, with denominators bounded in terms of g, of these endomorphisms, one constructs endomorphisms
$$\pi_i : Rf_*\mathbb{Q}_\ell \longrightarrow Rf_*\mathbb{Q}_\ell$$
such that $H^j(\pi_i) = \delta_{ij}$. Applying (1.11), one concludes, without any projectivity hypothesis, that
$$Rf_*\mathbb{Q}_\ell \simeq \bigoplus_i R^i f_*\mathbb{Q}_\ell[-i].$$
The idea used here is due to Lieberman.

3. The infinitesimal form of the semicontinuity theorem

I recall in this paragraph some results which one manages to find in EGA III, ğ 7. The formulation (3.4) was pointed out to me by L. Illusie.

(3.0) Throughout this paragraph, A denotes a fixed noetherian local ring, $\mathfrak{m}$ its maximal ideal, and k its residue field. Let $D^-_{\mathrm{coh}}(A)$ denote the subcategory of the derived category of A-modules formed by bounded-above complexes of finite-type A-modules.

Proposition (3.1). Let M be a finite-type module over A.

(i) For every homomorphism, not necessarily local, from A to an Artinian ring B, and every finite-type B-module N, one has

$$\lg_B(M \otimes_A N) \leq \lg_B(N) \cdot \dim_k(M \otimes_A k). \tag{3.1.1}$$

(ii) The following conditions on M are equivalent:

a) M is a free module;

b) for all B and N as in (i), one has

$$\lg_B(M \otimes_A N) = \lg_B(N) \cdot \dim_k(M \otimes_A k); \tag{3.1.2}$$

c) for every $n \in \mathbb{N}$, condition (3.1.2) is satisfied for $B = N = A/\mathfrak{m}^n$;

d) for every $\mathfrak{p} \in \operatorname{Ass}(A)$, $M_\mathfrak{p}$ is flat over $A_\mathfrak{p}$, and condition (3.1.2) is satisfied for $B = N = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$.

If A has no embedded components, these conditions are also equivalent to:

e) for every maximal point $\mathfrak{p}$ of $\operatorname{Spec}(A)$ one has

$$\lg_{A_\mathfrak{p}}(M_\mathfrak{p}) = \lg(A_\mathfrak{p}) \cdot \dim_k(M \otimes_A k). \tag{3.1.2'}$$

The proof, an application of Nakayama's lemma, is left to the reader.

(3.2) For every complex K and every integer n, let $\tau_{\geq n}(K)$ denote the complex defined by

$$\tau_{\geq n}(K)^i = \begin{cases} 0, & i < n, \\ \operatorname{coker}(d^{n-1}), & i = n, \\ K^i, & i > n, \end{cases} \tag{3.2.1}$$

so that

$$H^i(\tau_{\geq n}(K)) = \begin{cases} 0, & i < n, \\ H^i(K), & i \geq n. \end{cases} \tag{3.2.2}$$

Let K be an object of $D^-_{\mathrm{coh}}(A)$ (cf. (3.0)). Replacing K, if necessary, by an isomorphic complex in the derived category, one may suppose the components of K to be free of finite type.

For all B and N as in (3.1) (i), one has at the level of complexes

$$\tau_{\geq n}(K \otimes^L_A N) \simeq \tau_{\geq n}(K) \otimes_A N.$$

Taking the Euler–Poincaré characteristics of the two sides gives

$$\sum_{i \geq 0} (-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) = \lg_B(\operatorname{coker}(d^{n-1}) \otimes_A N) + \sum_{i > 0} (-1)^i \lg_B(K^{n+i} \otimes_A N).$$

Subtract from this identity the analogous identity for $B = N = k$, multiplied by $\lg_B(N)$. By (3.1) (ii), $a \Leftrightarrow b$, and since the K^i are free, one obtains

$$\begin{aligned} &\sum_{i \geq 0} (-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) - \lg_B(N) \sum_{i \geq 0} (-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big) \\ &\quad = \lg_B(\operatorname{coker}(d^{n-1}) \otimes_A N) - \lg_B(N) \dim_k(\operatorname{coker}(d^{n-1}) \otimes_A k), \end{aligned} \tag{3.2.3}$$

which allows one to apply (3.1) to the first member of (3.2.3).

Theorem (3.3) (exchange and semicontinuity theorems). Let $K \in \operatorname{Ob} D^-_{\mathrm{coh}}(A)$ (cf. (3.0)) and $n \in \mathbb{Z}$.

(i) For every homomorphism from A to an Artinian ring B and every finite-type B-module N, one has

$$\sum_{i\geq 0}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) \leq \lg_B(N)\sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.1}$$

(ii) The following conditions on K and n are equivalent:

a) The cohomological functor in the A-module N, $H^*(K \otimes^L_A N)$, satisfies the exchange property in degree $* = n$, i.e. the following equivalent conditions on K and n are satisfied:

a.1) the functor $H^n(K \otimes^L_A N)$ is left exact;
a.2) there is a module Q such that this functor is isomorphic to $\operatorname{Hom}_A(Q, N)$;
a.3) the functor $H^{n-1}(K \otimes^L_A N)$ is right exact;
a.4) there is a module P such that this functor is isomorphic to $P \otimes_A N$;
a.5) the map $H^{n-1}(K) \to H^{n-1}(K \otimes^L_A k)$ is surjective;
a.6) there exists a bounded-above complex K', quasi-isomorphic to K and with flat components, respectively free components of finite type, such that $d'^{\,n-1} = 0$.

b) For all B and N as in (i), one has

$$\sum_{i\geq 0}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) = \lg_B(N)\sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.2}$$

If A has no embedded components, these conditions are also equivalent to:

c) for every maximal point $\mathfrak{p}$ of $\operatorname{Spec}(A)$ one has

$$\sum_{i\geq 0}(-1)^i \lg_{A_\mathfrak{p}} H^{n+i}(K_\mathfrak{p}) = \lg(A_\mathfrak{p})\sum_{i\geq 0}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.3.2'}$$

Remark (3.3.3). One could have lengthened the list by adding the conditions analogous to (3.1) (ii) c) and d).

Assertion (i) follows from (3.2.3) and (3.1) (i). By EGA III, (7.4.2), and its proof, conditions a.1), a.3), and a.6) are equivalent and, if K is represented by a bounded-above complex of finite-type free modules, they also mean that $\operatorname{coker}(d^{n-1})$ is free; taking (3.1) (ii) $a \Leftrightarrow b \Leftrightarrow e$ and (3.2.3) into account proves the equivalence of a.1), a.3), a.6), b), and c). Trivially, a.6) $\Rightarrow$ a.2) $\Rightarrow$ a.1) and a.6) $\Rightarrow$ a.4) $\Rightarrow$ a.3). The equivalence of condition a.5) with the others is contained in EGA III, (7.5.2).

Corollary (3.4). Let A, K, and n be as in (3.3), and let $k \in \mathbb{N}$.

(i) For all B and N as in (3.3) (i), one has

$$\sum_{0\leq i\leq 2k}(-1)^i \lg_B\big(H^{n+i}(K \otimes^L_A N)\big) \leq \lg_B(N)\sum_{0\leq i\leq 2k}(-1)^i \dim_k\big(H^{n+i}(K \otimes^L_A k)\big). \tag{3.4.1}$$

(ii) The following conditions on K, n, and k are equivalent:

a) The cohomological functor in N, $H^*(K \otimes^L_A N)$, satisfies the exchange property in degrees $* = n$ and $* = n + 2k + 1$.

a' The complex K is quasi-isomorphic to a bounded-above complex K' with flat components, respectively free components of finite type, such that $d'^{\,n-1} = d'^{\,n+2k} = 0$.

b) For all B and N as in (i), equality holds in (3.4.1).

If A has no embedded components, these conditions are also equivalent to:

c) for every maximal point $\mathfrak{p}$ of $\operatorname{Spec}(A)$, equality holds in (3.4.1) for $B = N = A_\mathfrak{p}$.

(3.4.2) For $k = 0$, these are properties only of the single hypertor $H^n(K \otimes^L_A N)$, considered as a functor in the A-module N. They can also be expressed by saying that $H^n(K)$ is free and that, for all N, the canonical map from $H^n(K) \otimes_A N$ to $H^n(K \otimes^L_A N)$ is an isomorphism.

(3.4.3) On the other hand, if K is a perfect complex, one deduces from (3.4), for $n \ll 0$, a result analogous to (3.3), with the summation over $i \geq 0$ in (3.3.1, 2, 2') replaced by a summation over $i \leq 0$.

Assertion (i) and the equivalence of a), b), and c) follow from (3.3) and from the identity

$$\sum_{0\leq i\leq 2k}(-1)^i x_i = \sum_{0\leq i}(-1)^i x_i + \sum_{0\leq i}(-1)^i x_{i+2k+1}.$$

To prove a'), one successively applies (3.3) (ii) $a \Leftrightarrow a.6$) to (K, n) and to $(\tau_{\geq n}(K), n+2k+1)$.

(3.5) When A is Artinian, (3.4) gives, for $k = 0$, the inequality

$$\lg_A H^n(K) \leq \lg(A)\dim_k\bigl(H^n(K\otimes^L_A k)\bigr), \tag{3.5.1}$$

and, if equality holds in (3.5.1), then the exchange condition is satisfied in degrees n and $n+1$, and $H^n(K)$ is a free A-module.

4. The Gysin morphism in Hodge cohomology

This paragraph is written from notes of A. Grothendieck.

(4.0) Let $f : X \to Y$ be a proper morphism between schemes X and Y which are smooth and purely of relative dimensions n and m over a base S. We suppose that S is noetherian and has a dualizing complex, in order to refer to [4] for the definition of $\mathrm{R}f^!$; this restriction is probably unnecessary. Finally, put $d = n - m$.

(4.1) Every relative differential form on Y defines, by inverse image, a relative differential form on X, hence a morphism

$$f^*\Omega^p_{Y/S} = Lf^*\Omega^p_{Y/S} \longrightarrow \Omega^p_{X/S}. \tag{4.1.1}$$

By the duality theory for smooth morphisms and the transitivity of $\mathrm{R}f^!$, one has canonically

$$\mathrm{R}f^!\Omega^m_{Y/S} = \Omega^n_{X/S}[d]. \tag{4.1.2}$$

Apply to $\Omega^p_{Y/S}$ and $\Omega^m_{Y/S}$ the induction formula

$$\mathrm{R}f^!\,\mathrm{RHom}(K, L) = \mathrm{RHom}(Lf^*K, \mathrm{R}f^!L),$$

which is obtained readily from the definition of $\mathrm{R}f^!$ by biduality (Hartshorne [4], chap. VII, ğ 3). Since

$$\mathrm{RHom}(\Omega^p_{Y/S}, \Omega^m_{Y/S}) \simeq \Omega^{m-p}_{Y/S},$$

one obtains

$$\mathrm{R}f^!\Omega^{m-p}_{Y/S} = \mathrm{RHom}(Lf^*\Omega^p_{Y/S}, \Omega^n_{X/S})[d].$$

The arrow (4.1.1) therefore defines, by transposition, an arrow[5]

$$\Omega^{n-p}_{X/S} = \mathrm{RHom}(\Omega^p_{X/S}, \Omega^n_{X/S}) \longrightarrow \mathrm{R}f^!\Omega^{m-p}_{Y/S}[d]. \tag{4.1.3}$$

By duality theory, giving an arrow (4.1.3) amounts to giving an arrow

$$\mathrm{R}f_*\Omega^{n-p}_{X/S} \longrightarrow \Omega^{m-p}_{Y/S}[-d]. \tag{4.1.4}$$

Definition (4.2). Let X, Y, f, S, n, m, and d be as in (4.0). The arrow (4.1.4)

$$\mathrm{Tr}_f : \mathrm{R}f_*\Omega^p_{X/S} \longrightarrow \Omega^{p-d}_{Y/S}[-d]$$

is called the Gysin morphism, as are the induced maps

$$\mathrm{Tr}_f : H^q(X, \Omega^p_{X/S}) \longrightarrow H^{q-d}(Y, \Omega^{p-d}_{Y/S}).$$

[5] In the first member of (4.1.3), the printed source switches once from p to q; the index p used in (4.1.1), (4.1.4), and (4.2) is retained here.

Proposition (4.3). Let $f : X \to Y$ be a proper birational morphism of smooth schemes over a field k. The maps
$$f^* : H^q(Y, \Omega^p_Y) \longrightarrow H^q(X, \Omega^p_X)$$
are injective.

One reduces to the case where X and Y are purely of dimension n. The composite morphism

$$\mathrm{Tr}_f \circ f^* : \Omega^p_Y \longrightarrow \mathrm{R}f_* f^* \Omega^p_Y \longrightarrow \mathrm{R}f_* \Omega^p_X \longrightarrow \Omega^p_Y$$

is the identity, because it coincides with the identity on a dense open subset. Applying the functor $H^q(Y, \ )$, one finds that the composite morphism

$$\mathrm{Tr}_f \circ f^* : H^q(Y, \Omega^p_Y) \longrightarrow H^q(X, \Omega^p_X) \longrightarrow H^q(Y, \Omega^p_Y)$$

is the identity, which proves (4.3).

5. Hodge and De Rham cohomology

(5.1) Let $f : X \to S$ be a smooth morphism of schemes. By definition, the relative De Rham cohomology sheaves of X over S are given by
$$H^*_{\mathrm{DR}}(X/S) = \mathrm{R}^* f_*(\Omega^\bullet_{X/S}),$$
where $\Omega^\bullet_{X/S}$, the relative De Rham complex, is an S-linear differential complex. There is a spectral sequence

$$E_1^{pq} = \mathrm{R}^q f_* \Omega^p_{X/S} \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/S). \tag{5.1.1}$$

(5.2) When $S = \mathrm{Spec}(\mathbb{C})$ and X is proper and smooth, GAGA ([7]) shows that the cohomology of each coherent sheaf Ω^p_X, and hence the hypercohomology of $\Omega^\bullet_X$, are the same in the algebraic sense (Zariski topology) and in the transcendental sense (usual topology of X^{an}); in particular, since $\Omega^\bullet_{X^{\mathrm{an}}}$ is a resolution of the constant sheaf $\mathbb{C}$, one has
$$H^n_{\mathrm{DR}}(X) = H^n(X^{\mathrm{an}}, \mathbb{C}),$$
and complex conjugation on the constant sheaf $\mathbb{C}$ induces a complex conjugation, of transcendental nature, on $H^n_{\mathrm{DR}}(X)$. The spectral sequence (5.1.1) is here

$$E_1^{pq} = H^q(X, \Omega^p_X) \Longrightarrow H^{p+q}_{\mathrm{DR}}(X); \tag{5.2.1}$$

let $F^p(H^n(X))$ denote the decreasing filtration which it defines on its abutment. By Weil [10], chap. IV, no. 4, th. 2, the spectral sequence (5.2.1) can be computed as the spectral sequence of the double complex $H^0(X^{\mathrm{an}}, \Omega^{p,q})$, filtered by the first degree, where $\Omega^{p,q}$ denotes the sheaf of C^∞ differential forms bihomogeneous of bidegree (p, q) (Weil [10], chap. II, no. 1).

Suppose now that X is projective, so that X^{an} is a Kähler variety. The harmonic forms then form a double subcomplex (Weil [10], chap. II, no. 6, cor. 1 to th. 2) of the preceding double complex, on which d' and d'' vanish. The inclusion of this subcomplex induces an isomorphism on the E_1 terms of the spectral sequences defined by these double complexes, filtered by the first degree, as one sees by considering the operator H (harmonic component), a retraction of the double complex $H^0(X^{\mathrm{an}}, \Omega^{p,q})$ onto the subcomplex of harmonic forms and satisfying

$$1 - H = d'(2\delta' G) + (2\delta' G)d'$$

(Weil [10], chap. IV, no. 1: (I) and lemma 3; no. 3: cor. 2 and chap. II, no. 5, th. 2 (IX)).

If $H^{p,q}(X)$ denotes the image in $H^{p+q}(X^{\mathrm{an}}, \mathbb{C})$ of the space of harmonic forms of type (p, q), one therefore has

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{p,q}(X), \tag{5.2.2}$$

$$F^p(H^n(X)) = \bigoplus_{i \geq p} H^{i,n-i}(X), \tag{5.2.3}$$

$$\overline{H^{p,q}(X)} = H^{q,p}(X) \quad \text{(complex conjugation)}, \tag{5.2.4}$$

and the spectral sequence (5.2.1) degenerates.

Returning to the general case where X is only assumed proper and smooth over $\mathbb{C}$, put

$$H^{p,q}(X) = F^p(H^{p+q}(X)) \cap \overline{F^q(H^{p+q}(X))}, \tag{5.2.5}$$

so that

$$\overline{H^{p,q}(X)} = H^{q,p}(X). \tag{5.2.6}$$

By (5.2.3), this notation is compatible with the one used when X is projective. Put also

$$h^{p,q} = \dim H^q(X, \Omega_X^p). \tag{5.2.7}$$

Proposition (5.3). Let X be a proper smooth scheme over $\mathbb{C}$:

(i) the spectral sequence (5.2.1) degenerates;

(ii) one has

$$F^p(H^n(X)) = \bigoplus_{i \geq p} H^{i,n-i}(X), \tag{5.3.1}$$

and in particular

$$H^n(X) = \bigoplus_{p+q=n} H^{p,q}(X) \quad \text{(direct sum)}, \tag{5.3.2}$$

$$h^{p,q} = h^{q,p} = \dim H^{p,q}(X). \tag{5.3.3}$$

One reduces to the case where X is connected. Let N be its dimension. By Chow's lemma and resolution of singularities (Hironaka [5]), there exists a projective birational morphism $g : X' \to X$, with X' projective and smooth over $\mathbb{C}$. The spectral sequence

$$E_1^{pq} = H^q(X, \Omega_X^p) \Longrightarrow H_{\mathrm{DR}}^{p+q}(X) \tag{5.3.4}$$

maps, by inverse image, into the analogous spectral sequence

$$E_1'^{\,pq} = H^q(X', \Omega_{X'}^p) \Longrightarrow H_{\mathrm{DR}}^{p+q}(X'), \tag{5.3.5}$$

which degenerates by (5.2). By (4.3), the morphism $g^* : E_1 \to E_1'$ is injective. One concludes, by induction on $r \geq 1$, that $d_r = 0$, that $E_r = E_{r+1}$, and that g^* injects E_{r+1} into E_{r+1}':

$$\begin{tikzcd} E_r^{pq} \arrow[r, "d_r"] \arrow[d, "g^*"'] & E_r^{p+r,q-r+1} \arrow[d, "g^*"] \\ E_r'^{\,pq} \arrow[r, "0"] & E_r'^{\,p+r,q-r+1}. \end{tikzcd}$$

Since the spectral sequences (5.3.4) and (5.3.5) degenerate, and since g^* is injective on their initial term, it is still injective on their abutment. From formulas (5.2.3), (5.2.4), applied to X', it follows that

$$F^p(H^n(X')) \cap \overline{F^{n-p+1}(H^n(X'))} = \{0\};$$

hence one obtains the analogous formula

$$F^p(H^n(X)) \cap \overline{F^{n-p+1}(H^n(X))} = \{0\}, \tag{5.3.6}$$

because g^* sends $F^p(H^n(X))$ to $F^p(H^n(X'))$ and commutes with complex conjugation. In particular,

$$\sum_{i \geq p} h^{i,n-i} + \sum_{i \geq n-p+1} h^{i,n-i} \leq \dim H^n(X) \leq \sum_i h^{i,n-i},$$

which may be written

$$\sum_{i \geq p} h^{i,n-i} \leq \sum_{i \leq n-p} h^{i,n-i}. \tag{5.3.7}$$

By Serre duality, $h^{i,j} = h^{N-i,N-j}$, inequality (5.3.7), for $N-n$, implies the opposite inequality, so that

$$\dim F^p(H^n(X)) + \dim \overline{F^{n-p+1}(H^n(X))} = \dim H^n(X).$$

By (5.3.6), therefore,

$$H^n(X) = F^p(H^n(X)) \oplus \overline{F^{n-p+1}(H^n(X))}. \tag{5.3.8}$$

The decomposition (5.3.8) induces on $F^{p-1}(H^n(X))$, which contains one of the factors, a decomposition

$$F^{p-1}(H^n(X)) = F^p(H^n(X)) \oplus H^{p-1,n-p+1}(X),$$

and formula (5.3.1) follows by descending induction on p; (5.3.2) and (5.3.3) follow at once from (5.3.1) and (5.2.6), completing the proof of (5.3).

Corollary (5.4). Under the hypotheses of (5.3), one has:

(i) A complex cohomology class a on X lies in $H^{p,q}(X)$ if and only if it can be represented by a closed form α of type (p,q).

(ii) Every cohomology class can be represented by a form α satisfying $d'\alpha = d''\alpha = 0$.

(iii) If the form α satisfies $d'\alpha = d''\alpha = 0$, the following conditions are equivalent:

a) $(\exists\beta)$ $\alpha = d\beta$;
b) $(\exists\beta)$ $\alpha = d'\beta$;
c) $(\exists\beta)$ $\alpha = d''\beta$.[6]

By definition, $a \in H^{p,q}(X)$ if and only if in the class of a there exist closed forms α_1 and α_2 whose components of type (p',q') are zero for $p' < p$ (respectively $q' < q$). Then there exists β such that $\alpha_1 - \alpha_2 = d\beta$. Let β_1 (respectively β_2) be the sum of the components of β of type (p',q') with $p' \geq p$ (respectively $q' \geq q$); one has $\beta = \beta_1 + \beta_2$ and

$$\alpha = \alpha_1 - d\beta_1 = \alpha_2 + d\beta_2$$

is a closed form of type (p,q) in the class of a. This proves (i). It suffices to prove (ii) for a bihomogeneous, hence represented by a closed bihomogeneous form α by (i). Such a form automatically satisfies $d'\alpha = d''\alpha = 0$.

If α satisfies $d'\alpha = d''\alpha = 0$, its bihomogeneous components are closed, and each of the conditions a), b), c) is equivalent to the conjunction of the analogous conditions for the various components of α, trivially for b) and c), and by (5.3) (ii) and (5.4) (i) for a). To prove (iii), suppose therefore that α is closed of type (p,q). The cohomology class a defined by α lies in $H^{p,q}(X)$, hence in $F^p(H^{p+q}(X))$, and is zero if and only if it already lies in $F^{p+1}(H^{p+q}(X))$, i.e. if its image in $H^q(X, \Omega^p_X)$ is zero. This proves a) $\Leftrightarrow$ c), and the equivalence a) $\Leftrightarrow$ b) follows by conjugation.

Theorem (5.5). Let S be a scheme of characteristic 0 and let $f : X \to S$ be a proper smooth morphism. Then:

(i) the sheaves $\mathrm{R}^q f_* \Omega^p_{X/S}$ are locally free of finite type and their formation is compatible with arbitrary base change;

(ii) the spectral sequence (5.1.1)

$$E_1^{pq} = \mathrm{R}^q f_* \Omega^p_{X/S} \Longrightarrow H^{p+q}_{\mathrm{DR}}(X/S)$$

degenerates;

(iii) at every point of S, the sheaves $\mathrm{R}^q f_* \Omega^p_{X/S}$ and $\mathrm{R}^p f_* \Omega^q_{X/S}$ have the same rank (Hodge symmetry).

The second assertion of (i) follows from the first and from EGA III, (7.8.5), applied to each of the $\Omega^p_{X/S}$; it implies that it suffices to verify (iii) for S the spectrum of a field.

Standard arguments allow one to reduce successively to the case where S is affine, since the question is local on S; noetherian, by passage to an inductive limit of rings; the spectrum of a noetherian local ring, since the question is local; the spectrum of a complete noetherian local ring, by faithful flatness of completion; and finally the spectrum of an Artinian local ring. To make this last reduction, write the complete noetherian local ring R as the projective limit of its Artinian quotients $R/\mathfrak{m}^n$. By the comparison theorem EGA III, (4.1.5), the E_1 term

[6](Added in proofs.) One can show that these conditions are also equivalent to the existence of a form β such that $\alpha = d'd''\beta$.

of the spectral sequence (5.1.1) is the projective limit of the E_1 terms of the analogous spectral sequences over the $R/\mathfrak{m}^n$, so that if these degenerate, then (5.1.1) also degenerates. If assertion (i) is true over the $R/\mathfrak{m}^n$, these E_1 terms form a projective system of free modules, obtained from one another by reduction modulo $\mathfrak{m}^n$, so that their projective limit is a free module.

Suppose therefore that $S = \operatorname{Spec}(A)$ with A an Artinian local ring with residue field k of characteristic 0. This ring admits a field of representatives (EGA, 0_{IV}, (19.8.6), (i), for $W = k$, $u =$ identity) and can therefore be regarded as a local k-algebra of finite dimension. The Lefschetz principle permits one to suppose $k = \mathbb{C}$, in which case (iii) follows from (i) and (5.3.3).

To complete the proof of (5.5), it remains to prove the first assertion of (i), and (ii), under the additional hypothesis:

$$S = \operatorname{Spec}(A) \quad \text{for a finite-dimensional local } \mathbb{C}\text{-algebra } A \text{ with maximal ideal } \mathfrak{m}. \tag{5.5.1}$$

The spectral sequence (5.1.1) becomes

$$E_1^{pq} = H^q(X, \Omega^p_{X/S}) \Longrightarrow \mathrm{R}^{p+q}\Gamma(\Omega^\bullet_{X/S}). \tag{5.5.2}$$

Lemma (5.5.3). Under the hypotheses (5.5) and (5.5.1), the complex $\Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of the constant sheaf A on the topological space X^{an}.

Since the complex $\Omega^{\mathrm{an},\bullet}_{X/S}$ is A-linear, its $\mathfrak{m}$-adic filtration is a filtration by subcomplexes. For this filtration,

$$\operatorname{Gr} \Omega^{\mathrm{an},\bullet}_{X/S} = \operatorname{Gr} A \otimes_{\mathbb{C}} \operatorname{Gr}^0 \Omega^{\mathrm{an},\bullet}_{X/S}.$$

The complex $\operatorname{Gr}^0 \Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of the constant sheaf $\mathbb{C}$, since it is just the De Rham complex of X_{red}. Thus the complex $\operatorname{Gr} \Omega^{\mathrm{an},\bullet}_{X/S}$ is a resolution of $\operatorname{Gr} A$, and the assertion follows.

The space X is compact, so by GAGA (cf. (5.2)) the algebraic and transcendental hypercohomologies of $\Omega^\bullet_{X/S}$ coincide, and by (5.5.3) they also coincide with the cohomology of X^{an} with values in A. Hence

$$\lg_A \mathrm{R}^n\Gamma(\Omega^\bullet_{X/S}) = \lg(A) \dim_{\mathbb{C}} \mathrm{R}^n\Gamma(\Omega^\bullet_{X_{\mathrm{red}}}). \tag{5.5.4}$$

By (3.5.1) and EGA III, (6.10.5), one knows that

$$\lg_A H^q(X, \Omega^p_{X/S}) \leq \lg(A) \dim_{\mathbb{C}} H^q(X_{\mathrm{red}}, \Omega^p_{X_{\mathrm{red}}}), \tag{5.5.5}$$

with equality possible only if $H^q(X, \Omega^p_{X/S})$ is A-free.

From (5.5.2) one obtains

$$\sum_{p+q=n} \lg_A H^q(X, \Omega^p_{X/S}) \geq \lg_A \mathrm{R}^n\Gamma(\Omega^\bullet_{X/S}), \tag{5.5.6}$$

equality being possible for all n only if the spectral sequence (5.5.2) degenerates. Chaining these inequalities, one finds

$$\lg(A) \sum_{p+q=n} \dim_{\mathbb{C}} H^q(X_{\mathrm{red}}, \Omega^p_{X_{\mathrm{red}}}) \geq \lg(A) \mathrm{R}^n\Gamma(\Omega^\bullet_{X_{\mathrm{red}}}), \tag{5.5.7}$$

equality being possible only if the conclusions of (5.4) are satisfied. This equality follows from (5.3) (i) applied to X_{red}.

6. Application to coherent cohomology

Theorem (6.1). Let S be a scheme of characteristic 0, and let $f : X \to S$ be a projective smooth morphism of relative dimension n. For every p, for example $p = 0$, one has in $D^b(S, \mathcal{O}_S)$

$$\mathrm{R}f_*\Omega^p_{X/S} \simeq \bigoplus_q \mathrm{R}^q f_*\Omega^p_{X/S}[-q].$$

Let u denote the first Chern class of a relatively ample invertible sheaf on X, with values in $H^1(X, \Omega^1_{X/S})$. By cup product, the class u defines homomorphisms in $D^b(S)$,

$$u\wedge : \mathrm{R}f_*\Omega^p_{X/S} \longrightarrow \mathrm{R}f_*\Omega^{p+1}_{X/S}[1],$$

and a degree-two endomorphism

$$u\wedge : \bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p] \longrightarrow \left(\bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p]\right)[2]. \tag{6.1.1}$$

Lemma (6.2). The endomorphism (6.1.1) satisfies the hypotheses of (1.5).

By (5.4) (i) and the Lefschetz principle, it suffices to prove (6.2) when $S = \operatorname{Spec}(\mathbb{C})$. The assertion then follows from the Lefschetz theorem (see (2.6.2) or (2.6.3)) and from the Hodge decomposition.

By (1.5), the complex $\bigoplus_p \mathrm{R}f_*\Omega^p_{X/S}[-p]$ is isomorphic, in the derived category, to the sum of its cohomology objects, and the same is true for each of the $\mathrm{R}f_*\Omega^p_{X/S}$, which, up to a shift, are direct factors of it (1.12).

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Ordinary abelian varieties over a finite field

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We give a down-to-earth description of the category of ordinary abelian varieties over a finite field $\mathbb{F}_q$. The result obtained was inspired by Ihara [2], ch. V (see also [3]).

1. Let p be a prime number, let $\mathbb{F}_p$ be the field $\mathbb{Z}/(p)$, let $\overline{\mathbb{F}_p}$ be an algebraic closure of $\mathbb{F}_p$, and, for each power q of p, let $\mathbb{F}_q$ be the subfield with q elements of $\overline{\mathbb{F}_p}$. For every algebraic extension k of $\mathbb{F}_p$, let $W_0(k)$ denote the henselian discrete valuation ring, essentially of finite type over $\mathbb{Z}$, absolutely unramified, with residue field k; let $W(k)$ be the ring of Witt vectors over k, i.e. the completion of $W_0(k)$, let $W = W(\overline{\mathbb{F}_p})$, and let φ be an embedding of W in the field $\mathbb{C}$ of complex numbers. We denote by $\mathbb{Z}(1)$ the subgroup $2\pi i\mathbb{Z}$ of $\mathbb{C}$. The exponential map defines an isomorphism between $\mathbb{Z}(1) \otimes \mathbb{Z}_\ell$ and

$$\mathbb{Z}_\ell(1)(\mathbb{C}) = \varprojlim_n \mu_{\ell^n}(\mathbb{C}).$$

We denote by A^* the abelian variety dual to an abelian variety A. For every field k, let $\bar{k}$ denote an algebraic closure of k.

2. Let A be an abelian variety of dimension g, defined over a field k of characteristic p. Recall that A is called ordinary if the following equivalent conditions are satisfied:

(I) A has p^g points of order dividing p with values in $\bar{k}$.

(II) The "Hasse–Witt matrix"

$$F^* : H^1(A^{(p)}, \mathcal{O}_{A^{(p)}}) \longrightarrow H^1(A, \mathcal{O}_A)$$

is invertible.

(III) The neutral component of the group scheme A_p, the kernel of multiplication by p, is of multiplicative type, hence geometrically isomorphic to a power of μ_p.

If $k = \mathbb{F}_q$, if F is the Frobenius endomorphism of A, and if $P_{cA}(F; x)$ is its characteristic polynomial, these conditions are further equivalent to:

(IV) At least half of the roots of $P_{cA}(F; x)$ in $\overline{\mathbb{Q}}_p$ are p-adic units. In other words, if $n = \dim A$, the reduction modulo p of the polynomial $P_{cA}(F; x)$ is not divisible by x^{n+1}.

3. Let A be an ordinary abelian variety over $\overline{\mathbb{F}_p}$. Let $\widetilde{A}$ denote the canonical Serre–Tate lift [4] of A over W. Recall that $\widetilde{A}$ depends functorially on A and is characterized by the fact that the p-divisible group $T_p(\widetilde{A})$ over W attached to $\widetilde{A}$ [5] is the product of the p-divisible groups, uniquely determined by 2. (III), lifting respectively the neutral component and the largest étale quotient of $T_p(A)$. The canonical lift $\widetilde{A}$ is also the unique lift of A such that every endomorphism of A lifts to $\widetilde{A}$. Let $T(A)$ denote the integral homology of the complex abelian variety $A_{\mathbb{C}}$ deduced from $\widetilde{A}$ and φ by extension of scalars from W to $\mathbb{C}$:

$$T(A) = H_1(\widetilde{A} \otimes_\varphi \mathbb{C}).$$

It is known that $\widetilde{A}$ descends uniquely to $W_0(\overline{\mathbb{F}_p})$, so that $A_{\mathbb{C}}$ depends only on A and on the restriction of φ to $W_0(\overline{\mathbb{F}_p})$. The free $\mathbb{Z}$-module $T(A)$ has rank $2\dim(A)$; it is functorial in A. Moreover, if ℓ is a prime number distinct from p, one has functorially

$$T(A) \otimes \mathbb{Z}_\ell = T_\ell(A). \tag{3.1}$$

The canonical lift of the abelian variety A^* dual to A is the dual of $\widetilde{A}$, so that $(A_{\mathbb{C}})^* = A^*_{\mathbb{C}}$ and $T(A)$ and $T(A^*)$ are in perfect duality with values in $\mathbb{Z}(1)$:

$$T(A) \otimes T(A^*) \longrightarrow \mathbb{Z}(1). \tag{3.2}$$

It is indispensable to use $\mathbb{Z}(1)$ rather than $\mathbb{Z}$ in order to obtain a theory invariant under complex conjugation. The pairings (3.2) are compatible, via (3.1), with the pairings

$$T_\ell(A) \otimes T_\ell(A^*) \longrightarrow \mathbb{Z}_\ell(1);$$

a morphism $\xi : A \to A^*$ defines a polarization of A if and only if $\xi_{\mathbb{C}} : A_{\mathbb{C}} \to A^*_{\mathbb{C}}$ defines a polarization of $A_{\mathbb{C}}$. Put

$$T'_p(A) = \mathrm{Hom}(\mathbb{Q}_p/\mathbb{Z}_p, A(\overline{\mathbb{F}_p}))$$

and

$$T''_p(A) = \mathrm{Hom}_{\mathbb{Z}_p}(T'_p(A^*), \mathbb{Z}(1) \otimes \mathbb{Z}_p).$$

These $\mathbb{Z}_p$-modules are covariant functors of A.

By definition of the canonical lift, the p-divisible group $T_p(\widetilde{A})$ is the sum of the constant pro-étale group $T'_p(A)$ and of the Cartier dual of $T'_p(A^*)$. For every morphism $u : A \to B$, the induced morphism

$$u : T_p(\widetilde{A}) \to T_p(\widetilde{B})$$

is identified with the sum of $u|T'_p(A) : T'_p(A) \to T'_p(B)$ and the Cartier transpose of ${}^t u|T'_p(B^*) : T'_p(B^*) \to T'_p(A^*)$. Over $\mathbb{C}$, one has canonically $\mathbb{Z}(1)/(p^n) \sim \mu_{p^n}$, whence an isomorphism of functors:

$$T_{(p)}(A) = T(A) \otimes \mathbb{Z}_p = T'_p(A) \oplus T''_p(A). \tag{3.2}$$

4. Recall that, if $\varphi : X \to Y$ is an isogeny between complex abelian varieties, the exact homotopy sequence reduces to a short exact sequence

$$0 \longrightarrow H_1(X) \longrightarrow H_1(Y) \longrightarrow \mathrm{Ker}(\varphi) \longrightarrow 0.$$

The abelian varieties which are quotients of X by a finite subgroup, and these finite subgroups of X, correspond bijectively to the sublattices of $H_1(X) \otimes \mathbb{Q}$ containing $H_1(X)$.

Let A be an ordinary abelian variety over $\overline{\mathbb{F}_p}$. If n is an integer prime to p, the finite group subschemes of order n of A, of $\widetilde{A}$, and of $A_{\mathbb{C}}$ correspond bijectively, and still correspond to the overlattices R of $T(A)$ such that $[R : T(A)] = n$.

Put

$$V'_p = T'_p(A) \otimes \mathbb{Q}_p, \qquad V''_p(A) = T''_p(A) \otimes \mathbb{Q}_p.$$

The finite group subschemes of order p^k of A are products of an étale subgroup and an infinitesimal subgroup. The étale subgroups of order p^k of A correspond to those subgroups of order p^k of $A_{\mathbb{C}}$ for which the corresponding overlattice R of $T(A)$ is contained in $T_{(p)}(A) + V'_p(A)$. By duality, the infinitesimal subgroups of A correspond to the overlattices R of $T(A)$, p-isogenous to $T(A)$, i.e. such that $[R : T(A)]$ is a power of p, and contained in $T_{(p)}(A) + V''_p(A)$.

In all, the finite subgroups of A, or the abelian varieties quotient of A, correspond bijectively to the overlattices R of $T(A)$ such that

$$R \otimes \mathbb{Z}_p = (R \otimes \mathbb{Z}_p \cap V'_p) + (R \otimes \mathbb{Z}_p \cap V''_p). \tag{4.1}$$

5. In particular, $A^{(p)}$, the quotient of A by the largest infinitesimal subgroup of A killed by p for A ordinary, is defined by the overlattice $T(A)^{(p)}$ of $T(A)$, p-isogenous to $T(A)$, such that

$$T(A)^{(p)} \otimes \mathbb{Z}_p = T'_p(A) + \frac{1}{p} T''_p(A).$$

6. Let A be an abelian variety over $\mathbb{F}_q$ and let $F : x \mapsto x^q$ be its Frobenius endomorphism. Recall that A is uniquely determined by the pair $(\overline{A}, F)$ deduced from (A, F) by extension of scalars from $\mathbb{F}_q$ to $\overline{\mathbb{F}_q}$; the endomorphism F of $\overline{A}$ factors through the relative Frobenius

$$\mathrm{Fr}^{(q)} : \overline{A} \longrightarrow \overline{A}^{(q)}$$

and an isomorphism

$$F' : \overline{A}^{(q)} \longrightarrow \overline{A}.$$

If A is ordinary, $T(A)$ will denote the $\mathbb{Z}$-module $T(\overline{A})$ endowed with the endomorphism F induced by the Frobenius endomorphism of A. By 5, by what precedes, and by (3.2), the lattices $T(A)$ and $F(T(A))$ are p-isogenous and one has

$$F(T_p'(A)) = T_p'(A), \tag{6.1}$$

$$F(T_p''(A)) = qT_p''(A). \tag{6.2}$$

7. Theorem. The functor

$$A \longmapsto (T(A), F)$$

is an equivalence of categories between the category of ordinary abelian varieties over $\mathbb{F}_q$ and the category of finite-type free $\mathbb{Z}$-modules T endowed with an endomorphism F satisfying the following conditions:

(a) F is semisimple, and its eigenvalues have complex absolute value $q^{1/2}$.

(b) At least half of the roots in $\overline{\mathbb{Q}}_p$ of the characteristic polynomial of F are p-adic units; in other words, if T has rank d, the reduction modulo p of the polynomial $P_{cT}(F; x)$ is not divisible by $x^{\lfloor d/2 \rfloor + 1}$.

(c) There exists an endomorphism V of T such that $FV = q$.

If condition (a) is satisfied, conditions (b) and (c) are equivalent to:

(d) The module $T \otimes \mathbb{Z}_p$ admits a decomposition, stable under F, into two $\mathbb{Z}_p$-submodules T_p' and T_p'' of the same dimension, such that $F|T_p'$ is invertible and $F|T_p''$ is divisible by q.

Proof. (A) We prove that (a)+(b)+(c) imply (d). If α is a complex eigenvalue of F, then $\bar{\alpha}$ is another, with the same multiplicity, and $\alpha\bar{\alpha} = q$. Apart from those equal to $\pm q^{1/2}$, the eigenvalues of F in $\mathbb{C}$, hence in $\overline{\mathbb{Q}}_p$, fall into pairs of roots α and q/α. The roots α and q/α cannot both be p-adic units; it follows from (b) that $\pm q^{1/2}$ is not an eigenvalue of F, that half of the eigenvalues of F in $\overline{\mathbb{Q}}_p$ are p-adic units, say $\alpha_1, \ldots, \alpha_{d/2}$, and that the other half consists of $\beta_1 = q/\alpha_1, \ldots, \beta_{d/2} = q/\alpha_{d/2}$. Let

$$T_{(p)} = T \otimes \mathbb{Z}_p, \qquad V_p = T \otimes \mathbb{Q}_p,$$

let V_p' be the subspace of V_p which is the kernel of $\prod_i (F - \alpha_i)$, and let V_p'' be the kernel of the endomorphism $\varphi = \prod_i (F - \beta_i)$. One has

$$V_p = V_p' \oplus V_p''.$$

Let T_p' be the projection of $T_{(p)}$ onto V_p' and let $T_p'' = T_{(p)} \cap V_p''$. Since φ kills V_p'' and preserves T, it sends T_p' into $T_{(p)} \cap V_p' \subset T_p'$. On the other hand,

$$\det(\varphi | V_p') = \prod_{i,j} (\alpha_i - \beta_j)$$

is a p-adic unit, hence $\varphi(T_p') = T_p'$ and $T_{(p)} \cap V_p' = T_p'$, so that

$$T_{(p)} = T_p' \oplus T_p''.$$

(B) *Full faithfulness.* Let A and B be two abelian varieties over $\mathbb{F}_q$, and let ψ be the map

$$\psi : \mathrm{Hom}(A, B) \longrightarrow \mathrm{Hom}_F(T(A), T(B)).$$

By Tate's theorem [7] and (3.1), the map

$$\psi_\ell : \mathrm{Hom}(A, B) \otimes \mathbb{Z}_\ell \longrightarrow \mathrm{Hom}_F(T(A), T(B)) \otimes \mathbb{Z}_\ell$$

is an isomorphism for $(\ell, p) = 1$, so that $\psi \otimes \mathbb{Q}$ is an isomorphism. It is known that $\mathrm{Hom}(A, B)$ is torsion-free, hence ψ is injective. Let $u : A \to B$ be a morphism such that $T(u)$ is divisible by n. The induced morphism

$$u_{\mathbb{C}} : \overline{A}_{\mathbb{C}} \longrightarrow \overline{B}_{\mathbb{C}}$$

is then divisible by n, hence so is $\widetilde{u} : \widetilde{\overline{A}} \to \widetilde{\overline{B}}$ at the generic point of W. The kernel of multiplication by n is flat over W; thus $\widetilde{u}$ vanishes on this kernel, $\widetilde{u}$ and u are divisible by n, and ψ is bijective.

(C) *Necessity.* That $(T(A), F)$ satisfies (a) follows from Weil; condition (d), which implies (b) and (c), follows from 6.

(D) *Isogenies.* Let (T_0, F) satisfy (a) (d) and let T be a lattice in $T_0 \otimes \mathbb{Q}$, stable under F, still satisfying (d). Suppose that (T_0, F) is the image of an abelian variety A over $\mathbb{F}_q$ and let us prove that (T, F) comes from an isogenous abelian variety. Replacing T by $\frac{1}{k}T$, which is isomorphic to it, one may suppose that $T \supset T_0$. Condition (d) implies that T satisfies (4.1), and T defines a subgroup H of $\overline{A}$, which is defined over $\mathbb{F}_q$, and

$$(T, F) = T(A/H).$$

(E) *Surjectivity.* The functor T induces a functor $T_{\mathbb{Q}}$ from the category of ordinary abelian varieties up to isogeny over $\mathbb{F}_q$ to the category of finite-rank $\mathbb{Q}$-vector spaces endowed with an automorphism F satisfying (a) (b). By (D), it is enough to prove that this functor $T_{\mathbb{Q}}$ is essentially surjective. It is even enough to show that every simple object (V, F) of the image category is attained. By Honda [1] (see also [6]), there exists an abelian variety A over $\mathbb{F}_q$ such that the characteristic polynomial of the Frobenius F_A of A is a power of that of F. The third characterization in 2.[1] of ordinary abelian varieties shows that A is ordinary. Moreover, $(T(A) \otimes \mathbb{Q}, F)$ is a sum of copies of (V, F), hence, by (B), the abelian variety up to isogeny $A \otimes \mathbb{Q}$ is a sum of copies of an abelian variety B satisfying

$$T(B) \otimes \mathbb{Q} = (V, F).$$

8. Let (T, F) be a pair satisfying the hypotheses of the theorem, let $2g$ be the rank of T, let A be the corresponding abelian variety over $\mathbb{F}_q$, and let $A_{\mathbb{C}}$ be the complex abelian variety deduced from it (3.). One has

$$T = H_1(A_{\mathbb{C}}),$$

so that $T \otimes \mathbb{R}$ is identified with the Lie algebra of $A_{\mathbb{C}}$ and, as such, is endowed with a complex structure. Here is, according to J.-P. Serre, how to reconstruct this complex structure in terms of T, F, and the restriction of φ to $W_0(\overline{\mathbb{F}_p})$.

Proposition. The complex structure defined above on $T \otimes \mathbb{R}$ is characterized by the following properties:

(I) The endomorphism F is $\mathbb{C}$-linear.

(II) If v is the valuation of the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$ extending the valuation of $W_0(\overline{\mathbb{F}_p})$, the valuations of the g eigenvalues of this endomorphism are strictly positive.

Condition (I) is evident, and condition (II) follows from the fact that the action of F on the Lie algebra of A is congruent to zero modulo p. The uniqueness of the complex structure satisfying (I) and (II) follows easily from condition (b) satisfied by (T, F).

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[1] The printed source refers here to 1.; the characterizations of ordinarity are stated in 2., and that reference is restored here.

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THE IRREDUCIBILITY
OF THE SPACE OF CURVES OF GIVEN GENUS

by P. DELIGNE and D. MUMFORD[1]

Fix an algebraically closed field k. Let M_g be the moduli space of curves of genus g over k. The main result of this note is that M_g is irreducible for every k. Of course, whether or not M_g is irreducible depends only on the characteristic of k. When the characteristic is 0, we can assume that $k = \mathbb{C}$, and then the result is classical. A simple proof appears in Enriques–Chisini [E, vol. 3, chap. 3], based on analyzing the totality of coverings of $\mathbf{P}^1$ of degree n, with a fixed number d of ordinary branch points. This method has been extended to char. p by William Fulton [F], using specializations from char. 0 to char. p provided that $p > 2g + 1$. Unfortunately, attempts to extend this method to all p seem to get stuck on difficult questions of wild ramification. Nowadays, the Teichmüller theory gives a thoroughly analytic but very profound insight into this irreducibility when $k = \mathbb{C}$. Our approach however is closest to Severi's incomplete proof ([Se], Anhang F; the error is on pp. 344–345 and seems to be quite basic) and follows a suggestion of Grothendieck for using the result in char. 0 to deduce the result in char. p. The basis of both Severi's and Grothendieck's ideas is to construct families of curves X, some singular, with $p_a(X) = g$, over non-singular parameter spaces, which in some sense contain enough singular curves to link together any two components that M_g might have.

The essential thing that makes this method work now is a recent "stable reduction theorem" for abelian varieties. This result was first proved independently in char. 0 by Grothendieck, using methods of etale cohomology (private correspondence with J. Tate), and by Mumford, applying the easy half of Theorem (2.5), to go from curves to abelian varieties (cf. [M_2]). Grothendieck has recently strengthened his method so that it applies in all characteristics (SGA 7, 1968). Mumford has also given a proof using theta functions in char. $\neq 2$. The result is this:

Stable Reduction Theorem. — Let R be a discrete valuation ring with quotient field K. Let A be an abelian variety over K. Then there exists a finite algebraic extension L of K such that, if $R_L =$ integral closure of R in L, and if $\mathcal{A}_L$ is the Néron model of $A \times_K L$ over R_L, then the closed fibre $\mathcal{A}_{L,s}$ of $\mathcal{A}_L$ has no unipotent radical.

We shall give two related proofs of our main result. One of these is quite elementary, and follows by quite standard techniques once the Stable Reduction Theorem for abelian varieties is applied, in § 2, to deduce an analogous stable reduction theorem for curves. The other proof is more powerful, and is based on the use of a larger category than the category of schemes, and on proving for the objects of this category many of the standard theorems for schemes, especially the Enriques–Zariski connectedness theorem (EGA 3, (4.3)). Unfortunately, this larger category is not quite a category — it is a simple type of 2-category; in fact, if X, Y are objects, then $\mathrm{Hom}(X, Y)$ is itself a category, but one in which all morphisms are isomorphisms. The objects of this 2-category we call algebraic stacks[2]. The moduli space M_g is just the "underlying coarse variety" of a more fundamental object, the moduli stack $\mathcal{M}_g$, studied in [M_3]. Full details on the basic properties and theorems for algebraic stacks will be given elsewhere. In this paper, we will only give definitions and state without proof the general theorems which we apply. Using the method of algebraic stacks, we can prove not only the irreducibility of M_g itself, but of all higher level moduli spaces of curves too (cf. § 5 below).

§ 1. Stable curves and their moduli.

The key definition of the whole paper is this:

Definition (1.1). — Let S be any scheme. Let $g \geq 2$. A stable curve of genus g over S is a proper flat morphism $\pi : C \to S$ whose geometric fibres are reduced, connected, 1-dimensional schemes C_s such that:

(i) *C_s has only ordinary double points;*

(ii) *if E is a non-singular rational component of C_s, then E meets the other components of C_s in more than 2 points;*

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[2] A slightly less general category of objects, called algebraic spaces, has been introduced and studied very deeply by M. Artin [A_1] and D. Knutson [K]. The idea of enlarging the category of varieties for the study of moduli spaces is due originally, we believe, to A. Weil.

(iii) $\dim H^1(\mathcal{O}_{C_s}) = g$.

We will study in this section three aspects of the theory of stable curves: their pluri-canonical linear systems, their deformations, and their automorphisms.

Suppose $\pi : C \to S$ is a stable curve. Since π is flat and its geometric fibres are local complete intersections, the morphism π is locally a complete intersection (i.e., locally, C is isomorphic as S-scheme to $V(f_1, \ldots, f_{n-1}) \subset \mathbf{A}^n \times U$, where $U \subset S$ is open, and $f_1, \ldots, f_{n-1} \in \Gamma(\mathcal{O}_{\mathbf{A}^n \times U})$ are a regular sequence). Therefore, by the theory of duality of coherent sheaves [H], there is a canonical invertible sheaf $\omega_{C/S}$ on C — the unique non-zero cohomology group of the complex of sheaves $f^!(\mathcal{O}_S)$. We need to know the following facts about $\omega_{C/S}$:

a) for all morphisms $f : T \to S$, $\omega_{C \times_S T/T}$ is canonically isomorphic to $f^*(\omega_{C/S})$;

b) if $S = \operatorname{Spec}(k)$, k algebraically closed, let $f : C' \to C$ be the normalization of C, $x_1, \ldots, x_n, y_1, \ldots, y_n$ the points of C' such that the $z_i = f(x_i) = f(y_i)$, $1 \le i \le n$, are the double points of C. Then $\omega_{C/S}$ is the sheaf of 1-forms η on C' regular except for simple poles at the x's and y's and with $\operatorname{Res}_{x_i}(\eta) + \operatorname{Res}_{y_i}(\eta) = 0$;

c) if $S = \operatorname{Spec}(k)$, and $\mathcal{F}$ is a coherent sheaf on C, then

$$\operatorname{Hom}(H^1(C, \mathcal{F}), k) \cong \operatorname{Hom}_{\mathcal{O}_C}(\mathcal{F}, \omega_{C/S}).$$

Theorem (1.2). — If $g \ge 2$ and C is a stable curve of genus g over an algebraically closed field k, then $H^1(C, \omega_{C/k}^{\otimes n}) = (0)$ if $n \ge 2$, and $\omega_{C/k}^{\otimes n}$ is very ample if $n \ge 3$.

Proof. — Since C is stable, of genus $g \ge 2$, every irreducible component E of C either 1) has (arithmetic) genus ≥ 2 itself, 2) has genus 1, but meets other components of C in at least one point, or 3) is non-singular, rational and meets other components of C in at least three points. But by b) above, $\omega_{C/k} \otimes \mathcal{O}_E$ is isomorphic to $\omega_{E/k}(\sum Q_i)$, where $\{Q_i\}$ are the points where E meets the rest of C. Since the degree of $\omega_{E/k}$ is $2g_E - 2$, it follows that in any of the cases 1, 2 or 3, $\omega_{C/k} \otimes \mathcal{O}_E$ has positive degree. This shows immediately that $\omega_{C/k}$ is ample on each component E of C, hence is ample.

Next, by c) above, $H^1(\omega_{C/k}^{\otimes n})$ is dual to $H^0(\omega_{C/k}^{\otimes(1-n)})$. Since $\omega_{C/k} \otimes \mathcal{O}_E$ has positive degree, $\omega_{C/k}^{\otimes(1-n)} \otimes \mathcal{O}_E$ has no sections for any E, any $n \ge 2$; therefore $H^0(\omega_{C/k}^{\otimes(1-n)}) = (0)$ if $n \ge 2$, and so $H^1(\omega_{C/k}^{\otimes n}) = (0)$ if $n \ge 2$.

To prove that an invertible sheaf $\mathcal{L}$ on any scheme C, proper over k, is very ample, it suffices to show

a) for all closed points $x \ne y$
$$H^0(C, \mathcal{L}) \to H^0(C, \mathcal{L} \otimes k(x)) \oplus H^0(C, \mathcal{L} \otimes k(y))$$
is surjective,

b) for all closed points x,
$$H^0(C, \mathcal{L}) \to H^0(C, \mathcal{L} \otimes \mathcal{O}_x/\mathfrak{m}_x^2)$$
is surjective.

Using the exact sequence of cohomology, these both follow if $H^1(C, \mathfrak{m}_x \cdot \mathfrak{m}_y \cdot \mathcal{L}) = (0)$ for all closed points $x, y \in C$. In our case, $\mathcal{L} = \omega_C^{\otimes n}$, $n \ge 3$, so if we use duality, we must show:

$$(*) \qquad \operatorname{Hom}(\mathfrak{m}_x \cdot \mathfrak{m}_y, \omega_C^{\otimes -n}) = (0), \qquad \text{if } n \ge 2.$$

Source note. The printed formula has $\omega_X^{\otimes -n}$; the surrounding argument is entirely about C, so the required $\omega_C^{\otimes -n}$ is retained here. If x is a non-singular point, $\mathfrak{m}_x$ is an invertible sheaf. If x is a double point, let $\pi : C' \to C$ be the result of blowing up x, and let $x_1, x_2 \in C'$ be the two points in $\pi^{-1}(x)$. Then it is easy to check that for any invertible sheaf $\mathcal{L}$ on C:

$$\operatorname{Hom}(\mathfrak{m}_x, \mathcal{L}) \cong H^0(C', \pi^*\mathcal{L}),$$
$$\operatorname{Hom}(\mathfrak{m}_x^2, \mathcal{L}) \cong H^0(C', \pi^*\mathcal{L}(x_1 + x_2)).$$

Therefore, we have 3 cases of (*) to check:

Case 1. — x, y non-singular points of C, then

$$H^0(\omega_C^{-n}(x + y)) = (0), \qquad \text{if } n \ge 2.$$

Case 2. — x double point of C, $\pi : C' \to C$ blowing up x, $\{x_1, x_2\} = \pi^{-1}(x)$, and y a non-singular point of C. Then

$$H^0(\pi^*(\omega_C)^{-n}(y)) = (0), \qquad H^0(\pi^*(\omega_C)^{-1}(x_1 + x_2)) = (0), \qquad \text{if } n \geq 2.$$

Case 3. — x, y double points of C, $\pi : C' \to C$ blowing up x and y. Then

$$H^0(\pi^*(\omega_C)^{-n}) = (0) \qquad \text{if } n \geq 2.$$

Now since the degree of ω_C^{-n} $(n \geq 2)$ on all components E of C is less than or equal to -2, all of this is clear, except in those cases where $n = 2$, the degree of ω_C on some E is 1, and in which two poles are allowed on E. This occurs if:

(i) case 1, $p_a(E) = 1$, E meets $C - E$ at only one point, $x, y \in E$.

(ii) case 1, $p_a(E) = 0$, E meets $C - E$ at only three points, $x, y \in E$.

(iii) case 2, E a rational curve with one double point meeting $C - E$ at one point, $x =$ double point of E.

But in all these cases, C has components besides E and a section in the H^0 in question must definitely vanish on all these other components. So at the points where E meets $C - E$, the section has extra zeroes. Since the sheaf in question has degree 0 on E, the section is zero on E too. Q.E.D.

Corollary. — Let $\pi : C \to S$ be any stable curve of genus $g \geq 2$. Then $\omega_{C/S}^{\otimes n}$ is relatively very ample if $n \geq 3$ and $\pi_*(\omega_{C/S}^{\otimes n})$ is a locally free sheaf on S of rank $(2n-1)(g-1)$.

Proof. — In fact, since for all $s \in S$, $H^1(\omega_{C/S}^{\otimes n} \otimes \mathcal{O}_{C_s}) = (0)$, it follows from [EGA, chap. 3, § 7], that $\pi_*(\omega_{C/S}^{\otimes n})$ is locally free and that $\pi_*(\omega_{C/S}^{\otimes n}) \otimes k(s) \cong H^0(\omega_{C/S}^{\otimes n} \otimes \mathcal{O}_{C_s})$. Therefore the corollary follows. Q.E.D.

Taking $n = 3$, it follows that every stable curve C/S can be realized as a family of curves in $\mathbf{P}^{5g-6}$ with Hilbert polynomial:

$$P_g(n) = (6n-1)(g-1).$$

Following standard arguments ([M$_1$], p. 99), it is easy to prove that there is a subscheme

$$H_g \subset \mathrm{Hilb}^{P_g}_{\mathbf{P}^{5g-6}}$$

of "all" tri-canonically embedded stable curves. (Hilb is the Hilbert scheme over $\mathbb{Z}$.) To be precise, there is an isomorphism of functors:

$$\mathrm{Hom}(S, H_g) \cong \left\{ \begin{array}{l} \text{set of stable curves } \pi : C \to S, \text{ plus isomorphisms:} \\ \mathbf{P}(\pi_*(\omega_{C/S}^{\otimes 3})) \cong \mathbf{P}^{5g-6} \times S\,, \quad \text{(modulo isomorphism).} \end{array} \right.$$

We will denote by $Z_g \subset H_g \times \mathbf{P}^{5g-6}$ the universal tri-canonically embedded stable curve. The functor of stable curves itself is the sheafification of the quotient of functors:

$$H_g / \mathrm{PGL}(5g-6).$$

We now consider the deformation theory of stable curves. Let k be any ground field. The deformation theory of X's smooth over k can be found in [SGA, 60–61]; for singular S's, the theory has been worked out in [Sc]. We shall indicate here the results of this theory for a scheme X which is

(i) one-dimensional;

(ii) generically smooth over k;

(iii) locally a complete intersection.

The advantages of this case are two-fold: first, the "cotangent complex" of Grothendieck, Lichtenbaum and Schlessinger reduces, in view of (ii) and (iii), to the single coherent sheaf $\Omega_{X/k}$, the Kähler differentials. Secondly, we have:

Lemma (1.3). — $\mathrm{Ext}^2(\Omega_{X/k}, \mathcal{O}_X) = (0)$.

Proof. — Use the spectral sequence:

$$H^p(X, \mathcal{E}xt^q(\Omega_{X/k}, \mathcal{O}_X)) \Rightarrow \mathrm{Ext}^{p+q}(\Omega_{X/k}, \mathcal{O}_X).$$

Then (i) $H^2(X, \mathcal{E}xt^0) = (0)$ since $\dim X = 1$. (ii) Since $\Omega_{X/k}$ is locally free except at a finite number of points, $\mathcal{E}xt^1(\Omega_{X/k}, \mathcal{O}_X)$ has 0-dimensional support, hence $H^1(X, \mathcal{E}xt^1) = (0)$. (iii) Locally, if we embed $X \subset \mathbf{A}^n$, then $\Omega_{X/k}$ has a free resolution of length 2:

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_{\mathbf{A}^n} \otimes \mathcal{O}_X \to \Omega_{X/k} \to 0$$

where $\mathcal{I}$ = sheaf of ideals defining X. Therefore $\mathcal{E}xt^2 = (0)$. Q.E.D.

In Schlessinger's theory, the significance of Lemma (1.3) is that all obstructions vanish, i.e., deformations of X over base schemes $\operatorname{Spec}(A/\mathcal{J})$ (A = local Artin ring with residue field k) can always be embedded in deformations over $\operatorname{Spec}(A)$. Moreover, the theory says that there is a canonical one-one correspondence between $\operatorname{Ext}^1(\Omega_{X/k}, \mathcal{O}_X)$ and the first order deformations of X, i.e., proper, flat morphisms p, and isomorphisms α as follows:

$$\begin{array}{ccccc} X_1 & \supset & X_1 \times_{\operatorname{Spec} k[\varepsilon]} \operatorname{Spec} k & \xrightarrow{\sim_\alpha} & X \\ \downarrow p & & \downarrow & & \downarrow \\ \operatorname{Spec} k[\varepsilon]/\varepsilon^2 & \supset & \operatorname{Spec} k & = & \operatorname{Spec} k. \end{array}$$

Since the obstructions vanish, there is a versal formal deformation $\mathfrak{X}$ of X over the base scheme

$$\mathfrak{M} = \operatorname{Spec} o_k[[t_1, \ldots, t_N]],$$

where $o_k = k$ if the char. is 0, or o_k is the complete regular local ring, with maximal ideal $p \cdot o_k$ and residue field k, if $\operatorname{char}(k) = p$, unique (by Cohen's structure theorem), and

$$N = \dim_k \operatorname{Ext}^1(\Omega_X, \mathcal{O}_C).$$

This means that $\mathfrak{X}$ is a formal scheme, proper and flat over $\mathcal{M}$, with fibre X over $\operatorname{Spec}(k)$ and the two properties:

a) Every deformation of X is induced from $\mathfrak{X}$, i.e., if A is a local Artin o_k-algebra with residue field k, and $p : Y \to \operatorname{Spec}(A)$ is proper and flat with fibre X over $\operatorname{Spec}(k)$, then there is a commutative diagram:

$$\begin{tikzcd} Y \simeq \mathfrak{X} \times_{\mathcal{M}} \operatorname{Spec}(A) \arrow[rr] \arrow[dd] & & \mathfrak{X} \arrow[dd] \\ & X \arrow[ul] \arrow[ur] \arrow[dd] & \\ \operatorname{Spec}(A) \arrow[rr] & & \mathcal{M} \\ & \operatorname{Spec}(k) \arrow[ul] \arrow[ur] & \end{tikzcd}$$

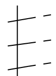
b) If $A = k[\varepsilon]/(\varepsilon^2)$, the above morphism f is uniquely determined by the diagram.

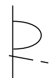
This implies that the tangent space to $\mathcal{M} \times_{o_k} k$ at its closed point is canonically isomorphic to $\operatorname{Ext}^1(\Omega_{X/k}, \mathcal{O}_X)$.

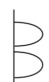
In case $\operatorname{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$, $\mathfrak{X}/\mathcal{M}$ is, in fact, *universal*: i.e., in property a), f is always unique, which means that the functor represented by $\mathcal{M}$ in the category of artin, local o_k-algebras is isomorphic to the functor of deformations Y/A of X. This fortunately holds for stable curves:

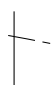
Lemma (1.4). — If X is a stable curve, $\operatorname{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$.

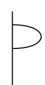
Proof. — We may assume that k is algebraically closed. Now a homomorphism from Ω_X to $\mathcal{O}_X$ is given by an everywhere regular vector field D on X. Such a vector field is given, in turn, by a regular vector field D' on the normalization X' of X which vanishes at all points of X' lying over the double points of X. In particular, D' and hence D vanishes identically on all components E of X whose normalization E' has genus ≥ 2. There remain the following possibilities for E:


E non-singular rational


E' rational, E one double pt.


E' rational, $E \geq 2$ double pts.


E non-singular elliptic


E' elliptic, $E \geq 1$ double points

In all cases where E' is rational, note that D' has to have at least 3 zeroes; and where E' is elliptic, D' has to have at least one zero. So D' vanishes on all components E'. This proves that $\mathrm{Ext}^0(\Omega_{X/k}, \mathcal{O}_X) = (0)$. Q.E.D.

Schlessinger's theory also allows us to trace what happens to the singularities of X in this deformation $\mathcal{M}$. For each closed point $x \in X$, he studies deformations of the complete local ring $\widehat{\mathcal{O}}_{x,X}$ alone, i.e., flat A-algebras $\mathcal{O}$ plus isomorphisms:

$$\mathcal{O} \otimes_A k \simeq \widehat{\mathcal{O}}_{x,X}.$$

This is a functor of A exactly as before. Since $\mathrm{Ext}^2_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x) = (0)$, there are no obstructions to extending deformations. First order deformations with $A = k[\varepsilon]/(\varepsilon^2)$ are classified by $\mathrm{Ext}^1_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x)$. And there is a versal formal deformation $\widehat{\mathcal{O}}$, which is a complete local ring and a flat $o_k[[t_1, \ldots, t_N]]$-algebra, $N = \dim_k \mathrm{Ext}^1_{\widehat{\mathcal{O}}_x}(\widehat{\Omega}_{x/k}, \widehat{\mathcal{O}}_x)$, such that

$$\widehat{\mathcal{O}}/(p, t_1, \ldots, t_N) = \widehat{\mathcal{O}}_{x,X}.$$

Whenever X is smooth over k at x, $N = 0$, and the theory is uninteresting. The first non-trivial example is a k-rational ordinary double point with k-rational tangent lines:

$$\widehat{\mathcal{O}}_{x,X} \simeq k[[u,v]]/(u \cdot v).$$

Then $N = 1$, and

$$\widehat{\mathcal{O}} \simeq o_k[[u, v, t_1]]/(uv - t_1).$$

In other words, if $\mathcal{O}$ is any deformation of $\widehat{\mathcal{O}}_{x,X}$ and u, v are lifted suitably into $\mathcal{O}$, then $u \cdot v = w \in A$ and $\mathcal{O}$ is induced from $\widehat{\mathcal{O}}$ via the homomorphism $o_k[[t_1]] \to A$ taking t_1 to w. This is easy to prove.

Finally, Schlessinger's theory connects global deformations to local ones. Let $\mathfrak{X}/\mathcal{M}_{gl}$ with $\mathcal{M}_{gl} = \mathrm{Spec}(A)$ be the versal global deformation, let $x_1, \ldots, x_k \in X$ be the points where X is not smooth over k, and let $\widehat{\mathcal{O}}_i$ as A_i-algebra be the versal deformation of the local ring $\widehat{\mathcal{O}}_{x_i,X}$. Let $\mathcal{M}_{lo} = \mathrm{Spec}(A_1 \widehat{\otimes}_{o_k} \cdots \widehat{\otimes}_{o_k} A_k)$. Then we may consider the local rings

$$\widehat{\mathcal{O}}_{x_i,\mathfrak{X}}$$

of $\mathfrak{X}$ at x_i. There are o_k-homomorphisms $\varphi_i : A_i \to A$ such that $\widehat{\mathcal{O}}_{x_i,\mathfrak{X}} \simeq \widehat{\mathcal{O}}_i \otimes_{A_i} A$. Dualizing, we obtain a morphism

$$\Phi = \prod_i \mathrm{Spec}(\varphi_i) : \mathcal{M}_{gl} \to \mathcal{M}_{lo}$$

which describes exactly how the various singularities of X behave in the versal deformation $\mathfrak{X}$. The final fact that we need is:

Proposition (1.5). — $\Phi : \mathcal{M}_{gl} \to \mathcal{M}_{lo}$ is formally smooth, i.e., there are isomorphisms

$$\mathcal{M}_{gl} \simeq \mathrm{Spec}\, o_k[[t_1, \ldots, t_{N+M}]], \qquad \mathcal{M}_{lo} \simeq \mathrm{Spec}\, o_k[[t_1, \ldots, t_N]]$$

such that $\Phi^*(t_i) = t_i$, $1 \le i \le N$.

Proof. — In view of the functorial significance of $\mathcal{M}_{gl}$ and $\mathcal{M}_{lo}$, this follows if we prove that the natural map:

$$(*) \qquad \mathrm{Ext}^1_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X) \longrightarrow \prod_{i=1}^k \mathrm{Ext}^1_{\widehat{\mathcal{O}}_{x_i}}(\widehat{\Omega}_{x_i}, \widehat{\mathcal{O}}_{x_i})$$

is surjective; $(*)$ is the induced map $d\Phi$ on the tangent spaces to $\mathcal{M}_{gl}$ and $\mathcal{M}_{lo}$. Since Ω_X is an invertible $\mathcal{O}_X$-Module outside the x_i's, it follows that:

$$\prod_{i=1}^k \mathrm{Ext}^1_{\widehat{\mathcal{O}}_{x_i}}(\widehat{\Omega}_{x_i}, \widehat{\mathcal{O}}_{x_i}) \simeq \prod_{i=1}^k \mathrm{Ext}^1_{\mathcal{O}_{x_i}}(\Omega_{x_i}, \mathcal{O}_{x_i}) \simeq H^0\big(X, \mathcal{E}xt^1_{\mathcal{O}_X}(\Omega_X, \mathcal{O}_X)\big).$$

Therefore, $(*)$ is surjective by the spectral sequence used in Lemma (1.3) and the fact that $H^2(X, \mathcal{E}xt^0(\Omega_X, \mathcal{O}_X)) = (0)$. Q.E.D.

In particular, suppose C is a stable curve over an algebraically closed ground field k, and let $x_1, \ldots, x_k$ be the double points of C. Let $N = \dim \mathrm{Ext}^1(\Omega_X, \mathcal{O}_X)$. Then C has a universal formal deformation $\mathcal{C}/\mathcal{M}$ where

$\mathcal{M} = \operatorname{Spec} o_k[[t_1, \dots, t_N]]$. Note that since in this case, the invertible sheaf $\omega_{\mathcal{C}/\mathcal{M}}$ is relatively ample, $\mathcal{C}$ is not only a formal scheme over $\mathcal{M}$, but also the formal completion of a unique scheme proper and flat over $\mathcal{M}$, which we will also denote by $\mathcal{C}$. $\mathcal{C}$ is clearly a stable curve over $\mathcal{M}$. Now each double point x_i has one modulus (cf. our example above) so the versal deformation space of the rings $\mathcal{O}_{x_i,C}$ is $o_k[[t'_1, \dots, t'_k]]$. By Proposition (1.5), we may identify t_i with t'_i, and we conclude that for suitable u_i, v_i:

$$\widehat{\mathcal{O}}_{x_i,\mathcal{C}} \simeq o_k[[u_i, v_i, t_1, \dots, t_N]]/(u_i v_i - t_i).$$

In particular, $t_i = 0$ is the locus in $\mathcal{M}$ where "x_i remains a double point".

The relation between the formal moduli space $\mathcal{M}$ of C and the local structure of H_g at a point x with $\kappa(x) = k$ corresponding to some tri-canonical model of C is exactly the same as in the case of non-singular curves ([M1], chap. 5, § 2). Let $\widehat{\mathcal{O}}_x$ be the completion of the local ring $\mathcal{O}_{x,H_g}$, and let $T = \operatorname{Spec}(\widehat{\mathcal{O}}_x)$. Let x denote the closed point of T too. The universal family of stable curves $Z_g \subset H_g \times \mathbf{P}^{5g-6}$ induces a family $Z' \subset T \times \mathbf{P}^{5g-6}$, whose fibre Z'_x over x is C. Then there is a unique morphism $f : T \to \mathcal{M}$ such that $Z' \simeq \mathcal{C} \times_{\mathcal{M}} T$, with this isomorphism restricting to the identity on the fibres over x, both of which are C. I claim that via f, T is formally smooth over $\mathcal{M}$, i.e., $\widehat{\mathcal{O}}_x \simeq o_k[[t_1, \dots, t_N, t_{N+1}, \dots, t_m]]$. In fact, by choosing an isomorphism $\mathbf{P}(\pi_*(\omega^{\otimes 3}_{\mathcal{C}/\mathcal{M}})) \simeq \mathbf{P}^{5g-6} \times \mathcal{M}$, we obtain a tri-canonical embedding $\mathcal{C} \subset \mathbf{P}^{5g-6} \times \mathcal{M}$ of $\mathcal{C}$, hence a morphism $s : \mathcal{M} \to H_g$ such that $\mathcal{C}$, with this embedding, is the pull-back of Z_g. Then s factors through T and $s : \mathcal{M} \to T$ is a section of f. On the other hand, consider the action of $\mathrm{PGL}(5g-6)$ on H_g. Let S_x be the stabilizer of the k-valued point x. Then S_x is finite and reduced. Because if it were not, S_x would have a non-trivial tangent space at the origin, i.e., there would be a $k[\varepsilon]/(\varepsilon^2)$-valued point of $\mathrm{PGL}(5g-6)$ centered at the identity, which maps the embedded stable curve $C \subset \mathbf{P}^{5g-6}$ corresponding to x into itself. But this action is given by an everywhere regular derivation on C, and we have

seen that all such vanish. This means that this $k[\varepsilon]/(\varepsilon^2)$-valued automorphism is the identity at all points of C and, since C is connected and spans $\mathbf{P}^{5g-6}$, the automorphism is the identity everywhere. Thus S_x is finite and reduced. It follows that the action of $\mathrm{PGL}(5g-6)$ on T is formally free, and hence that T is formally a principal fibre bundle over $\mathcal{M}$ with group $\mathrm{PGL}(5g-6)$. Therefore T is formally smooth over $\mathcal{M}$ as required.

Putting this together with what we know about $\mathcal{M}$, we conclude the following:

Let k be any algebraically closed field,
let $H'_g = H_g \times \operatorname{Spec}(o_k)$, $Z'_g = Z_g \times \operatorname{Spec}(o_k)$,
let $x \in H'_g$ be a closed point,
let $C \subset Z'_g$ be the stable curve over x,
let $x_1, \dots, x_k \in C$ be its double points.

Then

Theorem (1.6). — There are isomorphisms

$$\widehat{\mathcal{O}}_{x,H'_g} \simeq o_k[[t_1, \dots, t_N]],$$

$$\widehat{\mathcal{O}}_{x_i,Z'_g} \simeq o_k[[u_i, v_i, t_1, \dots, t_N]]/(u_i v_i - t_i).$$

Corollary (1.7). — H_g is smooth over $\mathbb{Z}$. In particular, for all algebraically closed fields k, $H_g \times \operatorname{Spec}(k)$ is a disjoint union of a finite number of non-singular algebraic varieties over k.

Let

$$H^0_g = \{x \in H_g \mid \text{the corresponding stable curve } (Z_g)_x \text{ is non-singular}\},$$
$$S = \{x \in Z_g \mid \text{the projection } \pi : Z_g \to H_g \text{ is not smooth at } x\}.$$

Definition (1.8). — Let $p : X \to Y$ be a smooth morphism of finite type, with Y a noetherian scheme, and let $D \subset X$ be a relative Cartier divisor. Then D has normal crossings relative to Y if for all $x \in D$, the local equation $d = 0$ of D decomposes in the strict completion[3] $\widetilde{\mathcal{O}}_{x,X}$ of $\mathcal{O}_{x,X}$ as $d = d_1 \cdots d_k$, where $d_1, \dots, d_k$ are linearly independent in

$$\widetilde{\mathfrak{m}}_{x,X} / \big(\widetilde{\mathfrak{m}}^2_{x,X} + \mathfrak{m}_{y,Y} \widetilde{\mathcal{O}}_{x,X}\big),$$

[3] The complete local ring, formally etale over $\mathcal{O}_{x,X}$ with residue field the separable closure of $\mathcal{O}_{x,X}/\mathfrak{m}_{x,X}$.

with $y = p(x)$.

Corollary (1.9). — $H_g^0 = H_g - S^*$, where S^* is a divisor with normal crossings relative to $\mathbb{Z}$. Z_g and S are smooth over $\mathbb{Z}$, and the projection $p : S \to S^*$ is finite and an isomorphism at all points where S^* is smooth over $\mathbb{Z}$, i.e., S is the normalization of S^*.

Proof. — In the notation of Theorem (1.6), S^* is defined in $\widehat{\mathcal{O}}_{x,H'_g}$ by the local equation $t_1 \cdots t_k = 0$. And

$$\widehat{\mathcal{O}}_{x_i,Z'_g} \simeq o_k[[u_i, v_i, t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_N]],$$

$$\widehat{\mathcal{O}}_{x_i,S} \simeq \widehat{\mathcal{O}}_{x_i,Z'_g}/(u_i, v_i) \simeq o_k[[t_1, \ldots, t_{i-1}, t_{i+1}, \ldots, t_N]].$$

Q.E.D.

Next we take up the isomorphisms and automorphisms of stable curves. Suppose $p : X \to S$, $q : Y \to S$ are two stable curves:

Definition (1.10). — *$\underline{\mathrm{Isom}}_S(X, Y)$ is the functor on (Sch/S) associating to each S-scheme S' the set of S'-isomorphisms between $X \times_S S'$ and $Y \times_S S'$. If $X = Y$, we denote $\underline{\mathrm{Isom}}_S(X, X)$ by $\underline{\mathrm{Aut}}_S(X)$.*

Since both X and Y have the canonical polarizations $\omega_{X/S}$, $\omega_{Y/S}$ respectively, any isomorphism $f : X \to Y$ must satisfy $f^*(\omega_{Y/S}) \simeq \omega_{X/S}$. Therefore, by Grothendieck's results on the representability of the Hilbert scheme and related functors [Gr_1], we conclude that $\underline{\mathrm{Isom}}_S(X, Y)$ is represented by a scheme $\mathrm{Isom}_S(X, Y)$, quasi-projective over S. Concerning this scheme, we have:

Theorem (1.11). — *$\mathrm{Isom}_S(X, Y)$ is finite and unramified over S.*

Proof. — To check that $\mathrm{Isom}_S(X, Y)$ is unramified, we may take S to be the spectrum of an algebraically closed field k, in which case $\mathrm{Isom}_S(X, Y)$ is either empty or isomorphic to $\mathrm{Aut}_k(X)$. A point of $\mathrm{Aut}_k(X)$ with values in $k[\varepsilon]/(\varepsilon^2)$ with image the identity may be identified with a vector field on X. By Lemma (1.4), stable curves have no non-zero vector fields. This proves that $\mathrm{Isom}_S(X, Y)$ is unramified over S, and since it is also of finite type over S, it is quasi-finite over S. It remains to check that $\mathrm{Isom}_S(X, Y)$ is proper over S.

Locally over S, X and Y are the pull-backs of the universal tri-canonically embedded stable curve by some morphisms from S to H_g, so that it suffices to prove the properness of $\mathrm{Isom}_S(X, Y)$ in the "universal" case where $S = H_g \times H_g$, X and Y being the two inverse images of the universal curve on H_g. In that case, the open subset of $\mathrm{Isom}_S(X, Y)$ corresponding to smooth curves is dense, so that the Theorem follows from the valuative criterion of properness which holds by:

Lemma (1.12). — *Let X and Y be two stable curves over a discrete valuation ring R, with algebraically closed residue field. Denote by η and s the generic and closed points of $\mathrm{Spec}(R)$, and assume that the generic fibres X_η and Y_η of X and Y are smooth. Then any isomorphism φ_η between X_η and Y_η extends to an isomorphism φ between X and Y.*

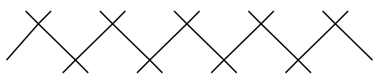
(A posteriori, it follows from Theorem (1.11) that the lemma holds for any valuation ring R and without assuming X_η or Y_η smooth.)

Proof. — Another way to put the lemma is that if we start with a smooth curve X_η of genus $g \geq 2$ over the quotient field K of R, there is, up to canonical isomorphism, at most one stable curve X over R with X_η as its generic fibre. We shall deduce this from the analogous uniqueness assertion for minimal models ([L] and [S]): given a smooth curve X_η of genus $g \geq 1$ over K, there is, up to canonical isomorphism, at most one regular 2-dimensional scheme $\widetilde{X}$, proper and flat over R, with X_η as its generic fibre, without exceptional curves of the first kind in $\widetilde{X}_s$.

Let z denote a generator of the maximal ideal of R and consider the affine plane curve C_n over R given by:

$$xy = z^n.$$

Let $\widetilde{C}_n$ denote the scheme obtained by: 1) blowing up the maximal ideal at the unique singularity of C_n; 2) blowing up the maximal ideal at the unique singularity of this scheme..., and so on $\left[\frac{n}{2}\right]$ times. It is easy to check that $\widetilde{C}_n$ is a regular scheme whose special fibre is the same as that of C_n except that the singular point is replaced by a sequence of $n-1$ projective lines as follows:



Now suppose x is a singular point of the stable curve X over R. At x, X is formally isomorphic as scheme over R to one of the schemes C_n, so we may blow up X the same way we blew up C_n. If we do this for all singular points of X, we get a regular scheme $\widetilde{X}$ with generic fibre X_η. In addition, any non-singular rational component of $\widetilde{X}_s$ is linked to the other irreducible components by at least two points, hence it is not exceptional of first kind. Therefore $\widetilde{X}$ is the minimal model of X_η. Note finally that C_n is a normal scheme, hence so is X; therefore X is the unique normal scheme obtained from $\widetilde{X}$ by contracting all non-singular rational components of $\widetilde{X}_s$ linked to the other irreducible components by exactly two points. This proves that X is essentially unique. Q.E.D.

Another important fact about the automorphisms of stable curves is:

Theorem (1.13). — Let k be an algebraically closed field and X a stable curve over k. Let $\mathrm{Pic}^0(X)$ *denote the group of invertible sheaves on X of degree 0 on each component. Then the map (of ordinary groups):*

$$\mathrm{Aut}_k(X) \longrightarrow \mathrm{Aut}_k(\mathrm{Pic}^0(X))$$

is injective.

Proof. — Let φ be an automorphism of X inducing the identity on $\mathrm{Pic}^0 X$.

Lemma (1.14). — If X is smooth, then φ is the identity.

Proof. — If not, by the Lefschetz-Weil fixed point formula, the number n of fixed points of φ, counted with their multiplicities, is

$$n = 1 - \mathrm{Tr}(\varphi, T_l(\mathrm{Pic}^0(X))) + 1 = 2 - 2g < 0,$$

which is absurd. Q.E.D.

Lemma (1.15). — If X is irreducible, then φ is the identity.

Proof. — Let φ' be the action of φ on the normalization X' of X. Each singular point of X, together with an ordering of its two inverse images in X', defines a distinct morphism from $\mathbb{G}_m$ to $\mathrm{Pic}^0(X)$, so that the inverse image S of the singular locus of X is pointwise fixed by φ'. One has either

a) $\mathrm{genus}(X') \geq 2$: then conclude by Lemma (1.14);

b) $\mathrm{genus}(X') = 1$, $|S| \geq 2$, then φ' is a translation on X' leaving a point fixed, and so φ' is the identity;

c) X' is the projective line, $|S| \geq 4$ and φ' is a projectivity leaving more than three points fixed, so is the identity. Q.E.D.

Let Γ be the following (unoriented) graph:

(i) The set of vertices of Γ is the set Γ^0 of irreducible components of X,

(ii) the set of edges of Γ is the set Γ^1 of the singular points of X which lie on two distinct irreducible components,

(iii) an edge $x \in \Gamma^1$ has for extremities the irreducible components on which x lies.

Lemma (1.16). — If φ induces the identity on Γ, then φ is the identity.

Proof. — If X_1 is an irreducible component of X, then $\varphi(X_1) = X_1$ and φ leaves fixed the points of intersection of X_1 with the other components. In addition, $\mathrm{Pic}^0(X)$ maps onto $\mathrm{Pic}^0(X_1)$ so that φ acts trivially on $\mathrm{Pic}^0(X_1)$. Either:

a) $\mathrm{genus}(X_1) \geq 2$ and $\varphi|X_1$ is the identity by Lemma (1.15);

b) $\mathrm{genus}(X_1) = 1$, φ acts by a translation and leaves a point fixed, so is the identity on X_1;

c) X_1 is the projective line and φ leaves fixed at least three points, so is the identity on X_1. Q.E.D.

Lemma (1.17). — (i) Any edge in Γ has distinct extremities.

(ii) Any vertex which is the extremity of 0, 1 *or* 2 *edges is fixed by φ.*

(iii) φ acts trivially on $H^1(\Gamma, \mathbb{Z})$.

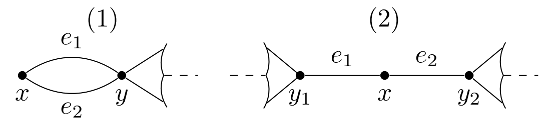
Proof. — It is easy to check that the subgroup of $\mathrm{Pic}^0(X)$ corresponding to invertible sheaves whose restriction to each irreducible component of X is trivial is canonically isomorphic to

$$H^1(\Gamma, \mathbb{Z}) \otimes \mathbb{G}_m.$$

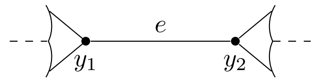
This implies (iii), and (i) is trivial.

The morphism from $\mathrm{Pic}^0(X)$ to the product $\prod_i \mathrm{Pic}^0(X_i)$, extended over the irreducible components of X, is surjective, so that if $\mathrm{Pic}^0(X_i) \neq \{e\}$, then $\varphi(X_i) = X_i$. This is the case, unless X_i is a projective line, linked to the other components in at least three points. Q.E.D.

We prove now that if an automorphism φ of any finite graph Γ has the properties stated in Lemma (1.17), it is the identity. Make induction on the sum of the number of vertices and edges of Γ. If Γ has an isolated point x, then $\varphi(x) = x$ so let $\Gamma^* = \Gamma - \{x\}$. Then φ = identity on Γ^* by induction, so φ = identity on Γ too. If Γ has an extremity x, then $\varphi(x) = x$, and again let Γ^* be Γ minus x and the edge abutting at x. Then Γ^* has all the properties Γ has, so φ = identity on Γ^*, hence φ = identity on Γ. If Γ has a vertex x on which only 2 edges abut, we have one of the two cases:



In the first case, $\varphi(y) = y$, and let Γ^* be Γ minus x, e_1 and e_2. Then φ = identity on Γ^* and $\varphi(e_i) = e_i$ too, since if φ reverses the e_i's, this contradicts (iii). In the second case, let Γ^* be Γ minus x, and with e_1 and e_2 identified:



Then φ = identity on Γ^*, so φ = identity on Γ. Next, say Γ has an edge e with extremities x and y such that $\varphi(x) = x$, $\varphi(y) = y$, $\varphi(e) = e$. Let Γ^* be Γ minus e. Then φ = identity on Γ^*, so φ = identity on Γ. If none of these reductions are possible, we must be in a situation where a) every vertex is the abutment of at least three edges and b) no edge is left fixed. It is easily seen that the first Betti number b_1 of any of the connected components of Γ is at least 2. Let

$$n_0 = \text{number of fixed vertices},$$
$$n_1 = \text{number of edges reversed by } \varphi.$$

Then, unless $\Gamma = \varnothing$, the Lefschetz fixed point formula reads:

$$n_0 + n_1 = b_0 - b_1 < 0,$$

which is impossible. Q.E.D.

§2. Degenerations of curves and their jacobians.

We consider the situation:

K = discretely-valued field;
R = integers in K, $\quad k = R/\mathfrak{M}$ = residue field (assumed algebraically closed);
$S = \mathrm{Spec}(R)$, $\quad \eta$ and s its generic and closed points respectively;
C = a curve, smooth, geometrically irreducible and proper over K, of genus $g \geq 2$;
J = the jacobian variety of C;
$\mathscr{J}$ = the Néron model of J over R (cf. $[N]$);
$\mathscr{J}^0 \subset \mathscr{J}$ the open subgroup scheme with $\mathscr{J}_s^0$ = identity component of $\mathscr{J}_s$;
$\mathscr{C}$ = the minimal model of C over R.

A word about the existence and uniqueness of $\mathscr{C}$ is needed. We recall that $\mathscr{C}$ is to be a regular scheme, flat and proper over R, with generic fibre $\mathscr{C}_\eta = C$ such that for any other regular scheme $\mathscr{C}'$, flat over R, with generic fibre $\mathscr{C}'_\eta = C$, the birational map $\mathscr{C}' \to \mathscr{C}$ is a morphism. Šafarevič in [Š] and Lichtenbaum [L] have proven that such a $\mathscr{C}$ (which is obviously unique) exists, provided that there is some regular $\mathscr{C}'$, proper and flat over R, with generic fibre C. And, in fact, that $\mathscr{C}$ is projective over R. To construct such a $\mathscr{C}$, proceed as follows: first let $\mathscr{C}''$ be any scheme, projective and flat over R with generic fibre C. Let $\widehat{R}$ be the completion of R, and let

$$\widehat{\mathscr{C}}'' = \mathscr{C}'' \times_{\operatorname{Spec} R} \operatorname{Spec} \widehat{R}.$$

Then $\widehat{\mathscr{C}}''$ is an excellent surface, so by [Ab] and by unpublished results of Hironaka, there is a sheaf of ideals $\widehat{\mathcal{J}}$ with support in the singular locus of $\widehat{\mathscr{C}}''$ such that blowing up $\widehat{\mathcal{J}}$ leads to a regular surface $\widehat{\mathscr{C}}'$. But then

$$\widehat{\mathcal{J}} \supset \mathfrak{M}^n \cdot \mathcal{O}_{\widehat{\mathscr{C}}''}$$

for some n, so $\widehat{\mathcal{J}}$ is induced by a unique sheaf of ideals $\mathcal{J} \subset \mathcal{O}_{\mathscr{C}''}$. Let $\mathscr{C}'$ be obtained by blowing up $\mathcal{J}$. Then

$$\widehat{\mathscr{C}}' \simeq \mathscr{C}' \times_{\operatorname{Spec} R} \operatorname{Spec} \widehat{R}$$

so $\mathscr{C}'$ is regular. $\mathscr{C}'$ is also projective over R, hence a projective $\mathscr{C}$ exists.

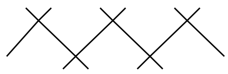
Definition (2.1). — J has stable reduction if $\mathscr{J}_s$ has no unipotent radical.

Definition (2.2). — C has stable reduction in sense 1 if $\mathscr{C}_s$ is reduced and has only ordinary double points. C has stable reduction in sense 2 if there is a stable curve $\mathscr{C}'$ over R with generic fibre $\mathscr{C}'_\eta = C$.

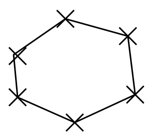
Note that if a stable $\mathscr{C}'$ exists, then by Theorem (1.11) it is unique.

Proposition (2.3). — The two senses of stable reduction for C are equivalent.

Proof. — Say a stable $\mathscr{C}'$ exists. Blowing up the singularities of $\mathscr{C}'$ as in Lemma (1.12), we obtain the minimal model $\mathscr{C}$ of C and it is seen that $\mathscr{C}_s$ is reduced with only ordinary double points. Conversely, suppose the minimal model $\mathscr{C}$ has this property. Let $E_1, \dots, E_n$ be the non-singular rational components of $\mathscr{C}_s$ which meet the other components in only two points. Then the E_i's divide into several chains of the type:



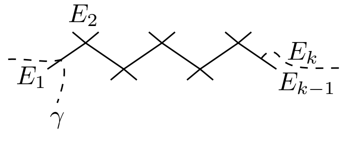
unless the entire fibre consisted of E_i's and has the type:



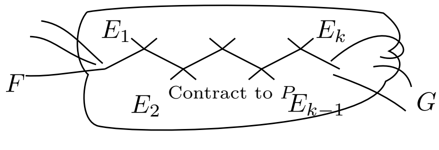
$n \geq 2$
$\mathscr{C}_s =$ loop of E_i's
$(E_i^2)_{\text{on}\,\mathscr{C}} = -2.$

But in this case genus(C) = genus$(\mathscr{C}_s) = 1$, which contradicts our assumption. Now, according to Theorem (27.1) of Lipman [Li] (generalizing a result of Artin [A$_2$], which works for surfaces of finite type over a field) any set of k non-singular rational curves connected in a chain as above on a regular surface X, with self-intersection -2 on X, can be blown down to a rational double point P of type A_k on a normal surface X_0. Reversing the process and blowing up a rational double point of type A_k, it is easy to see that non-singular branches γ on X, crossing transversally only the first or the last rational curve in the chain:

Source note. The printed prose says "with self-intersection 2 on X"; the displayed loop immediately above and the contraction argument require self-intersection -2, which is retained here.



are still non-singular branches when mapped to X_0; and if γ_1, γ_2 intersect E_1 or E_k in distinct points, then they cross transversally on X_0. Therefore, suppose we blow down all the chains of E_i's on $\mathscr{C}$. Let $\mathscr{C}'$ be the normal surface so obtained. Then if one of these chains fits into $\mathscr{C}_s$ like this:



then on $\mathscr{C}'_s$, the images of F and G still have only ordinary double points, each has one non-singular branch through the singular point P, and these branches cross transversally. Therefore $\mathscr{C}'$ is a stable curve over R. Q.E.D.

We are now ready to prove the key result on which our proof of irreducibility depends:

Theorem (2.4). — J has stable reduction if and only if C has stable reduction.

Proof. — The connection between $\mathscr{C}$ and $\mathscr{J}$ is based on the following result of Raynaud [R]:

Theorem (2.5). — If $\mathscr{C}$ and $\mathscr{J}$ are as above, and the greatest common denominator d of the multiplicities of the components of $\mathscr{C}_s$ is 1, then $\mathscr{J}^0$ represents the functor $\underline{\mathrm{Pic}}^0(\mathscr{C}/S)$.

(This result is not stated as such in [R]. It comes out like this, in the terminology of his paper:

a) Condition (N) is verified and $p_*(\mathcal{O}_{\mathscr{C}}) = \mathcal{O}_S$, so

b) $\mathscr{C}$ is cohomologically flat over S in dim. 0 by Theorem 4;

c) therefore $\underline{\mathrm{Pic}}^0$ is representable and separated over S by Theorem 3;

d) since $E = \overline{\{0\}}$ and $Q = P/E$, we find $P^0 = Q^0$, and since $\mathscr{C}$ is 1-dimensional over S, P^0 and Q^0 are smooth over S; therefore $R^0 = Q^0$ and $\widetilde{R} = R$.

e) Then by Theorem 5, Pic^0 is the identity component of the Néron model of $\mathrm{Pic}^0(\mathscr{C}_\eta) = J$.)

Source note. The printed text says "in the terminology of is paper"; the evident omitted letter is restored as "his paper" above.

Now assume that C has stable reduction. Then $\mathscr{C}_s$ is reduced so $d = 1$. By Theorem (2.5), $\mathscr{J}^0 = \mathrm{Pic}^0(\mathscr{C}/S)$. Therefore $\mathscr{J}^0_s = \mathrm{Pic}^0(\mathscr{C}_s/k)$. Since $\mathscr{C}_s$ is reduced with only ordinary double points, its generalized jacobian $\mathrm{Pic}^0(\mathscr{C}_s/k)$ is an extension of an abelian variety by a torus, i.e., has no unipotent radical. Therefore J has stable reduction.

The converse is more difficult. Assume J has stable reduction. We first prove that C has stable reduction under the additional hypothesis that C has a K-rational point.[4] In this case, $\mathscr{C}$ has an R-rational point, and since $\mathscr{C}$ is regular, sections of $\mathscr{C}$ over R pass through components of $\mathscr{C}_s$ of multiplicity one. Therefore $d = 1$, and

Theorem (2.5) applies. In particular, $\mathrm{Pic}^0(\mathscr{C}_s/k) = \mathscr{J}^0_s$, so $\mathrm{Pic}^0(\mathscr{C}_s/k)$ has no unipotent radical. We apply:

Lemma (2.6). — Let D be a complete 1-dimensional scheme over k such that $H^0(\mathcal{O}_D) \simeq k$, and such that the generalized jacobian of D has no unipotent radical. Then

(i) $\chi(\mathcal{O}_D) = \chi(\mathcal{O}_{D_{\mathrm{red}}})$;

(ii) *the singularities of D_{red} are all transversal crossings of a set of non-singular branches (i.e., analytically isomorphic to the union of the coordinate axes in $\mathbf{A}^n$).*

Proof. — Let $\mathcal{J} \subset \mathcal{O}_D$ be the ideal of nilpotent elements. Filtering $\mathcal{J}$ by a chain of ideals $\mathcal{J}_k$ such that $\mathcal{J}\mathcal{J}_k \subset \mathcal{J}_{k+1}$, and using the exact sequence

$$0 \longrightarrow \mathcal{J}_k/\mathcal{J}_{k+1} \xrightarrow{a \mapsto 1+a} (\mathcal{O}_D/\mathcal{J}_{k+1})^* \longrightarrow (\mathcal{O}_D/\mathcal{J}_k)^* \longrightarrow 0,$$

it is easy to deduce that $\mathrm{Pic}^0(D/k)$ is an extension of $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$ by a unipotent group. Therefore, since by assumption $\mathrm{Pic}^0(D/k)$ has no unipotent subgroups,

$$\mathrm{Pic}^0(D/k) \simeq \mathrm{Pic}^0(D_{\mathrm{red}}/k).$$

Since $H^1(\mathcal{O}_D)$, resp. $H^1(\mathcal{O}_{D_{\mathrm{red}}})$, is naturally isomorphic to the Zariski tangent space to $\mathrm{Pic}^0(D/k)$, resp. $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$, it follows that

$$H^1(\mathcal{O}_D) \simeq H^1(\mathcal{O}_{D_{\mathrm{red}}}),$$

hence (i) is proven. Let $\pi : C \to D_{\mathrm{red}}$ be the normalization of D_{red} and let D^* be the local ringed space which, as topological space is D, and whose structure sheaf is given by

$$\Gamma(U, \mathcal{O}_{D^*}) = \{f \in \Gamma(U, \pi_*(\mathcal{O}_C)) \mid x_1, x_2 \in \pi^{-1}(U),\ f(x_1) = f(x_2) \text{ if } \pi(x_1) = \pi(x_2)\}.$$

It is easy to check that D^* is a 1-dimensional scheme whose singularities are all transversal crossings of a set of non-singular branches and that π factors:

$$C \xrightarrow{\pi'} D^* \xrightarrow{\pi''} D_{\mathrm{red}}.$$

[4] This is in fact the only case which will be needed in our application to questions of irreducibility.

I claim that $\pi'' : D^* \to D_{\mathrm{red}}$ is an isomorphism. Filter $\mathcal{O}_{D^*}/\mathcal{O}_{D_{\mathrm{red}}}$ so as to obtain a chain of coherent $\mathcal{O}_{D_{\mathrm{red}}}$-algebras

$$\mathcal{O}_{D^*} = \mathcal{O}^{(0)} \supset \mathcal{O}^{(1)} \supset \cdots \supset \mathcal{O}^{(k)} = \mathcal{O}_{D_{\mathrm{red}}}$$

such that $\ell(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)}) = 1$. Equivalently, this factors π'':

$$D^* = D_0 \longrightarrow D_1 \longrightarrow \cdots \longrightarrow D_k = D_{\mathrm{red}},$$

where $\mathcal{O}^{(n)} \simeq \mathcal{O}_{D_n}$, and all arrows are homeomorphisms. If $\{x_n\} = \mathrm{Supp}(\mathcal{O}^{(n)}/\mathcal{O}^{(n+1)})$, then we get an exact sequence:

$$0 \longrightarrow \mathcal{O}^*_{D_{n+1}} \longrightarrow \mathcal{O}^*_{D_n} \longrightarrow k_{x_n} \longrightarrow 0$$

(k_y denotes the residue field at y, as sheaf on D), and hence:

$$0 \longrightarrow k \longrightarrow H^1(\mathcal{O}^*_{D_{n+1}}) \longrightarrow H^1(\mathcal{O}^*_{D_n}) \longrightarrow 0.$$

It follows easily that $\mathrm{Pic}^0(D_{n+1}/k)$ is an extension of $\mathrm{Pic}^0(D_n/k)$ by G_a. But $\mathrm{Pic}^0(D_{\mathrm{red}}/k)$ has no unipotent subgroups, and this can only happen if $D_{\mathrm{red}} = D^*$. Q.E.D. for lemma.

We apply the lemma to $\mathscr{C}_s$. According to Lichtenbaum [L] and Šafarevič [Š], there is a divisor K on $\mathscr{C}$ such that for all positive divisors D lying over the closed point of $\mathrm{Spec}(R)$, we have:

$$\chi(\mathcal{O}_D) = -\frac{(D \cdot (D + K))}{2}.$$

Source note. The printed display has the extraneous lower-case factor kK in place of K. The standard adjunction formula, and every immediately following use of it, require $D + K$ as delivered above.

Let $E_1, \ldots, E_n$ be the components of $\mathscr{C}_s$, $d_1, \ldots, d_n$ their multiplicities. Then conclusion (i) of the lemma implies that:

$$\Big(\sum_i d_i E_i \cdot \big(\sum_i d_i E_i + K\big)\Big) = \Big(\sum_i E_i \cdot \big(\sum_i E_i + K\big)\Big).$$

But $((\sum_i d_i E_i) \cdot E_k) = 0$, all k, since $\sum_i d_i E_i$ is the divisor of a function $\pi \in R$ if $(\pi) =$ maximal ideal of R. Therefore:

$$(*) \qquad \Big(\big(\sum_i (d_i - 1) E_i\big) \cdot K\Big) = \Big(\sum_i E_i \cdot \sum_i E_i\Big).$$

Note that at least one d_i equals 1 since $\mathscr{C}$ has a section over $\mathrm{Spec}(R)$ and every section must pass through a component of $\mathscr{C}_s$ of multiplicity 1. Moreover the intersection matrix $(E_i \cdot E_j)$ is negative semidefinite, with one-dimensional degenerate subspace generated by $\sum_i d_i E_i$, hence if some $d_i > 1$, it follows that $(\sum_i E_i \cdot \sum_i E_i) < 0$. Therefore, by $(*)$, $(E_{i_0} \cdot K) < 0$ for some i_0. Then we have:
Source note. The printed text says "negative indefinite" while simultaneously specifying a one-dimensional degenerate subspace. The mathematically consistent reading "negative semidefinite" is delivered above.

a) $(E_{i_0} \cdot K) < 0$;

b) $(E_{i_0} \cdot E_{i_0}) < 0$;

c) $(E_{i_0} \cdot (E_{i_0} + K)) = -2\chi(\mathcal{O}_{E_{i_0}}) \geq -2$;

hence in fact $(E_{i_0} \cdot E_{i_0}) = (E_{i_0} \cdot K) = -1$, so E_{i_0} is an exceptional curve of the first kind. This contradicts our assumption that on $\mathscr{C}$ all possible curves have been blown down. Therefore $d_i = 1$, all i.

This proves that $\mathscr{C}_s$ is reduced. By conclusion (ii) of the lemma, plus the fact that the dimension of its Zariski tangent-space is everywhere one or two (since $\mathscr{C}_s$ lies on a regular surface $\mathscr{C}$), we deduce that $\mathscr{C}_s$ has only double points. This proves that C has stable reduction in the case that $C(K) \neq \varnothing$.

In the general case, C will acquire a rational point in a finite extension K' of K. Let S' be the spectrum of the localisation at some maximal ideal of the integral closure of R in K'; S' is the spectrum of a valuation ring and is faithfully flat over S. We put $S'' = S' \times_S S'$ and denote by C' and C'' the inverse images of C on $S'_\eta = \mathrm{Spec}(K')$ and S''_η respectively.

$$\begin{tikzcd}
C'' \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & C' \arrow[r] \arrow[d] & C \arrow[d] \\
S'' \arrow[r, shift left, "\mathrm{pr}_1"] \arrow[r, shift right, "\mathrm{pr}_2"'] & S' \arrow[r, "p"] & S
\end{tikzcd}$$

Let $\overline{C}'$ be the stable curve on S' having C' as generic fibre. The restriction, C', of $\overline{C}'$ to S'_η carries a descent datum with respect to p, i.e. an isomorphism, φ_η, between the restrictions of $\mathrm{pr}_1^*(\overline{C}')$ and $\mathrm{pr}_2^*(\overline{C}')$ to S''_η. It remains only to extend the isomorphism φ_η to an isomorphism, φ, between the $\mathrm{pr}_i^*(\overline{C}')$. Then φ will be a descent datum for $\overline{C}'$ with respect to p. As $\overline{C}'$ is canonically polarized (1.2), this descent datum will be effective, and so define a stable curve, $\overline{C}$, over S with generic fibre C.

Because $\mathscr{J}_s^0$ has no unipotent radical, the inverse image, $p^*\mathscr{J}^0$, of $\mathscr{J}^0$ on S' is the identity component of the Néron Model of the jacobian of C', so that, defining $q = p \circ \mathrm{pr}_1 = p \circ \mathrm{pr}_2$, one has

$$\mathrm{Pic}^0(\overline{C}'/S') = p^*\mathscr{J}^0,$$

$$\mathrm{Pic}^0(\mathrm{pr}_1^*\overline{C}') = \mathrm{Pic}^0(\mathrm{pr}_2^*\overline{C}') = q^*\mathscr{J}^0.$$

We denote by T the closed subscheme of $\mathrm{Isom}(S'', \mathrm{pr}_1^*(\overline{C}'), \mathrm{pr}_2^*(\overline{C}'))$ corresponding to those isomorphisms which induce (via the preceding identifications) the identity on the inverse image of $\mathscr{J}^0$.

By (1.11), T is finite and unramified over S'', and by (1.13) T is radicial over S''. We conclude that the morphism from T to S'' identifies T with a closed subscheme X of S''. As X contains S''_η, which is schematically dense in S'', we have that X is S'' and T "is" the desired section, φ, of $\mathrm{Isom}(S'', \mathrm{pr}_1^*(\overline{C}'), \mathrm{pr}_2^*(\overline{C}'))$ over S'' which extends φ_η. Q.E.D.

Combining Proposition (2.3), Theorem (2.4), and the stable reduction theorem for abelian varieties quoted in the introduction, we obtain the most important consequence:

Corollary (2.7). — Let R be a discrete valuation ring with quotient field K. Let C be a smooth geometrically irreducible curve over K of genus $g \geq 2$. Then there exists a finite algebraic extension L of K and a stable curve $\mathscr{C}_L$ over R_L, the integral closure of R in L, with generic fibre $\mathscr{C}_{L,\eta} = C \times_K L$.

§3. Elementary derivation of the theorem.

Let k be an algebraically closed field of char. $p \neq 0$. We use the notation of § 1, except that we will now denote by H_g the product $H_g \times \mathrm{Spec}(k)$ of the previous H_g with $\mathrm{Spec}(k)$: it is a disjoint union of non-singular varieties $H_{g,1}, \ldots, H_{g,n}$ over k and is the subscheme of $\mathrm{Hilb}^{P_g}_{\mathbf{P}^{5g-6}/k}$ of tri-canonical stable curves. Similarly, H_g^0 is the open dense subset of H_g of tri-canonical non-singular curves. By the results of [M_1], we know that a coarse geometric quotient

$$M_g^0 = H_g^0 / \mathrm{PGL}(5g-6)$$

exists, that it is a disjoint union of normal varieties over k and is the coarse moduli space for non-singular curves of genus g. Let $S^* = H_g - H_g^0$. Then everything decomposes into the same set of components.

Let

$$\begin{array}{rcl} H_{g,i} & = & \text{components of } H_g, \\ \text{then } H_{g,i}^0 & = & H_{g,i} \cap H_g^0 = \text{components of } H_g^0, \\ \text{and } M_{g,i}^0 & = & H_{g,i}^0/\mathrm{PGL}(5g-6) = \text{components of } M_g^0, \\ \text{and } S^* & = & \text{disjoint union of } S_1^*, \ldots, S_n^*, \quad S_i^* = H_{g,i} \cap S^*. \end{array}$$

We want to prove that M_g^0, or equivalently H_g^0, or equivalently H_g is irreducible. We shall use: (i) the fact that these statements are true in char. 0; (ii) the inductive assumption that these statements are true for smaller genus.

Step I. — No component of M_g^0 is complete (i.e., proper over k).

Proof. — Here we use the char. 0 result. By [M_1], there is a scheme X, quasi-projective over $\mathrm{Spec}(W(k))$, $W(k)$ the Witt vectors, whose closed fibre is M_g^0, and whose generic fibre X_η is the char. 0 coarse moduli space over the quotient field of $W(k)$. In particular, X_η is known to be connected. Since X is quasi-projective over $W(k)$, we can embed X as an open dense subset of a scheme $\overline{X}$ projective over $W(k)$. $\overline{X}_\eta$ is still connected, hence by the connectedness theorem of Enriques-Zariski [EGA 3], the closed fibre $\overline{X}_0$ of $\overline{X}$ is connected. But if Y were a complete variety which is a component of M_g^0, then: a) Y is an open subset of M_g^0, which is an open subset of $\overline{X}_0$, and: b) since Y is proper/k, Y would be a closed subset of $\overline{X}_0$ too. Therefore, $Y = \overline{X}_0$, hence M_g^0 is itself irreducible and complete. On the other hand, if A_g is the coarse moduli space of principally polarized g-dimensional abelian varieties, then the map associating to each curve its jacobian defines a morphism:

Source note. The printed text has the typographical form “principally pclarized”; “principally polarized” is restored above.

$$\theta : M_g^0 \to A_g.$$

If M_g^0 were complete, the image of θ would be closed. But it is well known that the closure of the image of θ contains all products of lower dimensional jacobians too, so it is not closed. Q.E.D.

Step II. — No component of H_g consists entirely of non-singular curves, i.e., $S_i^* \neq \varnothing$ for all i.
Proof. — Here we combine Step I with the result of § 2. Take any i. Let $T = \operatorname{Spec} k[[t]]$. Since $M_{g,i}^0$ is not complete, there is a morphism φ of the generic point T_η of T into $M_{g,i}^0$ which does not extend to a morphism of T into $M_{g,i}^0$. Now we replace T by its normalization T' in a finite algebraic extension and let $\varphi' : T'_\eta \to M_{g,i}^0$ be the induced morphism, which still does not extend to a morphism from T' to $M_{g,i}^0$. By the results of § 2, if T' is chosen suitably there exists a stable curve $\pi : C' \to T'$ over T' whose generic fibre C'_η is a non-singular curve corresponding to the morphism $\varphi' : T'_\eta \to M_g^0$ via the functorial properties of the moduli space. Since T' is the spectrum of a local ring, we can choose an isomorphism

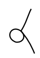
$$\mathbf{P}\big(\pi_*(\omega_{C'/T'}^{\otimes 3})\big) \simeq \mathbf{P}^{5g-6} \times T'$$

and get a tri-canonical embedding $C' \subset \mathbf{P}^{5g-6} \times T'$. C', with this embedding, is then induced from the universal tri-canonically embedded stable curve by a morphism $\psi : T' \to H_g$. Since the generic fibre of C' is C'_η, we get a commutative diagram:

$$\begin{array}{ccc}
T' & \xrightarrow{\psi} & H_g \\
\cup & & \cup \\
T'_\eta & \xrightarrow{\operatorname{res}(\psi)} & H_g^0 \\
\varphi' \searrow & & \downarrow \\
& M_{g,i}^0 & \subset M_g^0.
\end{array}$$

If $\operatorname{Im}(\psi) \subset H_g^0$, then φ' would extend to a morphism from T' to M_g^0, hence from T' to $M_{g,i}^0$, and this is a contradiction. Moreover, $\psi(T_\eta)$ must be a point of $H_{g,i}^0$, since its image in M_g^0 is in $M_{g,i}^0$. Therefore the image x of the closed point of T' by ψ is in the closure of $H_{g,i}^0$, i.e., in $H_{g,i}$, but not in $H_{g,i}^0$ itself. Q.E.D.

Step III. — S^* is connected.
Proof. — This will follow using only the induction assumption of irreducibility for lower genera. Let $Z \subset H_g \times \mathbf{P}^{5g-6}$ be the universal tri-canonically embedded stable curve. Let $S \subset Z$ be the set of points where Z is not smooth over H_g. As we proved in § 1, S is non-singular and is the normalization of S^*. In particular, this shows that if $x \in S^*$, then the corresponding curve Z_x has exactly one double point if and only if x is a non-singular point of S^*. Stable curves C of genus g with exactly one double point belong to one of the following types:

type 0 :  C irreducible
normalization C' of C has genus $g-1$;

type k :  C has two non-singular components C_1, C_2
genus$(C_1) = k$
genus$(C_2) = g - k$

where $1 \leq k \leq \left[\frac{g}{2}\right]$. If $0 \leq k \leq \left[\frac{g}{2}\right]$, let

$$S^*(k) = \{x \in S^* \mid Z_x \text{ has one double point and is of type } k\}.$$

Then the open dense subset of S^* of non-singular points is the disjoint union of open subsets $S^*(0), \ldots, S^*(\left[\frac{g}{2}\right])$. We first check:

$(*)$ Each set $S^*(k)$ is irreducible.
Proof of $(*)$. — Take the case $k = 0$: the cases $1 \leq k \leq \left[\frac{g}{2}\right]$ are similar. Let

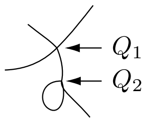
$$T = \{\{x_1, x_2\} \mid x_1 \neq x_2 \text{ and } \pi(x_1) = \pi(x_2) \in H_{g-1}^0\} \subset Z_{g-1} \times_{H_{g-1}} Z_{g-1}.$$

T is smooth with irreducible fibres over H_{g-1}^0, hence T is irreducible. Consider the correspondence relating $S^*(0)$ and T:

$$W = \left\{ \begin{array}{l} \text{set of pairs } x \in S^*(0),\ \{x_1, x_2\} \in T \text{ such that if } y = \pi(x_1) = \pi(x_2), \\ \text{then there exists a birational morphism } f : (Z_{g-1})_y \to (Z_g)_x \\ \text{such that } f(x_1) = f(x_2). \end{array} \right.$$

It is easy to check that W is Zariski-closed. Moreover, for any $\{x_1, x_2\} \in T$, $W \cap (S^*(0) \times \{x_1, x_2\})$ is an orbit in $S^*(0)$ under $\mathrm{PGL}(5g-6)$: so these are non-empty irreducible subsets all of the same dimension. It follows that W itself is irreducible. But the projection from W to $S^*(0)$ is surjective, so $S^*(0)$ is irreducible. Q.E.D. for $(*)$.

Now for any k, $1 \leq k \leq \left[\frac{g}{2}\right]$, choose any stable curve $C(k)$ of the type:



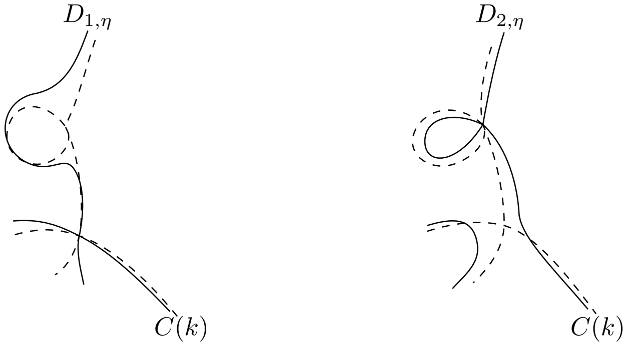
one non-singular component $C(k)'$ of genus k,
one component $C(k)''$ with one double point, normalization of genus $g-k-1$.

Let $P(k)$ be a point of H_g such that $Z_{P(k)} \simeq C(k)$. Step III will be completed if we prove:

$(**)$ $P(k)$ is in the closure of $S^*(0)$ and of $S^*(k)$, hence $S^*(0)$, $S^*(k)$ both lie in the same topological component of S^*.

Proof of $(**)$. — Let $T = \operatorname{Spec} k[[t]]$. Using the fact that S^* has two branches through $P(k)$, one for each of the double points of $C(k)$, we see that there exist two morphisms

$$f_1, f_2 : T \to S^*,$$
$$f_1(T_s) = f_2(T_s) = P(k), \qquad T_s = \text{closed point of } T,$$



such that if $\pi_1 : D_1 \to T$, $\pi_2 : D_2 \to T$ are the two stable curves over T induced by f_1 and f_2, then: a) the closed fibres $D_{1,s}$, $D_{2,s}$ are $C(k)$; b) there are sections $s_1 : T \to D_1$, $s_2 : T \to D_2$ whose images are non-smooth points of π_1, π_2 and such that $s_1(T_s)$ and $s_2(T_s)$ are the two double points Q_1 and Q_2 of $C(k)$ respectively; and c) the generic fibres $D_{1,\eta}$, $D_{2,\eta}$ have only one double point (cf. figure). I claim:

A) $D_{1,\eta}$ is of type (k);

B) $D_{2,\eta}$ is of type (0).

To prove A), let D_1' be the result of blowing up the subscheme $s_1(T)$ of D_1. Then D_1' is still flat and proper over T and its special fibre is the special fibre of D_1 with Q_1 blown up (use the fact that formally at Q_1, D_1 is isomorphic to $k[[t,x,y]]/(xy)$ with the section given by $x = y = 0$). Therefore the special fibre of D_1' is the disjoint union of $C(k)'$, $C(k)''$, so the general fibre of D_1' is the disjoint union of two irreducible curves which specialize to $C(k)'$, $C(k)''$ respectively. Since $D_{1,\eta}$ has only one double point, $D_{1,\eta}'$ is non-singular, so $D_{1,\eta}'$ is the disjoint union of two non-singular irreducible curves which must then have the same genera as $C(k)'$ and $C(k)''$, i.e., k, $g-k$. Thus $D_{1,\eta}$ has type (k).

To prove B), it suffices to check that $D_{2,\eta}$ is geometrically irreducible. If not, $D_{2,\eta}$ would have two components meeting at the single point $s_2(T_\eta)$. Since D_2 is smooth over T at each generic point of its special fibre, distinct geometric components of $D_{2,\eta}$ have to have specializations which are distinct components of $(D_2)_s$. Then $(D_2)_s$ would have two components meeting at the point $Q_2 = s_2(T_s)$. This is false, so B) is proven.

Now because of A), $f_1(T_\eta) \in S^*(k)$, hence $P(k) = f_1(T_s)$ is in the closure of $S^*(k)$. And because of B), $f_2(T_\eta) \in S^*(0)$, hence $P(k) = f_2(T_s)$ is in the closure of $S^*(0)$ too. Q.E.D. for $(**)$.

This completes the proof of Step III since we now see that all irreducible components of S^* are part of the same topological component. Finally, from Steps II and III, we see that

a) S^*, being connected, is part of a single component of H_g, while

b) each component of H_g contains part of S^*.

Thus H_g is irreducible, as was to be proven.

§4. Some results on algebraic stacks.

The proofs of the results stated in this section will be given elsewhere.

Let C be a category and let $p : \mathscr{S} \to C$ be a category over C. For each $U \in \operatorname{Ob} C$, we denote by $\mathscr{S}_U$ the fibre $p^{-1}(U)$. The category $\mathscr{S}$ is *fibered in groupoids* over C if the following two conditions are verified:

a) For all $\varphi : U \to V$ in C and $y \in \operatorname{Ob} \mathscr{S}_V$ there is a map $f : x \to y$ in $\mathscr{S}$ with $p(f) = \varphi$.

b) Given a diagram

$$\begin{tikzcd} x \arrow[dr, "f"] & \\ & z \\ y \arrow[ur, "g"'] & \end{tikzcd}$$

in $\mathscr{S}$, let

$$\begin{tikzcd} U \arrow[dr, "\varphi"] & \\ & W \\ V \arrow[ur, "\psi"'] & \end{tikzcd}$$

be its image in C. Then for all $\chi : U \to V$ such that $\varphi = \psi\chi$, there is a unique $h : x \to y$ such that $f = g \cdot h$ and $p(h) = \chi$.

Source note. The printed first diagram labels its lower-left object Y; the surrounding notation requires the y retained here.

Condition b) implies that the $f : x \to y$ whose existence is asserted in a) is unique up to canonical isomorphism.

Assume that for each $\varphi : U \to V$ in C and each $Y \in \operatorname{Ob} \mathscr{S}_V$, such an $f : x \to y$ has been chosen. This x will be written as $\varphi^* y$. Then, φ^* "is" a functor from $\mathscr{S}_V$ to $\mathscr{S}_U$ and if $\varphi\psi$ is a composite morphism in C, the functors $(\varphi\psi)^*$ and $\psi^*\varphi^*$ are canonically isomorphic.

We propose the terminology "stack" for the French word "champ" of non-abelian cohomology (Giraud [G]).

Definition (4.1). — Let C be a category with a Grothendieck topology. We assume products and fibre products exist in C. A stack in groupoids over C is a category over C, $p : \mathscr{S} \to C$ such that:

(i) *$\mathscr{S}$ is fibered in groupoids over C.*

(ii) *For any $U \in \operatorname{Ob} C$ and any objects x, y in $\mathscr{S}_U$ the functor from C/U to (sets) which to any $\varphi : V \to U$ associates $\operatorname{Hom}_{\mathscr{S}_V}(\varphi^* x, \varphi^* y)$ is a sheaf.*

(iii) *If $\varphi_i : V_i \to U$ is a covering family in C, any descent datum relative to the φ_i, for objects in $\mathscr{S}$, is effective.*

For each $x \in \operatorname{Ob} \mathscr{S}_U$, there are given isomorphisms between the inverse images of $x_i = \varphi_i^* x$ and $x_j = \varphi_j^* x$ over $V_{ij} = V_i \times_U V_j$, and the pull-backs of these isomorphisms on $V_{ijk} = V_i \times_U V_j \times_U V_k$ satisfy a "cocycle" condition. In (iii) it is required that reciprocally, any such "descent datum" be defined by some $x \in \mathscr{S}_U$.

In what follows, for the sake of brevity, we will use "stack" to mean "stack in groupoids".

If $U \in \operatorname{Ob} C$ and if $\mathscr{S}$ is a stack over C, the fibre $\mathscr{S}_U$ will be called the category of sections of $\mathscr{S}$ over U.

Let C be as in (4.1). The stacks over C are the objects of a 2-category [B] (stacks/C): 1-morphisms are functors from one stack to another, compatible with the projection into C; and 2-morphisms are morphisms of functors. In this 2-category every 2-morphism is an isomorphism. Products and 2-fibre products exist in this 2-category.

To each $X \in \operatorname{Ob} C$ is associated the "representable" stack over C whose category of sections over U is the discrete category whose objects are the morphisms from U to X. This stack will be denoted simply X. For any stack $\mathscr{S}$, the category

$$\operatorname{Hom}(X, \mathscr{S})$$

is canonically equivalent to the category of sections of $\mathscr{S}$ over X. Because of this, $\mathscr{S}$ is sometimes said to "classify" its sections over variable $X \in \operatorname{Ob} C$. In the category $\operatorname{Hom}(\mathscr{S}, X)$, all morphisms are identities, i.e., $\operatorname{Hom}(\mathscr{S}, X)$ is just a set.

Let us denote also by C the 2-category having the same objects and morphisms as C, and in which the identities are the only 2-morphisms. The above construction then identifies C with a full sub-2-category of (stacks/C).

For each $S \in \operatorname{Ob} C$, the category C/S satisfies the assumptions of (4.1). Any stack $\mathscr{S}_0$ over C/S (an "S-stack") gives rise to a stack $\mathscr{S}$ over C; a section (φ, η) of $\mathscr{S}$ over $U \in \operatorname{Ob} C$ consists of

(i) a morphism $\varphi : U \to S$;

(ii) a section η of $\mathscr{S}_0$ over (U, φ).

Definition (4.2). — A 1-morphism of stacks over C, $F : \mathscr{S}_1 \to \mathscr{S}_2$, will be called representable if for any X in C and any 1-morphism $x : X \to \mathscr{S}_2$, the fibre product $X \times_{\mathscr{S}_2} \mathscr{S}_1$ is a representable stack.

In down to earth terms, this means the following:

(i) for any $f : Y \to X$ in C, the category whose objects are pairs

$$\{\text{a section, } y, \text{ of } \mathscr{S}_1 \text{ over } Y; \text{ an isomorphism } F(y) \xrightarrow{\sim} f^*(x)\}$$

is equivalent to a category $S(f)$ in which all morphisms are identities;

(ii) the functor $f \mapsto \operatorname{Ob} S(f)$ is representable by some $g : Z \to X$. Such a Z represents the fibre product $X \times_{\mathscr{S}_2} \mathscr{S}_1$.

Let P be a property of morphisms in C, stable by change of base and of a local nature on the target.

Definition (4.3). — A representable morphism $F : \mathscr{S}_1 \to \mathscr{S}_2$ of stacks over C has property P if for any 1-morphism $x : X \to \mathscr{S}_2$ the morphism in C deduced by base change: $F' : X \times_{\mathscr{S}_2} \mathscr{S}_1 \to X$ has that property.

Proposition (4.4). — Let $\mathscr{S}$ be a stack. The diagonal map

$$\mathscr{S} \to \mathscr{S} \times \mathscr{S}$$

is representable if and only if for all $X, Y \in \operatorname{Ob} C$ and 1-morphisms $x : X \to \mathscr{S}$, $y : Y \to \mathscr{S}$, the fibre product $X \times_{\mathscr{S}} Y$ is representable.

If $X \in \operatorname{Ob} C$ and x, y are sections of $\mathscr{S}$ over X, we denote by $\underline{\operatorname{Isom}}(X, x, y)$ the sheaf on C/X which to every Z over X associates the set of isomorphisms between the inverse images of x and y over Z. Then the object representing $\mathscr{S} \times_{(\mathscr{S} \times \mathscr{S})} X$ (the product taken with the map $(x, y) : X \to \mathscr{S} \times \mathscr{S}$) is just $\operatorname{Isom}(X, x, y)$. If $X \to \mathscr{S}$, $Y \to \mathscr{S}$ are 1-morphisms, then the object representing $X \times_{\mathscr{S}} Y$ is just $\operatorname{Isom}(X \times Y, p_1^* x, p_2^* y)$.

Henceforth, C will be the category of schemes with the etale topology (SGAD, IV, (6.3)).

Definition (4.5). — A stack $\mathscr{S}$ is quasi-separated if the diagonal morphism from $\mathscr{S}$ to $\mathscr{S} \times \mathscr{S}$ is representable, quasi-compact and separated.

Source note. The printed definition reads "the diagonal morphism from to $\mathscr{S}$ to $\mathscr{S} \times \mathscr{S}$"; the duplicated "to" is omitted here.

Definition (4.6). — A stack $\mathscr{S}$ is an algebraic stack[5] *if*

(i) *$\mathscr{S} \to \mathscr{S} \times \mathscr{S}$ is representable;*

(ii) *there exists a 1-morphism $x : X \to \mathscr{S}$ such that for all $y : Y \to \mathscr{S}$, the projection morphism $X \times_{\mathscr{S}} Y \to Y$ is surjective and etale (i.e., x is etale and surjective).*

The 2-category of algebraic stacks contains the representable stacks and is stable under products and fibre products. If $\mathscr{S}$ is a quasi-separated algebraic stack, the diagonal map is unramified and quasi-affine.

Definition (4.7). — An algebraic stack $\mathscr{S}$ is separated if $\mathscr{S} \to \mathscr{S} \times \mathscr{S}$ is proper (or, equivalently, finite). A 1-morphism $f : \mathscr{S}_1 \to \mathscr{S}_2$ is separated (resp. quasi-separated) if for any morphism $x : X \to \mathscr{S}_2$ from a separated scheme X to $\mathscr{S}_2$, the fibre product $\mathscr{S}_1 \times_{\mathscr{S}_2} X$ is separated (resp. quasi-separated).

Example (4.8). — Let X be a scheme over S. Let G be a group scheme over S, etale, separated and of finite type over S, which operates on X. We will denote by $[X/G]$ the S-stack whose category of sections over an S-scheme T is the category of principal homogeneous spaces (p.h.s.) E over T, with structural group G (i.e., a p.h.s. under G_T), provided with a G-morphism $\varphi : E \to X$. The principal homogeneous space $G \times X$ over X (G acting only on the first factor) plus the G-morphism $G \times X \to X$ (given by the action of G on X) is a section of $[X/G]$ over X. The corresponding morphism $q : X \to [X/G]$ is etale and surjective, so that $[X/G]$ is an algebraic stack. In addition, X is a principal homogeneous space over $[X/G]$; the stack $[X/G]$ is representable if and only if X is a principal homogeneous space over a scheme Y, in which case

$$[X/G] \sim Y.$$

[5] This definition is the "right" one only for quasi-separated stacks. It will however be sufficient for our purposes.

If $X = S$, then $[X/G] = [S/G]$ might be called the "classifying stack" of G over S.

Example (4.9). — Suppose a stack $\mathscr{S}$ has the property that in each category $\mathscr{S}_X$ the only morphisms are the identity morphisms. Then $\mathscr{S}_X$ is just a set $\mathscr{F}(X) = \mathrm{Ob}\,\mathscr{S}_X$, and this set, under pull-back, is a contravariant functor $\mathscr{F}$ in X. Conditions (ii) and (iii) of (4.1) assert that the functor $\mathscr{F}$ is a sheaf on C. Artin and Knutson [K] have defined an *algebraic space* to be a sheaf $\mathscr{F}$ such that:

(i) for any morphisms $X \to \mathscr{F}$, $Y \to \mathscr{F}$ of representable functors to $\mathscr{F}$, the fibre product $X \times_{\mathscr{F}} Y$ is representable;

(ii) there exists a morphism $X \to \mathscr{F}$, represented by surjective, etale morphisms of schemes.

This is exactly what we have called an algebraic stack in this case.

Definition (4.10). — Let $\mathscr{S}$ be an algebraic stack. The etale site $\mathscr{S}_{et}$ of $\mathscr{S}$ is the category with objects the etale morphisms

$$x : X \to \mathscr{S}$$

and where a morphism from (X, x) to (Y, y) is a morphism of schemes $f : X \to Y$ plus a 2-morphism between the 1-morphism $x : X \to \mathscr{S}$ and $y \cdot f : X \to \mathscr{S}$. A collection of morphisms $f_i : (X_i, x_i) \to (X, x)$ is a covering family if the underlying family of morphisms of schemes is surjective.

The site $\mathscr{S}_{et}$ is in a natural way ringed. When we speak of sheaves on $\mathscr{S}$ we mean sheaves on $\mathscr{S}_{et}$.

We now explain how many concepts from the theory of schemes may be applied to algebraic stacks.

Let $\mathcal{P}$ be a property of morphisms of schemes, stable by etale change of base, and of a local nature (for the etale topology) on the target.

For instance: being an open immersion with dense image, being dominant, birational...

A representable morphism of algebraic stacks $f : T_1 \to T_2$ is said to have property $\mathcal{P}$ if for one (and hence for every) surjective etale morphism $x : X \to T_2$, the morphism of schemes deduced by base change

$$f' : X \times_{T_2} T_1 \longrightarrow X$$

has that property.

Let $\mathcal{P}$ be a property of morphisms of schemes, which, at source and target, is of a local nature for the etale topology. This means that, for any family of commutative squares

$$\begin{array}{ccc} X_i & \xrightarrow{g_i} & X \\ {\scriptstyle f_i}\downarrow & & \downarrow{\scriptstyle f} \\ Y_i & \xrightarrow[h_i]{} & Y \end{array}$$

where the g_i (resp. h_i) are etale and cover X (resp. Y):

$$\mathcal{P}(f) \iff \forall i\, \mathcal{P}(f_i).$$

For instance: f flat, smooth, etale, unramified, normal, locally of finite type, locally of finite presentation.

If $f : T_1 \to T_2$ is a morphism of algebraic stacks, we say f has property $\mathcal{P}$ if for one, then necessarily for every, commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{x} & T_1 \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y & \xrightarrow[y]{} & T_2 \end{array}$$

where X and Y are schemes and x, y are etale and surjective, f' has property $\mathcal{P}$.

Similarly, if $\mathcal{P}$ is a property of schemes, of a local nature for the etale topology, an algebraic stack T will be said to have property $\mathcal{P}$ if for one (and hence for every) surjective etale morphism $x : X \to T$, X has property $\mathcal{P}$. This applies to, for instance, the properties of being regular, normal, locally noetherian, of characteristic p, reduced, Cohen–Macaulay...

An algebraic stack T will be called *quasi-compact* if there exists a surjective etale morphism $x : X \to T$ with X quasi-compact. A morphism $f : T_1 \to T_2$ of algebraic stacks will be called *quasi-compact* if for any quasi-compact scheme X over T_2, the fibre product $T_1 \times_{T_2} X$ is quasi-compact. It is enough to test the condition for a surjective family $f_i : X_i \to T_2$. We define a morphism $f : T_1 \to T_2$ to be of *finite type*, if it is quasi-compact, and locally of finite type; of *finite presentation*, if it is quasi-compact, quasi-separated, and locally of finite presentation. An algebraic stack is *noetherian*, if it is quasi-compact, quasi-separated, and locally noetherian.

The key point in what follows will be the definition of a "proper morphism" and the analogue of Chow's lemma.

A morphism $f : T_1 \to T_2$ is said to have a property $\mathcal{P}$, locally on T_2, if there exists a surjective etale morphism $x : X \to T_2$ such that the morphism f' deduced from f by the change of base by x has property $\mathcal{P}$.

Definition (4.11). — A morphism $f : T_1 \to T_2$ is proper if it is separated, of finite type and if, locally over T_2, there exist commutative diagrams

$$\begin{tikzcd} T_3 \arrow[rr, "g"] \arrow[dr, "h"'] & & T_1 \arrow[dl, "f"] \\ & T_2 & \end{tikzcd}$$

with g surjective and h representable and proper.

Source note. The printed definition reads "there exists commutative diagrams"; the verb is corrected here without changing the plural construction.

The following form of Chow's lemma will be sufficient for our purposes.

Theorem (4.12). — Let S be a noetherian scheme and f be a morphism from an etale site T to S. We assume f to be separated and of finite type. Then, there exists a commutative diagram

$$\begin{tikzcd} T \arrow[d, "f"'] & T' \arrow[l, "g"'] \arrow[r, "j"] \arrow[dl, "f'"'] & T'' \arrow[dll, "f''"] \\ S & & \end{tikzcd}$$

in which T' and T'' are schemes and such that

(i) *g is proper, surjective and generically finite;*

(ii) *j is an open immersion;*

(iii) *f'' is projective.*

Using (4.12), it is easy to extend the cohomological theory of coherent sheaves to the present situation. In fact if $f : T_1 \to T_2$ is a proper morphism of noetherian algebraic stacks and if $\mathscr{F}$ is a coherent sheaf on T_1, then the $R^i f_*(\mathscr{F})$ are coherent sheaves on T_2.

However, the $R^i f_*(\mathscr{F})$ don't need to be zero for i large enough. Let S be a scheme and G a finite group. We denote by p the projection $p : [S/G] \to S$. Quasi-coherent (resp. coherent) sheaves of modules on $[S/G]$ may (and will) be identified with quasi-coherent (resp. coherent) sheaves of modules on S, on which G acts. One has

$$R^i p_*(\mathscr{F}) \simeq H^i(G, \mathscr{F}).$$

In general, the (quasi-coherent sheaf) cohomology of algebraic stacks appears as a mixture of finite group cohomology and of scheme cohomology.

The *disjoint sum* T of a family $(T_i)_{i \in I}$ of stacks is the stack a section of which over a scheme X consists of

(i) a decomposition $X = \coprod_i X_i$ of X;

(ii) a section of T_i over X_i for each i.

The *void stack* $\varnothing$ is the one represented by the void scheme.

A stack is *connected* if it is non-void and is not the disjoint sum of two non-void stacks.

Proposition (4.13). — A locally noetherian algebraic stack is in one and only one way the disjoint sum of a family of connected algebraic stacks (called its connected components).

We denote by $\pi_0(T)$ the set of connected components of a locally noetherian algebraic stack T. If $x : X \to T$ is surjective and etale, $\pi_0(T)$ is the cokernel of the two maps

$$\pi_0(X \times_T X) \rightrightarrows \pi_0(X) \longrightarrow \pi_0(T).$$

Proposition (4.14). — Let T be an algebraic stack of finite type over a field k. Then, T is connected if and only if there exists a connected scheme X, of finite type over k, and a surjective morphism from X to T.

An open subset U of an algebraic stack T is a full subcategory $U \subset T$ which is an algebraic stack, which contains together with any $t \in \mathrm{Ob}(T)$ all isomorphic t' and such that the inclusion $j : U \to T$ is representable by open immersions. The open subsets of T correspond bijectively to the open subsets of its etale site.

Source note. The printed sentence has "The open subsets of T corresponds"; subject–verb agreement is corrected here.

For each open subset U of T, there exists one and only one full subcategory $T - U$ of T, which is an algebraic stack, which contains together with any $t \in \mathrm{Ob}(T)$ all isomorphic t' and such that

(i) $T - U$ is reduced;

(ii) the inclusion map $i : T - U \to T$ is representable by closed immersions;

(iii) for any etale surjective morphism $x : X \to T$, the inverse image of $T - U$ on X is the complement of the inverse image of U.

An algebraic stack F in T satisfying (i) and (ii) is a *closed subset* of T, and the functor $U \mapsto T - U$ is an isomorphism of the set of open and the set of closed subsets of T. If F satisfies only (ii), F_{red} satisfies (i) and (ii) so that F defines a closed subset of T.

An algebraic stack T is *irreducible* if it is not the union of two closed subsets, non-void and distinct from T.

Proposition (4.15). — A noetherian algebraic stack T is in one and only one way the union of irreducible closed subsets, none of which contains any other. They are called the irreducible components of T. If U is an open dense subset of T, the irreducible components of U are the non-void intersections of U with the irreducible components of T.

Each irreducible component of T is contained in a connected component of T. Conversely:

Proposition (4.16). — The connected components of a normal noetherian algebraic stack are irreducible.

Theorem (4.17). — Let f be a morphism of finite type from an algebraic stack T to a noetherian scheme S. For $s \in S$, let $n(s)$ be the number of connected components of the geometric fibre of T at s. Then

(i) *$n(s)$ is a constructible function of s;*

(ii) *if f is proper and flat, then $n(s)$ is lower-semi-continuous;*

(iii) *if f is proper flat, and has geometrically normal fibres, then $n(s)$ is constant.*

Let $f : T \to S$ be a morphism of finite type from an algebraic stack T to a noetherian scheme S. Assume that the diagonal map $T \to T \times_S T$ is separated and quasi-compact.

Theorem (4.18) (Valuative criterion for separation). — The morphism f is separated if and only if, for any complete discrete valuation ring with algebraically closed residue field and any commutative diagram

$$\begin{tikzcd} & T \arrow[d, "f"] \\ \mathrm{Spec}(V) \arrow[ur, shift left, "g_1"] \arrow[ur, shift right, "g_2"'] \arrow[r] & S \end{tikzcd}$$

any isomorphism between the restrictions of g_1 and g_2 to the generic point of $\mathrm{Spec}(V)$ *can be extended to an isomorphism between g_1 and g_2.*

This criterion is nothing other but the valuative criterion of properness (EGA, II, 7.3.8) applied to the (representable) diagonal morphism.

Theorem (4.19) (Valuative criterion for properness). — If f is separated, then f is proper if and only if, for any discrete valuation ring V with field of fractions K and any commutative diagram

$$\begin{tikzcd} & & T \arrow[d, "f"] \\ \mathrm{Spec}(K) \arrow[urr, "g"] \arrow[r] & \mathrm{Spec}(V) \arrow[r] & S \end{tikzcd}$$

there exists a finite extension K' of K such that g extends to $\mathrm{Spec}(V')$*, where V' is the integral closure of V in K'*

$$\begin{tikzcd}
& & T \arrow[dd, "f"] \\
\mathrm{Spec}(K') \arrow[r] & \mathrm{Spec}(V') \arrow[ur, dashed] \arrow[d] & \\
\mathrm{Spec}(K) \arrow[uurr, "g"] \arrow[r] & \mathrm{Spec}(V) \arrow[r] & S
\end{tikzcd}$$

To prove a given f is proper, it suffices to verify the above criterion under the additional hypothesis that V is complete and has an algebraically closed residue field. Further, given a dense open subset U of T, it is enough to test only g's which factor through U.

Proposition (4.20). — Let $\mathscr{S}$ be an algebraic stack. The functor which, to any algebraic stack over $\mathscr{S}$, $f : \mathscr{T} \to \mathscr{S}$, associates the $\mathcal{O}_{\mathscr{S}}$-sheaf of algebras $f_*\mathcal{O}_{\mathscr{T}}$ induces an equivalence of categories between:

(i) *the category of algebraic stacks representable and affine over $\mathscr{S}$;*

(ii) *the dual of the category of quasi-coherent $\mathcal{O}_{\mathscr{S}}$-algebras.*

Let $\mathscr{A}$ be a quasi-coherent sheaf of $\mathcal{O}_{\mathscr{S}}$-algebras on an algebraic stack $\mathscr{S}$. For each etale morphism $x : U \to \mathscr{S}$, with U affine, let $\mathscr{A}'(U)$ be the integral closure of $\Gamma(U, \mathcal{O}_{\mathscr{S}})$ in $\mathscr{A}(U)$. By (EGA, II, 6.3.4), the $\mathscr{A}'(U)$ for variable U are the sections over U of a quasi-coherent sheaf $\mathscr{A}'$ on $\mathscr{S}$, which will be called the integral closure of $\mathcal{O}_{\mathscr{S}}$ in $\mathscr{A}$.

Let $f : \mathscr{T} \to \mathscr{S}$ be representable and affine. The algebraic stack which is associated by (4.20) to the integral closure of $\mathcal{O}_{\mathscr{S}}$ in $f_*\mathcal{O}_{\mathscr{T}}$ will be called the normalization of $\mathscr{S}$ with respect to $\mathscr{T}$. Its formation is compatible with any etale change of basis.

Theorem (4.21). — Let $\mathscr{S}$ be a quasi-separated stack over a noetherian scheme S. Assume that

(i) *the diagonal map $\mathscr{S} \to \mathscr{S} \times_S \mathscr{S}$ is representable and unramified;*

(ii) *there exists a scheme X of finite type over S and a smooth and surjective S-morphism from X to $\mathscr{S}$.*

Then, $\mathscr{S}$ is an algebraic stack of finite type over S.

M. Artin has developed powerful methods to relate pro-representability of a stack to the existence of etale surjective maps $x : X \to \mathscr{S}$.

§ 5. Second proof of the irreducibility theorem.

Let $\mathcal{M}_g$ ($g \geq 2$) be the stack whose category of sections over a scheme S is the category of stable curves of genus g over S, the morphisms being the isomorphisms of schemes over S. By (1.11), the diagonal morphism

$$\Delta : \mathcal{M}_g \longrightarrow \mathcal{M}_g \times \mathcal{M}_g$$

is representable, finite and unramified.

We saw in § 1 that the stack classifying the tricanonically embedded stable curves of genus g is represented by a scheme H_g, smooth and of finite type over $\mathrm{Spec}(\mathbb{Z})$. The "forgetful" morphism

$$H_g \longrightarrow \mathcal{M}_g$$

is representable, smooth and surjective. Indeed, if $p : C \to S$ is a stable curve over S, defining a morphism

$$\gamma : S \longrightarrow \mathcal{M}_g,$$

then the fibre product $H_g \times_{\mathcal{M}_g} S$ is the scheme, smooth over S, of isomorphisms between the standard projective space of dimension $5g - 6$ over S and $\mathbf{P}(p_*(\omega_{C/S}^{\otimes 3}))$. We deduce from this and (4.21) that:

Proposition (5.1). — $\mathcal{M}_g$ is a separated algebraic stack of finite type over $\mathrm{Spec}(\mathbb{Z})$.

Let us denote by $\mathcal{M}_g^0$ the open subset of $\mathcal{M}_g$ which "consists of" smooth curves, and by $\mathcal{Z}_g$, the "universal curve" over $\mathcal{M}_g$, the algebraic stack classifying pointed stable curves.

Theorem (5.2). — The algebraic stacks $\mathcal{M}_g$ and $\mathcal{Z}_g$ are proper and smooth over $\mathrm{Spec}(\mathbb{Z})$ *and the complement of $\mathcal{M}_g^0$ in $\mathcal{M}_g$ is a divisor with normal crossings relative to* $\mathrm{Spec}(\mathbb{Z})$.

Proof. — Let $x : X \to \mathcal{M}_g$ be etale and consider the following commutative diagram of algebraic stacks of finite type over $\mathrm{Spec}(\mathbb{Z})$:

$$\begin{tikzcd}
X \times_{\mathcal{M}_g} \mathcal{Z}_g \arrow[rrr] \arrow[dr] \arrow[dd] & & & \mathcal{Z}_g \arrow[dr] & \\
& X \times_{\mathcal{M}_g} H_g \arrow[rrr, "x'"] \arrow[dd, "q'"] & & & H_g \arrow[dd, "q"] \\
X \times_{\mathcal{M}_g} \mathcal{Z}_g \arrow[rrr] \arrow[dr] & & & \mathcal{Z}_g \arrow[dr, "p"] & \\
& X \arrow[rrr, "x"] & & & \mathcal{M}_g
\end{tikzcd}$$

In this diagram, the four horizontal arrows are etale and the four vertical arrows are smooth and surjective. As H_g and $\mathcal{Z}_g$ (p. 78) are smooth over $\mathrm{Spec}(\mathbb{Z})$, so are X and $X \times_{\mathcal{M}_g} \mathcal{Z}_g$. In addition,

$$q'^{-1} x^{-1}(\mathcal{M}_g^0) = x'^{-1}(H_g^0)$$

is the complement of a divisor with normal crossings relative to $\mathrm{Spec}(\mathbb{Z})$, so that the inclusion of $x^{-1}(\mathcal{M}_g^0)$ in X has the same property. This being true for any x, $\mathcal{M}_g$ and $\mathcal{Z}_g$ are smooth over $\mathrm{Spec}(\mathbb{Z})$ and $\mathcal{M}_g^0$ is the complement of a divisor with normal crossings relative to $\mathrm{Spec}(\mathbb{Z})$. In particular $\mathcal{M}_g^0$ is dense in $\mathcal{M}_g$.

We may now use the valuative criterion for properness in its modified form (4.19) to deduce the properness of $\mathcal{M}_g$ from the stable reduction theorem. The properness of $\mathcal{Z}_g$ then results from that of p.

(5.3) Let $p : X \to S$ be a stable smooth curve of genus $g \geq 2$ over S. If $k \in \mathbb{N}$ is invertible in $\mathcal{O}_S$, the sheaf $R^1 p_*(\mathbb{Z}/k\mathbb{Z})$ on S_{et} is locally free over $\mathbb{Z}/k\mathbb{Z}$, of rank $2g$, and the cup-product is a non degenerate alternating form

$$R^1 p_*(\mathbb{Z}/k\mathbb{Z}) \otimes R^1 p_*(\mathbb{Z}/k\mathbb{Z}) \longrightarrow R^2 p_*(\mathbb{Z}/k\mathbb{Z}) \simeq \mu_k^{\otimes -1}.$$

Locally on S_{et}, μ_k is isomorphic to $\mathbb{Z}/k\mathbb{Z}$, and thus, locally, $R^1 p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with a non degenerate symplectic form with values in $\mathbb{Z}/k\mathbb{Z}$, which is determined up to a unit in $\mathbb{Z}/k\mathbb{Z}$, or, as we shall say, $R^1 p_*(\mathbb{Z}/k\mathbb{Z})$ is provided with an homogeneous symplectic structure. The constant sheaf $(\mathbb{Z}/k\mathbb{Z})^{2g}$ will be provided with the homogeneous symplectic structure induced by the standard symplectic structure of $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

Definition (5.4). — A Jacobi structure of level k on X is an isomorphism (respecting their homogeneous symplectic structures) between $R^1 p_*(\mathbb{Z}/k\mathbb{Z})$ and $(\mathbb{Z}/k\mathbb{Z})^{2g}$.

(5.5) Given a section s of p, and a set of prime numbers $\mathbf{P}$ including all residue characteristics of S, the specialisation theorem for the fundamental group enables one to construct a pro-object $\pi_1(X/S, s)^{(\mathbf{P})}$ of the category (l.c. $\mathrm{gr}^{(\mathbf{P})}$) of locally constant sheaves of finite groups of order prime to $\mathbf{P}$, with the properties

(i) $\mathrm{Hom}_S(\pi_1(X/S, s)^{(\mathbf{P})}, G) = R^1 p_*(X \bmod s, p^* G) = R^1 p_*(\mathrm{Ker}(p^* G \to s_* G))$ functorially in $G \in$ (l.c. $\mathrm{gr}^{(\mathbf{P})}$);

(ii) the formation of $\pi_1(X/S, s)^{(\mathbf{P})}$ is compatible with any change of base.

If G and H are two sheaves of groups on S_{et}, we define the sheaf of exterior morphisms from H to G, $\mathrm{Hom}^{\mathrm{ext}}(H, G)$, as the quotient of $\mathrm{Hom}(H, G)$ by the action of H induced by its action on itself by inner automorphism.

The sheaf

$$\mathrm{Hom}_S^{\mathrm{ext}}(\pi_1(X/S, s)^{(\mathbf{P})}, G) \simeq R^1 p_*(p^* G) \qquad (\text{for } G \in (\text{l.c. } \mathrm{gr}^{(\mathbf{P})}))$$

is "independent" of the choice of s.

We shall denote it as

$$\mathrm{Hom}_S^{\mathrm{ext}}(\pi_1(X/S)^{(\mathbf{P})}, G).$$

As $p : X \to S$ admits sections locally for the etale topology, this sheaf makes sense without assuming p to have a global section. It is independent of the choice of $\mathbf{P}$, so long as $\mathbf{P}$ is prime to the order of G and includes all residue characteristics of S.

Definition (5.6). — Let G be a finite group of order n prime to $\mathbf{P}$. A Teichmüller structure of level G on X is a surjective exterior homomorphism from $\pi_1(X/S)$ to G.

The finite generation of $\pi_1(X/S)$ (SGA, 60/61, exp. 10) implies:

Lemma (5.7). — The sheaf on (Sch/S) *of the Teichmüller structures of level G on X is represented by an etale covering of S.*

We denote by ${}_G\mathcal{M}_g^0$ the stack classifying the stable smooth curves of genus g and characteristic prime to n, with a Teichmüller structure of level G.

For any algebraic stack $\mathscr{M}$, we denote by $\mathscr{M}[1/n]$ its open subset $\mathscr{M} \times \mathrm{Spec}(\mathbb{Z}[1/n])$.

Lemma (5.7) may now be rephrased:

Proposition (5.8). — The "forgetful" morphism

$$ {}_G\mathcal{M}_g^0 \longrightarrow \mathcal{M}_g^0[1/n] $$

is representable, finite and etale.

The stack ${}_G\mathcal{M}_g^0$ is thus an algebraic stack. Let ${}_G\mathcal{M}_g$ be the normalisation of $\mathcal{M}_g[1/n]$ with respect to ${}_G\mathcal{M}_g^0$. The stack ${}_G\mathcal{M}_g$, being representable and finite over $\mathcal{M}_g[1/n]$, is proper over $\mathrm{Spec}(\mathbb{Z}[1/n])$.

Theorem (5.9). — The geometric fibres of the projection of ${}_G\mathcal{M}_g$ onto $\mathrm{Spec}(\mathbb{Z}[1/n])$ *are normal, and, fibre by fibre, ${}_G\mathcal{M}_g^0$ is dense in ${}_G\mathcal{M}_g$.*

Proof. — We will use Abhyankar–Artin's lemma, in its "absolute" form:

Lemma (5.10). — Let D be a divisor with normal crossings on an excellent regular scheme X, Y an etale covering of $X - D$ and $\overline{Y}$ the normalisation of X with respect to Y. Assume that the generic points of the irreducible components of D are all of characteristic 0*. Then every geometric point of X has an etale neighbourhood $x : X' \to X$ such that, on X':*

(i) *D becomes a union of regular divisors $(D_i)_{i \in I}$, D_i of equation $t_i = 0$.*

(ii) *There exists an integer k prime to residue characteristics of X' such that $\overline{Y}$ becomes isomorphic to a disjoint union of quotients (by subgroups of μ_k^I) of the covering of X' obtained by extracting the k^{th}-roots of the t_i's.*

If $x : X \to \mathcal{M}_g[1/n]$ is an etale morphism, then $X^0 = x^{-1}(\mathcal{M}_g^0)$ is the complement in X of a divisor with normal crossing relative to $\mathrm{Spec}(\mathbb{Z})$, $X_1^0 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g^0)$ is an etale covering of X^0 and $X_1 = X \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g)$ is the normalisation of X with respect to this covering of X^0.

Source note. The printed formula has $X_1 = S \times_{\mathcal{M}_g} ({}_G\mathcal{M}_g)$; since the morphism under discussion is $x : X \to \mathcal{M}_g[1/n]$ and X_1 is its base change, the required factor X is retained above.

By the explicit local description (5.10), for any prime number l prime to n, $X_1 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ is the normalisation of $X \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ with respect to $X_1^0 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$. As this is true for any modular family, we get (5.9).

Corollary (5.11). — The geometric fibres of the projection

$$ {}_G\mathcal{M}_g \longrightarrow \mathrm{Spec}(\mathbb{Z}[1/n]) $$

all have the same number of connected components.

Proof. — By (4.17), all geometric fibres $\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ of $\mathcal{M}_g$ over $\mathrm{Spec}(\mathbb{Z}[1/n])$ have the same number of connected components. These connected components are irreducible (4.16). Furthermore ${}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ is dense in ${}_G\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$ and thus has the same number of connected (or irreducible) components as ${}_G\mathcal{M}_g \times \mathrm{Spec}(\overline{\mathbb{F}}_l)$, and this number is independent of l.

(5.12) Let us denote by Π the fundamental group of an oriented closed differentiable surface S_0 of genus g. The group Π may be defined by generators and relations as follows:

(i) generators: x_i for $1 \leq i \leq 2g$;

(ii) relation: $(x_1, x_{g+1}) \cdots (x_i, x_{g+i}) \cdots (x_g, x_{2g}) = e$, where $(a, b) = aba^{-1}b^{-1}$.

Let (e_i) be the standard basis of $\mathbb{Z}^{2g}$. The morphism φ from Π to $\mathbb{Z}^{2g}$ with $\varphi(x_i) = e_i$ identifies $\Pi/(\Pi, \Pi) = H_1(\Pi, \mathbb{Z})$ with $\mathbb{Z}^{2g}$.

Source note. The printed formula has $\varphi(X_i) = e_i$ with an upper-case X_i; the generators immediately above are x_i, which is retained here.

The surface S_0 is a $K(\Pi, 1)$ and thus

$$ H^2(\Pi, \mathbb{Z}) = H^2(S_0, \mathbb{Z}) = \mathbb{Z}, $$

and the cup product defines a symplectic structure on $\Pi/(\Pi,\Pi)$. This structure is identified by φ with the standard symplectic structure of $\mathbb{Z}^{2g}$.

We denote by $\mathrm{Aut}^0(\Pi)$ the subgroup of $\mathrm{Aut}(\Pi)$ which acts trivially on $H^2(\Pi,\mathbb{Z})$. Dehn has proved that each exterior automorphism of Π is induced by a diffeomorphism of S_0 onto itself and that the map induced by φ:

$$\mathrm{Aut}^0(\Pi) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}),$$

is surjective (see [Ma]).

Theorem (5.13). — The number of connected components of any geometric fibre of the projection of ${}_G\mathcal{M}_g^0$ onto $\mathrm{Spec}(\mathbb{Z}[1/n])$ *is equal to the number of orbits of* $\mathrm{Aut}^0(\Pi)$ *in the set of exterior epimorphisms from* Π *to* G.

Proof. — By (5.11), it suffices to prove that ${}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\mathbb{C})$ has the said number of connected components. As follows from (4.14),

$$\pi_0({}_G\mathcal{M}_g^0 \times \mathrm{Spec}(\mathbb{C})) = \pi_0({}_GM_g)$$

where ${}_GM_g$ is the coarse moduli scheme classifying stable smooth curves of genus g over $\mathbb{C}$ with a Teichmüller structure of level G.

Source note. The printed proof begins "By (5.13)", a self-reference to the theorem being proved; Corollary (5.11) is the result that reduces the claim to one geometric fibre, and is cited above. The printed phrase "As results from (4.14)" is corrected to "As follows from (4.14)".

Recall that a Teichmüller curve of genus g is a stable smooth curve C of genus g over $\mathbb{C}$ together with an exterior isomorphism φ of the transcendental fundamental group $\pi_1(C)$ with Π, which induces a symplectic isomorphism[6] between

$$H_1(C,\mathbb{Z}) \simeq \pi_1(C)/(\pi_1(C),\pi_1(C))$$

and $\Pi/(\Pi,\Pi)$. By Teichmüller's theory [W], the analytic space T_g classifying Teichmüller curves of a given genus $g \geq 2$ is homeomorphic to a ball, and hence connected.

If ψ is a surjective homomorphism from Π to G, the map

$$(C,\varphi) \longmapsto (C,\psi\varphi)$$

defines a morphism $t_\psi : T_g \to {}_GM_g$. Two such maps t_ψ and $t_{\psi'}$ have the same image if and only if $\psi = \psi'\sigma$ for $\sigma \in \mathrm{Aut}^0(\Pi)$, and

$$ {}_GM_g = \coprod_{\psi \bmod \mathrm{Aut}^0(\Pi)} t_\psi(T_g)$$

which implies (5.13).

(5.14) Let us denote by ${}_n\mathcal{M}_g^0$ the algebraic stack classifying stable smooth curves with a Jacobi structure of level n. This algebraic stack "is" a true scheme for $n \geq 3$ (by [S]).

If φ is a Jacobi structure of level n on a stable smooth curve $p : X \to S$, we define the "multiplicator" $\mu(\varphi)$ of φ by the commutative diagram

$$\begin{tikzcd}
\Lambda^2(\mathbb{Z}/n\mathbb{Z})^{2g} \arrow[r] \arrow[d, "\Lambda^2\varphi"] & \mathbb{Z}/n\mathbb{Z} \arrow[d, "\mu(\varphi)"] \\
\Lambda^2 R^1p_*(\mathbb{Z}/n\mathbb{Z}) \arrow[r, "\Lambda"] & \mu_n^{\otimes -1} \simeq R^2p_*(\mathbb{Z}/n\mathbb{Z})
\end{tikzcd}$$

The scheme of isomorphisms between $\mathbb{Z}/n\mathbb{Z}$ and $\mu_n^{\otimes -1}$ is $\mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$; thus $\varphi \mapsto \mu(\varphi)$ defines a morphism μ from S to $\mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This being true for any X and S, μ is induced by

$$\mu : {}_n\mathcal{M}_g^0 \longrightarrow \mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n]).$$

Theorem (5.15). — The geometric fibres of the morphism μ are connected.

Proof. — By definition, ${}_n\mathcal{M}_g^0$ is open and closed in ${}_G\mathcal{M}_g^0$ for $G = (\mathbb{Z}/n\mathbb{Z})^{2g}$. The group $\mathrm{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})$ acts on G, and thus on ${}_G\mathcal{M}_g^0$. One has:

[6] An arbitrary isomorphism φ induces an isomorphism between $H_1(C,\mathbb{Z})$ and $\Pi/(\Pi,\Pi)$ which is always symplectic up to sign.

(i) the open subset ${}_n\mathcal{M}_g^0$ of ${}_G\mathcal{M}_g^0$ is stable under the subgroup $H = \mathrm{CSp}_{2g}(\mathbb{Z}/n\mathbb{Z})$ of symplectic similitudes;

(ii) ${}_G\mathcal{M}_g^0 = \coprod_{\sigma \in \mathrm{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})/H} \sigma({}_n\mathcal{M}_g^0)$.

Source note. The printed coproduct is indexed by G/H, although $G = (\mathbb{Z}/n\mathbb{Z})^{2g}$ and H has just been defined as a subgroup of $\mathrm{GL}_{2g}(\mathbb{Z}/n\mathbb{Z})$. The required coset space of the acting group is written explicitly above.

It now follows from (5.11) that all geometric fibres of μ have the same number of connected components.

Source note. The printed phrase "It now results from (5.11)" is corrected to "It now follows from (5.11)".

Consider the geometric fibre of μ at the standard complex place of $\mathrm{Spec}(\mathbb{Z}[e^{2\pi i/n}, 1/n])$. This fibre is the algebraic stack classifying complex stable smooth curves C provided with a symplectic isomorphism

$$H^1(C, \mathbb{Z}/n\mathbb{Z}) \xrightarrow{\sim} (\mathbb{Z}/n\mathbb{Z})^{2g}.$$

Reasoning as in (5.13), we are reduced to proving

Lemma (5.16). — The homomorphism

$$\mathrm{Aut}^0(\Pi) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

Proof. — This follows from Dehn's theory (5.12) and from the fact that the groups Sp_{2g}, being split semi-simple simply connected groups, are generated by their unipotent elements, so that the reduction map

$$\mathrm{Sp}_{2g}(\mathbb{Z}) \longrightarrow \mathrm{Sp}_{2g}(\mathbb{Z}/n\mathbb{Z})$$

is surjective.

Source note. The printed proof says "This results from Dehn's theory" and names the groups Sp_n. The English construction is corrected to "This follows", and the group is restored to Sp_{2g}, as required by both the lemma and the displayed reduction map.

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MODULAR FORMS AND ℓ-ADIC REPRESENTATIONS

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Pierre Deligne

Editorial note. The controlling source for this English edition is an English retype dated 28 June 2004. The published French Bourbaki original, pp. 139–172, controls the French edition and adjudicates damaged English-retype passages. Evident non-mathematical typographical slips are regularized in the running text; substantive or non-obvious restorations are disclosed in place and registered in the source-alignment ledger delivered with this edition.

1 Introduction

Let

$$D(q) = q\prod_{n=1}^{\infty}(1-q^n)^{24} = \sum_{n=1}^{\infty}\tau(n)q^n \qquad (|q|<1)$$

and

$$\Delta(z) = D(e^{2\pi iz}) \qquad (\operatorname{Im} z > 0).$$

It is known that the function Δ is, up to a constant factor, the unique cusp form of weight 12 for the group $\mathrm{SL}_2(\mathbb{Z})$. Put, for p prime,

$$H_p(X) = 1 - \tau(p)X + p^{11}X^2.$$

Following Hecke's theory, the Dirichlet series

$$L_\tau(s) = \sum \tau(n)n^{-s} = \prod_p \frac{1}{H_p(p^{-s})}$$

extends to an entire function of s, and the function

$$(2\pi)^{-s}\Gamma(s)L_\tau(s)$$

is invariant under $s \mapsto 12-s$. Ramanujan's conjecture asserts that the roots of the polynomial H_p have absolute value $p^{-11/2}$; equivalently, $|\tau(p)| < 2p^{11/2}$.

These properties, proven or conjectural, are similar to conjectural properties of zeta functions of algebraic varieties over $\mathbb{Q}$. Suggested by this, as a first approximation, one is tempted to interpret L_τ as the zeta function of such a variety.

For each prime number ℓ, let K_ℓ be the maximal extension of $\mathbb{Q}$ unramified away from ℓ and, for $p \neq \ell$, let F_p be the inverse in $\mathrm{Gal}(K_\ell/\mathbb{Q})$ of the Frobenius element φ_p relative to p. This object is well defined up to conjugation.

Translating the preceding in terms of ℓ-adic cohomology, Serre conjectured the existence, for each ℓ, of a representation of $\mathrm{Gal}(K_\ell/\mathbb{Q})$ in a $\mathbb{Q}_\ell$-vector space W_ℓ of rank 2 such that, for each $p \neq \ell$,

$$H_p(X) = \det(1 - F_pX; W_\ell).$$

Moreover, the representation W_ℓ should lie in the range of application of the Weil conjectures, and Ramanujan's conjecture should be a particular case of them.

This program has been carried out by Kuga–Shimura [4] in the analogous case of modular forms relative to certain subgroups of $\mathrm{SL}_2(\mathbb{R})$ with compact quotient. Reduced to the present case, the fundamental idea of Sato–Kuga–Shimura is the following: if E is the universal elliptic curve over the moduli scheme S of elliptic curves (forgetting that it does not exist) and if E^k is the fiber product of E with itself over S, repeated k times, then $L_\tau(s)$ is essentially the zeta function of E^k for $k = 10 = 12-2$.

We show below how to resolve the difficulties created by the points at infinity, and how to construct the representations W_ℓ having the properties indicated above. For more historical details and for applications, see Serre [6].

Source note. The retyped source has "the difficulties creeted by the points." The evident spelling is regularized, and "at infinity" is restored from the subject of the construction that follows.

Notation.

- We denote by $\mathbb{A}$ the ring of adeles of $\mathbb{Q}$, by $\mathbb{A}_f$ the ring of finite adeles, the restricted product over all prime numbers of the fields $\mathbb{Q}_p$, and, for S a finite set of prime numbers, put

$$\mathbb{A}_f^S = \prod_{p \in S} \mathbb{Q}_p \times \prod_{p \notin S} \mathbb{Z}_p \subset \mathbb{A}_f.$$

  Source note. In the second product the retyped source prints $\mathbb{Q}_p$. The restricted-product definition and the immediately following identity $\widehat{\mathbb{Z}} = \mathbb{A}_f^{\varnothing}$ require $\mathbb{Z}_p$, as delivered here. For $S = \varnothing$, one writes $\widehat{\mathbb{Z}} = \mathbb{A}_f^{\varnothing}$.
- If X is a topological space, or the étale site of a scheme, and G is a set, G also denotes the constant sheaf on X defined by G.
- We denote by $\mathbb{G}_a$ and $\mathbb{G}_m$ the additive and multiplicative groups.
- An elliptic curve is an abelian variety of dimension one, and in particular is equipped with an origin.
- If L is an invertible sheaf and $n \in \mathbb{Z}$, L^n denotes the nth tensor power $L^{\otimes n}$.
- We denote by $\overline{\mathbb{Q}}$ the algebraic closure of $\mathbb{Q}$ in $\mathbb{C}$.
- The symbol $\square$ marks the end of a proof, or its absence.

2 The Shimura Isomorphism

(2.1) An elliptic curve over a complex analytic space S is a proper and flat morphism of analytic spaces $f : E \to S$, equipped with a section e, whose fibers are elliptic curves. An elliptic curve over S admits one and only one S-group law

$$\mu : E \times_S E \to E$$

whose unit section is e. Associated with an elliptic curve are the following.

(a) The invertible sheaf

$$\omega_E = e^* \Omega^1_{E/S}.$$

The relative Lie algebra $\mathrm{Lie}_S(E)$ is the invertible sheaf ω^{-1} dual to ω. One has

$$f_* \Omega^1_{E/S} \xrightarrow{\sim} \omega.$$

(b) The local system of free rank-2 $\mathbb{Z}$-modules $R^1 f_* \mathbb{Z}$. One puts

$$T_{\mathbb{Z}}(E) = (R^1 f_* \mathbb{Z})^\vee, \qquad T_{\mathbb{Q}}(E) = T_{\mathbb{Z}}(E) \otimes \mathbb{Q},$$

the local system of homology of E over S.

The exponential mapping defines an exact sequence of sheaves of sections

$$0 \longrightarrow T_{\mathbb{Z}}(E) \xrightarrow{\alpha} \omega^{-1} \longrightarrow E \longrightarrow 0,$$

so that the elliptic curve E can be reconstructed from the map α.

The local system $\bigwedge^2 R^1 f_* \mathbb{Z} \simeq R^2 f_* \mathbb{Z}$ is canonically isomorphic to $\mathbb{Z}$. An isomorphism between $\mathbb{Z}^2$ and $R^1 f_* \mathbb{Z}$ is said to be admissible if it induces 1 on second exterior powers. Denote by $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$ the set of $\mathbb{R}$-vector-space isomorphisms between $\mathbb{R}^2$ and $\mathbb{C}$ which respect the natural orientations of $\mathbb{R}^2$ and $\mathbb{C}$, defined by $e_1 \wedge e_2 > 0$ and $1 \wedge i > 0$. Such a homomorphism is determined by its restriction to $\mathbb{Z}^2$, and one puts

$$\mathrm{Hom}^+(\mathbb{Z}^2, \mathbb{C}) = \mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C}).$$

Source note. The French original prints that an admissible isomorphism "induces -1 (sic)" and that the elements of Hom^+ "do not respect (sic)" the orientations. The plus sign, the displayed orientations, the Poincaré half-plane in 2.2(ii), and the ensuing construction require respectively 1 and "respect," as delivered here. This space is equipped with the complex structure obtained from its inclusion in the complex vector space $\mathrm{Hom}(\mathbb{Z}^2, \mathbb{C})$. Over this space one arranges a "universal" exact sequence

$$0 \longrightarrow \mathbb{Z}^2 \xrightarrow{\alpha} \mathbb{G}_a \longrightarrow E_0 \longrightarrow 0.$$

Proposition (2.2)**.**

(i) The functor which associates with each analytic space S the set of isomorphism classes of elliptic curves E over S, equipped with isomorphisms

$$\omega_E \simeq \mathbb{G}_a, \qquad R^1 f_* \mathbb{Z} \simeq \mathbb{Z}^2$$

(the last being admissible), is represented by the analytic space $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$*, equipped with a universal elliptic curve E_0.*

(ii) The functor which associates with each analytic space S the set of isomorphism classes of elliptic curves over S, equipped with an admissible isomorphism $R^1 f_* \mathbb{Z} \simeq \mathbb{Z}^2$, is represented by the analytic space

$$X = \mathbb{C}^\times \backslash \mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C}),$$

the Poincaré half-plane.

(iii) The space $\mathrm{Hom}^+(\mathbb{R}^2, \mathbb{C})$ *is a principal homogeneous space of the group $\mathbb{G}_m$ over X.*

One can again regard X as the set of complex structures on $\mathbb{R}^2$. This space is, by (ii), equipped with a universal elliptic curve E_X, whose local system of real cohomology is canonically isomorphic to $\mathbb{R}^2$. Let ω be the invertible sheaf associated with E_X.

The coherent analytic sheaf

$$R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X$$

is the relative de Rham cohomology sheaf of E_X over X, and as such it is inserted into an exact sequence, the Hodge filtration,

$$0 \longrightarrow \omega \longrightarrow R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X \xrightarrow{q} \omega^{-1} \longrightarrow 0$$

since, by Serre duality, $\omega^{-1} \simeq R^1 f_* \mathcal{O}_{E_X}$.

The functorial description in 2.2(ii) evidently gives a right action of the group $\mathrm{SL}_2(\mathbb{Z})$ on (X, E_X): to $\gamma \in \mathrm{SL}_2(\mathbb{Z})$ and to the elliptic curve E equipped with $\alpha : \mathbb{Z}^2 \xrightarrow{\sim} R^1 f_* \mathbb{Z}$, one associates $(E, \alpha \circ \gamma)$. If one regards X, equipped with

$$q : \mathbb{R}^2 \otimes_{\mathbb{R}} \mathcal{O}_X \simeq R^1 f_* \mathbb{R} \otimes_{\mathbb{R}} \mathcal{O}_X \longrightarrow \omega^{-1},$$

as classifying the complex structures on $\mathbb{R}^2$, one has the same evident action of the group $\mathrm{GL}_2^+(\mathbb{R})$ on $(X, \mathbb{R}^2, \omega, q)$.

(2.3) Choose a basis (x_1, x_2) of $\mathbb{R}^2$ such that $x_1 \wedge x_2 > 0$. A point

$$(f : \mathbb{R}^2 \to \mathbb{C}) \quad \mod \mathbb{C}^\times$$

of X is located by

$$z = \frac{f(x_1)}{f(x_2)} \qquad (\operatorname{Im} z > 0),$$

and the map q is identified with

$$q : \mathbb{R}^2 \to \mathbb{G}_a, \qquad a x_1 + b x_2 \longmapsto az + b.$$

This gives an evident trivialization of ω^{-1} over X, which is not equivariant. Relative to this trivialization, a section $f(z)$ of ω^k over X is transformed by an element

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} \in \mathrm{GL}^+(\mathbb{R}^2)$$

(with respect to the basis (x_1, x_2)) by

$$f \cdot \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} (z) = (cz+d)^{-k} f\left(\frac{az+b}{cz+d}\right).$$

Source note. In the first matrix on the printed line, both the French original and the English retype have $\left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right)^{-1}$. The fractional-linear formula on that same line and the determinant matrix immediately below both require $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)^{-1}$, as delivered here. From the identity

$$dz = (cz+d)^2 \, d\left(\frac{az+b}{cz+d}\right) \begin{vmatrix} a & b \\ c & d \end{vmatrix}^{-1},$$

one deduces that dz is a section of $\omega^{-2}\Omega_X^1$ invariant under $\mathrm{SL}_2(\mathbb{R})$. This section is everywhere nonzero and defines an equivariant isomorphism of $\mathrm{SL}_2(\mathbb{R})$-sheaves between ω^2 and Ω_X^1.

(2.4) Let Γ be a discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$ which has no elements of finite order and has quotient of finite volume. One then knows that the quotient space X/Γ is identified with a smooth projective curve $\overline{X/\Gamma}$ minus a finite number of points. The group Γ acts on X without fixed points. The equivariant local system $\mathbb{R}^2$ on X, as well as the exact sequence

$$0 \to \omega \to \mathbb{R}^2 \otimes_{\mathbb{R}} \mathcal{O} \to \omega^{-1} \to 0,$$

therefore define a local system U over X/Γ and an exact sequence

$$0 \to \omega \to U \otimes_{\mathbb{R}} \mathcal{O} \to \omega^{-1} \to 0. \tag{2.5}$$

In the particular case where $\Gamma \subset \mathrm{SL}_2(\mathbb{Z})$, these structures are obtained from an elliptic curve E over X/Γ whose inverse image is the equivariant elliptic curve E_X over X.

(2.6) The points at infinity of X/Γ are described as follows (see [9]).

(a) They correspond to conjugacy classes in Γ of nontrivial subgroups of Γ which are maximal among the subgroups of Γ consisting of unipotent elements.

(b) Let $\Gamma_0 \subset \Gamma$ be one such subgroup and choose a basis (x_1, x_2) of $\mathbb{R}^2$ such that, in this basis, Γ_0 is represented as the set of matrices

$$\left\{ \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix} \,\middle|\, n \in \mathbb{Z} \right\}.$$

Let z be the coordinate (2.3) on X defined by (x_1, x_2). There exists N such that the subset

$$X_N = \{ z \mid \operatorname{Im} z > N \}$$

of X is disjoint from all of its conjugates γX_N for $\gamma \in \Gamma/\Gamma_0$, so that $X_N/\Gamma_0 \hookrightarrow X/\Gamma$. The function $q = e^{2\pi i z}$ establishes an isomorphism between X_N/Γ_0 and the punctured disk

$$0 < |q| < e^{-2\pi N}.$$

If P_{Γ_0} is the point of $\overline{X/\Gamma} - X/\Gamma$ associated with Γ_0, this isomorphism extends to an isomorphism of a neighborhood of P_{Γ_0} with the disk $0 \le |q| < e^{-2\pi N}$.

Source note. The retyped source prints the two disk inequalities without absolute-value bars around the complex coordinate q. The disk description requires $|q|$, which is restored in both inequalities above.

By virtue of (2.3), the sections of ω over X_N which are invariant under Γ_0 are identified with periodic holomorphic functions of period one on X_N. One again denotes by ω the invertible sheaf over $\overline{X/\Gamma}$ which extends ω and such that, in a neighborhood of a point P_{Γ_0}, the section of ω over X_N/Γ_0 defined by the function 1 extends to an invertible section on X_N/Γ_0.

(2.7) Over X/Γ one has the two invertible sheaves Ω^1 and ω^2, and an isomorphism φ (2.3) between their restrictions to X/Γ. From the formula

$$dq = d(e^{2\pi i z}) = 2\pi i e^{2\pi i z}\, dz = 2\pi i q\, dz$$

it follows that the map

$$\varphi : \Omega^1 \to \omega^2$$

extends to $\overline{X/\Gamma}$, and has a simple zero at each of the points at infinity.

Definition (2.8)**.** *The space of cusp forms of weight $k+2$, relative to the group Γ, is the space of global sections*

$$H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k).$$

By virtue of (2.7), this space is again identified with the space of global sections of ω^{k+2} vanishing at infinity, that is, at each "point."

(2.9) Denote by U^k the kth symmetric power of the local system U over X/Γ. The map (2.5) induces a map

$$\iota^k : \omega^k \to U^k \otimes_{\mathbb{R}} \mathcal{O},$$

hence a map, again denoted by ι^k,

$$\iota^k : \Omega^1 \otimes \omega^k \to \Omega^1(U^k),$$

where the target is the sheaf of holomorphic differential forms over X/Γ with coefficients in the local system U^k.

The de Rham resolution of $U^k \otimes_{\mathbb{R}} \mathbb{C}$,

$$0 \to U^k \otimes_{\mathbb{R}} \mathbb{C} \to U^k \otimes_{\mathbb{R}} \mathcal{O} \xrightarrow{d} U^k \otimes_{\mathbb{R}} \Omega^1 \to 0,$$

induces a map

$$\delta : H^0(X/\Gamma, \Omega^1(U^k)) \to H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

Moreover, the cohomology space $H^1(X/\Gamma, U^k \otimes \mathbb{C})$ is equipped with a natural complex conjugation, such that δ defines a conjugate-linear mapping $\overline{\delta}$ from the complex conjugate of $H^0(X/\Gamma, \Omega^1(U^k))$ into $H^1(X/\Gamma, U^k \otimes \mathbb{C})$. One thus obtains a map

$$\mathrm{sh}_0 = \delta \circ H^0(\iota^k) \oplus \overline{\delta} \circ H^0(\overline{\iota^k})$$

$$\mathrm{sh}_0 : H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} \longrightarrow H^1(X/\Gamma, U^k \otimes \mathbb{C}).$$

For an arbitrary sheaf F on a space Y, one denotes by $\widetilde{H}^i(Y, F)$ the image of cohomology with compact supports $H^i_c(Y, F)$ in cohomology without supports $H^i(Y, F)$.

Theorem 4.2.6 of [9] is essentially equivalent to the following theorem; in loc. cit. k is supposed even, but the same proof works in general.

Theorem (2.10, Shimura [7])**.** *There exists an isomorphism* sh *making the following diagram commutative:*

$$\begin{array}{ccc}
H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} & \xrightarrow{\mathrm{sh}} & \widetilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \\
\cap & & \cap \\
H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{X/\Gamma}, \Omega^1 \otimes \omega^k)} & \xrightarrow{\mathrm{sh}_0} & H^1(X/\Gamma, U^k \otimes \mathbb{C}).
\end{array}$$

One calls sh *the Shimura isomorphism.*

(2.11) In the particular case where Γ is a subgroup of finite index of $\mathrm{SL}_2(\mathbb{Z})$, the elliptic curve E over X/Γ furnishes an elliptic curve scheme over the algebraic curve X/Γ; equivalently, its modular invariant is meromorphic at infinity. It therefore admits a Néron model $\overline{E}$ over $\overline{X/\Gamma}$. One can show that the fibers of $\overline{E}$ at the points at infinity are of multiplicative type, and that over all of $\overline{X/\Gamma}$ one has

$$\omega = e^* \Omega^1_{\overline{E}/\overline{X/\Gamma}}.$$

In this particular case, one has $U = R^1 f_* \mathbb{Z} \otimes \mathbb{R}$, so that the target of the Shimura isomorphism can be rewritten as

$$\widetilde{H}^1(X/\Gamma, U^k \otimes \mathbb{C}) \simeq \widetilde{H}^1(X/\Gamma, \mathrm{Sym}^k(R^1 f_* \mathbb{Z})) \otimes_{\mathbb{Z}} \mathbb{C}.$$

Source note. The printed text twice has the typographical form "fournishes"; both occurrences are delivered as "furnishes."

3 Hecke Operators and the Fundamental ℓ-adic Representation

(3.1) Recall (cf. [3]) that the category of "locally constant" constructible $\mathbb{Z}_\ell$-sheaves, abbreviated lcc, over a scheme S is the category of projective systems $(F_n)_{n \in \mathbb{N}}$ on the étale site $S_{\text{ét}}$ of S satisfying:

(i) F_n is a locally constant sheaf of $\mathbb{Z}/(\ell^n)$-modules of finite type;

(ii) if $n \leq m$, then

$$F_m \otimes_{\mathbb{Z}/(\ell^m)} \mathbb{Z}/(\ell^n) \xrightarrow{\sim} F_n.$$

The lcc $\mathbb{Z}_\ell$-sheaves form a stack of abelian categories over S; the stack of lcc $\mathbb{Q}_\ell$-sheaves is the quotient of this stack by the thick substack of lcc $\mathbb{Z}_\ell$-sheaves killed by a power of ℓ. One denotes by $\otimes\mathbb{Q}_\ell$ the canonical functor from the category of lcc $\mathbb{Z}_\ell$-sheaves into that of lcc $\mathbb{Q}_\ell$-sheaves.

If S is connected and equipped with a geometric point s, the category of lcc $\mathbb{Z}_\ell$-sheaves, respectively $\mathbb{Q}_\ell$-sheaves, over S is equivalent, by the functor "fiber over s," to the category of continuous representations of the fundamental group $\pi_1(S,s)$ on $\mathbb{Z}_\ell$-modules of finite type, respectively on finite-dimensional $\mathbb{Q}_\ell$-vector spaces.

If T is a finite set of prime numbers, an lcc $\mathbb{A}^f_T$-sheaf consists of the giving, for each prime number ℓ, of an lcc $\mathbb{Z}_\ell$-sheaf if $\ell \notin T$, and an lcc $\mathbb{Q}_\ell$-sheaf if $\ell \in T$. For $T = \varnothing$, one speaks of lcc $\widehat{\mathbb{Z}}$-sheaves rather than of lcc $\mathbb{A}^f_\varnothing$-sheaves.

For general T, the category of lcc $\mathbb{A}^f_T$-sheaves is the inductive limit of the categories of lcc $\mathbb{A}^f_{T'}$-sheaves for $T' \subset T$ finite. One puts

$$\mathbb{Z}_\ell = \varprojlim_n \mathbb{Z}/(\ell^n), \qquad \mathbb{Q}_\ell = \mathbb{Z}_\ell \otimes \mathbb{Q}_\ell, \qquad \widehat{\mathbb{Z}} = (\mathbb{Z}_\ell)_\ell, \qquad \mathbb{A}^f_T = \widehat{\mathbb{Z}} \otimes \mathbb{A}^f_T.$$

Source note. The English retype's obvious spelling slips on this page—including "Shiumra," "particluar," "schme," "constent," and "numbesr"—are regularized without changing their mathematical content.

The stack of elliptic curves up to isogeny over S is the stack obtained from the stack of elliptic curves over S by "formally inverting isogenies." One denotes by $\otimes\mathbb{Q}$ the functor associating to an elliptic curve its underlying elliptic curve up to isogeny. For S quasi-compact, one has

$$\mathrm{Hom}(E,F)\otimes\mathbb{Q} \xrightarrow{\sim} \mathrm{Hom}(E\otimes\mathbb{Q}, F\otimes\mathbb{Q}),$$

and, for S normal, every elliptic curve up to isogeny over S underlies an elliptic curve over S.

(3.2) Let $f : E \to S$ be an elliptic curve over a scheme S. One denotes by $T_\ell(E)$ the projective system of kernels E_{ℓ^n} of multiplication by ℓ^n on E, the transition maps from E_{ℓ^n} to E_{ℓ^m}, for $n \geq m$, being multiplication by ℓ^{n-m}. Proceeding in the same way for $\mathbb{G}_m$, one puts $T_\ell(\mathbb{G}_m) = \mathbb{Z}_\ell(1)$. If ℓ is invertible on S, then $T_\ell(E)$ and $\mathbb{Z}_\ell(1)$ are $\mathbb{Z}_\ell$-sheaves on S. One defines $T_\infty(E)$ to be the relative Lie algebra of E over S, that is, the invertible dual of the invertible sheaf ω of (2.1(a)).

Suppose S is of characteristic 0. One then defines the $\widehat{\mathbb{Z}}$-sheaf $T_f(E)$ over S to be the system of the $T_\ell(E)$, and puts

$$V_f(E) = T_f(E) \otimes \mathbb{A}_f.$$

If $u : E \to F$ is an isogeny, then u induces an isomorphism of $V_f(E)$ onto $V_f(F)$ and of $T_\infty(E)$ onto $T_\infty(F)$; the functors V_f and T_∞ therefore factor through the category of elliptic curves up to isogeny over S.

Proposition (3.3)**.** *Let S be a scheme of characteristic* 0*; let $E_1(S)$ be the category of elliptic curves over S; and let $E_2(S)$ be the category of triples composed of an elliptic curve up to isogeny E over S, a $\widehat{\mathbb{Z}}$-sheaf T which is a twisted form of $\widehat{\mathbb{Z}}^2$, and an isomorphism*

$$\beta : V_f(E) \xrightarrow{\sim} T \otimes \mathbb{A}_f.$$

Then the functor

$$I : E \longmapsto \big(E\otimes\mathbb{Q},\ T_f(E),\ V_f(E) \simeq T_f(E)\otimes\mathbb{A}_f\big)$$

from $E_1(S)$ to $E_2(S)$ is an equivalence of categories.

The question is local on S, which one may suppose quasi-compact. If $f : E \to F$ is a morphism of S-elliptic curves, and if f is an isogeny, then one has an exact sequence

$$0 \to T_f(E) \to T_f(F) \to \ker(f) \to 0. \tag{3.4}$$

The morphism f is divisible by n if and only if it kills the kernel E_n of multiplication by n, because the "multiplication by n" map $E/E_n \to E$ is an isomorphism. By (3.4), this holds if and only if $T_f(f)$ is divisible by n, and one deduces from this that $\mathrm{Hom}_S(E,F)$ is the subgroup of $\mathrm{Hom}_S(E\otimes\mathbb{Q}, F\otimes\mathbb{Q})$ consisting of the morphisms f such that $V_f(f)$ sends $T_f(E)$ into $T_f(F)$. The functor I is thus fully faithful.

Source note. The canonical French original says "pleinement fidèle," whereas the inherited English retype says "faithfully flat." Since I is a functor between categories and the preceding argument identifies its Hom sets, the

French reading "fully faithful" is retained. The English retype's obvious case slips $u : e \to F$ and $E_1(s)$ are normalized to $u : E \to F$ and $E_1(S)$.

Let $X \in \mathrm{Ob}(E_2(S))$. Locally on S, X is defined by an elliptic curve up to isogeny $E \otimes \mathbb{Q}$, and by a "lattice" T in $V_f(E)$ which, for almost all ℓ, coincides with $T_\ell(E)$. For $q \in \mathbb{Q}$, $(E \otimes \mathbb{Q}, T)$ is isomorphic to $(E \otimes \mathbb{Q}, qT)$, which allows us to suppose that $T_f(E) \subset T$. The quotient $K = T/T_f(E)$ is then canonically isomorphic to a finite subgroup of E, and X is the image of E/K under I (cf. 3.4). □

Corollary (3.5)**.** *The functor $F_1(S)$, respectively $F_1'(S)$, which associates with each scheme S of characteristic* 0 *the set of isomorphism classes of elliptic curves over S equipped with an isomorphism*

$$\alpha : T_f(E) \xrightarrow{\sim} \widehat{\mathbb{Z}}^2$$

respectively also with an isomorphism

$$\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathbb{G}_a,$$

is isomorphic to the functor $F_2(S)$, respectively $F_2'(S)$, which associates with S the set of isomorphism classes of elliptic curves up to isogeny over S, equipped with an isomorphism

$$\beta : V_f(F) \xrightarrow{\sim} \mathbb{A}_f^2$$

respectively also with an isomorphism

$$\beta_\infty : T_\infty(F) \xrightarrow{\sim} \mathbb{G}_a.$$

Proposition (3.6)**.** *The functor F_1 (respectively F_1') is represented by a scheme M_∞ (respectively M_∞') over $\mathbb{Q}$.*

Let $n \geq 3$ be an integer. The functor which associates with each scheme S the set of isomorphism classes of elliptic curves, equipped with an isomorphism

$$\alpha_n : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$$

respectively also with $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, is represented by an affine curve M_n (respectively by an affine surface M_n') over $\mathrm{Spec}(\mathbb{Z}[1/n])$. For $n \mid m$, the morphism from M_m to M_n defined by

$$(E, \alpha_m : E_m \xrightarrow{\sim} (\mathbb{Z}/m)^2) \longmapsto (E, (n/m)\alpha_m : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2)$$

is finite and étale over $\mathrm{Spec}(\mathbb{Z}[1/m])$, and one has

$$M_\infty = \varprojlim M_n.$$

One proceeds in the same way to represent F_1'. □

Source note. The canonical French original gives $\mathcal{O}_S$ and the factor $(n/m)\alpha_m$. The inherited readings $\mathcal{O}_X$ and $(m/n)\alpha_m$, incompatible with the functor's base and with restriction of level for $n \mid m$, are corrected in both editions. The English retype's "schem" and duplicated phrase "One goes one proceeds" are also regularized to "scheme" and "One proceeds."

(3.7) The scheme M_∞ (respectively M_∞') is equipped with a universal elliptic curve $f_\infty : E_\infty \to M_\infty$ and an isomorphism

$$\alpha : T_f(E_\infty) \xrightarrow{\sim} \widehat{\mathbb{Z}}^2,$$

respectively also with an isomorphism

$$\alpha_\infty : T_\infty(E_\infty) \xrightarrow{\sim} \mathbb{G}_a.$$

Following (3.5), the scheme M_∞ (respectively M_∞') represents the functor F_2 (respectively F_2'), which gives an evident left action of the adelic group $\mathrm{GL}_2(\mathbb{A}_f)$ on

$$(M_\infty, E_\infty \otimes \mathbb{Q}, \alpha \otimes \mathbb{A}_f)$$

respectively on

$$(M_\infty', E_\infty \otimes \mathbb{Q}, \alpha \otimes \mathbb{A}_f, \alpha_\infty).$$

On the functor this action is given, for $g \in \mathrm{GL}_2(\mathbb{A}_f)$, by

$$g : (F, \beta : V_f(F) \xrightarrow{\sim} \mathbb{A}_f^2, \beta_\infty) \longmapsto (F, g \circ \beta : V_f(F) \xrightarrow{\sim} \mathbb{A}_f^2, \beta_\infty).$$

Source note. Both printed witnesses abbreviate the induced trivialization as $\alpha \otimes \mathbb{A}_f^2$ and write $V_f(E)$ inside a tuple whose elliptic curve is F. Since $\alpha : T_f(E_\infty) \to \widehat{\mathbb{Z}}^2$, scalar extension is $\alpha \otimes \mathbb{A}_f$, and the action's domain is $V_f(F)$; these type-required readings are delivered above. The French original also uses the same universal curve E_∞ in both tuples; the inherited prime on E'_∞ is therefore removed in both editions. Shafarevich first noted this fact.

Let Y be a scheme over $\mathbb{C}$, equal to the projective limit of schemes Y_i of finite type over $\mathbb{C}$, with finite transition maps. The locally compact ringed space Y^{an}, equal to the projective limit of the Y_i^{an}, depends only on Y and not on its representation as a projective limit. If Y is a scheme over $\mathbb{Q}$, equal to the projective limit of schemes Y_i of finite type over $\mathbb{Q}$ with finite transition maps, one puts $Y^{\mathrm{an}} = (Y \otimes \mathbb{C})^{\mathrm{an}}$. This applies to M_∞ and M'_∞.

Proposition (3.8)**.** *One has canonically*

$$M'^{\mathrm{an}}_\infty \simeq \mathrm{Hom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2)/\,\mathrm{GL}_2(\mathbb{Q})$$

and

$$M^{\mathrm{an}}_\infty \simeq \mathbb{C}^\times \backslash \mathrm{Hom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2)/\,\mathrm{GL}_2(\mathbb{Q}),$$

and, less canonically,

$$M'^{\mathrm{an}}_\infty \simeq \mathrm{GL}_2(\mathbb{A})/\,\mathrm{GL}_2(\mathbb{Q}), \qquad M^{\mathrm{an}}_\infty \simeq K_\infty \backslash \mathrm{GL}_2(\mathbb{A})/\,\mathrm{GL}_2(\mathbb{Q}),$$

where K_∞ is the maximal compact subgroup at infinity, enlarged by the real homotheties. These isomorphisms are compatible with the action of $\mathrm{GL}_2(\mathbb{A}_f)$*.*

Source note. The IAS English retype contains unresolved "[a/the, bad copy]" and "[bad copy]" placeholders here. The canonical French original reads exactly that K_∞ is the maximal compact subgroup at infinity, enlarged by the real homotheties; that source-established wording is delivered above.

The notion of elliptic curve up to isogeny, and its variants, extends to the complex analytic case. On the other hand, an isogeny $\varphi : E \to F$ induces an isomorphism φ^* between the local systems of rational cohomology of E and F, which allows us to define this last object for an elliptic curve up to isogeny. Let S be a complex analytic space. With the aid of (2.1), one sees that to give an elliptic curve up to isogeny over S is equivalent to giving an invertible sheaf T_∞, a local system $T_\mathbb{Q}$ of $\mathbb{Q}$-vector spaces, and a morphism $\alpha : T_\mathbb{Q} \to T_\infty$ inducing an isomorphism $T_\mathbb{Q} \otimes \mathbb{R} \to T_\infty$ point by point.

Let n be an integer and let K_n be the kernel of the natural mapping

$$\prod_\ell \mathrm{GL}_2(\mathbb{Z}_\ell) \to \mathrm{GL}_2(\mathbb{Z}/n).$$

Let G_1 be the functor which associates with S the set of isomorphism classes of elliptic curves $f : E \to S$ over S, equipped with an isomorphism

$$\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E),$$

an isomorphism $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, and an isomorphism $\alpha_n : E_n \xrightarrow{\sim} (\mathbb{Z}/n)^2$. As in (3.3), one sees that G_1 is isomorphic to the functor G_2 which associates with S the set of isomorphism classes of elliptic curves up to isogeny E over S, equipped with $\varphi : \mathbb{Q}^2 \xrightarrow{\sim} T_\mathbb{Q}(E)$, $\alpha_\infty : T_\infty(E) \xrightarrow{\sim} \mathcal{O}_S$, and an isomorphism $V_f(E) \xrightarrow{\sim} \mathbb{A}_f^2$ given locally over S up to composition with an element of K_n.

Such an object is determined by the composite map φ' (given locally modulo K_n) obtained from

$$\varphi' : \mathbb{Q}^2 \xrightarrow{\varphi} T_\mathbb{Q}(E) \longrightarrow T_\infty(E) \times V_f(E) \xrightarrow{\sim} \mathcal{O}_S \times \mathbb{A}_f^2.$$

One has

$$E = \mathcal{O}_S/\varphi'\big(\mathbb{Q}^2 \cap \varphi'^{-1}(T_\infty(E) \times T_f(E))\big) = \widehat{\mathbb{Z}}^2 \backslash (\mathcal{O}_S \times \mathbb{A}_f^2)/\varphi'(\mathbb{Q}^2),$$

so that, cf. 2.2, G_1 and G_2 are represented by

$$K_n \backslash \mathrm{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2).$$

Suppose still that $n \geq 3$, so that $\mathrm{GL}_2(\mathbb{Q})$ acts freely on the preceding space. The analytic space M_n^{an}, respectively $M_n'^{\mathrm{an}}$, represents the functor analogous in analytic geometry to the functor represented by M_n, respectively M'_n, because this functor, say X, is representable and the arrow $X \to M_n^{\mathrm{an}}$, respectively $X \to M_n'^{\mathrm{an}}$, induces a bijection on sets of points with values in any finite-rank $\mathbb{C}$-algebra. Consequently one deduces from the preceding that

$$M_n'^{\mathrm{an}} \simeq K_n \backslash \operatorname{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2)/\operatorname{GL}_2(\mathbb{Q}).$$

Proceeding in the same way for M_n, one obtains the first assertion of (3.8) by passage to the limit on n.

A point x of

$$\operatorname{Isom}(\mathbb{Q}^2 \otimes \mathbb{A}, \mathbb{C} \times \mathbb{A}_f^2)/\operatorname{GL}_2(\mathbb{Q})$$

is identified with a "lattice" L_x of $\mathbb{C} \times \mathbb{A}_f^2$, and the corresponding curve is

$$E_x \simeq \widehat{\mathbb{Z}}^2 \backslash (\mathbb{C} \times \mathbb{A}_f^2)/L_x,$$

equipped with

$$V_f(E_x) \simeq L_x \otimes \mathbb{A}_f \xrightarrow{\sim} \mathbb{A}_f^2.$$

The last assertion of (3.8) follows easily. □

Denote by $f_n : E \to M_n$ the universal elliptic curve over M_n. The integer k being fixed, one makes the following definition.

Definition (3.9)**.** *One denotes by W (or by ${}^kW$ if confusion is possible) the $\mathbb{Q}$-vector space*

$$W = \varinjlim_n \widetilde{H}^1(M_n^{\mathrm{an}}, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Q})) = \varinjlim_n {}_nW.$$

Source note. The retype's page-level slips "beause," "arros," and "Once deduces" are regularized. In the description of the curve corresponding to x, the printed $V_f(E)$ is delivered as $V_f(E_x)$, the only object just defined there. The English retype then says that W "doesn't depend on" the universal curve, reversing the canonical French "ne dépend que de." The source-coherent reading "depends only on" is delivered below in both editions. The level index n, visible in the French f_{n*}, is also restored in the first formula defining W.

This vector space depends only on the universal elliptic curve, up to isogeny, $f_\infty : E \to M_\infty$, so that, by transport of structure, it is equipped with a left action of the adelic group $\operatorname{GL}_2(\mathbb{A}_f)$.

If ℓ is a prime number, the vector space $W_\ell = W \otimes \mathbb{Q}_\ell$ admits a purely algebraic definition, in terms of the ℓ-adic cohomology of the scheme over the algebraic closure $\overline{\mathbb{Q}}$ of $\mathbb{Q}$ obtained from M_n by extension of scalars:

$$W_\ell = \varinjlim_n \widetilde{H}^1(M_n \otimes \overline{\mathbb{Q}}, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Q}_\ell)) = \varinjlim_n {}_nW_\ell. \tag{3.10}$$

Thus the Galois group of $\overline{\mathbb{Q}}$ over $\mathbb{Q}$ acts, by transport of structure, on W_ℓ and on the finite-level spaces.

Finally, the space M_n^{an} is the disjoint union of quotients of the Poincaré half-plane by congruence subgroups of $\operatorname{SL}_2(\mathbb{Z})$, so that, denoting by ω the invertible sheaf on M_n defined by E, Shimura's theory (2.10) gives

$$W_\infty = W \otimes \mathbb{C} = \varinjlim_n \left(H^0(\overline{M}_n^{\mathrm{an}}, \Omega^1 \otimes \omega^k) \oplus \overline{H^0(\overline{M}_n^{\mathrm{an}}, \Omega^1 \otimes \omega^k)} \right). \tag{3.11}$$

Source note. The inherited transcription dropped the bars on the compactified curves $\overline{M}_n^{\mathrm{an}}$ in both summands of (3.11); they are restored above. The retype's opening "if is equipped" is regularized to "it is equipped." This decomposition of $W \otimes \mathbb{C}$ into two complex-conjugate subspaces, one of which is the space of all cusp forms of weight $k+2$ relative to a general congruence subgroup of $\operatorname{SL}_2(\mathbb{Z})$, is analogous to a Hodge decomposition, of type

$$(0, k+1) + (k+1, 0).$$

The action of the adelic group commutes with the action of the Galois group and respects the preceding decomposition.

While the ℓ-adic local system $R^1 f_* \mathbb{Q}_\ell$ is trivial on M_∞, I do not know whether there is a relation between W_ℓ and

$$\varinjlim_n \big(\widetilde{H}^1(M_n \otimes \overline{\mathbb{Q}}, \mathbb{Q}_\ell)\big) \otimes \operatorname{Sym}^k(\mathbb{Q}_\ell^2).$$

(3.12) Let $n \geq 3$ be an integer and let K_n be as in (3.8). Then one has

$$W^{K_n} = {}_nW.$$

This follows by passage to the limit and from the fact that, in rational cohomology, the cohomology of the quotient of a space by a finite group is obtained by taking the invariants of that group in the cohomology.

Put, for p prime,

$$W^{(p)} = W^{\mathrm{GL}_2(\mathbb{Z}_p)}.$$

By passage to the limit, one obtains

$$W^{(p)} = \varinjlim_{(n,p)=1} {}_nW.$$

On this cohomology space there again act:

(i) the subgroup $\prod_{\ell \neq p} \mathrm{GL}_2(\mathbb{Q}_\ell)$ of $\mathrm{GL}_2(\mathbb{A}_f)$, because this subgroup centralizes $\mathrm{GL}_2(\mathbb{Z}_p)$;

(ii) the Hecke algebra $\mathcal{H}(\mathrm{GL}_2(\mathbb{Q}_p), \mathrm{GL}_2(\mathbb{Z}_p))$, which is the algebra of integral measures on the discrete space $\mathrm{GL}_2(\mathbb{Q}_p)/\mathrm{GL}_2(\mathbb{Z}_p)$ invariant on the left by $\mathrm{GL}_2(\mathbb{Z}_p)$. This subalgebra of the group algebra of $\mathrm{GL}_2(\mathbb{Q}_p)$ acts on W and respects $W^{(p)}$. It acts already on each ${}_nW$ for n prime to p.

The Hecke algebra admits as a basis the measures associated with characteristic functions of the double cosets of $\mathrm{GL}_2(\mathbb{Z}_p)$ in $\mathrm{GL}_2(\mathbb{Q}_p)$, and one knows that

$$\mathcal{H}(\mathrm{GL}_2(\mathbb{Q}_p), \mathrm{GL}_2(\mathbb{Z}_p)) = \mathbb{Z}[T_p, R_p, R_p^{-1}],$$

where T_p and R_p are the double cosets of

$$\begin{pmatrix} 1 & 0 \\ 0 & p^{-1} \end{pmatrix} \qquad \text{and} \qquad \begin{pmatrix} p^{-1} & 0 \\ 0 & p^{-1} \end{pmatrix}.$$

(3.13) Let p be a prime number, let $n \geq 3$ be an integer prime to p, and let $F_{n,p}$ be the functor which associates to each scheme S the set of isomorphism classes of commutative diagrams of S-schemes

$$\begin{array}{ccccc} & & \mathbb{Z}/(n)^2 & & \\ & \alpha \nearrow & & \nwarrow \alpha' & \\ E_n & \longrightarrow & F_n & & \\ \cap & & \cap & & \\ E & \xrightarrow{\varphi} & F & & \end{array} \tag{3.14}$$

where φ is a p-isogeny between elliptic curves and α is an isomorphism. One denotes by q_1 and $q_2 : F_{n,p} \to M_n$ the morphisms of functors associating to a diagram (3.14) the subdiagrams (E, E_n, α) and (F, F_n, α').

Proposition (3.15)**.** *The functor $F_{n,p}$ is represented by a scheme $M_{n,p}$, and the morphisms $q_1, q_2 : M_{n,p} \to M_n$ are finite.*

The automorphism σ of $F_{n,p}$ sending $\varphi : E \to F$ to ${}^t\varphi : F \to E$ exchanges q_1 and q_2; it suffices therefore to consider q_1. This morphism identifies $F_{n,p}$ with the functor of subgroups of order p of the universal elliptic curve E over M_n, so that, by the theory of Hilbert schemes, $F_{n,p}$ is representable and $M_{n,p}$ is proper over M_n. If s is a geometric point of M_n, $q_1^{-1}(s)$ is the set of subgroups of order p of E_s, and has $p+1$ elements if $\mathrm{char}(k(s)) \neq p$, and only one, the kernel of Frobenius, if $\mathrm{char}(k(s)) = p$. □

One can show that $M_{n,p}$ is regular, and that q_1 and q_2 are finite flat; we do not use this delicate result, but content ourselves here to note that over $\mathrm{Spec}(\mathbb{Z}[1/p])$, each q_i makes $M_{n,p}$ an étale covering of degree $p+1$ of M_n.

The morphisms q_i can be inserted into a commutative diagram

$$\begin{tikzcd} & q_1^*E \arrow[rr, "\varphi"] \arrow[dl] \arrow[dr, "u"] & & q_2^*E \arrow[dl, "v"'] \arrow[dr] & \\ E \arrow[dr, "f_n"'] & & M_{n,p} \arrow[dl, "q_1"] \arrow[dr, "q_2"'] & & E \arrow[dl, "f_n"] \\ & M_n & & M_n & \end{tikzcd} \tag{3.16}$$

where (φ, u, v) is part of the universal diagram (3.14).

Source note. In the inherited TeX, diagram (3.16) attached u and v to the lower object $M_{n,p}$ and collapsed the two diagonal pullback maps. The source has $u : q_1^* E \to M_{n,p}$ and $v : q_2^* E \to M_{n,p}$, with separate unlabeled diagonal maps from each pullback to E; the native diagram above restores that topology.

One denotes by I_p the morphism from M_n to M_n corresponding to the morphism of functors $(E, \alpha) \mapsto (E, \alpha/p)$:

$$\begin{array}{ccc} E & \longrightarrow & E \\ \downarrow & & \downarrow \\ M_n & \xrightarrow{I_p} & M_n \end{array} \qquad I_p^*(E, \alpha) = (E, \alpha/p). \tag{3.17}$$

I_p^* is an automorphism of $\widetilde{H}^i(M_n^{\mathrm{an}}, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}))$. It is tiresome, but routine, to prove the following.

Proposition (3.18)**.**

(i) The endomorphism T_p of ${}_nW$ is expressed, with the aid of (3.16), as the composite map

$$\begin{aligned} &\widetilde{H}^1(M_n^{\mathrm{an}}, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q})) \xrightarrow{q_2^*} \widetilde{H}^1(M_{n,p}^{\mathrm{an}}, \mathrm{Sym}^k(R^1 v_*\mathbb{Q})) \\ &\xrightarrow{\varphi^*} \widetilde{H}^1(M_{n,p}^{\mathrm{an}}, \mathrm{Sym}^k(R^1 u_*\mathbb{Q})) \xrightarrow{q_{1*}} \widetilde{H}^1(M_n^{\mathrm{an}}, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q})). \end{aligned}$$

where q_{1*} is the trace morphism for the covering q_1.

(ii) Similarly, $R_p = p^k I_p^*$.

□

The distrustful reader could forget the adelic preliminaries and define T_p by (i). When $n = 1$ or 2, one puts ${}_nW = W^{K_n}$, so that

$${}_1W = {}_nW^{\mathrm{GL}_2(\mathbb{Z}/(n))}.$$

If S_{k+2} denotes the space of cusp forms, for the group $\mathrm{SL}_2(\mathbb{Z})$, of weight $k+2$, the Shimura isomorphism (3.11) induces an isomorphism

$${}^k_1W_\infty = {}^k_1W \otimes \mathbb{C} = S_{k+2} \oplus \overline{S}_{k+2}.$$

It is tiresome, but routine, to prove the following.

Proposition (3.19)**.** *The endomorphism T_p of ${}^k_1W_\infty$ is identified, via the Shimura isomorphism, with the direct sum of the Hecke operator on S_{k+2}, including the factor p^{k-1}, and its conjugate.*

□

(3.20) One has canonically

$$\bigwedge^2 R^1 f_{n*}\mathbb{Z}_\ell \simeq R^2 f_{n*}\mathbb{Z}_\ell \simeq \mathbb{Z}_\ell(-1),$$

so that $\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)$ is equipped with a bilinear form, symmetric for k even and alternating for k odd, with values in $\mathbb{Z}_\ell(-k)$. The form induced by tensoring with $\mathbb{Q}_\ell$ is nondegenerate.

If F is an lcc $\mathbb{Q}_\ell$-sheaf over a smooth scheme X of pure dimension n over an algebraically closed field k, then Poincaré duality gives

$$\begin{aligned} H^i(X, F)^\vee &\simeq H_c^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))), \\ H_c^i(X, F)^\vee &\simeq H^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))), \\ \widetilde{H}^i(X, F)^\vee &\simeq \widetilde{H}^{2n-i}(X, \mathrm{Hom}(F, \mathbb{Q}_\ell(n))). \end{aligned}$$

Taking $X = \overline{M}_n$ and $F = \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q}_\ell)$ in these considerations, one defines a nondegenerate bilinear form ${}_n(\ ,\ )$ on ${}^k_nW_\ell$ with values in $\mathbb{Q}_\ell(-k-1)$. This form is symmetric for k odd and alternating for k even. This is the ℓ-adic analogue of the Petersson inner product. If $n \mid m$, and if the covering $\psi : M_m \to M_n$ is of degree d, then one has

$${}_m(\psi^* x, \psi^* y) = d \cdot {}_n(x, y).$$

Source note. The printed text spells the name "Peterson" here and again in the proof of (5.6), and "Pertersson" at the end. The standard name "Petersson" is restored throughout. The same page's evident spelling slip "isomprhism" is regularized to "isomorphism."

4 The Congruence Formula

In this section we fix integers $k \geq 0$ and $n \geq 3$, and prime numbers p and ℓ. We suppose that p is prime to ℓ and n. We denote by $f_n : E \to M_n$ the universal elliptic curve over M_n, equipped with $\alpha : E_n \xrightarrow{\sim} \mathbb{Z}/(n)^2$.

Whatever the scheme Y, one denotes by a the unique morphism of Y to $\mathrm{Spec}(\mathbb{Z})$ or, if necessary, to a subscheme of $\mathrm{Spec}(\mathbb{Z})$. If Y is separated and of finite type over $\mathrm{Spec}(\mathbb{Z})$, and if F is a $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaf on Y, one denotes by $R^i a_*(Y, F)$, respectively $R^i a_!(Y, F)$, respectively $R^i \widetilde{a}(Y, F)$, the $\mathbb{Z}_\ell$- or $\mathbb{Q}_\ell$-sheaf over $\mathrm{Spec}(\mathbb{Z})$ that is the ith higher direct image of F under a, respectively the ith direct image with proper supports, respectively $\mathrm{Im}(R^i a_!(Y, F) \to R^i a_*(Y, F))$. For $m \in \mathbb{N}$, one puts

$$Y[1/m] = Y \times \mathrm{Spec}(\mathbb{Z}[1/m]).$$

Theorem (4.1, Igusa [1])**.** *The scheme M_n can be compactified to a scheme of curves M_n^* that is projective and smooth over* $\mathrm{Spec}(\mathbb{Z}[1/n])$, *such that $M_n^* \setminus M_n$ is an étale covering of* $\mathrm{Spec}(\mathbb{Z}[1/n])$.

The schemes M_n are formally smooth, thus smooth over $\mathrm{Spec}(\mathbb{Z})$. The modular invariant j of the universal curve over M_n is a morphism of M_n to the affine line $\mathbb{A}^1$ over $\mathrm{Spec}(\mathbb{Z}[1/n])$. The morphism j is finite, and is an étale covering away from the sections 0 and 1728 of $\mathbb{A}^1$; in fact:

(a) Two elliptic curves over an algebraically closed field with the same j-invariant are isomorphic (e.g. [8] 6.3), thus the geometric fibers of j are finite. The schemes M_n and $\mathbb{A}^1$ being smooth of the same dimension over $\mathrm{Spec}(\mathbb{Z})$, j is quasi-finite and flat.

(b) If E is an elliptic curve over the field of fractions K of a discrete valuation ring R, of j-invariant in R, and whose points of order n are rational over K, then E has good reduction. The valuative criterion for properness shows thus that j is proper.

(c) If E and F are two elliptic curves over a scheme S, of the same j-invariant, and if j and $j - 1728$ are invertible, then the scheme $\mathrm{Isom}(S; E, F)$ of isomorphisms between E and F is étale over S ([8] 6.3). In the diagram

$$\begin{array}{ccc} \mathrm{Isom}(M_n \times_{\mathbb{A}^1} M_n; \mathrm{pr}_1^* E, \mathrm{pr}_2^* E) & \sim & M_n \times \mathrm{GL}_2(\mathbb{Z}/(n)) \\ \downarrow u & & \downarrow v \\ M_n \times_{\mathbb{A}^1} M_n & \xrightarrow{\mathrm{pr}_1} & M_n, \end{array}$$

where $j \neq 0, 1728$, u and v are étale surjective, thus pr_1 is étale and, by flat descent, j is étale.

The section at infinity of the projective line $\mathbb{P}^1 \supset \mathbb{A}^1$ over $\mathrm{Spec}(\mathbb{Z}[1/n])$ is a regular divisor whose generic point is of characteristic 0 in a regular scheme. It follows from a theorem of Abhyankar (see [5]) that along this divisor $j = \infty$, M_n is tamely ramified over $\mathbb{P}^1$, and that the normalization M_n^* of $\mathbb{P}^1$ in M_n satisfies (4.1). $\square$

Source note. The English retype's "The schemes M_n is" is regularized to "are." Both printed witnesses spell Abhyankar's name "Abyankhar," and the English retype renders the standard French term *modérément ramifié* as the unresolved "moderately (tamely?) ramified." The standard name and English term "tamely ramified" are delivered above.

It follows from the same theorem that the lcc $\mathbb{Z}_\ell$-sheaves $R^i f_{n*}\mathbb{Z}_\ell$ on $M_n[1/\ell]$ are tamely ramified at infinity. Hence, from (4.1) and the specialization theorems for ℓ-adic cohomology (see [5]), it follows that $R^i a_*(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$, $R^i a_!(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$, and therefore $R^i \widetilde{a}(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$ are lcc $\mathbb{Z}_\ell$-sheaves over $\mathrm{Spec}(\mathbb{Z}[1/n, 1/\ell])$, whose formation is compatible with all base changes.

Corollary (4.2)**.** *The Galois module ${}_nW_\ell$ is the fiber of the lcc $\mathbb{Q}_\ell$-sheaf*

$$R^1 \widetilde{a}(M_n, \mathrm{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell)) \otimes \mathbb{Q}_\ell$$

over $\mathrm{Spec}(\mathbb{Z}[1/n, 1/\ell])$ *at the geometric point $\overline{\mathbb{Q}}$. It is unramified away from n and ℓ.*

$\square$

Let the two commutative diagrams over $M_n \otimes \mathbb{F}_p$

$$\begin{tikzcd} & \mathbb{Z}/(n)^2 & \\ E_n \arrow[ur, "\alpha"] \arrow[rr] & & E_n^{(p)} \arrow[ul, "\alpha^{(p)}"'] \\ \cap & & \cap \\ E \arrow[rr, "F"] & & E^{(p)} \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} & \mathbb{Z}/(n)^2 & \\ E_n^{(p)} \arrow[ur, "p\alpha^{(p)}"] \arrow[rr] & & E_n, \arrow[ul, "\alpha"'] \\ \cap & & \cap \\ E^{(p)} \arrow[rr, "V"] & & E \end{tikzcd}$$

be written more briefly

$$F : (E, \alpha) \to (E^{(p)}, \alpha^{(p)}) \qquad \text{and} \qquad V : (E^{(p)}, p\alpha^{(p)}) \to (E, \alpha),$$

where F is the Frobenius morphism and V, its transpose, is the Verschiebung. These diagrams define morphisms Φ_1 and Φ_2 of $M_n \otimes \mathbb{F}_p$ to $M_{n,p}$. These morphisms are finite, considered as sections of q_1 and q_2, and define a morphism

$$\Phi = \Phi_1 \coprod \Phi_2 : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \to M_{n,p} \otimes \mathbb{F}_p.$$

Source note. The inherited arrays incorrectly repeated the labels F and V on the induced arrows between the n-torsion subgroup schemes. In the source those upper horizontal arrows are unlabeled; only the maps between the elliptic curves carry F and V. The native diagrams above restore that distinction. The printed English witness also has four bare inclusion signs $\cap$, the central conjunction "and," F and V above the lower arrows, and a terminal comma after the right-hand E_n; those details are retained here. Let Φ^h be the restriction of Φ to the opens M_n^h and $M_{n,p}^h$ of $M_n \otimes \mathbb{F}_p$ and $M_{n,p} \otimes \mathbb{F}_p$ corresponding to curves of nonzero Hasse invariant h.

Proposition (4.3)**.** *Φ^h is an isomorphism.*

Let $\varphi : E_1 \to E_2$ be a p-isogeny between elliptic curves of invertible Hasse invariant over a scheme S of characteristic p. At each geometric point of S, either the kernel $\ker(\varphi)$ of φ is étale over S, or its Cartier dual, isomorphic to $\ker({}^t\varphi)$, is étale over S. The property "$\ker(\varphi)$ is étale" is an open property, so that locally on S, either $\ker(\varphi)$ is purely infinitesimal, or $\ker({}^t\varphi)$ is. The only infinitesimal subgroup of order p of E_1 or E_2 being the kernel of Frobenius, in the first case φ is isomorphic to $F : E_1 \to E_1^{(p)}$, and in the second case ${}^t\varphi$ is isomorphic to $F : E_2 \to E_2^{(p)}$ and φ to $V : E_2^{(p)} \to E_2$. $\square$

Proposition (4.4)**.**

(i) The scheme $M_{n,p}$ is smooth over $\operatorname{Spec}(\mathbb{Z})$ *away from points of characteristic p where $h = 0$.*

(ii) The morphisms q_1 and q_2 induce finite flat morphisms q_1' and q_2' from the normalization $M_{n,p}'$ of $M_{n,p}$ to M_n.

(iii) The morphism Φ can be factored through a surjective morphism

$$\Phi' : M_n \otimes \mathbb{F}_p \coprod M_n \otimes \mathbb{F}_p \to M_{n,p}' \otimes \mathbb{F}_p.$$

The automorphism σ of (3.15) exchanges φ and ${}^t\varphi$, so that it suffices to prove (i) at the points of characteristic p of $M_{n,p}$ where the kernel of φ is infinitesimal: there is no obstruction to infinitesimally lifting an elliptic curve and the infinitesimal part of the kernel of multiplication by p.

Where $p = h = 0$, the fiber of the finite morphism (3.15) $q_i : M_{n,p} \to M_n$ is reduced to a point, so that the open of smoothness of $M_{n,p}$ is dense in $M_{n,p}$ and $M_{n,p}'$ is everywhere of dimension 2. The scheme M_n being regular, following EGA 0_{IV} 16.5.1 and 17.3.5(ii), the morphism $q_i : M_{n,p}' \to M_n$ is flat. Finally, (iii) results from the fact that Φ is finite and $M_n \otimes \mathbb{F}_p$ is a normal curve. $\square$

The Hecke endomorphism T_p of ${}_nW_\ell$, such as is made explicit in (3.18), is the tensorization with $\mathbb{Q}_\ell$ of the fiber at the geometric point $\overline{\mathbb{Q}}$ of $\operatorname{Spec}(\mathbb{Z}[1/n, 1/\ell])$ of the endomorphism, again denoted by T_p, of $R^1\tilde{a}(M_n, \operatorname{Sym}^k(R^1 f_{n*}\mathbb{Z}_\ell))$ defined by the correspondence

$$\begin{tikzcd}
 & q_1'^*E \arrow[rr, "\varphi"] \arrow[dl] \arrow[dr, "u"] & & q_2'^*E \arrow[dl, "v"'] \arrow[dr] & \\
E \arrow[dr, "f_n"'] & & M_{n,p}' \arrow[dl, "q_1'"] \arrow[dr, "q_2'"'] & & E, \arrow[dl, "f_n"] \\
 & M_n & & M_n &
\end{tikzcd} \tag{4.5}$$

with

$$T_p = q_{1*}' \, \varphi^* \, q_2'^* \qquad \text{(cf. 3.18)}.$$

Source note. As in the inherited version of (3.16), the inherited array for (4.5) attached u and v to the lower moduli object and collapsed two distinct pullback-to-curve maps. The native diagram restores the source's full nine-arrow topology and retains the printed English witness's terminal comma after the right-hand E. The endomorphisms R_p and I_p can be interpreted in the same way.

Lemma (4.6)**.** *Let X, Y, Z_1, and Z_2 be four schemes that are separated and of finite type over a noetherian scheme S, let F be a $\mathbb{Z}_\ell$-sheaf on X, let G be a $\mathbb{Z}_\ell$-sheaf on Y, and denote by a each of the structural maps from X, Y, Z_1 or Z_2 to S. Suppose we have the following commutative diagram of S-schemes and morphisms of sheaves:*

$$y_1^*G \xrightarrow{z_1} x_1^*F \qquad\qquad y_2^*G \xrightarrow{z_2} x_2^*F$$

$$\begin{tikzcd}
& Z_1 \arrow[rr, "f"] \arrow[dl, "x_1"'] \arrow[drrr, "y_1"] & & Z_2 \arrow[dlll, "x_2"'] \arrow[dr, "y_2"] & \\
X & & & & Y
\end{tikzcd}$$

Suppose that $f^*z_2 = z_1$, that y_1 and y_2 are proper, that x_1 and x_2 are finite and flat, and that, for every geometric point s of Z_2, the multiplicity of s in its fiber $x_2^{-1}(x_2(s))$ is equal to the sum of the multiplicities in their fibers, for x_1, of the geometric points of Z_1 in the inverse image of s under f. Then the diagram

$$\begin{tikzcd}
R^i\widetilde{a}(Y,G) \arrow[r, "y_1^*"] \arrow[d, equal] & R^i\widetilde{a}(Z_1,G) \arrow[r, "z_1"] & R^i\widetilde{a}(Z_1,F) \arrow[r, "x_{1*}"] & R^i\widetilde{a}(X,F) \arrow[d, equal] \\
R^i\widetilde{a}(Y,G) \arrow[r, "y_2^*"'] & R^i\widetilde{a}(Z_2,G) \arrow[r, "z_2"'] \arrow[u, "f^*"] & R^i\widetilde{a}(Z_2,F) \arrow[r, "x_{2*}"'] \arrow[u, "f^*"] & R^i\widetilde{a}(X,F)
\end{tikzcd}$$

commutes.

Source note. The inherited arrays lost the shared $Z_1 \to Z_2$ object pair in the structural diagram and misaligned the vertical maps in the two cohomological diagrams. The diagrams are rebuilt natively with the source's shared objects and its distinct diagonal, vertical, equality, pullback, and trace arrows. An earlier repair duplicated $Z_1 \to Z_2$ into two disconnected copies; direct replay of the French original at printed p. 164 exposed that remaining topology error, which is now repaired in both editions.

This lemma results from the analogous lemmas for $R^ia_!$ and R^ia_*. The commutativity of the first squares is trivial. The last is rewritten

$$\begin{tikzcd}
R^i\widetilde{a}(Z_1, x_1^*F) & R^i\widetilde{a}(X, x_{1*}x_1^*F) \arrow[l, "\sim"'] \arrow[r, "\mathrm{Tr}"] & R^i\widetilde{a}(X,F) \arrow[d, equal] \\
R^i\widetilde{a}(Z_2, x_2^*F) \arrow[u] & R^i\widetilde{a}(X, x_{2*}x_2^*F) \arrow[l, "\sim"'] \arrow[r, "\mathrm{Tr}"'] \arrow[u] & R^i\widetilde{a}(X,F)
\end{tikzcd}$$

and one recalls the definition of the trace in verifying that the square

$$\begin{array}{ccc}
x_{1*}x_1^*F & \xrightarrow{\mathrm{Tr}} & F \\
\uparrow & & \| \\
x_{2*}x_2^*F & \xrightarrow{\mathrm{Tr}} & F
\end{array}$$

is commutative. $\square$

(4.7) One denotes by $T_p/\mathbb{F}_p$ the endomorphism induced by T_p on the restriction to $\mathrm{Spec}(\mathbb{F}_p)$ of the lcc $\mathbb{Z}_\ell$-sheaf $R^1\widetilde{a}(M_n, \mathrm{Sym}^k R^1f_{n*}\mathbb{Z}_\ell)$. One has

$$R^1\widetilde{a}(M_n, \mathrm{Sym}^k R^1f_{n*}\mathbb{Z}_\ell)|\,\mathrm{Spec}(\mathbb{F}_p) \xrightarrow{\sim} R^1\widetilde{a}(M_n\otimes\mathbb{F}_p, \mathrm{Sym}^k R^1f_{n*}\mathbb{Z}_\ell);$$

the formation of the trace morphism for a finite flat morphism is compatible with change of base, so that $T_p/\mathbb{F}_p$ can be constructed, over the model (3.18), starting from the fiber over $\mathbb{F}_p$ of the "correspondence" (4.5). Lemma (4.6), applied to the commutative diagram

$$\begin{tikzcd}
& M_n\otimes\mathbb{F}_p \coprod M_n\otimes\mathbb{F}_p \arrow[rr, "\Phi'"] \arrow[dl] \arrow[drrr] & & M'_{n,p}\otimes\mathbb{F}_p \arrow[dlll, "q'_1"] \arrow[dr, "q'_2"] & \\
M_n\otimes\mathbb{F}_p & & & & M_n\otimes\mathbb{F}_p,
\end{tikzcd}$$

furnishes then a decomposition of $T_p/\mathbb{F}_p$ into the sum of endomorphisms defined by the two following correspondences:

$$\begin{tikzcd}
 & & (E,\alpha) \arrow[rr, "F"] \arrow[dl, equal] \arrow[dr] & & (E^{(p)},\alpha^{(p)}) = F^*(E,\alpha) \arrow[dl] \arrow[dr] & \\
\text{(a)} & (E,\alpha) \arrow[dr] & & M_n\otimes\mathbb{F}_p \arrow[dl, equal] \arrow[dr, "F"] & & (E,\alpha) \arrow[dl] \\
 & & M_n\otimes\mathbb{F}_p & & M_n\otimes\mathbb{F}_p &
\end{tikzcd}$$

where F is the absolute Frobenius. One recognizes in this correspondence the geometric Frobenius

$$\begin{tikzcd}
 & & x^*(E,\alpha) = (E^{(p)},p\alpha^{(p)}) \arrow[rr, "V"] \arrow[dl] \arrow[dr, "f_n^{(p)}"] & & (E,\alpha) \arrow[dl, "f_n"'] \arrow[dr, equal] & \\
\text{(b)} & (E,\alpha) \arrow[dr] & & M_n\otimes\mathbb{F}_p \arrow[dl, "x"'] \arrow[dr, equal] & & (E,\alpha). \arrow[dl] \\
 & & M_n\otimes\mathbb{F}_p & & M_n\otimes\mathbb{F}_p &
\end{tikzcd}$$

Source note. The later English retype and the inherited arrays duplicated the shared upper row of the base-change diagram into two disconnected triangles; the native diagram restores the two unique upper objects and their four arrows from the canonical French original. The inherited arrays for correspondences (a) and (b) retained only four of the eight descending source relations and rendered two equality relations as arrows. The native diagrams restore every diagonal, both equality relations in each correspondence, and the labels $f_n^{(p)}$, f_n, x, and F at their printed attachment sites. The canonical French original places the terminal period for (b) in the cell to the right of the lower-right object; the later English retype places it after the middle-right object, and each independent edition preserves its printed punctuation. Both printed witnesses likewise place the final period of the composite x ladder in a separate terminal cell to the right of the lower-right object; both editions preserve it there. The map x is the composite map $I_p^{-1}\circ F$:

$$\begin{tikzcd}
(E,\alpha) \arrow[d] & (E,p\alpha) \arrow[l] \arrow[d] & (E^{(p)},p\alpha^{(p)}) \arrow[l] \arrow[d] & \\
M_n\otimes\mathbb{F}_p & M_n\otimes\mathbb{F}_p \arrow[l, "I_p^{-1}"'] & M_n\otimes\mathbb{F}_p \arrow[l, "F"'] & .
\end{tikzcd}$$

The corresponding endomorphism is thus the composite of

$$V : R^1\widetilde{a}(M_n\otimes\mathbb{F}_p,\operatorname{Sym}^k(R^1f_{n*}\mathbb{Z}_\ell)) \xrightarrow{V^*} R^1\widetilde{a}(M_n\otimes\mathbb{F}_p,\operatorname{Sym}^k(R^1f_{n*}^{(p)}\mathbb{Z}_\ell)) \xrightarrow{\operatorname{Tr}_F} R^1\widetilde{a}(M_n\otimes\mathbb{F}_p,\operatorname{Sym}^k(R^1f_{n*}\mathbb{Z}_\ell))$$

and of

$$I_p^* = \operatorname{Tr}_{I_p^{-1}} : \quad \text{endomorphism of } R^1\widetilde{a}(M_n\otimes\mathbb{F}_p,\operatorname{Sym}^k R^1f_{n*}\mathbb{Z}_\ell).$$

Proposition (4.8)**.** *One has $T_p/\mathbb{F}_p = F + I_p^*V$, and:*

(i) F can be identified with the inverse of the "arithmetic" Frobenius element φ_p of the Galois group $\operatorname{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$ *acting on $\widetilde{H}^1(M_n\otimes\overline{\mathbb{F}}_p,\operatorname{Sym}^k(R^1f_{n*}\mathbb{Z}_\ell))$;*

(ii) F and V are transposes of one another, relative to the scalar product (3.20);

(iii) $FV = VF = p^{k+1}$.

For the relation (i) between the geometric and arithmetic Frobenius, one is referred to the exposé of C. Houzel (SGA 5.XV). The composite VF is the composite of the homomorphisms obtained from the following maps:

$$\begin{tikzcd}
E \arrow[d] & E^{(p)} \arrow[l] \arrow[r, "F_E"] \arrow[dr] & E \arrow[r, "V_E"] \arrow[d] & E^{(p)} \arrow[r] \arrow[dl] & E \arrow[d] \\
M_n \arrow[rr, "F"] & & M_n \arrow[rr, "F"] & & M_n
\end{tikzcd}$$

Source note. The inherited array collapsed the printed five-object $E \leftarrow E^{(p)} \to E \to E^{(p)} \to E$ diagram to three objects and thereby omitted both intermediate Frobenius twists and their diagonal structure maps. The complete native diagram above restores all five objects and all eleven source arrows. Thus

$$VF = \mathrm{Tr}_F \circ F_E^* \circ V_E^* \circ F^*.$$

The map $F_E^* V_E^* = (F_E V_E)^* = (p \cdot 1_E)^*$ acts by multiplication by p^k on $\mathrm{Sym}^k R^1 f_{n*}\mathbb{Z}_\ell$, so that

$$VF = p^k \,\mathrm{Tr}_F \circ F^* = p^k \cdot p = p^{k+1},$$

because $F : M_n \to M_n$ is of degree p.

By transport of structure, φ_p respects the scalar product (3.20) with values in $\mathbb{Q}_\ell(-k-1)$, a group on which φ_p acts by multiplication by p^{-k-1}. One has thus

$$(Fx, y) = p^{k+1}(\varphi_p Fx, \varphi_p y) = (x, p^{k+1}F^{-1}y) = (x, Vy).$$

□

The following theorem, synonymous with (4.8), goes back to Eichler.

Theorem (4.9, The Congruence Formula)**.** *Let $K_{n,\ell}$ be the largest subextension of $\overline{\mathbb{Q}}$ that is unramified away from n and ℓ, let φ_p be a Frobenius element relative to p in $\mathrm{Gal}(K_{n,\ell}/\mathbb{Q})$, let F be the endomorphism φ_p^{-1} of ${}_nW_\ell$, and let V be the transpose of F relative to the scalar product (3.20). Then*

$$T_p = F + I_p^* V, \qquad FV = p^{k+1}, \qquad 1 - T_pX + pR_pX^2 = (1 - FX)(1 - I_p^*VX).$$

□

5 Weil implies Ramanujan

If p is a prime number and X a scheme over $\mathbb{F}_p$, then one denotes by $\overline{\mathbb{F}}_p$ an algebraic closure of $\mathbb{F}_p$, by $F : X \to X$ the "geometric" Frobenius endomorphism, and one puts $\overline{X} = X \otimes \overline{\mathbb{F}}_p$; ℓ always denotes a prime number different from p.

By the "Weil conjectures" one means to claim the following: let X be a projective smooth scheme over $\mathbb{F}_p$ and let ℓ be a prime number different from p. Then the eigenvalues of the endomorphism F^* of $H^i(\overline{X}, \mathbb{Q}_\ell)$ are algebraic integers all of whose complex conjugates have absolute value $p^{i/2}$.

With the hypotheses and notations of (4.9), one has, recalling that $(p, n) = 1$:

Theorem (5.1)**.** *If the Weil conjectures are true, then the eigenvalues of the endomorphism F of ${}_n^kW_\ell$ are algebraic integers, all of whose complex conjugates have absolute value $p^{(k+1)/2}$.*

Source note. The printed statement says "the absolute values of the endomorphism" and prints $p^{k+1/2}$. The theorem concerns the eigenvalues of F, and the weight is $k+1$; the mathematically required reading $p^{(k+1)/2}$ is delivered above.

Assume the Weil conjectures.

Lemma (5.2, modulo Weil)**.** *Let X be a smooth scheme over $\mathbb{F}_p$, which can be represented as an open in a smooth projective scheme X^*. Then the eigenvalues of the endomorphism F^* of $\widetilde{H}^i(X, \mathbb{Q}_\ell)$ are algebraic integers of absolute value $p^{i/2}$.*

Source note. The printed phrase "the absolute values of [the] endomorphism" is corrected to "the eigenvalues of the endomorphism"; the displayed absolute value $p^{i/2}$ is unchanged.

The natural map of $H_c^i(X, \mathbb{Q}_\ell)$ to $H^i(X, \mathbb{Q}_\ell)$ can be factored through $H^i(X^*, \mathbb{Q}_\ell)$:

$$H_c^i(X, \mathbb{Q}_\ell) \to H^i(X^*, \mathbb{Q}_\ell) \to H^i(X, \mathbb{Q}_\ell),$$

so that, as a $\mathrm{Gal}(\overline{\mathbb{F}}_p/\mathbb{F}_p)$-module, $\widetilde{H}^i(X, \mathbb{Q}_\ell)$ is the quotient of a subobject of $H^i(X^*, \mathbb{Q}_\ell)$. □

Lemma (5.3, modulo Weil)**.** *Let S be a smooth scheme over $\mathbb{F}_p$ and $f : A \to S$ an abelian scheme over S. Suppose that A can be represented as an open in a projective smooth scheme A^* over $\mathbb{F}_p$. Then the geometric Frobenius F^* of $\widetilde{H}^i(S, R^j f_* \mathbb{Q}_\ell)$ has eigenvalues that are algebraic integers of absolute value $p^{(i+j)/2}$.*

Source note. The printed exponent is $i + j/2$. Since the cohomological weight is $i + j$, the required exponent is $(i + j)/2$, as delivered above and used in the proof via (5.2).

Let m be an integer > 1, and consider the Leray spectral sequences

$$E : E_2^{ij} = H^i(\overline{S}, R^j f_*\mathbb{Q}_\ell) \Longrightarrow H^{i+j}(\overline{A}, \mathbb{Q}_\ell), \qquad {}_cE : {}_cE_2^{ij} = H_c^i(\overline{S}, R^j f_*\mathbb{Q}_\ell) \Longrightarrow H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell).$$

The endomorphism of multiplication by m, $\psi_m = m \cdot 1_A$, defines endomorphisms of E and ${}_cE$ that fit into a commutative diagram

$$\begin{array}{ccccc} {}_cE & \longrightarrow & E & \\ \psi_m^* \downarrow & & \psi_m^* \downarrow & \\ {}_cE & \longrightarrow & E & . \end{array}$$

ψ_m^* acts on $R^j f_*\mathbb{Q}_\ell$ by multiplication by m^j, so that ψ_m^* acts by multiplication by m^j on the terms ${}_cE_r^{ij}$ and E_r^{ij} of ${}_cE$ and E. The maps d_r for $r \geq 2$ commute with ψ_m^*, and send E_r^{ij}, respectively ${}_cE_r^{ij}$, to $E_r^{i'j'}$, respectively ${}_cE_r^{i'j'}$, with $j \neq j'$. These maps are zero, and E_2^{ij}, respectively ${}_cE_2^{ij}$, can be identified with a subspace of $H^{i+j}(\overline{A}, \mathbb{Q}_\ell)$, respectively $H_c^{i+j}(\overline{A}, \mathbb{Q}_\ell)$, where $\psi_m^* = m^j$. Consequently, $\widetilde{H}^i(S, R^j f_*\mathbb{Q}_\ell)$ is identified, as a Galois module, with a subspace of $\widetilde{H}^{i+j}(A, \mathbb{Q}_\ell)$ where $\psi_m^* = m^j$, and one applies (5.2). The trick used here is due to Lieberman. $\square$

Source note. The printed proof says that the differentials are nonzero. Since they commute with ψ_m^* but go between the distinct eigenspaces m^j and $m^{j'}$ with $j \neq j'$, they are zero; this is precisely the degeneration used in the following identification. Both printed witnesses place the final period of the commutative diagram in a separate cell to the right of the lower-right object; this edition preserves that placement.

Let $f_n : E \to M_n \otimes \mathbb{F}_p$ be the universal elliptic curve over $M_n \otimes \mathbb{F}_p$, and let $f_{n,k} : E_k \to M_n \otimes \mathbb{F}_p$ be its k-fold fiber product with itself. The Künneth formula shows that the $\mathbb{Q}_\ell$-sheaf $R^k f_{n,k*}\mathbb{Q}_\ell$ admits as a direct factor the k-fold tensor power of $R^1 f_{n*}\mathbb{Q}_\ell$; the latter in turn contains as a direct factor the $\mathbb{Q}_\ell$-sheaf $\mathrm{Sym}^k(R^1 f_{n*}\mathbb{Q}_\ell)$. Theorem 5.1 results thus from (5.3) and the following lemma.

Lemma (5.4)**.** *The scheme E_k is an open in a smooth projective scheme E_k^* over $\mathbb{F}_p$.*

Let E^* be the Néron minimal model of E over $M_n^* \otimes \mathbb{F}_p$ (4.1). The scheme E^* is smooth and projective over $\mathbb{F}_p$. Since $n \geq 3$ and the points of order n of E form a trivial covering of $M_n \otimes \mathbb{F}_p$, the Néron model is "semistable," case a or b_m in Néron's classification. In particular, the projection $f_n : E^* \to M_n^*$ has only a finite number of nonsmooth points, and at these points f_n is nondegenerate, presenting an ordinary quadratic singularity.

Let E_k^{**} be the k-fold fiber product of E^* over M_n^*. To prove (5.4), it suffices to resolve the singularities of E_k^{**} without touching the open E_k. We first prove:

Lemma (5.5)**.** *Let V be the subvariety of affine space over a field k, with coordinates $(X_i)_{0\leq i\leq r}$, $(Y_i)_{0\leq i\leq r}$, and $(T_i)_{1\leq i\leq s}$, with equation*

$$X_0Y_0 = X_1Y_1 = \cdots = X_rY_r.$$

Let $\mathfrak{m}$ be the ideal of $\mathcal{O}_V$ generated by the monomials obtained from the monomial $\prod_{i=0}^r X_i^i$ by a permutation of the coordinates which respects the set of pairs $\{X_i, Y_i\}$, $0 \leq i \leq r$. Then $\mathfrak{m} = \mathcal{O}_V$ away from the singular locus of V, and the variety $\widetilde{V}$ obtained from V by blowing up $\mathfrak{m}$ is smooth over k.

The singular locus is the place where the four coordinates X_i, Y_i, X_j, Y_j for $i \neq j$ vanish. The open affine of $\widetilde{V}$ defined by the element $\prod_{i=0}^r X_i^i$ of the ideal $\mathfrak{m}$ of blowing up is the spectrum of the regular ring

$$k[Y_0/X_1, X_0/X_1, X_1/X_2, \ldots, X_{r-1}/X_r, X_r, T_1, \ldots, T_s]$$

(for the proof, note that $X_i/X_{i+1} = Y_{i+1}/Y_i$), and 5.5 results. $\square$

One still shows that, locally for the étale topology, the singularities of E_k^{**} are isomorphic to those of V for $r = k - 1$, and that this allows the definition on E_k^{**} of an ideal $\mathfrak{m}$ analogous to the ideal $\mathfrak{m}$ of (5.5). Blowing up this ideal, one obtains E_k^*. $\square$

An approximation to the following theorem has been proven by Ihara [2].

Theorem (5.6)**.** *The Weil conjectures imply the Ramanujan conjecture.*

We note first that (5.1) is true for $n = 1$, because ${}_1^kW_\ell$ is the Galois submodule of ${}_m^kW_\ell$ invariant under $\mathrm{GL}_2(\mathbb{Z}/(m))$. On ${}_1^kW_\ell$, I_p^* induces the identity, and (4.8) reduces to

$$1 - T_pX + p^{k+1}X^2 = (1 - FX)(1 - VX).$$

The endomorphisms F and V are transposes of each other, so that

$$\det(1 - FX; {}^k_1W_\ell) = \det(1 - VX; {}^k_1W_\ell).$$

The action of T_p on ${}^k_1W_\ell$ is induced by its action on ${}^k_1W$, and is compatible with the decomposition of ${}^k_1W \otimes \mathbb{C}$ into the sum of the space S_{k+2} of cusp forms relative to $\mathrm{SL}_2(\mathbb{Z})$ of weight $k+2$, and of the complex conjugate space. From the Hermitian property for T_p, for the Petersson inner product, and from (3.19), one deduces then that

$$\det(1 - T_pX + p^{k+1}X^2; {}^k_1W_\ell) = \det(1 - T_pX + p^{k+1}X^2; S_{k+2})^2,$$

and

$$\det(1 - T_pX + p^{k+1}X^2; S_{k+2})^2 = \det(1 - FX; {}^k_1W_\ell)^2,$$

so that

$$\det(1 - T_pX + p^{k+1}X^2; S_{k+2}) = \det(1 - FX; {}^k_1W_\ell). \tag{5.7}$$

Resume the notations of section 1 and make $k = 10$. By virtue of Hecke's theory and of (3.19), (5.7) can be rewritten

$$H_p(X) = \det(1 - FX; {}^{10}_1W_\ell),$$

and one applies (5.1). $\square$

One verifies in the same way that the Weil conjectures imply Petersson's generalization of the Ramanujan conjecture.

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GRIFFITHS'S WORK

by Pierre DELIGNE

This talk contains part of Griffiths's fundamental results on families of Hodge structures. It contains none of his results on algebraic cycles.

1. Hodge structures.

Let $H_{\mathbb{R}}$ be a real vector space of finite dimension.

Definition 1.1.- *A real Hodge structure on* $H_{\mathbb{R}}$ *is a bigrading of*

$$H_{\mathbb{C}} = H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C}$$

$$H_{\mathbb{C}} = \bigoplus_{p,q} H^{pq},$$

such that H^{pq} *and* H^{qp} *are complex conjugates.*

The Hodge numbers of a real Hodge structure H are the integers

$$h^{pq} = \dim_{\mathbb{C}}(H^{pq}).$$

One says that H has weight n if $h^{pq} = 0$ for $p + q \neq n$. A variant of 1.1 is the following.

Definition 1.2.- *A real Hodge structure of weight* n *on* $H_{\mathbb{R}}$ *is a finite decreasing filtration* F *of* $H_{\mathbb{C}}$ *(the Hodge filtration) such that, for* $p + q = n + 1$*, one has*

(1.2.1) $$F^p(H_{\mathbb{C}}) \oplus \overline{F}^q(H_{\mathbb{C}}) \xrightarrow{\sim} H_{\mathbb{C}}.$$

One passes from 1.1 to 1.2 by putting

(1.2.2) $$F^p(H_{\mathbb{C}}) = \sum_{p' \geq p} H^{p'q'}$$

and one passes from 1.2 to 1.1 by putting, for $p + q = n$,

(1.2.3) $$H^{pq} = F^p(H_{\mathbb{C}}) \cap \overline{F}^q(H_{\mathbb{C}}).$$

1.3. Let $\mathbb{S}$ denote the real algebraic group obtained from the multiplicative group $\mathbb{G}_m$ by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$. We have $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$; there is a natural morphism w from $\mathbb{G}_m$ to $\mathbb{S}$ which, on real points, is the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$, and there is a natural morphism t from $\mathbb{S}$ to $\mathbb{G}_m$ which, on real points, is the norm. The composite tw is the squaring map. A new variant of 1.1 is the following.

Definition 1.4.- *A real Hodge structure on* $H_{\mathbb{R}}$ *is an action (defined over* $\mathbb{R}$*) of the algebraic group* $\mathbb{S}$ *on* $H_{\mathbb{R}}$*.*

If z denotes the defining isomorphism $z : \mathbb{S}(\mathbb{R}) \xrightarrow{\sim} \mathbb{C}^*$, one passes from 1.1 to 1.4 by defining H^{pq} as the subspace of $H_{\mathbb{C}}$ on which $\mathbb{S}(\mathbb{R})$ acts by multiplication by $z^p \overline{z}^q$.

If $H_{\mathbb{R}}$ is endowed with a real Hodge structure, the Weil automorphism C is defined as multiplication by i^{p-q} on H^{pq}. This automorphism is real and is simply the action of the element i of $\mathbb{S}(\mathbb{R}) \subset \mathbb{C}^*$.

Definition 1.5.- *A polarization of a real Hodge structure* H *of weight* n *is a bilinear form* Ψ *on* $H_{\mathbb{R}}$*, invariant under the compact subgroup* $\mathrm{Ker}(t)$ *of* $\mathbb{S}$*, and such that the form* $\Psi(x, Cy)$ *is symmetric and positive definite.*

From the identity $C^2 = (-1)^n$, one obtains

$$\Psi(x, y) = (-1)^n \Psi(x, C^2 y) = (-1)^n \Psi(Cy, Cx) = (-1)^n \Psi(y, x)$$

so that Ψ is alternating or symmetric according to the parity of n. The variance property under $\mathbb{S}$ and the positivity property are written as follows:

(1.5.1) the orthogonal complement of $F^p(H_{\mathbb{C}})$ with respect to Ψ is $F^{n-p}(H_{\mathbb{C}})$;

(1.5.2) for nonzero x in H^{pq}, one has $i^{p-q}\Psi(x, \overline{x}) > 0$

(Riemann bilinear relations).

Definition 1.6.- *(i) A Hodge structure H of weight n consists of a finitely generated free $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice") and a real Hodge structure of weight n on $H_{\mathbb{R}} = H_{\mathbb{Z}} \otimes_{\mathbb{Z}} \mathbb{R}$.*

(ii) A polarization of a Hodge structure H of weight n is a polarization Ψ of the underlying real Hodge structure that takes integral values on the integral lattice $H_{\mathbb{Z}}$.

2. Hodge theory.

2.1. Let $X \subset \mathbb{P}^r(\mathbb{C})$ be a nonsingular projective complex algebraic variety, pure of dimension d, and let $\eta \in H^2(X, \mathbb{Z})$ be the cohomology class of a hyperplane section of X, i.e., the first Chern class of $\mathcal{O}(1)$.

The holomorphic de Rham complex Ω^*_X is a resolution of the constant sheaf $\mathbb{C}$ (the holomorphic Poincaré lemma), whence a spectral sequence

(2.1.1) $$E_1^{pq} = H^q(X, \Omega^p_X) \Rightarrow H^{p+q}(X, \mathbb{C}).$$

2.2. Hodge theory asserts that

(A) the spectral sequence (2.1.1) degenerates ($E_1 = E_\infty$).

(B) the filtration of $H^n(X, \mathbb{C})$ to which it abuts is a Hodge structure of weight n (1.2) on $H^n(X, \mathbb{R})$.

By (A), one has

$$H^{pq} \simeq H^q(\Omega^p_X).$$

(C) For $n \leq d$, let $P^n_{\mathbb{Q}}$ denote the kernel of the iterated cup-product

$$\eta^{d-n+1} : H^n(X, \mathbb{Q}) \longrightarrow H^{2d-n+2}(X, \mathbb{Q})$$

(the primitive part of cohomology), and let $P^n_{\mathbb{Z}}$ be the intersection of $P^n_{\mathbb{Q}}$ with the image of $H^n(X, \mathbb{Z})$ in $H^n(X, \mathbb{Q})$. Since the class η is of type $(1,1)$, $P^n_{\mathbb{R}} = P^n_{\mathbb{Z}} \otimes \mathbb{R}$ is a Hodge substructure of $H^n(X, \mathbb{R})$. Up to a sign depending only on d and n, the form

$$\Psi(x, y) = \int \eta^{d-n} \wedge x \wedge y$$

is a polarization of the Hodge structure P^n.

(D) The natural map

$$\bigoplus_{n-2k \leq d} \eta^k \wedge : \bigoplus P^{n-2k}_{\mathbb{Q}} \longrightarrow H^n(X, \mathbb{Q})$$

is an isomorphism (the Hodge-Lepage decomposition).

3. Families of Hodge structures.

3.1. Let $f : X \to S$ be a projective and smooth morphism of analytic spaces, of pure relative dimension d. For simplicity, we suppose that S is nonsingular and that f is provided with a factorization through $\mathbb{P}^r_S = S \times \mathbb{P}^r(\mathbb{C})$. The fibers $X_s = f^{-1}(s)$ therefore form an analytic family, parametrized by S, of nonsingular subvarieties of $\mathbb{P}^r(\mathbb{C})$.

3.2. From the C^∞ point of view, f is a locally trivial fibration. The cohomology algebra of the fibers of f therefore forms a local system on S, denoted

$$R^*_{\mathbb{Z}}(f) = \sum R^n f_* \mathbb{Z}.$$

For every abelian sheaf F on S (for example $\mathbb{Q}$, $\mathbb{C}$, $\mathcal{O}$), put

$$R^n_F(f) = F \otimes R^n_{\mathbb{Z}}(f). \tag{3.2.1}$$

Since f is proper, one also has

$$R^n_F(f) \xrightarrow{\sim} R^n f_*(f^* F).$$

The cohomology classes of the hyperplane sections of the X_s define a <u>locally constant</u> section

$$\eta \in H^0(S, R^2_{\mathbb{Z}}(f)). \tag{3.2.2}$$

For $n \leq d$, the $P^n_{\mathbb{Z}}(X_s)$ therefore form a local subsystem $P^n_{\mathbb{Z}}(f)$ of $R^n_{\mathbb{Q}}(f)$. Putting $P^n_F(f) = F \otimes P^n_{\mathbb{Z}}(f)$, one has by definition

$$P^n_{\mathbb{Q}}(f) = \operatorname{Ker}\bigl(\eta^{d-n+1} \wedge : R^n_{\mathbb{Q}}(f) \to R^{2d-n+2}_{\mathbb{Q}}(f)\bigr), \tag{3.2.3}$$

$$P^n_{\mathbb{Z}}(f) = P^n_{\mathbb{Q}}(f) \cap \operatorname{Im}\bigl(R^n_{\mathbb{Z}}(f) \to R^n_{\mathbb{Q}}(f)\bigr). \tag{3.2.4}$$

The polarization form of 2.2 (C)

$$\Psi(x, y) = \pm \int_{X_s} \eta^{d-n} \wedge x \wedge y \tag{3.2.5}$$

is locally constant on S, i.e., it defines

$$\Psi : P^n_{\mathbb{Z}}(f) \otimes P^n_{\mathbb{Z}}(f) \longrightarrow \mathbb{Z}. \tag{3.2.6}$$

3.3. For F a coherent analytic sheaf on S, $f^{\cdot} F$ will denote its inverse image as a sheaf, and

$$f^* F = \mathcal{O}_X \otimes_{f^{\cdot}\mathcal{O}_S} f^{\cdot} F$$

its inverse image as a sheaf of modules. For $s \in S$, $F_{(s)}$ denotes the $\mathcal{O}_{S,s}$-module of germs of sections of F at s; the fiber of F at s is the vector space

$$F_s = F_{(s)} \otimes_{\mathcal{O}_{S,s}} \mathbb{C}.$$

Recall finally that the relative de Rham complex $\Omega^*_{X/S}$ is the quotient of Ω^*_X with components $\Omega^1_{X/S} = \Omega^1_X / f^*\Omega^1_S$ and $\Omega^p_{X/S} = \bigwedge^p \Omega^1_{X/S}$. If $T^1_{X/S}$, the dual of $\Omega^1_{X/S}$, is the relative tangent bundle, then one has "contraction" pairings

$$\lrcorner : T^1_{X/S} \otimes \Omega^p_{X/S} \longrightarrow \Omega^{p-1}_{X/S}. \tag{3.3.1}$$

3.4. By the relative holomorphic Poincaré lemma, $\Omega^*_{X/S}$ is a resolution of $f^{\cdot}\mathcal{O}_S$. Hence one has a spectral sequence

$$E_1^{pq} = R^q f_* \Omega^p_{X/S} \Rightarrow R^n f_*(f^{\cdot}\mathcal{O}_S) = R^n_{\mathcal{O}}(f). \tag{3.4.1}$$

It follows from 2.2 (A) that the E_1^{pq} are locally free, that the spectral sequence (3.4.1) degenerates ($E_1 = E_\infty$) and that its formation commutes with arbitrary base change. The filtration of $R^n_{\mathcal{O}}(f)$ to which it abuts (<u>the Hodge filtration</u>) therefore induces on $R^n_{\mathcal{O}}(f)_s \simeq H^n(X_s, \mathbb{C})$ the Hodge filtration of 2.2: the latter varies holomorphically with $s \in S$.

Definition 3.5.- *<u>The Gauss-Manin connection on the holomorphic vector bundle</u> $R^n_{\mathcal{O}}(f)$ (<u>identified here with its sheaf of holomorphic sections</u>) <u>is the holomorphic and integrable connection</u>*

$$\nabla : R^n_{\mathcal{O}}(f) \longrightarrow \Omega^1_S \otimes R^n_{\mathcal{O}}(f)$$

<u>whose horizontal local sections</u> ($\nabla v = 0$) <u>are the local sections of</u> $R^n_{\mathbb{C}}(f)$.

Theorem of transversality 3.6.- *The Hodge filtration* F *on* $R^n_{\mathcal{O}}(f)$ *satisfies*

$$\nabla F^p(R^n_{\mathcal{O}}(f)) \subset \Omega^1_S \otimes F^{p-1}(R^n_{\mathcal{O}}(f)).$$

Proof (following Katz-Oda [10] and a personal communication from Katz). The question is local on S. Let $\mathcal{U} = (U_i)_{0\leq i\leq N}$ be a finite covering of X by Stein opens. For $Q \subset [0,N]$, let $U_Q = \bigcap_{i\in Q} U_i$, let j_Q be the inclusion of U_Q in X, and put

$$f_*(U_Q, \Omega^p_{X/S}) = (fj_Q)_*(\Omega^p_{X/S}|U_Q).$$

We denote by $f_*(\mathcal{U}, \Omega^*_{X/S})$ the double complex of sheaves on S with components

$$f_*(\mathcal{U}, \Omega^*_{X/S})^{pq} = \bigoplus_{\#Q=q+1} f_*(U_Q, \Omega^p_{X/S}) \tag{3.6.1}$$

whose first differential is induced by the exterior differential and whose second differential is the "Čech" differential. The spectral sequence 3.4.1 is the spectral sequence of this double complex obtained from the filtration by the first degree.

Let v be a (holomorphic) vector field on S, and suppose that on each U_i a vector field v_i lifting v is given. We denote by $\theta(v_i)$ the endomorphism of $f_*(\mathcal{U}, \Omega^*_{X/S})$ satisfying, for $Q = (i_1, \ldots, i_q)$ with $i_1 < \cdots < i_q$,

$$\theta(v_i) f_*(U_Q, \Omega^p_{X/S}) \subset f_*(U_Q, \Omega^p_{X/S}) \oplus \bigoplus_{i_0<i_1} f_*(U_{\{i_0\}\cup Q}, \Omega^{p-1}_{X/S}),$$

and having as coordinates the Lie derivative

$$\mathcal{L}_{v_{i_1}} : f_*(U_Q, \Omega^p_{X/S}) \longrightarrow f_*(U_Q, \Omega^p_{X/S})$$

and the contractions

$$(-1)^p(v_{i_1} - v_{i_0}) \lrcorner : f_*(U_Q, \Omega^p_{X/S}) \longrightarrow f_*(U_{\{i_0\}\cup Q}, \Omega^{p-1}_{X/S}).$$

A simple calculation shows that $\theta(v_i)$ commutes with d.

Lemma 3.7.- *On the cohomology sheaves* $R^n_{\mathcal{O}}(f)$ *of* $f_*(\mathcal{U}, \Omega^*_{X/S})$*, the endomorphism* $\theta(v_i)$ *induces* ∇_v.

Let $\mathcal{C}^\infty$ be the sheaf of C^∞ functions on S. If $f_*(\mathcal{U}, \Omega^*_{\infty X/S})$ is the C^∞ analogue of (3.6.1), constructed in terms of the complex of relative C^∞ differential forms with complex values on X, then

$$R^n_{\mathcal{C}^\infty}(f) = \mathcal{H}^n(f_*(\mathcal{U}, \Omega^*_{\infty X/S})).$$

Since the sheaves $\Omega^p_{\infty X/S}$ are soft, this formula remains valid for every covering $\mathcal{U}'$.

If $(v'_i)_{0\leq i\leq N}$ are C^∞ liftings of v on the U_i, the construction 3.6 still defines endomorphisms $\theta(v'_i)$ of $R^n_{\mathcal{C}^\infty}(f)$. If (v'_i) and (v''_i) are two systems of liftings, a simple calculation shows that, at the level of complexes,

$$\theta(v'_i) - \theta(v''_i) = dH - Hd, \tag{3.7.1}$$

where the only nonzero coordinates of H are the contractions

$$v'_{i_1} - v''_{i_1} \lrcorner : f_*(U_Q, \Omega^p_{\infty X/S}) \longrightarrow f_*(U_Q, \Omega^{p-1}_{\infty X/S}),$$

$(Q = \{i_1, \ldots, i_q\},\ i_1 < \cdots < i_q)$.

Thus the endomorphism $\theta(v'_i)$ of $R^n_{\mathcal{C}^\infty}(f)$ is independent of the choice of the v'_i. Via the injection of $R^n_{\mathcal{O}}(f)$ into $R^n_{\mathcal{C}^\infty}(f)$, it is compatible with the endomorphism $\theta(v_i)$ of 3.7. It therefore remains only to verify that $\theta(v'_i) = \nabla_v$ on $R^n_{\mathcal{C}^\infty}(f)$. One verifies that $\theta(v'_i)$, which is independent of the v'_i, is also independent of $\mathcal{U}$. Taking $\mathcal{U} = \{X\}$ and a lift v' of v, one then has

$$f_*(\mathcal{U}, \Omega^*_{\infty X/S}) = f_*(\Omega^*_{\infty X/S}),$$

and $\theta(v')$ is the Lie derivative along v', visibly equal to ∇_v.

3.8. We finish the proof of 3.6. By construction, one has

$$\theta(v_i)\left(\bigoplus_{p'\geq p} f_*(\mathcal{U},\Omega^*_{X/S})^{p',q'}\right) \subset \bigoplus_{p'\geq p-1} f_*(\mathcal{U},\Omega^*_{X/S})^{p',q'}. \tag{3.8.1}$$

Since the spectral sequence (3.4.1) is obtained from the double complex $f_*(\mathcal{U},\Omega^*_{X/S})$, it follows from 3.7 and (3.8.1) that

$$\theta(v_i)F^p(R^n_{\mathcal{O}}(f)) = \nabla_v F^p(R^n_{\mathcal{O}}(f)) \subset F^{p-1}(R^n_{\mathcal{O}}(f)). \tag{3.8.2}$$

Since this is true locally on S and for every v, it implies 3.6.

3.9. By the Leibniz formula

$$\nabla(fh) = df\cdot h + f\cdot\nabla h$$

the map induced by ∇

$$\operatorname{def}(\nabla) : \operatorname{Gr}^p_F(R^n_{\mathcal{O}}(f)) \longrightarrow \Omega^1_S \otimes \operatorname{Gr}^{p-1}_F(R^n_{\mathcal{O}}(f))$$

is $\mathcal{O}$-linear. By 3.4.1 it is identified with

$$\operatorname{def}(\nabla) : R^q f_*\Omega^p_{X/S} \longrightarrow \Omega^1_S \otimes R^{q+1}f_*\Omega^{p-1}_{X/S}. \tag{3.9.1}$$

It follows from formula 3.7 for ∇_v that

Proposition 3.10.-
The homomorphism (3.9.1) is the cup product, via the pairing (3.3.1), with the Kodaira–Spencer class

$$c \in \Omega^1_S \otimes R^1 f_*(T^1_{X/S})$$

which expresses how X_s deforms with s.

Definition 3.11.- *(i) A family H of real Hodge structures of weight n on S consists of*

a) a local system of real vector spaces $H_{\mathbb{R}}$ on S;

b) a finite holomorphic filtration F of the locally free analytic sheaf

$$H_{\mathcal{O}} = H_{\mathbb{R}} \otimes_{\mathbb{R}} \mathcal{O},$$

these data satisfying the following conditions:

(H.1) The natural connection ∇ on $H_{\mathcal{O}}$ is such that

$$\nabla F^p(H_{\mathcal{O}}) \subset \Omega^1_S \otimes F^{p-1}(H_{\mathcal{O}});$$

(H.2) At every point $s\in S$, F defines on $(H_{\mathbb{R}})_s$ a Hodge structure of weight n.

(ii) A polarization of H is a locally constant bilinear form Ψ on $H_{\mathbb{R}}$ which at every point $s\in S$ induces a polarization of $(H_{\mathbb{R}})_s$.

3.12. A family of Hodge structures H of weight n on S is a local system $H_{\mathbb{Z}}$ of finitely generated free $\mathbb{Z}$-modules on S, together with a family of real Hodge structures on $H_{\mathbb{R}} = \mathbb{R}\otimes H_{\mathbb{Z}}$. A polarization of H is a polarization Ψ of $H_{\mathbb{R}}$ taking integral values on $H_{\mathbb{Z}}$.

With these definitions, one can reformulate 2.2 (C) and 3.6 by saying that, under the hypotheses of 3.1, $P^n_{\mathbb{Z}}(f)$ is a polarized family of Hodge structures on S.

4. The regularity theorem.

4.1. When one starts with a projective and smooth morphism $f : X \to S$ of schemes of finite type over $\mathbb{C}$, such that $f^{\mathrm{an}} : X^{\mathrm{an}} \to S^{\mathrm{an}}$ satisfies the hypotheses of 3.1, some of the objects constructed in no. 3 are obtained from purely algebraic objects by applying the "passage to the analytic category" functor. They are:

a) the sheaves of modules $R^n_{\mathcal{O}}(f)$, their Hodge filtration, the spectral sequence 3.4.1, and the Gauss–Manin connection;

b) the algebra structure (cup-product) on $R^*_{\mathcal{O}}(f)$, and the "primitive part" $P^n_{\mathcal{O}}(f)$;

c) the product of the polarization form Ψ by a suitable power of $2\pi i$.

By contrast, the "integral lattice" $R^*_{\mathbb{Z}}(f)$, and the real structure thereby obtained on the complex cohomology of the fibers of f, are transcendental in nature.

4.2. Indeed, let S be a smooth scheme of finite type over $\mathbb{C}$, and let X be a closed subscheme of $\mathbb{P}^r(\mathbb{C}) \times S$, such that the projection $f : X \to S$ is a smooth morphism of pure relative dimension d.

By the relative variant of GAGA, for every coherent algebraic sheaf F on X one has

$$(R^n f_* F)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*(F^{\mathrm{an}}).$$

More generally, if K^* is a complex of coherent algebraic sheaves whose

differentials d_i are (algebraic) differential operators which are $f^{\cdot}\mathcal{O}_S$-linear, then the direct images in hypercohomology satisfy

$$\big(R^n f_*(K^*)\big)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*(K^{*\mathrm{an}})$$

(this case reduces to the preceding one via one of the hypercohomology spectral sequences).

Take K to be the relative algebraic de Rham complex $\Omega^*_{X/S}$. One obtains (in the notation of 3.2)

$$\big(R^n f_* \Omega^*_{X/S}\big)^{\mathrm{an}} \xrightarrow{\sim} R^n f^{\mathrm{an}}_*\big((\Omega^*_{X/S})^{\mathrm{an}}\big) \xleftarrow{\sim} R^n f^{\mathrm{an}}_*\big(f^{\cdot\mathrm{an}}\mathcal{O}^{\mathrm{an}}_S\big) = R^n_{\mathcal{O}}(f).$$

Moreover, the spectral sequence (3.4.1) comes from a purely algebraic spectral sequence

$$E_1^{pq} = R^q f_* \Omega^p_{X/S} \Longrightarrow R^{p+q} f_* \Omega^*_{X/S}. \tag{4.2.1}$$

According to 3.4, the E_1^{pq} are locally free (since the $E_1^{pq\,\mathrm{an}}$ are), the spectral sequence (4.2.1) degenerates, and its formation is compatible with every change of base. The Hodge filtration to which it abuts is therefore, ipso facto, algebraic.

The cup-product is defined in terms of the exterior product in $\Omega^*_{X/S}$. The de Rham cohomology class of a hyperplane section can be defined algebraically (up to multiplication by a power of $2\pi i$), from the logarithmic differential

$$\mathrm{d}f/f : \mathcal{O}^*_X \longrightarrow \Omega^1_{X/S},$$

the primitive part $P^n_{\mathcal{O}}(f)$ and Ψ are also algebraic in nature.

Finally, the construction 3.6–3.7 of the Gauss–Manin connection carries over easily to the algebraic case (Katz–Oda [10]).

4.3. Let D be a Riemann surface isomorphic to the unit disk, let 0 be a point of D, put $D^* = D - \{0\}$, let V be a holomorphic vector bundle on D^*, let ∇ be a holomorphic connection on V, and let V_o be the local system of horizontal sections of V. One has

$$V \xrightarrow{\sim} V_o \otimes_{\mathbb{C}} \mathcal{O}.$$

The (commutative) fundamental group $\pi_1(D^*) \simeq \mathbb{Z}$ acts on V_o. We denote by T the monodromy transformation, i.e. the action on V_o or on V of the positive generator of $\pi_1(D^*)$.

There exists U such that

$$T = \exp(2\pi i U).$$

Let $z : D \to \mathbb{C}$ be a coordinate, with $z(0) = 0$, and let $\widetilde{D}^*$ be the (universal) covering of D^* on which $\log z$ is defined. For v a section of V_o on $\widetilde{D}^*$,

$$\exp(-\log z \cdot U)(v)$$

is the inverse image of a holomorphic section of V on D^*.

This construction defines an isomorphism

$$m_{U,z} : \mathcal{O} \otimes_{\mathbb{C}} H^0(\widetilde{D}^*, V_o) \xrightarrow{\sim} V,$$

and hence a natural extension V_U of V to D, depending only on U. A section v of V on D^* will be called ∇-moderate if, as a section of V_U, it is meromorphic at 0. This notion does not depend on the choice of U.

4.4. Let S be the complement, in a smooth projective curve $\overline{S}$, of a finite set T, and let V be an algebraic vector bundle on S. An (algebraic) connection ∇ on V is called regular if, for every $t \in T$, the meromorphic sections of V in a neighborhood of t are ∇-moderate.

Theorem 4.5.- *For $f : X \to S$ a projective and smooth morphism of schemes, the Gauss–Manin connection on $R^n f_* \Omega^*_{X/S}$ is regular.*

This theorem is proved in [2]. A completely different proof, proceeding by arithmetic methods, is due to Katz [9].

4.6. Let D^* be as in 4.3, let H be a family of real Hodge structures on D^*, let T be the monodromy transformation of $H_{\mathbb{C}}$, let U be an endomorphism of $H_{\mathbb{C}}$ such that $T = \exp(2\pi i U)$, and let $H_{\mathcal{O},U}$ be the corresponding locally free extension of $H_{\mathcal{O}}$ to D.

One says that H is regular at 0 if the Hodge filtration of $H_{\mathcal{O}}$ extends to $H_{\mathcal{O},U}$. This condition does not depend on U. Theorem 4.5 has the following corollary, whose conclusion is purely analytic.

Corollary 4.7.- *Under the hypotheses of 4.5, the families of Hodge structures $R^n(f)$ on S^{an} are regular in a neighborhood of every point $t \in T$.*

One may hope that, in fact, every polarized family of Hodge structures on D^* is automatically regular.

5. Moduli Spaces.

5.1. Let G be a real algebraic group, let $C \in G(\mathbb{R})$, and let V be a real representation of G. A C-polarization of V is a bilinear form Ψ on V, invariant under G, and such that $\Psi(x, Cy)$ is symmetric and positive definite.

Lemma 5.2.- *If G is connected (by which one means that $G(\mathbb{C})$ is connected), the following conditions are equivalent:*

(i) G admits a representation $\rho : G \to \mathrm{GL}(V)$, with finite kernel $\mathrm{Ker}(\rho)$, which is C-polarizable.

(ii) Every representation of G is C-polarizable.

(iii) G is reductive and $\mathrm{ad}\, C$ is a Cartan involution (of G).

The proof is left to the reader.

Remark 5.3.- There exist in $G(\mathbb{R})$ elements C satisfying the conditions of 5.2 if and only if G is reductive and admits a compact maximal torus. In this case:

a) Such an element C is contained in a single maximal compact subgroup, namely its centralizer.

b) For G adjoint, the correspondence $C \longmapsto$ (centralizer of C) is a bijection between the set of these elements and the set of maximal compact subgroups of $G(\mathbb{R})$.

5.4. Let G be a real algebraic group, endowed with homomorphisms

(5.4.1) $$\mathbb{G}_m \xrightarrow{w} G \xrightarrow{t} \mathbb{G}_m$$

such that w is central and $tw(x) = x^2$. A representation $\rho : G \to \mathrm{GL}(V)$ will be called homogeneous of weight n if $\rho w(x) = x^n$.

By a morphism from $\mathbb{S}$ to G we shall always mean a homomorphism $h : \mathbb{S} \to G$ making the diagram

$$\begin{array}{ccccc} \mathbb{G}_m & \xrightarrow{w} & \mathbb{S} & \xrightarrow{t} & \mathbb{G}_m \\ \| & & \downarrow h & & \| \\ \mathbb{G}_m & \xrightarrow{w} & G & \xrightarrow{t} & \mathbb{G}_m \end{array}$$

commutative.

If $h : \mathbb{S} \to G$ is a morphism and $\rho : G \to \mathrm{GL}(V)$ a real representation of G, we shall denote by h_V the real Hodge structure (1.4) $\rho \circ h$ on V. For ρ homogeneous of weight n, a polarization of V is a polarization Ψ of (V, h_V) which satisfies

$$\Psi(gx, gy) = t(g)^n \Psi(x, y). \tag{5.4.2}$$

We shall say that h is positive if every representation of G is polarizable.

5.5. The group G, endowed with (5.4.1), is uniquely determined by $G^o = \mathrm{Ker}(t)$, endowed with the central element, of order dividing 2, $\varepsilon = w(-1)$. Indeed G is identified with the quotient of $\mathbb{G}_m \times G^o$ by the subgroup generated by $(-1, \varepsilon)$.

From this point of view, a morphism $h : \mathbb{S} \to G$ is identified with $h^o : \mathbb{S}^o \to G^o$ such that $h^o(-1) = \varepsilon$. A homogeneous representation of weight n of G is identified with a representation ρ of G^o such that $\rho(\varepsilon) = (-1)^n$, and a polarization of this representation is identified with an $h^o(i)$-polarization (5.1). According to 5.2, if G^o is connected, h is a positive morphism if and only if G is reductive and $\mathrm{ad}\, h^o(i)$ is a Cartan involution of G^o.

5.6. We shall henceforth suppose that G^o is reductive and connected.

Let h be a morphism from $\mathbb{S}$ to G. For $\rho : G \to \mathrm{GL}(V)$ a representation of G, denote by F_h the Hodge filtration of $V_{\mathbb{C}}$ deduced from h_V. For the adjoint representation $\mathrm{Lie}(G)$,

$$F_h^0(\mathrm{Lie}(G)_{\mathbb{C}}) = \bigoplus_{p \geq 0} \mathrm{Lie}(G_{\mathbb{C}})^{p,-p} \tag{5.6.1}$$

is the Lie algebra of a parabolic subgroup $P(h)$ of $G_{\mathbb{C}}$. For every representation $\rho : G \to \mathrm{GL}(V)$ of G, $P(h)$ respects the Hodge filtration F_h of $V_{\mathbb{C}}$; if $\mathrm{Ker}(\rho) \cap G^o$ is central, then $P(h)$ is exactly the subgroup of G which respects this filtration.

Finally, denote by $H(h)$ the centralizer of h in $G(\mathbb{R})$.

5.7. Let X be a conjugacy class of morphisms from $\mathbb{S}$ to G. We shall denote by $\check{X}$ the set of conjugates in $G(\mathbb{C})$ of $P(h)$, for $h \in X$.

Lemma 5.8.- *The $G(\mathbb{R})$-equivariant map $P : h \mapsto P(h)$ identifies X with an open subset of the flag space $\check{X}$.*

Let $h \in X$ and let $\rho : G \to \mathrm{GL}(V)$ be a homogeneous representation of G such that $\mathrm{Ker}(\rho) \cap G^o$ is finite. The set X is identified with the set of Hodge structures on V, conjugate under $G(\mathbb{R})$ to h_V, while $\check{X}$ is identified with the set of filtrations on $V_{\mathbb{C}}$, conjugate under $G(\mathbb{C})$ to F_h. Since a homogeneous Hodge structure is determined by its Hodge filtration, the map P is injective. It is identified with the map

$$G(\mathbb{R})/H(h) \longrightarrow G(\mathbb{C})/P(h)$$

whose differential at the origin is

$$\mathrm{Lie}(G)/\mathrm{Lie}(H) \longrightarrow \mathrm{Lie}(G_{\mathbb{C}})/\mathrm{Lie}(P(h)),$$

i.e.

$$\mathrm{Lie}(G)/(F^0 \cap \overline{F^0})(\mathrm{Lie}(G)) \longrightarrow F^{-\infty}/F^0(\mathrm{Lie}(G_{\mathbb{C}})).$$

This differential is bijective, whence 5.8.

We have seen in passing that

$$H(h) = G(\mathbb{R}) \cap P(h). \tag{5.8.1}$$

5.9. Let V be a real representation of G, let $V(\check{X},\mathbb{R})$ be the constant sheaf on $\check{X}$ with value V, put $V(\check{X},\mathbb{C}) = \mathbb{C}\otimes V(\check{X},\mathbb{R})$ and $V(\check{X},\mathcal{O}) = \mathcal{O}\otimes_{\mathbb{C}} V(\check{X},\mathbb{C})$, and let ∇ be the natural integrable connection of $V(\check{X},\mathcal{O})$. The $G(\mathbb{R})$-equivariant map $h \mapsto h_V$ from X to the Hodge structures on V extends uniquely to a $G(\mathbb{C})$-equivariant, and hence holomorphic, map from $\check{X}$ to the filtrations of $V_{\mathbb{C}}$. This map corresponds to a filtration F of $V(\check{X},\mathcal{O})$, the Hodge filtration. The system $(V(\check{X},\mathcal{O}),\nabla,F)$ is $G(\mathbb{C})$-equivariant. The real structure $V(\check{X},\mathbb{R})$ of $V(\check{X},\mathbb{C})$ is $G(\mathbb{R})$-equivariant. Finally, at $h\in X$, F induces on $V_{\mathbb{C}}$ the Hodge filtration F_h.

5.10. The action of $G(\mathbb{C})$ on $\check{X}$ defines a morphism

(5.10.1)
$$\mathrm{Lie}(G)(\check{X},\mathcal{O}) \longrightarrow T_{\check{X}},$$

where $T_{\check{X}}$ is the tangent bundle. At $h\in X$, the kernel of (5.10.1) is $\mathrm{Lie}(P(h)) = F_h^0(\mathrm{Lie}(G_{\mathbb{C}}))$ (5.6), so that (5.10.1) factors through a $G(\mathbb{C})$-equivariant isomorphism

(5.10.2)
$$F^{-\infty}/F^0\big(\mathrm{Lie}(G)(\check{X},\mathcal{O})\big) \xrightarrow{\sim} T_{\check{X}}.$$

Let $T^1_{\check{X}}$ denote the holomorphic subbundle of $T_{\check{X}}$ which is the image of $F^{-1}(\mathrm{Lie}(G)(\check{X},\mathcal{O}))$. It is easily verified that if v is a local section of $T^1_{\check{X}}$, then

$$\nabla_v\big(F^p(V(\check{X},\mathcal{O}))\big) \subset F^{p-1}(V(\check{X},\mathcal{O})),$$

whatever the representation $\rho : G\to \mathrm{GL}(V)$ may be. The converse is true when $\mathrm{Ker}(\rho)\cap G^o$ is central.

5.11. Suppose now that X is a conjugacy class of positive morphisms from $\mathbb{S}$ to G. For $h\in X$, put $C_h = h(i)$, and denote by $K(h)$ the centralizer of C_h in $G(\mathbb{R})$. The group $K^o(h) = K(h)\cap G^o(\mathbb{R})$ is compact. Let V be a representation of G and let Ψ be a G-invariant bilinear form on V. If Ψ is a polarization of (V,h_V) for one $h\in X$, then Ψ is a polarization of (V,h_V) for every $h\in X$. Indeed, putting $\Phi_h(x,y) = \Psi(x,C_h y)$, one has

(5.11.1)
$$\Phi_{gh} = g(\Phi_h).$$

We shall then say that Ψ is a polarization of V. By hypothesis, every representation of G is polarizable.

5.12. Let Ψ be a polarization of the adjoint representation of G on $\mathrm{Lie}(G)$. By 5.10.2, Ψ induces, at each point $h\in X$, a positive definite Hermitian form on

(5.12.1)
$$(T_{\check{X}})_h \simeq F_h^{-\infty}/F_h^0(\mathrm{Lie}(G)_{\mathbb{C}}) \simeq \bigoplus_{p<0}\mathrm{Lie}(G)^{p,-p}.$$

The polarization Ψ therefore defines a Hermitian structure, invariant under $G(\mathbb{R})$, on X.

If Z is the center of G, one already has

(5.12.2)
$$F^{-\infty}/F^0\big(\mathrm{Lie}(G/Z)(\check{X},\mathcal{O})\big) \xrightarrow{\sim} T_{\check{X}},$$

and a polarization Ψ of $\mathrm{Lie}(G/Z)$ already defines a structure of Hermitian variety on X. One may take for Ψ the negative of the Killing form of G/Z.

Put

(5.12.3)
$$\begin{cases} (T^v_{\check{X}})_h = \bigoplus_{p<0}(\mathrm{Lie}(G/Z))^{2p,-2p}, \\ (T^h_{\check{X}})_h = \bigoplus_{p<0}(\mathrm{Lie}(G/Z))^{2p-1,-2p+1}. \end{cases}$$

Thus one obtains an orthogonal C^∞ decomposition of the tangent bundle into two complex subbundles. One has

(5.12.4)
$$(T^1_{\check{X}})_h \subset (T^h_{\check{X}})_h \qquad (h\in X).$$

5.13. The subgroup $K(h)_{\mathbb{C}} \cap P(h)$ of $K(h)_{\mathbb{C}}$ is parabolic, so that $K(h)_{\mathbb{C}} \subset K(h) \cdot P(h)$ and $K(h) \cdot h = K(h)_{\mathbb{C}} \cdot h$ is a compact complex analytic subvariety of X. These varieties are the fibers of the C^∞, $G(\mathbb{R})$-equivariant fibration

$$X = G/H(h) \longrightarrow G/K(h).$$

It is easily verified that T_X^v is the tangent bundle to this fibration.

5.14. Recall that on a holomorphic Hermitian bundle F there exists a unique Hermitian connection ∇ such that $\nabla'' = \partial''$. The curvature R of the bundle is, by definition, the curvature of this connection. It is a form of type $(1,1)$ with values in the Lie algebra of the unitary group of the bundle.

If F is the constant bundle defined by a vector space F_o, and if the Hermitian form is written

$$\Phi(x, y) = \Phi_o(hx, y), \tag{5.14.1}$$

then, for $x : S \to F_o$ a C^∞ section of F, one has

$$\nabla x = \partial x + h^{-1} \partial' h \cdot x. \tag{5.14.2}$$

It follows that, for u and v two tangent fields,

$$R(u, v) = \partial_u (h^{-1} \partial'_v h) - \partial_v (h^{-1} \partial'_u h) - h^{-1} \partial'_{[uv]} h + [h^{-1} \partial'_u h, h^{-1} \partial'_v h]. \tag{5.14.3}$$

5.15. Let F be the tangent bundle of a Hermitian complex analytic variety. For u a tangent vector to S, denote by $u_{1,0}$ and $u_{0,1} = \overline{u_{1,0}}$ its components of type $(1,0)$ and $(0,1)$ in the complexification of the tangent bundle. The holomorphic curvature of S in the direction u is the real number

$$R(u) = \Phi(R(u_{1,0}, u_{0,1})(u), u) / \Phi(u, u)^2. \tag{5.15.1}$$

If X is endowed with the Hermitian structure 5.12, one has the fundamental result ([8], §9):

Theorem 5.16.- *There exists* $\lambda < 0$ *such that* $R(u) \leq \lambda$ *in every direction belonging to* T_X^h.

Proof (a sketch). By homogeneity it is enough to prove 5.16 at a point $e \in X$. Put $H = H(e)$, $P = P(e)$, let N^+ be the unipotent radical of P, and let N^- be the conjugate complex subgroup whose Lie algebra is

$$\mathfrak{n}^- = \bigoplus_{p<0} \mathrm{Lie}(G)^{p,-p}.$$

In the calculations which follow, near e, we identify X with N^- by the maps

$$X = G/H \hookrightarrow G(\mathbb{C})/P \hookleftarrow N^-.$$

Left translations on N^- permit one to identify the tangent bundle with $\mathfrak{n}^-$. We shall denote by t^+, t^o, and t^- the orthogonal projections of $\mathrm{Lie}(G)_{\mathbb{C}}$ onto $\mathfrak{n}^-$, $\mathrm{Lie}(H_{\mathbb{C}})$, and $\mathfrak{n}^+ = \mathrm{Lie}(N^+)$. For $g = np$ ($g \in G(\mathbb{R})$, $n \in N^-$, $p \in P$), the differential, at e, of the action of g on X is identified with

$$t^- \operatorname{ad} p : \mathfrak{n}^- \longrightarrow \mathfrak{n}^-.$$

The Hermitian structure is therefore written

$$\Phi_n(x, y) = \Phi_e(t^- \operatorname{ad} p^{-1} x, t^- \operatorname{ad} p^{-1} y).$$

Denoting by Ψ the polarization form and by C the Cartan involution associated to e, one has

$$\begin{aligned}
\Phi_n(x, y) &= \Psi(t^- \operatorname{ad} p^{-1} x,\ t^- C\, \overline{\operatorname{ad} p^{-1} y}) \\
&= \Psi(t^- \operatorname{ad} p^{-1} x,\ \operatorname{ad} C(\overline{p})^{-1} \cdot C\overline{y}) \\
&= \Psi(\operatorname{ad} C(\overline{p}) \cdot t^- \cdot \operatorname{ad} p^{-1} \cdot x,\ C\overline{y}) \\
&= \Phi_e(\operatorname{ad} C(\overline{p}) \cdot t^- \cdot \operatorname{ad} p^{-1} \cdot x,\ y).
\end{aligned}$$

One verifies that, for $n = \exp(u)$ with $u \sim 0$, to third order one has

$$p = \exp\bigl(\overline{u} - \tfrac{1}{2}t^o[u,\overline{u}] - t^+[u,\overline{u}]\bigr)$$

and

$$h = \operatorname{ad} C(\overline{p})\, t^- \ \operatorname{ad} p^{-1} = \exp(H),$$

for

$$H = t^- \circ \operatorname{ad}(Cu - \overline{u} + [u,\overline{u}]) - \tfrac{1}{2}[t^- \operatorname{ad} Cu, t^- \operatorname{ad} \overline{u}].$$

Under the hypotheses of 5.14, if $h = \exp(H)$ and if $H = 0$ at $s_o \in S$, one deduces from 5.14.3 that, at s_o,

$$\begin{aligned} R(u,v) &= (\partial_u\partial'_v - \partial_v\partial'_u)H - \tfrac{1}{2}[\partial''_u H, \partial'_v H] + \tfrac{1}{2}[\partial''_v H, \partial'_u H] \\ &\quad + (\partial'_u\partial'_v - \partial'_v\partial'_u - \partial'_{[uv]})H. \end{aligned}$$

Applying this formula, one sees that the curvature is given, at e, by

$$R(u,v)_e = S(u,\overline{v}) - S(v,\overline{u}),$$

with

$$\begin{aligned} S(u,\overline{v}) &= -t^- \circ \operatorname{ad}[u,\overline{v}] + \tfrac{1}{2}[t^- \operatorname{ad} Cu, t^- \operatorname{ad} \overline{v}] + \tfrac{1}{2}[-t^- \operatorname{ad} \overline{v}, t^- \operatorname{ad} Cu] \\ &= [t^- \operatorname{ad} Cu, t^- \operatorname{ad} \overline{v}] - t^- \circ \operatorname{ad}[u,\overline{v}]. \end{aligned}$$

For any $u, v \in \mathfrak{n}^-$, if u_{10} and u_{01} are the components of type (10) and (01) of u, as in 5.15, then, by the invariance of Ψ,

$$\begin{aligned} \Phi(R(u_{10},u_{01})(v),v) &= \Psi(S(u,\overline{u})(v), C\overline{v}) \\ &= \Psi\bigl([Cu, t^-[\overline{u},v]] - t^-[\overline{u}[Cu,v]] - t^-[[u\overline{u}],v], C\overline{v}\bigr) \\ &= -\Psi\bigl(t^-[\overline{u},v],[Cu,C\overline{v}]\bigr) + \Psi\bigl([Cu,v],[\overline{u},C\overline{v}]\bigr) - \Psi\bigl([u,\overline{u}],[v,C\overline{v}]\bigr). \end{aligned}$$

For $u = v \in T^h_X$, one has $Cu = -u$ and

$$\begin{aligned} \Phi(R(u_{10},u_{01})(u),u) &= \Psi\bigl(t^-[u,\overline{u}],[u,\overline{u}]\bigr) + \Psi\bigl([u,\overline{u}],[v,\overline{v}]\bigr) \\ &= -\Phi\bigl(t^-[u,\overline{u}],t^-[u,\overline{u}]\bigr) - \Phi\bigl([u,\overline{u}],[u,\overline{u}]\bigr) < 0, \end{aligned}$$

which proves 5.16.

A generalization of Schwarz's lemma allows one to deduce from 5.16 (see [1], III, no. 10):

Corollary 5.17.- *Let $f : D \to X$ be a holomorphic map from the unit disk into X. Endow D with the hyperbolic metric d for which it has curvature -1, and X with the metric 4.12. Then, if $f(D)$ is tangent to T^h_X, one has*

$$d(f(x), f(y)) \leq |\lambda|\, d(x,y).$$

6. The Monodromy Theorem, after A. Borel.

6.1. Let S be a connected complex analytic variety, endowed with a base point $s_o \in S$, and let $p : \widetilde{S} \to S$ be its universal covering. Let H be a polarized family of Hodge structures of weight n on S. Denote by H_o the polarized Hodge structure which is the fiber of H at s_o.

The polarization Ψ is a bilinear form on $H_{o\mathbb{Q}}$. Denote by G^o the algebraic group, defined over $\mathbb{Q}$, which is the neutral component of the group of automorphisms of $(H_{o\mathbb{Q}}, \Psi)$; by $\varepsilon \in G^o$ the homothety of ratio $(-1)^n$; by G the group deduced, by the procedure 5.5, from (G^o, ε); and by Γ the discrete group of automorphisms of $(H_{o\mathbb{Z}}, \Psi)$. One has

(6.1.1) $$T : \pi_1(S, s_o) \longrightarrow \Gamma.$$

6.2. For each point $s \in \widetilde{S}$, one has a natural isomorphism

$$H_{o\mathbb{Z}} \simeq H_{p(s),\mathbb{Z}},$$

and the Hodge structure $H_{p(s)}$ is identified with a Hodge structure on $H_{o,\mathbb{Z}}$, defined by $h(s) : \mathbb{S} \to G_{\mathbb{R}}$. The morphisms $h(s)$ are positive and mutually conjugate, because they form a connected continuous family and G is reductive. If X is the corresponding moduli space, defined in § 5, one therefore has a map

$$h : \widetilde{S} \longrightarrow X.$$

The family H is the pullback along h of the system of Hodge structures on X defined in (5.9) by the natural weight-n representation of G on H_o.

The morphism h is equivariant, via (5.5.1), for the actions of $\pi_1(S, s_o)$ and Γ on $\widetilde{S}$ and X.

By 3.4 and 3.6, the map h is holomorphic and $\mathrm{Im}(dh) \subset T_X^1$ (cf. 5.10).

6.3. Suppose that S is the punctured unit disk

$$D^* = \{z \mid 0 < |z| < 1\}$$

and take for the universal covering of D^* the Poincaré half-plane Y, endowed with

$$\exp(2\pi i y) : Y \longrightarrow D^*.$$

If $T \in \Gamma$ is the monodromy transformation (4.3) of $H_{\mathbb{Z}}$, the map h satisfies

$$h(y+1) = Th(y). \tag{6.3.1}$$

Choose $h_o \in X$. If $h(y) = g_y \cdot h_o$, the distance in X, for a metric (5.12), satisfies

$$d(h(y+1), h(y)) = d(Tg_y h_o, g_y h_o) = d(g_y^{-1} T g_y h_o, h_o). \tag{6.3.2}$$

By 5.17, if Y is endowed with its natural hyperbolic distance, the map h decreases distances (suitably normalized). In Y, one has

$$d(y, y+1) \sim (\mathrm{Im}(y))^{-1}$$

and therefore

$$d(g_y^{-1} T g_y h_o, h_o) \longrightarrow 0 \qquad \text{for } \mathrm{Im}(y) \to \infty. \tag{6.3.3}$$

The same argument applies to T^2; moreover, $T^2 \in G(\mathbb{R})$, and (6.2.3) implies that a limit of conjugates of T^2 lies in the compact subgroup $H(h_o)$ of $G(\mathbb{R})$. The eigenvalues of T^2 (and of T) are therefore all of absolute value 1. These eigenvalues form a set of algebraic integers stable under conjugation. Now,

Lemma 6.4.- *If all the complex conjugates of an algebraic integer α have absolute value 1, then α is a root of unity.*

The eigenvalues of T are therefore roots of unity, which proves

Monodromy Theorem 6.5 (A. Borel).- *Let H be a polarized family of Hodge structures on the punctured disk D^*. There exist integers N and M such that the monodromy transformation T satisfies*

$$(T^N - I)^M = 0.$$

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SHIMURA'S WORK(*)

by Pierre DELIGNE

Let X/Γ be the quotient of a Hermitian symmetric domain X by a discrete arithmetic group Γ (assumed to be defined by congruence conditions; see 1.7). By Baily and Borel, X/Γ is a complex algebraic variety. In papers [10] through [14], Shimura shows, among other things, that many of these varieties can be defined over number fields which he explicitly identifies. This is the subject of the present report.

To obtain clean statements, one is forced to use adelic language. This leads one to consider, for a given "congruence condition", a scheme over $\mathbb{C}$, a disjoint union of finitely many varieties of the type X/Γ (see ğğ 1 and 2). In many cases, this scheme can be defined over a number field E, independent of the congruence condition under consideration. Its geometric connected components, however, will be defined over class fields of E. This phenomenon, although essential, will be neglected in the rest of this introduction.

In a small number of cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties, endowed with some additional algebraic structures (polarizations, endomorphisms, structures on the points of order n). The solution, over $\mathbb{Q}$, of a corresponding moduli problem then provides a scheme M over an explicit number field F, whose complex points are X/Γ. We shall call M a "model" of X/Γ. The fundamental case is that of polarized abelian varieties of the principal series, endowed with a level-n structure (1.6, 1.11, 4.16, 4.17 and 4.21).

In other cases, X/Γ can be interpreted as the set of isomorphism classes of complex abelian varieties A endowed with some structures as above, and with some integral cohomology classes of type (p,p) $a_i \in H^{2p}(A\times\cdots\times A,\mathbb{C})$ (cf. Mumford [7]). In this case, the idea will be to construct a model M of X/Γ as a subscheme of a model M' already constructed for a variety X'/Γ'. Roughly speaking, one first constructs in M' the points of M corresponding to abelian varieties of C.M. type (= maximal complex multiplication). One then defines M as the closure of the set of these points. This construction rests on the detailed knowledge of abelian varieties of C.M. type provided by [16].

In the cases considered above, one has much information on the fields of definition of the points of M corresponding to abelian varieties of C.M. type. This makes it possible to give partial solutions to Hilbert's twelfth problem (cf. the introduction of [15] and 3.15, 3.16).

Looking at what these cases have in common, one arrives at the notion of a "canonical model" (ğ 3). One shows that there is at most one "canonical model" for X/Γ (5.5). Descent techniques then make it possible to construct some canonical models which seem far from being solutions of moduli problems ([12] [13] [14] and [5]). For a description of these "strange models", see ğğ 5 and 6.

We shall denote by $\mathbb{Z}/n(1)$ both the group scheme μ_n of nth roots of unity and the group of nth roots of unity in a fixed algebraically closed field $\bar{k}$. For $\bar{k}=\mathbb{C}$, the exponential map identifies $\mathbb{Z}/n(1)$ with $\mathbb{Z}/n$. The $\mathbb{Z}/n(1)$ form a projective system: the transition maps are $\varphi_{n,m} : x \longmapsto x^{n/m}$. We put

$$\widehat{\mathbb{Z}} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \simeq \prod_p \mathbb{Z}_p, \qquad \widehat{\mathbb{Z}}(1) = \varprojlim_n \mathbb{Z}/n\mathbb{Z}(1),$$

$$\mathbb{A}^f = \mathbb{Q}\otimes\widehat{\mathbb{Z}} \simeq {\prod_p}'\mathbb{Q}_p \quad \text{(restricted product)}, \qquad \mathbb{A}^f(1) = \mathbb{Q}\otimes\widehat{\mathbb{Z}}(1) = \mathbb{A}^f \otimes_{\widehat{\mathbb{Z}}} \widehat{\mathbb{Z}}(1),$$

and we denote by $\mathbb{A}$ the ring of adeles of $\mathbb{Q}$.

We shall use the following notation:

$\pi_0(X)$ (for X a topological space): the set of connected components of X. In what follows, $\pi_0(X)$, endowed with the quotient topology from X, will be discrete or compact totally disconnected.

G^o: the connected component of the identity of a topological group G (or of the base point of a pointed topological space).

(*) Text written in March 1971

$G(K)$, G_K, $G \otimes_F K$: for G a scheme over F (for instance an algebraic group over F) and K an F-algebra, $G(K)$ denotes the set of K-valued points of G, and G_K or $G \otimes_F K$ denotes the scheme over K obtained from G by extension of scalars.

E^*: for E an algebra of finite rank over a field F, E^* denotes the algebraic group over F of invertible elements of E. For E a number field and $F = \mathbb{Q}$, $E^*(\mathbb{A})/E^*(\mathbb{Q})$ is the idele class group of E.

We shall say that an algebraic group G over $\mathbb{Q}$ satisfies the Hasse principle if $H^1(\mathbb{Q}, G) \underset{\text{dfn}}{=} H^1(\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), G(\overline{\mathbb{Q}}))$ injects into the product, over all places, of the $H^1(\mathbb{Q}_v, G_{\mathbb{Q}_v})$. We shall use the following theorems.

(0.1) Hasse principle: a simply connected semisimple group, without a factor of type E_8, satisfies the Hasse principle [3].

(0.2) A simply connected semisimple group G over a non-Archimedean local field K satisfies $H^1(K, G) = 0$ (a proof independent of the classification is announced in Bruhat–Tits [1]).

(0.3) Strong approximation theorem: let G be a simply connected semisimple group over a global field K. If G is K-simple, and if v is a place of K such that $G(K_v)$ is noncompact, then $G(K_v)G(K)$ is dense in $G(\mathbb{A})$ (for a proof independent of the classification, see Platonov [9]).

(0.4) Real approximation theorem: let G be a connected (linear) algebraic group over $\mathbb{Q}$. Then $G(\mathbb{Q})$ is dense in $G(\mathbb{R})$ (to prove this, one reduces easily to the case of tori, which is a corollary of ([4] 5.1)).

ğ 1. The adelic language.

1.1. Let G be a reductive group over $\mathbb{Q}$. We denote by G' the derived group of the identity component of G, by C the center of this identity component, and we put $T = G/G'$. There are exact sequences of algebraic groups

$$0 \longrightarrow C \longrightarrow G \longrightarrow G/C \longrightarrow 0, \tag{1.1.1}$$

$$0 \longrightarrow G' \longrightarrow G \xrightarrow{\nu} T \longrightarrow 0, \tag{1.1.2}$$

and the canonical maps from C to T and from G' to G/C are isogenies with kernel $C \cap G'$.

1.2. Let $\mathbb{S}$ be the real algebraic group of invertible elements of the $\mathbb{R}$-algebra $\mathbb{C}$. It is also the real algebraic group obtained from the multiplicative group $\mathbb{G}_m$ over $\mathbb{C}$ by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$. One has $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, and $\mathbb{S}_{\mathbb{C}} \simeq \mathbb{G}_{m,\mathbb{C}} \times \mathbb{G}_{m,\mathbb{C}}$. The group $\mathrm{Hom}(\mathbb{S}_{\mathbb{C}}, \mathbb{G}_{m,\mathbb{C}})$ of characters of $\mathbb{S}_{\mathbb{C}}$ has as a basis the characters z and $\bar{z}$ such that the composite

$$\mathbb{C}^* = \mathbb{S}(\mathbb{R}) \longrightarrow \mathbb{S}(\mathbb{C}) \xrightarrow{z, \bar{z}} \mathbb{C}^*$$

is respectively the identity and complex conjugation.

Let V be a real vector space. For any representation $h : \mathbb{S} \to \mathrm{GL}(V)$, we denote by V^{pq} the subspace of $V_{\mathbb{C}}$ on which $\mathbb{S}$ acts by the character $z^p \bar{z}^q$. The V^{pq} form a bigrading of $V_{\mathbb{C}}$, and the construction $h \mapsto V^{pq}$ identifies the representations of $\mathbb{S}$ in $\mathrm{GL}(V)$ with Hodge bigradings of $V_{\mathbb{C}}$, i.e. with bigradings of $V_{\mathbb{C}}$ such that $V^{pq} = \overline{V^{qp}}$. We shall denote by $F(h)$ the Hodge filtration of $V_{\mathbb{C}}$:

$$F(h)^p = \bigoplus_{p' \geq p} V^{p'q'}.$$

1.3. We denote by $r : \mathbb{G}_{m,\mathbb{C}} \to \mathbb{S}_{\mathbb{C}}$ the $\mathbb{C}$-homomorphism such that $(z^p \bar{z}^q) \circ r = (x \longmapsto x^p)$, and by $w : \mathbb{G}_{m,\mathbb{R}} \to \mathbb{S}$ the $\mathbb{R}$-homomorphism such that $(z^p \bar{z}^q) \circ w = (x \longmapsto x^{p+q})$. For a representation $h : \mathbb{S} \to \mathrm{GL}(V)$ and $v^{pq} \in V^{pq}$, one has

$$hr(x) \cdot v^{pq} = x^p \cdot v^{pq} \qquad \text{and}$$
$$hw(x) \cdot v^{pq} = x^{p+q} \cdot v^{pq}.$$

The composite $w : \mathbb{R}^* = \mathbb{G}_m(\mathbb{R}) \xrightarrow{w} \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ is the evident inclusion.

1.4. Let $h : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism. For any representation $\rho : G \to \mathrm{GL}(V)$ of G on a $\mathbb{Q}$-vector space V, ρh defines a bigrading of $V_{\mathbb{C}}$ (1.2). If V is faithful, and if G is the subgroup of $\mathrm{GL}(V)$ fixing some tensors s_i, the construction $h \mapsto (V^{pq})$ identifies the homomorphisms h with Hodge bigradings of $V_{\mathbb{C}}$ such that the s_i are of type $(0, 0)$.

1.5. Let $h_o : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism satisfying the following conditions (cf. [7] and [2]; (1.5.2) is motivated by the Griffiths transversality theorem).

(1.5.1) The image of $h_o w : \mathbb{G}_m \to G_{\mathbb{R}}$ is central. We shall call $h_o w$ the weight of h_o;

(1.5.2) the Hodge bigrading on the complexification $\mathrm{Lie}(G)_{\mathbb{C}}$ of the adjoint representation space (1.2) is of type $(-1,1)+(0,0)+(1,-1)$;

(1.5.3) the automorphism $\mathrm{ad}\, h_o(i)$ of G, an involution by (1.5.1), induces a Cartan involution of G'.

Let K_∞ be the centralizer of h_o in $G(\mathbb{R})$; it contains the center of $G(\mathbb{R})$, and $K_\infty \cap G'(\mathbb{R})^o$ is a maximal compact subgroup of $G'(\mathbb{R})^o$, equal to the centralizer of $h_o(i)$ in $G'(\mathbb{R})^o$. The space $K_\infty \backslash G(\mathbb{R})$ is identified with the set X of conjugates of h_o by elements of $G(\mathbb{R})$. One checks that it has a unique complex structure such that, for every representation $\rho : G \to \mathrm{GL}(V)$, the filtration $F(\rho h)$ of $V_{\mathbb{C}}$ depends holomorphically on $h \in X$. This structure is right invariant under $G(\mathbb{R})$, and the connected components of X are Hermitian symmetric domains. We shall denote by X^o the connected component of h_o.

1.6. Example I. Let $\mathrm{Gp}(V)$ be the group of symplectic similitudes of a real vector space V endowed with a non-degenerate alternating form ψ. There is then a unique conjugacy class X of homomorphisms $h : \mathbb{S} \to \mathrm{Gp}(V)$ satisfying (1.5.1) and having weight -1, i.e. weight $hw : \mathbb{G}_m \to \mathrm{Gp}(V) : x \longmapsto$ the homothety with ratio x^{-1}. The space X is identified with the union of two Siegel half-spaces.

Construction 1.5 identifies the elements $h \in X$ with the decompositions $V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1}$ of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$ such that $V^{pq} = \overline{V^{qp}}$, $V^{0,-1}$ is totally isotropic for ψ, and, if C denotes the endomorphism of V which induces multiplication by i^{p-q} on V^{pq}, the symmetric form $\psi(x, Cy) = \psi(x, h(i)y)$ on V is definite (> 0 or < 0).

1.7. Let Γ be an arithmetic subgroup of $G(\mathbb{Q})$. We are interested in the case where Γ is defined by congruence conditions. If $V_{\mathbb{Z}}$ is an integral lattice in a faithful representation V of G, this means that there exists n such that Γ contains, with finite index, the group $\Gamma_n \subset G(\mathbb{Q})$ of $\gamma \in G(\mathbb{Q})$ such that $\gamma \in G(\mathbb{R})^o$, $\gamma V_{\mathbb{Z}} = V_{\mathbb{Z}}$, and $\gamma \equiv \mathrm{Id} \pmod{n}$. By Baily and Borel, the quotient X^o/Γ carries a natural quasi-projective scheme structure over $\mathbb{C}$. Let K_n be the compact open subgroup of $G(\mathbb{A}^f)$ consisting of those $k = (k_p)$ such that $k_p \cdot V_{\mathbb{Z}} \otimes \mathbb{Z}_p = V_{\mathbb{Z}} \otimes \mathbb{Z}_p$ and $k_p \equiv 1 \pmod{n}$ for $p \mid n$. The group Γ_n is then the intersection, inside $G(\mathbb{A})$, of $G(\mathbb{Q})$ and $G(\mathbb{R})^o \times K_n$.

1.8. Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The space $K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}) = K\backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = X \times (K\backslash G(\mathbb{A}^f))/G(\mathbb{Q})$ is then a finite disjoint union of spaces of the type considered in 1.7. It is therefore a quasi-projective scheme over $\mathbb{C}$. We shall denote it by ${}_K M_{\mathbb{C}}(G, h_o)$ (or simply ${}_K M_{\mathbb{C}}(G)$ or ${}_K M_{\mathbb{C}}$).

As K becomes smaller, these schemes form a projective system of schemes with finite and surjective transition morphisms

$$ {}_K M_{\mathbb{C}}(G, h_o) \qquad (K \to 0). \tag{1.8.1} $$

We shall denote by $M_{\mathbb{C}}(G, h_o)$ (or simply $M_{\mathbb{C}}(G)$, or $M_{\mathbb{C}}$) the quasi-compact and separated scheme (not of finite type for $G \neq \{1\}$) that is the projective limit of (1.8.1)

$$ M_{\mathbb{C}}(G, h_o) = \varprojlim_K K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}). \tag{1.8.2} $$

It should be regarded as an avatar of the projective system (1.8.1). The group $G(\mathbb{A}^f)$ acts on $M_{\mathbb{C}}$, or, if one prefers, on the pro-object defined by the projective system (1.8.1), but of course not on the individual ${}_K M_{\mathbb{C}}$ (cf. 3.2). One has

$$ K\backslash M_{\mathbb{C}} = {}_K M_{\mathbb{C}}, $$

so that

$$ M_{\mathbb{C}} = \varprojlim_K K\backslash M_{\mathbb{C}} \qquad (K \text{ compact open in } G(\mathbb{A}^f)). \tag{1.8.3} $$

This amounts to saying that $G(\mathbb{A}^f)$ acts continuously on $M_{\mathbb{C}}$.

The following construction will be useful for interpreting the ${}_K M_{\mathbb{C}}$ as moduli schemes.

1.9. Let H be an algebraic group over $\mathbb{Q}$ and V a faithful representation of H. We define an H-structure $\bar{\iota}$ on a vector space W to be an isomorphism from V to W, given modulo $H(\mathbb{Q})$: $\bar{\iota} \in \mathrm{Isom}(V, W)/H(\mathbb{Q})$. If $H(\mathbb{Q})$ is the group of automorphisms of a structure s of type Σ on V, then an H-structure on W is identified with a structure t of the same type on W, isomorphic to s (with $\bar{\iota} = \mathrm{Isom}((V, s), (W, t))$). Let W be endowed with an H-structure $\bar{\iota}$. The algebraic group $\iota H \iota^{-1} \subset \mathrm{GL}(W)$ ($\iota \in \bar{\iota}$) depends only on $\bar{\iota}$. We denote it by $H^{\bar{\iota}}$. Let A be a $\mathbb{Q}$-algebra. The set $\mathrm{Isom}_{\bar{\iota}}(W \otimes A, V \otimes A)$ of permissible isomorphisms from $W \otimes A$ to $V \otimes A$ is the set of isomorphisms k such that, for $\iota \in \bar{\iota}$, one has $k\iota \in H(A)$.

1.10. Let (G, h_o) be as in 1.5, let V be a faithful linear representation of G, let K be a compact open subgroup of $G(\mathbb{A}^f)$, and consider objects H of the following type: H consists of

(a) a vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$, endowed with a G-structure $\bar{\iota}$ (1.9);

(b) a homomorphism $h : \mathbb{S} \to G_{\mathbb{R}}^{\bar{\iota}}$ such that, for $\iota \in \bar{\iota}$, $\iota^{-1} h \iota : \mathbb{S} \to G_{\mathbb{R}}$ is conjugate to h_o;

(c) a class $\bar{k} \in K \backslash \operatorname{Isom}_{\bar{\iota}}(H \otimes \mathbb{A}^f, V \otimes \mathbb{A}^f)$ (1.9).

1.11. Example I (continued from 1.6). Let V be a $\mathbb{Q}$-vector space endowed with a non-degenerate alternating form ψ, let G be the group of symplectic similitudes of V, let $V_{\mathbb{Z}}$ be an integral lattice on which ψ has integral values and discriminant 1, and let h_o be as in 1.6. Take $K = \prod_\ell \mathrm{Gp}(V_{\mathbb{Z}} \otimes \mathbb{Z}_\ell)$. A datum (a) is interpreted as a vector space H of the same dimension as V, endowed with an alternating form ψ given up to a rational factor. A datum (b) is interpreted as a Hodge bigrading of $H_{\mathbb{C}}$, of type 1.6. A datum (c) is interpreted as an integral lattice $H_{\mathbb{Z}}$ in H, on which a rational multiple of ψ has integral values and discriminant 1: $\bar{k}$ is the set of k such that, for every ℓ, $V_{\mathbb{Z}} \otimes \mathbb{Z}_\ell = k_\ell(H_{\mathbb{Z}} \otimes \mathbb{Z}_\ell)$. We may normalize ψ by imposing the conditions that it have integral values and discriminant 1 on $H_{\mathbb{Z}}$, and that $\psi(x, Cx) > 0$ (cf. 1.6).

Let n be an integer, and let K_n be the subgroup of K consisting of those k such that $k \equiv 1 \pmod{n}$. For K_n, a datum (c) is interpreted as $H_{\mathbb{Z}}$ as above, together with a symplectic similitude $H_{\mathbb{Z}}/nH_{\mathbb{Z}} \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$.

1.12. Under the hypotheses of 1.10, the points of ${}_K M_{\mathbb{C}}(G, h_o)$ are in bijective correspondence with the isomorphism classes of objects H of type 1.10. Indeed, let $H = (H_{\mathbb{Q}}, \bar{\iota}, h, \bar{k})$. Given H and $\iota \in \bar{\iota}$, we associate the morphism $\iota^{-1} h \iota : \mathbb{S} \to G_{\mathbb{R}}$, an element of $X \simeq K_\infty \backslash G(\mathbb{R})$, and $\bar{k}\iota$ in $K \backslash G(\mathbb{A}^f)$, hence an element of $K \backslash X \times G(\mathbb{A}^f)$; to H alone we associate only an element of ${}_K M_{\mathbb{C}}(G, h_o) = K \backslash X \times G(\mathbb{A}^f)/G(\mathbb{Q}) = K_\infty \times K \backslash G(\mathbb{A})/G(\mathbb{Q})$, which uniquely determines the isomorphism class of H.

In certain cases (see 4.11), to give an object H as above amounts to giving an abelian variety endowed with some additional structures. One then obtains an interpretation of ${}_K M_{\mathbb{C}}(G)$ (and of $M_{\mathbb{C}}(G)$) as the coarse moduli scheme for abelian varieties of this type.

1.13. Let G^i $(i = 1, 2)$ be two reductive groups, let $h^i : \mathbb{S} \to G_{\mathbb{R}}^i$ be homomorphisms satisfying the conditions of 1.5, and let X^i be the set of conjugates of h^i by an element of $G^i(\mathbb{R})$. Let $G = G^1 \times G^2$, let $h : \mathbb{S} \to G_{\mathbb{R}}$ be the morphism (h^1, h^2), and let $X = X^1 \times X^2$ be the set of conjugates of h. There is an evident isomorphism

$$M_{\mathbb{C}}(G^1, h^1) \times M_{\mathbb{C}}(G^2, h^2) = M_{\mathbb{C}}(G, h). \tag{1.13.1}$$

For K^i compact open in $G^i(\mathbb{A}^f)$ and $K = K^1 \times K^2$, one likewise has

$${}_{K^1} M_{\mathbb{C}}(G^1, h^1) \times {}_{K^2} M_{\mathbb{C}}(G^2, h^2) = {}_K M_{\mathbb{C}}(G, h). \tag{1.13.2}$$

1.14. Let G^1, G^2, h^1, h^2 be as in 1.13. We shall denote by $u : (G^1, h^1) \to (G^2, h^2)$ a homomorphism $u : G^1 \to G^2$ such that $uX^1 \subset X^2$. Such a homomorphism defines

$$u : M_{\mathbb{C}}(G^1, h^1) \longrightarrow M_{\mathbb{C}}(G^2, h^2). \tag{1.14.1}$$

For K^i compact open in $G^i(\mathbb{A}^f)$, with $u(K^1) \subset K^2$, it defines

$$u(K^1, K^2) : {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \longrightarrow {}_{K^2} M_{\mathbb{C}}(G^2, h^2). \tag{1.14.2}$$

Proposition 1.15. With the notation of 1.14, suppose that G^1 is a subgroup of G^2. Then, for every compact open subgroup K^1 of $G^1(\mathbb{A}^f)$, there exists a compact open subgroup $K^2 \supset K^1$ of $G^2(\mathbb{A}^f)$ such that $u(K^1, K^2)$ identifies ${}_{K^1} M_{\mathbb{C}}(G^1, h^1)$ with a closed subscheme of ${}_{K^2} M_{\mathbb{C}}(G^2, h^2)$.

If the K^i are small enough that the map from $X^i \times (K^i \backslash G^i(\mathbb{A}^f))$ to ${}_{K^i} M_{\mathbb{C}}(G^i, h^i)$ is étale, then the map $u(K^1, K^2)$ is finite and unramified. It suffices to treat this case, and to show that $u(K^1, K^2)$ is injective for $K^2 \supset K^1$ sufficiently small. The graph of the equivalence relation $u(K^1, K^2)(x) = u(K^1, K^2)(y)$ is a closed subscheme of $({}_{K^1} M_{\mathbb{C}}(G^1, h^1))^2$, decreasing with K^2, hence stationary for $K^2 \supset K^1$ sufficiently small. It is enough to show that

$$u(K^1) : {}_{K^1} M_{\mathbb{C}}(G^1, h^1) \longrightarrow \varprojlim_{K^2 \supset K^1} {}_{K^2} M_{\mathbb{C}}(G^2, h^2) = \mathrm{dfn}\, {}_{K^1} M_{\mathbb{C}}(G^2, h^2)$$

is injective. One has

$${}_{K^1} M_{\mathbb{C}}(G^1, h^1) = K^1 \backslash M_{\mathbb{C}}(G^1, h^1),$$
$${}_{K^1} M_{\mathbb{C}}(G^2, h^2) = K^1 \backslash M_{\mathbb{C}}(G^2, h^2),$$

so it is enough to prove the

Variant 1.15.1. $u : M_{\mathbb{C}}(G^1, h^1) \longrightarrow M_{\mathbb{C}}(G^2, h^2)$ is injective.

This assertion follows easily from the following two lemmas.

Lemma 1.15.2. Let $U^i \subset C^i(\mathbb{Q})$ be the group of units of the centre of G^i, and put $G^i(\mathbb{Q})^\wedge = \varprojlim_n \big(G^i(\mathbb{Q})/(U^i)^n\big)$. Then

$$M_{\mathbb{C}}(G^i, h^i) = K^i_\infty \backslash G^i(\mathbb{A})/G^i(\mathbb{Q})^\wedge.$$

This is a corollary of Chevalley's theorem, according to which the $(U^i)^n$ are congruence subgroups of U^i.

Lemma 1.15.3. $u : G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge \longrightarrow G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge$ is injective.

By left translation, it is enough to show that $u^{-1}(\{e\}) = \{e\}$: consider the diagram

$$\begin{array}{ccccccc}
0 & \longrightarrow & G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{Q})^\wedge & \longrightarrow & (G^2/G^1)(\mathbb{Q})^\wedge \\
 & & \downarrow & & \downarrow & & \downarrow\ ! \\
0 & \longrightarrow & G^1(\mathbb{A}^f) & \longrightarrow & G^2(\mathbb{A}^f) & \longrightarrow & (G^2/G^1)(\mathbb{A}^f) \\
 & & \downarrow & & \downarrow & & \\
 & & G^1(\mathbb{A}^f)/G^1(\mathbb{Q})^\wedge & \longrightarrow & G^2(\mathbb{A}^f)/G^2(\mathbb{Q})^\wedge & &
\end{array}$$

where $(G^2/G^1)(\mathbb{Q})^\wedge = \varprojlim_n (G^2/G^1)(\mathbb{Q})/(U^2/U^1)^n$.

§ 2. Connected components.

2.1. Let G be a connected reductive group over $\mathbb{Q}$. We use the notation of 1.1 and assume that G' admits no non-trivial invariant subgroup H (defined over $\mathbb{Q}$) and such that $H(\mathbb{R})$ is compact. This hypothesis does not exclude the possibility that $G'_{\mathbb{R}}$ has compact factors. We denote by $\widetilde{G}'$ the simply connected covering of the derived group G'.

Proposition 2.2. (i) The group $G(\mathbb{A}^f)$ acts transitively on the set $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$ of connected components of $G(\mathbb{A})/G(\mathbb{Q})$.

(ii) The group $\widetilde{G}'(\mathbb{A})$ acts trivially on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.

(iii) The quotient of $G(\mathbb{A})$ by the image of $\widetilde{G}'(\mathbb{A})$ is a commutative group.

Assertion (i) follows from (0.4): $G(\mathbb{A}) = G(\mathbb{A}^f)G(\mathbb{R})^0G(\mathbb{Q})$. Since $\widetilde{G}'$ is simply connected, $\widetilde{G}'(\mathbb{R})$ is connected. By (0.3), $\widetilde{G}'(\mathbb{R}) \cdot \widetilde{G}'(\mathbb{Q})$ is dense in $\widetilde{G}'(\mathbb{A})$, so that $\widetilde{G}'(\mathbb{A})/\widetilde{G}'(\mathbb{Q})$ is connected. For $x \in G(\mathbb{A})$ one has $\widetilde{G}'(\mathbb{A}) \cdot x = x \cdot \widetilde{G}'(\mathbb{A})$; hence the image of $\widetilde{G}'(\mathbb{A}) \cdot x$ in $G(\mathbb{A})/G(\mathbb{Q})$ is an image of $\widetilde{G}'(\mathbb{A})/\widetilde{G}'(\mathbb{Q})$: it is connected, and (ii) follows.

Let Z be the centre of $\widetilde{G}'$. Assertion (iii) follows from the fact that the commutator map $xyx^{-1}y^{-1} : G \times G \longrightarrow G$ has a factorization

$$G \times G \longrightarrow G/C \times G/C \longleftarrow \widetilde{G}'/Z \times \widetilde{G}'/Z \longrightarrow \widetilde{G}' \longrightarrow G.$$

Definition 2.3. We denote by $\pi(G)$ the commutative quotient of $G(\mathbb{A})$ (or $G(\mathbb{A}^f)$) that acts faithfully on $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$.

In other words, $\pi(G)$ is $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$, endowed with its "evident" commutative group structure. We recall the following theorem.

Theorem 2.4. Under the hypotheses of 2.1, if G' is simply connected, then the canonical homomorphism

$$\pi(G) = \pi_0(G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q})) \tag{2.4.1}$$

is bijective.

To prove the surjectivity of (2.4.1), by 2.2(i) applied to T it is enough to show that $G(\mathbb{A}^f)$ maps onto $T(\mathbb{A}^f)$. Since the kernel G' of $\nu : G \to T$ is connected, one knows that, for almost all p, $\nu(G(\mathbb{Z}_p)) = T(\mathbb{Z}_p)$; this formula has a meaning for almost all p, and is a corollary of Lang's theorem applied to G' reduced modulo p. It is therefore enough to prove that, for every p, $\nu(G(\mathbb{Q}_p)) = T(\mathbb{Q}_p)$, which follows from (0.2) applied to G'.

Theorem 2.4 is equivalent to the following more concrete statement.

Variant 2.5. For every compact open subgroup K of $G(\mathbb{A}^f)$, the map

$$\nu : \pi_0(K \backslash G(\mathbb{A})/G(\mathbb{Q})) \longrightarrow \pi_0(\nu(K) \backslash T(\mathbb{A})/T(\mathbb{Q})) \tag{2.5.1}$$

is bijective.

Indeed, $\pi_0(G(\mathbb{A})/G(\mathbb{Q})) = \varprojlim_K \pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q}))$, and similarly for T. Also, $\pi_0(K\backslash G(\mathbb{A})/G(\mathbb{Q})) = G(\mathbb{R})^0\backslash K\backslash G(\mathbb{A})/G(\mathbb{Q})$. We show that if $x, y \in G(\mathbb{A})$ have the same image in the right-hand side of (2.5.1), then they have the same image in the left-hand side. Left translation by y^{-1}, which replaces K by $y^{-1}Ky$, allows us to suppose that $y = e$. Modifying x by an element of $G(\mathbb{R})^0 \times K$, we may suppose that $\nu(x) = \tau$ with $\tau \in T(\mathbb{Q})$. By the Hasse principle (0.1) for G', the equation $\nu(y) = \tau$ has a solution in $G(\mathbb{Q})$. Correcting x by y, we may suppose that $x \in G'(\mathbb{A})$. By 2.2(ii), the image of x in $G(\mathbb{A})/G(\mathbb{Q})$ is in the neutral component, and a fortiori x and y have the same image in the left-hand side of 2.5.1.

This proof seems to use, via (0.1), that G' has no factor of type E_8. In fact, the class in $H^1(\mathbb{Q}, G')$ which obstructs the equation $\nu(y) = \tau$ comes from the centre of G', and the E_8 factors, being adjoint, count for nothing.

Proposition 2.6. One has $K_\infty = C(\mathbb{R}) \cdot \big(K_\infty \cap G'(\mathbb{R})^0\big)$. The group $K_\infty \cap G'(\mathbb{R})^0$ is connected, and

$$\pi_0(K_\infty) \xrightarrow{\sim} \mathrm{Im}\big(\pi_0(C(\mathbb{R})) \longrightarrow \pi_0(G(\mathbb{R}))\big),$$

Let $h' : \mathbb{S} \to G_\mathbb{R} \to (G/C)_\mathbb{R}$. The centralizer of h' in $(G/C)(\mathbb{R})$ is connected, because it is a compact group whose complexification is connected (as the centralizer of a torus). In particular, the image of K_∞ in $G/C(\mathbb{R})$ lies in the neutral component, the image of $G'(\mathbb{R})^0$, and $K_\infty = C(\mathbb{R}) \cdot \big(K_\infty \cap G'(\mathbb{R})^0\big)$. The group $K_\infty \cap G'(\mathbb{R})^0$ is connected as a maximal compact subgroup of a connected group. This proves 2.6.

2.7. The image of $\pi_0(K_\infty)$ in $\pi_0(G(\mathbb{A})/G(\mathbb{Q}))$ depends only on the conjugacy class of h, and $\pi_0(M_\mathbb{C}(G,h)) = \varprojlim_K \pi_0(K_\infty \times K\backslash G(\mathbb{A})/G(\mathbb{Q}))$ is identified with $\pi(G)/\pi_0(K_\infty)$.

In particular, if G' is simply connected, one has

$$\pi_0(M_\mathbb{C}(G,h)) \xrightarrow{\sim} \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_\infty). \tag{2.7.1}$$

More concretely, for every compact open subgroup K of $G(\mathbb{A}^f)$, $\pi_0({}_K M_\mathbb{C}(G,h))$ is then a principal homogeneous space under $\nu(K_\infty \times K)\backslash T(\mathbb{A})/T(\mathbb{Q})$.

§ 3. Models.

Let G be a reductive group over $\mathbb{Q}$, let $h : \mathbb{S} \to G_\mathbb{R}$ satisfy the conditions of 1.5, and let E be a field endowed with a complex embedding $\rho : E \to \mathbb{C}$. Put $M_\mathbb{C}(G) = M_\mathbb{C}(G,h)$.

Definition 3.1. A model over E of $M_\mathbb{C}(G)$ consists of

(a) a scheme M over E, endowed with a continuous action of $G(\mathbb{A}^f)$;

(b) an isomorphism m from $M \otimes_{E,\rho} \mathbb{C}$ to $M_\mathbb{C}(G)$ compatible with the action of $G(\mathbb{A}^f)$.

Let F be a finite extension of E, endowed with a complex embedding extending that of E. If $M_E(G,h)$ is a model of $M_\mathbb{C}(G,h)$ over E, we denote by $M_F(G,h)$ the model $M_E(G,h) \otimes_E F$ of $M_\mathbb{C}(G,h)$ over F.

Remark 3.2. Giving a scheme M over E, endowed with a continuous action of $G(\mathbb{A}^f)$, is equivalent to giving

(a) for every compact open subgroup K of $G(\mathbb{A}^f)$, a scheme ${}_K M$ over E;

(b) for compact open subgroups K, L of $G(\mathbb{A}^f)$ and for $x \in G(\mathbb{A}^f)$ such that $xKx^{-1} \subset L$, a homomorphism $J_{L,K}(x) : {}_K M \longrightarrow {}_L M$, these homomorphisms satisfying

$$\begin{array}{ll} (\alpha) & J_{M,L}(y)J_{L,K}(x) = J_{M,K}(yx), \\ (\beta) & J_{K,K}(x) = \mathrm{Id} \qquad \text{if } x \in K, \\ (\gamma) & \text{for } K \text{ normal in } L,\ J \text{ defines an action of } L/K \text{ on } {}_K M, \\ & \text{and } J_{L,K}(e) \text{ defines } (L/K)\backslash {}_K M \xrightarrow{\sim} {}_L M. \end{array}$$

One has

$$M = \varprojlim_K {}_K M \qquad \text{and} \tag{3.2.1}$$

$${}_K M = K\backslash M. \tag{3.2.2}$$

3.3. Let $M(G)$ be a model over E of $M_\mathbb{C}(G)$. For K compact open in $G(\mathbb{A}^f)$, put ${}_K M(G) = K\backslash M(G)$. The ${}_K M(G)$ are quasi-projective schemes over E, not necessarily geometrically connected. The structural isomorphism m induces isomorphisms

$${}_K M(G) \otimes_{E,\rho} \mathbb{C} \xrightarrow{\sim} {}_K M_\mathbb{C}(G). \tag{3.3.1}$$

3.4. Suppose that G satisfies the hypotheses of 2.1, and let (M, m) be a model of $M_{\mathbb{C}}(G,h)$ over E. Let $\overline{E}$ be the algebraic closure of E in $\mathbb{C}$. Since M is defined over E, the Galois group $\mathrm{Gal}(\overline{E}/E)$ acts on the profinite set

$$\pi_0(M \otimes_E \overline{E}) = \varprojlim_K \pi_0({}_K M \otimes_E \overline{E}) \xrightarrow[m]{\sim} \pi_0(M_{\mathbb{C}}(G,h)).$$

The group $G(\mathbb{A}^f)$ acts on $\pi_0(M \otimes_E \overline{E})$ through its quotient $\pi(G)/\pi_0(K_\infty)$, and $\pi_0(M \otimes_E \overline{E})$ is a principal homogeneous space under the commutative group $\pi(G)/\pi_0(K_\infty)$ (2.6). The action of $\mathrm{Gal}(\overline{E}/E)$ commutes with that of $G(\mathbb{A}^f)$, hence with that of $\pi(G)/\pi_0(K_\infty)$. The action of an element σ of $\mathrm{Gal}(\overline{E}/E)$ can therefore only be the translation defined by an element $\lambda(\sigma)$ of $\pi(G)/\pi_0(K_\infty)$, and λ is a homomorphism

(3.4.1) $$\lambda_M : \mathrm{Gal}(\overline{E}/E) \longrightarrow \pi(G)/\pi_0(K_\infty).$$

Assume that E is a number field. Class field theory identifies the largest abelian quotient of $\mathrm{Gal}(\overline{E}/E)$ with $\pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q}))$, and identifies (3.4.1) with

(3.4.2) $$\lambda_M : \mathrm{Gal}(\overline{E}/E)^{\mathrm{ab}} = \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \longrightarrow \pi(G)/\pi_0(K_\infty).$$

In the special case where G' is simply connected, the homomorphism (3.4.2) is identified, by (2.7.1), with

(3.4.3) $$\lambda_M : \pi_0(E^*(\mathbb{A})/E^*(\mathbb{Q})) \longrightarrow \pi_0(T(\mathbb{A})/T(\mathbb{Q}))/\pi_0(K_\infty).$$

Definition 3.5. The homomorphisms (3.4.1), (3.4.2), and (3.4.3) are called the reciprocity law of the model M. If G' is simply connected, and if λ_M comes from a homomorphism of algebraic groups over $\mathbb{Q}$ from E^* to T, the latter will also be called the "reciprocity law" of M.

3.6. There is a rule, applicable to each of the models constructed by Shimura, for determining E and the reciprocity law λ_M from G and h. The field E, for arbitrary G, and λ_M, for simply connected G', are described as follows.

3.7. The composite homomorphism hr (1.3)

$$hr : \mathbb{G}_m \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}}$$

is a homomorphism over $\mathbb{C}$ between algebraic groups defined over $\mathbb{Q}$. The subfield E of $\mathbb{C}$ is the field of definition $E(G,h)$ of the conjugacy class of this homomorphism.

For G semisimple adjoint, the field $E(G,h)$ can be described as follows. Let Δ be the Dynkin diagram of $G_{\mathbb{C}}$, and let $|\Delta|$ be the set of vertices of Δ. The Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on Δ.

Let H be a maximal torus of $G_{\mathbb{C}}$, endowed with a system of simple roots $(\alpha_i)_{i\in|\Delta|}$. The α_i identify H with $\mathbb{G}_{m,\mathbb{C}}^{|\Delta|}$. For every homomorphism $u : \mathbb{G}_{m,\mathbb{C}} \to G_{\mathbb{C}}$, there is then a unique homomorphism $u' : \mathbb{G}_{m,\mathbb{C}} \to H = \mathbb{G}_{m,\mathbb{C}}^{\Delta}$: $x \longmapsto (x^{n_i})_{i\in|\Delta|}$ with $n_i \geq 0$, which is conjugate to u. The construction $u \mapsto \eta(u) = (n_i)_{i\in|\Delta|}$ identifies the set of conjugacy classes of maps from $\mathbb{G}_{m,\mathbb{C}}$ to $G_{\mathbb{C}}$ with the set of functions from $|\Delta|$ to $\mathbb{N}$. It follows that $E(G,h)$ is defined by the subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which stabilizes $\eta(hr)$.

Proposition 3.8. Let $G/C = \prod_{i=1}^{g} G_i$ be the decomposition of the adjoint group G/C into $\mathbb{Q}$-simple factors, and let h_i be the composite $h_i : \mathbb{S} \xrightarrow{h} G_{\mathbb{R}} \longrightarrow (G_i)_{\mathbb{R}}$.

(i) The subfield $E(G,h)$ of $\mathbb{C}$ is the compositum of the subfields $E(T, \nu \circ h)$ and $E(G_i, h_i)$ $(1 \leq i \leq g)$ of $\mathbb{C}$.

(ii) If the opposition involution of the Dynkin diagram of G_i respects $\eta(h_i r)$, then $E(G_i, h_i)$ is totally real. Otherwise, $E(G_i, h_i)$ is a totally imaginary quadratic extension of a totally real number field.

Assertion (i) is easy to check, and it is enough to prove (ii) for G $\mathbb{Q}$-simple and adjoint. Let Δ be its Dynkin diagram. The hypotheses 1.5 imply that $G_{\mathbb{R}}$ has a compact maximal torus; this implies that complex conjugation acts on Δ by the opposition involution ι. This involution is in the centre of the automorphism group of Δ.

Let E' be the Galois extension of $\mathbb{Q}$ defined by the subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which acts trivially on Δ. Then the image σ of complex conjugation is central in $\mathrm{Gal}(E'/\mathbb{Q})$; hence E' is totally real or a totally imaginary quadratic extension of a totally real field. Finally, σ is the identity on $E(G,h) \subset E'$ if and only if ι respects $\eta(hr)$, and the assertion follows.

3.9. We use the notation of 3.6. Suppose that G' is simply connected, that E contains $E(G,h)$, and that the complex embedding of E induces that of $E(G,h)$. The composite morphism

$$r''(h) : \nu\, h\, r : \mathbb{G}_{m,\mathbb{C}} \xrightarrow{r} \mathbb{S}_{\mathbb{C}} \xrightarrow{h} G_{\mathbb{C}} \xrightarrow{\nu} T_{\mathbb{C}}$$

depends only on the conjugacy class of hr. It is therefore defined over E, i.e. it arises, by extension of scalars from E to $\mathbb{C}$, from a homomorphism of algebraic groups over E, again denoted $r''(h) : \mathbb{G}_m \longrightarrow T_E$. Apply Weil restriction of scalars, and let $r'(h) : E^* \to T$ be the composite

$$E^* = R_{E/\mathbb{Q}}(\mathbb{G}_{m,E}) \xrightarrow{R_{E/\mathbb{Q}}(r''(h))} R_{E/\mathbb{Q}}(T_E) \xrightarrow{N_{E/\mathbb{Q}}} T.$$

The reciprocity law of M is the inverse $r(G,h)$ of $r'(h)$:

$$r(G,h) = r'(h)^{-1} : E^* \longrightarrow T. \tag{3.9.1}$$

3.10. Example II. Let H be a torus and let $h : \mathbb{S} \to H_{\mathbb{R}}$. The conditions of 1.5 are automatically satisfied. The ${}_KM_{\mathbb{C}}(H,h)$ are finite sets. A finite reduced scheme over the field $E(H,h)$ (3.7) is identified with a Galois set. It follows that, up to unique isomorphism, there is a unique model of $M_{\mathbb{C}}(H,h)$ over $E(H,h)$ whose reciprocity law is given by 3.9.1. It is denoted $M(H,h)$.

3.11. Let F be a finite extension of E, endowed with a complex embedding extending that of E. The diagram

$$\begin{tikzcd} F^* \arrow[rr, "N_{F/E}"] \arrow[dr, "r(G,h)"'] & & E^* \arrow[dl, "r(G,h)"] \\ & T & \end{tikzcd}$$

is commutative. If $M_E(G,h)$ is a model over E of $M_{\mathbb{C}}(G,h)$ whose reciprocity law is (3.9.1), it follows that $M_F(G,h)$ (3.1) again has (3.9.1) as its reciprocity law.

3.12. Let $u : (G^1,h^1) \to (G^2,h^2)$ be as in 1.14, and let $M^i_{E^i}(G^i,h^i)$ be a model of $M_{\mathbb{C}}(G^i,h^i)$ over E^i. Let E be the compositum of the subfields E^1 and E^2 of $\mathbb{C}$. With the notation of 3.1 one has

$$M^i_E(G^i,h^i) \otimes_E \mathbb{C} = M_{\mathbb{C}}(G^i,h^i) \qquad (i = 1,2).$$

It therefore makes sense to ask whether

$$u : M_{\mathbb{C}}(G^1,h^1) \longrightarrow M_{\mathbb{C}}(G^2,h^2)$$

is defined over E.

Definition 3.13. Let G be a connected reductive group over $\mathbb{Q}$ satisfying the condition of 2.1, $h : \mathbb{S} \to G_{\mathbb{R}}$ be a homomorphism satisfying the conditions of 1.5, and E be an extension of $E(G,h)$ (3.7) endowed with a complex embedding extending that of $E(G,h)$. A model $M_E(G,h)$ of $M_{\mathbb{C}}(G,h)$ over E is called weakly canonical if, for every torus $u : H \to G$ in G, endowed with $h' : \mathbb{S} \to H_{\mathbb{R}}$ such that uh' is the conjugate of h by an element of $G(\mathbb{R})$, the morphism (1.14)

$$M_{\mathbb{C}}(H,h') \longrightarrow M_{\mathbb{C}}(G,h)$$

is defined, in the sense of (3.10) and (3.12), over the compositum $E(H,h') \cdot E \subset \mathbb{C}$. A model $M_E(G,h)$ is called canonical if it is weakly canonical and if $E = E(G,h)$.

3.14. Translation. Let M be a canonical model of $M_{\mathbb{C}}(G,h)$, with reciprocity law λ_M. If G' is simply connected, then λ_M is given by 3.9.1 (5.6). For every compact open subgroup K of $G(\mathbb{A}^f)$, again denote by $\nu(K)$ the image of K in $\pi(G)/\pi_0(K_\infty)$, and let $E(K)$ be the extension of $E(G,h)$ defined by the idèle class group $\lambda_M^{-1}(\nu(K))$. One verifies that $E(K)$ is the field of definition of the neutral component ${}_KM^0_{\mathbb{C}}$ of ${}_KM_{\mathbb{C}}$, considered as a subscheme of ${}_KM_{\mathbb{C}} = {}_KM \otimes_{E,\rho} \mathbb{C}$.

By definition, ${}_KM^0_{\mathbb{C}}$ is the scalar extension to $\mathbb{C}$ of a subscheme ${}_KM'$ of ${}_KM \otimes_E E(K)$. The scheme ${}_KM'/E(K)$ is geometrically connected over $E(K)$, and one can also describe ${}_KM'$ as the union of the connected components (over E) of ${}_KM$ that, after extension of scalars from E to $\mathbb{C}$, contain the origin:

$$\begin{tikzcd} {}_KM \arrow[dd] & {}_KM' \arrow[l, hook'] \arrow[d] & {}_KM^0_{\mathbb{C}} \arrow[l] \arrow[d] \\ & \operatorname{Spec} E(K) \arrow[d] & \operatorname{Spec} \mathbb{C} \arrow[l] \\ \operatorname{Spec} E \arrow[r, equal] & \operatorname{Spec} E & \end{tikzcd} .$$

Shimura usually states his results in terms of the system of varieties ${}_K M'/E(K)$ and the homomorphisms of type 2.2 between them. For a typical statement, see [14, p. 146]. For the translation, cf. [14, 2.7].

3.15. For $u : (H, h') \longrightarrow (G, h)$ as in 3.13, the image of $M_{\mathbb{C}}(H, h')$ in ${}_K M_{\mathbb{C}}(G, h)$ is a finite set. The hypothesis is that this subset of ${}_K M_{\mathbb{C}}(G, h)$ be defined over $E(H, h')$, that its points be defined over an abelian extension of $E(H, h')$, and that $\mathrm{Gal}(\overline{E(H, h')}/E(H, h'))$ act on them in the prescribed manner.

We shall call "special" the points of ${}_K M_{\mathbb{C}}(G, h)$ lying in the image of some $M_{\mathbb{C}}(H, h')$.

Suppose given a projective embedding $q : {}_K M^0_{\mathbb{C}} \longrightarrow \mathbb{P}^r$ defined over $E(K)$. It may be interpreted as $\widetilde{q} : X^0 \longrightarrow \mathbb{P}^r$ (cf. 1.7, 1.8). If $uh' \in X$ lies in X^0, then the coordinates of $\widetilde{q}(uh')$ generate over $E(K)E(H, h')$ an abelian extension of $E(H, h')$ that can be described explicitly as a class field.

3.16. The classical example of such a situation is provided by the moduli scheme of elliptic curves (corresponding to $G = \mathrm{GL}_2$, $K = \mathrm{GL}_2(\widehat{\mathbb{Z}})$, h as in 1.6), and by its points corresponding to elliptic curves with complex multiplication.

Let j be the modular invariant, viewed as a function on the Poincaré upper half-plane X^0. If $\tau \in X^0$ lies in an imaginary quadratic field K, let $L(\tau) = \mathbb{Z} \oplus \mathbb{Z}\tau \subset \mathbb{C}$, and let $\mathcal{O}(\tau) = \{\, k \in K \mid kL(\tau) \subset L(\tau) \,\}$. The field $K(j(\tau))$ is the abelian extension of K whose Galois group is the ideal class group of $\mathcal{O}(\tau)$. If σ in this Galois group corresponds to an invertible $\mathcal{O}(\tau)$-module L, and if $L(\tau') \simeq L(\tau) \otimes_{\mathcal{O}(\tau)} L$, then $j(\tau) = j(\tau')^\sigma$.

For more exotic explicit examples, see [15], pp. 45–46.

ğ 4. Abelian varieties.

4.1. Let k be an algebraically closed field of characteristic 0. Let A be an abelian variety over k. For $n \in \mathbb{N}^+$, denote by A_n the kernel of multiplication by n. The A_n form a projective system, with $\varphi_{nm,n} : A_{nm} \longrightarrow A_n$ given by $x \longmapsto x^m$, and we put

$$\widehat{T}(A) = \varprojlim A_n, \qquad \widehat{V}(A) = \mathbb{A}^f \otimes_{\widehat{\mathbb{Z}}} \widehat{T}(A).$$

The $\widehat{\mathbb{Z}}$-module $\widehat{T}(A)$ is the product of the Tate modules (= Weil representations) $T_\ell(A)$. If $k = \mathbb{C}$, then

$$\widehat{T}(A) = \widehat{\mathbb{Z}} \otimes H_1(A, \mathbb{Z}), \tag{4.1.1}$$

$$\widehat{V}(A) = \mathbb{A}^f \otimes H_1(A, \mathbb{Q}). \tag{4.1.2}$$

4.2. The category of abelian varieties up to isogeny over k is the category whose objects are the abelian varieties over k and in which $\mathrm{Hom}_{\mathrm{iso}}(A, B) = \mathrm{Hom}(A, B) \otimes \mathbb{Q}$. We denote by $A \otimes \mathbb{Q}$ the abelian variety up to isogeny "underlying" an abelian variety A. Every additive functor from the category of abelian varieties to a $\mathbb{Q}$-linear additive category factors through the functor $A \mapsto A \otimes \mathbb{Q}$. Thus the contravariant functor "dual variety" $A \mapsto A^*$ extends to abelian varieties up to isogeny. The functor $A \mapsto \widehat{V}(A)$ extends in the same way. Moreover, if A_0 is an abelian variety up to isogeny, it is equivalent to give B together with an isomorphism $B \otimes \mathbb{Q} = A_0$, or to give $\widehat{T}(B) \subset \widehat{V}(A_0)$.

4.3. Let A be an abelian variety over k. Let $\mathrm{NS}(A)$ denote the group of algebraic-equivalence classes of invertible sheaves on A. An effective polarization (resp. a polarization) of A is defined to be an element of $\mathrm{NS}(A)$, respectively of $\mathrm{NS}(A) \otimes \mathbb{Q}$, a positive multiple of which is defined by a projective embedding of A. A homogeneous polarization is an element of $(\mathrm{NS}(A) \otimes \mathbb{Q})/\mathbb{Q}^*$ which is the class of a polarization.

Recall that $\mathrm{NS}(A) \otimes \mathbb{Q}$ is identified, by a bijection $p \mapsto p'$, with the set of elements of $\mathrm{Hom}(A, A^*) \otimes \mathbb{Q}$ equal to their transpose. An isomorphism $u \in \mathrm{Hom}(A \otimes \mathbb{Q}, B \otimes \mathbb{Q})$ induces isomorphisms from $\mathrm{Hom}(A, A^*) \otimes \mathbb{Q}$ to $\mathrm{Hom}(B, B^*) \otimes \mathbb{Q}$, and from $\mathrm{NS}(A) \otimes \mathbb{Q}$ to $\mathrm{NS}(B) \otimes \mathbb{Q}$, and carries polarizations to polarizations. This makes it possible to speak of a polarization of an abelian variety up to isogeny.

4.4. A polarization p of A defines a positive involution on $\mathrm{End}(A) \otimes \mathbb{Q}$, $u \longmapsto p'^{-1} u^* p'$, which depends only on the homogeneous polarization $\mathbb{Q}^*.p$.

Let F be a product of totally real fields, and let $\rho : F \longrightarrow \mathrm{End}(A) \otimes \mathbb{Q}$.

Let $\mathrm{NS}_\rho(A) \subset \mathrm{NS}(A)$ be the set of p such that $p'.\rho(f) = \rho(f)^* p'$ for $f \in F$.

The vector space $\mathrm{NS}_\rho(A) \otimes \mathbb{Q}$ is endowed, for $f \in F$, with an action of F: $(f.p)' = p' \circ \rho(f) = \rho(f)^* \circ p'$. A weak polarization of A (relative to ρ, F) is an element of $(\mathrm{NS}_\rho(A) \otimes \mathbb{Q})/F^*$ which is the class of a polarization. The set of weak polarizations of A depends only on $A \otimes \mathbb{Q}$. The restriction to the centralizer of F of the involution on $\mathrm{End}(A) \otimes \mathbb{Q}$ defined by a polarization $p \in \mathrm{NS}_\rho(A)$ depends only on the weak polarization $F^*.p$.

4.5. If A_0^* is the dual of an abelian variety up to isogeny A_0, then $\widehat{V}(A_0)$ and $\widehat{V}(A_0^*)$ are in duality with values in $\mathbb{A}^f(1)$. If p is a polarization of A_0, the polarization form $\psi_p(x,y) = \langle x, p'(y)\rangle$ is a nondegenerate alternating form on the free $\mathbb{A}^f$-module $\widehat{V}(A_0)$, with values in $\mathbb{A}^f(1)$.

The facts recalled in 4.1 and 4.5 extend to abelian schemes over an arbitrary base.

4.6. Set $k = \mathbb{C}$. The additive functor $A \mapsto H_1(A, \mathbb{Q})$ extends to abelian varieties up to isogeny over $\mathbb{C}$ (4.2). For an abelian variety up to isogeny A_0, $H_1(A_0, \mathbb{Q})$ and $H_1(A_0^*, \mathbb{Q})$ are in duality. Via 4.1.2 and the isomorphism of $\mathbb{A}^f(1)$ with $\mathbb{A}^f$ defined by the exponential, this duality is compatible with the one considered in 4.5. The alternating form $\psi_p(x,y) = \langle x, p'(y)\rangle$ on $H_1(A_0, \mathbb{Q})$ defined by a polarization p of A_0 is again called the polarization form.

The Hodge structure on $H_1(A_0, \mathbb{Q})$ is the homomorphism $h : \mathbb{S} \longrightarrow \mathrm{GL}(H_1(A_0, \mathbb{Q}))_{\mathbb{R}}$, such that $\mathbb{S}$ acts on $H_1(A_0, \mathbb{Q}) \otimes \mathbb{C}$ through the characters z^{-1} and $\bar{z}^{-1}$, and such that $F(h)^\circ$ (1.5) is the kernel of the exponential map $H_1(A_0, \mathbb{Q}) \otimes \mathbb{C} \longrightarrow \mathrm{Lie}(A)$.

Theorem 4.7. (Riemann). The constructions

$$(A, p) \longmapsto (H_1(A, \mathbb{Q}), \psi_p, H_1(A, \mathbb{Z}), h),$$

$$(V, \psi, V_{\mathbb{Z}}, h) \longmapsto F(h)^\circ \backslash V \otimes \mathbb{C}/V_{\mathbb{Z}}$$

establish an equivalence between

(a) polarized abelian varieties over $\mathbb{C}$;

(b) systems consisting of a vector space V over $\mathbb{Q}$, a nondegenerate alternating form ψ on V, an integral lattice $V_{\mathbb{Z}}$ in V and a homomorphism h from $\mathbb{S}$ into the group $\mathrm{Gp}(V)_{\mathbb{R}}$ of symplectic similitudes of $V \otimes \mathbb{R}$, of type 1.6 and such that

$$\psi(x, h(i)x) > 0 \qquad (x \in V \otimes \mathbb{R},\ x \neq 0).$$

Variant 4.8. It is equivalent to give an abelian variety up to isogeny A over $\mathbb{C}$, endowed with a homogeneous polarization, or to give a vector space over $\mathbb{Q}$, endowed with a nondegenerate alternating form ψ given up to scalar, together with $h : \mathbb{S} \longrightarrow \mathrm{Gp}(V)$ of type 1.6.

4.9. Let L be a semisimple algebra with involution over $\mathbb{Q}$, and let V be a vector space over $\mathbb{Q}$, endowed with a faithful L-module structure and with a nondegenerate alternating form ψ such that

$$\psi(\ell x, y) = \psi(x, \ell^* y). \tag{4.9.1}$$

Let G denote the algebraic group over $\mathbb{Q}$ of L-linear symplectic similitudes of V. The group $G(\mathbb{Q})$ is the set of $g \in \mathrm{GL}_L(V)$ for which there exists $\mu(g) \in \mathbb{Q}^*$ satisfying

$$\psi(gx, gy) = \mu(g)\psi(x, y). \tag{4.9.2}$$

Suppose given a homomorphism $h_0 : \mathbb{S} \longrightarrow G_{\mathbb{R}}$ such that, via h_0, $\mathbb{S}$ acts on $V_{\mathbb{C}}$ through the characters z^{-1} and $\bar{z}^{-1}$ and such that the form $\psi(x, h_0(i)y)$ is symmetric and positive definite. The conditions (1.5.1) to (1.5.3) are then satisfied by (G, h_0), and the involution of L is positive.

4.10. Apply (1.11) to (G, h_0) and to the representation V of G. A G-structure $\bar{\iota}$ on a vector space H may be interpreted, by 1.8, as specifying on H an L-module structure and an alternating form ψ, given up to scalar, such that H, endowed with these structures, is isomorphic to V. Let $\bar{\iota}$ be a G-structure on H. A datum 1.9 (b), $h : \mathbb{S} \longrightarrow G^1_{\mathbb{R}}$ on H, then defines, by 4.8 applied to (H, ψ, h), an abelian variety up to isogeny A, endowed with a homogeneous polarization $\bar{p}$, such that H is $H_1(A, \mathbb{Q})$, h is the Hodge structure on $H_1(A, \mathbb{Q})$, and ψ is the polarization form. Moreover, the L-module structure on H comes from $\rho : L \longrightarrow \mathrm{End}(A)$, and H, endowed with $\bar{\iota}$ and h, is determined by $(A, \bar{p}, \rho)$.

For a triple $(A, \bar{p}, \rho)$ to arise from such an H, it is necessary and sufficient that

(4.10.1) $H_1(A, \mathbb{Q})$, endowed with its L-module structure and with the polarization form (given up to scalar), is isomorphic to V;

(4.10.2) For an isomorphism $t : V \longrightarrow H_1(A, \mathbb{Q})$, $t^{-1}h_1 t$ is conjugate to h_0.

Let t be the map from L to $\mathbb{C}$ given by

$$t(\ell) = \mathrm{Tr}(\ell; V_{\mathbb{C}}/F^0(h_0)). \tag{4.10.3}$$

Scholium 4.11. Let K be a compact open subgroup of $G(\mathbb{A}^f)$. The points of ${}_K M_{\mathbb{C}}(G, h_0)$ correspond bijectively to isomorphism classes of abelian varieties up to isogeny A, endowed with

(a) $\rho : L \longrightarrow \mathrm{End}(A)$ such that

$$\mathrm{Tr}(\rho(\ell), \mathrm{Lie}(A)) = t(\ell) \qquad (\ell \in L),$$

(b) a homogeneous polarization $\bar{p}$ that induces the given involution of L,

(c) a class mod K, $\bar{k}$, of symplectic similitudes L-linear

$k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$,

and such that

(*) the conditions (4.10.1) and (4.10.2) are satisfied.

We apply 1.11 and 4.10, observing that a datum 1.9 (c) is identified with a datum (c), and that the conditions in (b) and (a) follow from (4.10.1) and (4.10.2), respectively.

4.12. Let K be a compact open subgroup of $G(\mathbb{A}^f)$, let $V_{\mathbb{Z}}$ be an integral lattice in V such that $V_{\widehat{\mathbb{Z}}} = V_{\mathbb{Z}} \otimes \widehat{\mathbb{Z}}$ is K-invariant, let $L_{\mathbb{Z}}$ be an order in L such that $L_{\mathbb{Z}} V_{\mathbb{Z}} \subset V_{\mathbb{Z}}$, let $\psi_{\mathbb{Z}}$ be a positive rational multiple of the alternating form of V, with integral values on $V_{\mathbb{Z}}$, and let $V'_{\mathbb{Z}}$ be the largest lattice in V such that $\psi_{\mathbb{Z}}(V_{\mathbb{Z}}, V'_{\mathbb{Z}}) \subset \widehat{\mathbb{Z}}$. Let n be an integer, and put $K_n = \{\, g \in G(\mathbb{A}^f) \mid (g-1)V_{\widehat{\mathbb{Z}}} \subset nV_{\widehat{\mathbb{Z}}} \}$. Assume that n is large enough that $K \supset K_n$.

If A is as in 4.11, then $k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A)$ does not depend on $k \in \bar{k}$, and defines an abelian variety B with $B \otimes \mathbb{Q} \simeq A$ and $\widehat{T}(B) = k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A)$. The action of L on A induces an action of $L_{\mathbb{Z}}$ on B. There exists a unique effective polarization p of B, with $p \in \bar{p}$, such that $k^{-1}(V'_{\widehat{\mathbb{Z}}})$ is the largest "lattice" $\widehat{T}'(B) \supset \widehat{T}(B)$ satisfying $\psi_p(\widehat{T}(B), \widehat{T}'(B)) \subset \widehat{\mathbb{Z}}(1)$. Finally, the set $\bar{k}$ of L-linear symplectic isomorphisms from $\widehat{T}(B)$ to $V_{\widehat{\mathbb{Z}}}$ is the inverse image of its image $\bar{k}_n$ in $\mathrm{Isom}(B_n, V_{\mathbb{Z}}/nV_{\mathbb{Z}})$.

This again permits the points of ${}_K M_{\mathbb{C}}(G, h_0)$ to be interpreted as corresponding to isomorphism classes of systems $(B, p, \rho, \bar{k}_n)$ consisting of

(a) a polarized abelian variety (B, p), with complex multiplication by $L_{\mathbb{Z}}$, satisfying 4.11 (a) and (b);

(b) a class mod K/K_n, $\bar{k}_n$, of isomorphisms $k_n : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$, which can be lifted to L-linear symplectic isomorphisms $k : \widehat{T}(B) \longrightarrow V_{\widehat{\mathbb{Z}}}$, and which satisfy (4.10.1) and (4.10.2).

4.13. Let F denote the subalgebra of the center of L fixed by $*$. It is a product of totally real fields. There exists a unique alternating F-bilinear form ${}_F\psi$ on V such that $\mathrm{Tr}_{F/\mathbb{Q}}\, {}_F\psi = \psi$.

Let G_1 denote the group of L-linear symplectic similitudes of ${}_F\psi$: $G_1(\mathbb{Q})$ is the set of $g \in \mathrm{GL}_L(V)$ for which there exists $\mu(g) \in F^*$ satisfying

$$ {}_F\psi(gx, gy) = \mu(g)\, {}_F\psi(x, y), \quad \text{i.e.} \tag{4.13.1}$$

$$ \psi(gx, gy) = \psi(\mu(g)x, y). \tag{4.13.2}$$

As above, we verify the

Variant 4.14. Let K be a compact open subgroup of $G_1(\mathbb{A}^f)$. The points of ${}_K M_{\mathbb{C}}(G_1, h_0)$ correspond bijectively to isomorphism classes of abelian varieties up to isogeny A, endowed with

(a) $\rho : L \longrightarrow \mathrm{End}(A)$ as in 4.11,

(b) a weak polarization (relative to F) $\bar{p}$ which induces the given involution of L,

(c) a class modulo K of L-linear symplectic similitudes $k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$ (for the $F \otimes \mathbb{A}^f$-bilinear forms on the two sides),

and such that

(*) conditions analogous to (4.10.1) and (4.10.2) are satisfied.

4.15. One can make 4.11 and 4.14 more precise by showing that ${}_K M_{\mathbb{C}}(G, h)$ is the coarse moduli scheme of the evident functor on schemes of finite type over $\mathbb{C}$.

4.16. Example I (continuation of 1.6, 1.11). Consider the special case of 4.9 in which $L = \mathbb{Q}$ and $\dim(V) = 2g$. Then $G = \mathrm{Gp}(V)$ (the group of symplectic similitudes). For $h_0 : \mathbb{S} \longrightarrow G_{\mathbb{R}}$ of type 1.6, one has $E(\mathrm{Gp}, h_0) = \mathbb{Q}$.

Let $V_{\mathbb{Z}}$, n, and K_n be as in 1.11. For every scheme S, let $F(S)$ be the set of isomorphism classes of principally polarized abelian schemes A/S endowed with a symplectic similitude $k_n : A_n \xrightarrow{\sim} (V_{\mathbb{Z}}/nV_{\mathbb{Z}})_S$. For $n \geq 3$, the classified objects have no nontrivial automorphisms, and the functor F is represented by a $\mathbb{Q}$-scheme ${}_{K_n}M$ [6]. By 1.11, 4.11, and 4.12, there is an isomorphism

$$m_n : {}_{K_n}M(\mathbb{C}) \xrightarrow{\sim} {}_{K_n}M_{\mathbb{C}}(\mathrm{Gp}, h_0).$$

In the language of 3.1, one has

Proposition 4.17. The $\mathbb{Q}$-scheme $M(\mathrm{Gp}, h_0) = \varprojlim_{K_n} {}_{K_n}M$ is a model over $\mathbb{Q} = E(\mathrm{Gp}, h_0)$ of $M_{\mathbb{C}}(\mathrm{Gp}, h_0)$.

4.18. Consider the special case of 4.9 in which V is a cyclic L-module. The groups G and G_1 are then contained in the center of L^*.

Let $\bar{E}$ be an algebraic closure of $E(G,h)$, and let M^+ be the set of isomorphism classes $[A, \rho, k, \bar{p}]$ of objects $(A, \rho, k, \bar{p})$ consisting of

(a) an abelian variety up to isogeny $A/\bar{E}$, endowed with a homogeneous polarization $\bar{p}$ and with $\rho : L \to \mathrm{End}(A)$ such that, with the notation of 4.10.3, for $\ell \in L$ one has

$$\mathrm{Tr}(\rho(\ell), \mathrm{Lie}(A)) = t(\ell) \in E(G,h);$$

(b) an $L \otimes \mathbb{A}^f$-linear isomorphism $k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$.

The abelian variety A is therefore of CM type. We refer to [16] for the proof of the following fundamental theorem.

Theorem 4.19. (Shimura–Taniyama). The Galois group $\mathrm{Gal}(\bar{E}/E(G,h))$ acts on M^+ through its largest abelian quotient $\pi_0(E(G,h)^*(\mathbb{A})/E(G,h)^*(\mathbb{Q}))$. For $\underline{e} \in E(G,h)^*(\mathbb{A})$, with image $\varphi(\underline{e})$ in abelianized Galois and with component $\underline{e}^f$ in $E(G,h)^*(\mathbb{A}^f)$, one has

$$\varphi(\underline{e})\,([A, \rho, k, \bar{p}]) = [A, \rho, r(G,h)(e^f) \cdot k, \bar{p}]. \tag{4.19.1}$$

Let τ be an embedding of $\bar{E}$ into $\mathbb{C}$ extending that of $E(G,h)$, let K be a compact open subgroup of $G(\mathbb{A}^f)$, and let ${}_KM(G,h)(\bar{E})$ be the set of isomorphism classes of objects $(A, \rho, \bar{k}, \bar{p})$ where

(a) A, ρ, and $\bar{p}$ are as above;

(b) $\bar{k}$ is a class modulo K of $L \otimes \mathbb{A}^f$-linear symplectic similitudes $k : \widehat{V}(A) \xrightarrow{\sim} V \otimes \mathbb{A}^f$;

(c) the vector space $H_1(A_{\mathbb{C}}, \mathbb{Q})$, where $A_{\mathbb{C}}$ is defined via τ, endowed with the action of L and with the polarization form given up to scalar, is isomorphic to V.

By 4.19, condition (c) is independent of the choice of τ, so that the finite set ${}_KM(G,h)(\bar{E})$ is endowed with an action of $\mathrm{Gal}(\bar{E}/E(G,h))$; as a Galois set, it is identified with the set of $\bar{E}$-points of a finite étale $E(G,h)$-scheme ${}_KM(G,h)$. By 4.14, ${}_KM(G,h)(\mathbb{C}) = G(\mathbb{R}) \times K\backslash G(\mathbb{A})/G(\mathbb{Q})$, and by 4.19 one has:

Proposition 4.20. The $E(G,h)$-scheme $M(G,h) = \varprojlim_K {}_KM(G,h)$ is a canonical model of $M_{\mathbb{C}}(G,h)$.

Theorem 4.21. The model $M(\mathrm{Gp}, h_0)$ constructed in 4.17 is a canonical model.

Let (V, ψ) be as in 4.16, and let L, $\rho : L \to \mathrm{End}(V)$, G, and $h : \mathbb{S} \to G_{\mathbb{R}}$ be as in 4.18. We have $u : (G,h) \longrightarrow (\mathrm{Gp}, h_0)$. Let $K \subset G(\mathbb{A}^f)$ be such that $u(K) \subset K_n$ (4.16). If $\bar{E}$ is an algebraic closure of $E(G,h)$, define

$$u : {}_KM(G,h)(\bar{E}) \longrightarrow {}_{K_n}M(\mathrm{Gp}, h_0)(\bar{E})$$

by "forgetting the complex multiplication", associating to $[A, \rho, \bar{k}, \bar{p}]$ the triple consisting of

(a) the abelian variety B, equipped with an isomorphism $B \otimes \mathbb{Q} \simeq A$ and such that

$$\widehat{T}(B) = k^{-1}(V_{\widehat{\mathbb{Z}}}) \subset \widehat{V}(A) \qquad (k \in \bar{k});$$

(b) the polarization $\bar{p}$ of B

(c) the isomorphism $k : B_n \xrightarrow{\sim} V_{\mathbb{Z}}/nV_{\mathbb{Z}}$ $(k \in \bar{k})$.

The maps u define a morphism of models

$$u : M(G,h) \longrightarrow M(\mathrm{Gp}, h_0) \otimes E(G,h).$$

We complete the proof by showing ([8]) that for every $u : (H, h') \longrightarrow (\mathrm{Gp}, h_0)$ as in 3.13, there exists $v : (G, h) \longrightarrow (\mathrm{Gp}, h_0)$ as above (for a suitable L) such that $uh' = vh$.

Variant 4.22. Let F be a totally real number field, let $n = [F : \mathbb{Q}]$, and let $\mathrm{Gp}_{2g}(F)$ be the group of symplectic similitudes of a symplectic F-vector space of dimension $2g$. Let $h_0 : \mathbb{S} \longrightarrow \mathrm{Gp}_{2g}(F)_{\mathbb{R}} \simeq \mathrm{Gp}_{2g}(\mathbb{R})^n$ be as in 1.6. One can construct a canonical model of $M_{\mathbb{C}}(\mathrm{Gp}_{2g}(F), h_0)$ from moduli of weakly polarized abelian schemes.

5. Construction techniques.

Definition 3.13 is usable only because there are "many" special points.

Theorem 5.1. Let G and h be as in 3.13. For every finite extension F of $E(G, h)$, there exists $(H, h', u : H \hookrightarrow G)$ as in 3.13 such that the extension $E(H, h')$ of $E(G, h)$ is linearly disjoint from F.

Let $Y = \underline{\mathrm{Hom}}(\mathbb{G}_m, G)/G$ be the scheme of conjugacy classes of morphisms from $\mathbb{G}_m$ to G. The Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ acts on the (infinite discrete) set $Y(\mathbb{C})$ of conjugacy classes of morphisms from $\mathbb{G}_{m\mathbb{C}}$ to $G_{\mathbb{C}}$. Let Y_o be the finite subscheme of Y such that $Y_o(\mathbb{C})$ is the Galois orbit of the class of hr. It is the spectrum of $E(G, h)$.

Let $V' = \mathrm{Lie}(G)$, regarded as an affine space over $\mathbb{Q}$, and let V be the open subscheme of regular elements of the Lie algebra. For v in V, let T_v denote the centralizer of v (a maximal torus). Let W be the moduli scheme of maximal tori T of G, endowed with $s : \mathbb{G}_m \longrightarrow T$ and $v \in \mathrm{Lie}(T)$ such that

(a) v is regular in $\mathrm{Lie}(G)$;

(b) the class of $s : \mathbb{G}_m \to G$ lies in Y_o.

The morphism $f : W \longrightarrow V : (T, s, v) \longmapsto v$ is finite étale and surjective (note that $T = T_v$). Let p be the morphism from W to $Y_o : (T, s, v) \longmapsto$ (the class of s).

$$\begin{tikzcd} W \arrow[r, "f"] \arrow[d, "p"'] & V \arrow[d] \\ \mathrm{Spec}(E(G,h)) \arrow[r] & \mathrm{Spec}(\mathbb{Q}) \end{tikzcd} \tag{5.1.1}$$

Lemma 5.1.2. The scheme W is irreducible, and the geometric fibers of p are (geometrically) irreducible.

Since Y_o is irreducible (the spectrum of an extension of $\mathbb{Q}$), it suffices to prove the second assertion. This follows from the fact that

(a) since G is connected, the scheme over $\mathbb{C}$ of homomorphisms from $\mathbb{G}_{m\mathbb{C}}$ to $G_{\mathbb{C}}$ conjugate to a given homomorphism is irreducible;

(b) for a given $i : \mathbb{G}_{m\mathbb{C}} \to G_{\mathbb{C}}$, by the theorem on conjugacy of maximal tori applied to the (connected) centralizer of $i(\mathbb{G}_{m\mathbb{C}})$, the scheme of maximal tori of $G_{\mathbb{C}}$ containing $i(\mathbb{G}_{m\mathbb{C}})$ is irreducible.

Let $U \subset V(\mathbb{R})$ be the set of $v \in V(\mathbb{R})$ such that $(T_v/C)(\mathbb{R})$ is compact. For the usual topology, U is open.

For $v \in U$ and $w \in W(\mathbb{C})$ such that $f(w) = v \in V(\mathbb{R}) \subset V(\mathbb{C})$, there exists a unique homomorphism $h(w) : \mathbb{S} \longrightarrow T_v$ such that $h(w) \circ r = s_w$: after extension of scalars to $\mathbb{C}$

$$h(w) = s_w \circ z \cdot \overline{s}_w \circ \overline{z}.$$

Let $v \in U$. By the theorem on conjugacy of maximal compact subgroups in $G/C(\mathbb{R})^\circ$, there exists $g \in G(\mathbb{R})$ (indeed one may take $g \in G'(\mathbb{R})^\circ$) such that $gT_vg^{-1} \subset K_\infty$. Since T_v is a maximal torus, gT_vg^{-1} contains the center of K_∞. It follows that there exists $w \in W(\mathbb{C})$ such that $f(w) = v$ and such that $h(w)$ is conjugate to h.

We conclude the proof of 5.1 by applying to diagram (5.1.1) and to U the following variant of Hilbert's irreducibility theorem, whose proof I owe to J.-P. Serre.

Lemma 5.13. Let E be a finite extension of $\mathbb{Q}$, and let there be a commutative diagram

$$\begin{tikzcd} W \arrow[rr, "f"] \arrow[d, "p"'] & & V \arrow[d] \\ \mathrm{Spec}(E) \arrow[rr] & & \mathrm{Spec}(\mathbb{Q}) \end{tikzcd}$$

in which

(i) V is a Zariski open subset of an affine space over $\mathbb{Q}$

(ii) the geometric fibers of p are irreducible, and W is reduced.

(iii) f is quasi-finite and dominant.

Then, for every open set $U \subset V(\mathbb{R})$ (in the usual topology) and every extension F of E, there exists $v \in V(\mathbb{Q}) \cap U$ such that $f^{-1}(v)$ is the spectrum of an extension of E linearly disjoint from F.

Let $W' = W \otimes_E F$. The hypotheses of 4.12 are again satisfied for W'/F and $f' : W' \longrightarrow W \longrightarrow V$, and one must construct $v \in V(\mathbb{Q}) \cap U$ such that $f'^{-1}(v) = f^{-1}(v) \otimes_E F$ is the spectrum of a field. This reduces us to the case $E = F$, which follows from Hilbert's irreducibility theorem (cf. Lang, *Diophantine Geometry*, Chap. VIII).

Proposition 5.2. For $x \in M_{\mathbb{C}}(G,h)$ and K compact open in $G(\mathbb{A}^f)$, the image in ${}_K M_{\mathbb{C}}(G,h)$ of $G(\mathbb{A}^f) \cdot x \subset M_{\mathbb{C}}(G,h)$ is dense.

This is a consequence of (0.4). The reader will verify:

Proposition 5.3. Let E be a field endowed with a complex embedding ρ, X a scheme over E, and Y a reduced closed subscheme of $X_{\mathbb{C}}$. Suppose there exist finite extensions E_i of E in $\mathbb{C}$, schemes $M_{i,\alpha}/E_i$ and E_i-morphisms $v_{i,\alpha} : M_{i,\alpha} \longrightarrow X \otimes_E E_i$, or $v_i : \coprod_\alpha M_{i,\alpha} \longrightarrow X \otimes_E E_i$, such that $E = \bigcap_i E_i$, that $v_i(\coprod_\alpha M_{i,\alpha} \otimes_{E_i} \mathbb{C}) \subset Y$ and that $v_i(\coprod_\alpha M_{i,\alpha} \otimes_{E_i} \mathbb{C})$ is dense in Y. Then Y is defined over E.

Corollary 5.4. Let $u : (G^1, h^1) \longrightarrow (G^2, h^2)$ be as in 1.14. Let $M_E(G^1,h^1)$ and $M_E(G^2,h^2)$ be weakly canonical models of the $M_{\mathbb{C}}(G^i,h^i)$ over a number field E ($E(G^i,h^i) \subset E \subset \mathbb{C}$). Then $v : M_{\mathbb{C}}(G^1,h^1) \longrightarrow M_{\mathbb{C}}(G^2,h^2)$ is defined over E (cf. 3.12)

Let $K^i \subset G^i(\mathbb{A}^f)$ be compact open subgroups such that $v(K^1) \subset K^2$. We prove that $v(K^1,K^2) : {}_{K^1}M_{\mathbb{C}}(G^1,h^1) \longrightarrow {}_{K^2}M_{\mathbb{C}}(G^2,h^2)$ is defined over E. Let $u : (H,h') \to (G^1,h^1)$ be as in 3.13 and set $F = E \cdot E(H,h')$. For $g \in G^1(\mathbb{A}^f)$, one has F-morphisms

$$\begin{aligned} a(g):&\quad M_F(H,h') \longrightarrow M_F(G^1,h^1) \xrightarrow{g} M_F(G^1,h^1) \longrightarrow {}_{K^1}M_F(G^1,h^1), \\ b(g):&\quad M_F(H,h') \longrightarrow M_F(G^2,h^2) \xrightarrow{v(g)} M_F(G^2,h^2) \longrightarrow {}_{K^2}M_F(G^2,h^2), \end{aligned}$$

and, after extension of scalars to $\mathbb{C}$, $v(K^1,K^2) \circ a(g) = b(g)$.

By 5.1 and 5.2, one can apply 5.3 to $X = {}_{K^1}M_E(G^1,h^1) \times {}_{K^2}M_E(G^2,h^2)$, to the graph Y of v, to the $E_i = E \cdot E(H,h')$ for variable (H,h',u), and to $(a(g),b(g)) : M_{E_i}(H,h') \longrightarrow X_{E_i}$: it follows that the graph of v is defined over E.

In the special case $(G^1,h^1) = (G^2,h^2)$ and $u = \mathrm{Id}$, 5.4 says:

Corollary 5.5. Up to (unique) isomorphism, there exists at most one weakly canonical model of $M_{\mathbb{C}}(G,h)$ over E ($E(G,h) \subset E \subset \mathbb{C}$).

Corollary 5.6. If G' is simply connected, the reciprocity law of a weakly canonical model of $M_{\mathbb{C}}(G,h)$ over E ($E(G,h) \subset E \subset \mathbb{C}$) is given by 3.9.1.

The proof is the same as in the special case $v : (G,h) \to (T, vh)$ of 5.4.

Corollary 5.7. Let $v : (G^1,h^1) \hookrightarrow (G^2,h^2)$ be as in 1.15. If $M_{\mathbb{C}}(G^2,h^2)$ admits a weakly canonical model over E, with $E(G^2,h^2) \subset E(G^1,h^1) \subset E \subset \mathbb{C}$, then $M_{\mathbb{C}}(G^1,h^1)$ admits a weakly canonical model over E.

Let K^i be a compact open subgroup of $G^i(\mathbb{A}^f)$. Using 1.15, we are reduced to showing only that if $v(K^1) \subset K^2$ and ${}_{K^1}M_{\mathbb{C}}(G^1,h^1) \hookrightarrow {}_{K^2}M_{\mathbb{C}}(G^2,h^2)$, then the subscheme ${}_{K^1}M_{\mathbb{C}}(G^1,h^1)$ of ${}_{K^2}M_{\mathbb{C}}(G^2,h^2) = {}_{K^2}M_E(G^2,h^2) \otimes_E \mathbb{C}$ is defined over E.

Let $u : (H, h') \to (G^1, h^1)$ be as in 3.13, and for $g \in G^1(\mathbb{A}^f)$ let $a(g)$ be the composite homomorphism defined over $F = E \cdot E(H, h')$:

$$a(g) : M_F(H, h') \longrightarrow M_F(G^1, h^1) \xrightarrow{v(g)} M_F(G^2, h^2) \longrightarrow {}_{K^2}M_F(G^2, h^2).$$

By 5.1 and 5.2, one concludes by applying 5.3 to $X = {}_{K^2}M_E(G^2, h^2)$, to $Y = {}_{K^1}M_{\mathbb{C}}(G^1, h^1)$, to the $E_i = E \cdot E(H, h')$ for variable (H, h', u), and to $a(g) : M_{E_i}(H, h') \longrightarrow X_{E_i}$.

Example 5.8. For (G, h_0) as in 4.9, one constructs, by 5.7 and 4.21, a canonical model $M(G^\circ, h_0) \subset M(\mathrm{Gp}(V, \psi), h_0) \otimes_{\mathbb{Q}} E(G^\circ, h^\circ)$.

Variant 5.9. Let (G_1, h_0) be such that, with the notation of 4.22, there exists $u : (G_1, h_0) \hookrightarrow (\mathrm{Gp}_{2g}^{(F)}, h_0)$ (for example (G_1°, h_0) with (G_1, h_0) as in 4.13). An application of 5.7 yields a canonical model of $M(G_1, h_0)$.

Proposition 5.10. Let (G, h) be as in 3.13, let the $E_i : E(G, h) \subset E_i \subset \mathbb{C}$ be number fields, let E be the intersection of the E_i, and let $M_{E_i}(G, h)$ be a weakly canonical model of $M_{\mathbb{C}}(G, h)$ over E_i. Then there exists a unique model $M_E(G, h)$ of $M_{\mathbb{C}}(G, h)$ over E, such that for every i the model $M_E(G, h) \otimes_E E_i$ over E_i is isomorphic to $M_{E_i}(G, h)$.

In view of the uniqueness assertion, one may suppose the E_i finite in number. Let F be the extension of E in $\mathbb{C}$ generated by them, and let $\mathfrak{E}$ be the set of subextensions of F containing one of the E_i. Using 5.5, one obtains for every $F' \in \mathfrak{E}$ a model $M_{F'}(G, h)$, and for every inclusion $F' \subset F''$ $(F', F'' \in \mathfrak{E})$ an isomorphism of models $\alpha(F'', F') : M_{F'}(G, h) \otimes_{F'} F'' \longrightarrow M_{F''}(G, h)$.

Let $u : (H, h') \to (G, h)$ be as in 3.13, with $E(H, h')$ linearly disjoint from F over $E(G, h)$ (5.1). Put $E_1 = E(H, h') \cdot E$ and $F_1' = E_1 \cdot F' = E(H, h') \cdot F'$ $(F' \in \mathfrak{E})$.

Let $g \in G(\mathbb{A}^f)$ and let $K \subset G(\mathbb{A}^f)$ be a compact open subgroup. Since $M_{F'}(G, h)$ is weakly canonical, one has an F_1'-morphism

$$a'(g) : M_{F_1'}(H, h') \longrightarrow M_{F_1'}(G, h) \xrightarrow{g} M_{F_1'}(G, h) \longrightarrow {}_K M_{F_1'}(G, h).$$

Since $F_1' = E_1 \otimes_E F'$, one may “interpret” this as a morphism of F'-schemes

$$a(g) : M_{E_1}(H, h') \otimes_E F' \longrightarrow {}_K M_{F'}(G, h).$$

The $a(g)$ satisfy $\alpha(F'', F')\, a(g)_{F''} = a(g)$. One concludes by 5.2 and the following lemma, applied to $X^1 = \coprod_{g \in G(\mathbb{A}^f)} M_{E_1}(H, h')$. Its proof is left to the reader as an exercise.

Lemma 5.10.1. Let F/E be a finite extension of fields and let $\mathfrak{E}$ be a set of subfields of F, whose intersection is reduced to E, such that

$$F \in \mathfrak{E} \text{ and } F' \subset F'' \subset F \Longrightarrow F'' \in \mathfrak{E}.$$

(i) The functor that sends a scheme X/E to the system $\mathcal{X}$ of schemes $X_{F'}/F'$ $(F' \in \mathfrak{E})$ and the isomorphisms $(X_{F'}) \otimes_{F'} F'' \xrightarrow{\sim} X_{F''}$ $(F' \subset F'',\ F', F'' \in \mathfrak{E})$ is fully faithful.

(ii) For a system $\mathcal{X} = \{X(F')/F'\ (F' \in \mathfrak{E});\ \alpha(F'', F') : X(F') \otimes_{F'} F'' \xrightarrow{\sim} X(F'')\ (F' \subset F'',\ F', F'' \in \mathfrak{E})\}$ to come from a scheme X/E, it is enough that the $X(F')/F'$ be quasi-projective and that there exist a scheme X^1/E and a morphism $u : X^1_{\mathfrak{E}} \longrightarrow \mathcal{X}$ such that the $u(F') : X^1_{F'} \longrightarrow X(F')$ have dense image.

Remark 5.10.2. Under the hypotheses of 5.10, for $u : (H, h') \hookrightarrow (G, h)$ as in 3.13, the field of definition of $u : M_{\mathbb{C}}(H, h') \longrightarrow M_{\mathbb{C}}(G, h)$ is contained in the extension $\bigcap_i E(H, h') \cdot E_i$ of $E(H, h') \cdot E$.

Proposition 5.11. Let $h : \mathbb{S} \longrightarrow G_{\mathbb{R}}$ be as in 3.13 and let $\delta : \mathbb{S} \longrightarrow C_{\mathbb{R}}^\circ$ (C the center of G). If $E(G, h), E(C, \delta) \subset E \subset \mathbb{C}$ and if $M_{\mathbb{C}}(G, h)$ admits a weakly canonical model over E, then $M_{\mathbb{C}}(G, h.\delta)$ does as well.

Let K be a compact open subgroup of $G(\mathbb{A}^f)$ and let $L = K \cap C^\circ(\mathbb{A}^f)$. Over $\mathbb{C}$, the morphism

$$u : (G \times C^\circ, (h, \delta)) \longrightarrow (G, h\delta) : (g, c) \longmapsto gc$$

induces morphisms

$$u_K : {}_K M_{\mathbb{C}}(G, h) \times {}_L M_{\mathbb{C}}(C^\circ, \delta) \longrightarrow {}_K M_{\mathbb{C}}(G, h\delta).$$

Let d be the morphism $d : C^\circ \longrightarrow G \times C^\circ : c \longmapsto (c, c^{-1})$. The subgroup $d(C^\circ(\mathbb{A}^f))$ of $G \times C^\circ(\mathbb{A}^f)$ acts on $M_{\mathbb{C}}(G, h) \times M_{\mathbb{C}}(C^\circ, \delta)$ through the quotient $\pi_0(C^\circ(\mathbb{A}))$ of $C^\circ(\mathbb{A}^f)$, and u_K induces an isomorphism

$$\bar{u}_K : \big({}_K M_{\mathbb{C}}(G, h) \times {}_L M_{\mathbb{C}}(C^\circ, \delta)\big) / \big(L \backslash \pi_0(C^\circ(\mathbb{A}))\big) \xrightarrow{\sim} {}_K M_{\mathbb{C}}(G, h\delta).$$

Define ${}_KM_E(G,h\delta)$ to be the quotient

$$ {}_KM_E(G,h)\times {}_LM_E(C^\circ,\delta)/\big(L\backslash\pi_0(C^\circ(\mathbb{A}))\big)$$

and one verifies that this gives a weakly canonical model. By construction, one has

$$M_E(G,h)\times_E M_E(C^\circ,\delta)\longrightarrow M_E(G,h\delta). \tag{5.11.1}$$

Conjecture 5.12. For every (G,h_0) as in 3.13, there exists a canonical model of $M_{\mathbb{C}}(G,h_0)$.

Suppose—this is a typical case—that G is $\mathbb{Q}$-simple and adjoint. Let $\widetilde{G}$ be the universal covering of G and let ω be a weight of $\widetilde{G}_{\mathbb{C}}$. If, with the notation of 3.7, the conjugacy class of h_0r is described by integers $(n_i)_{i\in|\Delta|}$, and if $\omega=\sum m_i\alpha_i$, put $\langle\omega,h_0r\rangle=\sum n_im_i$. So far, it has only been possible to attack 5.12 when $\widetilde{G}$ admits a nontrivial representation V such that all irreducible components V' of $V_{\mathbb{C}}$ satisfy:

(∗) as ω runs through the weights of V', $\langle\omega,h_0r\rangle$ takes at most two values.

Let σ be the opposition involution. Condition (∗) also means that the dominant weight ω of V' satisfies $\langle\omega,h_0r\rangle+\langle\sigma\omega,h_0r\rangle=0$ or 1. There is no such representation V if G is exceptional (the noncompact factors of $G_{\mathbb{R}}$ are then of type $E_{6(-14)}$ or $E_{7(-25)}$), or of trialitarian type D_4, nor if G is of type D_n and $G_{\mathbb{R}}$ has noncompact factors of both types $D_n^{\mathbb{H}}$ and $D_n^{\mathbb{R},2}$.

§ 6. Strange models.

6.1. Let F be a totally real number field, and let

$$ {}_F\Sigma:\quad 0\longrightarrow {}_FG'\longrightarrow {}_FG\longrightarrow \mathbb{G}_m\longrightarrow 0 \tag{6.1.1}$$

be a form over F of the exact sequence

$$ {}_F\Sigma_0:\quad 0\longrightarrow \mathrm{Sp}_{2n}\longrightarrow \mathrm{Gp}_{2n}\xrightarrow{\mu} \mathbb{G}_m\longrightarrow 0, \tag{6.1.2}$$

where Gp denotes a group of symplectic similitudes.

Assume that, for every real place τ of F, $\Sigma\otimes_{F,\tau}\mathbb{R}$ is of one of the following types:

(a) ${}_F\Sigma\otimes_{F,\tau}\mathbb{R}$ is isomorphic to ${}_F\Sigma_0\otimes_{F,\tau}\mathbb{R}$, or

(b) $G'\otimes_{F,\tau}\mathbb{R}$ is compact.

Let $\tau_1,\dots,\tau_g$ denote the real places of F, and suppose that the places of type (a) are $\tau_1,\dots,\tau_r$, with $r>0$.

6.2. The group ${}_FG$ may be described as follows:

(a) Choose a quaternion algebra B over F, indefinite at the places τ_i for $i\le r$, and definite for $i>r$. Let $x\mapsto\overline{x}$ denote its canonical involution.

(b) Choose a free B-module V, endowed with a nondegenerate F-bilinear form Φ, symmetric and such that

$$\Phi(bx,y)=\Phi(x,\overline{b}\,y).$$

(c) Assume that, for $i>r$, the form deduced from Φ by extension of scalars through $\tau_i:F\to\mathbb{R}$ is positive definite.

(d) Take ${}_FG$ to be the group of B-linear similitudes of Φ; in other words, the elements $g\in G(\mathbb{Q})$ are the $g\in\mathrm{GL}_B(V)$ for which there exists $\mu(g)\in F^*$ satisfying

$$\Phi(gx,gy)=\mu(g)\Phi(x,y).$$

6.3. Let

$$\Sigma:\quad 0\longrightarrow G'\longrightarrow G\longrightarrow F^*\longrightarrow 0$$

denote the exact sequence of algebraic groups over $\mathbb{Q}$ deduced from ${}_F\Sigma$ by Weil restriction of scalars from F to $\mathbb{Q}$. We have

$$G_{\mathbb{R}}=\prod_\tau {}_FG\otimes_{F,\tau}\mathbb{R}\qquad (\tau \text{ a real place}). \tag{6.3.1}$$

A morphism $h : \mathbb{S} \to G_{\mathbb{R}}$ is defined by its coordinates $h_i : \mathbb{S} \longrightarrow {}_F G \otimes_{F,\tau_i} \mathbb{R}$. By 1.6, there exists a unique conjugacy class of morphisms h such that, for $i \leq r$, h_i is of type 1.6, and for $i > r$, h_i is trivial. Let h_0 be in this class.

The field $E(G, h_0)$ is the totally real subfield of $\mathbb{C}$ generated by the elements $\displaystyle\sum_{i=1}^{r} \tau_i(f)$ for $f \in F$.

Theorem 6.4. With the preceding notation, $M_{\mathbb{C}}(G, h_0)$ admits a canonical model $M(G, h_0)$.

Let Z be a totally imaginary quadratic extension of F. Let $V' = V \otimes_F Z$, $L = B \otimes_F Z$, and let G' be the algebraic subgroup of $\mathrm{GL}_L(V')$ generated by G and Z^*:

$$0 \longrightarrow F^* \xrightarrow{(f, f^{-1})} G \times Z^* \xrightarrow{q} G' \longrightarrow 0. \tag{6.4.1}$$

Let $x \mapsto \widetilde{x}$ be the involution of L given by the tensor product of the involutions of B and Z. Up to a factor in F, there exists on Z a unique alternating F-bilinear form ψ_Z. If ${}_F\psi$ is the alternating F-bilinear form on V' which is the tensor product of Φ and ψ_Z, then

$${}_F\psi(\ell x, y) = {}_F\psi(x, \widetilde{\ell}\, y).$$

Put $\psi(x, y) = \mathrm{Tr}_{F/\mathbb{Q}}\big({}_F\psi(x, y)\big)$.

Choose, for every real place of F, a complex embedding of Z inducing it. Then $Z^*(\mathbb{R}) \simeq \displaystyle\prod_{i=1}^{g} \mathbb{C}^*$. Let h_Z be the homomorphism from $\mathbb{S}$ into $Z^*_{\mathbb{R}}$ with coordinates $h_i : \mathbb{S}(\mathbb{R}) = \mathbb{C}^* \longrightarrow \mathbb{C}^*$ $(1 \leq i \leq g)$ equal to $(z \mapsto 1)$ for $i \leq r$, and to $(z \mapsto z^{-1})$ for $i > r$.

For every $h : \mathbb{S} \to G_{\mathbb{R}}$ as in 6.3, put $h' = q \circ (h \times h_Z) : \mathbb{S} \longrightarrow G'_{\mathbb{R}}$.

Lemma 6.4.2. There exists $b \in L$ such that, for a suitable choice of ψ_Z, the form

$$\psi\big(x, b\, h'_0(i) y\big) \qquad \text{on} \quad V' \otimes_{\mathbb{Q}} \mathbb{R}$$

is symmetric and positive definite.

This is verified after extension of scalars from $\mathbb{Q}$ to $\mathbb{R}$.

This lemma and 5.9 make it possible to construct a canonical model of $M(G', h'_0)$ (since G' is contained in the group of symplectic similitudes of ${}_F\psi(x, by)$). Applying 5.11 and 5.7, one obtains a weakly canonical model of $M_{\mathbb{C}}(G, h_0)$ over the extension $E(Z^*, h_Z)$ of $E(G, h_0)$. The conclusion follows from 5.10, 5.10.2 and the following lemma, which can be deduced from 5.13.

Lemma 6.5. Let F be a totally real number field, endowed with a set S of real embeddings. Let F^* be the number field corresponding to the subgroup of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ which stabilizes S. For Z a totally imaginary quadratic extension of F, endowed with a set T of complex embeddings, one above each element of S, let likewise $Z^* \supset F^*$ be the field of invariants of the stabilizer of T. Then, for every extension E of F^*, there exist Z and T such that Z^* is linearly disjoint from E.

Remark 6.6. The canonical map 1.13, 1.14

$$M_{\mathbb{C}}(G, h_0) \times M_{\mathbb{C}}(Z^*, h_Z) \longrightarrow M_{\mathbb{C}}(G', h'_0) \tag{6.6.1}$$

is interpreted as follows. Each conjugate of h defines on $V_{\mathbb{C}}$ a Hodge bigrading invariant under F:

$$V_{\mathbb{C}} = V^{-1,0} \oplus V^{0,-1} \oplus V^{0,0} \qquad (\overline{V^{pq}} = V^{qp}).$$

One should beware that the grading (by weight) of $V_{\mathbb{C}}$ by the $V^n = \displaystyle\sum_{p+q=n} V^{pq}$ is not defined over $\mathbb{Q}$ if $r \neq g$.

Similarly, $Z_{\mathbb{C}}$ is bigraded:

$$Z_{\mathbb{C}} = Z^{-1,0} \oplus Z^{0,-1} \oplus Z^{0,0}.$$

The miracle is that the $\mathbb{Q}$-vector space $V' = V \otimes_F Z$, endowed with the tensor-product Hodge bigrading $V'_{\mathbb{C}} = V'^{-1,0}_{\mathbb{C}} \oplus V'^{0,-1}_{\mathbb{C}}$ (!), is the H_1 of an abelian variety.

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HODGE THEORY I

by PIERRE DELIGNE

We propose to give a heuristic dictionary between statements in ℓ-adic cohomology and statements in Hodge theory. This dictionary draws in particular on [3] and on Grothendieck's conjectural theory of motives [2]. So far it has mainly served to formulate conjectures in Hodge theory, and has sometimes suggested proofs of them.

Definition 1.1. A mixed Hodge structure H consists of

(a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice");

(b) a finite increasing filtration W on $H_{\mathbb{Q}} = H_{\mathbb{Z}} \otimes \mathbb{Q}$ (the "weight filtration");

(c) a finite decreasing filtration F on $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$ (the "Hodge filtration").

These data are subject to the axiom:

There exists on $\mathrm{Gr}_W(H_{\mathbb{C}})$ a (unique) bigrading by subspaces $H^{p,q}$ such that

(i)
$$\mathrm{Gr}_W^n(H_{\mathbb{C}}) = \bigoplus_{p+q=n} H^{p,q};$$

(ii) the filtration F induces on $\mathrm{Gr}_W(H_{\mathbb{C}})$ the filtration
$$\mathrm{Gr}_W(F)^p = \bigoplus_{p' \geq p} H^{p',q'};$$

(iii) $\overline{H^{p,q}} = H^{q,p}$.

A morphism $f : H \to H'$ is a homomorphism $f_{\mathbb{Z}} : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{Q}} : H_{\mathbb{Q}} \to H'_{\mathbb{Q}}$ and $f_{\mathbb{C}} : H_{\mathbb{C}} \to H'_{\mathbb{C}}$ are respectively compatible with the filtrations W and F.

The Hodge numbers of H are the integers
$$h^{pq} = \dim H^{p,q} = h^{qp}. \tag{1.2}$$

One says that H is pure of weight n if $h^{pq} = 0$ for $p + q \neq n$ (i.e. if $\mathrm{Gr}_W^i(H) = 0$ for $i \neq n$). One also says that H is a Hodge structure of weight n.

The Tate Hodge structure $\mathbb{Z}(1)$ is the Hodge structure of weight -2, purely of type $(-1, -1)$, for which $\mathbb{Z}(1)_{\mathbb{C}} = \mathbb{C}$ and $\mathbb{Z}(1)_{\mathbb{Z}} = 2\pi i\mathbb{Z} = \mathrm{Ker}(\exp : \mathbb{C} \to \mathbb{C}^*) \subset \mathbb{C}$. Put $\mathbb{Z}(n) = \mathbb{Z}(1)^{\otimes n}$.

One can show that mixed Hodge structures form an abelian category. If $f : H \to H'$ is a morphism, then $f_{\mathbb{Q}}$ and $f_{\mathbb{C}}$ are strictly compatible with the filtrations W and F ([1], 2.3.5).

2. Let A be a normal integral domain of finite type over $\mathbb{Z}$, let K be its field of fractions, and let $\overline{K}$ be an algebraic closure of K. Let K_{nr} be the largest subextension of $\overline{K}$ unramified at every prime ideal of A. It is known—or one may take this as the definition—that
$$\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{nr}/K).$$

For every closed point x of $\mathrm{Spec}(A)$, defined by a maximal ideal m_x of A, the residue field $k_x = A/m_x$ is finite; the point x defines a conjugacy class of "Frobenius substitutions" $\varphi_x \in \pi_1(\mathrm{Spec}(A), \overline{K})$. Put $q_x = \#k_x$ and $F_x = \varphi_x^{-1}$.

Let K be a field of finite type over the prime field of characteristic p, let $\overline{K}$ be an algebraic closure of K, let ℓ be a prime number $\neq p$, and let H be a finitely generated $\mathbb{Z}_\ell$-module (or $\mathbb{Q}_\ell$-module) endowed with a continuous action ρ of $\mathrm{Gal}(\overline{K}/K)$. In what follows we shall always suppose that there exists A as above, with ℓ invertible in A, such that ρ factors through

$$\pi_1(\mathrm{Spec}(A), \overline{K}) = \mathrm{Gal}(K_{nr}/K).$$

We shall say that H is pure of weight n if, for every closed point x of a nonempty open subset of $\mathrm{Spec}(A)$, the eigenvalues α of F_x acting on H are algebraic integers all of whose complex conjugates have absolute value

$$|\alpha| = q_x^{n/2}.$$

Principle 2.1. If the Galois module H "comes from algebraic geometry", there exists on $H_{\mathbb{Q}_\ell} = H \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$ a (unique) increasing filtration W (the "weight filtration"), invariant under Galois, such that $\mathrm{Gr}_n^W(H)$ is pure of weight n.

One may think that $\mathrm{Gr}_n^W(H)$ is moreover semisimple.

When resolution of singularities is available, one can often give a conjectural definition of W, whose correctness follows from the Weil conjectures [5] (cf. 6).

Let μ be the subgroup of $\overline{K}^*$ formed by the roots of unity. The Tate module $\mathbb{Z}_\ell(1)$, defined by

$$\mathbb{Z}_\ell(1) = \mathrm{Hom}(\mathbb{Q}_\ell/\mathbb{Z}_\ell, \mu),$$

is pure of weight -2. Put $\mathbb{Z}_\ell(n) = \mathbb{Z}_\ell(1)^{\otimes n}$.

It is trivial that every morphism $f : H \to H'$ is strictly compatible with the weight filtration.

Principle 2.1 agrees with the fact that every extension of $\mathbb{G}_m$ ("weight -2") by an abelian variety ("weight $-1 > -2$") is trivial.

3. Translation. The Galois modules which appear in ℓ-adic cohomology have, over $\mathbb{C}$, mixed Hodge structures as analogues. In addition one has the dictionary

pure module of weight n	Hodge structure of weight n
weight filtration	weight filtration
Galois-equivariant homomorphism	morphism
Tate module $\mathbb{Z}_\ell(1)$	Tate Hodge structure $\mathbb{Z}(1)$.

4. Let X be a complex algebraic variety (= a scheme of finite type over $\mathbb{C}$, which we shall suppose separated). There exists a subfield K of $\mathbb{C}$, of finite type over $\mathbb{Q}$, such that X can be defined over K (i.e. arises by extension of scalars from K to $\mathbb{C}$ from a K-scheme X'). Let $\overline{K}$ be the algebraic closure of K in $\mathbb{C}$. The Galois group $\mathrm{Gal}(\overline{K}/K)$ then acts on the ℓ-adic cohomology groups $H^*(X, \mathbb{Z}_\ell)$; one has

$$H^*(X(\mathbb{C}), \mathbb{Z}) \otimes \mathbb{Z}_\ell = H^*(X, \mathbb{Z}_\ell) = H^*(X'_{\overline{K}}, \mathbb{Z}_\ell).$$

According to 3, one should expect the cohomology groups $H^n(X(\mathbb{C}), \mathbb{Z})$ to carry natural mixed Hodge structures. This can be proved (see [1], 3.2.5, for the case where X is smooth; the proof is algebraic, starting from classical Hodge theory [6]). For X projective and smooth, the Weil conjectures imply that $H^n(X, \mathbb{Z}_\ell)$ is pure of weight n, while classical Hodge theory endows $H^n(X, \mathbb{Z})$ with a Hodge structure of weight n. For every morphism $f : X \to Y$ and for K large enough, $f^* : H^*(Y, \mathbb{Z}_\ell) \to H^*(X, \mathbb{Z}_\ell)$ commutes with Galois (by transport of structure); likewise, $f^* : H^*(Y, \mathbb{Z}) \to H^*(X, \mathbb{Z})$ is a morphism of mixed Hodge structures. For X smooth, the cohomology class in $H^{2n}(X, \mathbb{Z}_\ell(n))$ of an algebraic cycle of codimension n, Z, defined over K, is Galois-invariant, i.e. defines

$$c(Z) \in \mathrm{Hom}_{\mathrm{Gal}}(\mathbb{Z}_\ell(-n), H^{2n}(X, \mathbb{Z}_\ell)).$$

Similarly, the cohomology class $c(Z) \in H^{2n}(X(\mathbb{C}), \mathbb{Z})$ is purely of type (n, n), i.e. corresponds to

$$c(Z) \in \mathrm{Hom}_{\mathrm{H.M.}}(\mathbb{Z}(-n), H^{2n}(X(\mathbb{C}), \mathbb{Z})).$$

5. If $f : H \to H'$ is a Galois-compatible morphism between $\mathbb{Q}_\ell$-vector spaces of different weights, then $f = 0$. Likewise, if $f : H \to H'$ is a morphism of pure mixed Hodge structures of different weights, then f is torsion. A more useful remark is the following.

Scholium 5.1. Let H and H' be Hodge structures of weights n and n', with $n > n'$. Let $f : H_{\mathbb{Q}} \to H'_{\mathbb{Q}}$ be a homomorphism such that $f : H_{\mathbb{C}} \to H'_{\mathbb{C}}$ respects F. Then $f = 0$.

6. Let X be a smooth projective variety over $\mathbb{C}$, let $D = \sum_{i=1}^{n} D_i$ be a normal-crossings divisor in X, a sum of smooth divisors, and let j be the inclusion into X of $U = X - D$. For $Q \subset [1, n]$, put

$$D_Q = \bigcap_{i \in Q} D_i.$$

In ℓ-adic cohomology, one has canonically

$$R^q j_* \mathbb{Z}_\ell = \bigoplus_{\#Q = q} \mathbb{Z}_\ell(-q)_{D_Q}, \tag{6.1}$$

and the Leray spectral sequence for j is

$$E_2^{pq} = \bigoplus_{\#Q = q} H^p(D_Q, \mathbb{Q}_\ell) \otimes \mathbb{Z}_\ell(-q) \Rightarrow H^{p+q}(U, \mathbb{Q}_\ell). \tag{6.2}$$

According to the Weil conjectures [5], $H^p(D_Q, \mathbb{Q}_\ell)$ is pure of weight p, so that E_2^{pq} is pure of weight $p + 2q$. As a quotient of a subobject of E_2^{pq}, E_r^{pq} is also pure of weight $p + 2q$. By 5, $d_r = 0$ for $r \geq 3$, since the weights $p + 2q$ and $p + 2q - r + 2$ of E_r^{pq} and $E_r^{p+r, q-r+1}$ are different. Hence $E_3^{pq} = E_\infty^{pq}$. Up to reindexing, the weight filtration of $H^*(U, \mathbb{Q}_\ell)$ is the abutment of (6.2):

$$\mathrm{Gr}_n^W(H^k(U, \mathbb{Q}_\ell)) = E_3^{2k-n, n-k}. \tag{6.3}$$

7. In integral cohomology, for the usual topology, the Leray spectral sequence for j is

$${}'E_2^{pq} = \bigoplus_{\#Q = q} H^p(D_Q, \mathbb{Z}) \Rightarrow H^{p+q}(U, \mathbb{Z}). \tag{7.1}$$

Since every D_Q is a nonsingular projective variety, ${}'E_2^{pq}$ is endowed with a Hodge structure of weight p. Put $E_2^{pq} = {}'E_2^{pq} \otimes \mathbb{Z}(-q)$ (a Hodge structure of weight $p + 2q$). As an abelian group, $E_2^{pq} = {}'E_2^{pq}$; it is useful to regard (7.1) as a spectral sequence with initial term E_2^{pq}. According to 3, one should expect

$$d_2 : E_2^{pq} \to E_2^{p+2, q-1}$$

to be a morphism of Hodge structures. This is proved by interpreting d_2 as a Gysin morphism. Thus E_3^{pq} is endowed with a Hodge structure of weight $p + 2q$. According to 3, one expects that, modulo torsion, the spectral sequence (6.4)[1] degenerates at E_3 ($E_3 = E_\infty$), and that the vanishing of the d_r ($r \geq 3$) is an application of 5.1. This program is carried out in [1] 3.2. There the weight filtration of $H^*(U, \mathbb{Q})$ is defined as the abutment of (7.1), up to the reindexing (6.3).

In fact, in order to endow cohomology groups H^* with a mixed Hodge structure, the key point up to now has always been to find a spectral sequence E abutting to H^* such that the ℓ-adic analogue of E_2^{pq} is conjecturally pure (of weight $p + 2q$); E_2^{pq} must then carry a natural Hodge structure (of weight $p + 2q$), and the filtration W is the abutment of E.

8. Let $\mathrm{Spec}(V)$ be the spectrum of a henselian discrete valuation ring (a henselian trait), with fraction field K and residue field k of finite type over the prime field of characteristic p. Let $\overline{K}$ be an algebraic closure of K and let H be a finite-dimensional vector space over $\mathbb{Q}_\ell$ ($\ell \neq p$), on which $\mathrm{Gal}(\overline{K}/K)$ acts continuously. According to Grothendieck, it is known ([4], appendix) that a finite-index subgroup of the inertia group I acts unipotently. Replacing V by a finite extension, one reduces to the case in which the action of all of I is unipotent (the semistable case); it then factors through the largest pro-ℓ quotient I_ℓ of I, canonically isomorphic to $\mathbb{Z}_\ell(1)$.

[1] The printed text refers to (6.4); the intended reference appears to be (7.1).

Principle 8.1. In the semistable case, if the Galois module H "comes from algebraic geometry", there exists a (unique) increasing filtration W of H (the "weight filtration") such that I acts trivially on $\mathrm{Gr}_n^W(H)$ and such that $\mathrm{Gr}_n^W(H)$, as a Galois module under $\mathrm{Gal}(\bar{k}/k) \simeq \mathrm{Gal}(\overline{K}/K)/I$, is pure of weight n.

Compare 2.1 and the appendix of [4].

When resolution of singularities is available, one can sometimes give a conjectural definition of W, whose validity follows from the Weil conjectures. With the help of resolution and Weil, it is often easy to show that in any case H has a filtration whose successive quotients are pure Galois modules under $\mathrm{Gal}(\bar{k}/k)$.

Suppose H is semistable. For $T \in I_\ell$, define $\log T$ as the finite sum

$$-\sum_{n>0} (\mathrm{Id} - T)^n/n.$$

The map $(T, x) \mapsto \log T(x)$ is identified with a homomorphism

$$M : \mathbb{Z}_\ell(1) \otimes H \longrightarrow H. \tag{8.2}$$

Since $\mathbb{Z}_\ell(1)$ has weight -2, one necessarily has (cf. 5)

$$M(\mathbb{Z}_\ell(1) \otimes W_n(H)) \subset W_{n-2}(H) \tag{8.3}$$

and M induces

$$\mathrm{Gr}(M) : \mathbb{Z}_\ell(1) \otimes \mathrm{Gr}_n^W(H) \longrightarrow \mathrm{Gr}_{n-2}^W(H). \tag{8.4}$$

8.5. If X is a nonsingular projective variety over an algebraically closed field k_0, define

$$L : \mathbb{Z}_\ell(-1) \otimes H^*(X, \mathbb{Z}_\ell) \longrightarrow H^*(X, \mathbb{Z}_\ell)$$

to be cup product with the cohomology class of a hyperplane section. One will note a formal analogy between L and M: just as M is defined by an action of $\mathbb{Z}_\ell(1)$, one may regard L as defined by an action of $\mathbb{Z}_\ell(-1)$; L raises degree by 2, and $\mathrm{Gr}\, M$ (8.4) lowers it by 2.

9. Let D be the unit disk, $D^* = D - \{0\}$, and let X

$$\begin{tikzcd} X \arrow[r, hook] \arrow[dr, "f"'] & \mathbb{P}^r(\mathbb{C}) \times D \arrow[d, "\mathrm{pr}_2"] \\ & D \end{tikzcd}$$

be a family of projective varieties parametrized by D, with f proper and $f|D^*$ smooth. Keep the notation of 8, and recall that in the analogy between a henselian trait and a small neighborhood of 0 in the complex line, one has the following dictionary (note that the spectrum of the ring of germs of holomorphic functions at 0 is a henselian trait):

D	$\mathrm{Spec}(V)$
D^*	$\mathrm{Spec}(K)$
a universal covering $\widetilde{D}^*$ of D^*	$\mathrm{Spec}(\overline{K})$
fundamental group $\pi_1(D^*)$	inertia group I
(with $\pi_1(D^*) = \mathbb{Z} \simeq \mathbb{Z}(1)_{\mathbb{Z}}$)	(with $I_\ell = \mathbb{Z}_\ell(1)$)
X	projective scheme X over $\mathrm{Spec}(V)$
$X^* = f^{-1}(D^*)$	X_K
$\widetilde{X} = X \times_D \widetilde{D}^*$	$X_{\overline{K}}$
local system $R^i f_* \mathbb{Z} \vert D^*$	Galois module $H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$
$H^i(\widetilde{X}, \mathbb{Z})$	$H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$.

(9.1)

Note that $\widetilde{X}$ is homotopy equivalent to each of the fibers $X_t = f^{-1}(t)$ $(t \in D^*)$: $H^i(X_{\overline{K}}, \mathbb{Z}_\ell)$ again has $H^i(X_t, \mathbb{Z})$ as analogue, and the action of I corresponds to the monodromy transformation T.

Here again, it is known that a finite-index subgroup of $\pi_1(D^*)$ acts unipotently on $H^i(\widetilde{X}, \mathbb{Q}) = H^i(X_t, \mathbb{Q})$. Assume we are in the semistable case where all of $\pi_1(D^*)$ acts unipotently (this amounts to replacing D by a finite covering), and let T be the action of the canonical generator of $\pi_1(D^*)$.

By 3 and 8, one expects $H^i(\widetilde{X}, \mathbb{Q}) \simeq H^i(X_t, \mathbb{Q})$ to be endowed with an increasing filtration W, $\mathrm{Gr}_n^W(H^i(\widetilde{X}, \mathbb{Q}))$ to be endowed with a Hodge structure of weight n, $\log T(W_n) \subset W_{n-2}$, and $\log T$ to induce a morphism of Hodge structures

$$M_n : \mathbb{Z}(-1) \otimes \mathrm{Gr}_n^W(H^i) \longrightarrow \mathrm{Gr}_{n-2}^W(H^i).$$

One would moreover like (8.2), and not only (8.3) and (8.4), to have an analogue.

In fact, for each vector u in the tangent space to D at 0, one can define a mixed Hodge structure $\mathcal{H}_u$ on $H^i(\widetilde{X}, \mathbb{Z})$. The filtration W and the Hodge structures on the $\mathrm{Gr}_n^W(H^i)$ are independent of u, and the dependence of $\mathcal{H}_u$ on u is expressed simply in terms of T. In analogy with (8.2), one finds that, for every u, $\log T$ induces a homomorphism of mixed Hodge structures

$$M : \mathbb{Z}(1) \otimes H^i(\widetilde{X}, \mathbb{Z}) \longrightarrow H^i(\widetilde{X}, \mathbb{Z}).$$

Finally, the analogy 8.5 is not misleading (but here the fact that $f|D^*$ is supposed proper and smooth is no doubt essential). One proves that

$$(\log T)^k : \mathrm{Gr}_{n+k}^W(H^n(\widetilde{X}, \mathbb{Q})) \longrightarrow \mathrm{Gr}_{n-k}^W(H^n(\widetilde{X}, \mathbb{Q}))$$

is an isomorphism for every k (cf. [6], IV 6, cor. to th. 5). This characterizes the filtration W. Up to now, an analogue of Hodge's positivity theorem (cf. [6], IV 7, cor. to th. 7) is available only in very special cases. One hopes that the mixed structures $\mathcal{H}_u$ determine, as $t \to 0$, the asymptotic behavior of the family of pure structures $H^i(X_t, \mathbb{Z})$ $(t \in D^*)$.

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HODGE THEORY, II

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CONTENTS

Introduction

According to Hodge, the cohomology space $H^n(X, \mathbb{C})$ of a compact Kähler variety X is endowed with a *Hodge structure* of weight n, that is, with a natural bigrading

$$H^n(X, \mathbb{C}) = \bigoplus_{p+q=n} H^{pq}$$

satisfying $\overline{H^{pq}} = H^{qp}$. We show here that the complex cohomology of a nonsingular algebraic variety, not necessarily compact, is endowed with a somewhat more general kind of structure, which presents $H^n(X, \mathbb{C})$ as a successive extension of Hodge structures of decreasing weights, lying between $2n$ and n, whose Hodge numbers $h^{pq} = \dim H^{pq}$ are zero for $p > n$ or $q > n$.

The reader will find in [14] an exposition of the yoga underlying this construction.

The proof, essentially algebraic, rests on the one hand on Hodge theory and on the other hand on Hironaka's resolution of singularities, which permits one, via a spectral sequence, to *express* the cohomology of a nonsingular quasi-projective algebraic variety in terms of the cohomology of nonsingular projective varieties.

Section 1 contains, besides a few elementary facts about filtrations collected for the reader's convenience, two key results:

a) Theorem (1.2.10), which will be used only through its corollary (2.3.5), giving the fundamental properties of mixed Hodge structures;

b) the *lemma on two filtrations* (1.3.16).

Section 2 recalls Hodge theory and introduces mixed Hodge structures. The heart of the paper is no. 3.2, which defines the mixed Hodge structure on $H^n(X, \mathbb{C})$ and establishes certain degeneracies of spectral sequences.

Section 4 gives various applications, all deduced from (4.1.1) and from the theory of the (K/k)-trace for Hodge structures which follows from it (4.1.2). The principal ones are (4.2.6) and (4.4.15).

[1] Work presented as a doctoral thesis at the University of Orsay.

1. Filtrations

1.1. Filtered objects

(1.1.1) Let $\mathcal{A}$ be an abelian category. We shall have to consider filtrations of type $\mathbb{Z}$, usually finite, on the objects of $\mathcal{A}$.

Definition (1.1.2). A decreasing, respectively increasing, filtration F of an object A of $\mathcal{A}$ is a family $(F^n(A))_{n\in\mathbb{Z}}$, respectively $(F_n(A))_{n\in\mathbb{Z}}$, of subobjects of A, satisfying

$$\forall n, m \qquad n \leq m \Rightarrow F^n(A) \supset F^m(A)$$

(resp. $\forall n, m \quad n \leq m \Rightarrow F_n(A) \subset F_m(A)$).

A filtered object is an object endowed with a filtration. When no confusion is possible, the same letter will often denote filtrations on different objects of $\mathcal{A}$.

If F is a decreasing, respectively increasing, filtration on A, put $F^{\infty}(A) = 0$ and $F^{-\infty}(A) = A$ (resp. $F_{-\infty}(A) = 0$ and $F_{\infty}(A) = A$).

The shifted filtrations of a decreasing filtration W are defined by

$$W[n]^p(A) = W^{n+p}(A).$$

(1.1.3) If F is a decreasing, respectively increasing, filtration of A, then the $F_n(A) = F^{-n}(A)$, respectively the $F^n(A) = F_{-n}(A)$, form an increasing, respectively decreasing, filtration of A. Thus in principle it is enough to consider decreasing filtrations; unless explicitly stated otherwise, the word *filtration* will henceforth mean *decreasing filtration*.

(1.1.4) A filtration F of A is called finite if there exist n and m such that $F^n(A) = A$ and $F^m(A) = 0$.

(1.1.5) A morphism from a filtered object (A, F) to a filtered object (B, F) is a morphism f from A to B satisfying

$$f(F^n(A)) \subset F^n(B) \qquad \text{for } n \in \mathbb{Z}.$$

The filtered objects (respectively, the objects with finite filtrations) of $\mathcal{A}$ form an additive category in which finite inductive and projective limits exist (and hence so do the kernels, cokernels, images, and coimages of a morphism).

A morphism $f : (A, F) \to (B, F)$ is called strict, or strictly compatible with the filtrations, if the canonical arrow from $\mathrm{Coim}(f)$ to $\mathrm{Im}(f)$ is an isomorphism of filtered objects (cf. (1.1.11)).

(1.1.6) Let $^\circ$ be the contravariant identity functor from $\mathcal{A}$ to the dual category $\mathcal{A}^\circ$. If (A, F) is a filtered object of $\mathcal{A}$, the objects $(A/F^{1-n}(A))^\circ$ are identified with subobjects of A°. The filtration on A° dual to F is defined by

$$F^n(A^\circ) = (A/F^{1-n}(A))^\circ.$$

The bidual of (A, F) is identified with (A, F). This construction identifies the dual of the category of filtered objects of $\mathcal{A}$ with the category of filtered objects of $\mathcal{A}^\circ$.

(1.1.7) If (A, F) is a filtered object of $\mathcal{A}$, its associated graded object is the object of $\mathcal{A}^{\mathbb{Z}}$ defined by

$$\mathrm{Gr}^n(A) = F^n(A)/F^{n+1}(A).$$

The convention (1.1.6) is justified by the simple formula

$$\mathrm{Gr}^n(A^\circ) = \mathrm{Gr}^{-n}(A)^\circ,$$

which is verified on the autodual diagram

$$\begin{tikzcd}
A/F^n(A) & A/F^{n+1}(A) \arrow[l] & 0 & \\
& A \arrow[ul] \arrow[u] & \mathrm{Gr}^n(A) \arrow[ul] \arrow[u] & \\
F^{n+1}(A) \arrow[r] \arrow[ur] & F^n(A) \arrow[u] \arrow[ur] & & 0 \arrow[ul]
\end{tikzcd} \tag{1.1.7.1}$$

(1.1.8) Let (A, F) be a filtered object and let $j : X \hookrightarrow A$ be a subobject of A. The filtration induced on X by F is the unique filtration on X such that j is strictly compatible with the filtrations; one has

$$F^n(X) = j^{-1}(F^n(A)) = X \cap F^n(A).$$

Dually, the quotient filtration on A/X, the unique filtration such that $p : A \to A/X$ is strictly compatible with the filtrations, is given by

$$F^n(A/X) = p(F^n(A)) \simeq (X + F^n(A))/X \simeq F^n(A)/(X \cap F^n(A)).$$

Lemma (1.1.9). If X and Y are two subobjects of A, with $X \subset Y$, then on $Y/X \simeq \mathrm{Ker}(A/X \to A/Y)$ the quotient filtration from Y coincides with the filtration induced from A/X. In the diagram

$$\begin{tikzcd}
A/Y & & A/X \arrow[ll] & \\
& A \arrow[ul] \arrow[ur] & & Y/X \arrow[ul] \\
X \arrow[rr] \arrow[ur] & & Y \arrow[uu] \arrow[ul] \arrow[ur] &
\end{tikzcd}$$

the arrows are strict.

(1.1.10) The filtration of (1.1.9) on Y/X will be called the filtration induced by that of A. By (1.1.9), its definition is autodual.

In particular, if $\Sigma : A \xrightarrow{f} B \xrightarrow{g} C$ is a 0-sequence and if B is filtered, then

$$H(\Sigma) = \mathrm{Ker}(g)/\mathrm{Im}(f) = \mathrm{Ker}(\mathrm{Coker}(f) \to \mathrm{Coim}(g))$$

is endowed with a canonical induced filtration.

The reader will verify the following.

Proposition (1.1.11).

(i) Let $f : (A, F) \to (B, F)$ be a morphism of finitely filtered objects. For f to be strict, it is necessary and sufficient that the sequence

$$0 \to \mathrm{Gr}(\mathrm{Ker}(f)) \to \mathrm{Gr}(A) \to \mathrm{Gr}(B) \to \mathrm{Gr}(\mathrm{Coker}(f)) \to 0$$

be exact.

(ii) Let $\Sigma : (A, F) \to (B, F) \to (C, F)$ be a 0-sequence of strict morphisms. Then one has canonically

$$H(\mathrm{Gr}(\Sigma)) \simeq \mathrm{Gr}(H(\Sigma)).$$

In particular, if Σ is exact in $\mathcal{A}$, then $\mathrm{Gr}(\Sigma)$ is exact in $\mathcal{A}^{\mathbb{Z}}$.

In a category of modules, to say that a morphism $f : (A, F) \to (B, F)$ is strict means that every $b \in B$ of filtration $\geq n$ ($b \in F^n(B)$) which is in the image of A is already in the image of $F^n(A)$:

$$f(F^n(A)) = f(A) \cap F^n(B).$$

(1.1.12) If $\otimes : \mathcal{A}_1 \times \cdots \times \mathcal{A}_n \to \mathcal{B}$ is a multiadditive right exact functor, and if A_i is a finitely filtered object of $\mathcal{A}_i$ ($1 \le i \le n$), one defines a filtration on $\bigotimes_{i=1}^n A_i$ by

$$F^k\left(\bigotimes_{i=1}^n A_i\right) = \sum_{\sum k_i = k} \operatorname{Im}\left(\bigotimes_{i=1}^n F^{k_i}(A_i) \to \bigotimes_{i=1}^n A_i\right)$$

(sum of subobjects).

Dually, if H is left exact, put

$$F^k(H(A_i)) = \bigcap_{\sum k_i = k} \operatorname{Ker}\left(H(A_i) \to H(A_i/F^{k_i}(A_i))\right).$$

For exact H, the two definitions are equivalent. These definitions are extended to functors contravariant in certain variables by (1.1.6). In particular, for the left exact functor Hom, one puts

$$F^k(\operatorname{Hom}(A,B)) = \{f : A \to B \mid \forall n,\ f(F^n(A)) \subset F^{n+k}(B)\}.$$

Thus

$$\operatorname{Hom}((A,F),(B,F)) = F^0(\operatorname{Hom}(A,B)).$$

Under the preceding hypotheses, one has evident morphisms

$$\bigotimes_{i=1}^n \operatorname{Gr}(A_i) \to \operatorname{Gr}\left(\bigotimes_{i=1}^n A_i\right)$$

and

$$\operatorname{Gr} H(A_i) \to H(\operatorname{Gr}(A_i)).$$

For exact H, these are mutually inverse isomorphisms.

These constructions are compatible with the composition of functors, in a sense which we leave to the reader not to make explicit.

1.2. Opposite filtrations

(1.2.1) Let A be an object of $\mathcal{A}$ endowed with two filtrations F and G. By definition, $\operatorname{Gr}_F^m(A)$ is a quotient of a subobject of A, and as such is endowed with a filtration induced by G (1.1.10). Passing to the associated graded object, one defines a bigraded object $(\operatorname{Gr}_G^n \operatorname{Gr}_F^m(A))_{n,m\in\mathbb{Z}}$. By a lemma of Zassenhaus, $\operatorname{Gr}_G^n \operatorname{Gr}_F^m(A)$ and $\operatorname{Gr}_F^m \operatorname{Gr}_G^n(A)$ are canonically isomorphic: if one defines the induced filtrations (1.1.10) as quotient filtrations of filtrations induced on a subobject, one has

$$\begin{array}{rcl} \operatorname{Gr}_G^n \operatorname{Gr}_F^m(A) & \approx & (F^m(A)\cap G^n(A))/((F^{m+1}(A)\cap G^n(A)) + (F^m(A)\cap G^{n+1}(A))) \\ & & \| \\ \operatorname{Gr}_F^m \operatorname{Gr}_G^n(A) & \approx & (G^n(A)\cap F^m(A))/((G^{n+1}(A)\cap F^m(A)) + (G^n(A)\cap F^{m+1}(A))). \end{array}$$

(1.2.2) Let H be a third filtration of A. It induces a filtration on $\operatorname{Gr}_F(A)$, and hence on $\operatorname{Gr}_G \operatorname{Gr}_F(A)$. It also induces a filtration on $\operatorname{Gr}_F \operatorname{Gr}_G(A)$. One should note that these filtrations do not in general correspond under the isomorphism (1.2.1). In the expression $\operatorname{Gr}_H \operatorname{Gr}_G \operatorname{Gr}_F(A)$, therefore, G and H play symmetric roles, but F and G do not.

Definition (1.2.3). Two finite filtrations F and $\overline{F}$ on A are called n-opposite if

$$\operatorname{Gr}_F^p \operatorname{Gr}_{\overline{F}}^q(A) = 0 \qquad \text{for } p+q \neq n.$$

(1.2.4) If $A^{p,q}$ is a bigraded object of $\mathcal{A}$ such that:

a) $A^{p,q} = 0$ except for finitely many pairs (p, q);

b) $A^{p,q} = 0$ for $p + q \neq n$,

then one defines two finite n-opposite filtrations of $A = \sum_{p,q} A^{p,q}$ by putting

$$F^p(A) = \sum_{p' \geq p} A^{p',q'}, \tag{1.2.4.1}$$

$$\overline{F}^q(A) = \sum_{q' \geq q} A^{p',q'}. \tag{1.2.4.2}$$

One has

$$\mathrm{Gr}^p_F \mathrm{Gr}^q_{\overline{F}}(A) = A^{p,q}. \tag{1.2.4.3}$$

Conversely:

Proposition (1.2.5).

(i) Let F and $\overline{F}$ be two finite filtrations on A. For F and $\overline{F}$ to be n-opposite, it is necessary and sufficient that

$$\forall p, q \qquad p + q = n + 1 \Rightarrow F^p(A) \oplus \overline{F}^q(A) \xrightarrow{\sim} A.$$

(ii) If F and $\overline{F}$ are n-opposite, and if one puts

$$\begin{cases} A^{p,q} = 0 & \text{for } p + q \neq n, \\ A^{p,q} = F^p(A) \cap \overline{F}^q(A) & \text{for } p + q = n, \end{cases}$$

then A is the direct sum of the $A^{p,q}$, and F and $\overline{F}$ are deduced from the bigrading $A^{p,q}$ of A by the procedure (1.2.4).

Proof. (i) The condition $\mathrm{Gr}^p_F \mathrm{Gr}^q_{\overline{F}}(A) = 0$ for $p + q > n$ means that

$$F^p \cap \overline{F}^q = (F^{p+1} \cap \overline{F}^q) + (F^p \cap \overline{F}^{q+1}) \qquad (p + q > n).$$

By hypothesis, $F^p \cap \overline{F}^q$ is zero for $p + q$ sufficiently large; by descending induction one deduces that the condition $\mathrm{Gr}^p_F \mathrm{Gr}^q_{\overline{F}}(A) = 0$ for $p + q > n$ is equivalent to the condition $F^p(A) \cap \overline{F}^q(A) = 0$ for $p + q > n$. Dually ((1.1.6), (1.1.7), (1.1.10)) the condition $\mathrm{Gr}^p_F \mathrm{Gr}^q_{\overline{F}}(A) = 0$ for $p + q < n$ is equivalent to $A = F^p(A) + \overline{F}^q(A)$ for $(1 - p) + (1 - q) > -n$, that is, for $p + q \leq n + 1$, and (i) follows.

(ii) If F and $\overline{F}$ are n-opposite, we prove by descending induction on p that

$$\bigoplus_{p' \geq p} A^{p',q'} \xrightarrow{\sim} F^p(A). \tag{1.2.5.1}$$

a) For $F^p(A) = 0$, the assertion is evident.

b) The decomposition $A = F^{p+1}(A) \oplus \overline{F}^{n-p}(A)$ induces on $F^p(A) \supset F^{p+1}(A)$ a decomposition

$$F^p(A) = F^{p+1}(A) \oplus (F^p(A) \cap \overline{F}^{n-p}(A)),$$

and the induction concludes the proof. For p sufficiently small, $F^p(A) = A$. By (1.2.5.1), the $A^{p,q}$ therefore form a bigrading of A, and F satisfies (1.2.4.1). That $\overline{F}$ satisfies (1.2.4.2) follows by symmetry.

(1.2.6) The constructions (1.2.4) and (1.2.5) establish quasi-inverse equivalences of categories between objects of $\mathcal{A}$ endowed with two finite n-opposite filtrations and bigraded objects of $\mathcal{A}$ of the type considered in (1.2.4).

Definition (1.2.7). Three finite filtrations W, F, and $\overline{F}$ on an object A of $\mathcal{A}$ are called opposite if

$$\mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^n(A) = 0 \qquad \text{for } p+q+n \neq 0.$$

This condition is symmetric in F and $\overline{F}$. It means that F and $\overline{F}$ induce on $W^n(A)/W^{n+1}(A)$ two $(-n)$-opposite filtrations. Put

$$A^{p,q} = \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_W^{-p-q}(A),$$

whence decompositions (1.2.4), (1.2.5)

$$W^n(A)/W^{n+1}(A) = \bigoplus_{p+q=-n} A^{p,q} \tag{1.2.7.1}$$

which make $\mathrm{Gr}_W(A)$ into a bigraded object.

Lemma (1.2.8). Let W, F, and $\overline{F}$ be three finite opposite filtrations on A, and let σ be a sequence $(p_i, q_i)_{i\geq 0}$ of pairs of integers satisfying:

a) $p_i \leq p_j$ and $q_i \leq q_j$ for $i \geq j$;

b) For $i > 0$, $p_i + q_i = p_0 + q_0 - i + 1$.

Put $p = p_0$, $q = q_0$, $n = -p-q$, and

$$A_\sigma = \left(\sum_{0\leq i}(W^{n+i}(A)\cap F^{p_i}(A))\right) \cap \left(\sum_{0\leq i}(W^{n+i}(A)\cap \overline{F}^{q_i}(A))\right).$$

Then the projection from $W^n(A)$ to $\mathrm{Gr}_W^n(A)$ induces an isomorphism

$$A_\sigma \xrightarrow{\sim} A^{p,q} \subset \mathrm{Gr}_W^n(A).$$

We prove by induction on k the following assertion:
$(*_k)$ The projection from $W^n(A)/W^{n+k}(A)$ to $\mathrm{Gr}_W^n(A)$ induces an isomorphism from

$$(((\sum_{i<k}(W^{n+i}(A)\cap F^{p_i}(A))) + W^{n+k}(A)) \cap ((\sum_{i<k}(W^{n+i}(A)\cap \overline{F}^{q_i}(A))) + W^{n+k}(A)))/W^{n+k}(A)$$

to $A^{p,q} \subset \mathrm{Gr}_W^n(A)$.

For $k=1$, this is exactly the definition of $A^{p,q}$. By (1.2.5)(i), one has

$$F^{p_k}(\mathrm{Gr}_W^{n+k}(A)) \oplus \overline{F}^{q_k}(\mathrm{Gr}_W^{n+k}(A)) \xrightarrow{\sim} \mathrm{Gr}_W^{n+k}(A). \tag{1.2.8.1}$$

Put

$$\begin{aligned}
B &= \sum_{i<k}(W^{n+i}(A)\cap F^{p_i}(A))\\
C &= \sum_{i<k}(W^{n+i}(A)\cap \overline{F}^{q_i}(A))\\
B' &= (W^{n+k}(A)\cap F^{p_k}(A)) + W^{n+k+1}(A)\\
C' &= (W^{n+k}(A)\cap \overline{F}^{q_k}(A)) + W^{n+k+1}(A)\\
D &= W^{n+k}(A)\\
E &= W^{n+k+1}(A).
\end{aligned}$$

Formula (1.2.8.1) becomes

$$B' + C' = D,$$

and

$$B' \cap C' = E.$$

Moreover, since $p_k \leq p_i$ $(i \leq k)$,

$$B \cap D \subset F^{p_k}(A) \cap W^{n+k}(A) \subset B'$$

and since $q_k \leq q_i$ $(i \leq k)$,

$$C \cap D \subset \overline{F}^{q_k}(A) \cap W^{n+k}(A) \subset C'.$$

The assertion $(*_{k+1})$ therefore follows from $(*_k)$ and from the following lemma.

Lemma (1.2.9). Let B, C, B', C', D, and E be subobjects of A. Assume that

$$B' + C' = D, \qquad B' \cap C' = E,$$

$$B \cap D \subset B', \qquad C \cap D \subset C'.$$

Then,

$$((B + B') \cap (C + C'))/E \xrightarrow{\sim} ((B + D) \cap (C + D))/D.$$

To prove surjectivity, one writes

$$\begin{aligned}((B + B') \cap (C + C')) + D &= (((B + B') \cap (C + C')) + B') + C' \\ &= ((B + B') \cap (C + C' + B')) + C' \\ &= (B + B' + C') \cap (C + C' + B') \\ &= (B + D) \cap (C + D).\end{aligned}$$

To prove injectivity, one writes

$$(B + B') \cap (C + C') \cap D = ((B + B') \cap D) \cap ((C + C') \cap D).$$

Since $B' \subset D$, one has

$$(B + B') \cap D = (B \cap D) + B' = B';$$

similarly,

$$(C + C') \cap D = C',$$

and

$$(B + B') \cap (C + C') \cap D = B' \cap C' = E.$$

Finally, the proof of (1.2.8) is completed by observing that (1.2.8) is equivalent to $(*_k)$ for large k.

Theorem (1.2.10). Let $\mathcal{A}$ be an abelian category, and provisionally denote by $\mathcal{A}'$ the category of objects of $\mathcal{A}$ endowed with three opposite filtrations W, F, and $\overline{F}$. The morphisms in $\mathcal{A}'$ are the morphisms in $\mathcal{A}$ compatible with the three filtrations.

(i) $\mathcal{A}'$ is an abelian category.

(ii) The kernel (respectively, the cokernel) of an arrow $f : A \to B$ in $\mathcal{A}'$ is the kernel (respectively, the cokernel) of f in $\mathcal{A}$, endowed with the filtrations induced from those of A (respectively, the quotient filtrations induced from those of B).

(iii) Every morphism $f : A \to B$ in $\mathcal{A}'$ is strictly compatible with the filtrations W, F, and $\overline{F}$; the morphism $\mathrm{Gr}_W(f)$ is compatible with the bigradings of $\mathrm{Gr}_W(A)$ and $\mathrm{Gr}_W(B)$; the morphisms $\mathrm{Gr}_F(f)$ and $\mathrm{Gr}_{\overline{F}}(f)$ are strictly compatible with the filtration induced by W.

(iv) The functors *forget the filtrations*, Gr_W, Gr_F, $\mathrm{Gr}_{\overline{F}}$, and

$$\begin{aligned}\mathrm{Gr}_W \mathrm{Gr}_F \simeq \mathrm{Gr}_F \mathrm{Gr}_W &\simeq \mathrm{Gr}_{\overline{F}} \mathrm{Gr}_F \mathrm{Gr}_W \\ &\simeq \mathrm{Gr}_{\overline{F}} \mathrm{Gr}_W \simeq \mathrm{Gr}_W \mathrm{Gr}_{\overline{F}}\end{aligned}$$

from $\mathcal{A}'$ to $\mathcal{A}$ are exact.

Let $\sigma_0(p,q)$ and $\sigma_1(p,q)$ denote the sequences

$$\sigma_0(p,q) = (p,q), (p,q), (p,q-1), (p,q-2), (p,q-3), \ldots$$

$$\sigma_1(p,q) = (p,q), (p,q), (p-1,q), (p-2,q), (p-3,q), \ldots$$

and, with the notation of (1.2.8), put

$$A_i^{p,q} = A_{\sigma_i(p,q)} \qquad (i = 0, 1).$$

If $f : A \to B$ is compatible with W, F, and $\overline{F}$, then

$$f(A_i^{p,q}) \subset B_i^{p,q} \qquad (i = 0, 1). \tag{1.2.10.1}$$

Assertion (iii) therefore follows from the following lemma.

Lemma (1.2.11). The $A_i^{p,q}$ form a bigrading of A. One has

$$W^n(A) = \sum_{n+p+q\leq 0} A_i^{p,q} \qquad (i=0,1), \tag{1.2.11.1}$$

$$F^p(A) = \sum_{p'\geq p} A_0^{p',q'}, \tag{1.2.11.2}$$

$$\overline{F}^q(A) = \sum_{q'\geq q} A_1^{p',q'}. \tag{1.2.11.3}$$

By symmetry, it is enough to prove the assertions for $i=0$. Put $A_0 = \bigoplus A_0^{p,q}$, and define on A_0 filtrations W and F by the formulae of (1.2.11). The canonical map i from A_0 to A is compatible with the filtrations W and F. Moreover, by (1.2.8), $\mathrm{Gr}_W(i)$ is an isomorphism, and it induces isomorphisms of graded objects

$$\sum_{p+q=n} A_0^{p,q} \xrightarrow{\sim} \mathrm{Gr}_W^{-n}(A) = \sum_{p+q=n} A^{p,q}. \tag{1.2.11.4}$$

The morphism i is therefore an isomorphism, and the $A_0^{p,q}$ form a bigrading of A. Formula (1.2.11.1) then expresses that $\mathrm{Gr}_W(i)$ is an isomorphism. By (1.2.11.4), $\mathrm{Gr}_F\,\mathrm{Gr}_W(i)$ is an isomorphism, hence so are $\mathrm{Gr}_W\,\mathrm{Gr}_F(i)$ and $\mathrm{Gr}_F(i)$. Formula (1.2.11.2) follows.

(1.2.12) We prove (1.2.10). Let $f : A \to B$ be in $\mathcal{A}'$ and endow $K = \mathrm{Ker}(f)$ with the filtrations induced from those of A. By (1.2.11), $\mathrm{Gr}_W(K) \hookrightarrow \mathrm{Gr}_W(A)$; moreover, the filtration F (respectively, $\overline{F}$) of K induces on $\mathrm{Gr}_W(K)$ the inverse-image filtration of the filtration F on $\mathrm{Gr}_W(A)$. Finally, the subobject $\mathrm{Gr}_W(K)$ of $\mathrm{Gr}_W(A)$ is compatible with the bigrading of $\mathrm{Gr}_W(A)$:

$$\mathrm{Gr}_W(K) = \bigoplus_{p,q} (\mathrm{Gr}_W(K) \cap A^{p,q}).$$

It follows that

$$\mathrm{Gr}_F^p\,\mathrm{Gr}_{\overline{F}}^q\,\mathrm{Gr}_W^n(K) \hookrightarrow \mathrm{Gr}_F^p\,\mathrm{Gr}_{\overline{F}}^q\,\mathrm{Gr}_W^n(A);$$

the filtrations W, F, and $\overline{F}$ of K are therefore opposite, and K is a kernel of f in $\mathcal{A}'$. Together with the dual result, this proves (ii).

If f is an arrow of $\mathcal{A}'$, the canonical morphism from $\mathrm{Coim}(f)$ to $\mathrm{Im}(f)$ is an isomorphism in $\mathcal{A}$; by (iii), it is also an isomorphism in $\mathcal{A}'$, which is therefore abelian.

The functor *forget the filtrations* is exact by (ii). The exactness of the other functors in (iv) follows at once from (ii), (iii), and (1.1.11)(i) or (ii).

(1.2.13) Let A be an object of $\mathcal{A}$ endowed with a finite increasing filtration W, and with two finite decreasing filtrations F and $\overline{F}$. The construction (1.1.3) associates to $W_\bullet$ a decreasing filtration $W^\bullet$. We shall say that the filtrations $W_\bullet$, F, and $\overline{F}$ are opposite if the filtrations $W^\bullet$, F, and $\overline{F}$ are opposite, that is, if for every n the filtrations induced by F and $\overline{F}$ on

$$\mathrm{Gr}_n^W(A) = W_n(A)/W_{n-1}(A)$$

are n-opposite.

Theorem (1.2.10) carries over trivially to this variant.

1.3. The lemma on two filtrations

(1.3.1) Let K be a differential complex of objects of $\mathcal{A}$, endowed with a filtration F. This filtration is called biregular if it induces on each component of K a finite filtration. Recall the definition of the terms $E_r^{pq}(K,F)$, or simply E_r^{pq}, of the spectral sequence defined by F. Put

$$Z_r^{pq} = \mathrm{Ker}\left(d : F^p(K^{p+q}) \to K^{p+q+1}/F^{p+r}(K^{p+q+1})\right),$$

and define B_r^{pq} dually by the formula

$$K^{p+q}/B_r^{pq} = \operatorname{coker}\left(d : F^{p-r+1}(K^{p+q-1}) \to K^{p+q}/F^{p+1}(K^{p+q})\right).$$

These formulae still make sense for $r = \infty$.

One should note that the use made here of the notation B_r^{pq} is different from that of Godement [TF].

By definition one has:

$$E_r^{pq} = \operatorname{Im}\left(Z_r^{pq} \to K^{p+q}/B_r^{pq}\right) \tag{1.3.1.1}$$
$$= Z_r^{pq}/(B_r^{pq} \cap Z_r^{pq}) \tag{1.3.1.2}$$
$$= \operatorname{Ker}\left(K^{p+q}/B_r^{pq} \to K^{p+q}/(Z_r^{pq} + B_r^{pq})\right). \tag{1.3.1.3}$$

One can also write

$$B_r^{p*} \cap Z_r^{p*} \underset{\text{dfn}}{=} (dF^{p-r+1} + F^{p+1}) \cap (d^{-1}F^{p+r} \cap F^p)$$
$$= (dF^{p-r+1} \cap F^p) + (F^{p+1} \cap d^{-1}F^{p+r}), \tag{1.3.1.4}$$

since $dF^{p-r+1} \subset d^{-1}F^{p+r}$ and $F^{p+1} \subset F^p$.

For $r < \infty$, the E_r form a complex graded by the degree $p - r(p+q)$, and E_{r+1} is expressed as the cohomology of this complex:

$$E_{r+1}^{p,q} = H\big(E_r^{p-r,q+r-1} \xrightarrow{d_r} E_r^{p,q} \xrightarrow{d_r} E_r^{p+r,q-r+1}\big). \tag{1.3.1.5}$$

For $r = 0$ one has

$$E_0^{**} = \operatorname{Gr}_F^*(K^*). \tag{1.3.1.6}$$

Proposition (1.3.2). Let K be a complex endowed with a biregular filtration F. The following conditions are equivalent:

(i) the spectral sequence defined by F degenerates ($E_1 = E_\infty$).

(ii) the morphisms $d : K^i \to K^{i+1}$ are strictly compatible with the filtrations.

We check this when $\mathcal{A}$ is a category of modules. For fixed p and q, the hypothesis that the arrows d_r with source $E_r^{p,q}$ are zero for $r \geq 1$ means that, if $x \in F^p(K^{p+q})$ satisfies $dx \in F^{p+1}(K^{p+q+1})$, then there exists y in K^{p+q} such that $dy = 0$ and such that x and y have the same image in $E_1^{p,q}$. Modifying y by a boundary and putting $z = x - y$, one obtains

$$\forall x \in F^p(K^{p+q}) \ \big(dx \in F^{p+1}(K^{p+q+1}) \Rightarrow \exists z \in F^{p+1}(K^{p+q}),\ dz = dx\big)$$

or, in other words,

$$F^{p+1}(K^{p+q+1}) \cap dF^p(K^{p+q}) = dF^{p+1}(K^{p+q}). \tag{1}$$

If this condition holds for all p and q, then by induction on r one has

$$F^{p+r} \cap dF^p = dF^{p+r},$$

which, for $p + r$ large, becomes

$$F^p \cap dK = dF^p. \tag{2}$$

Assertion (2) trivially implies (i), and is equivalent to (ii). This proves (1.3.2).

(1.3.3) If (K, F) is a filtered complex, we shall denote by $\operatorname{Dec}(K)$ the complex K endowed with the shifted filtration

$$\operatorname{Dec}(F)^p K^n = Z_1^{p+n,-p}.$$

This filtration is compatible with the differentials:

$$dZ_1^{p+n,-p} \subset F^{p+n+1}(K^{n+1}) \cap \operatorname{Ker}(d) \subset Z_\infty^{p+n+1,-p} \subset Z_1^{p+n+1,-p}.$$

Since

$$Z_1^{p+1+n,-p-1} \subset F^{p+1+n}(K^n) \subset B_1^{p+n,-p} \subset Z_1^{p+n,-p}, \tag{1.3.3.1}$$

the evident map from $Z_1^{p+n,-p}/Z_1^{p+1+n,-p-1}$ to $Z_1^{p+n,-p}/B_1^{p+n,-p}$ is a morphism

$$u : E_0^{p,n-p}(\operatorname{Dec} K) \longrightarrow E_1^{p+n,-p}(K). \tag{1.3.3.2}$$

Proposition (1.3.4).

(i) The morphisms (1.3.3.2) form a morphism of graded complexes from $E_0(\operatorname{Dec} K)$ to $E_1(K)$.

(ii) This morphism induces an isomorphism on cohomology.

(iii) Step by step, via (1.3.1.5), it induces isomorphisms of graded complexes

$$E_r(\operatorname{Dec}(K)) \xrightarrow{\sim} E_{r+1}(K) \qquad (r \geq 1).$$

Proof. Let F' be the filtration of K defined by

$$F'^p(K^n) = \operatorname{Dec}(F)^{p-n}(K^n) = Z_1^{p,n-p}.$$

There are trivial isomorphisms, compatible with the d_r and with (1.3.1.5),

$$E_r^{p,n-p}(\operatorname{Dec} K) = E_{r+1}^{p+n,-p}(K, F'). \tag{1.3.4.1}$$

The map u is deduced from (1.3.4.1) and from the identity morphism

$$(K, F') \longrightarrow (K, F).$$

This proves (i), and it remains to verify that for $r \geq 2$,

$$E_r^{p,q}(K, F') \simeq E_r^{p,q}(K, F).$$

Indeed one has

$$Z_r^{p,q}(K, F') = Z_r^{p,q}(K, F) \qquad (r \geq 1),$$

and

$$Z_r^{p,q}(K, F') \cap B_r^{p,q}(K, F') = Z_r^{p,q}(K, F) \cap B_r^{p,q}(K, F) \qquad (r \geq 2),$$

and then applies (1.3.1.2).

(1.3.5) The construction (1.3.3) is not self-dual. The dual construction consists in defining

$$\operatorname{Dec}^*(F)^p K^n = B_1^{p+n-1,-p+1}.$$

One then has morphisms

$$E_0^{p,n-p}(\operatorname{Dec} K) \longrightarrow E_1^{p+n,p}(K) \longrightarrow E_0^{p,n-p}(\operatorname{Dec}^* K)$$

and, for $r \geq 1$, isomorphisms

$$E_r^{p,n-p}(\operatorname{Dec} K) \xrightarrow{\sim} E_{r+1}^{p+n,p}(K) \xrightarrow{\sim} E_r^{p,n-p}(\operatorname{Dec}^* K).$$

Recall that a morphism of complexes is called a quasi-isomorphism if it induces an isomorphism on cohomology.

Definition (1.3.6).

(i) A morphism $f : (K, F) \to (K', F')$ of filtered complexes with biregular filtration is a filtered quasi-isomorphism if $\operatorname{Gr}_F(f)$ is a quasi-isomorphism, i.e. if the $E_1^{p,q}(f)$ are isomorphisms.

(ii) A morphism $f : (K, F, W) \to (K', F', W')$ between biregular bifiltered complexes is a bifiltered quasi-isomorphism if $\operatorname{Gr}_F \operatorname{Gr}_W(f)$ is a quasi-isomorphism.

(1.3.7) Let K be a differential complex of objects of $\mathcal{A}$, endowed with two filtrations F and W. Let $E_r^{p,q}$ be the spectral sequence defined by W. The filtration F induces several filtrations on the $E_r^{p,q}$, which we shall compare.

(1.3.8) Formula (1.3.1.2) identifies $E_r^{p,q}$ with a quotient of a subobject of K^{p+q}. Thus the term $E_r^{p,q}$ receives a filtration F_d induced by F, called the first direct filtration.

(1.3.9) Dually, formula (1.3.1.3) identifies $E_r^{p,q}$ with a subobject of a quotient of K^{p+q}, and hence gives a new filtration F_{d*} induced by F, the second direct filtration.

Lemma (1.3.10). On E_0 and E_1 one has $F_d = F_{d*}$. For $r = 0$ or 1, one has $B_r^{p,q} \subset Z_r^{p,q}$, and one applies (1.1.9).

(1.3.11) Formula (1.3.1.5) identifies $E_{r+1}^{p,q}$ with a quotient of a subobject of $E_r^{p,q}$. The recurrent filtration F_r of the $E_r^{p,q}$ is defined by the following conditions:

(i) on $E_0^{p,q}$, $F_r = F_d = F_{d*}$.

(ii) on $E_{r+1}^{p,q}$, the recurrent filtration is the one induced by the recurrent filtration of $E_r^{p,q}$.

(1.3.12) The definitions (1.3.8) and (1.3.9) still make sense for $r = \infty$. If the filtration of K is biregular, the direct filtrations of $E_\infty^{p,q}$ coincide with those of $E_r^{p,q} = E_\infty^{p,q}$ for r large enough, and the recurrent filtration of $E_\infty^{p,q}$ is defined by requiring it to coincide with that of $E_r^{p,q}$ for r large enough.

The filtrations F and W each induce a filtration of $H^*(K)$, and $E_\infty^{**} = \mathrm{Gr}_W^*(H^*(K))$. The filtration F of $H^*(K)$ therefore induces a new filtration on $E_\infty^{p,q}$.

Proposition (1.3.13).

(i) For the first direct filtration, the morphisms d_r are compatible with filtrations. If $E_{r+1}^{p,q}$ is regarded as a quotient of a subobject of $E_r^{p,q}$, then the first direct filtration on $E_{r+1}^{p,q}$ is finer than the filtration F' induced by the first direct filtration on $E_r^{p,q}$; one has $F_d(E_{r+1}^{p,q}) \subset F'(E_{r+1}^{p,q})$.

(ii) Dually, the morphisms d_r are compatible with the second direct filtration, and the second direct filtration on $E_{r+1}^{p,q}$ is less fine than the filtration induced by that of $E_r^{p,q}$.

(iii) $F_d(E_r^{p,q}) \subset F_r(E_r^{p,q}) \subset F_{d*}(E_r^{p,q})$.

(iv) On $E_\infty^{p,q}$, the filtration induced by the filtration F of $H^*(K)$ (1.3.12) is finer than the first direct filtration and less fine than the second.

(i) is evident, (ii) is its dual, and (iii) follows by induction. The first assertion of (iv) is easy to check, and the second is dual to it.

(1.3.14) Denote by $\mathrm{Dec}(K)$, respectively by $\mathrm{Dec}^*(K)$, the complex K endowed with the filtrations $\mathrm{Dec}(W)$ and F, respectively $\mathrm{Dec}^*(W)$ and F.

It is clear from (1.3.4.1) that the isomorphism (1.3.4) transforms the first direct filtration of $E_r(\mathrm{Dec}\,K)$ into the first direct filtration of $E_{r+1}(K)$, for $r \geq 1$. The dual isomorphism (1.3.5) transforms the second direct filtration of $E_r(\mathrm{Dec}^*\,K)$ into the second direct filtration of $E_{r+1}(K)$.

Lemma (1.3.15). If the filtration F is biregular, and if, on the $\mathrm{Gr}_W^p(K)$, the morphisms d are strictly compatible with the filtration induced by F, then:

(i) the morphism (1.3.3.2) of graded complexes filtered by F

$$u : \mathrm{Gr}_{\mathrm{Dec}(W)}(K) \longrightarrow E_1(K, W)$$

is a filtered quasi-isomorphism;

(ii) dually, the morphism (1.3.5)

$$u : E_1(K, W) \longrightarrow \mathrm{Gr}_{\mathrm{Dec}^*(W)}(K)$$

is a filtered quasi-isomorphism.

By duality it is enough to prove (i). By (1.3.3) and (1.3.4), the complex $E_1(K, W)$, filtered by F, is a quotient of the filtered complex $\mathrm{Gr}_{\mathrm{Dec}(W)}(K)$. Let U be the filtered kernel complex; it is acyclic by (1.3.4) (ii). The long exact cohomology sequence associated with the exact sequence of complexes

$$0 \to \mathrm{Gr}_F(U) \to \mathrm{Gr}_F(\mathrm{Gr}_{\mathrm{Dec}(W)}(K)) \to \mathrm{Gr}_F(E_1(K, W)) \to 0$$

shows that u is a filtered quasi-isomorphism if and only if $\mathrm{Gr}_F(U)$ is an acyclic complex. By (1.3.2), and because U is acyclic, this amounts to requiring that the differentials of U be strictly compatible with the filtration F. From (1.3.3.1) one obtains that U is the sum, over p, of the complexes

$$(U^p)^n = B_1^{p+n,-p} / Z_1^{p+1+n,-p-1},$$

endowed with the filtration induced by F.

Each differential d of each of the complexes U^p is included in a commutative diagram of filtered objects of the following type, where, for simplicity, the total or complementary degree has been omitted:

$$\begin{tikzcd}
B^p/Z^{p+1} \arrow[rrr, "d"] \arrow[dr] & & & B^{p+1}/Z^{p+1} \arrow[dd] \\
& \mathrm{Coim}(d) \arrow[r, "5"] & \mathrm{Im}(d) \arrow[ur] \arrow[dr] \arrow[d, phantom, "4"] & \\
W^{p+1}/Z^{p+1} \arrow[uu] \arrow[ur] \arrow[rrr, "3"] & & & B^{p+1}/W^{p+2} \arrow[d] \\
W^{p+1}/W^{p+2} \arrow[u] \arrow[rrr, "1"] \arrow[urrr, phantom, "2"] & & & W^{p+1}/W^{p+2}
\end{tikzcd}$$

By hypothesis, the morphism (1) is strict. Since the square (2) is just the canonical decomposition of (1), the arrow (3) is a filtered isomorphism. The arrows of the trapezium (4) are isomorphisms; hence they are filtered isomorphisms, because (3) is one. That (5) is a filtered isomorphism means that d is strict. This proves (1.3.15).

Theorem (1.3.16). Let K be a complex endowed with two filtrations W and F, with F biregular. Let $r_0 \geq 0$ be an integer, and suppose that, for $0 \leq r < r_0$, the differentials of the graded complex $E_r(K, W)$ are strictly compatible with the filtration F_r. Then, for $r \leq r_0 + 1$, one has

$$F_d = F_r = F_{d*} \quad \text{on } E_r^{p,q}.$$

The theorem is proved by induction on r_0. For $r_0 = 0$ the hypothesis is empty, and one applies (1.3.10) and (1.3.13), (iii). For $r_0 \geq 1$, by the induction hypothesis, one has $F_d = F_r = F_{d*}$ on $E_r^{p,q}$ for $r \leq r_0$.

By (1.3.15), the morphism $u : E_0(\operatorname{Dec} K) \to E_1(K)$ is a filtered quasi-isomorphism. It therefore induces a filtered isomorphism from $H^*(\operatorname{Dec} K)$ to $H^*(E_1(K))$:

$$u : (E_1(\operatorname{Dec} K), F_r) \xrightarrow{\sim} (E_2(K), F_r).$$

Step by step, it follows that the canonical isomorphism from $E_s(\operatorname{Dec} K)$ to $E_{s+1}(K)$ $(s \geq 1)$ is a filtered isomorphism for the recurrent filtration. On $E_1(\operatorname{Dec}(K))$, $F_r = F_d$ (1.3.10), and we already know (1.3.14) that u' is a filtered isomorphism

$$u' : (E_1(\operatorname{Dec} K), F_d) \xrightarrow{\sim} (E_2(K), F_d).$$

Thus on $E_2(K)$ one has $F_d = F_r$. Together with the dual result, this proves (1.3.16) for $r_0 = 1$.

Assume now that $r_0 \geq 2$. Then the arrows d_1 of $E_1(K)$ are strictly compatible with the filtrations; hence so are the arrows d_0 of $E_0(\operatorname{Dec} K)$ (indeed, u induces an isomorphism of spectral sequences, and one applies the criterion (1.3.2)).

For $0 < s < r_0 - 1$, the isomorphism $(E_s(\operatorname{Dec} K), F_r) \simeq (E_{s+1}(K), F_r)$ shows that the d_s are strictly compatible with the recurrent filtrations. By the induction hypothesis one therefore has $F_d = F_r$ on $E_s(\operatorname{Dec} K)$ for $s \leq r_0$. The isomorphism $(E_s(\operatorname{Dec} K), F_d) \simeq (E_{s+1}(K), F_d)$ (1.3.13) then shows that $F_d = F_r$ on $E_r(K)$ for $r \leq r_0 + 1$. Together with the dual result, this proves (1.3.16).

Corollary (1.3.17). Under the general hypotheses of (1.3.16), suppose that, for every r, the differentials d_r are strictly compatible with the recurrent filtrations of the E_r. Then, on E_∞, the filtrations F_d, F_r, and F_{d*} coincide, and coincide with the filtration induced by the filtration F of $H^*(K)$.

This follows at once from (1.3.16) and (1.3.13) (iv).

1.4. Hypercohomology of filtered complexes.

In this section we recall some standard constructions in hypercohomology. We do not use the language of derived categories, which would be the most natural here.

Throughout this section, by a "complex" we shall mean a "complex bounded below".

(1.4.1) Let T be a left exact functor from an abelian category $\mathcal{A}$ to an abelian category $\mathcal{B}$. Assume that every object of $\mathcal{A}$ injects into an injective object; the derived functors $R^iT : \mathcal{A} \to \mathcal{B}$ are therefore defined. An object A of $\mathcal{A}$ will be called acyclic for T if $R^iT(A) = 0$ for $i > 0$.

(1.4.2) Let (A, F) be a filtered object with a finite filtration, and let TF be the filtration of TA by its subobjects TF^pA (these are subobjects because T is left exact). If $\mathrm{Gr}_F(A)$ is T-acyclic, then the $F^p(A)$ are T-acyclic as successive extensions of T-acyclic objects. The image under T of the sequence

$$0 \to F^{p+1}(A) \to F^p(A) \to \mathrm{Gr}^p(A) \to 0$$

is therefore exact, and

$$\mathrm{Gr}_{TF}\, TA \xrightarrow{\sim} T\,\mathrm{Gr}_F\, A. \tag{1.4.2.1}$$

(1.4.3) Let A be an object with two finite filtrations F and W such that $\mathrm{Gr}_F \mathrm{Gr}_W A$ is T-acyclic. The objects $\mathrm{Gr}_F A$ and $\mathrm{Gr}_W A$ are then T-acyclic, as are the $F^q(A) \cap W^p(A)$. The sequences

$$0 \to T(F^q \cap W^{p+1}) \to T(F^q \cap W^p) \to T\big((F^q \cap W^p)/(F^q \cap W^{p+1})\big) \to 0$$

are therefore exact, and $T(F^q(\mathrm{Gr}^p_W(A)))$ is the image in $T(\mathrm{Gr}^p_W(A))$ of $T(F^p \cap W^q)$. The diagram

$$\begin{tikzcd}
T(F^q \cap W^p) \arrow[r] \arrow[d] & T(F^q\,\mathrm{Gr}^p_W A) \arrow[r] & T\,\mathrm{Gr}^p_W A \arrow[d] \\
TF^q \cap TW^p \arrow[rr] & & \mathrm{Gr}^p_{TW} TA
\end{tikzcd}$$

then shows that the isomorphism (1.4.2.1) relative to W carries the filtration $\mathrm{Gr}_{TW}(TF)$ to the filtration $T(\mathrm{Gr}_W(F))$.

(1.4.4) Let K be a complex of objects of $\mathcal{A}$. The *hypercohomology objects* $R^iT(K)$ are computed as follows:

a) Choose a quasi-isomorphism $i : K \to K'$ such that the components of K' are acyclic for T. For example, one may take for K' the simple complex associated with a Cartan–Eilenberg injective resolution of K.

b) Put
$$R^iT(K) = H^i(T(K')).$$

One checks that $R^iT(K)$ does not depend on the choice of K', depends functorially on K, and that a quasi-isomorphism $f : K_1 \to K_2$ induces *isomorphisms*

$$R^iT(f) : R^iT(K_1) \longrightarrow R^iT(K_2).$$

(1.4.5) Let F be a biregular filtration of K. A *filtered T-acyclic resolution* of K is a filtered quasi-isomorphism $i : K \to K'$ from K to a biregular filtered complex such that the $\mathrm{Gr}^p(K'^n)$ are acyclic for T. If K' is such a resolution, the K'^n are acyclic for T and the filtered complex (cf. (1.4.2)) $T(K')$ defines a spectral sequence

$$E_1^{p,q} = R^{p+q}T(\mathrm{Gr}^p K) \Rightarrow R^{p+q}T(K).$$

This is independent of the choice of K'. It is called the *hypercohomology spectral sequence of the filtered complex* K. It depends functorially on K, and a filtered quasi-isomorphism induces an isomorphism of spectral sequences.

The differentials d_1 of this spectral sequence are the connecting morphisms defined by the short exact sequences

$$0 \to \mathrm{Gr}^{p+1} K \to F^pK/F^{p+2}K \to \mathrm{Gr}^p K \to 0.$$

(1.4.6) Let K be a complex. Denote by $\tau_{\leq p}(K)$ the following subcomplex:

$$\tau_{\leq p}(K)^n = \begin{cases} K^n & \text{for } n < p \\ \operatorname{Ker}(d) & \text{for } n = p \\ 0 & \text{for } n > p. \end{cases}$$

The *filtration*, called *canonical*, of K by the $\tau_{\leq p}(K)$ is obtained by shifting the trivial filtration G for which $G^0(K) = K$ and $G^1(K) = 0$. For the canonical filtration one has

$$\begin{array}{ll} E_1^{p,q} = 0 & \text{if } p+q \neq -p \\ H^{-p} & \text{if } p+q = -p. \end{array}$$

A quasi-isomorphism $f : K \to K'$ is automatically a filtered quasi-isomorphism for the canonical filtrations.

(1.4.7) The subcomplexes $\sigma_{\geq p}(K)$ of K:

$$\sigma_{\geq p}(K)^n = \begin{cases} 0 & \text{if } n < p \\ K^n & \text{if } n \geq p \end{cases}$$

define a biregular filtration, the *stupid filtration* of K.

The hypercohomology spectral sequences attached to the stupid or canonical filtrations of K are the two *hypercohomology spectral sequences* of K.

Example (1.4.8). — Let $f : X \to Y$ be a continuous map between topological spaces, and let $\mathcal{F}$ be an abelian sheaf on X. Let $\mathcal{F}^*$ be a resolution of $\mathcal{F}$ by f_*-acyclic sheaves. Then $R^i f_* \mathcal{F} \simeq \mathcal{H}^i(f_* \mathcal{F}^*)$. Take as T the functor $\Gamma(Y, \cdot)$. The hypercohomology spectral sequence of the complex $f_*\mathcal{F}^*$, endowed with its canonical filtration (1.4.6),

$$E_1^{p,q} = H^{2p+q}(Y, R^{-p} f_* \mathcal{F}) \Rightarrow H^{p+q}(X, \mathcal{F})$$

is, up to the renumbering $E_r^{p,q} \mapsto E_{r+1}^{2p+q,-p}$, the *Leray spectral sequence* for f and $\mathcal{F}$.

(1.4.9) Let (K, W, F) be a biregular bifiltered complex. To this complex one associates:

a) A spectral sequence

$$_W E_1^{p,n-p} = H^n(\operatorname{Gr}_W^p(K)) \Rightarrow H^n(K),$$

whose differentials $_W d_1$ are the connecting morphisms deduced from the short exact sequences

$$0 \to \operatorname{Gr}_W^{p+1}(K) \to W^p(K)/W^{p+2}(K) \to \operatorname{Gr}_W^p(K) \to 0;$$

b) An analogous spectral sequence for the filtration F;

c) Exact squares

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \operatorname{Gr}_F^{p+1}\operatorname{Gr}_W^{q+1} K & \longrightarrow & F^p/F^{p+2}(\operatorname{Gr}_W^{q+1} K) & \longrightarrow & \operatorname{Gr}_F^p \operatorname{Gr}_W^{q+1} K & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \operatorname{Gr}_F^{p+1}(W^q/W^{q+2}(K)) & \longrightarrow & F^p/F^{p+2}(W^q/W^{q+2}(K)) & \longrightarrow & \operatorname{Gr}_F^p(W^q/W^{q+2}(K)) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & \operatorname{Gr}_F^{p+1}\operatorname{Gr}_W^q K & \longrightarrow & F^p/F^{p+2}\operatorname{Gr}_W^q K & \longrightarrow & \operatorname{Gr}_F^p\operatorname{Gr}_W^q K & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & 0 & & 0 & & 0 & &
\end{array}$$

The exterior rows and columns of this square define connecting morphisms

$$ {}_{F,W}d_1 : H^n \operatorname{Gr}_F^p \operatorname{Gr}_W^q K \longrightarrow H^{n+1} \operatorname{Gr}_F^{p+1} \operatorname{Gr}_W^q K, $$

$$ {}_{W,F}d_1 : H^n \operatorname{Gr}_F^p \operatorname{Gr}_W^q K \longrightarrow H^{n+1} \operatorname{Gr}_F^{p} \operatorname{Gr}_W^{q+1} K. $$

These morphisms satisfy

$$ {}_{FW}d_1 \circ {}_{WF}d_1 + {}_{WF}d_1 \circ {}_{FW}d_1 = 0 $$
$$ {}_{FW}d_1^2 = 0 \qquad {}_{WF}d_1^2 = 0. $$

The morphisms ${}_{FW}d_1$ are the d_1 morphisms of the spectral sequences $E^{(q)}$, with term $E_1^{(q)p,n-p}$ equal to

$$ E_1^{p,q,n-p-q} \underset{\text{dfn}}{=} H^n(\operatorname{Gr}_F^p \operatorname{Gr}_W^q K) \Rightarrow H^n(\operatorname{Gr}_W^q K) = {}_W E_1^{q,n-q}, \tag{1.4.9.1} $$

defined by the filtered complex $\operatorname{Gr}_W^q(K)$. This spectral sequence abuts to the filtration induced by F on $H^* \operatorname{Gr}_W^q K$. Similarly, the ${}_{WF}d_1$ are the d_1 of spectral sequences with the same initial terms

$$ E_1^{p,q,n-p-q} = H^n(\operatorname{Gr}_F^p \operatorname{Gr}_W^q K) \Rightarrow H^n(\operatorname{Gr}_F^p K). \tag{1.4.9.2} $$

$$ \begin{tikzcd} & H^n \operatorname{Gr}_W^q K \arrow[dr, Rightarrow, "({}_W d_1)"] & \\ H^n \operatorname{Gr}_F^p \operatorname{Gr}_W^q K \arrow[ur, Rightarrow, "({}_{F,W} d_1)"] \arrow[dr, Rightarrow, "({}_{W,F} d_1)"'] & & H^n K \\ & H^n \operatorname{Gr}_F^p K \arrow[ur, Rightarrow, "({}_F d_1)"'] & \end{tikzcd} $$

These constructions are symmetric in F and W, *via* the isomorphism

$$ \operatorname{Gr}_F^p \operatorname{Gr}_W^q \simeq \operatorname{Gr}_W^q \operatorname{Gr}_F^p . $$

(1.4.10) One may also interpret the ${}_{W,F}d_1$ as the initial morphisms of a morphism of spectral sequences (1.4.9.2), abutting to ${}_W d_1$. Indeed, let C^q be the cone of the morphism

$$ W^q(K)/W^{q+2}(K) \longrightarrow \operatorname{Gr}_W^q(K). $$

In the diagram

$$ \Sigma : \operatorname{Gr}_W^{q+1}(K)[1] \xrightarrow{u} C^q \xleftarrow{i} \operatorname{Gr}_W^q(K), $$

u is a quasi-isomorphism, and one has

$$ {}_W d_1 = H(u)^{-1} \circ H(i). $$

In fact, u is even a filtered quasi-isomorphism (for F), and the preceding construction defines a morphism from the spectral sequence defined by $(\operatorname{Gr}_W^q(K), F)$ to the one defined by $(\operatorname{Gr}_W^{q+1}(K)[1], F)$, a morphism which abuts to ${}_W d_1$. The initial term of this morphism, deduced from $\operatorname{Gr}_F(\Sigma)$, is precisely ${}_{W,F}d_1$.

(1.4.11) These constructions pass unchanged to hypercohomology. Let K be a complex endowed with two biregular filtrations F and W.

A *bifiltered T-acyclic resolution* of K is a bifiltered quasi-isomorphism $i : K \to K'$ such that the $\operatorname{Gr}_F^p \operatorname{Gr}_W^n(K'^m)$ are T-acyclic. Such resolutions always exist. In the special case where $\mathcal{A}$ is the category of sheaves of A-modules on a topological space X, and where T is the functor Γ from $\mathcal{A}$ to A-modules, one example of a bifiltered T-acyclic resolution of K is the simple complex associated with the Godement double-resolution complex $\mathcal{C}^*(K)$ of K, filtered by the $\mathcal{C}^*(F^p(K))$ and the $\mathcal{C}^*(W^n(K))$. Since $\mathcal{C}^*$ is exact, one indeed has

$$ \operatorname{Gr}_F \operatorname{Gr}_W(\mathcal{C}^*(K)) \simeq \mathcal{C}^*(\operatorname{Gr}_F \operatorname{Gr}_W(K)). $$

No other case will be needed here.

If K' is a bifiltered T-acyclic resolution of K, the complex TK' is filtered by the TF^pK' and the TW^qK' (1.4.3). Moreover, $\mathrm{Gr}^n_W(K')$ is a filtered T-acyclic resolution (for F) of $\mathrm{Gr}^n_W(K)$, $\mathrm{Gr}^n_F(K')$ is a filtered T-acyclic resolution (for W) of $\mathrm{Gr}^n_F(K)$, $\mathrm{Gr}^n_F \mathrm{Gr}^m_W(K')$ is a T-acyclic resolution of $\mathrm{Gr}^n_F \mathrm{Gr}^m_W(K)$, and

$$T\,\mathrm{Gr}_F K' \approx \mathrm{Gr}_F TK' \qquad \text{(as a } W\text{-filtered complex)}$$
$$T\,\mathrm{Gr}_W K' \approx \mathrm{Gr}_W TK' \qquad \text{(as an } F\text{-filtered complex)}$$
$$T\,\mathrm{Gr}_F \mathrm{Gr}_W K' \approx \mathrm{Gr}_F \mathrm{Gr}_W TK'.$$

Lemma (1.4.12). — Under the hypotheses of (1.4.11):

(i) The initial terms of the hypercohomology spectral sequences

$$_W E_1^{q,n-q} = R^nT(\mathrm{Gr}^q_W K) \Rightarrow R^nT(K) \tag{1}$$

$$_F E_1^{p,n-p} = R^nT(\mathrm{Gr}^p_F K) \Rightarrow R^nT(K) \tag{2}$$

are the abutments of the hypercohomology spectral sequences of the filtered complexes $\mathrm{Gr}^q_W K$ and $\mathrm{Gr}^p_F K$, with E_1 term given by

$$E_1^{p,q,n-p-q} \underset{\mathrm{dfn}}{=} R^nT(\mathrm{Gr}^p_F \mathrm{Gr}^q_W K) \Rightarrow {_W E_1^{q,n-q}} \qquad (q \text{ fixed}) \tag{3}$$

$$E_1^{p,q,n-p-q} \underset{\mathrm{dfn}}{=} R^nT(\mathrm{Gr}^p_F \mathrm{Gr}^q_W K) \Rightarrow {_F E_1^{p,n-p}} \qquad (p \text{ fixed}). \tag{4}$$

(ii) The filtration of $_W E_1^{p,n-p}$, the abutment of the spectral sequence (3), is the filtration of $_W E_1^{p,n-p}(TK')$ induced by the filtration F of TK'.

(iii) The differentials of the complexes $\mathrm{Gr}^n_W(T(K))$ are strictly compatible with the filtration F if and only if the hypercohomology spectral sequences (3) degenerate at E_1.

(iv) The morphisms d_1 of the spectral sequence (4) are the initial terms of degree-one morphisms of spectral sequences (3) abutting to the morphisms d_1 of the spectral sequence (1).

Assertions (i) and (iv) follow from (1.4.9) and (1.4.10) applied to TK', *via* the isomorphisms (1.4.11). Assertion (ii) is then immediate from the definition of the recurrent filtration F (identical with the direct filtrations by (1.3.10) and (1.3.13) (iii)), and assertion (iii) follows from (1.3.2).

2. Hodge structures

2.1. Pure structures.

(2.1.1) In what follows, $\mathbb{C}$ will denote an algebraic closure of $\mathbb{R}$: we do not assume that a root i of the equation $x^2+1=0$ has been chosen. The theory will be invariant under complex conjugation (cf. (2.1.14)).

(2.1.2) Let S denote the real algebraic group $\mathbb{C}^*$, obtained by Weil restriction of scalars from $\mathbb{C}$ to $\mathbb{R}$ of the group $\mathbb{G}_m$:

$$S = \prod_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m.$$
$$S(\mathbb{R}) = \mathbb{C}^*.$$

The group S is a torus, i.e. it is connected and of multiplicative type. It is therefore described by the finitely generated free abelian group

$$X(S) = \mathrm{Hom}(S_{\mathbb{C}}, \mathbb{G}_m) = \mathrm{Hom}(S, \mathbb{G}_m)(\mathbb{C})$$

of its complex characters, endowed with the action of $\mathrm{Gal}(\mathbb{C}/\mathbb{R}) = \mathbb{Z}/(2)$.

The group $X(S)$ has generators z and $\bar{z}$, inducing respectively the identity and complex conjugation:

$$\mathbb{C}^* = S(\mathbb{R}) \longrightarrow S(\mathbb{C}) \longrightarrow \mathbb{G}_m(\mathbb{C}) = \mathbb{C}^*.$$

Complex conjugation exchanges z and $\bar{z}$.

(2.1.3) There is a canonical map

$$w : \mathbb{G}_m \longrightarrow S \tag{2.1.3.1}$$

which, on real points, induces the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$. One has

$$zw = \bar{z}w = \mathrm{Id}\,. \tag{2.1.3.2}$$

There is also a map

$$N : S \longrightarrow \mathbb{G}_m \tag{2.1.3.3}$$

which, on real points, is identified with the norm $N_{\mathbb{C}/\mathbb{R}} : \mathbb{C}^* \to \mathbb{R}^*$. One has

$$N = z\bar{z} \tag{2.1.3.4}$$

and

$$N \circ w = (x \mapsto x^2). \tag{2.1.3.5}$$

Definition (2.1.4). — *A real Hodge structure is a finite-dimensional real vector space V endowed with an action of the real algebraic group S.*

(2.1.5) By the general theory of groups of multiplicative type, giving a real Hodge structure on V is equivalent to giving a bigrading V^{pq} of $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ such that $\overline{V^{pq}} = V^{qp}$. The action of S and the bigrading determine one another *via* the condition:

$$\text{On } V^{pq}, \quad S \text{ acts by multiplication by } z^p \bar{z}^q. \tag{2.1.5.1}$$

(2.1.6) Let $(\mathbb{C}, \times)$ be the multiplicative monoid, and put

$$\overline{S} = \prod_{\mathbb{C}/\mathbb{R}} (\mathbb{C}, \times).$$

If V is a real vector space, giving an action of $\overline{S}$ on V is equivalent to giving on V a bigrading such that $\overline{V^{pq}} = V^{qp}$ and $V^{pq} = 0$ for $p < 0$ or $q < 0$.

(2.1.7) Let V be a real Hodge structure, defined by a representation σ of S and by a bigrading V^{pq}. The grading of $V_{\mathbb{C}}$ by the $V^n_{\mathbb{C}} = \sum_{p+q=n} V^{pq}$ is then defined over $\mathbb{R}$. It is called the *weight grading*. On $V^n = V \cap V^n_{\mathbb{C}}$, the representation σw of $\mathbb{G}_m$ is multiplication by x^n.

We shall say that V is *of weight n* if $V^{pq} = 0$ for $p + q \neq n$, i.e. if σw is multiplication by x^n.

(2.1.8) Let V be a real Hodge structure. The *Hodge filtration* on $V_{\mathbb{C}}$ is defined by

$$F^p(V_{\mathbb{C}}) = \sum_{p' \geq p} V^{p'q'}.$$

By (1.2.6), one has

Proposition (2.1.9). — *Let n be an integer. The construction (2.1.8) establishes an equivalence of categories between:*

a) the category of real Hodge structures of weight n;

b) the category of pairs consisting of a finite-dimensional real vector space V and a filtration F on $V_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{R}} V$ which is n-opposed to its complex conjugate $\overline{F}$.

Definition (2.1.10). — *A Hodge structure H of weight n consists of*

a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice");

b) a real Hodge structure of weight n on $H_{\mathbb{R}} = \mathbb{R} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$.

(2.1.11) A morphism $f : H \to H'$ is a homomorphism $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ such that $f_{\mathbb{R}} : H_{\mathbb{R}} \to H'_{\mathbb{R}}$ is compatible with the action of S (i.e. such that $f_{\mathbb{C}}$ is compatible with the bigrading, or with the Hodge filtration).

Hodge structures of weight n form an abelian category. If H has weight n and H' has weight n', one defines a Hodge structure $H \otimes H'$ of weight $n + n'$ by the formulas:

a) $(H \otimes H')_{\mathbb{Z}} = H_{\mathbb{Z}} \otimes H'_{\mathbb{Z}}$;

b) the action of S on $(H \otimes H')_{\mathbb{R}} = H_{\mathbb{R}} \otimes H'_{\mathbb{R}}$ is the tensor product of the actions of S on $H_{\mathbb{R}}$ and $H'_{\mathbb{R}}$.

The bigrading (respectively, the Hodge filtration) of $(H \otimes H')_{\mathbb{C}} = H_{\mathbb{C}} \otimes H'_{\mathbb{C}}$ is the tensor product of the bigradings (respectively, of the Hodge filtrations (cf. (1.1.12))) of $H_{\mathbb{C}}$ and $H'_{\mathbb{C}}$.

Similarly one defines the Hodge structure $\mathrm{Hom}(H, H')$ (of weight $n' - n$), the Hodge structures $\bigwedge^p H$ (of weight pn), and the dual Hodge structure H^*.

The preceding internal *Hom*, and the group of homomorphisms, are related by the *Remark* **(2.1.11.1).** — $\mathrm{Hom}(H, H')$ is the subgroup of

$$\mathrm{Hom}(H, H')_{\mathbb{Z}} = \mathrm{Hom}_{\mathbb{Z}}(H_{\mathbb{Z}}, H'_{\mathbb{Z}})$$

formed by the elements of type $(0, 0)$.

The actions of S on $H_{\mathbb{R}}$, $H'_{\mathbb{R}}$, and $\mathrm{Hom}(H, H')_{\mathbb{R}} = \mathrm{Hom}(H_{\mathbb{R}}, H'_{\mathbb{R}})$ are indeed related by

$$s(f(x)) = s(f)(s(x)).$$

For f to be of type $(0, 0)$, i.e. invariant under S, therefore means that it commutes with the action of S.

(2.1.12) For A a noetherian subring of $\mathbb{R}$, an *A-Hodge structure of weight n* is defined as consisting of a finite-type A-module H_A and of a real Hodge structure of weight n on $H_{\mathbb{R}} = \mathbb{R} \otimes_A H_A$. These definitions are used mainly for $A = \mathbb{Q}$. An A-Hodge structure consists of a finite-type A-module H_A and of a real Hodge structure on $H_{\mathbb{R}} = H_A \otimes_A \mathbb{R}$, such that the weight grading is defined over the field of fractions of A.

Definition (2.1.13). — *The Tate Hodge structure $\mathbb{Z}(1)$ is the Hodge structure of weight -2, of rank 1, purely of bidegree $(-1, -1)$, with integral lattice $2\pi i\mathbb{Z} \subset \mathbb{C}$.*

The action of S is therefore multiplication by the inverse of the norm (2.1.3.3). For $n \in \mathbb{Z}$, define $\mathbb{Z}(n)$ to be the n-th tensor power of $\mathbb{Z}(1)$: $\mathbb{Z}(n)$ is the Hodge structure of weight $-2n$, rank 1, purely of bidegree $(-n, -n)$, with integral lattice $(2\pi i)^n\mathbb{Z} \subset \mathbb{C}$. The action of S is multiplication by $N(x)^{-n}$.

(2.1.14) The choice in $\mathbb{C}$ of a solution i of the equation $x^2 + 1 = 0$ determines on every complex variety X purely of dimension n an orientation $\mathrm{or}_i(X)$. When i is changed to $-i$, one has

$$\mathrm{or}_{-i}(X) = (-1)^n \, \mathrm{or}_i(X).$$

The choice of i also defines an element C of order 4 in $S(\mathbb{R})$: the image of i under the isomorphism $S(\mathbb{R}) \simeq \mathbb{C}^*$.

Finally it defines an isomorphism between $\mathbb{Z}$ and the integral lattice of $\mathbb{Z}(n)$: multiplication by $(2\pi i)^n$.

Whenever i, an orientation of X, C, or an identification $\mathbb{Z} \simeq \mathbb{Z}(n)_{\mathbb{Z}}$ occurs in a formula, it will in principle be understood that they are subordinated to the same choice of i, and that changing i to $-i$ would give an equivalent definition or formula.

Definition (2.1.15). — *A polarization of a Hodge structure H of weight n is a homomorphism*

$$(x, y) : H \otimes H \longrightarrow \mathbb{Z}(-n)$$

such that the real bilinear form $(2\pi i)^n(x, Cy)$ on $H_{\mathbb{R}}$ is symmetric and positive definite.

(2.1.16) The *real Tate Hodge structure* is the real Hodge structure $\mathbb{R}(1)$ underlying $\mathbb{Z}(1)$. Similarly one defines $\mathbb{R}(n)$ underlying $\mathbb{Z}(n)$. A polarization of a *real* Hodge structure H of weight n is a homomorphism

$$(x, y) : H \otimes H \longrightarrow \mathbb{R}(-n)$$

such that the real bilinear form $(2\pi i)^n(x, Cy)$ on $H_\mathbb{R}$ is symmetric and positive definite. A polarization is entirely determined by the positive-definite quadratic form $(2\pi i)^n(x, Cy)$ on $H_\mathbb{R}$, subject only to the condition that it be invariant under the compact subtorus of S, the kernel of N.

One has
$$(x, y) = (Cx, Cy) = (y, C^2x) = (-1)^n(y, x).$$
The form (x, y) is therefore symmetric or alternating according to the parity of n.

(2.1.17) The reader will generalize these definitions to A-Hodge structures of weight n (2.1.12).

2.2. Hodge theory.

(2.2.1) Let X be a compact Kähler variety, for example smooth projective. By the holomorphic Poincaré lemma, the De Rham complex Ω^*_X is a resolution of the constant sheaf $\mathbb{C}$. Thus one has an isomorphism
$$H^*(X, \mathbb{C}) \sim \mathbb{H}^*(X, \Omega^*_X),$$
and the bête filtration of Ω^*_X (1.4.5) defines the hypercohomology spectral sequence

(2.2.1.1)
$$E_1^{pq} = H^q(X, \Omega^p_X) \Rightarrow H^{p+q}(X, \mathbb{C}),$$
whose abutment is the *Hodge filtration* of $H^*(X, \mathbb{C})$.

By Hodge theory [9], [12], one has:

(A) The spectral sequence (2.2.1.1) degenerates: $E_1 = E_\infty$.

(B) The Hodge filtration of $H^n(X, \mathbb{C})$ is n-opposed to the complex conjugate filtration.

(2.2.2) Let V be a local system of real vector spaces on X, i.e. a sheaf of $\mathbb{R}$-vector spaces locally isomorphic to a constant sheaf $\mathbb{R}^n$. Suppose that on V there exists a bilinear form
$$Q : V \otimes V \longrightarrow \mathbb{R}$$
which is locally constant and *definite*. For X connected, this is the case if V is defined by a representation of a finite quotient of the fundamental group of X.

The assertions (A) and (B) above remain valid exactly as stated, without any change in the proofs, for cohomology with coefficients in $V_\mathbb{C} = V \otimes_\mathbb{R} \mathbb{C}$: the spectral sequence
$$E_1^{pq} = H^q(X, \Omega^p_X(V)) \Rightarrow H^{p+q}(X, V_\mathbb{C})$$
deduced from the De Rham resolution of $V_\mathbb{C}$ by $\Omega^*_X(V_\mathbb{C})$ degenerates and abuts to a filtration on $H^n(X, V_\mathbb{C})$ which is n-opposed to the complex conjugate filtration.

The vector space $H^n(X, V)$ is therefore endowed with a canonical real Hodge structure of weight n.

(2.2.3) It is shown in [2] that the statements (2.2.1) remain valid for X a complete nonsingular algebraic variety, not necessarily Kähler. The proof in *loc. cit.*, based on a reduction to the projective case *via* Chow's lemma and resolution of singularities, extends to the setting (2.2.2).

(2.2.4) Let $\mathcal{L}$ be an invertible sheaf. Here are two ways to define the class $c_1(\mathcal{L})$.

(2.2.4.1) The sheaf $\mathcal{L}$ defines an element c in $H^1(X, \mathcal{O}^*)$. Its image under $df/f : \mathcal{O}^* \to \Omega^1_X$ lies in $H^1(X, \Omega^1_X)$. More precisely, df/f defines a morphism of complexes
$$d\log : \mathcal{O}^*[-1] \longrightarrow [\,0 \to \Omega^1_X \to \Omega^2_X \to \cdots\,] = \sigma_{\geq 1}(\Omega^*_X).$$
This complex maps to Ω^*_X, whence
$$d\log : \mathcal{O}^*[-1] \longrightarrow \Omega^*_X,$$
and the image of c under $d\log$ lies in $H^2(\Omega^*_X)$. This construction would still make sense for an algebraic variety over an arbitrary field k. For $k = \mathbb{C}$, one also has
$$H^2(X, \mathbb{C}) \sim H^2(X, \Omega^*_X)$$
and hence a class
$$c_1'(\mathcal{L}) \in H^2(X, \mathbb{C}).$$

(2.2.4.2) The exponential exact sequence

$$0 \to \mathbb{Z}(1) \to \mathcal{O} \to \mathcal{O}^* \to 0$$

defines a homomorphism

$$\partial : H^1(X, \mathcal{O}^*) \longrightarrow H^2(X, \mathbb{Z}(1)),$$

hence a class

$$\partial c = c_1''(\mathcal{L}) \in H^2(X, \mathbb{Z}(1)).$$

If i is chosen, this class is identified with

$$c_1'''(\mathcal{L}) \in H^2(X, \mathbb{Z}),$$

and for $\mathcal{L} = \mathcal{O}(D)$, $c_1'''(\mathcal{L})$ is just the integral cohomology class defined by D (the orientations being defined by i).

(2.2.5) We prove that:

(2.2.5.1) For α the natural injection of $\mathbb{Z}(1)$ into $\mathbb{C}$, one has

$$-\alpha c_1''(\mathcal{L}) = c_1'(\mathcal{L}).$$

(2.2.5.2) For β the natural injection of $\mathbb{Z}$ into $\mathbb{C}$, one has

$$-\beta c_1'''(\mathcal{L}) = \frac{1}{2\pi i} c_1'(\mathcal{L}).$$

This is verified by contemplating the following diagram of complexes, in which $\simeq$ denotes a quasi-isomorphism, and whose upper rectangle is anticommutative up to homotopy

$$\begin{tikzcd}
\mathbb{C} \arrow[r, "\simeq"] \arrow[d, equal] & \Omega_X^* & \sigma_{\geq 1}\Omega^* \arrow[l] \arrow[d, equal] \\
\mathbb{C} & {[\mathbb{C} \to \mathcal{O}]} \arrow[l] \arrow[r, "\simeq", "d"'] & \sigma_{\geq 1}\Omega^* \\
\mathbb{Z}(1) \arrow[u] & {[\mathbb{Z}(1) \to \mathcal{O}]} \arrow[l] \arrow[u] \arrow[r, "\simeq", "\exp"'] & \mathcal{O}^*[-1] \arrow[u, "d\log"']
\end{tikzcd}$$

(2.2.6) Let X be a nonsingular projective variety purely of dimension n. A choice of i defines an orientation of X and an isomorphism $\mathbb{Z}(1) \simeq \mathbb{Z}$. The corresponding trace morphism:

$$H^{2n}(X, \mathbb{Z}(n)) \longrightarrow \mathbb{Z}$$

which follows from it does not depend on the choice of i.

According to Hodge, for $i \leq n$ the morphism

$$L^{n-i} = \wedge c_1''(\mathcal{O}(1))^{n-i} : H^i(X, \mathbb{Z}) \longrightarrow H^{2n-i}(X, \mathbb{Z}(n-i)),$$

is an isomorphism and, combined with Poincaré duality

$$H^i(X, \mathbb{Z}) \otimes H^{2n-i}(X, \mathbb{Z}(n-i)) \longrightarrow H^{2n}(X, \mathbb{Z}(n-i)) \longrightarrow \mathbb{Z}(-i),$$

it furnishes a polarization on the *primitive part* $\mathrm{Ker}(L^{n-i+1})$ of $H^i(X, \mathbb{Z})$. It follows that the rational Hodge structures $H^i(X, \mathbb{Q})$ are polarizable.

2.3. Mixed structures.

Definition (2.3.1). — *A mixed Hodge structure H consists of:*

a) a finitely generated $\mathbb{Z}$-module $H_{\mathbb{Z}}$ (the "integral lattice").

b) a finite increasing filtration W_n of $H_{\mathbb{Q}} = \mathbb{Q} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$, called the weight filtration.

c) a finite filtration F^p of $H_{\mathbb{C}} = \mathbb{C} \otimes_{\mathbb{Z}} H_{\mathbb{Z}}$, called the Hodge filtration.

It is required that on $H_{\mathbb{C}}$, the filtration $W_{\mathbb{C}}$ deduced from W by extension of scalars, the filtration F, and its complex conjugate $\overline{F}$ form a system $(W_{\mathbb{C}}, F, \overline{F})$ of three opposed filtrations ((1.2.7) and (1.2.13)).

(2.3.2) Let W again denote the filtration of $H_{\mathbb{Z}}$ which is the inverse image of the filtration W of $H_{\mathbb{Q}}$. The axiom of mixed Hodge structures means that, for every n, the filtration F induces on $\mathbb{C} \otimes_{\mathbb{Z}} \mathrm{Gr}_n^W(H_{\mathbb{Z}})$ a filtration n-opposed to its complex conjugate. By (2.1.9), $\mathrm{Gr}_n^W(H_{\mathbb{Z}})$ is endowed with a Hodge structure of weight n, with Hodge filtration induced by F.

Example (2.3.3). — If H is a Hodge structure of weight n, one defines a mixed Hodge structure with the same integral lattice and the same Hodge filtration by putting

$$W_i(H_{\mathbb{Q}}) = 0 \quad \text{for } i < n \qquad \text{and} \qquad W_i(H_{\mathbb{Q}}) = H_{\mathbb{Q}} \quad \text{for } i \geq n.$$

(2.3.4) A *morphism* $f : H \to H'$ of Hodge structures is a homomorphism $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$ compatible with the filtrations W and F (and therefore compatible with $\overline{F}$). One immediately deduces from (1.2.10) the following theorem.

Theorem (2.3.5). —

(i) *The category of mixed Hodge structures is abelian.*

(ii) *The kernel (respectively, cokernel) of a morphism $f : H \to H'$ has as integral lattice the kernel (respectively, cokernel) K of $f : H_{\mathbb{Z}} \to H'_{\mathbb{Z}}$, with $K \otimes \mathbb{Q}$ and $K \otimes \mathbb{C}$ endowed with the induced (respectively, quotient) filtrations from the filtrations W and F of $H_{\mathbb{Q}}$ and $H_{\mathbb{C}}$ (respectively, of $H'_{\mathbb{Q}}$ and $H'_{\mathbb{C}}$).*

(iii) *Every morphism $f : H \to H'$ is strictly compatible with the filtration W of $H_{\mathbb{Q}}$ and $H'_{\mathbb{Q}}$, and with the filtration F of $H_{\mathbb{C}}$ and $H'_{\mathbb{C}}$. It induces morphisms of $\mathbb{Q}$-Hodge structures:*

$$\mathrm{Gr}_n^W(f) : \mathrm{Gr}_n^W(H_{\mathbb{Q}}) \longrightarrow \mathrm{Gr}_n^W(H'_{\mathbb{Q}})$$

and morphisms strictly compatible with the filtration induced by $W_{\mathbb{C}}$:

$$\mathrm{Gr}_F^p(f) : \mathrm{Gr}_F^p(H_{\mathbb{C}}) \longrightarrow \mathrm{Gr}_F^p(H'_{\mathbb{C}}).$$

(iv) *The functor Gr_n^W is an exact functor from the category of mixed Hodge structures to the category of $\mathbb{Q}$-Hodge structures of weight n.*

(v) *The functor Gr_F^p is exact.*

(2.3.6) Let H be a mixed Hodge structure. The $W_n(H_{\mathbb{Z}})$, endowed with the filtrations induced by W and F, then form mixed Hodge substructures $W_n(H)$ of H. The quotient $W_n(H)/W_{n-1}(H)$ is identified with $\mathrm{Gr}_n^W(H_{\mathbb{Z}})$ endowed with its Hodge structure (2.3.2), (2.3.3). This Hodge structure will be denoted by $\mathrm{Gr}_n^W(H)$.

(2.3.7) Put $H^{pq} = \mathrm{Gr}_F^p \mathrm{Gr}_{\overline{F}}^q \mathrm{Gr}_{p+q}^W(H_{\mathbb{C}}) = \big(\mathrm{Gr}_{p+q}^W(H)\big)^{p,q}$. The *Hodge numbers* of H are the integers

$$h^{pq} = \dim_{\mathbb{C}} H^{pq}.$$

Thus the Hodge number h^{pq} of H is the Hodge number h^{pq} of the Hodge structure $\mathrm{Gr}_{p+q}^W(H)$.

(2.3.8) A $\mathbb{Q}$-*mixed Hodge structure* H is defined as consisting of a finite-dimensional vector space $H_{\mathbb{Q}}$ over $\mathbb{Q}$, a finite increasing filtration W of $H_{\mathbb{Q}}$, and a finite decreasing filtration F of $H_{\mathbb{C}}$, the filtrations $W_{\mathbb{C}}$, F, and $\overline{F}$ being opposed. Theorem (2.3.5) generalizes trivially to this variant.

3. Hodge theory of nonsingular algebraic varieties

3.1. Logarithmic poles and residues.

(3.1.1) We recall some classical properties of "logarithmic poles" of holomorphic differential forms. The reader will find proofs in [3], II, (3.1) to (3.7), for example.

(3.1.2) A divisor Y in a smooth complex analytic variety X will be said *to have normal crossings* if the inclusion of Y into X is locally isomorphic to the inclusion of a union of coordinate hyperplanes in $\mathbb{C}^n$; this does not imply that Y is a union of smooth divisors. Let Y be a normal-crossings divisor in X, and let j be the inclusion of $X^* = X - Y$ in X. Let $\Omega^1_X\langle Y\rangle$ denote the locally free $\mathcal{O}$-submodule of $j_*\Omega^1_{X^*}$ generated by Ω^1_X and by the $\frac{dz_i}{z_i}$, where z_i is a local equation of a local irreducible component of Y. The *sheaf* $\Omega^p_X\langle Y\rangle$ *of differential p-forms on* X *with logarithmic pole along* Y is, by definition, the locally free subsheaf $\bigwedge^p \Omega^1_X\langle Y\rangle$ of $j_*\Omega^p_{X^*}$.

Proposition (3.1.3). —

(i) A section α of $j_*\Omega^p_{X^*}$ belongs to $\Omega^p_X\langle Y\rangle$ if and only if α and $d\alpha$ have at worst simple poles along the divisor Y.

(ii) The $\Omega^p_X\langle Y\rangle$ form the smallest subcomplex of $j_*\Omega^*_{X^*}$, stable under exterior product, containing Ω^*_X, and containing the logarithmic differential df/f of every local section, meromorphic along Y, of $j_*\mathcal{O}^*_{X^*}$.

The complex $\Omega^*_X\langle Y\rangle$ is called the *logarithmic De Rham complex* of X along Y. By (3.1.3), (ii), this complex is contravariant in the pair (X, X^*).

(3.1.4) Locally on X, Y is a union of smooth divisors Y_i, and Y^n (resp. $\widetilde{Y}^n$) denotes the union (resp. the disjoint sum) of the intersections n at a time of the Y_i. The Y^n glue to a subspace Y^n of X, and the $\widetilde{Y}^n$ glue to the normalization of Y^n. One has $\widetilde{Y}^0 = Y^0 = X$ and puts $\widetilde{Y} = \widetilde{Y}^1$.

The two-element set of *orientations* of a finite set E with n elements is defined to be the set of generators of $\bigwedge^n \mathbb{Z}^E$. For $n \geq 2$, this set is the set of conjugacy classes, under the alternating group, of total orderings of E.

If to each point y of $\widetilde{Y}^n$ one associates the set of the n local components of Y which contain the image in X of a neighborhood of y in $\widetilde{Y}^n$, then one defines on $\widetilde{Y}^n$ a local system E_n of n-element sets. The local system of orientations of these sets is a torsor under $\mathbb{Z}/(2)$. This torsor defines, *via* the inclusion of $\mathbb{Z}/(2)$ into $\mathbb{C}^*$, a rank-one complex local system ε^n on $\widetilde{Y}^n$, endowed with an isomorphism

$$(\varepsilon^n)^{\otimes 2} \simeq \mathbb{C}.$$

One has

$$\varepsilon^n \simeq \bigwedge^n \mathbb{C}^{E_n}.$$

Locally on $\widetilde{Y}^n$, ε^n is endowed with two opposite isomorphisms $\pm\alpha : \varepsilon^n \xrightarrow{\sim} \mathbb{C}$. Put

$$\varepsilon^n_{\mathbb{Z}} = \alpha^{-1}\big((2\pi i)^{-n}\mathbb{Z}\big).$$

The local system ε^n, endowed with $\varepsilon^n_{\mathbb{Z}}$, should be regarded as a twisted form of $\mathbb{Z}(-n)$ on $\widetilde{Y}^n$.

Let ε^n_X (resp. $(\varepsilon^n_X)_{\mathbb{Z}}$) denote the direct image of ε^n (resp. $\varepsilon^n_{\mathbb{Z}}$) under the map from $\widetilde{Y}^n$ to X. One has

(3.1.4.1) $$\varepsilon^n_X \simeq \bigwedge^n \varepsilon^1_X \qquad (n \geq 0).$$

If Y is a union of distinct smooth divisors $(Y_i)_{i\in I}$, the choice of a total order on I trivializes the ε^n.

(3.1.5) Let $W_n(\Omega^p_X\langle Y\rangle)$ denote the submodule of $\Omega^p_X\langle Y\rangle$ formed by linear combinations of products

$$\alpha \wedge \frac{dt_{i(1)}}{t_{i(1)}} \wedge \cdots \wedge \frac{dt_{i(m)}}{t_{i(m)}} \qquad (m \leq n),$$

with α holomorphic and with the $t_{i(j)}$ local equations of distinct local components Y_j of Y. The *weight filtration* of $\Omega^*_X\langle Y\rangle$ is the *increasing* filtration by the subcomplexes $W_n(\Omega^*_X\langle Y\rangle)$. One has

(3.1.5.1) $$W_n(\Omega^p_X\langle Y\rangle) \wedge W_m(\Omega^q_X\langle Y\rangle) \subset W_{n+m}(\Omega^{p+q}_X\langle Y\rangle).$$

Letting i_n denote the map from $\widetilde{Y}^n$ to X, one checks that the correspondence

$$\alpha \wedge \frac{dt_{i(1)}}{t_{i(1)}} \wedge \cdots \wedge \frac{dt_{i(n)}}{t_{i(n)}} \longmapsto (\alpha \mid Y_{i(1)} \cap \cdots \cap Y_{i(n)}) \otimes (\text{orientation } i(1)\ldots i(n))$$

defines isomorphisms of complexes

(3.1.5.2) $$\mathrm{Res} : \mathrm{Gr}_n^W(\Omega_X^*\langle Y\rangle) \simeq i_{n*}\Omega^*_{\widetilde{Y}^n}(\varepsilon^n)[-n]$$

(the *"Poincaré residue"*).

(3.1.6) The following interpretation, in terms of (3.1.5.2), of the Leray spectral sequence for the inclusion of X^* in X was pointed out to me by N. Katz. It will make it possible to verify a point which I had initially considered evident ((3.2.5), (ii), first part).

(3.1.7) Every point of X admits a fundamental system of Stein open neighborhoods whose trace on X^* is Stein. For $\mathcal{F}$ a coherent analytic sheaf on X^*, one therefore has $R^i j_*\mathcal{F} = 0$ for $i > 0$. The De Rham complex $\Omega^*_{X^*}$ is therefore a resolution of the constant sheaf $\mathbb{C}$ by sheaves acyclic for the functor j_*. Hence

(3.1.7.1) $$H^*(X^*, \mathbb{C}) \xrightarrow{\sim} \mathbb{H}^*(X^*, \Omega^*_{X^*}) \xleftarrow{\sim} \mathbb{H}^*(X, j_*\Omega^*_{X^*})$$

and the Leray spectral sequence for the morphism j is identified with the hypercohomology spectral sequence for $j_*\Omega^*_{X^*}$, corresponding to the filtration τ by the subcomplexes $\tau_{\leq -n}(j_*\Omega^*_{X^*})$ (1.4.6).

Proposition (3.1.8). — *The morphisms of filtered complexes*

$$(\Omega_X^*\langle Y\rangle, W) \xleftarrow{\alpha} (\Omega_X^*\langle Y\rangle, \tau) \xrightarrow{\beta} (j_*\Omega^*_{X^*}, \tau)$$

are filtered quasi-isomorphisms. They define an isomorphism between the Leray spectral sequence for j in complex cohomology and the hypercohomology spectral sequence of the filtered complex $(\Omega_X^*\langle Y\rangle, W)$ on X.

By (1.4.5) and (3.1.7), it suffices to prove the first assertion. In [3], II, (6.9), or in [1], one finds the proof that β is a quasi-isomorphism, hence a filtered quasi-isomorphism. One can also compute directly the cohomology sheaves of the two sides: those of $\Omega_X^*\langle Y\rangle$ are determined by (3.1.5.2), whereas those of $j_*\Omega^*_{X^*}$ are the $R^i j_*\mathbb{C}$, which can be computed topologically.

For $n \geq p$ one has

$$W_n(\Omega_X^p\langle Y\rangle) = \Omega_X^p\langle Y\rangle,$$

so that α is a morphism from $\Omega_X^*\langle Y\rangle$, endowed with τ, to $\Omega_X^*\langle Y\rangle$, endowed with the decreasing filtration associated with W (1.1.3). By (3.1.5.2), one has

(3.1.8.1) $$\mathcal{H}^i(\mathrm{Gr}_n^W(\Omega_X^*\langle Y\rangle)) = \begin{cases} 0 & \text{for } i \neq n \\ \varepsilon_X^n & \text{for } i = n, \end{cases}$$

from the first line of this formula one deduces that α is a filtered quasi-isomorphism. This proves (3.1.8).

By (3.1.7), the isomorphism (3.1.8.1) defines an isomorphism

(3.1.8.2) $$R^n j_*\mathbb{C} \simeq \mathcal{H}^n(j_*\Omega^*_{X^*}) \simeq \mathcal{H}^n(\Omega_X^*\langle Y\rangle) \simeq \varepsilon_X^n.$$

The isomorphisms (3.1.4.1) correspond, *via* (3.1.8.2), to the cup product.

Proposition (3.1.9). — *The canonical morphism from $R^n j_*\mathbb{Z}$ to $R^n j_*\mathbb{C}$ identifies, via (3.1.8.2), the sheaf $R^n j_*\mathbb{Z}$ with $(\varepsilon_X^n)_\mathbb{Z}$ (3.1.4).*

The question is local on X. One can therefore suppose that X is an open polydisc D^m, with $D = \{z \in \mathbb{C} \mid |z| < 1\}$, and that $Y = \bigcup_{k=1}^{\ell} Y_k$, $Y_k = \mathrm{pr}_k^{-1}(0)$. The fiber at 0 of $R^n j_*\mathbb{Z}$ is then the integral cohomology of $X^* = D^{*\ell} \times D^{(m-\ell)}$, with $D^* = \{z \in \mathbb{C} \mid 0 < |z| < 1\}$. The space X^* has the homotopy type of a torus; its cohomology is therefore torsion-free, and the cup product defines isomorphisms

$$\bigwedge^n (R^1 j_*\mathbb{Z})_0 \xrightarrow{\sim} (R^n j_*\mathbb{Z})_0.$$

It is therefore enough to prove (3.1.9) for $n = 1$.

The integral homology $H_1(X^*)$ is generated by the loops γ_k winding around the various Y_k. One has

$$\oint_{\gamma_k} \frac{dz_k}{z_k} = \pm 2\pi i;$$

the integral cohomology is therefore generated by the $\frac{1}{2\pi i}\frac{dz_k}{z_k}$, and this proves (3.1.9).

(3.1.10) Let $\mathcal{F}$ be a coherent analytic sheaf on X^*, given as the restriction to X^* of a coherent analytic sheaf $\mathcal{F}'$ on X. The meromorphic direct image $j_*^m\mathcal{F}$ of $\mathcal{F}$ is the inductive limit

$$j_*^m\mathcal{F} = \varinjlim_n \mathcal{F}'(nY).$$

Locally on X, Y is the sum of a finite family $(Y_i)_{i\in I}$ of smooth divisors, and the *filtration by order of pole* P on $j_*^m\mathcal{O}_{X^*}$ is defined by the formula

(3.1.10.1)
$$P^p(j_*^m\mathcal{O}_{X^*}) = \sum_{\mathbf{n}\in A_p} \mathcal{O}_X\Big(\sum_i (n_i+1)Y_i\Big)$$

where

$$A_p = \{(n_i)_{i\in I} \mid \sum_i n_i \leq -p \ \text{ and } \ \forall i,\ n_i \geq 0\}.$$

This construction globalizes and provides on $j_*^m\mathcal{O}_{X^*}$ an exhaustive filtration such that $P^p = 0$ for $p > 0$.

The *filtration by order of pole* of the complex $j_*^m\Omega^*_{X^*} = j_*^m\mathcal{O}_{X^*} \otimes \Omega^*_X$ is the filtration

(3.1.10.2)
$$P^p(j_*^m\Omega^k_{X^*}) = P^{p-k}(j_*^m\mathcal{O}_{X^*}) \otimes \Omega^k_X.$$

The filtration P induces on the subcomplex $\Omega^*_X\langle Y\rangle$ of $j_*^m\Omega^*_{X^*}$ the bête filtration by the $\sigma_{\geq p}(\Omega^*_X\langle Y\rangle)$, a filtration also called the *Hodge filtration* F.

Proposition (3.1.11). — *The inclusion morphism*

$$(\Omega^*_X\langle Y\rangle, F) \longrightarrow (j_*^m\Omega^*_{X^*}, P)$$

is a filtered quasi-isomorphism.

This statement was suggested to me by [4]. A proof appears in [3], II, (3.13).

3.2. Mixed Hodge theory.

Recall that from now on, by a scheme we mean a scheme of finite type over $\mathbb{C}$, and by a sheaf on S a sheaf on S^{an}.

(3.2.1) Let X be a smooth separated scheme. By Nagata [11], X is a Zariski open subset of a complete scheme $\overline{X}$. By Hironaka [8], one can take $\overline{X}$ smooth, and such that $Y = \overline{X} - X$ is a normal-crossings divisor.

The reader who would like to avoid the reference to Nagata may suppose X quasi-

projective. The smooth completion $\overline{X}$ can then be chosen projective and such that Y is a union of smooth divisors. When one restricts oneself to such compactifications, Hodge theory is needed only in its standard form (2.2.1).

(3.2.2) By (3.1.7) and (3.1.8), one has

$$H^*(X,\mathbb{C}) \simeq \mathbb{H}^*(\overline{X}, \Omega^*_{\overline{X}}\langle Y\rangle).$$

The *Hodge filtration* F on the complex $\Omega^*_{\overline{X}}\langle Y\rangle$ is defined to be the filtration $F^p = \sigma_{\geq p}$ by bête truncations (1.4.5). Thus $\Omega^*_{\overline{X}}\langle Y\rangle$ carries two filtrations: F and W (3.1.5).

(3.2.3) We shall use the existence of bifiltered resolutions $i : \Omega^*_{\overline{X}}\langle Y\rangle \longrightarrow K^*$ such that the $\mathrm{Gr}^p_F\,\mathrm{Gr}^W_n(K^j)$ are sheaves acyclic for the functor Γ:

$$H^i(\overline{X}, \mathrm{Gr}^p_F\,\mathrm{Gr}^W_n(K^j)) = 0 \qquad \text{for } i > 0.$$

Here are two ways to construct them:

a) One may take for K^* the canonical Godement resolution $\mathcal{C}^*(\Omega^*_{\overline{X}}\langle Y\rangle)$, filtered by the $\mathcal{C}^*(W_n(\Omega^*_{\overline{X}}\langle Y\rangle))$ and the $\mathcal{C}^*(F^p(\Omega^*_{\overline{X}}\langle Y\rangle))$. This is a bifiltered resolution because $\mathcal{C}^*$ is an exact functor.

b) One may take for K^* the d''-resolution of $\Omega^*_{\overline{X}}\langle Y\rangle$. Let $\Omega^{pq}_{\overline{X}}$ be the sheaf of C^∞ forms of type (p,q); then K^* is the simple complex associated with the double complex of the $\Omega^p_{\overline{X}}\langle Y\rangle \otimes_{\mathcal{O}} \Omega^{0,q}_{\overline{X}}$ (a subcomplex of the $j_*\Omega^{**}_X$). This complex is filtered by the $F^p(\Omega^*_{\overline{X}}\langle Y\rangle)\otimes\Omega^{0,*}_{\overline{X}}$ and by the $W_n(\Omega^*_{\overline{X}}\langle Y\rangle)\otimes\Omega^{0,*}_{\overline{X}}$; to prove that this is a bifiltered resolution, one uses the fact that the sheaf $\mathcal{O}_\infty$ of complex-valued C^∞ functions on $\overline{X}$ is flat over $\mathcal{O}$ (a corollary of Malgrange's C^∞ preparation theorem). The sheaves $\mathrm{Gr}_F\,\mathrm{Gr}_W(K^*)$ are fine, because they are sheaves of modules over the soft sheaf $\mathcal{O}_\infty$.

(3.2.4) With the notation of (3.2.3), the complex cohomology of X appears as the cohomology of the bifiltered complex $\Gamma(\overline{X},K^*)$. Thus there are two spectral sequences abutting to $H^*(X,\mathbb{C})$. With the notation of (3.1.4), they are

(3.2.4.1) $$_WE_1^{pq} = H^{p+q}(\overline{X},\varepsilon^{-p}_{\overline{X}}[p]) = H^{2p+q}(\widetilde{Y}^p,\varepsilon^{-p}) \Rightarrow H^n(X,\mathbb{C})$$

(3.2.4.2) $$_FE_1^{pq} = H^q(\overline{X},\Omega^p_{\overline{X}}\langle Y\rangle) \Rightarrow H^n(X,\mathbb{C}).$$

The first of these, up to the renumbering $_WE_1^{pq}\mapsto E_2^{2p+q,-p}$, is just the Leray spectral sequence of the inclusion j.

Theorem (3.2.5). —

(i) On the terms $_WE_r^{pq}$ of the spectral sequence (3.2.4.1), the first direct filtration, the second direct filtration, and the recurrent filtration defined by F coincide.

(ii) The filtration on $H^n(X,\mathbb{C})$ which is the abutment of the spectral sequence $_WE$ is deduced from a filtration W of $H^n(X,\mathbb{Q})$. Neither it, nor the filtration F which is the abutment of the spectral sequence $_FE$, depends on the chosen compactification $\overline{X}$ of X or on the choice of K^*.

(iii) The filtrations $W[n]$ (1.1.2) and F define on $H^n(X,\mathbb{Z})$ a mixed Hodge structure, functorial in X.

By (3.1.7), the spectral sequence $_WE$ is the Leray spectral sequence for j_* (up to renumbering). It is therefore obtained by tensoring with $\mathbb{C}$ a spectral sequence of $\mathbb{Q}$-vector spaces, and the first assertion of (ii) is true. Complex conjugation acts on the $_WE_r$; it can be computed via (3.1.10).

Lemma (3.2.6). — *The hypercohomology spectral sequences of the filtered complexes $\mathrm{Gr}^W_n(\Omega^*_{\overline{X}}\langle Y\rangle)$, endowed with the filtration induced by the Hodge filtration, degenerate at E_1.*

Define Y^n and $\widetilde{Y}^n$ as in (3.1.4), and let $i_n:\widetilde{Y}^n\to\overline{X}$. By (3.1.5.2), one has

$$\mathrm{Gr}^W_n(\Omega^*_{\overline{X}}\langle Y\rangle)\simeq i_{n*}\Omega^*_{\widetilde{Y}^n}(\varepsilon^n)[-n].$$

Moreover, the Hodge filtration induces the bête filtration (1.4.5), so that the spectral sequence (3.2.6) is obtained, by translations in the degrees, from the classical spectral sequence

$$E_1^{pq} = H^q(\widetilde{Y}^n,\Omega^p_{\widetilde{Y}^n}(\varepsilon^n)) \Rightarrow H^{p+q}(\widetilde{Y}^n,\varepsilon^n).$$

If $\overline{X}$ is projective and Y is a union of smooth divisors, then $\widetilde{Y}^n$ is projective, ε^n is a trivial local system, and classical Hodge theory (2.2.1) gives the degeneration (3.2.6). For the general case, one must refer to [2] (see (2.2.2), (2.2.3)).

Hodge theory further gives that the Hodge filtration on $H^k(\widetilde{Y}^n,\varepsilon^n)$ is k-opposed to its complex conjugate (complex conjugation being defined in terms of $\varepsilon^n_{\mathbb{Z}}$ (3.1.4)). Taking account here of the translations in the degrees, one has:

Lemma (3.2.7). — *The filtration on*

$$_WE_1^{-n,k+n} = H^k(\overline{X},\mathrm{Gr}^W_n(\Omega^*_{\overline{X}}\langle Y\rangle)) \approx H^{k-n}(\widetilde{Y}^n,\varepsilon^n),$$

which is the abutment of the spectral sequence (3.2.6), is $(k+n)$-opposed to its complex conjugate.

Lemma (3.2.8). — *The differentials d_1 of the spectral sequence ${}_W E$ are strictly compatible with the filtration F.*

On the terms E_1, there is only one filtration induced by F to consider ((1.3.10) and (1.3.13), (iii)), and d_1 is compatible with this filtration ((1.3.13), (i)). This filtration is the abutment of the spectral sequence (3.2.6) ((1.4.8), (iii)). By (3.2.7), the arrow d_1

$$d_1 : H^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^*_{\overline{X}}\langle Y\rangle)) \longrightarrow H^{k+1}(\overline{X}, \mathrm{Gr}_{n-1}^W(\Omega^*_{\overline{X}}\langle Y\rangle))$$

that is,

(3.2.8.1)
$$d_1 : H^{k-n}(\widetilde{Y}^n, \varepsilon^n) \longrightarrow H^{k-n+2}(\widetilde{Y}^{n-1}, \varepsilon^{n-1})$$

is compatible with filtrations that are $(k+n)$-opposed to their complex conjugates. Since d_1 commutes with complex conjugation, it respects the bigrading, and hence the filtration defined by F and $\overline{F}$; this proves (3.2.8). Moreover, the cohomology of the complex E_1 will still be bigraded.

Lemma (3.2.9). — *On ${}_W E_2^{pq}$, the recurrent filtration F is q-opposed to its complex conjugate.*

We prove by induction on r that:

Lemma (3.2.10). — *For $r \geq 0$, the differentials d_r of the spectral sequence ${}_W E$ are strictly compatible with the recurrent filtration F. For $r \geq 2$, they are zero.*

For $r = 0$ (resp. $r = 1$), one applies (3.2.6) and (1.4.8), (iii) (resp. (3.2.8)). For $r \geq 2$, it suffices to prove that $d_r = 0$. By induction and by (1.3.16), one may suppose that on the terms ${}_W E_s$ ($s \geq r+1$) one has $F_d = F_r = F_{d*}$, and that ${}_W E_r = {}_W E_2$. By (1.3.13), (i), d_r is therefore compatible with the filtration F_r.

On ${}_W E_r^{pq} = {}_W E_2^{pq}$, the filtration F_r is q-opposed to its complex conjugate. The morphism

$$d_r : {}_W E_r^{pq} \longrightarrow {}_W E_r^{p+r,q-r+1}$$

therefore satisfies, for $r-1 > 0$,

$$\begin{aligned} d_r({}_W E_r^{pq}) &= d_r\left(\sum_{a+b=q} \left(F^a({}_W E_r^{pq}) \cap \overline{F}^b({}_W E_r^{pq})\right)\right) \\ &\subset \sum_{a+b=q} \left(F^a({}_W E_r^{p+r,q-r+1}) \cap \overline{F}^b({}_W E_r^{p+r,q-r+1})\right) = 0. \end{aligned}$$

This proves (3.2.10), which by (1.3.16) implies (3.2.5) (i).

By (1.3.17), the filtration of ${}_W E_\infty^{pq}$ induced by the filtration F of $H^{p+q}(X, \mathbb{C})$ is q-opposed to its complex conjugate. Since $q = -p + (p+q)$, this proves the first part of (3.2.5) (iii).

(3.2.11) We prove (ii) and (iii), which will complete the proof.

A. *Independence of the choice of K^*.* The filtrations F and W of $H^*(X, \mathbb{C})$ are the abutments of the hypercohomology spectral sequences of $\Omega^*_{\overline{X}}\langle Y\rangle$ for the filtrations F and W. These entire spectral sequences do not depend on the choice of K^*.

B. *Functoriality.* Let $f : X_1 \to X_2$ be a morphism of schemes. Suppose given a morphism of smooth compactifications

(3.2.11.1)
$$\begin{tikzcd} X_1 \arrow[r, "f"] \arrow[d, hook, "j_1"'] & X_2 \arrow[d, hook, "j_2"'] \\ \overline{X}_1 \arrow[r, "\overline{f}"'] & \overline{X}_2, \end{tikzcd}$$

where $Y_i = \overline{X}_i - X_i$ are normal-crossings divisors. The canonical morphism (see (3.1.3)) from $\overline{f}^*\Omega^*_{\overline{X}_2}\langle Y_2\rangle$ to $\Omega^*_{\overline{X}_1}\langle Y_1\rangle$ is then a morphism of bifiltered complexes; on hypercohomology it induces a morphism compatible with F and W, and therefore $f^* : H^n(X_2, \mathbb{Z}) \longrightarrow H^n(X_1, \mathbb{Z})$ is a morphism of mixed Hodge structures, for the structures defined by the compactifications $\overline{X}_i$.

C. *Independence of the compactification.* With the notation of B, if f is an isomorphism, then f^* is a bijective morphism of mixed Hodge structures, hence an isomorphism (2.3.5).

If $\overline{X}_1$ and $\overline{X}_2$ are two smooth compactifications of X, with $Y_i = \overline{X}_i - X_i$ a normal-crossings divisor, there exists a third smooth compactification $\overline{X}$, with $Y = \overline{X} - X$ a normal-crossings divisor, fitting into a commutative diagram

$$\begin{tikzcd} & X \arrow[dl, hook'] \arrow[d, hook] \arrow[dr, hook] & \\ \overline{X}_1 & \overline{X} \arrow[l] \arrow[r] & \overline{X}_2 \end{tikzcd}$$

Namely, one takes for $\overline{X}$ a resolution of the singularities of the closure of the diagonal image of X in $\overline{X}_1 \times \overline{X}_2$. The identity map from $H^n(X, \mathbb{Z})$, endowed with the mixed Hodge structure defined by $\overline{X}_1$, to $H^n(X, \mathbb{Z})$, endowed with that defined by $\overline{X}_2$, is therefore a composition of two isomorphisms.

To finish the proof of (ii) and (iii), one observes that every morphism f fits into a diagram (3.2.11.1): choose compactifications $\overline{X}_1'$ and $\overline{X}_2$ of X_1 and X_2, and then take for $\overline{X}_1$ a resolution of the singularities of the closure of the image of X_1 in $\overline{X}_1' \times \overline{X}_2$.

Definition (3.2.12). — *The mixed Hodge structure on the cohomology of a smooth separated algebraic variety is the mixed Hodge structure (3.2.5), (iii).*

Corollary (3.2.13). — *With the preceding notation:*

(i) The spectral sequence (3.2.4.1) degenerates at E_2, i.e. the Leray spectral sequence for the inclusion $j : X^* \hookrightarrow \overline{X}$ degenerates at E_3 ($E_3 = E_\infty$).

(ii) The spectral sequence (3.2.4.2)

$$_F E_1^{pq} = H^q(\overline{X}, \Omega^p_{\overline{X}}\langle Y\rangle) \Longrightarrow H^{p+q}(X, \mathbb{C})$$

degenerates at E_1.

(iii) The spectral sequence defined by the sheaf $\Omega^p_{\overline{X}}\langle Y\rangle$, endowed with the filtration W:

$$\begin{aligned} E_1^{-n,k+n} &= H^k(\overline{X}, \mathrm{Gr}_n^W(\Omega^p_{\overline{X}}\langle Y\rangle)) \\ &\approx H^k(\widetilde{Y}^n, \Omega^{p-n}_{\widetilde{Y}^n}(\varepsilon^n)) \Longrightarrow H^k(\overline{X}, \Omega^p_{\overline{X}}\langle Y\rangle) \end{aligned}$$

degenerates at E_2.

Assertion (i) was proved in (3.2.10).

Consider the four spectral sequences of (1.4.12), relative to the bifiltered complex $\Omega^*_{\overline{X}}\langle Y\rangle$. By (i), one has

$$\sum_n \dim H^n(X, \mathbb{C}) = \sum_{p,q} \dim {}_W E_2^{p,q}.$$

The terms $_W E_1^{n,k-n}$ are, for fixed n, the abutment of a spectral sequence *degenerate at* E_1 (3.2.6), with initial terms $E_1^{p,n,k-p-n}$ (notation of (1.4.12)), the initial terms of spectral sequence (iii). Since $_W d_1$ is strictly compatible with the filtration F that is the abutment of this spectral sequence (3.2.7), and is (1.4.12) the abutment of a morphism between these spectral sequences, beginning with the differentials of (iii), one has

$$\mathrm{Gr}_F^p({}_W E_2^{n,k-n}) \approx H^*(E_1^{p,n-1,k-n-p} \to E_1^{p,n,k-n-p} \to E_1^{p,n+1,k-n-p})$$

and $\mathrm{Gr}_F^*({}_W E_2^{**})$ is the sum of the terms E_2^{***} of the spectral sequences (iii). Therefore

(3.2.13.1)
$$\sum_n \dim H^n(X, \mathbb{C}) = \sum \dim E_2^{***}.$$

On the other hand,

$$\sum_n \dim H^n(X, \mathbb{C}) \leq \sum \dim {}_F E_1^{**},$$

with equality if and only if the spectral sequence (ii) degenerates at E_1, and

$$\sum \dim {}_F E_1^{**} \leq \sum \dim E_2^{***},$$

with equality if and only if the spectral sequences (iii) degenerate at E_2. Comparing with (3.2.13.1), one obtains (3.2.13).

Corollary (3.2.14). — *Let ω be a meromorphic differential p-form on $\overline{X}$, holomorphic on X and having at worst logarithmic poles along Y. Then the restriction $\omega|X$ of ω to X is closed, and if the cohomology class in $H^p(X, \mathbb{C})$ defined by ω is zero, then $\omega = 0$.*

This is the special case ${}_F E_1^{p0} = {}_F E_\infty^{p0}$ of (3.2.13), (ii).

Corollary (3.2.15). —

(i) If X is a complete smooth algebraic variety, the mixed Hodge structure on $H^n(X, \mathbb{Z})$ is the classical Hodge structure of weight n.

(ii) The Hodge numbers h^{pq} of the mixed Hodge structure of $H^n(X, \mathbb{Z})$ (X smooth algebraic) can be nonzero only for $p \leq n$, $q \leq n$, and $p + q \geq n$.

Assertion (i) is clear. To prove (ii), one observes that, for Y a union of smooth divisors, the rational Hodge structure $\mathrm{Gr}_k^W(H^n(X, \mathbb{Q}))$ is a quotient of a subobject of a Hodge structure of weight $n + k$, namely

$$H^{n-k}(\widetilde{Y}^n, \mathbb{Q}) \otimes \mathbb{Q}(-k).$$

(3.2.16) Let X be a smooth separated scheme. It is known that X admits smooth compactifications $\overline{X}$, and that the subgroup of $H^n(X, \mathbb{Z})$ which is the image of $H^n(\overline{X}, \mathbb{Z})$ is independent of the choice of $\overline{X}$ (cf. [6], (9.1) to (9.4)).

Corollary (3.2.17). — *Under the hypotheses (3.2.16), the image of $H^n(\overline{X}, \mathbb{Q})$ in $H^n(X, \mathbb{Q})$ is $W_n(H^n(X, \mathbb{Q}))$* (W denoting the weight filtration (3.2.12)).

One may suppose that $\overline{X} - X$ is a normal-crossings divisor. The assertion then follows from the fact that $W[-n]$ is the abutment of the Leray spectral sequence for the inclusion $j : X \hookrightarrow \overline{X}$.

Corollary (3.2.18). — *Let f be a morphism from a proper smooth scheme Y to a smooth scheme X admitting a smooth compactification $\overline{X}$:*

$$Y \xrightarrow{f} X \xhookrightarrow{j} \overline{X}.$$

Then the groups $H^n(X, \mathbb{Q})$ and $H^n(\overline{X}, \mathbb{Q})$ have the same image in $H^n(Y, \mathbb{Q})$.

Since f^* and $(jf)^*$ are strictly compatible with the weight filtration ((3.2.5), (iii)), it is enough to prove that $\mathrm{Gr}^W(f^*)$ and $\mathrm{Gr}^W(f^* j^*)$ have the same image in $\mathrm{Gr}^W(H^n(Y, \mathbb{Q}))$. By (3.2.17) and (3.2.15), $\mathrm{Gr}_n^W(j^*)$ is an isomorphism, while $\mathrm{Gr}_m^W(f^*) = 0$ for $m \neq n$ because $\mathrm{Gr}_m^W(H^n(Y, \mathbb{Q})) = 0$ for $m \neq n$.

***Remark* (3.2.19).** — By (3.1.11) and (1.4.5), under the hypotheses of (3.2.5), the Hodge filtration on $H^n(X, \mathbb{C})$ is the abutment of the hypercohomology spectral sequence of the complex $j_*^m \Omega_{X^*}^*$, endowed with the filtration by order of pole (3.1.10): this spectral sequence coincides with (3.2.4.2).

4. Applications and complements

4.1. The fixed-part theorem.

***Theorem* (4.1.1).** — *Let S be a smooth separated scheme, and let $f : X \to S$ be a proper smooth morphism.*

(i) In rational cohomology, the Leray spectral sequence

$$E_2^{pq} = H^p(S, R^q f_* \mathbb{Q}) \Longrightarrow H^{p+q}(X, \mathbb{Q})$$

degenerates ($E_2 = E_\infty$).

(ii) If $\overline{X}$ is a nonsingular compactification of X, the canonical morphism

$$H^n(\overline{X}, \mathbb{Q}) \longrightarrow H^0(S, R^n f_* \mathbb{Q})$$

is surjective.

For f projective and smooth, assertion (i) is proved in [2]. The proof of [2] is rewritten more readably in [5]; it does not use the smoothness of S.

Fix f, and prove that (i) $\Rightarrow$ (ii). We reduce to the case where S is connected and nonempty. If $s \in S$ is a point of S, the local system $R^n f_* \mathbb{Q}$ is entirely described by its fiber $(R^n f_* \mathbb{Q})_s$ at s, and by the action of the fundamental group $\pi = \pi_1(S, s)$ on this fiber. One has

$$H^0(S, R^n f_* \mathbb{Q}) \xrightarrow{\sim} [(R^n f_* \mathbb{Q})_s]^\pi.$$

If $X_s = f^{-1}(s)$, the composite arrow

$$H^0(S, R^n f_* \mathbb{Q}) \longrightarrow (R^n f_* \mathbb{Q})_s \simeq H^n(X_s, \mathbb{Q})$$

is therefore injective. Consider the arrows

$$H^n(\overline{X}, \mathbb{Q}) \xrightarrow{a} H^n(X, \mathbb{Q}) \xrightarrow{b} H^0(S, R^n f_* \mathbb{Q}) \overset{c}{\hookrightarrow} H^n(X_s, \mathbb{Q}).$$

The arrows cb and cba have the same image by (3.2.18). The arrow b is an edge homomorphism of the Leray spectral sequence, hence is surjective by hypothesis. Since c is injective, b and ba have the same image and ba is surjective. This proves (ii) for f projective. We now deduce the general case. Again reduce to the case where S is connected and nonempty.

By Chow's lemma and resolution of singularities, there exists a scheme X' which is quasi-projective and smooth, and a projective birational morphism

$$p : X' \longrightarrow X.$$

There then exists (by Bertini, or Sard) a nonempty Zariski open subset S_1 of S such that $X'_1 = (fp)^{-1}(S_1)$ is smooth over S_1. Finally, let $X_1 = f^{-1}(S_1)$, let $\overline{X}$ be a smooth compactification of X, and let $\overline{X}'_1$ be a smooth compactification of X'_1 which dominates $\overline{X}$.

$$\begin{tikzcd}
\overline{X} & & \overline{X}'_1 \arrow[ll, "\overline{p}"'] \\
X \arrow[u, hook] & X_1 \arrow[l, hook'] \arrow[d, "f_1"'] & X'_1 \arrow[l, "p"'] \arrow[u, hook] \arrow[d, "f'_1"'] \\
S \arrow[from=u, "f"'] & S_1 \arrow[l, hook', "i"] \arrow[r, equal] & S_1
\end{tikzcd}$$

If $s \in S_1$, the morphism $i_* : \pi_1(S_1, s) \to \pi_1(S, s)$ is surjective because the topological codimension of $S - S_1$ is ≥ 2. Therefore

$$H^0(S, R^n f_* \mathbb{Q}) \xrightarrow{\sim} H^0(S_1, R^n f_{1*} \mathbb{Q}),$$

and it suffices to prove that

$$H^n(\overline{X}, \mathbb{Q}) \longrightarrow H^0(S_1, R^n f_{1*} \mathbb{Q})$$

is surjective.

The vertical arrows of the diagram

$$\begin{tikzcd}
H^n(\overline{X}, \mathbb{Q}) \arrow[r] \arrow[d, "\overline{p}^*"] & H^n(X_1, \mathbb{Q}) \arrow[r] \arrow[d, "p^*"] & H^0(S_1, R^n f_{1*} \mathbb{Q}) \arrow[d, "p^*"] \\
H^n(\overline{X}'_1, \mathbb{Q}) \arrow[r] & H^n(X'_1, \mathbb{Q}) \arrow[r] & H^0(S_1, R^n f'_{1*} \mathbb{Q})
\end{tikzcd}$$

admit as left inverses the *Gysin morphisms* $\overline{p}_!$, $p_!$, and $p_!$, defined by Poincaré duality as transposes of the arrows analogous to the preceding ones in cohomology with proper support. Moreover, the diagram

$$\begin{array}{ccc} H^n(\overline{X}, \mathbb{Q}) & \xrightarrow{u} & H^0(S_1, R^n f_{1*}\mathbb{Q}) \\ \uparrow{\scriptstyle \overline{p}_!} & & \uparrow{\scriptstyle p_!} \\ H^n(\overline{X}'_1, \mathbb{Q}) & \xrightarrow[v]{} & H^0(S_1, R^n f'_{1*}\mathbb{Q}) \end{array}$$

is commutative. Thus u is a direct factor of v. The latter being surjective (since f'_1 is projective), the same is true of u.

We now prove (i) in the general case. We reduce to the case where f is purely of relative dimension n. Let $f \times f$ be the projection of $X \times_S X$ onto S.

Let δ be the image of the cohomology class of the diagonal of $X \times_S X$ in $H^0(S, R^n(f \times f)_*\mathbb{Q})$. By Künneth,

$$R^n(f \times f)_*\mathbb{Q} = \sum_{p+q=n} (R^p f_*\mathbb{Q} \otimes R^q f_*\mathbb{Q}).$$

Let δ'_{pq} $(p + q = n)$ denote the components of δ in this decomposition, and let δ_{pq} be classes in $H^n(X \times_S X)$ whose images are the δ'_{pq}.

The δ_{pq} define in the derived category $D^+(S)$ homomorphisms

$$\delta_p : Rf_*\mathbb{Q} \longrightarrow Rf_*\mathbb{Q}$$

such that $\mathcal{H}^q(\delta_p)$ is zero for $p \neq q$, and is the identity for $p = q$. By [2], one therefore has in $D^+(S)$

(4.1.1.1) $$Rf_*\mathbb{Q} \approx \sum_p R^p f_*\mathbb{Q}[-p],$$

and the Leray spectral sequence degenerates.

***Corollary* (4.1.2).** — *Let $f : X \to S$ be a proper smooth morphism whose target is a reduced connected separated scheme S. Let $(R^n f_*\mathbb{Q})^0$ be the largest constant local subsystem of $R^n f_*\mathbb{Q}$, with fiber $H^0(S, R^n f_*\mathbb{Q})$. Then, for each $s \in S$, $(R^n f_*\mathbb{Q})^0_s$ is a Hodge substructure of $(R^n f_*\mathbb{Q})_s \simeq H^n(X_s, \mathbb{Q})$, and the Hodge structure induced on $H^0(S, R^n f_*\mathbb{Q})$ is independent of s.*

Let $s \in S$ and $X_s = f^{-1}(s)$. If S is smooth, and if $\overline{X}$ is a smooth compactification of X, then by (4.1.1) the subspace $(R^n f_*\mathbb{Q})^0_s$ of $(R^n f_*\mathbb{Q})_s$ is the image of $H^n(\overline{X}, \mathbb{Q})$. Since the restriction map

$$H^n(\overline{X}, \mathbb{Q}) \longrightarrow H^n(X_s, \mathbb{Q})$$

is a morphism of Hodge structures, its image is a Hodge substructure, and the Hodge structure induced on $H^0(S, R^n f_*\mathbb{Q})$, a quotient of that of $H^n(\overline{X}, \mathbb{Q})$, is independent of s.

In the general case, (4.1.2) also means that if a is a global section of $R^n f_*\mathbb{C}$, then its components $a^{p,q}$ of type (p, q), *a priori* only continuous sections of the complex vector bundle defined by $R^n f_*\mathbb{C}$, are in fact locally constant, i.e. are sections of $R^n f_*\mathbb{C}$. This is so because $a^{p,q}$ is continuous and locally constant on the smooth locus (which is dense) of S, by what precedes.

(4.1.3) In the case where S is compact, a generalization of (4.1.2) is proved analytically in Griffiths [5].

As corollaries of (4.1.2), let us cite:

(4.1.3.1) (Griffiths [5]) Let $f : X \to S$ be a proper smooth morphism as in (4.1.2). If a global section a of $R^n f_*\mathbb{C}$ is of Hodge type (p, q) at one point, then a is of type (p, q) everywhere. In particular, if $n = 2$ and if, at one point s, a is the cohomology class of a divisor of X_s, then a is at every point the cohomology class of a divisor and, for S smooth, is even defined by a divisor D on X.

(4.1.3.2) (Grothendieck [7]) Let S be a reduced connected scheme of finite type over $\mathbb{C}$, and let $f_1 : X_1 \to S$ and $f_2 : X_2 \to S$ be two abelian schemes over S. If a morphism

$$u : R^1 f_{2*}\mathbb{Z} \longrightarrow R^1 f_{1*}\mathbb{Z}$$

comes at one point s of S from a morphism of abelian varieties

$$\widetilde{u}_s : (X_1)_s \longrightarrow (X_2)_s,$$

then u comes from one (and only one) morphism of abelian schemes

$$\widetilde{u} : X_1 \longrightarrow X_2.$$

(4.1.3.3) (Cf. Katz [10]) Let S be a smooth connected scheme, let s be a point of S, let $f : X \to S$ be a proper smooth morphism, and let P be a direct factor of the local system $R^i f_*\mathbb{Q}$ which is pointwise a Hodge substructure. The following conditions are equivalent:

a) The Hodge structure of P is locally constant.

b) The representation P_s of $\pi_1(S, s)$ factors through a finite quotient of $\pi_1(S, s)$.

c) There exists a nonempty finite étale covering $u : S' \to S$ such that u^*P is a constant family of Hodge structures.

4.2. The semisimplicity theorem.

(4.2.1) Let S be a topological space. A *continuous family of Hodge structures* on S consists of:

a) A local system $H_{\mathbb{Z}}$ of finitely generated $\mathbb{Z}$-modules on S.

b) For every point $s \in S$, a Hodge structure on the fiber $(H_{\mathbb{Z}})_s$, this structure varying continuously with s.

A continuous family H of Hodge structures on S is said to be *of weight n* if the fibers H_s $(s \in S)$ have weight n.

Similarly, a *continuous family of $\mathbb{Q}$-Hodge structures* is defined as a local system of $\mathbb{Q}$-vector spaces endowed at each point with a continuously varying $\mathbb{Q}$-Hodge structure.

A *polarization* of a continuous family H of $\mathbb{Q}$-Hodge structures of weight n is a morphism of local systems from $H_{\mathbb{Q}} \otimes H_{\mathbb{Q}}$ into the constant local system $\mathbb{Q}(-n)_{\mathbb{Q}}$, which at every point $s \in S$ defines a polarization of H_s.

(4.2.2) Suppose S is connected, and let $\mathcal{C}$ be a strictly full subcategory of the category of continuous families of $\mathbb{Q}$-Hodge structures on S. We shall consider the following conditions:

(4.2.2.1) $\mathcal{C}$ is stable under direct factors, direct sums, and tensor products; the constant Tate families $\mathbb{Q}(n)$ $(n \in \mathbb{Z})$ belong to $\mathcal{C}$.

(4.2.2.2) Every homogeneous Hodge structure (= of some weight n) in $\mathcal{C}$ is polarizable.

(4.2.2.3) For every $H \in \operatorname{Ob}\mathcal{C}$, there exists a local system $H'_{\mathbb{Z}}$ of free $\mathbb{Z}$-modules on S such that $H'_{\mathbb{Z}} \otimes \mathbb{Q} \approx H_{\mathbb{Q}}$.

(4.2.2.4) For every H in $\mathcal{C}$, the largest constant local subsystem H' of H is a constant family of Hodge substructures of H.

***Lemma* (4.2.3).** — *If $\mathcal{C}$ satisfies conditions (4.2.2.1) and (4.2.2.2), then:*

(i) $\mathcal{C}$ is a semisimple abelian subcategory of the abelian category of continuous families of $\mathbb{Q}$-Hodge structures on S.

(ii) If $H \in \operatorname{Ob}\mathcal{C}$, then its dual H^* and the $\bigwedge^p H$ belong to $\mathcal{C}$; if $H_1, H_2 \in \operatorname{Ob}\mathcal{C}$, then $\operatorname{Hom}(H_1, H_2) \in \operatorname{Ob}\mathcal{C}$.

If $H \in \operatorname{Ob}\mathcal{C}$ and H_1 is a subobject of H, in the category of continuous families of $\mathbb{Q}$-Hodge structures, we prove that H_1 is a direct factor of H in that category. We may suppose H homogeneous. If ψ is a polarization form for H, the orthogonal complement of H_1 for ψ is indeed a subobject of H complementary to H_1. This proves (i).

If $H \in \operatorname{Ob}\mathcal{C}$ has weight n, a polarization of H defines an isomorphism between H^* and $H \otimes \mathbb{Q}(n)$. By (4.2.2.1), one therefore has $H^* \in \operatorname{Ob}\mathcal{C}$. For arbitrary H in $\mathcal{C}$, decomposing H into its homogeneous components, one has

$$H^* = \bigoplus_n (H^n)^*,$$

whence again $H^* \in \operatorname{Ob}\mathcal{C}$. Finally, $\bigwedge^p H$ is a direct factor in $\bigotimes^p H$, and $\operatorname{Hom}(H_1, H_2) \simeq H_1^* \otimes H_2$.

Let $f : X \to S$ be a proper smooth morphism of reduced schemes. The sheaf $R^i f_* \mathbb{Q}$ is then a local system, and for $s \in S$,

$$(R^i f_* \mathbb{Q})_s \simeq H^i(X_s, \mathbb{Q})$$

is endowed with a $\mathbb{Q}$-Hodge structure. This varies continuously with s.

***Definition* (4.2.4).** — *Let S be a smooth connected scheme. A continuous family H of $\mathbb{Q}$-Hodge structures on S^{an} will be called algebraic if there exist a nonempty Zariski open subset U of S, an integer k, and a projective smooth morphism $f : X \to U$ such that $H|U$ is a direct factor of $Rf_* \mathbb{Q} \otimes \mathbb{Q}(k)$.*

***Proposition* (4.2.5).** —

(i) The category of algebraic continuous families of $\mathbb{Q}$-Hodge structures on S satisfies the conditions of (4.2.2).

(ii) If a continuous family H of Hodge structures on S is such that its restriction to a dense Zariski open subset U of S is algebraic, then H is algebraic.

(iii) If $f : X \to S$ is proper and smooth, then $Rf_* \mathbb{Q}$ is algebraic.

Assertion (ii) is evident. We prove (i). Let $\mathcal{C}_0$ be the set of continuous families of Hodge structures on S that are of the form $Rf_* \mathbb{Q}$ (f projective and smooth). Then:

a) By the Künneth formula,

$$R(f \times g)_* \mathbb{Q} \simeq Rf_* \mathbb{Q} \otimes Rg_* \mathbb{Q},$$

$\mathcal{C}_0$ is stable under tensor product.

b) $\mathcal{C}_0$ is stable under direct sum:

$$R(f \amalg g)_* \mathbb{Q} \simeq Rf_* \mathbb{Q} \oplus Rg_* \mathbb{Q}.$$

c) The homogeneous components of every $H \in \mathcal{C}_0$ are polarizable (cf. (2.2.6)).

d) The objects $H \in \mathcal{C}_0$ satisfy (4.2.2.3), since

$$Rf_* \mathbb{Q} \simeq Rf_* \mathbb{Z} \otimes \mathbb{Q}.$$

e) The objects $H \in \mathcal{C}_0$ satisfy (4.2.2.4), by (4.1.2).

Let $\mathcal{C}_1$ be the set of direct factors of objects of $\mathcal{C}_0$. Then $\mathcal{C}_1$ is stable under tensor product, direct sum, and direct factor, and satisfies (4.2.2.2) to (4.2.2.4). Moreover $\mathbb{Q}(-1)$ belongs to $\mathcal{C}_1$, since it is of the form $R^2 f_* \mathbb{Q}$ for the projective bundle $f : \mathbb{P}^1_S \to S$. The category $\mathcal{C}_2$ of the $H \otimes \mathbb{Q}(k)$ ($k \in \mathbb{N}$) therefore satisfies (4.2.2.1) to (4.2.2.4).

To deduce (i), it suffices to note that if H is a continuous family of Hodge structures, and U is a dense Zariski open subset of S, then a subobject, a polarization, or an integral lattice of $H|U$ extends uniquely to H.

We prove (iii). If $f : X \to S$ is projective and smooth, then there exist a dense Zariski open subset U of S and a commutative diagram

$$\begin{tikzcd}
X' \arrow[rr, "p"] \arrow[dr, "f'"'] & & X|U \arrow[dl, "f|U"] \\
& U &
\end{tikzcd}$$

with f' projective and smooth and p birational and surjective (apply Chow's lemma and resolution of singularities to the generic fiber of f). The restriction map

$$p^* : (Rf_* \mathbb{Q})|U \longrightarrow Rf'_* \mathbb{Q}$$

is then a split injection of continuous families of Hodge structures, with left inverse the Gysin morphism p_*; hence $Rf_* \mathbb{Q}$ is algebraic.

Theorem (4.2.6)[(1)]. — *Let S be a connected, locally connected and locally simply connected topological space, endowed with a base point s. Let $\mathcal{C}$ be a category of continuous families of Hodge structures on S satisfying the conditions (4.2.2). Then, if $H \in \operatorname{Ob}\mathcal{C}$, the representation of $\pi_1(S,s)$ on the fiber $(H_{\mathbb{Q}})_s$ is semisimple.*

Let $H \in \operatorname{Ob}\mathcal{C}$. The local system $H_{\mathbb{Q}}$ defines a complex local system $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes \mathbb{C}$. For every complex local system V on S, denote by V^c the complex vector bundle which it defines, identified with the sheaf of its continuous sections.

By definition, the group S (2.1.2) acts on $H^c_{\mathbb{C}}$. A subbundle of $H^c_{\mathbb{C}}$ will be called *horizontal*, or *locally constant*, if it is defined by a local subsystem of $H_{\mathbb{C}}$.

Lemma (4.2.7). — *Let V be a rank-one local subsystem of $H_{\mathbb{C}}$. Suppose that a tensor power $V^{\otimes n}$ $(n \geq 1)$ of V is a trivial local system, i.e. that $\pi_1(S,s)$ acts on V_s through a finite group (necessarily cyclic). Then, for $t \in S$, tV^c is again locally constant.*

For tV^c to be locally constant, it suffices that

$$(tV^c)^{\otimes n} = tV^{c\otimes n} \subset \bigotimes^n H_{\mathbb{C}}$$

be locally constant. But $V^{\otimes n}$ is generated by a global horizontal section v, and by hypothesis tv is again horizontal (4.2.2.4).

We proceed by induction on $\dim(H_{\mathbb{Q}})_s$. We may suppose H homogeneous and nonzero. Let d be the minimal dimension of the nonzero complex local subsystems of $H_{\mathbb{C}}$. The sum W of all local subsystems of $H_{\mathbb{C}}$ of dimension d (which are automatically simple) is "defined over $\mathbb{Q}$", i.e. is of the form $W_{\mathbb{Q}} \otimes \mathbb{C}$ for $W_{\mathbb{Q}}$ a local subsystem of $H_{\mathbb{Q}}$. By construction, W_s is a semisimple complex $\pi_1(S,s)$-module, hence $(W_{\mathbb{Q}})_s$ is a semisimple $\pi_1(S,s)$-module over $\mathbb{Q}$.

Let $H_{\mathbb{Z}}$ be a local system of free $\mathbb{Z}$-modules such that $H_{\mathbb{Z}} \otimes \mathbb{Q} \simeq H_{\mathbb{Q}}$, and let $W_{\mathbb{Z}} = H_{\mathbb{Z}} \cap W_{\mathbb{Q}}$. If W has dimension e, the local system $(\bigwedge^e W)^{\otimes 2}$ is trivial. Indeed:

$$\bigwedge^e W \simeq \bigwedge^e W_{\mathbb{Z}} \otimes \mathbb{C},$$

and $\pi_1(S,s)$ can act on $(\bigwedge^e W_{\mathbb{Z}})_s$ only by ± 1.

Let V be a complex local subsystem of dimension d of $H_{\mathbb{C}}$, and let V' be a complement of V in W (semisimplicity of W). One has

$$\bigwedge^e W \simeq \bigwedge^d V \otimes \bigwedge^{e-d} V'.$$

Apply (4.2.7) to the local subsystem $\bigwedge^d V \otimes \bigwedge^{e-d} V'$ of $\bigwedge^d H_{\mathbb{C}} \otimes \bigwedge^{e-d} H_{\mathbb{C}}$. We find that, for $t \in S$,

$$t\Big(\bigwedge^d V \otimes \bigwedge^{e-d} V'\Big)^c = \bigwedge^d tV^c \otimes \bigwedge^{e-d} tV'^c \subset \bigwedge^d H_{\mathbb{C}} \otimes \bigwedge^{e-d} H_{\mathbb{C}}$$

is locally constant. Hence $\bigwedge^d tV^c \subset \bigwedge^d H_{\mathbb{C}}$ and $tV^c \subset H_{\mathbb{C}}$ are locally constant.

By definition of W, one has $tV^c \subset W^c$: W^c is stable under S, and $W_{\mathbb{Q}}$ is a Hodge substructure of H. If ψ is a polarization of H, H is therefore the direct sum of W and of its orthogonal complement, both in $\mathcal{C}$, and the induction concludes.

Corollary (4.2.8). — *Under the hypotheses of (4.2.6):*

(i) *The algebra A of endomorphisms of the local system $H_{\mathbb{Q}}$ is semisimple. It admits one (and only one) $\mathbb{Q}$-Hodge structure such that, at each point s, $A \otimes H_s \to H_s$ is a morphism of Hodge structures.*

---

(1) *(Added in proof.)* Let S be a smooth connected scheme. A *family of Hodge structures on S* is a continuous family H of Hodge structures on S satisfying the following conditions:

a) The Hodge filtration of $(H_{\mathbb{C}})_s$ varies holomorphically with s, i.e. corresponds to a filtration F of $H_{\mathcal{O}} = H_{\mathbb{Z}} \otimes \mathcal{O}$.

b) The covariant derivative ∇ satisfies $\nabla F^p \subset \Omega^1_S \otimes F^{p-1}$.

Let $\mathcal{C}$ be the category of those continuous families of $\mathbb{Q}$-Hodge structures on S which underlie a family of Hodge structures, and whose homogeneous direct factors are polarizable. It is clear that $\mathcal{C}$ satisfies (4.2.2.1) to (4.2.2.3). From results of W. Schmid (August 1970, unpublished) and P. A. Griffiths [5] one can deduce that $\mathcal{C}$ satisfies (4.2.2.4). This theorem, of which (4.1.2) is a corollary, makes it possible to apply (4.2.6) and its corollaries to objects of $\mathcal{C}$.

(ii) *The center of A is of type $(0,0)$ and the composition law $\cdot : A \otimes A \to A$ is a morphism of Hodge structures.*

(iii) *Let W be a complex local subsystem of dimension d of $H_{\mathbb{C}}$:*

a) *For $t \in S$, $tW^c \subset H_{\mathbb{C}}$ is locally constant and defines a local subsystem tW isomorphic to W.*

b) *A power $(\bigwedge^d W)^{\otimes n}$ $(n \geq 1)$ of the local system $\bigwedge^d W$ is a trivial local system.*

The algebra A is the commutant of $\pi_1(S,s)$ in $(H_{\mathbb{Q}})_s$. Its semisimplicity therefore follows from (4.2.6) and Bourb., *Alg.*, chap. 8, § 5, no. 2, prop. 4. Moreover, A is the fiber of the largest constant local subsystem of $\mathrm{Hom}(H,H)$. Since $\mathrm{Hom}(H,H) \in \mathcal{C}$ ((4.2.3), (ii)), the existence of the Hodge structure in (i) follows from hypothesis (4.2.2.4). It is clear that the composition law is a morphism of Hodge structures. It follows that the center Z of A is a Hodge substructure: $Z_{\mathbb{C}}$ is bigraded, as the intersection of the kernels of the bihomogeneous morphisms $[x, \ ]$ for x bihomogeneous. Finally Z, and hence $Z_{\mathbb{C}}$, is semisimple, and for $(p,q) \neq (0,0)$, Z^{pq} is zero because it is nilpotent.

A complex local subsystem W of $H_{\mathbb{C}}$ is defined by a projector e of $A_{\mathbb{C}}$. For $t \in S$, tW^c is defined by the projector $t.e$, and is therefore locally constant. Since the group S is connected, to verify that tW is isomorphic to W, it suffices to verify that for t in a sufficiently small neighborhood of 1, the $\mathbb{C}[\pi_1(S,s)]$-module tW_s is isomorphic to W_s. Let H_i be the isotypic components of the $\mathbb{C}[\pi_1(S,s)]$-module $(H_{\mathbb{C}})_s$. One has

$$W_s = \bigoplus_i (W_s \cap H_i)$$

and

$$tW_s = \bigoplus_i (tW_s \cap H_i).$$

Moreover, for t sufficiently close to 1, tW_s is close to W_s in the Grassmannian, and hence

(4.2.8.1) $$\dim(tW_s \cap H_i) \leq \dim(W_s \cap H_i).$$

In fact, since $\dim(tW_s) = \dim(W_s)$, equality holds in (4.2.8.1). The respective isotypic components of W_s and tW_s therefore have the same length, and W_s is isomorphic to tW_s.

We prove (iii), *b*). It suffices to treat the case where H is homogeneous and W_s is a simple $\mathbb{C}[\pi_1(S,s)]$-module. Let H_1 be the isotypic component of $(H_{\mathbb{C}})_s$ which contains W_s. By *a*), H_1 is stable under S. If W_s has dimension d and H_1 has length k, one has

$$\bigwedge^{kd} H_1 \simeq (\bigwedge^{d} W_s)^{\otimes k}$$

as a $\mathbb{C}[\pi_1(S,s)]$-module. If χ is the character of $\pi_1(S,s)$ defined by $\bigwedge^d W$, the character χ_1 defined by $\bigwedge^{kd} H_1$ is therefore χ^k. Put $H_2 = H_1 + \overline{H_1}$. Since H_1 is stable under S, H_2 is defined by a real Hodge substructure of $(H_{\mathbb{R}})_s$. Every polarization form on H therefore induces on H_2 a nondegenerate bilinear form invariant under $\pi_1(S,s)$.

If $\dim(H_2) = e$, the representation $(\bigwedge^e H_2)^{\otimes 2}$ of $\pi_1(S,s)$ is therefore trivial. Thus:

a) For H_1 real: χ^{2k} is trivial;

b) For H_1 nonreal $(H_1 \neq \overline{H_1})$: $\chi^{2k} \cdot \bar{\chi}^{2k}$ is trivial.

In both cases one has $|\chi| = 1$.

The representation of $\pi_1(S,s)$ on $(H_{\mathbb{C}})_s$ is obtained by extension of scalars from a representation defined over $\mathbb{Q}$ (namely $(H_{\mathbb{Q}})_s$). The conjugate representations of W therefore occur in $(H_{\mathbb{C}})_s$; there are at most $N = \dim(H)$ of them, and for every automorphism σ of $\mathbb{C}$ one has

$$|\sigma(\chi)| = 1.$$

For $\gamma \in \pi_1(S,s)$, $\chi(\gamma)$ is an algebraic integer with at most N complex conjugates, all of absolute value 1; $\chi(\gamma)$ is therefore a k-th root of unity, with $k \leq N$. Hence $\chi^{N!} = 1$, which proves b).

Corollary (4.2.9). — *Let S be a smooth connected separated scheme, endowed with a base point $s \in S$. Let $f : X \to S$ be a morphism of schemes such that $R^n f_* \mathbb{Q}$ is a local system on S, let G be the Zariski closure of the image of $\pi_1(S,s)$ in $\mathrm{Aut}_{\mathbb{C}}((R^n f_* \mathbb{C})_s)$, and let G^0 be the identity component of G. Then:*

a) *If f is proper and smooth, G^0 is semisimple.*

b) *In general, G^0 has no quotient of multiplicative type (i.e., the radical of G^0 is unipotent).*

We prove *a*). After replacing S by a finite étale covering, we reduce to the case $G = G^0$. By ((4.2.5), (iii)), the local system $R^n f_* \mathbb{Q}$ underlies an algebraic family H of Hodge structures, so (4.2.8) applies to it ((4.2.5), (i)). The group G is reductive, since it has a faithful semisimple representation, namely $(R^n f_* \mathbb{C})_s$. If G (assumed connected) were not semisimple, it would admit a nontrivial complex representation ρ of rank one; since H_s is faithful, ρ would be a direct summand of $((H_{\mathbb{C}} + H_{\mathbb{C}}^*)^{\otimes m})_s$ for some m. This contradicts (4.2.8), (iii), *b*), because $(H + H^*)^{\otimes m}$ is algebraic and no tensor power of ρ is trivial.

Let π be a group, and consider the following property of a representation ρ of π on a finite-dimensional complex vector space V:

> $(*)$ The identity component G^0 of the Zariski closure G of $\rho(\pi)$ in $\mathrm{Aut}(V)$ has no quotient of multiplicative type.

Lemma (4.2.10). — *Let $0 \to V' \to V \to V'' \to 0$ be an exact sequence of representations of the preceding kind. Then V satisfies $(*)$ if (and only if) V' and V'' satisfy $(*)$.*

Let G, G', and G'' be the groups defined by V, V', and V''. Since, like π, G leaves V' invariant, there is a morphism

$$u = (u', u'') : G \longrightarrow G' \times G'',$$

with unipotent kernel, and such that u' and u'' are surjective. Let N be the identity component of the kernel of u''. Since u' is surjective, $u'(N)$ is normal in G', and therefore has no quotient of multiplicative type. If χ is a character of G^0, then χ factors through $u(G^0)$, is trivial on N, a power of χ factors through G''^0, and χ is trivial.

We prove (4.2.9), *b)*, by dévissage and induction on the relative dimension of f. We may suppose X reduced.

First suppose that X is quasi-projective. Then there exist a nonempty open subset U of S, a dense open subset X' of $X_U = f^{-1}(U)$ smooth over U, and a compactification $\overline{X}'$ of X' over U, proper and smooth over U. One can choose $\overline{X}'$ so that an open X'' of $\overline{X}'$ contains X' and is proper over X_U

$$\begin{tikzcd}
X_U \arrow[ddr, "f_U"'] & & X'' \arrow[ll, "p"'] \arrow[r, hook, "k"] \arrow[ddl, "f''"] & \overline{X}' \arrow[ddll, "\bar{f}'"] \\
& X' \arrow[ul, hook, "i"'] \arrow[ur, hook, "j"] \arrow[d, "f'"] & & \\
& U & &
\end{tikzcd}$$

We have long exact sequences of sheaves

$$\begin{gathered}
\to R^i f_{U*}(i_! \mathbb{Q}) \to R^i f_{U*} \mathbb{Q} \to R^i f_{U*}(\mathbb{Q}/i_! \mathbb{Q}) \to \\
\to R^i f''_*(j_! \mathbb{Q}) \to R^i f''_* \mathbb{Q} \to R f''_*(\mathbb{Q}/j_! \mathbb{Q}) \to \\
\to R^i f''_! \mathbb{Q} \to R^i \bar{f}'_* \mathbb{Q} \to R^i \bar{f}'_*(\mathbb{Q}/k_! \mathbb{Q}) \to
\end{gathered}$$

For U sufficiently small, these sheaves are local systems and the induction hypothesis applies to $R^* f_{U*}(\mathbb{Q}/i_! \mathbb{Q})$, to $R^* f''_*(\mathbb{Q}/j_! \mathbb{Q})$, and to $R^* \bar{f}'_*(\mathbb{Q}/k_! \mathbb{Q})$. By *a)*, condition $(*)$ is satisfied by $R^* \bar{f}'_* \mathbb{Q}$. By (4.2.10), it is therefore satisfied by $R^* f''_! \mathbb{Q}$, and by the dual local system $R^* f''_* \mathbb{Q}$ (Poincaré duality). By (4.2.10), condition $(*)$ is satisfied by $R^* f''_*(j_! \mathbb{Q})$, which is isomorphic to $R^* f_{U*}(i_! \mathbb{Q})$ because p is proper, and also by $R^* f_{U*} \mathbb{Q}$. Since the morphism $\pi_1(U) \to \pi_1(S)$ is surjective, $R^i f_* \mathbb{Q}$ satisfies (4.2.9), *b)*.

To pass from the quasi-projective case to the general case, it suffices to use the Leray spectral sequence for a covering (U_i) of X by quasi-projective open subsets:

$$R^* f_*(\bigcap_i U_i, \mathbb{Q}) \Rightarrow R^* f_* \mathbb{Q},$$

and (4.2.10).

4.3. Complement to [2].

In [2], (5.4) (footnote), I assert without proof that

Proposition (4.3.1). — *Let X be a proper smooth scheme over $\mathbb{C}$. If a C^∞ form on X satisfies $d'\alpha = d''\alpha = 0$ and is cohomologous to zero, then there exists β such that $\alpha = d'd''\beta$.*

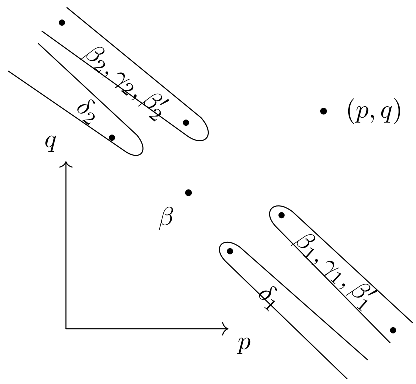
Here is the proof. As in *loc. cit.*, we reduce to the case where α is bihomogeneous, of bidegree (p, q). Since the spectral sequence of the double complex $H^0(X, \mathcal{A}^{pq})$ degenerates, the exterior differential d is strictly compatible with the filtration by first degree p (1.3.2). If $\alpha \sim 0$, there is therefore a form β_1 such that $d\beta_1 = \alpha$ and such that the components $\beta_1^{p'q'}$ of β_1 are zero for $p' < p$.

By complex conjugation, there similarly exists β_2 such that $d\beta_2 = \alpha$ and $\beta_2^{p'q'} = 0$ for $q' < q$. One has $d(\beta_1 - \beta_2) = 0$.

Let γ be a form satisfying $d'\gamma = d''\gamma = 0$ in the cohomology class of $\beta_1 - \beta_2$ (*loc. cit.*). Let γ_1 (resp. γ_2) be the sum of the components of γ of type (p', q') with $p' \geq p$ (resp. $q' \geq q$). Setting $\beta_1' = \beta_1 - \gamma_1$ and $\beta_2' = \beta_2 + \gamma_2$, one still has $d\beta_1' = d\beta_2' = \alpha$; moreover, $\beta_1' - \beta_2' = \beta_1 - \beta_2 - \gamma$ is cohomologous to zero: there exists δ such that

$$d\delta = \beta_1' - \beta_2'.$$

Let δ_1 be the sum of the components of type (p', q') of δ with $p' < p - 1$, let δ_2 be the sum of the components for which $q' > q - 1$, and let β be the component of type $(p - 1, q - 1)$.



One has $\beta_2' = d\delta_2 + d''\beta$ and $\alpha = d\beta_2' = dd\delta_2 + dd''\beta = d'd''\beta$.

4.4. Homomorphisms of abelian schemes.

We propose to show how, in some cases, one can remove from (4.1.3.2) the hypothesis of algebraicity at one point.

(4.4.1) For $f : X \to S$ a proper smooth morphism, we shall denote by $R_i f_* \mathbb{Z}$ the local system on S^{an} of the homology of the fibers of f. At each point $s \in S$, $(R_i f_* \mathbb{Z})_s \simeq H_i(X_s, \mathbb{Z})$ is endowed with a Hodge structure of weight $-i$, dual to the Hodge structure of $H^i(X_s, \mathbb{Z})$. This structure varies continuously with $s \in S$.

(4.4.2) If $f : X \to S$ is an abelian scheme, the exponential defines an exact sequence of sheaves on S^{an}

$$0 \to R_1 f_* \mathbb{Z} \to \mathrm{Lie}(X) \to X \to 0,$$

and the Hodge filtration is the short exact sequence

$$0 \to \mathrm{Ker}(\alpha) \to R_1 f_* \mathbb{Z} \otimes \mathcal{O} \xrightarrow{\alpha} \mathrm{Lie}(X) \to 0.$$

It is known that

Reminder (4.4.3). — *Let S be a smooth scheme of finite type over $\mathbb{C}$. The construction (4.4.2) establishes an equivalence of categories between:*

a) the category of abelian schemes over S;

b) the category of polarizable continuous families of Hodge structures H of type $(-1,0)+(0,-1)$ on S, such that $H_{\mathbb{Z}}$ is torsion-free and the Hodge filtration varies holomorphically on S.

Essential surjectivity of the functor a)$\mapsto$b) follows from Borel [13]; full faithfulness follows from the fact that, for X_1 and X_2 two abelian schemes over S, the S-scheme $\mathrm{Hom}_S(X_1,X_2)$ is unramified over S, and therefore has the same algebraic and holomorphic sections.

(4.4.4) Let Z be a commutative field, H a central simple algebra over Z, and $*$ a Z-linear anti-automorphism of H. After extension of scalars, one has $H \simeq \mathrm{End}(V)$ for a vector space V over Z, and $*$ is then transposition with respect to a bilinear form Φ on V, unique up to scalar. Suppose that $*$ is involutive; the forms $\Phi(X,Y)$ and $\Phi(Y,X)$ are then proportional: $\Phi(X,Y)=\lambda\Phi(Y,X)$, with $\lambda^2=1$. *Put* $\varepsilon_* = \lambda$. If $[H:Z]=d^2$, then

$$\mathrm{Tr}(*: H \to H) = \varepsilon_* \cdot d.$$

Every Z-linear automorphism C of H is inner: $Cx = cxc^{-1}$. If C is involutive and commutes with $*$, the elements c^2 and cc^* are central, so that $c^* = \lambda c$ with λ central. Since $c^{**}=c$, one has $\lambda^2=1$. *Put* $\varepsilon_C = \lambda$.

One checks easily that the involution $C \circ *$ satisfies

$$\varepsilon_{C\circ *} = \varepsilon_C \cdot \varepsilon_*.$$

Reminder (4.4.5). — *Let X be a simple abelian variety over a field K and let H be the division algebra $\mathrm{End}_K(X)\otimes\mathbb{Q}$. Every polarization of X defines a positive involution $*$ of H:*

$$\mathrm{Tr}(hh^*) > 0 \qquad \textit{if } h \neq 0.$$

If Z_0 is the subfield invariant under $*$ in the center Z of H, then Z_0 is totally real and one is in one of the following cases:

a) $Z = Z_0$; for every real place ρ of Z, $H\otimes_\rho \mathbb{R}$ is a matrix algebra over $\mathbb{R}$, and $\varepsilon_* = 1$.

b) $Z = Z_0$; for every real place ρ of Z, $H\otimes_\rho \mathbb{R}$ is a matrix algebra over $\mathbb{H}$, and $\varepsilon_* = -1$.

c) Z is a totally imaginary quadratic extension of Z_0.

(4.4.6) Let $f_i : X_i \to S$ $(i=1,2)$ be two abelian schemes over a smooth scheme S. The free abelian group

$$\mathrm{Hom}(R_1f_{1*}\mathbb{Z}, R_1f_{2*}\mathbb{Z}) \simeq \Gamma\big(S, \mathrm{Hom}(R_1f_{1*}\mathbb{Z}, R_1f_{2*}\mathbb{Z})\big)$$

is then endowed with a Hodge structure of weight 0 (4.2.8). Moreover, by (2.1.11.1) and (4.4.3), the group $\mathrm{Hom}_S(X_1,X_2)$ is the component of type $(0,0)$ of this group.

(4.4.7) Let S be a nonempty smooth connected scheme with generic point η, and let $f: X\to S$ be an abelian scheme over S. The center Z of $H = \mathrm{End}(R_1f_*\mathbb{Z})$ is then of type $(0,0)$ (4.2.8), and is therefore contained in $\mathrm{End}_S(X) \simeq \mathrm{End}_{k(\eta)}(X_\eta)$.

A polarization of X defines an involution $*$ of H. Its restriction to $\mathrm{End}_S(X)$ is positive, so that Z is a product of totally real fields or totally imaginary quadratic extensions of totally real fields.

Let C denote the actions of $i \in S(\mathbb{R})$ on $H_{\mathbb{R}} = H\otimes\mathbb{R}$ and on the $(R_1f_*\mathbb{R})_s$, for $s\in S$. On $H_{\mathbb{R}}$, $C^2=1$, and C commutes with $*$; on $(R_1f_*\mathbb{R})_s$, $C^2=-1$. If ψ is the alternating form on $R_1f_*\mathbb{Z}$, with integral values, deduced from the polarization of X, then

$$\psi(hx, Cy) = \psi(x, h^*Cy) = \psi(x, C\cdot C(h^*)\cdot y).$$

It follows that

$$\mathrm{Tr}(h\cdot C(h^*)) > 0 \qquad \text{for } h\neq 0. \tag{4.4.7.1}$$

(4.4.8) Now suppose that X_η is simple, so that Z is a field. Let $\rho : Z \to \mathbb{C}$ be a complex embedding. The algebra

$$H_\rho = H \otimes_{Z,\rho} \mathbb{C}$$

is then a matrix algebra and a bigraded direct factor in $H_{\mathbb{C}} = H \otimes_{\mathbb{Q}} \mathbb{C}$. Choose an isomorphism

$$H_\rho \simeq \mathrm{End}(V_\rho). \tag{4.4.8.1}$$

Since the infinitesimal generators of S act on H_ρ by derivations, necessarily inner, it follows that V_ρ admits a bigrading, unique up to translation, such that (4.4.8.1) is a bigraded isomorphism. Let $(R_1 f_* \mathbb{C})_\rho$ be the bigraded local system

$$(R_1 f_* \mathbb{C}) \otimes_{Z \otimes \mathbb{C}, \rho} \mathbb{C},$$

a direct factor of $R_1 f_* \mathbb{C}$. Denote by T_ρ the local system

$$T_\rho = \mathrm{Hom}_{H_\rho}(V_\rho, (R_1 f_* \mathbb{C})_\rho).$$

One has

$$V_\rho \otimes_{\mathbb{C}} T_\rho \xrightarrow{\sim} (R_1 f_* \mathbb{C})_\rho \tag{4.4.8.2}$$

(a bigraded isomorphism). Since $(R_1 f_* \mathbb{C})_\rho$ has type $(-1, 0) + (0, -1)$, one of the following conditions is satisfied:

a) V_ρ is bihomogeneous;

b) T_ρ is bihomogeneous.

It is clear that condition a) (respectively b)) is simultaneously satisfied for ρ and for $\bar{\rho}$. If condition a) is satisfied at every place, then H is of type $(0, 0)$. If condition b) is satisfied at every place, then the Hodge structure on $R_1 f_* \mathbb{Q}$ is locally constant.

(4.4.9) Now suppose that Z is totally real. For every real embedding φ of Z, put $\varepsilon_{H,\varphi} = +1$ if $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra and $\varepsilon_{H,\varphi} = -1$ otherwise. If a complex embedding ρ induces a real embedding φ, put $\varepsilon_{H,\rho} = \varepsilon_{H,\varphi}$.

Let $\rho : Z \to \mathbb{C}$ factor through $\varphi : Z \to \mathbb{R}$. Complex conjugation on H_ρ is induced by an antilinear automorphism σ_1 of V_ρ, unique up to a real scalar factor, with scalar square. Since σ_1 commutes with σ_1^2, σ_1^2 is real and, normalizing σ_1, one may suppose that $\sigma_1^2 = \pm 1$. Then

$$\sigma_1^2 = \varepsilon_{H,\rho}. \tag{4.4.9.1}$$

The automorphism of complex conjugation on

$$(R_1 f_* \mathbb{C})_\rho = ((R_1 f_* \mathbb{Q}) \otimes_{Z,\varphi} \mathbb{R}) \otimes_{\mathbb{R}} \mathbb{C}$$

is “compatible” with (4.4.8.2) and may be written

$$\sigma = \sigma_1 \otimes \sigma_2.$$

Since $\sigma^2 = 1$, by (4.4.9.1) one has

$$\sigma_1^2 = \sigma_2^2 = \varepsilon_{H,\rho}. \tag{4.4.9.2}$$

Let Φ be a polarization form on $R_1 f_* \mathbb{Z}$. This form is necessarily of the type

$$\Phi = \mathrm{Tr}_{Z/\mathbb{Q}}(\psi),$$

where ψ is a Z-bilinear alternating form on $R_1 f_* \mathbb{Q}$.

Let ψ_ρ be the induced form on $V_\rho \otimes_{\mathbb{C}} T_\rho$. The alternating form ψ_ρ is invariant under $\pi_1(S, s)$. Since T_ρ is irreducible, it can be written

$$\psi_\rho = A_\rho \otimes B_\rho, \tag{4.4.9.3}$$

with A_ρ on V_ρ, and B_ρ on T_ρ, invariant under $\pi_1(S, s)$.

Since T_ρ is irreducible, B_ρ is symmetric or alternating. Since ψ_ρ is alternating, A_ρ is then alternating or symmetric. Since the T_ρ are conjugate to one another, which case occurs is independent of ρ. Put

$$A_\rho(X, Y) = \varepsilon_V A_\rho(Y, X), \tag{4.4.9.4}$$

$$B_\rho(X,Y) = \varepsilon_T B_\rho(Y,X). \tag{4.4.9.5}$$

Thus

$$\varepsilon_V \varepsilon_T = -1 \tag{4.4.9.6}$$

and it is clear from (4.4.4) that if $*$ is the involution of H deduced from ψ,

$$\varepsilon_* = \varepsilon_V. \tag{4.4.9.7}$$

In case (4.4.8), a) (respectively b)), normalize the bigradings so that V_ρ (respectively T_ρ) has bidegree $(0,0)$. Then for nonzero x, y in V_ρ and T_ρ:

$$\begin{array}{ll} \text{case a)} & A_\rho(x, \sigma_1 x)\, B_\rho(y, C\sigma_2 y) > 0; \\ \text{case b)} & A_\rho(x, C\sigma_1 x)\, B_\rho(y, \sigma_2 y) > 0. \end{array}$$

Normalizing A_ρ and B_ρ (only their tensor product being given), we may assume that

$$\begin{array}{llll} \text{case a)} & A_\rho(x, \sigma_1 x) > 0 & \text{and} & B_\rho(y, C\sigma_2 y) > 0; \\ \text{case b)} & A_\rho(x, C\sigma_1 x) > 0 & \text{and} & B_\rho(y, \sigma_2 y) > 0. \end{array}$$

Moreover, A_ρ and B_ρ, like Φ, are invariant under the compact subtorus $\mathrm{Ker}(N)$ of S, and in particular under C. In case a) one has

$$0 < A_\rho(\sigma_1 x, \sigma_1\sigma_1 x) = \varepsilon_V \cdot \varepsilon_{H,\rho} \cdot A_\rho(x, \sigma_1 x),$$

whence $\varepsilon_V \cdot \varepsilon_{H,\rho} = 1$. Similarly, in case b), one finds $\varepsilon_T \cdot \varepsilon_{H,\rho} = 1$.

Lemma (4.4.10). If $\varepsilon_T = \varepsilon_{H,\rho}$, we are in case b); if $\varepsilon_T = -\varepsilon_{H,\rho}$, we are in case a).

Proposition (4.4.11). Let S be a nonempty smooth connected scheme with generic point η, let $f : X \to S$ be an abelian scheme over S, let $H = \mathrm{End}(R_1 f_* \mathbb{Q})$, and let Z be the center of H. Suppose that:

(a) X_η is simple over $k(\eta)$.

(b) X does not become constant over any finite covering of S.

(c) One of the following conditions is satisfied (cf. (4.4.5)):

- (c_1) Z is imaginary quadratic;
- (c_2) Z is totally real, and for every real embedding $\varphi : Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra over $\mathbb{R}$;
- (c_3) Z is totally real, and for every real embedding $\varphi : Z \to \mathbb{R}$, $H \otimes_{Z,\varphi} \mathbb{R}$ is a matrix algebra over $\mathbb{H}$.

Then

$$\mathrm{End}(X) \xrightarrow{\sim} \mathrm{End}(R_1 f_* \mathbb{Z}).$$

We show that we are in case (4.4.8), a), or (4.4.8), b) at all places of Z simultaneously. This is clear from (4.4.8) under hypothesis (c_1), and follows from (4.4.10) under hypotheses (c_2) or (c_3), because not only ε_T but also $\varepsilon_{H,\rho}$ is then independent of ρ.

Condition (4.4.8), b) cannot hold at every place of Z, for the Hodge structure would then be locally constant, which by (4.1.3.3) and (4.4.3) contradicts b). Hence condition a) holds at every place of Z, so that H is of type $(0,0)$ (4.4.8). In view of (4.4.3), this completes the proof.

Proposition (4.4.12). Let $f : X \to S$ be an abelian scheme over a smooth scheme S. The following conditions are equivalent:

(i) For every abelian scheme $g : Y \to S$, one has

$$\mathrm{Hom}_S(X, Y) \xrightarrow{\sim} \mathrm{Hom}_S(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z}).$$

(ii) Condition (i) holds for $X = Y$, and the center Z of $\mathrm{End}_S(X) \otimes \mathbb{Q}$ has no complex place $\rho : Z \to \mathbb{C}$ such that the direct summand $R_1 f_* \mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$ of $R_1 f_* \mathbb{C}$ is purely of Hodge type $(-1, 0)$.

One reduces to the case where S is nonempty and connected, with generic point η, and where X_η is a simple abelian variety over $k(\eta)$. If (i) or (ii) is satisfied, then $D = \mathrm{End}(X_\eta) \otimes \mathbb{Q}$ is a division algebra, and, for $s \in S$, $(R_1 f_* \mathbb{Q})_s$ is an irreducible representation of $\pi_1(S, s)$.

To prove that (ii) implies (i), one may suppose that Y_η is simple and that $(R_1 g_* \mathbb{Q})_s$ is isotypic of type $(R_1 f_* \mathbb{Q})_s$. One then has an isomorphism of continuous families of $\mathbb{Q}$-Hodge structures (with D of type $(0,0)$)

$$R_1 g_* \mathbb{Q} \approx R_1 f_* \mathbb{Q} \otimes_D \mathrm{Hom}(R_1 f_* \mathbb{Q}, R_1 g_* \mathbb{Q}).$$

One has $D \otimes \mathbb{C} \simeq M_n(Z \otimes \mathbb{C})$. If $e \in D \otimes \mathbb{C}$ is transformed, under such an isomorphism, into the idempotent e_{11}, one still has an isomorphism of bigraded vector spaces

$$(R_1 g_* \mathbb{C})_s \approx (R_1 f_* \mathbb{C})_s \otimes_{Z \otimes \mathbb{C}} e\big(\mathrm{Hom}(R_1 f_* \mathbb{C}, R_1 g_* \mathbb{C})\big).$$

If condition (ii) is satisfied, and if $e\big(\mathrm{Hom}(R_1 f_* \mathbb{C}, R_1 g_* \mathbb{C})\big)$ is not purely of type $(0,0)$, then $(R_1 g_* \mathbb{C})_s$ could not be purely of type $(-1,0)+(0,-1)$, which is absurd. It follows that $\mathrm{Hom}(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z})$ is purely of type $(0,0)$, and (i) follows from (4.4.3) and (2.1.11.1).

Let Z be a totally imaginary quadratic extension of a totally real number field Z^0. If $Z \otimes \mathbb{R} \simeq \mathbb{C} \oplus \cdots \oplus \mathbb{C}$, define an action of S on $Z \otimes \mathbb{R}$ by scalar multiplication, sending $s \in S(\mathbb{R}) \simeq \mathbb{C}^*$ to $(s^2 \cdot N(s)^{-1}, 1, \ldots, 1)$. The pair consisting of Z and this action of S on $Z \otimes \mathbb{R}$ is a polarizable Hodge structure Z_ρ, of type $(-1,1)+(0,0)+(1,-1)$, depending on Z and on the chosen place $\rho : Z \to \mathbb{C}$.

Let X be an abelian scheme over S, with X_η simple. If the center Z of $\mathrm{End}(R_1 f_* \mathbb{Q})$ is not totally real, it is a totally imaginary quadratic extension of a totally real number field. If the second hypothesis of (ii) is violated, the direct summand $R_1 f_* \mathbb{Q} \otimes_Z Z_\rho$ of the continuous family of polarizable Hodge structures $R_1 f_* \mathbb{Q} \otimes_{\mathbb{Q}} Z_\rho$ is purely of type $(-1,0)+(0,-1)$.

By (4.4.3), the choice of an integral lattice in $R_1 f_* \mathbb{Q} \otimes_Z Z_\rho$ defines an abelian scheme $g : Y \to S$, and $\mathrm{Hom}(R_1 f_* \mathbb{Q}, R_1 g_* \mathbb{Q})$ is not purely of type $(0,0)$, which violates (i).

***Corollary* (4.4.13).** — *Let S be a smooth connected scheme with generic point η, and let $f : X \to S$ be an abelian scheme over S of relative dimension $g \leq 3$. Suppose that on no finite étale covering of S does X admit a constant abelian subscheme. Then, for every abelian scheme $g : Y \to S$, one has*

$$\mathrm{Hom}(X, Y) \xrightarrow{\sim} \mathrm{Hom}(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z})$$

and

$$\mathrm{Hom}(Y, X) \xrightarrow{\sim} \mathrm{Hom}(R_1 g_* \mathbb{Z}, R_1 f_* \mathbb{Z}).$$

By duality, it suffices to prove the first assertion. We may also reduce to the case where X_η is simple. We verify that X satisfies one of the conditions (c) of (4.4.11). Put $H = M_c(D)$, where D is a division algebra of rank b^2 over its center Z, which is known to be either totally real or totally imaginary. Putting $a = [Z : \mathbb{Q}]$, one has

$$ab^2 c \mid 2g \leq 6. \tag{4.4.13.1}$$

1) If Z is totally imaginary but not an imaginary quadratic field, one must have $a = 2g = 4$. For $s \in S$, the commutant of the action of Z on $(R_1 f_* \mathbb{Q})_s$ is then commutative, so the action of $\pi_1(S, s)$ is abelian and therefore factors through a finite group ((4.2.8), (iii), b) or (4.2.9)). This is absurd ((4.1.3.3) and (4.4.3)).

   With the notation of (4.4.8), one could also note that the T_ρ are then of rank one, so condition (4.4.8), b) holds at every place and the Hodge structure is locally constant.

2) If Z is totally real and conditions (c_1) and (c_2) fail, then $a \geq 2$ and $b \geq 2$, which is absurd.

Let us verify that X satisfies the conditions of (4.4.12). If Z is imaginary quadratic and there exists $\rho : Z \to \mathbb{C}$ such that $R_1 f_* \mathbb{Q} \otimes_{Z,\rho} \mathbb{C}$ is purely of Hodge type $(-1,0)$, then the Hodge structure on $R_1 f_* \mathbb{Q}$ is locally constant, which is absurd. This proves (4.4.13).

(4.4.14) Let $f : X \to S$ be a polarized abelian scheme of dimension g over a smooth connected base S. If $s \in S$, the fundamental group $\pi_1(S, s)$ acts on the fiber $H^1(X_s, \mathbb{Z})$ of $R^1 f_* \mathbb{Z}$ at s. The polarization of X defines an alternating form on $H^1(X_s, \mathbb{Z})$, with values in $\mathbb{Z}(-1)$, nondegenerate over $\mathbb{Q}$, and the action of $\pi_1(S, s)$ respects this form. Consider the hypothesis:

(4.4.14.1) The morphism defined by X from S into one of the irreducible components of the corresponding moduli scheme of polarized abelian varieties is dominant.

The following result was suggested to me by J.-P. Serre.

***Corollary* (4.4.15).** — *Let S be a smooth connected scheme and let $f : X \to S$ be a polarized abelian scheme over S satisfying (4.4.14.1). Then, for every abelian scheme $g : Y \to S$, one has*

$$\mathrm{Hom}(X, Y) \xrightarrow{\sim} \mathrm{Hom}(R_1 f_* \mathbb{Z}, R_1 g_* \mathbb{Z}).$$

Let $s \in S$. By (4.4.11) and (4.4.12), it suffices to show that the representation of $\pi_1(S, s)$ on $(R_1 f_* \mathbb{Q})_s$ is absolutely irreducible, so that $H = \mathbb{Q}$. This follows from the following lemma.

***Lemma* (4.4.16).** — *If X satisfies (4.4.14.1), then the image of $\pi_1(S, s)$ in the group of symplectic automorphisms of $H^1(X_s, \mathbb{Z})$ has finite index.*

Let n be an integer. After replacing S by a surjective étale covering corresponding to a subgroup of finite index in $\pi_1(S, s)$, we may assume that the local system ${}_nX = \mathrm{Ker}(n \cdot \mathrm{id} : X \to X)$ is trivial on S. Replacing X by an isogenous abelian scheme, we may further assume that X is principally polarized. Suppose $n \geq 3$.

The abelian scheme X then defines a dominant morphism x from S to the "Jacobi" moduli scheme M_n of principally polarized abelian schemes equipped with a symplectic isomorphism between ${}_nX$ and $(\mathbb{Z}/n)^{2g}$; moreover, X is the pullback via x of the universal abelian scheme over M_n. It is known that $\pi_1(M_n, x(s))$ maps isomorphically onto the subgroup of $\mathrm{Sp}(H^1(X_s, \mathbb{Z}))$ which is the kernel of reduction modulo n.

It therefore remains to establish the following:

***Lemma* (4.4.17).** — *Let $p : S \to M$ be a dominant morphism of smooth connected schemes. The subgroup $p(\pi_1(S, s))$ of $\pi_1(M, p(s))$ then has finite index.*

Let t be a closed point of the generic fiber of p, and let T' be its closure in S. There exists a dense Zariski open subset U of M such that $T = p^{-1}(U) \cap T'$ is an étale cover of U.

There is no loss of generality in taking $s \in T$. In the diagram

$$\begin{tikzcd} \pi_1(T, s) \arrow[r] \arrow[d, "a"] & \pi_1(S, s) \arrow[d, "c"] \\ \pi_1(U, p(s)) \arrow[r, "b"'] & \pi_1(M, p(s)) \end{tikzcd}$$

the arrow a has image of finite index, because T is a finite étale cover, and the arrow b is surjective, because $M - U$ has real codimension ≥ 2. Thus the arrow c has image of finite index.

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A CONGRUENCE BETWEEN MULTINOMIAL COEFFICIENTS

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Let p be a prime. For every nonzero integer m, let $v(m)$ denote the exponent of the highest power of p dividing m. Set also $v(0) = \infty$.

For a k-tuple $\mathfrak{n} = (n_1, \dots, n_k)$ of nonnegative integers, set

$$|\mathfrak{n}| = \sum_1^k n_i, \qquad a\mathfrak{n} = (an_1, \dots, an_k), \qquad X^{\mathfrak{n}} = \prod_1^k X_i^{n_i}, \qquad \mathfrak{n}! = \prod_1^k n_i!$$

and denote by $C(\mathfrak{n})$ the coefficient of the monomial $X^{\mathfrak{n}}$ in

$$\left(\sum_1^k X_i \right)^{|\mathfrak{n}|}$$

(multinomial coefficients).

Thus

$$\left(\sum_1^k X_i \right)^n = \sum_{|\mathfrak{n}|=n} C(\mathfrak{n}) X^{\mathfrak{n}} \tag{1}$$

$$C(\mathfrak{n}) = \frac{|\mathfrak{n}|!}{\mathfrak{n}!} \tag{2}$$

For $k = 2$, these are the binomial coefficients.

$$C(n_1, n_2) = \binom{n_1 + n_2}{n_1} \tag{3}$$

Proposition.— *Let $\mathfrak{n}$ be a k-tuple of nonnegative integers, and let $n = |\mathfrak{n}|$. If p is a prime ≥ 5, then*

$$C(p^l \mathfrak{n}) \equiv C(\mathfrak{n}) \pmod{p^{v(n)+3}} \tag{4}$$

Example.— For $p = 5$, $k = 2$, $l = 1$, and $\mathfrak{n} = (1, 1)$, one has

$$\binom{10}{5} = 252 = \binom{2}{1} + 2 \cdot 5^3$$

Proof.— It is known, or follows from (2), that

$$\left(\sum_1^k X_i \right)^{p^l} = \left(\sum_1^k X_i^{p^l} \right) + pQ \tag{5}$$

where Q is a polynomial with *integer* coefficients, of degree $< p^l$ in each variable. The binomial formula then gives

$$\left(\sum_1^k X_i \right)^{p^l \cdot n} = \left(\sum_1^k X_i^{p^l} \right)^n + \sum_1^n p^i \binom{n}{i} R_i \tag{6}$$

with

$$R_i = \left(\sum_1^k X_i^{p^l} \right)^{n-i} \cdot Q^i \tag{7}$$

For two polynomials A and B with integer coefficients, write $A \doteq B \pmod r$ if the coefficients of $X^{p^l \mathfrak{n}}$ in A and B are congruent modulo r. With this notation, (4) becomes

$$\left(\sum_1^k X_i\right)^{p^l \cdot n} \doteq \left(\sum_1^k X_i^{p^l}\right)^n \pmod{p^{v(n)+3}} \tag{8}$$

Lemma 1.

$$v\left(\binom{n}{i}\right) \geq v(n) - v(i) \tag{9}$$

Let the additive group $\mathbb{Z}/(n)$ act on itself by translation[1]; if an i-element subset P of $\mathbb{Z}/(n)$ has as stabilizer[2] a subgroup H of $\mathbb{Z}/(n)$ $(H \cdot P = P)$, then P is a union of cosets of H, and $\#H \mid i$. The cardinality $r_p = n/\#H$ of the orbit of P under $\mathbb{Z}/(n)$, in $\mathcal{P}(\mathbb{Z}/(n))$, therefore satisfies

$$p^{v(n)-v(i)} \mid r_p,$$

hence, since the set of i-element subsets of $\mathbb{Z}/(n)$ is a disjoint union of orbits,

$$p^{v(n)-v(i)} \mid \binom{n}{i}$$

Lemma 2. — *For $p \neq 2, 3$ and $i \geq 3$, we have*

$$p^i \binom{n}{i} \equiv 0 \pmod{p^{v(n)+3}} \tag{10}$$

We have

$$v\left(p^i\binom{n}{i}\right) = i + v\binom{n}{i} \geq v(n) + (i - v(i)) \qquad \text{(Lemma 1)}$$

For $3 \leq i < p$, we have $v(i) = 0$ and $i - v(i) \geq 3$. For $i \geq p$, we have $i - v(i) \geq i - \log i/\log p$. This last function of i is increasing for $i \geq p$, and equals $p - 1 \geq 3$ when $i = p$.

Lemma 3. — *The coefficient of $X^{p^l \mathfrak{n}}$ in R_1 is zero.*

None of the monomials $X^{\mathbf{r}}$ occurring in Q has an exponent $\mathbf{r}$ divisible by p^l. The polynomial

$$R_1 = \left(\sum_1^k X_i^{p^l}\right)^{n-1} \cdot Q$$

inherits this property.

It follows from Lemmas 2 and 3 that

$$\left(\sum_1^k X_i\right)^{p^l n} \doteq \left(\sum_1^k X_i^{p^l}\right)^n + \binom{n}{2} p^2 R^2, \quad \pmod{p^{v(n)+3}} \tag{11}$$

It already follows from this congruence, by Lemma 1, that

$$C(p^l \mathfrak{n}) \equiv C(\mathfrak{n}) \pmod{p^{v(n)+2}} \tag{12}$$

The only monomials $X^{\mathbf{r}}$ with exponent divisible by p^l which occur in $(pQ)^2$ are the monomials

$$X_i^{p^l} \cdot X_j^{p^l} \qquad (i < j)$$

[1]To each $x \in \mathbb{Z}/(n)$ there is thus associated the permutation of $\mathbb{Z}/(n)$ that sends y to $x + y$.

[2]If a group G acts on a set E, the stabilizer of a subset P of E is the set of $g \in G$ for which P is stable.

The coefficient of each is

$$\sum_{i\neq 0,p^l}\binom{p^l}{i}\binom{p^l}{p^l-i}=\sum\binom{p^l}{i}^2-2=\binom{2p^l}{p^l}-2 \tag{13}$$

and

$$\binom{2p^l}{p^l}-2\equiv\binom{2p}{p}-2 \pmod{p^3}$$

by (12).

Thus

$$\left(\sum_1^k X_i\right)^{p^l\cdot n}\doteq\left(\sum_1^k X_i^{p^l}\right)^n+\binom{n}{2}\left[\binom{2p}{2}-2\right]\cdot R_2 \pmod{p^{v(n)+3}}$$

with R a polynomial with integer coefficients.

Since

$$v\left(\binom{n}{2}\right)\geq v(n) \qquad \text{(Lemma 1)},$$

the proposition follows from

Lemma 4. — *For a prime $p\geq 5$, one has:*

$$\binom{2p}{p}\equiv 2 \pmod{p^3}.$$

One has

$$p^{-1}\binom{p}{i}\equiv(-1)^{i-1}\cdot i^{-1} \pmod{p}$$

for $1\leq i\leq p-1$, and hence

$$p^{-2}\left[\binom{2p}{p}-2\right]=\sum_1^{p-1}p^{-2}\binom{p}{i}^2\equiv\sum_1^{p-1}(1/i)^2 \pmod{p} \tag{14}$$

Define $z\in\mathbb{Z}/(p)$ by

$$z=\sum_{i\in\mathbb{Z}/(p)^*}1/i^2$$

For $u\in\mathbb{Z}/(p)^*$, one has

$$z=\sum_{i\in\mathbb{Z}/(p)^*}1/(u^{-1}i)^2=u^2z$$

and

$$(u^2-1)z=0.$$

Since $p\neq 2,3$, there exists u such that $u^2\neq 1$, and $z=0$. Formula (14) therefore simplifies to

$$p^{-2}\left[\binom{2p}{p}-2\right]\equiv 0 \pmod{p},$$

which completes the proof.

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The Weil Conjecture for K3 Surfaces

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1. Statement of the Theorem

Let $\mathbb{F}_q$ be a field with q elements, let $\overline{\mathbb{F}}_q$ be an algebraic closure of $\mathbb{F}_q$, let $\varphi \in \mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ be the Frobenius substitution $x \mapsto x^q$, and let $F = \varphi^{-1}$ be the "geometric Frobenius."

Let X be a scheme (separated and of finite type) over $\mathbb{F}_q$, and let $\overline{X}$ be the scheme over $\overline{\mathbb{F}}_q$ obtained from it by extension of scalars. For every closed point x of X, let $\deg(x) = [k(x) : \mathbb{F}_q]$ be the degree of the residue-field extension over $\mathbb{F}_q$. The zeta function $Z(X,t) \in \mathbb{Z}[[t]]$ is defined by

$$Z(X,t) = \prod_{x \in X} (1 - t^{\deg(x)})^{-1} \qquad (x \text{ a closed point of } X),$$

$$\log Z(X,t) = \sum_{n>0} \# X(\mathbb{F}_{q^n}) \frac{t^n}{n}.$$

For every prime number ℓ prime to q, the ℓ-adic cohomology with proper supports $H^i_c(\overline{X}, \mathbb{Q}_\ell)$ of $\overline{X}$ is a finite-dimensional vector space over $\mathbb{Q}_\ell$, on which $\mathrm{Gal}(\overline{\mathbb{F}}_q/\mathbb{F}_q)$ acts by transport of structure. By Grothendieck, one has

$$Z(X,t) = \prod_i \det(1 - Ft, H^i_c(\overline{X}, \mathbb{Q}_\ell))^{(-1)^{i+1}}.$$

Take X to be a nonsingular projective surface. In this case it is known that the factors

$$P_i(t) = \det(1 - Ft, H^i(\overline{X}, \mathbb{Q}_\ell))$$

of $Z(X,t)$ are polynomials with integer coefficients independent of ℓ. The Weil conjecture asserts here that the roots of $P_i(t)$ have complex absolute value $q^{-i/2}$. This conjecture is invariant under finite extension of the finite base field. For X as above, and $i \neq 2$, it follows from Weil [9]. If, for simplicity, X is assumed geometrically connected, one indeed has

$$P_0(t) = 1 - t, \qquad P_4(t) = 1 - q^2 t,$$

$$P_1(t) = \det(1 - \varphi t; T_\ell(\mathrm{Pic}^0(X)_{\mathrm{red}})), \qquad P_3(t) = P_1(qt).$$

*This article was written during a stay at the University of Warwick, which I thank for its hospitality.

Let k be a field of characteristic 0, and let X be a nonsingular projective geometrically connected surface over k. We say that X is a K3 surface if X is regular and has trivial canonical divisor, that is, if $H^1(X, \mathcal{O}) = 0$ and Ω_X^2 is isomorphic to $\mathcal{O}_X$. For the remarkable properties of these surfaces, we refer to [8]. We shall mainly use the following facts.

1.1. Lemma. *(i) One has $h^{2,0} = h^{0,2} = 1$.*

(i) One has $H^2(X, T_X^1) = 0$ (infinitesimal deformations are unobstructed), and the map induced by interior product

$$H^1(X, T_X^1) \otimes H^0(X, \Omega_X^2) \xrightarrow{\sim} H^1(X, \Omega_X^1) \tag{1.1.1}$$

is bijective, while $H^0(X, T_X^1) = 0$ (X has no infinitesimal automorphisms).

One has

$$h^{2,0} =_{\mathrm{dfn}} \dim H^0(X, \Omega_X^2) = \dim H^0(X, \mathcal{O}_X) = 1,$$

and (i) follows by Hodge symmetry. Since $H^1(X, \mathcal{O}_X) = 0$, Hodge symmetry and Serre duality give

$$h^{0,1} = h^{1,0} = h^{2,1} = h^{1,2} = 0.$$

Since $\Omega_X^2 = \mathcal{O}_X$, the "interior product" isomorphism

$$T_X^1 \otimes \Omega_X^2 \xrightarrow{\sim} \Omega_X^1$$

induces isomorphisms $T_X^1 \simeq \Omega_X^1$. The vanishing of $H^2(X, T_X^1)$ therefore follows from that of $h^{1,2}$, and it is clear that (1.1.1) is bijective. The same argument gives $H^0(X, T_X^1) = 0$.

1.2. Lemma. *Let Y_0 be a nonsingular projective surface over a finite field $\mathbb{F}_q$. The following conditions are equivalent.*

(i) *There exists a complete discrete valuation ring V of unequal characteristic with finite residue field k, a proper smooth morphism $f_V : Y \to \mathrm{Spec}(V)$ whose generic fiber is a K3 surface, and $i : \mathbb{F}_q \to k$ such that $Y_0 \otimes_{\mathbb{F}_q} k \simeq Y \otimes_V k$.*

(ii) *There exists an integral scheme S whose fraction field has characteristic 0, a point $s \in S$, a proper smooth morphism $f : Y \to S$ whose generic fiber is a K3 surface, and $i : \mathbb{F}_q \to k(s)$ such that $Y_0 \otimes_{\mathbb{F}_q} k(s) \simeq Y_s$.*

The proof uses a standard passage to the limit. If the equivalent conditions of 1.2 are satisfied, we say that Y_0 lifts to a K3 surface.

1.3. Theorem. *A nonsingular projective surface over a finite field $\mathbb{F}_q$ which lifts to a K3 surface satisfies the Weil conjecture.*

This theorem was obtained independently by Pjateckii-Shapiro and Šafarevitch.

It applies, for example, to surfaces of degree 4 in $\mathbb{P}^3$. The Betti numbers of a K3 surface are $(1, 0, 22, 0, 1)$; the theorem therefore implies that

$$\left|\#X(\mathbb{F}_q) - \#\mathbb{P}^2(\mathbb{F}_q)\right| \le 21q.$$

Grothendieck's conjectural theory of motives (in particular his theory of the "motivic Galois group") was very useful to me in constructing the proof. In this language (which will not be used below), the idea is, roughly speaking, to construct, by reduction from $\mathbb{C}$ to $\mathbb{F}_q$, an abelian variety A and an injective "morphism" of motives

$$H^2(X) \hookrightarrow H^1(A) \otimes H^1(A),$$

in order to deduce the Weil conjecture for X from Weil's theorem for A. However, we do not define this "morphism" by an algebraic cycle; it is therefore not a morphism in the sense of [5]. In Hodge theory, its mode of definition has the following analogue (which will not be used below).

1.4. Lemma. *Let H be a variation of Hodge structures of weight n on a scheme S. If the local system $H_{\mathbb{Z}}$ has no global sections other than the section s and its multiples, then at every point of S the section s is of type $(n/2, n/2)$.*

1.5. Notations. 1.5.1. Scalar extensions will be denoted by subscripts: if M_A is a module over a ring (commutative with identity) A (or a scheme over A, for example an algebraic group over A), then M_B denotes the module over the A-algebra B (or the scheme over B) obtained from it by extension of scalars.

1.5.2. If a group G acts on a set X, we write $g * x$ for the transform of x by g.

1.5.3. When an equality is the definition of one of its sides, we write it $=_{\mathrm{dfn}}$.

2. Polarized Hodge Structures

To work with Hodge structures, we shall use the formalism set out in [4] and briefly recalled below.

2.1. Let $\underline{S}$ be the real algebraic group of invertible elements of the $\mathbb{R}$-algebra $\mathbb{C}$, let $w : \mathbb{G}_{m_{\mathbb{R}}} \to \underline{S}$ be the inclusion of $\mathbb{R}^*$ into $\mathbb{C}^*$, and let $t : \underline{S} \to \mathbb{G}_{m_{\mathbb{R}}}$ be the inverse of the norm; then $tw(x) = x^{-2}$. Let V be a finite-dimensional real vector space. It amounts to the same thing to give:

a) $\rho : \underline{S} \to \mathrm{GL}(V)$ of weight n: $\rho w(x) * v = x^n \cdot v$;

b) a Hodge bigrading of weight n of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$:

$$V_{\mathbb{C}} = \bigoplus_{p+q=n} V^{p,q} \qquad \text{with} \qquad \overline{V^{p,q}} = V^{q,p};$$

c) a Hodge filtration of weight n of $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$: this is a decreasing filtration such that

$$F^p(V_{\mathbb{C}}) \oplus \overline{F^{n-p+1}(V_{\mathbb{C}})} \xrightarrow{\sim} V_{\mathbb{C}}.$$

One has

$$F^p(V_{\mathbb{C}}) = \bigoplus_{p' \geq p} V^{p',q'};$$

for $z \in \underline{S}(\mathbb{R}) = \mathbb{C}^*$ and $v^{p,q} \in V^{p,q}$, one has

$$z * v^{p,q} = z^p \bar{z}^q v^{p,q}.$$

Put $C = \rho(i)$; this is the *Weil operator* C. One has $Cv^{p,q} = i^{p-q}v^{p,q}$. For $\mathscr{E}$ a subset of $\mathbb{Z} \times \mathbb{Z}$, a Hodge bigrading is said to be of *type* $\mathscr{E}$ if the *Hodge numbers* $h^{pq} = \dim V^{pq}$ vanish for $(p,q) \notin \mathscr{E}$.

2.2. Here a *Hodge structure of weight n H* will mean a free finite-type $\mathbb{Z}$-module $H_{\mathbb{Z}}$ together with a Hodge bigrading of weight n of $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$. The *Tate Hodge structure* $\mathbb{Z}(n)$ is the Hodge structure of type $\{(-n,-n)\}$ given by

$$\mathbb{Z}(n)_{\mathbb{C}} = \mathbb{C} \qquad \text{and} \qquad \mathbb{Z}(n)_{\mathbb{Z}} = (2\pi i)^n \mathbb{Z} \subset \mathbb{C}.$$

A polarization of a Hodge structure H of weight n is a morphism of Hodge structures

$$\psi : H \otimes H \longrightarrow \mathbb{Z}(-n)$$

such that, on $H_{\mathbb{R}} = H_{\mathbb{Z}} \otimes \mathbb{R}$, the real bilinear form

$$(2\pi i)^n \psi(x, Cy)$$

is symmetric and positive definite. The form ψ is then $(-1)^n$-symmetric.

A *rational Hodge structure of weight n* consists of a $\mathbb{Q}$-vector space $H_{\mathbb{Q}}$ and a Hodge bigrading of weight n of $H_{\mathbb{C}} = H_{\mathbb{Q}} \otimes \mathbb{C}$. We denote by $\mathbb{Q}(n)$ the rational Hodge structure underlying $\mathbb{Z}(n)$. The definition of polarizations is left to the reader.

The following statement reformulates a theorem of Riemann.

2.3. Scholium. *The functor $A \mapsto H^1(A, \mathbb{Z})$ identifies polarized abelian varieties with polarized Hodge structures of type* $\{(0,1),(1,0)\}$.

From the remainder of this subsection, we shall use only 2.11(iii$'$) $\Rightarrow$ (ii). To prove this implication, one may restrict to connected groups, which makes 2.5 unnecessary or obvious.

2.4. Let G be a real linear algebraic group. We say that G is *compact* if $G(\mathbb{R})$ is compact and every connected component of $G(\mathbb{C})$ has a real point. The obvious functor is then an equivalence from the category of algebraic linear representations of G to the category of linear representations of the compact group $G(\mathbb{R})$.

2.5. Lemma. *Every real algebraic subgroup H of a compact real algebraic group G is compact.*

It is known that the map

$$(g, \ell) \longmapsto g \cdot \exp(i\ell) : G(\mathbb{R}) \times \mathrm{Lie}(G) \longrightarrow G(\mathbb{C})$$

is bijective. It is enough to prove that, for every real subgroup H of G and every $h = g \cdot \exp(i\ell) \in H(\mathbb{C})$, one has $\ell \in \mathrm{Lie}(H)$. Since $\bar{h} \in H(\mathbb{C})$, one has

$$\exp(2i\ell) = \bar{h}^{-1}h \in H(\mathbb{C}).$$

Regard $G(\mathbb{C})$ as the set of real points of a real algebraic group. Let $\overline{J} \subset H(\mathbb{C})$ be, within this group, the Zariski closure of $J = \{\exp(ni\ell) \mid n \in 2\mathbb{Z}\}$. The elements $g \in J$, and hence the elements $g \in \overline{J}$, satisfy $g^{-1} = \bar{g}$, whence $\mathrm{Lie}(\overline{J}) \subset i\,\mathrm{Lie}(G)$. The Lie algebra $\mathrm{Lie}(\overline{J})$ is abelian, and $\exp(\mathrm{Lie}(\overline{J}))$ is a group, necessarily of finite index in $\overline{J}$. There are therefore n and $\ell' \in \mathrm{Lie}(\overline{J})$ such that

$$\exp(ni\ell) = \exp(\ell').$$

Thus $i\ell \in \mathrm{Lie}(\overline{J}) \subset \mathrm{Lie}(H)$ and $\ell \in \mathrm{Lie}(H)$, completing the proof.

2.6. Lemma. *Let G be a real algebraic group. The following conditions are equivalent:*

(i) *G is compact;*

(ii) *for every real (respectively complex) linear representation of G, there exists a positive-definite G-invariant symmetric bilinear (respectively hermitian sesquilinear) form;*

(iii) *for a faithful representation of G, there exists a form as in (ii).*

For G connected, these conditions are further equivalent to

(iii') *for a representation of G with finite kernel, there exists a form as in (ii).*

For G compact, G-invariance is equivalent to $G(\mathbb{R})$-invariance, and (ii) is classical. If (iii) holds, G is a subgroup of an orthogonal or unitary group, and 2.5 applies. Condition (iii') implies that $G(\mathbb{R})$ is compact (as a finite covering of a compact group); if G is connected, G is therefore compact.

2.7. Let G be a real linear algebraic group. For every involutive automorphism σ of G, let $G^{(\sigma)}$ be the real form of $G_{\mathbb{C}}$ defined by the antilinear involution $g \mapsto \sigma(\bar{g})$. We say that σ is a *Cartan involution* if $G^{(\sigma)}$ is compact.

Let C be an element of $G(\mathbb{R})$ whose square is central. A real (respectively complex) representation of G will be called C-polarizable if there exists on V a G-invariant bilinear (respectively sesquilinear) form ψ such that $\psi(x, Cy)$ is symmetric (respectively hermitian) and positive definite. We then say that ψ is a C-polarization of V. A special case of the following lemma is stated without proof in [4].

2.8. Lemma. *a) Whether ψ is a C-polarization of V depends only on the $G(\mathbb{R})$-conjugacy class of C.*

b) The following conditions are equivalent:

(i) ad C *is a Cartan involution of* G*;*

(ii) *every real (respectively complex) representation of* G *is* C*-polarizable;*

(iii) G *has a faithful real (respectively complex)* C*-polarizable representation.*

If G *is connected, these conditions are also equivalent to*

(iii') G *has a real (respectively complex)* C*-polarizable representation with finite kernel.*

To prove a), note that $\psi(x, gCg^{-1}y) = \psi(g^{-1}x, Cg^{-1}y)$. A sesquilinear form ψ on a complex representation of G is G-invariant if and only if

$$\psi(gx, \bar{g}y) = \psi(x, y) \qquad \text{for } g \in G(\mathbb{C}).$$

This condition is equivalent to

$$\psi(gx, CC^{-1}\bar{g}Cy) = \psi(x, Cy) \qquad \text{for } g \in G(\mathbb{C})$$

(replace y by Cy), i.e. to the $G^{(\operatorname{ad} C)}$-invariance of $\psi(x, Cy)$. The "respectively" version of 2.8 therefore follows from 2.6. The non-"respectively" version follows from it:

a) the complexification (respectively realification) of a real (respectively complex) C-polarizable representation is C-polarizable;

b) a polarization of V induces a polarization of every subrepresentation of V;

c) for V real (respectively complex), one has $V \hookrightarrow V \otimes_{\mathbb{R}} \mathbb{C}$.

2.9. Let G be a reductive group over $\mathbb{Q}$, and let t and w be morphisms defined over $\mathbb{Q}$

$$\mathbb{G}_m \xrightarrow{w} G \xrightarrow{t} \mathbb{G}_m,$$

with $tw(x) = x^{-2}$ and w central. Suppose also given a commutative diagram

$$\begin{array}{ccccc} \mathbb{G}_{m,\mathbb{R}} & \xrightarrow{w} & \underline{S} & \xrightarrow{t} & \mathbb{G}_{m,\mathbb{R}} \\ \| & & \downarrow h & & \| \\ \mathbb{G}_{m,\mathbb{R}} & \xrightarrow{w} & G_{\mathbb{R}} & \xrightarrow{t} & \mathbb{G}_{m,\mathbb{R}}. \end{array}$$

A representation defined over $\mathbb{Q}$, $V_{\mathbb{Q}}$, of G is called *homogeneous of weight* n if $w(x) * v = x^n \cdot v$ ($v \in V$). Every representation is a sum of homogeneous representations. Let $V_{\mathbb{Q}}$ be a representation of weight n. Then h endows $V_{\mathbb{Q}}$ with a rational Hodge structure of weight n.

We shall regard $\mathbb{Q}(n)_{\mathbb{Q}}$ as a representation of G by the rule

$$g * x = t(g)^n \cdot x.$$

This convention is reasonable, since $\underline{S}$ acts on $\mathbb{Q}(n)_{\mathbb{R}}$ by the same formula. A *polarization* of a weight-n representation $V_{\mathbb{Q}}$ of G is a G-invariant form

$$\psi : V_{\mathbb{Q}} \otimes V_{\mathbb{Q}} \longrightarrow \mathbb{Q}(n)$$

such that $\psi(x, h(i)y)$ is a symmetric positive-definite form on $V_{\mathbb{R}}$. Being G-invariant, a polarization ψ is also $\underline{S}$-invariant and hence a polarization of the rational Hodge structure $V_{\mathbb{Q}}$.

2.10. Lemma. *A representation $V_{\mathbb{Q}}$ is polarizable if and only if the representation $V_{\mathbb{R}}$ is $h(i)$-polarizable.*

Let $P_{\mathbb{Q}}$ (resp. $P_{\mathbb{R}}$) be the space of G-invariant $(-1)^n$-symmetric bilinear forms on $V_{\mathbb{Q}}$ (resp. $V_{\mathbb{R}}$). One has $P_{\mathbb{R}} = \mathbb{R} \otimes P_{\mathbb{Q}}$. The $h(i)$-polarizations form an open subset of $P_{\mathbb{R}}$, and the polarizations of $V_{\mathbb{Q}}$ are the intersection of this open subset with $P_{\mathbb{Q}}$. The lemma follows.

2.11. Proposition. *a) Whether ψ is a polarization of V depends only on the $G(\mathbb{R})$-conjugacy class of h.*

b) The following conditions are equivalent:

(i) $\operatorname{ad} h(i)$ *is a Cartan involution of* $\operatorname{Ker}(t)_{\mathbb{R}}$*;*

(ii) *every homogeneous representation of G is polarizable;*

(iii) *G has a faithful family of polarizable homogeneous representations.*

If G is connected, these conditions are equivalent to

(iii') *G has a polarizable representation ρ such that* $\operatorname{Ker}(\rho) \cap \operatorname{Ker}(t)$ *is finite.*

Assertion a) follows from 2.8.a). If G is connected, then $\operatorname{Ker}(t)$ is connected: otherwise t would be of the form t_0^n, with $n > 1$. Since $twx = x^{-2}$, one would have $n = 2$ and $w(-1) \notin \operatorname{Ker}(t)^0$. This is absurd, since

$$w(-1) \in h\big(\operatorname{Ker}(t : \underline{S} \longrightarrow \mathbb{G}_{m,\mathbb{R}})\big),$$

which is connected. With this established, b) follows from 2.10 and 2.8.b).

3. Review of the Spinor Group

For the material recalled in this section, see [3].

3.1. Let V be a free module over a ring A, equipped with a quadratic form Q. The module V is identified with a submodule of the Clifford algebra $C(V)$; in $C(V)$ one has

$$v \cdot v = Q(v).$$

We denote by $C^+(V)$ the even part of $C(V)$.

If V is a free module over A, equipped with a symmetric bilinear form ψ, we denote by $C(V)$ (or by $C(V, \psi)$) the Clifford algebra of V equipped with the quadratic form $\psi(x, x)$.

3.2. Suppose that A is a field of characteristic 0, and that Q is nondegenerate. We denote by $\mathrm{CSpin}(V)$ the *Clifford group*. It is the algebraic group of invertible elements g of $C^+(V)$ such that $gVg^{-1} = V$. We define a morphism from $\mathrm{CSpin}(V)$ to $\mathrm{SO}(V)$ by

$$g \longmapsto (v \longmapsto gvg^{-1}).$$

The kernel of this morphism consists only of the scalars: there is an exact sequence of algebraic groups

$$0 \longrightarrow \mathbb{G}_m \xrightarrow{w} \mathrm{CSpin}(V) \longrightarrow \mathrm{SO}(V) \longrightarrow 0.$$

The spin group $\mathrm{Spin}(V)$ is the algebraic subgroup of $\mathrm{CSpin}(V)$ that is the kernel of the spinor norm

$$N : \mathrm{CSpin}(V) \longrightarrow \mathbb{G}_m.$$

Let t denote the inverse of N; there is a commutative diagram

(3.2.1)
$$\begin{tikzcd}
 & & 0 \arrow[d] & & \\
 & & \mathrm{Spin}(V) \arrow[d] \arrow[dr] & & \\
0 \arrow[r] & \mathbb{G}_m \arrow[r, "w"] \arrow[dr, "x \mapsto x^{-2}"'] & \mathrm{CSpin}(V) \arrow[r] \arrow[d, "t"] & \mathrm{SO}(V) \arrow[r] & 0 \\
 & & \mathbb{G}_m \arrow[d] & & \\
 & & 0 & &
\end{tikzcd}$$

By definition, one also has

(3.2.2)
$$\alpha : \mathrm{CSpin}(V) \hookrightarrow C^+(V)^*.$$

3.3. We denote by $(C^+(V))_s$ the representation of $\mathrm{CSpin}(V)$ on $C^+(V)$ given by

(3.3.1)
$$g * x = \alpha(g) \cdot x.$$

We denote by $(C^+(V))_{\rm ad}$ the representation of $\mathrm{CSpin}(V)$ on $C^+(V)$ obtained from the action of $\mathrm{SO}(V)$ on $C^+(V)$ by transport of structure. One has

$$g *_{\rm ad} x = \alpha(g) \cdot x \cdot \alpha(g)^{-1}. \tag{3.3.2}$$

The action of $\mathrm{CSpin}(V)$ on $(C^+(V))_s$ is compatible with the right $C^+(V)$-module structure of $(C^+(V))_s$. There is an isomorphism of representations

$$\mathrm{End}_{C^+(V)}((C^+(V))_s) = (C^+(V))_{\rm ad}. \tag{3.3.3}$$

There is an isomorphism of representations

$$(C^+(V))_{\rm ad} = \bigoplus \bigwedge^{2i} V. \tag{3.3.4}$$

3.4. Suppose that A is algebraically closed. We distinguish two cases.

1) $\dim(V)$ *odd*: $C^+(V)$ is here a matrix algebra. Let W be a simple $C^+(V)$-module; W is a *spin representation* of $\mathrm{CSpin}(V)$:

$$g * W = \alpha(g) \cdot W.$$

There are isomorphisms of representations

$$(C^+(V))_s = \text{a sum of copies of } W, \tag{3.4.1}$$

$$(C^+(V))_{\rm ad} = \mathrm{End}_A(W). \tag{3.4.2}$$

2) $\dim(V)$ *even*: $C^+(V)$ is here a product of two matrix algebras. Let W_1 and W_2 be two nonisomorphic simple $C^+(V)$-modules, and let $W = W_1 + W_2$; W_1 and W_2 are the *half-spin representations.*

There are isomorphisms of representations

$$(C^+(V))_s = \text{a sum of copies of } W, \tag{3.4.3}$$

$$(C^+(V))_{\rm ad} = \mathrm{End}_A(W_1) \times \mathrm{End}_A(W_2). \tag{3.4.4}$$

In both cases, $C^+(V)$ is the bicommutant of the representation W.

3.5. Proposition. *Let Γ be a Zariski-dense subgroup of* $\mathrm{Spin}(V)$ *and let a be an automorphism of the A-algebra $C^+(V)$ that commutes with the action of Γ given by $\gamma \mapsto \alpha(\gamma) \cdot x \cdot \alpha(\gamma)^{-1}$. Then a is the identity.*

It is enough to treat the case in which A is an algebraically closed field and Γ is the whole spin group (indeed, commuting with a is a Zariski-closed condition).

With the notation of 3.4, let us identify $C^+(V)$ with a subalgebra of $\mathrm{End}_A(W)$.

There is an automorphism $\bar{a}$ of W such that

$$a(x) = \bar{a} x \bar{a}^{-1} \qquad (x \in C^+(V)).$$

For $\gamma \in \mathrm{Spin}(V)$, by hypothesis one has

$$\alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1} \cdot x \cdot \bar{a}\alpha(\gamma)\bar{a}^{-1}\alpha(\gamma)^{-1} = x \qquad (x \in C^+(V));$$

the commutator $(\alpha(\gamma), \bar{a}) = \alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1}$ therefore lies in the center of $C^+(V)$.

1) $\dim(V)$ *odd*: here $C^+(V) = \mathrm{End}_A(W)$, and $\det_W((\alpha(\gamma), \bar{a})) = 1$, so that $\alpha(\gamma)\bar{a}\alpha(\gamma)^{-1}\bar{a}^{-1}$ is a root of unity ν. Since $\mathrm{Spin}(V)$ is connected, and since for $\gamma = e$ one has $\nu = 1$, one even has

$$(\alpha(\gamma), \bar{a}) = 1, \qquad \text{i.e.} \qquad \alpha(\gamma)\bar{a} = \bar{a}\alpha(\gamma).$$

2) $\dim(V)$ *even*: here $C^+(V) = \mathrm{End}_A(W_1) \times \mathrm{End}_A(W_2)$, and $\bar{a}$ either permutes the W_i or preserves them. In both cases, from

$$\det_{W_i}(\alpha(\gamma)) = 1 \qquad (i = 1, 2)$$

one deduces that

$$\det_{W_i}((\alpha(\gamma), \bar{a})) = 1 \qquad (i = 1, 2)$$

and one concludes as above that

$$\alpha(\gamma)\bar{a} = \bar{a}\alpha(\gamma).$$

In both cases, $\bar{a}$ lies in the commutant of the representation, and therefore commutes with the bicommutant $C^+(V)$: a is the identity.

4. Construction of abelian varieties

In this paragraph and part of the next, we present results of [7].

4.1. Let $V_{\mathbb{Z}}$ be a free $\mathbb{Z}$-module of rank $n+2$ endowed with a bilinear form of nonzero discriminant

$$B : V_{\mathbb{Z}} \otimes V_{\mathbb{Z}} \longrightarrow \mathbb{Z}.$$

Assume that B has index $(n+, 2-)$. We are interested in weight-0 morphisms $h : \underline{S} \to \mathrm{SO}(V_{\mathbb{R}})$ such that

1) B is a polarization of the representation $V_{\mathbb{Q}}$ of $\mathrm{SO}(V_{\mathbb{Q}})$;

2) the Hodge numbers of $V_{\mathbb{C}}$ are
$$h^{-1,1} = h^{1,-1} = 1 \qquad \text{and} \qquad h^{0,0} = n.$$

4.2. A morphism h as above lifts uniquely to $\tilde{h} : \underline{S} \to \mathrm{CSpin}(V_{\mathbb{R}})$, making the diagram

$$\begin{tikzcd}
\mathbb{G}_{m,\mathbb{R}} \arrow[r, "w"] \arrow[d, equal] & \underline{S} \arrow[r, "t"] \arrow[d, "\tilde{h}"] & \mathbb{G}_{m,\mathbb{R}} \arrow[d, equal] \\
\mathbb{G}_{m,\mathbb{R}} \arrow[r] & \mathrm{CSpin}(V_{\mathbb{R}}) \arrow[r, "t"] & \mathbb{G}_{m,\mathbb{R}}.
\end{tikzcd}$$

commutative.

4.3. Lemma. *With respect to $\tilde{h}$, every representation of* $\mathrm{CSpin}(V_{\mathbb{Q}})$ *is polarizable (2.9).*

We apply criterion 2.11. b)(iii') to the representation $V_{\mathbb{Q}}$ of $\mathrm{CSpin}(V_{\mathbb{Q}})$.

4.4. Lemma. *The Hodge structure on $C^{+}(V_{\mathbb{Q}})_{\mathrm{ad}}$ defined by $\tilde{h}$ is of type* $\{(-1,1),(0,0),(1,-1)\}$.

We have

$$C^{+}(V_{\mathbb{Q}})_{\mathrm{ad}} \simeq \bigoplus \bigwedge^{2i} V_{\mathbb{Q}},$$

and the conclusion follows from $h^{-1,1}(V_{\mathbb{Q}}) = 1$.

4.5. Proposition. *The Hodge structure on $C^{+}(V_{\mathbb{Q}})_s$ defined by $\tilde{h}$ is of type* $\{(0,1),(1,0)\}$. *Thus the Hodge structure $C^{+}(V_{\mathbb{Q}})_s$, endowed with the lattice $C^{+}(V_{\mathbb{Z}})$, defines an abelian variety.*

First suppose that n is odd. After extension of scalars to $\mathbb{C}$, $C^{+}(V_{\mathbb{Q}})$ becomes a matrix algebra

$$C^{+}(V_{\mathbb{C}}) \simeq \mathrm{End}(W).$$

On W, $\mathrm{CSpin}(V_{\mathbb{Q}})_{\mathbb{C}}$ acts by $g \cdot w = \alpha(g) \cdot w$: this is the spin representation. The representation $C^{+}(V_{\mathbb{C}})_s$ is a sum of $\dim(W)$ copies of W, so it is enough to prove that the weight-one representation W is purely of type $\{(0,1),(1,0)\}$. Otherwise, the representation $\mathrm{End}(W) \simeq C^{+}(V_{\mathbb{C}})_{\mathrm{ad}}$ could not be purely of type $\{(-1,1),(0,0),(1,-1)\}$.

For n even, similarly,

$$C^{+}(V_{\mathbb{C}}) \simeq \mathrm{End}(W') \oplus \mathrm{End}(W''),$$

where W' and W'' are the half-spin representations. As above, one proves that W' and W'' are of type $\{(0,1),(1,0)\}$, and hence so is $C^{+}(V_{\mathbb{C}})_s$.

The second assertion follows from 2.3 and 3.3.

4.6. The construction given in this paragraph conceals the following fact (which we shall not use), visible from the Dynkin diagram and the real reason why everything works.

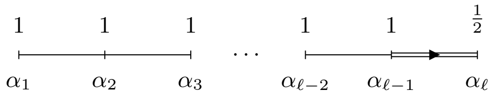
Let $V_{\mathbb{Z}}$ be a polarized Hodge structure of weight 0 and type $\{(1,-1),(0,0),(-1,1)\}$, with $h^{1,-1} = 0$. Let

$$r : \mathbb{G}_m \longrightarrow \mathrm{SO}(V_{\mathbb{C}})$$

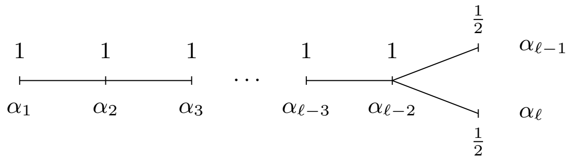
be the homomorphism such that $r(x) \cdot v^{pq} = x^p v^{pq}$ for v^{pq} of type (p,q). Let T be a maximal torus of $\mathrm{SO}(V_{\mathbb{C}})$ equipped with a system of simple roots Φ, let $\widetilde{T}$ be the inverse image of T in $\mathrm{Spin}(V_{\mathbb{C}})$, and let $(w_\alpha)_{\alpha\in\Phi}$ be the fundamental weights of $\widetilde{T}$. The homomorphism r has a unique conjugate

$$r_0 : \mathbb{G}_m \longrightarrow T$$

such that the rational numbers $\langle w_\alpha, r_0\rangle \in \frac{1}{2}\mathbb{Z}$ are all positive. One checks that r_0 is the homomorphism $H_{\alpha_1} : \mathbb{G}_m \to T$ corresponding to the root denoted below by α_1. The numbers $\langle w_\alpha, r_0\rangle$ are as follows:



for $\mathrm{SO}(2, 2\ell - 1)$,



for $\mathrm{SO}(2, 2\ell - 2)$.

Let w be a dominant weight of a spin or half-spin representation, and let σ be the opposition involution. The point is that

$$\langle w, r_0\rangle + \langle \sigma w, r_0\rangle = 1.$$

5. Construction of Families of Abelian Varieties

We shall use the following theorem of A. Borel (to appear).

5.1. Theorem (A. Borel). *Let X/Γ be the quotient of a Hermitian symmetric domain by a torsion-free arithmetic group. It is known [2] that X/Γ is naturally a quasi-projective algebraic variety. Let S be a reduced scheme over $\mathbb{C}$ and let $f : S^{\mathrm{an}} \to X/\Gamma$ be a morphism of analytic spaces. Then f is algebraic.*

Let S be smooth over $\mathbb{C}$. For the notion of a variation of Hodge structures $\underline{H}$ on S, and of a polarization of $\underline{H}$, we refer to Griffiths [6]. We write $\underline{H}_{\mathbb{Z}}$ for the local system of free $\mathbb{Z}$-modules on S defined by $\underline{H}$, and $\underline{H}_s$ for the fiber Hodge structure of $\underline{H}$ at $s \in S$.

For X the Siegel upper half-space, Theorem 5.1 has the following corollary, which generalizes 2.3.

5.2. Scholium. *The functor $(a : A \to S) \mapsto R^1 a_* \mathbb{Z}$ identifies polarized abelian schemes A over S with polarized variations of Hodge structures of type $\{(0,1),(1,0)\}$ on S.*

5.3. In the category of variations of Hodge structures on S, all the usual tensor operations make sense. Thus

a) if $\underline{H}$ and $\underline{H}'$ are variations of Hodge structures, there are variations of Hodge structures

$$\underline{H} \otimes \underline{H}', \qquad \underline{\mathrm{Hom}}(\underline{H}, \underline{H}'), \qquad \bigwedge^n \underline{H}, \qquad \underline{H}^\vee, \qquad \underline{\mathrm{End}}(\underline{H}), \ldots.$$

b) if $\underline{H}$ is a variation of Hodge structures of weight 0, and ψ is a polarization of $\underline{H}$ (or any symmetric morphism of Hodge structures $\underline{H} \otimes \underline{H} \to \mathbb{Z}(0)$), then $C(\underline{H}, \psi)$ and $C^+(\underline{H}, \psi)$ are defined. At every point $t \in S$, with the notation of 3.4, one has

$$C^+(\underline{H}, \psi)_t = C^+(\underline{H}_t, \psi)_{\mathrm{ad}}.$$

5.4. Let $(V_{\mathbb{Z}}, B)$ be as in 4.1, and let X be the set of $h : \underline{S} \to \mathrm{SO}(V_{\mathbb{R}})$ of the type considered in 4.1. Giving such an h is equivalent to giving a two-dimensional subspace $V_{\mathbb{R}}^-$ of $V_{\mathbb{R}}$, on which B is negative definite, together with an orientation of $V_{\mathbb{R}}^-$: for $z = \tau \cdot e^{i\theta} \in \underline{S}(\mathbb{R}) = \mathbb{C}^*$, $h(z)$ induces the identity on the orthogonal complement of $V_{\mathbb{R}}^-$ and the rotation through angle 2θ on $V_{\mathbb{R}}^-$. The group $\mathrm{O}(V_{\mathbb{R}})$ acts on X by transport of structure: $(g * h)(z) = g \cdot h(z) \cdot g^{-1}$, and the description above makes it geometrically evident that X is a homogeneous space under $\mathrm{SO}(V_{\mathbb{R}})$. The stabilizer of $h \in X$ is a subgroup $\mathrm{SO}(2) \times \mathrm{SO}(n)$, the identity component of a *maximal* compact subgroup.

5.5. For $h \in X$, let F_h be the corresponding Hodge filtration of $V_{\mathbb{C}}$. The function $h \mapsto F_h^1$ identifies X with an open subset of the space of isotropic lines in $V_{\mathbb{C}}$. This construction endows X with a complex structure for which F_h is a holomorphic function of h. It is known that, for this structure, the (two) connected components of X are Hermitian symmetric domains.

5.6. Let $W_{\mathbb{Q}}$ be a weight-n representation of the algebraic group $\mathrm{CSpin}(V_{\mathbb{Q}})$: $w(x) * y = x^n \cdot y$. Let $\underline{W}_{\mathbb{Q}}$ be the constant local system on X with fiber $W_{\mathbb{Q}}$. The group $\mathrm{CSpin}(V_{\mathbb{Q}})$ acts on $(X, \underline{W}_{\mathbb{Q}})$ by

$$\gamma * (w \text{ at } h \in X) = (\gamma w \text{ at } \gamma h \gamma^{-1} \in X).$$

Each $h \in X$ defines $\tilde{h} : \underline{S} \to \mathrm{CSpin}(V_{\mathbb{R}})$, hence a $\mathbb{Q}$-Hodge structure on the fiber of $\underline{W}_{\mathbb{Q}}$ at $h \in X$. Thus one obtains a variation of $\mathbb{Q}$-Hodge structures of weight n on X, again denoted $\underline{W}_{\mathbb{Q}}$:

a) the Hodge filtration is a holomorphic function of h;

b) the transversality axiom is satisfied (we shall use it only in a trivial case).

This construction is compatible with tensor products. In particular, a polarization $\psi : W_{\mathbb{Q}} \otimes W_{\mathbb{Q}} \to \underline{\mathbb{Q}}(-n)$ of the representation $W_{\mathbb{Q}}$ defines a polarization of $\underline{W}_{\mathbb{Q}}$.

Let $W_{\mathbb{Z}}$ be an integral lattice in $W_{\mathbb{Q}}$ and let Γ be a subgroup of $\mathrm{CSpin}(V_{\mathbb{Q}})$, discrete in $\mathrm{CSpin}(V_{\mathbb{R}})$, acting freely on X and satisfying $\Gamma W_{\mathbb{Z}} = W_{\mathbb{Z}}$. Then the constant local system $\underline{W}_{\mathbb{Z}}$ on X, with fiber $W_{\mathbb{Z}}$, underlies a Γ-equivariant variation of Hodge structure $\underline{W}$ on X as above. Passing to the quotient, it defines a variation of Hodge structures $\underline{W}$ on X/Γ.

If $W_{\mathbb{Q}}$ is a representation of $\mathrm{SO}(V_{\mathbb{Q}})$, the same construction applies to discrete subgroups of $\mathrm{SO}(V_{\mathbb{Q}})$.

5.7. Proposition. *Let $(\underline{H}, \psi)$ be a polarized variation of Hodge structures of type $\{(-1,1),(0,0),(1,-1)\}$, with $h^{1,-1}=1$, on a smooth connected scheme S. There exist*

a) *a finite surjective étale cover $u : S_1 \to S$;*

b) *an abelian scheme A over S_1;*

c) *a $\mathbb{Z}$-algebra C, and $\mu : C \hookrightarrow \mathrm{End}_{S_1}(A)$;*

d) *an isomorphism of variations of Hodge structures*

$$u : C^+(\underline{H}, \psi) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_* \mathbb{Z})$$

which induces an isomorphism of local systems of algebras

$$u_{\mathbb{Z}} : C^+(\underline{H}_{\mathbb{Z}}, \psi) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_* \mathbb{Z}).$$

Let $(V_{\mathbb{Z}}, B)$ be a $\mathbb{Z}$-module equipped with a symmetric bilinear form, isomorphic to $(H_{s\mathbb{Z}}, \psi)$, and retain the notation of 5.4. Let n be an integer,

$$\Gamma = \{\gamma \in \mathrm{SO}(V_{\mathbb{Z}}) \mid \gamma \equiv 1 \pmod{n}\}$$

and

$$\widetilde{\Gamma} = \{\gamma \in \mathrm{CSpin}(V_{\mathbb{Q}}) \mid \alpha(\gamma) \equiv 1 \pmod{n} \text{ in } C^+(V_{\mathbb{Z}})\}.$$

Assume n is large enough that Γ and $\widetilde{\Gamma}$ are torsion-free.

There is a finite étale cover $v : S_1' \to S$ such that the local system $v^* \underline{H}_{\mathbb{Z}/n\mathbb{Z}}$ is trivial. Choose an isomorphism σ_n between this local system and the constant local system $\underline{V}_{\mathbb{Z}/n\mathbb{Z}}$. Let $s \in S_1'$ and let $\sigma : V_{\mathbb{Z}} \xrightarrow{\sim} H_{s\mathbb{Z}}$ be an isometry such that $\sigma \equiv \sigma_n \pmod{n}$ (we choose σ_n so that such σ exist). The Hodge structure h_s on H_s defines $\sigma^{-1}(h_s) \in X$. The image of $\sigma^{-1}(h_s)$ in X/Γ depends only on s. We thus define

$$m : S_1' \longrightarrow X/\Gamma$$

and an isomorphism compatible with the polarizations

$$\underline{H} \simeq m^* \underline{V}.$$

Let

$$S_1 = S_1' \times_{X/\Gamma} X/\widetilde{\Gamma} :$$

$$\begin{tikzcd}
 & & S_1 \arrow[dl, "p_1"'] \arrow[dr, "p_2"] & \\
S & S_1' \arrow[l, "v"'] \arrow[dr, "m"'] & & X/\widetilde{\Gamma}. \arrow[dl] \\
 & & X/\Gamma &
\end{tikzcd}$$

On S_1, one has

a) an isomorphism $(vp_1)^*H \simeq (mp_1)^*\underline{V}$;

b) a variation of Hodge structure, the pullback by p_2 of the one on $X/\widetilde{\Gamma}$ defined by the representation $(C^+(V_{\mathbb{Q}}))_s$ and the lattice $C^+(V_{\mathbb{Z}})$.

By 5.2, 5.6 and 4.5, this variation of Hodge structures corresponds to an abelian scheme $a : A \to S_1$.

c) Let C be the algebra $C^+(V_{\mathbb{Z}})$. The abelian scheme constructed in b) has complex multiplication by C, and the variation of Hodge structures $\underline{\mathrm{End}}_C(R^1a_*\mathbb{Z})$ is

$$p_2^*((\underline{C}^+(V_{\mathbb{Z}}))_{\mathrm{ad}}).$$

Combining this with a), one obtains an isomorphism of local systems of algebras and of variations of Hodge structures

$$\underline{C}^+(vp_1^*H) \simeq \underline{\mathrm{End}}_C(R^1a_*\mathbb{Z}).$$

This proves 5.7.

6. $K3$ Surfaces

6.1. A polarized $K3$ surface over a field K of characteristic 0 is a $K3$ surface over K, say Y, endowed with an element $\eta \in \mathrm{NS}(Y) = \mathrm{Pic}(Y)$ which is the class of an ample invertible sheaf. For $K = \mathbb{C}$, we shall identify $\mathrm{NS}(Y)$ with a subset of $H^2(Y, \mathbb{Z})$.

6.2. A *family of* $K3$ *surfaces* parametrized by a scheme S of characteristic 0 is a proper flat morphism $f : Y \to S$ whose fibers are $K3$ surfaces. A *polarization* of $f : Y \to S$ is a section η of $\mathrm{Pic}_S(Y)$ which, for every $s \in S$, is a polarization of $Y_s = f^{-1}(s)$.

For every prime number ℓ, we identify η with a section of $R^2f_*\mathbb{Z}_\ell$; we denote by $P^2f_*\mathbb{Z}_\ell$ the orthogonal complement of η (for the cup product); it is a subsheaf of $R^2f_*\mathbb{Z}_\ell$ in $\mathbb{Z}_\ell$-modules. The cup product

$$R^2f_*\mathbb{Z}_\ell \otimes R^2f_*\mathbb{Z}_\ell \longrightarrow R^4f_*\mathbb{Z}_\ell$$

and the trace morphism

$$R^4f_*\mathbb{Z}_\ell \xrightarrow{\sim} \underline{\mathbb{Z}}_\ell(-2)$$

induce

(6.2.1) $$\psi_\ell : P^2f_*\mathbb{Z}_\ell(1) \otimes P^2f_*\mathbb{Z}_\ell(1) \longrightarrow \underline{\mathbb{Z}}_\ell.$$

6.3. Take S to be a scheme of finite type over $\mathbb{C}$. Then $R^2f_*\mathbb{Z}$ is a variation of Hodge structures on S; η is everywhere of type $(1,1)$ and its orthogonal complement $P^2f_*\mathbb{Z}$ is again a variation of Hodge structures. As above, the cup product defines

(6.3.1) $$\psi : P^2f_*\mathbb{Z}(1) \otimes P^2f_*\mathbb{Z}(1) \longrightarrow \underline{\mathbb{Z}}.$$

This time, the negative of ψ is a polarization of the variation of Hodge structures $P^2 f_* \mathbb{Z}(1)$. Moreover, ψ_ℓ is obtained from ψ via the isomorphism

$$(P^2 f_* \mathbb{Z}(1)_{\mathbb{Z}}) \otimes \mathbb{Z}_\ell \simeq P^2 f_* \mathbb{Z}_\ell(1).$$

For every $s \in S$, the fiber at s, $(P^2 f_* \mathbb{Z}(1))_s$, is the primitive part of the cohomology $H^2(Y_s, \mathbb{Z})$. The Hodge structure on $(P^2 f_* \mathbb{Z}(1))_s$ is of type $\{(-1,1),(0,0),(1,-1)\}$, with $h^{-1,1} = 1$ ((1.1)(i)).

The action of $\pi_1(S,s)$ on $H^2(Y_s, \mathbb{Z})$ induces

$$\pi : \pi_1(S,s) \longrightarrow \mathrm{O}(P^2(Y_s,\mathbb{Z})(1)_{\mathbb{Z}}, \psi). \tag{6.3.2}$$

6.4. Proposition. *For every complex polarized $K3$ surface Y_0, there exists a family of polarized $K3$ surfaces $f : Y \to S$ as above, with $Y_s \simeq Y_0$ for some $s \in S$, such that the image of π in (6.3.2) has finite index.*

Let $f_T^\wedge : Y_T^\wedge \to T^\wedge$ be the versal formal deformation of Y_0. Since $H^0(Y_0, T^1) = 0$, $T^\wedge$ is even a formal moduli scheme for Y_0, but this is immaterial. Since $H^2(Y_0, T^1) = 0$ ((1.1)(i)), $T^\wedge$ is the spectrum of a formal power-series ring over $\mathbb{C}$. From $f_T^\wedge$ one obtains a (non-polarized) variation of Hodge structures on $T^\wedge$. By (1.1.1), this is the universal deformation of {the Hodge structure $H^2(Y_0,\mathbb{Z})$, endowed with the cup product}.

Let $\bar{\eta}$ be the image of η under the composite map

$$H^2(Y_0,\mathbb{Z}) \longrightarrow H^2_{\mathrm{DR}}(Y_T^\wedge/T^\wedge) \longrightarrow R^2 f^\wedge_{T*}\mathcal{O}.$$

Let $S^\wedge$ be the formal subscheme of $T^\wedge$ with equation $\bar{\eta} = 0$, and let $f^\wedge : Y^\wedge \to S^\wedge$ be induced by $f_T^\wedge$. Since $h^{0,2} = 1$, $S^\wedge \subset T^\wedge$ is defined by one equation; this equation is part of a regular system of parameters, so $S^\wedge$ is a power-series ring. On $S^\wedge$, η comes from an ample invertible sheaf $\mathcal{O}(1)$ on $Y^\wedge$; hence $f^\wedge$ is algebraizable. Moreover, on $S^\wedge$, the polarized variation of Hodge structure $(P^2(Y^\wedge/S^\wedge,\mathbb{Z})(1), \psi)$ is a universal deformation of $(P^2(Y_0,\mathbb{Z})(1), \psi)$.

Put $S^\wedge = \mathrm{Spec}(A^\wedge)$, and express $A^\wedge$ as an inductive limit of rings of finite type over $\mathbb{C}$. Since $Y^\wedge$ is algebraizable, there is a Cartesian diagram

$$\begin{tikzcd} Y^\wedge \arrow[r] \arrow[d, "f^\wedge"] & Y \arrow[d, "f"] \\ S^\wedge \arrow[r] & S \end{tikzcd}$$

with S of finite type over $\mathbb{C}$ and Y a polarized family of $K3$ surfaces over S (we do not assume that $S^\wedge$ is a completion of S). We may take S normal and connected; let $s \in S$ be the image of the closed point of $S^\wedge$.

In fact, by Artin's approximation theorem [1], one can find S such that $S^\wedge$ is a completion of S. Then S is smooth in a neighborhood of s. The simplifications resulting from such a choice are mostly psychological.

Let us show that $f : Y \to S$ satisfies 6.2. After replacing S by a finite étale cover, one defines, as in 5.7, a holomorphic map

$$m : S \longrightarrow X/\Gamma$$

from S to a moduli space of polarized Hodge structures such that the inverse image of the universal family of Hodge structures on X/Γ is $(P^2(Y/S, \mathbb{Z})(1), \psi)$.

By Borel (5.1), m is a morphism of schemes. The morphism m is dominant, since the composite $m^\wedge : S^\wedge \to X/\Gamma$ is dominant. It follows (see, for example, P. Deligne, Théorie de Hodge II. Publ. Math. IHES 40, Lemma 4.4.17) that $m(\pi_1(S, s))$ has finite index in $\pi_1(X/\Gamma, m(s)) = \Gamma$, which itself has finite index in the orthogonal group.

6.5. Proposition. *Let Y_0 be a polarized $K3$ surface over a field K of characteristic 0. There exist:*

a) *a scheme S of finite type over $\mathbb{Q}$, and a family $f : Y \to S$ of polarized $K3$ surfaces, parametrized by S;*

b) *a finite extension K' of K and $v : \mathrm{Spec}(K') \to S$ such that v^*Y is isomorphic to $Y_0 \otimes_K K'$;*

c) *an abelian scheme $a : A \to S$ over S;*

d) *a $\mathbb{Z}$-algebra C and $\mu : C \to \mathrm{End}_S(A)$;*

e) *an isomorphism of sheaves of $\mathbb{Z}_\ell$-algebras on S*

$$u : C^+(P^2 f_*\mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_*\mathbb{Z}_\ell).$$

We reduce to the case where K is of finite type over $\mathbb{Q}$. Choose a complex embedding $\sigma : K \to \mathbb{C}$ and apply 6.4 to $Y_0 \otimes_K \mathbb{C}$ to obtain a family of polarized $K3$ surfaces $f_1 : Y_1 \to S_1$. Apply 5.7 to f_1; one obtains a family $f_2 : Y_2 \to S_2$ still satisfying the conclusion of 6.4, and, over S_2, an abelian scheme $a_2 : A_2 \to S_2$ with complex multiplication $\mu_2 : C \to \mathrm{End}_{S_2}(A_2)$ and an isomorphism

$$u_2 : C^+(P^2 f_{2*}\mathbb{Z}(1), \psi) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_{2*}\mathbb{Z})$$

of local systems of algebras. The system of complex schemes and morphisms

$$\overline{E}_2 = (S_2, f_2, Y_2, a_2, A_2, \mu_2)$$

is defined by finitely many equations. Hence there exists an extension K_3 of K of finite type, with $K \subset K_3 \subset \mathbb{C}$, and, over K_3, a system

$$\overline{E}_3 = (S_3; f_3, Y_3, a_3, A_3, \mu_3)$$

which, after extension of scalars from K_3 to $\mathbb{C}$, gives $\overline{E}_2$.

6.5.1. Lemma. *There exists a unique isomorphism of sheaves of $\mathbb{Z}_\ell$-algebras*

$$u_3 : C^+(P^2 f_{3*}\mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_{3*}\mathbb{Z}_\ell).$$

It is enough to verify that such an isomorphism exists and is unique after extending scalars to an algebraic closure of K_3, for example $\mathbb{C}$. Existence holds by construction, and uniqueness follows from 3.5. Indeed, for $s \in S_2$, the image of $\pi_1(S_2, s)$ in $\mathrm{O}(P^2(Y_{2s}, \mathbb{Z}), \psi)$ has finite index and therefore contains a Zariski-dense subgroup of SO.

Let T be an integral scheme of finite type over $\mathbb{Q}$, with function field K_3. For a suitable T, $\overline{E}_3$ is the generic fiber of a system $\overline{E} = (S, f, Y, a, A, \mu)$ over T, and u_3 extends to

$$u : C^+(P^2 f_*\mathbb{Z}_\ell(1), \psi_\ell) \xrightarrow{\sim} \underline{\mathrm{End}}_C(R^1 a_*\mathbb{Z}_\ell).$$

The existence of K' as in b) is easily verified, and this completes the proof.

6.6. We prove Theorem 1.3. Let $f_V : Y_V \to \mathrm{Spec}(V)$ be as in 1.2(i). Let K be the fraction field of V, and apply 6.5 to the surface $Y_K = Y \otimes_V K$ over K, endowed with any polarization. After replacing V by its normalization in a finite extension of K, one may suppose that there exists $\overline{E} = (S, f, Y, a, A, \mu, u)$ as in 6.5, such that Y_K is the inverse image of Y under $v : \mathrm{Spec}(K) \to S$. Taking the inverse image gives an abelian variety with complex multiplication (A_K, μ) over K.

Let $\overline{K}$ be an algebraic closure of K, and let $\overline{k}$ be the corresponding algebraic closure of k (the residue field of the normalization of V in $\overline{K}$). From u one obtains an isomorphism of $\mathrm{Gal}(\overline{K}/K)$-modules

$$C^+(P^2(Y_{\overline{K}}, \mathbb{Z}_\ell)(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(H^1(A_{\overline{K}}, \mathbb{Z}_\ell)). \tag{6.6.1}$$

Since Y_V is smooth over V, the inertia group I acts trivially on

$$H^2(Y_{\overline{K}}, \mathbb{Z}_\ell) \simeq H^2(Y_{\overline{k}}, \mathbb{Z}_\ell).$$

The polarization defines an invariant η in $H^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, whose orthogonal complement is $P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, and I acts trivially on

$$C^+(P^2(Y_{\overline{K}}, \mathbb{Z}_\ell)(1), \psi_\ell) \simeq C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell).$$

Since I acts trivially on $\mathrm{End}_C(H^1(A_{\overline{K}}, \mathbb{Z}_\ell))$ and commutes with the complex multiplication, it acts through the center of $C \otimes \mathbb{Q}_\ell$. It is also known that a subgroup of finite index in I acts unipotently on $H^1(A_{\overline{K}}, \mathbb{Z}_\ell)$; it follows that I acts through one of its finite quotients. After a further finite extension of K, one may therefore suppose that A_K has good reduction over V.

For $A_{\overline{k}}$, the special fiber of its reduction, (6.6.1) reduces to an isomorphism of $\mathrm{Gal}(\overline{k}/k)$-modules

$$C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell) \xrightarrow{\sim} \mathrm{End}_C(H^1(A_{\overline{k}}, \mathbb{Z}_\ell)). \tag{6.6.2}$$

By Weil's theory for A_k, the eigenvalues of Frobenius acting on $C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell)$ are algebraic numbers all of whose complex conjugates have absolute value 1. We have

$$\bigwedge^2 (P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)) \subset C^+(P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1), \psi_\ell),$$

and, since $\dim P^2(Y_{\overline{k}}, \mathbb{Z}_\ell) > 2$ (as is the case here), it follows that the eigenvalues of Frobenius acting on $P^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$ likewise have complex absolute value 1. The same assertion holds for $H^2(Y_{\overline{k}}, \mathbb{Z}_\ell)(1)$, and the theorem follows.

7. Remarks

7.1. Let H be a rational Hodge structure. We shall call the *Mumford–Tate group* of H the smallest algebraic subgroup G of $\mathrm{GL}(H_{\mathbb{Q}})$, defined over $\mathbb{Q}$, such that $G(\mathbb{R})$ contains the image of $h : \underline{S} \to \mathrm{GL}(H_{\mathbb{R}})$. For polarizable H, G is reductive. This definition differs from Mumford's: if H has weight $n \neq 0$, our G contains the scalar homotheties. By 2.9, every homogeneous representation of G carries a natural Hodge structure.

One of the key points in the proof of 1.3 was that the Hodge structure $H^2(X, \mathbb{Q})$, for X a $K3$ surface, "is expressible" in terms of abelian varieties, in the following sense.

7.2. Definition. A rational Hodge structure H is expressible in terms of abelian varieties if it belongs to the smallest category of rational Hodge structures stable under direct factors, direct sums, and tensor products, and containing the $H^1(A, \mathbb{Q})$ for A an abelian variety, as well as the $\mathbb{Q}(n)$.

In this paper, we show that this property of $K3$ surfaces is exceptional.

Consider the following property of a rational Hodge structure with Mumford–Tate group G.

> $(*)$ The Hodge structure on the adjoint representation $\mathrm{Lie}(G)$ of G is purely of type $\{(-1, 1), (0, 0), (1, -1)\}$.

7.3. Proposition. *If a rational Hodge structure H is expressible in terms of abelian varieties, then condition $(*)$ holds.*

Indeed,

(a) the Hodge structures $H^1(A, \mathbb{Q})$ and $\mathbb{Q}(n)$ satisfy $(*)$;

(b) condition $(*)$ is stable under direct factors, direct sums, and tensor products.

7.4. Analogous arguments show that if H is expressible in terms of abelian varieties, then the reductive group $G_{\mathbb{C}}$ has no "factors" of exceptional type.

7.5. Proposition. *Let $\underline{H}$ be a polarized variation of Hodge structures on an irreducible scheme S. For $s \in S$ outside a meagre subset of S, there exists a finite-index subgroup of $\pi_1(S, s)$ whose image in $\mathrm{GL}(H_{s\mathbb{Z}})$ is contained in the Mumford–Tate group of H_s.*

At every point of S, the Mumford–Tate group G_s of H_s may be described as follows. It is the algebraic subgroup of $\mathrm{GL}(H_{s\mathbb{Q}})$ that fixes all tensors $t \in H_{s\mathbb{Q}}^{\otimes n} \otimes H_{s\mathbb{Q}}^{*\otimes n}$ $(n > 0)$ that are rational of type $(0,0)$. From this description it follows readily that, outside a meagre subset M of S, G_s is locally constant: for $s \notin M$, the largest subspace of type $(0,0)$, $V(n)_s \subset H_{s\mathbb{Q}}^{\otimes n} \otimes H_{s\mathbb{Q}}^{*\otimes n}$, is locally constant and defines a local system $V(n)$ on S. This local system underlies a polarized variation of Hodge structures of type $(0,0)$; hence $V(n)_s$ carries a positive-definite quadratic form invariant under $\pi_1(S,s)$, and $\pi_1(S,s)$ acts on $V(n)_s$ through a finite group. One concludes by noting that there is a finite sequence of integers n_i such that G_s is the algebraic subgroup of $\mathrm{GL}(H_{s\mathbb{Q}})$ acting trivially on the $V(n_i)_s$.

7.6. Consider a Lefschetz pencil of hypersurfaces of degree d and odd dimension $2r-1$.

$$\begin{tikzcd} X \arrow[r, hook] \arrow[d, "f"'] & \mathbb{P}^{2r}(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C}) \arrow[dl, "pr_2"] \\ \mathbb{P}^1(\mathbb{C}). & \end{tikzcd}$$

Let $U \subset \mathbb{P}^1(\mathbb{C})$ be the complement of the finite set of $t \in \mathbb{P}^1(\mathbb{C})$ for which the hypersurface $X_t = f^{-1}(t)$ is singular, and let $s \in U$. The cup product is an alternating form ψ on $H^{2r-1}(X_s, \mathbb{Z})$, preserved by monodromy, whence

$$m : \pi_1(U, s) \longrightarrow \mathrm{Sp}(H^{2r-1}(X_s, \mathbb{Z}), \psi).$$

By a theorem of Kajdan and Margulis, the image of m is Zariski-dense in the symplectic group. By 7.5, for s outside a meagre (in fact, countable) subset of $\mathbb{P}^1(\mathbb{C})$, the Mumford–Tate group of the rational Hodge structure $H^{2r-1}(X_s, \mathbb{Q})$ is therefore the entire group of symplectic similitudes. If this Hodge structure is not of Hodge level at most one (i.e. if some h^{pq} with $|q-p| \neq 1$ is nonzero), it follows from 7.3 that it is not expressible in terms of abelian varieties.

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Buildings of Generalized Braid Groups

Pierre Deligne (Bures-sur-Yvette)

Introduction

The main result of this work is the following.

Theorem. Let V be a finite-dimensional real vector space, let $\mathcal{M}$ be a finite set of homogeneous hyperplanes of V, let V_C be the complexification of V, and let

$$Y = V_C - \bigcup_{M \in \mathcal{M}} M_C$$

Assume that the connected components of $V - \bigcup_{M\in\mathcal{M}} M$ are open simplicial cones. Then Y is a $K(\pi,\, 1)$.

Let V be as above, and let $W \subset \mathrm{GL}(V)$ be a finite group generated by reflections. Suppose that no nonzero vector of V is fixed by W :

$$V^W = 0$$

Let Φ be any Euclidean structure on V invariant under W, and let $\mathcal{M}$ be the set of hyperplanes M such that the orthogonal reflection with respect to M belongs to W. It is then known that $(V,\, \mathcal{M})$ satisfies the hypothesis of the theorem, and that W acts freely on the corresponding space Y_W. The quotient $X_W = Y_W/W$ is therefore also a $K(\pi,\, 1)$.

This result had been conjectured by Brieskorn. It is new only for W of type H_3, H_4, E_6, E_7, or E_8 (Brieskorn [2]). We also give a new proof that the fundamental group of X_W is the generalized braid group $\widetilde{W}$ corresponding to W (Brieskorn [3]).

Reduced to the particular case considered above, the plan of the proof is as follows.

a) We construct a building $I(\widetilde{W})$ on which $\widetilde{W}$ acts, and a set $\mathcal{S}$ of spheres in $I(\widetilde{W})$, isomorphic to the unit sphere of V. The group $\widetilde{W}$ acts strictly simply transitively on $\mathcal{S}$.

b) We show that, up to homotopy, $I(\widetilde{W})$ is the wedge of the spheres $S \in \mathcal{S}$.

c) We relate X_W and $I(\widetilde{W})$.

The description of $\mathcal{S}$ and the possibility of c) had emerged during a conversation with Brieskorn and Tits in the spring of 1970. The ideas required to establish b) were supplied to me by Garside [4], whom I follow very closely in many places.

In Section 4, we determine the center of $\widetilde{W}$ and solve the word problem and the conjugacy problem in $\widetilde{W}$. These results were obtained independently by E. Brieskorn and K. Saito, who, like us, use Garside's methods [4].

The geometric language used and the techniques of proof are essentially due to Tits; I was able to familiarize myself with his ideas by attending one of his seminars and through conversations.

I am pleased to express my gratitude to him here.

0. Notation

(0.1) $L^+(D)$: the free monoid with identity generated by a set D.

(0.2) L(D): the free group generated by a set D.

(0.3) A^- : the closure of a subset A of a topological space.

(0.4) For x, y in a monoid with identity and $n \geq 0$, set

$$\mathrm{prod}(n; x, y) = xyxy\ldots \quad (n \text{ factors});$$

one has $\mathrm{prod}(2n; x, y) = (xy)^n$, $\mathrm{prod}(2n+1; x, y) = (xy)^n x$.

1. Galleries

(1.1) Let $\mathcal{M}$ be a finite set of hyperplanes (the walls) in a finite-dimensional real vector space V. We shall use the terminology (walls, chambers, faces, facets) of [1] V § 1 (the facets form a partition of V). The support P of a facet F is the intersection of the walls containing F; F is open in P. For every chamber A and every wall M, let $D_M(A)$ denote the set of chambers on the same side of M as A. If A and B are two chambers, let $D(A, B)$ denote the intersection of the $D_M(A)$ for which A and B are on the same side of M. Departing from the terminology of [1] IV 1 ex 15, we shall say that two chambers A and B are adjacent if A $\neq$ B and A and B have a face in common. We shall use the following elementary facts throughout.

(1.2) Lemma. (i) Let M be a wall of a chamber B. There exists a unique chamber B' adjacent to B and having M as a wall. M is the only wall separating B from B'.
(ii) Let B_1, B_2, B_3 be three chambers, and let $\mathcal{M}(B_i, B_j)$ be the set of walls separating B_i from B_j. Then

$$\mathcal{M}(B_1, B_3) = (\mathcal{M}(B_1, B_2) - \mathcal{M}(B_2, B_3)) \cup (\mathcal{M}(B_2, B_3) - \mathcal{M}(B_1, B_2))$$

A gallery of length n ($n \geq 0$), with source A and target B, is a sequence of chambers (C_0, ..., C_n) with A = C_0, C_{i+1} adjacent to C_i ($0 \leq$ i $\leq n$), and $C_n = B$. The composite of two galleries $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_m)$

is defined if $C_n = C'_0$, and is then

$GG' = (C_0, \ldots, C_n, C'_1, \ldots, C'_m)$.

If G is a gallery with source A, A.G denotes its target.

The opposite gallery G* of a gallery $G = (C_0, \ldots, C_n)$ is the gallery $(C_n, \ldots, C_0)$. One has $(GG')^* = G'^*$ G*.

The antipodal gallery $-G$ of a gallery $G = (C_0, \ldots, C_n)$ is the sequence of antipodal chambers $-G = (-C_0, \ldots, -C_n)$.

One has $-(GG') = (-G)(-G')$, and $-(G^*) = (-G)^*$.

The distance d(A, B) from a chamber A to a chamber B is the least length of a gallery from A to B. A gallery from A to B is minimal if it has length d(A, B).

(1.3) Proposition. The distance d(A, B) is the number of walls separating A from B. For a gallery G from A to B to be minimal, it is necessary and sufficient that it cross once the walls separating A from B and zero times the others.

It is clear that every gallery from A to B crosses every wall separating A from B, and it is enough to prove the existence of a gallery G from A to B whose length equals the number k of walls separating A from B. We proceed by induction on k. The intersection of the $D_M(A)$, for M a wall of A, is reduced to A. Thus either A = B, in which case we take G = (A), or there is a wall M of A separating A from B. Let A' be the chamber adjacent to A having M as a wall. M is the only wall separating A from A', so a wall N separates A' from B if and only if $M \neq N$ and N separates A from B. By the induction hypothesis, there is a gallery G' of length k − 1 from A' to B, and we take $G = (AA')G'$.

(1.4) Corollary. Let A, B, C be three chambers. The following conditions are equivalent.
(i) d(A, C) = d(A, B) + d(B, C), i.e. there exists a minimal gallery from A to C passing through B.
(ii) For a wall to separate A from C, it is necessary and sufficient that it separate A from B or B from C. In other words, $\mathcal{M}(A, B) \subset \mathcal{M}(A, C)$ (cf. 1.2(ii)).
(iii) $B \in D(A, C)$.

(1.5) Proposition. Let P be an intersection of walls, let V_P = V/P, let $pr_P : V \longrightarrow V_P$ be the projection, and let $\mathcal{M}_P$ be the set of hyperplanes N of V_P such that $pr_P^{-1}(N)$ is a wall. Denote by π'_P the unique map from the facets of (V, $\mathcal{M}$) to those of (V_P, $\mathcal{M}_P$) such that $pr_P(F) \subset \pi'_P(F)$.
(i) π'_P respects the incidence relation $F_1 \subset F_2^-$: if $F_1 \subset F_2^-$, then
codim(F_1 in F_2^-) $\geq$ codim($\pi'_P[F_1]$ in $\pi'_P[F_2]$).
π'_P takes chambers to chambers; if A and B are adjacent, either $\pi'_P(A) = \pi'_P(B)$, or $\pi'_P(A)$ and $\pi'_P(B)$ are adjacent.
(ii) Let F be a facet with support P (1.1). The restriction of π'_P to the set of facets E of (V, $\mathcal{M}$) such that $F \subset E^-$ is bijective. Like its inverse π_F, it preserves codimension, the incidence $E_1 \subset E_2^-$, and hence adjacency of chambers. We shall also write π_F for the bijection pr_P^{-1} between intersections of walls in V_P and intersections of walls in V containing P.
(iii) If $C = \pi_F(C')$, then, for every chamber X of (V, $\mathcal{M}$),
$\pi_F^{-1}D(C, X) = D(C', \pi'_P(X))$ and $\pi_F^{-1}\mathcal{M}(C, X) = \mathcal{M}_P(C', \pi'_P(X))$.

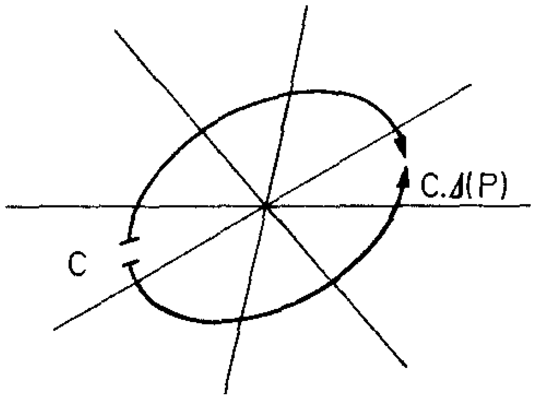
For $X = \pi_F(X')$, one has
$\pi_F(D(C', X')) = D(C, X)$ and $\pi_F(\mathcal{M}_P(C', X')) = \mathcal{M}(C, X)$;
π_F then induces a bijection between minimal galleries from C' to X' and from C to X.
The verification is left to the reader ((iii) follows from 1.4(ii) $\Longleftrightarrow$ (iii)).

(1.6) Notation. (i) For F a facet with support P, V_P, $\mathcal{M}_P$, pr_P, π'_P are also denoted V_F, $\mathcal{M}_F$, pr_F, π'_F.
(ii) Let P be an intersection of walls and C a chamber. Suppose that there is a facet of C (i.e. in C^-) with support P. This facet is then unique. Denote it by F(P), and set

$$C.\Delta(P) = \pi_{F(P)}(-\pi^{-1}_{F(P)}(C))$$

(1.7) Lemma. (i) A wall separates C from $C.\Delta(P)$ if and only if it contains P ((1.5)(iii)).
(ii) If M is a wall of C, then $C.\Delta(M)$ is the unique chamber adjacent to C and having M as a wall (cf. 1.5(iii), 1.2(i)).
(iii) Let $M \neq N$ be two walls of C whose intersection P contains a facet of C open in P. There are exactly two minimal galleries from C to $C.\Delta(P)$. One begins with $(C, C.\Delta(M))$; the other begins with $(C, C.\Delta(N))$.
Assertion (iii) is verified by reduction to rank two (cf. 1.5(iii)), where the drawing is
(1.7.1)

(1.7.1)



From now on we assume the following condition.

(1.8) Hypothesis. The chambers are open simplicial cones.
In other words, each chamber is the set of points with coordinates > 0 in a suitable basis of V.

(1.9.1) Remark. Let P be an intersection of walls.
(i) $(V_P, \mathcal{M}_P)$ still satisfies (1.8);
(ii) the set $\mathcal{M}^P$ of traces on P of the walls not containing P still satisfies (1.8).
If $\mathcal{M}$ is defined by a "Weyl group" W (see the introduction), $\mathcal{M}^P$ is in general no longer of this type.

(1.9.2) Remark. Hypothesis (1.8) ensures that if P is an intersection of walls of a chamber C, the condition of (1.6) is satisfied. The chamber $C.\Delta(P)$ is therefore defined.

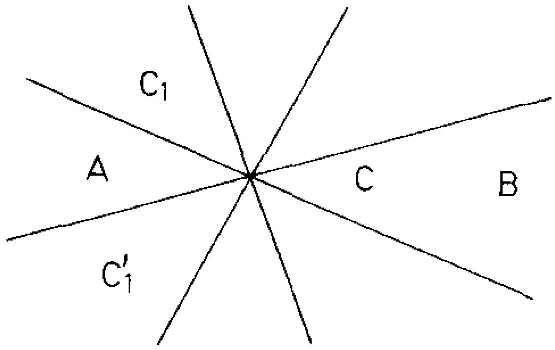
(1.10) Definition. Two galleries G and G' with the same endpoints are equivalent (notation: $G \sim G'$) if there exists a sequence of galleries $G = G_0, \ldots, G_n = G'$ ($n \geq 0$) such that G_{j+1} is obtained from G_j in the following way ($0 \leq$ j $< n$) :
a) there are decompositions $G_j = E_1 F E_2$, $G_{j+1} = E_1 F' E_2$;
b) there is a chamber C and two walls M, N of C such that F and F' are the two minimal galleries from C to $C.\Delta(M \cap N)$.

Composition of galleries, the operations $G \longrightarrow$ G* and $G \longrightarrow -G$, the source and target maps, and the "length" function are compatible with this equivalence relation and therefore pass to the quotient. We shall generally denote classes of galleries by lowercase letters. We say that a class of galleries e begins (respectively ends) with a class of galleries f if there exists g such that $e =$ fg (respectively $e =$ gf). We verify:

(1.11) Proposition. Equivalent galleries cross each wall the same number of times.

(1.12) Proposition. Two minimal galleries with the same endpoints are equivalent.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_n)$, with endpoints A and B. We proceed by induction on the length of the galleries under consideration. The case $n = 0$ is trivial, and the induction hypothesis allows us to suppose that $C_1 \neq C'_1$. Let M and M' be the walls separating $\mathrm{A} = C_0 = C'_0$ from C_1 and C'_1, and let $C = A.\Delta(M \cap M')$. The walls M and M' separate A from B. By reduction to rank two (1.5)(iii), it follows that every wall separating A from C separates A from B



(dessin dans $V_{M \cap M'}$; on omet d'écrire $\pi'_{M \cap M'}(\ )$).

(drawing in $V_{M\cap M'}$; we omit writing $\pi'_{M\cap M'}(\ )$).

Let F (respectively F') be the minimal gallery from A to C that passes through C_1 (respectively C'_1), and let E be a minimal gallery from C to B. The galleries FE and $F'E$ are minimal and equivalent. The minimal galleries G and FE (respectively G' and FE') begin with (A, C_1) (respectively with (A, C'_1)). The induction hypothesis implies that they are equivalent, and (1.12) follows.

(1.13) Notation. (i) Let $u(A, B)$ denote the equivalence class of minimal galleries from A to B.

(ii) For I a set of walls of a chamber C, with intersection P, set $\Delta(P) = u(C, C.\Delta(P))$ (cf. (1.9.2)). For I the set of all walls, set $\Delta = u(C, -C)$.

The interest of the notation lies in its ambiguity with respect to C.

(1.14) Proposition. Let A be a chamber and $\mathcal{G}$ a set of classes of galleries of bounded length with source A. Suppose that $\mathcal{G}$ satisfies the conjunction (i) of

(i_a) $(A) \in \mathcal{G}$.

(i_b) If $gh \in \mathcal{G}$, then $g \in \mathcal{G}$.

(i_c) Let g have target B, and let M, N be two walls of B. If $g\Delta(M)$ and $g\Delta(N)$ are in $\mathcal{G}$, then $g\Delta(M \cap N) \in \mathcal{G}$.

Then

(ii) There exists a unique class of galleries x with source A such that

$\mathcal{G} = \{\mathrm{g} \mid x \text{ begins with g}\}$.

Uniqueness is clear: if x begins with y and $x \neq y$, then y has strictly smaller length than x. Let x have maximal length in $\mathcal{G}$. To ensure that x has the required property, it is enough to prove the following assertion.

(*) Let g and M be such that (a) x begins with g and (b) x does not begin with $g\Delta(M)$, while $g\Delta(M) \in \mathcal{G}$. Then there exist g' and M' still satisfying (a) and (b), with g' strictly longer than g.

Since $g\Delta(M) \in \mathcal{G}$, maximality of x implies that $g \neq x$, and hence that x begins with $g\Delta(N)$ for a suitable N, necessarily distinct from M. By (i_c), one then has $g\Delta(M \cap N) \in \mathcal{G}$.

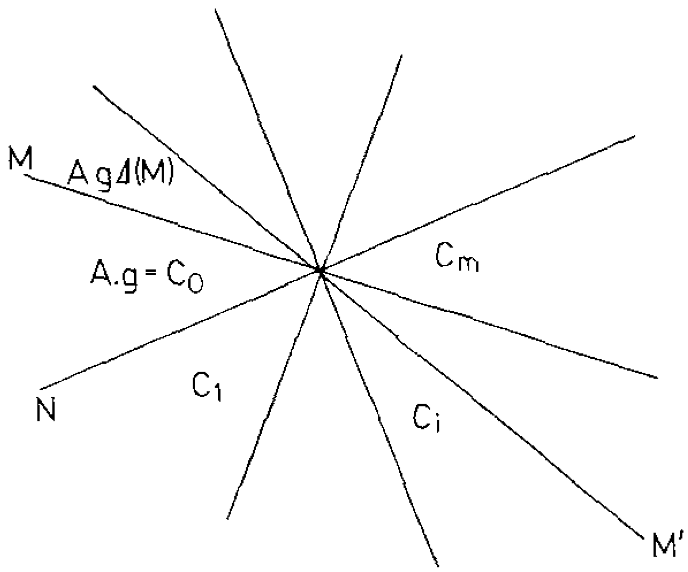
Since $g\Delta(M \cap N)$ begins with $g\Delta(M)$, x does not begin with $g\Delta(M \cap N)$.

Let $(C_0, \ldots, C_m)$ be the minimal gallery from A.g to $A.g\Delta(M \cap N)$ that begins with (A.g, $A.g\Delta(N)$). Let i ($0 <$ i $< m$) be the largest i such that x begins with $g' = g(C_0, \ldots, C_i)$. Then g' and the wall M' between C_i and C_{i+1} satisfy (a)(b).

(projection in $V_{M\cap N}$).

(1.15) Proposition. Let A be a chamber and $\mathcal{B}$ a set of chambers. For $\mathcal{B}$ to satisfy the conjunction (i) consisting of

(i_a) $A \in \mathcal{B}$.



(projection dans $V_{M \cap N}$).

(i_b) If $B \in \mathcal{B}$, then $D(A,\, B) \subset \mathcal{B}$.
(i_c) Let M and N be two walls of a chamber B. If A and B are on the same side of M and of N, and $B.\Delta(M)$ and $B.\Delta(N)$ belong to $\mathcal{B}$, then $B.\Delta(M \cap N) \in \mathcal{B}$.
it is necessary and sufficient that
(ii) There exists a unique chamber C such that $\mathcal{B} = D(A,\, C)$.

It is easy to verify that (ii) $\Rightarrow$ (i). To prove that (i) $\Rightarrow$ (ii), apply (1.14) to the set $\mathcal{G}$ of the $u(A,\, B)$ for $B \in \mathcal{B}$ (note that, by (1.3) and (1.11), every gallery G of class g such that some $u(A,\, B)$ begins with g is automatically minimal), and apply (1.4)(i) $\Longleftrightarrow$ (iii) to the gallery class $x = u(A,\, X)$ thereby obtained.

Applying criterion (1.15), we find:

(1.16) Corollary. Let A and B be two adjacent chambers separated by the wall M, and let C be on the same side of M as B. There exists C' such that

$$D(A, C) \cap D_M(B) = D(B, C')$$

(1.17) Corollary. (i) Let A, C_1, C_2 be three chambers. There exists C such that

(1.17.1) $D(A,\, C) = D(A,\, C_1) \cap D(A,\, C_2)$.
(ii) Let A and C_1 be two chambers, and let M be a wall of A. There exists C' such that

(1.17.2) $D(A,\, C') = D(A,\, C_1) \cap D_M(A)$.
In fact, (ii) is the special case of (i) with $C_2 = -(A.\Delta(M))$.

(1.18) Lemma. Let M, M', M'' be three distinct walls of a chamber A, $A_1 = A.\Delta(M)$, $B = A.\Delta(M \cap M' \cap M'')$, $B' = A.\Delta(M \cap M')$, and $B'' = A.\Delta(M \cap M'')$. For every chamber C, if $B' \in D(A_1,\, C)$ and $B'' \in D(A_1,\, C)$, then $B \in D(A_1,\, C)$.

The facet F of A open in $M \cap M' \cap M''$ is a facet of all the chambers A, A_1, B, B', and B''. Applying (1.5)(iii), we may therefore reduce to the case $\dim V = 3$. Applying (1.17)(ii), we may further suppose that A_1 and C are on the same side of M. Make these assumptions, and suppose that B', $B'' \in D(A_1,\, C)$. Since M' (respectively M'') separates A_1 from B' (respectively B''), A_1 and C are separated by M' and M'' (and are on the same side of M); hence A and C are separated by M, M', and M'', and $C = -A = B$.

The following key result is inspired by [4].

(1.19) Proposition. (i) Let A, B, C be three chambers, F a gallery from A to B, and G_1, G_2 two galleries from B to C. If $FG_1 \sim\ FG_2$, then $G_1 \sim\ G_2$.

(ii) Similarly, if $G_1E \sim G_2E$, then $G_1 \sim G_2$.

(iii) Let g be an equivalence class of galleries with source A. There exists a chamber C such that g begins with $u(A, B)$ if and only if $B \in D(A, C)$.

Statement (ii) follows from (i) by passing to opposite galleries.

Let $G = (C_0, \ldots, C_n)$ and $G' = (C'_0, \ldots, C'_n)$ be two galleries of length n with the same endpoints. If $n \geq 1$, set $G_1 = (C_1, \ldots, C_n)$ and $G'_1 = (C'_1, \ldots, C'_n)$. If moreover $C_1 \neq C'_1$, let M and M' be the walls separating $C_0 = C'_0$ from C_1 and C'_1, and set A $= C_0.\Delta(M \cap M')$. Consider the following relation $R_n(G, G')$ between galleries G and G' as above.

$R_n(G, G') \Longleftrightarrow$ one of the following holds:

α) $n = 0$;

β) $n \neq 0$, $C_1 = C'_1$, and $G_1 \sim G'_1$;

γ) $n \neq 0$, $C_1 \neq C'_1$, and there exists a gallery F from A to $C_n = C'_n$ such that $G_1 \sim u(C_1$, A).F and $G'_1 \sim u(C'_1$, A).F.

We shall prove by induction on n the following assertion:

(A_n) R_n is an equivalence relation.

Assume (A_i) for i $< n$, and prove (1.19.1) through (1.19.3) below.

(1.19.1) For galleries of length i $< n$, $R_i(G, G') \Longleftrightarrow G \sim G'$.

It is trivial that $R_i(G, G') \Rightarrow G \sim G'$; and, with the notation of (1.10), if $G \sim G'$, then $R_i(G_j, G_{j+1})$.

(1.19.2) Assertion (i) is true for FG_1 of length $< n$.

This follows from (1.19.1) and the second clause in the definition of the R_i.

(1.19.3) Assertion (iii) is true for g of length $< n$.

This follows from (1.15) applied to the set $\mathcal{B}$ of chambers B such that g begins with $u(A, B)$: hypothesis (i_c) of (1.15) is verified using (1.19.2), (1.18), and the third clause in the definition of the R_i.

It remains to prove that if G, G', and G'' are three galleries of length n from A to C, and $R_n(G, G')$ and $R_n(G, G'')$, then $R_n(G', G'')$. If $n = 0$, or $C_1 = C'_1$, or $C_1 = C''_1$, this is trivial.

For $n > 0$, denote by M, M', M'' the walls separating A $= C_0 = C'_0 = C''_0$ from C_1, C'_1, and C''_1. Distinguish two cases.

Case 1. $C_1 \neq C'_1 = C''_1$. Let $B = A.\Delta(M \cap M')$. By hypothesis,

$(C_1, \ldots, C_n) \sim u(C_1, B)F' \sim u(C_1, B)F''$,

$(C'_1, \ldots, C'_n) \sim u(C'_1, B)F'$,

$(C''_1, \ldots, C''_n) \sim u(C''_1, B)F''$.

It follows from (1.19.2) that $F' \sim F''$, and hence that $R_n(G', G'')$.

Case 2. $C_1 \neq C'_1 \neq C''_1 \neq C_1$. Let $B_1 = A.\Delta(M' \cap M'')$, $B' = A.\Delta(M \cap M')$, $B'' = A.\Delta(M \cap M'')$, and $B = A.\Delta(M \cap M' \cap M'')$. We have

$$G_1 = (C_1, \ldots, C_n) \sim u(C_1, B')F' \sim u(C_1, B'')F''$$

$$G'_1 = (C'_1, \ldots, C'_n) \sim u(C'_1, B')F'$$

$$G''_1 = (C''_1, \ldots, C''_n) \sim u(C''_1, B'')F''$$

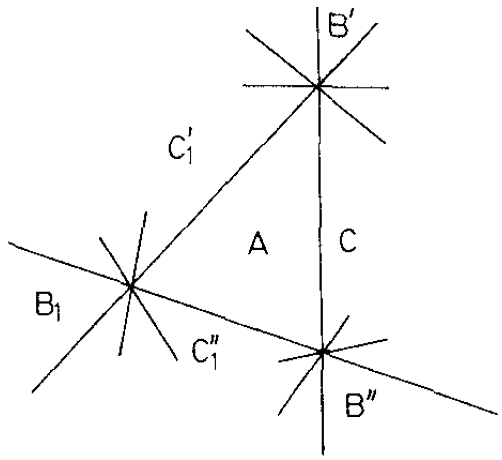
so that the class g_1 of G_1 begins with $u(C_1, B')$ and $u(C_1, B'')$. It then follows from (1.19.3) and Lemma (1.18) that g_1 begins with $u(C_1, B)$: $G_1 \sim u(C_1$, B)F. By (1.19.2), $F' \sim u(B'$, B)F and $F'' \sim u(B''$, B)F; hence $G_1 \sim u(C_1$, B)F, $G'_1 \sim u(C'_1$, B)F, and $G''_1 \sim u(C''_1$, B)F. Therefore,

$G'_1 \sim u(C'_1, B_1)u(B_1$, B)F

$G''_1 \sim u(C''_1, B_1)u(B_1$, B)F,

so that $R_n(G', G'')$.

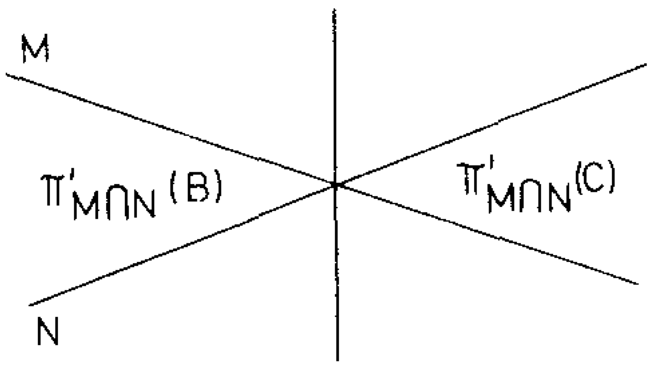
(drawing on the unit sphere of $V_{M \cap M' \cap M''}$).



(dessin sur la sphère unité de $V_{M\cap M'\cap M''}$).

(1.20) Corollary. Conditions (i) and (ii) of (1.14) are equivalent.

Assume (ii) and prove (i_c). Under the hypotheses of (i_c), if $x =$ gh (h with source B), then (1.19)(i) implies that h begins with $\Delta(M)$ and $\Delta(N)$. Let C be the chamber guaranteed by (1.19)(iii) for h. Since M and N separate B from C, $\pi'_{M\cap N}(C)$ can only be $\pi'_{M\cap N}(B).\Delta$, and (i_c) follows from (1.5)(iii).



(1.21) Corollary. For every chamber A, let n(A, i) be the number of classes of galleries with source A and length i. Set

$$f_A = \sum\nolimits_0^\infty n(A,i)t^i \in \mathbb{Z}[[t]]$$

For every chamber A, and for P running through the intersections of walls of A (including $\{0\}$ and V), one has

$$\sum\nolimits_P (-1)^{\operatorname{codim}(P)} t^{d(A,A.\Delta(P))} f_{A.\Delta(P)} = 1$$

This expresses that

a) the number of classes of galleries of length i that begin with $\Delta(P)$ is $n(A.\Delta(P)$, i − d(A, $A.\Delta(P)))$ ((1.19)(i));

b) classes of galleries that begin with $\Delta(P)$ and $\Delta(Q)$ begin with $\Delta(P\cap Q)$ (by (1.19)(iii)).

(1.22) Algorithm. Let $G = (A_0, \ldots, A_n)$ be a gallery of length $n \geq 1$, let $G_1 = (A_1, \ldots, A_n)$, let M be the wall separating A_0 from A_1, let C be the chamber whose existence is guaranteed by (1.19)(iii) (for G), and let C_1 be the analogous chamber for G_1. One can compute C recursively using the following formula (cf. (1.16)):

(1.22.1) $D(A_1, C) = D_M(A_1) \cap D(A_1, C_1)$.

Since $A_1 \in D(A, C)$, M indeed separates A from C and $C \in D_M(A_1)$. By (1.19)(i), one also has $D(A_1, C) \subset D(A_1, C_1)$, giving the inclusion $\subset$. Finally, if $B \in D_M(A_1) \cap D(A_1, C_1)$, then G_1 begins with $u(A_1, B)$ and G begins with $u(A, A_1).u(A_1, B) = u(A, B)$, which proves (1.22.1).

(1.23) Corollary. Let G and H be two composable galleries. Let A be the source of G, B that of H, and let C be guaranteed by (1.19)(iii): one has $H \sim\ u(B,\, C).H'$. Then, for every chamber D, for the class of GH to begin with $u(A,\, D)$, it is necessary (and sufficient) that the class of Gu(B, C) begin with $u(A,\, D)$.

(1.24) Proposition. Let A be a chamber, P an intersection of walls of A, F = F(P) (1.6), and g an equivalence class of galleries with source A. There exists a class of galleries g' of $(V_P,\, \mathcal{M}_P)$, with source $\pi_F^{-1}(A)$, such that, for every gallery H of $(V_P,\, \mathcal{M}_P)$, g begins with $\pi_F(H)$ if and only if g' begins with H.

Let $\mathcal{G}$ be the set of classes of galleries h with source $\pi_F^{-1}(A)$ in $(V_P,\, \mathcal{M}_P)$ such that g begins with $\pi_F(h)$. Apply (1.14) to $(V_P,\, \mathcal{M}_P)$ and to $\mathcal{G}$. By (1.19), condition (i) is satisfied (cf. the proof of (1.20)), and (1.24) follows from (1.14)(ii).

(1.25) One may regard the set of galleries as the set of arrows of a category $\mathrm{Gal}_0(V,\, \mathcal{M})$ whose objects are the chambers. Similarly, equivalence classes of galleries are the arrows of a quotient category $\mathrm{Gal}_+(V,\, \mathcal{M})$. Let A, B, and C be three chambers, E a gallery from A to B, and F a gallery from B to C. Although the general categorical conventions are different, we shall continue to write EF for the composite of E and F. The operation * (respectively $G \longrightarrow\ -G$) is an anti-equivalence (respectively an equivalence) of $\mathrm{Gal}_0(V,\, \mathcal{M})$ or $\mathrm{Gal}_+(V,\, \mathcal{M})$ with itself, inducing the identity (respectively $C \longrightarrow\ -C$) on the set of objects.

When there is no risk of confusion, we shall write simply Gal_0 and Gal_+ for $\mathrm{Gal}_0(V,\, \mathcal{M})$ and $\mathrm{Gal}_+(V,\, \mathcal{M})$ (or for a category $\mathrm{Gal}_0(V_P,\, \mathcal{M}_P)$ or $\mathrm{Gal}_+(V_P,\, \mathcal{M}_P)$). By abuse of language, we shall often call the arrows of Gal_+ galleries.

(1.26) Lemma. (i) In Gal_+, for every gallery g, one has

$$g\Delta = \Delta(-g)$$

(ii) For any g and h with source A in Gal_+, there exists n such that $g\Delta^n$ begins with h. If h is composed of k galleries $u(A_i,\, A_{i+1})$, one may take $n = k$.

Proceeding by induction, it is enough to prove (i) for g of length one. For $g = (B,\, C)$, one has

$$g\Delta = u(B,C)u(C,-C) = u(B,C)u(C,-B)u(-B,-C) = u(B,-B)u(-B,-C) = \Delta(-g)$$

For every chamber B, $\Delta = u(A,\, -A)$ begins with $u(A,\, B)$. Hence (ii) follows from (i) by induction on k.

The following proposition follows at once from (1.26), from its transform by *, and from (1.19)(i), (ii).

(1.27) Proposition. (i) In Gal_+, the set of all arrows admits a left and right calculus of fractions.

(ii) Let $\mathrm{Gal}(V,\, \mathcal{M})$, or simply Gal, be the category (a groupoid) obtained from Gal_+ by making all arrows invertible: call positive the arrows of Gal lying in the image of Gal_+. The canonical functor from Gal_+ to Gal is faithful, and every arrow g of Gal can be written in the forms $g = g_1\Delta^{-n} = \Delta^{-n}g_2$ with g_1 and g_2 positive.

If a category C has only one object A and the monoid Hom(A, A) is cancellative, saying that all arrows of C admit a left and right calculus of fractions means that Hom(A, A) satisfies the Öre condition on the left and on the right. The "theory of the calculus of fractions" used in (1.27) and below is an immediate generalization of the theory of Öre for embedding such a monoid in a group.

(1.28) For every wall M, the number of times that g in Gal_+ crosses M is well defined (1.11). This function of g extends additively to $g \in$ Gal. More generally, for every intersection of walls P, one could use the universal property of Gal to define functors

$$\pi'_P : \mathrm{Gal}(V,\mathcal{M}) \longrightarrow\ \mathrm{Gal}(V_P,\mathcal{M}_P)$$

(1.29) Let A be a chamber, P an intersection of walls of A, and F = F(P) (1.6). The function π_F induces a functor

(1.29.1) $\pi_F\ :\ \mathrm{Gal}_0(V_P,\, \mathcal{M}_P) \longrightarrow\ \mathrm{Gal}_0(V,\, \mathcal{M})$.

The image of this functor is stable under equivalence, and it induces a faithful functor

(1.29.2) $\pi_F\ :\ \mathrm{Gal}_+(V_P,\, \mathcal{M}_P) \longrightarrow\ \mathrm{Gal}_+(V,\, \mathcal{M})$.

From the theory of the calculus of fractions and (1.19)(i) follows the

(1.30) Proposition. Under the preceding hypotheses, the functor deduced from (1.29.2)

$$\pi_F : \mathrm{Gal}(V_P, \mathcal{M}_P) \longrightarrow \mathrm{Gal}(V, \mathcal{M})$$

is faithful.

(1.31) Proposition. Let C be a chamber, I and J two sets of walls of C, $K = \mathrm{I} \cap J$, P, Q, and R the intersections of the walls in I, J, and K, and F(P), F(Q), and F(R) the corresponding facets in C (1.6). The facet F(R) is the smallest facet containing F(P) and F(Q). In the groupoid Gal(V, $\mathcal{M}$), one has

$$\pi_{F(P)}(\mathrm{Gal}(V_P, \mathcal{M}_P)) \cap \pi_{F(Q)}(\mathrm{Gal}(V_Q, \mathcal{M}_Q)) = \pi_{F(R)}(\mathrm{Gal}(V_R, \mathcal{M}_R))$$

A chamber having F(P) and F(Q) as facets also has F(R) as a facet. This proves (1.31) for the objects.
Let $g \in \mathrm{Hom}_{\mathrm{Gal}}(A,\, B)$, and suppose that

$$g = \pi_{F(P)}(g_1') = \pi_{F(Q)}(g_2')$$

By (1.27)(ii), one has
$g_1' = \Delta(P)^{-n} g_1''$ and $g_2' = \Delta(Q)^{-n} g_2''$
with g_i'' positive, for $n \geq 0$ sufficiently large. Let $g_1 = \pi_{F(P)} g_1''$ and $g_2 = \pi_{F(Q)} g_2''$.
By (1.26)(ii) (transformed by *), $\Delta^n \Delta(P)^{-n}$ is positive; in $\mathrm{Gal}_+(V,\, \mathcal{M})$ one has

(1.31.1) $(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2$.

(1.31.2) Lemma. If $h \in \mathrm{Hom}_{\mathrm{Gal}_+}(A,\, B)$ begins both with $(\Delta^n \Delta(P)^{-n})$ and with $(\Delta^n \Delta(Q)^{-n})$, then h begins with $(\Delta^n \Delta(R)^{-n})$.

Let us deduce (1.31) from (1.31.2). By (1.31.2), one has
$(\Delta^n \Delta(P)^{-n}) g_1 = (\Delta^n \Delta(Q)^{-n}) g_2 = (\Delta^n \Delta(R)^{-n}) g_3$
with g_3 positive. Thus $g = \Delta(R)^{-n} g_3$. The positive gallery $g_3 = \Delta(R)^n\ g$ belongs to the image of $\pi_{F(P)}$ and of $\pi_{F(Q)}$. By (1.28), it therefore crosses zero times every wall not containing P or Q, i.e. not containing R. It consequently belongs to the image of $\pi_{F(R)}$.

Let us prove (1.31.2). Set A(P) = $A.\Delta.\Delta(P)$, and similarly for Q and R. The gallery $\Delta\Delta(P)^{-1}$ with source A is $u(A$, A(P)); the one with source A(P) is $u(A(P)$, A). It follows from (1.26)(i) that $\Delta\Delta(P) = \Delta(P)\Delta$. Hence

(1.31.3) $\Delta^n \Delta(P)^{-n} = u(A$, A(P)).u(A(P), A)u(A, A(P))... (n factors).

(1.31.4) Lemma. $A(R) \in D(A(P)$, A(Q)).

With the notation of (1.2), $\mathcal{M}(A\Delta$, A(P)) is the set of walls containing P, and $\mathcal{M}(A\Delta$, A(Q)) the set of those containing Q. By (1.2), $\mathcal{M}(A(P)$, A(Q)) is the set of those containing P or Q, but not R, namely $\mathcal{M}(A(P)$, A(R)) $\cup\ \mathcal{M}(A(Q)$, A(R)); apply (1.4).

By (1.19)(iii), a gallery class that begins with $\Delta\Delta(P)^{-1}$ and $\Delta\Delta(Q)^{-1}$ therefore also begins with $\Delta\Delta(R)^{-1}$, and (1.31.2) follows by induction from

(1.31.5) Lemma. Let P and R be intersections of walls of a chamber A, with $P \subset R$. Let h be in Gal_+ with source A. If h begins both with $\Delta^n \Delta(P)^{-n}$ ($n > 0$) and with $\Delta\Delta(R)^{-1}$, then h begins with $(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-1}$.

We may suppose that $n \geq 2$. Set $h = (\Delta\Delta(P)^{-1})h'$. With the preceding notation, h' then begins both with $\Delta\Delta(P)^{-1} = u(A(P)$, A) and with $u(A(P)$, A(R)). Apply (1.19)(iii).

The chamber C of loc. cit. is then separated from A(P) by every wall M that separates A(P) from A (i.e. $M \not\supset P$) or A(P) from A(R) (i.e. $M \supset P$ and $M \not\supset R$). Thus C is separated from $A(P)\Delta$ only by walls $M \supset R$, and $C \in D(A(P).\Delta,\ A(P).\Delta\Delta(R))$. Hence h' begins both with $(\Delta\Delta(R)^{-1})$ and with $(\Delta\Delta(P)^{-1})^{n-1}$. Proceeding by induction on n, one deduces that h' begins with $(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2}$, so that h begins with $(\Delta\Delta(P)^{-1})(\Delta\Delta(R)^{-1})(\Delta\Delta(P)^{-1})^{n-2}$; the conclusion follows by noting that

$$\Delta\Delta(P)^{-1}\Delta\Delta(R)^{-1} = \Delta\Delta(R)^{-1}\Delta\Delta(P)^{-1}$$

(1.32) Proposition. For F a facet of a chamber C, spanning an intersection of walls P, one has

$$\pi_F^{-1}(\mathrm{Gal}_+(V, \mathcal{M})) = \mathrm{Gal}_+(V_P, \mathcal{M}_P) \subset \mathrm{Gal}(V_P, \mathcal{M}_P)$$

Suppose that $\pi_F(\Delta^{-n} g) = h$ lies in $\mathrm{Gal}_+(V, \mathcal{M})$. Then $\pi_F(g) = \Delta(P)^n\ h$ and, by (1.28), every gallery H representing h crosses only walls passing through F. Thus $h = \pi_F(h_1)$, with h_1 in $\mathrm{Gal}_+(V_P,\ \mathcal{M}_P)$, and the conclusion follows from (1.30).

2. Buildings

(2.1) Let $(V,\ \mathcal{M})$ be as in § 1 (satisfying (1.8)), and let $r = \dim V$. Let S be the sphere of radius one in V, for any Euclidean structure on V (more intrinsically, one could set $S = V - \{0\}/\mathbb{R}_+^*$). The hyperplanes $M \in \mathcal{M}$ cut out a triangulation of S, and we shall also denote by S the corresponding simplicial space (simplicial scheme) and its geometric realization. We transfer to S the terminology used for V (chambers, facets, adjacency, gallery, . . .).

(2.2) Choose a "fundamental chamber" A_0 in S. We denote by I_+ the space constructed below; it depends on V, $\mathcal{M}$, and A_0, and is equipped with $q : I_+ \longrightarrow S$.

a) Let $\mathcal{L}_+$ be the set of equivalence classes of galleries g with source A_0,

$$\mathcal{L}_+ = \coprod_B \mathrm{Hom}_{\mathrm{Gal}_+}(A_0, B)$$

b) Let Z be the disjoint sum, indexed by $g \in \mathcal{L}_+$, of the closed simplex of S that is the closure of the open simplex of dimension $r-1$ which is the target of g :
$Z_+ = \coprod_{g\in\mathcal{L}_+}$ (target of $g)^-$.
Let q' be the evident map from Z_+ onto S.

c) I_+ (equipped with q) is a quotient of Z_+ (equipped with q'). If G is a gallery from A_0 to B and C is a chamber adjacent to B, one glues (target of $g)^-$ and (target of $g(BC))^-$ along the common closed face of their images in S.

(2.3) The space I_+ is decomposed into facets, images of facets of Z_+, each mapping bijectively onto a facet of S. Its chambers (respectively
faces and vertices) are its facets of dimension $r-1$ (respectively $r-2$ and 0).

The vertices of a facet F are the vertices contained in the closure $\overline{F}$ of F. The chambers of I_+ are indexed by $\mathcal{L}_+$; denote by $\widetilde{A}_0.g$ the one with index g and by $\widetilde{A}_0$ the one with index (A_0).

Let $g \in \mathrm{Hom}_{\mathrm{Gal}_+}(A_0,$ A), let $\widetilde{A} = \widetilde{A}_0.g$, and let $H = (C_0,\ \ldots,\ C_n)$ be a gallery from A to B, of class h.
Set
$\widetilde{A}.h = \widetilde{A}_0.gh$

The chambers $\widetilde{A}.(C_0,\ \ldots,\ C_i)$ $(0 \leq\ \mathrm{i} \leq\ n)$ form a gallery in I_+. Galleries obtained in this way will be called positive.

(2.4) Lemma. Let F be a facet of S, let A and B be two chambers of S having F as a facet, let $g \in \mathrm{Hom}_{\mathrm{Gal}_+}(A_0,$ A), and let $h = \mathrm{Hom}_{\mathrm{Gal}_+}(A_0,\ B)$. The following conditions are equivalent:

(a) in I_+, the facets $q^{-1}(F)$ of $\widetilde{A}_0.g$ and $\widetilde{A}_0.h$ coincide;

(b) $g^{-1}h \in \mathrm{Hom}_{\mathrm{Gal}}(A,\ B)$ lies in $\pi_F(\mathrm{Gal})$;

(c) there exist g_1 and h_1 in $\pi_F(\mathrm{Gal}_+)$ such that $gg_1 = hh_1$.

Condition (b) is an equivalence relation among galleries from A_0 to a chamber having F as a facet. It follows that (a) $\Rightarrow$ (b). That (b) $\Rightarrow$ (c) follows from (1.26)(i). Finally, (c) $\Rightarrow$ (a) is immediate from the definitions.

From (2.4)(a) $\iff$ (b) and (1.31), one immediately obtains the following result.

(2.5) Proposition. Each facet of I_+ is uniquely determined by the set of its vertices. Thus I_+ may be described as the geometric realization of the following simplicial scheme.

a) For x a vertex of S, let $\mathcal{G}(x)$ be the set of g in Gal_+ with source A_0 and target a chamber having x as a vertex. Let $\mathcal{G}(x)/\pi_x$ be the quotient of $\mathcal{G}(x)$ by the equivalence relation (2.4)(b) (for $F = x$). Then the set of vertices is

$$\coprod_x \mathcal{G}(x)/\pi_x$$

b) For a set E of vertices to span a simplex, it is necessary and sufficient that there exist a gallery g with source A_0 such that, for $y \in E$, g lies in $\mathcal{G}(qy)$ and y is the image of g in $\mathcal{G}(qy)/\pi_{qy}$.

(2.6) Proposition. Let F be a facet of I_+. There exists a gallery class g such that, for F to be a facet of a chamber $B = \widetilde{A}_0.h$, it is necessary and sufficient that

$$h = g\pi_F(h')$$

for a suitable h' in Gal_+ .

Take g to be a gallery of minimum length such that F is a facet of A $= \widetilde{A}_0.g$. Let $B = \widetilde{A}_0.h$ be a chamber having F as a facet. By (2.4)(a) $\iff$ (c), there exist galleries h_1 and h_2 in $(V_{qF}, \mathcal{M}_{qF})$ such that

$$g\pi_F(h_1) = h\pi_F(h_2)$$

Apply the transform by * of (1.24). In view of the minimality of g, one finds that h_1 ends with h_2; $h_1 = h'h_2$. By (1.19)(ii), one has

$$g\pi_F(h') = h,$$

which proves (2.6).

(2.7) Notation. Let A be a chamber of I_+. Denote by $S(A)$ the union of the closed chambers (A.(qA, B))$^-$, as B ranges over the chambers of S.

One verifies that $q|S(A)$ is an isomorphism between $S(A)$ and S.

(2.8) Definition. $\widehat{I}_+$ is the space obtained from I_+ by “filling in” the spheres $S(A)$.

More precisely, let B^r be a ball whose boundary is S. When working simplicially, take B^r to be the cone on the simplicial space S. Define $\widehat{I}_+$ from I_+ by attaching a family of copies $b(A)$ of B^r, indexed by the set of chambers of I_+. The attaching maps are

$$\partial b(A) \xrightarrow{\sim} S \xleftarrow{q} S(A).$$

The attaching maps are simplicial, so $\widehat{I}_+$ again appears as the geometric realization of a simplicial scheme (whose vertices are those of I_+ together with the “centers of the balls”). The space $\widehat{I}_+$ is equipped with an evident (simplicial) map

$$q : \widehat{I}_+ \longrightarrow B^r$$

(2.9) Proposition. The space $\widehat{I}_+$ is contractible.

The space I_+ is therefore the wedge of the spheres $S(A)$.

Let $(I_+)_n$ be the union of the closed chambers $(\widetilde{A}_0.g)^-$ for g of length $\leq n$, and let $(\widehat{I}_+)_n$ be the union of $(I_+)_n$ and the set of balls $b(A)$ for which $\partial b(A) \subset (I_+)_n$.

One has $(\widehat{I}_+)_0 = \widetilde{A}_0$ (contractible), and

$$\widehat{I}_+ = \varinjlim (\widehat{I}_+)_n$$

The proposition follows from

(2.9.1) Lemma. $(\widehat{I}_+)_n$ is a deformation retract of $(\widehat{I}_+)_{n+1}$.

We must construct a continuous family $(\varphi_t)_{0\leq t\leq 1}$ of continuous maps from $(\widehat{I}_+)_{n+1}$ to itself, with $\varphi_0 = \mathrm{Id}$, $\varphi_t|(\widehat{I}_+)_n = \mathrm{Id}$, and $\varphi_1((\widehat{I}_+)_{n+1}) = (\widehat{I}_+)_n$.

Let A $= \widetilde{A}_0.g$ be a chamber of $(I_+)_{n+1}$ that is not in $(I_+)_n$, and let F be a facet of A.

(2.9.2) Lemma. Suppose that there exists a chamber $B = \widetilde{A}_0.h$ in $(I_+)_{n+1}$, distinct from A, having F as a facet. Then there exists a chamber $B' = \widetilde{A}_0.h'$ in $(I_+)_n$ and a wall M of qB', containing qF, such that A $=$ $B'.\Delta(M)$.

Let g_0 be the gallery considered in (2.6). One has

$g = g_0\pi_F(g_1)$ and $h = g_0\pi_F(h_1)$.

The hypotheses imply that the length $\lg(g_1) \neq 0$ (otherwise, since A $\neq B$, one would have $\lg(h) > \lg(g) =$ n+1). For a suitable wall M of qA, one then has $g_1 = h'\Delta(M)$, and set $B' = \widetilde{A}_0.h'$.

For the closed chamber A^-, $A^- \cap (I_+)_n$ is therefore the union of a nonempty set of closed faces; and, in $(I_+)_{n+1}$, the points of $A^- - (A^- \cap (I_+)_n)$ belong only to the single closed chamber A^-.

We distinguish two cases.

Case 1. $A^- \cap (I_+)_n \neq \partial A^-$.

In this case, there is no sphere $S(B)$ with

$A \subset S(B) \subset (I_+)_{n+1}$.

We take $\varphi_t|A^-$ to be a collapse of A^- onto $A^- \cap (I_+)_n$.

Case 2. $A^- \cap (I_+)_n = \partial A^-$.

Set A $= \widetilde{A}_0.g$. For every wall M of qA, it then follows from (2.7.2) that g ends with $\Delta(M)$.

By ((1.19)(iii)), g therefore ends with Δ : one has A $= B.\Delta$, with B uniquely determined by A, by (1.19)(ii). In passing from $(\widehat{I}_+)_{n+1}$ to $(\widehat{I}_+)_n$, A and the interior of the ball $b(B)$ disappear. We take $\varphi_t|b(B)$ to be a collapse of $b(B)$ onto $S(B) - A$.

By what was seen in case 1, every ball that disappears is of the preceding type. This completes the construction of φ_t and proves (2.7).

(2.10) The space I, equipped with $q : \mathrm{I} \longrightarrow S$, is defined like I_+, replacing Gal_+ by Gal.

a) Set $\mathcal{L} = \coprod_B \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$.

b) Set $Z = \coprod_{g\in\mathcal{L}}$ (target of $g)^-$.

c) Let q' be the evident map from Z onto S. The space I, equipped with q, is a quotient of Z equipped with q'. For $g \in \mathcal{L}$ with target B and C a chamber adjacent to B, one glues (target of $g)^-$ and (target of $g(BC))^-$ along the common closed face of their images in S. In other words, if two

chambers B and C have a common facet F, if $g \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, B)$ and $h \in \mathrm{Hom}_{\mathrm{Gal}}(A_0, C)$, then in I the closed facets $q'^{-1}(F)^-$ of (target of $g)^-$ and of (target of $h)^-$ are identified if and only if

$hg^{-1} \in \pi_F\ \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}(B), \pi_F^{-1}(C))$.

(2.11) As for I_+, one defines the chambers, faces, and open or closed facets of I. One defines $\widetilde{A}_0$ and $\widetilde{A}.h$ (for $\widetilde{A}$ a chamber and h in Gal with source $q\widetilde{A}$) as in (2.3). The analogue of (2.4)(a) $\Longleftrightarrow$ (b) is here trivially true. As in (2.5), one then deduces from (1.31) that each facet of I is uniquely determined by the set of its vertices, and I appears as the geometric realization of a simplicial scheme admitting a description of type (2.5). For every chamber A of I, define a sphere $S(A)$ as in (2.7); q induces an isomorphism from $S(A)$ onto S. As in (2.8), define a building $\widehat{I}$, equipped with $q : \widehat{I} \longrightarrow B^r$, by "filling in" all the spheres $S(A)$.

(2.12) Besides q and its decomposition into chambers, I has the following additional structure:

— A "composition" $\widetilde{B}.g$ is defined which, to a chamber $\widetilde{B}$ of I with image B in S and to $g \in \mathrm{Hom}_{\mathrm{Gal}}(B, C)$, associates a chamber of I with image C in S. The construction $g \longmapsto \widetilde{B}.g$ defines an isomorphism from the space analogous to I obtained by taking B as fundamental chamber to the space I (for $A_0 = B$, one obtains automorphisms of I). In particular,

$\widetilde{B}.(gh) = (\widetilde{B}.g).h$.

(2.13) For every chamber A of I lying over A_0, define a map

$i_A : I_+ \longrightarrow \mathrm{I}$ (respectively $\widehat{I}_+ \longrightarrow \widehat{I}$),

compatible with the projection onto S (respectively B^r), by sending the closed chamber $(A.g)^-$ of I_+ (respectively the ball $b(A.g)$ of $\widehat{I}_+$) to the chamber (respectively the ball) of the same name in I (respectively $\widehat{I}$).

(2.14) Proposition. (i) i_A identifies I_+ with a closed subspace of I and $\widehat{I}_+$ with a closed subspace of $\widehat{I}$.

(ii) One has

$\mathrm{I} = \varinjlim_n i_{\widetilde{A}_0.\Delta^{-2n}}(I_+)$ and $\widehat{I} = \varinjlim_n i_{\widetilde{A}_0.\Delta^{-2n}}(\widehat{I}_+)$.

The assertions (i) or (ii) for $\widehat{I}_+$ and $\widehat{I}$ follow from the assertions (i) or (ii) for I_+ and I.

We prove (i) for I_+ and I. Let B and C be two chambers of S having a common facet F. Let $g \in \mathrm{Hom}_{\mathrm{Gal}_+}(A_0, B)$ and $h \in \mathrm{Hom}_{\mathrm{Gal}_+}(A, C)$. Suppose that the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are equal in I.

One then has

$g^{-1}h = \pi_F(e)$ for $e \in \mathrm{Hom}_{\mathrm{Gal}}(\pi_F^{-1}B, \pi_F^{-1}C)$.

By (1.26), one may write $e = e_1 e_2^{-1}$ with e_i in Gal_+ . By (1.29), one then has $g\pi_F(e_1) = h\pi_F(e_2)$ in Gal_+, and the facets $q^{-1}F$ of $(A.g)^-$ and $(A.h)^-$ are therefore already equal in I_+.

To prove (ii), it suffices to observe that every gallery g with source A_0 in Gal can be written $g = \Delta^{-2n}g'$ with g' in Gal_+ .

From (2.14) one obtains $\pi_i(\widehat{I}) = \varinjlim \pi_i(\widehat{I}_+)$. Since I is a CW complex, (2.9) gives the following result.

(2.15) Theorem. The space $\widehat{I}$ is contractible.

The space I is therefore the wedge of the spheres $S(A)$, as A ranges over the chambers of I.

3. Coverings

(3.1) Let V_C be the complexification of V, and M_C the complexification of M for $M \in \mathcal{M}$, and let

$$Y = V_C - \bigcup_{M \in \mathcal{M}} M_C$$

In this section we construct, by gluing, a space $\widetilde{Y}$ above Y. The gluing data will be described using I and $\widehat{I}$. The facets of S and the facets of V other than $\{0\}$ are in canonical bijection, and we shall constantly pass from one to the other. In general, we use the same notation for corresponding facets; when necessary, the correspondence will be denoted $F \longmapsto [F]$.

(3.2) Lemma. Let A, B, and C be three chambers of I, with $C \subset S(A) \cap S(B)$. Then q induces an isomorphism from $S(A) \cap S(B)$ onto the intersection of S with the closed half-spaces $D'_M(qC)$ containing qC and bounded by a wall M separating qA from qB.

The intersection of the $D'_M(qC)$ is a union of closed chambers. If C' is one of them, qA and qB lie on the same side of every wall separating qC from C'. It follows that the lifts to $S(A)$ and $S(B)$ of a minimal gallery from qC to C' coincide, and $(C')^- \subset q(S(A) \cap S(B))$.

Conversely, if a facet F is not in the intersection of the $D'_M(qC)$, there is a wall M such that $F \not\subset M$ which separates qA from qB and F from qC. Suppose, impossibly, that $F \subset q(S(A) \cap S(B))$. Let D be a chamber of S having F as a facet, and let $(C_0, C_1 \ldots C_{n+1})$ be a gallery from qC to D. Let M_i be the wall separating C_i from C_{i+1}, and let α_i (respectively β_i) $= \pm 1$ according as C_{i+1} and A (respectively B) are or are not on the same side of M_i. By hypothesis, F is a facet of $C\Delta(M_0)^{\alpha_0} \ldots \Delta(M_n)^{\alpha_n}$ and of $C.\Delta(M_0)^{\beta_0} \ldots \Delta(M_n)^{\beta_n}$.

By (2.4) and (1.24), one then has (since $M \not\supset F$)

$$\sum\nolimits_{M_i = M} \alpha_i = \sum\nolimits_{M_i = M} \beta_i,$$

which is absurd: one side is one, the other -1.

(3.3) Lemma. Let $\mathcal{R}$ and $\mathcal{I}$ be the "real part" and "imaginary part" maps from V_C to V. Let $v \in V_C$, and let F be the facet of V such that $\mathcal{R}v \in F$. For v to belong to Y, it is necessary and sufficient that $pr_F(\mathcal{I}v)$ lie in a chamber of $(V_F, \mathcal{M}_F)$.

This is clear.

What follows has as its intuitive basis the fact that to every "walk" in I (made by following a gallery of the form $\Delta(M_1)^{\varepsilon_1} \ldots \Delta(M_k)^{\varepsilon_k}$, $\varepsilon_i = \pm 1$) there corresponds a path ch (a path class) in Y. The image under $\mathcal{R}$ of ch in V passes from chamber to chamber, successively crossing the walls $M_1, \ldots, M_k$ at points m_i. For $\varepsilon_i = 1$ (respectively -1), the path passes through one of the two components R of $\mathcal{R}^{-1}(m_i) \cap Y$ (in bijection under $\mathcal{I}$ with $V - M$) : the one for which $\mathcal{I}R$ contains (respectively does not contain) the preceding chamber.

(3.4) Notations. (i) Let C be a chamber of $(V, \mathcal{M})$. Denote by Y(C) the open subset of Y consisting of the $x \in V_C$ satisfying the following condition:

— if F is the facet of V such that $\mathcal{R}x \in F$, then $pr_F\ \mathcal{I}x \in \pi'_F(C)$.

(ii) Let A and B be two chambers of I. If $S(A) \cap S(B)$ contains a chamber C, temporarily denote by $Y'(A, B)$ the intersection of the open half-spaces $D'_M(qC)^\circ$, for M as in 3.2. Set

Y(A, B) $= Y(qA) \cap Y(qB) \cap \mathcal{R}^{-1}Y'(A, B) \subset V_C$.

If $S(A) \cap S(B)$ contains no chamber, set Y(A, B) $= \varnothing$.

(iii) For a facet F of V, its star $\mathrm{Et}(F)$ is the union of the open facets G of $(V, \mathcal{M})$ such that $F \subset G^-$.

(iv) For a facet F of V and a chamber C of $(V_F, \mathcal{M}_F)$, set

V(F, C) $= \{v \in V_C \mid \mathcal{R}v \in \mathrm{Et}(F) \text{ and } pr_F\ \mathcal{I}v \in C\}$.

(3.5) Lemma. Let G be a facet of V. The V(F, C), for F a facet such that $G \subset F^-$ and C a chamber of V_F, form an open covering of $Y \cap \mathcal{R}^{-1}$(Et G). In particular (take $G = \{0\}$), the V(F, C) cover Y.

If $x \in Y \cap \mathcal{R}^{-1}$(Et G) and $\mathcal{R}x \in F$, then x lies in one of the V(F, C) (3.3).

(3.6) Let Y' be the disjoint sum, indexed by the balls $b(A)$ of $\widehat{I}$,

$Y' = \coprod_{b(A)}$ Y(qA).

Denote by $Y'(b(A))$ the component indexed by $b(A)$. There is an evident projection $p' : Y' \longrightarrow Y$. Let R be the following relation on Y' :

For $x \in Y'(b(A))$ and $y \in Y'(b(B))$, R(x, y) means that $p'(x) = p'(y)$ and that $p'(x), p'(y) \in Y(A, B)$.

It is an equivalence relation because (cf. (3.2)) Y(A, $B) \cap Y(B, C) \subset Y(A, C)$. One has

(3.6.1) The R-saturation of an open subset of Y' is again open.

Set $\widetilde{Y} = Y'/R$. There is an evident projection

$$p : \widetilde{Y} \longrightarrow Y$$

For every ball $b(A)$ of $\widehat{I}$, denote by $\widetilde{Y}(b(A))$ the image of $Y'(b(A))$ in $\widetilde{Y}$.

The following theorem implies the one with which the introduction begins.

(3.7) Theorem. $\widetilde{Y}$ is a contractible covering of Y.

A. $\widetilde{Y}$ is a covering of Y.

We shall show, for V(F, C) as in (3.4)(iv), that $p^{-1}V(F, C)$ is a sum of copies of V(F, C).

a) Let $b(A)$ be a ball of $\widehat{I}$. For B a chamber such that $F \subset B^-$, let the following be the chamber of I:

$$C'_B = A.u(A, B)u(\pi_F(C), B)^{-1}$$

By construction, $B \subset q(S(A) \cap S(C'_B))$, and in particular, if $F \neq \{0\}$,

(3.7.1) $[F] \subset q(S(A) \cap S(C'_B))$.

We prove that

(3.7.2) $\widetilde{Y}(b(A)) \cap p^{-1}V(F, C) \subset \bigcup_{F \subset B^-} \widetilde{Y}(b(C'_B))$.

Let x be in the left-hand side. Let G be the facet of V to which $\mathcal{R}px$ belongs. One has $F \subset G^-$. Choose B to be a chamber having G as a facet. By hypothesis, one has

$pr_G\ \mathcal{I}px \in \pi'_G(A)$ and $pr_F\ \mathcal{I}px \in C$.

Thus $\pi'_G(A) = \pi'_G(\pi_F\ C)$. It then follows from (3.2) that every chamber having G as a facet lies in $q(S(A) \cap S(C'_B))$, that px lies in Y(A, C'_B), and hence that $x \in \widetilde{Y}(b(C'_B))$.

b) Set $Y'(F, C) = \bigcup_{q(C')=C}(Y'(b(C')) \cap p'^{-1}(V(F, C)))$. By a), set-theoretically,

$p^{-1}V(F, C) = Y'(F,$ C)/R.

This is even an isomorphism of topological spaces, by (3.6.1), since $Y'(F, C)$ is open in Y'. By (3.2), the equivalence relation

induced by R on $\bigcup_{q(C')=C} Y'(b(C'))$ is trivial. We conclude by observing that, for every C' above C,

V(F, $C) \subset Y(b(C'))$.

B. **An open covering of $\widetilde{Y}$.**

(3.7.3) Notations. (i) For a facet F of I,

$\widetilde{Y}(F) = \bigcup_{F \subset S(C)}(\widetilde{Y}(b(C)) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(qF))$.

(ii) For $o(A)$, the center of a ball $b(A)$ of $\widehat{I}$,

$\widetilde{Y}(o(A)) = \widetilde{Y}(b(A)) \cap p^{-1}V(\{0\},$ A).

(3.7.4) Lemma. (i) The $\widetilde{Y}(s)$, for s a vertex of $\widehat{I}$, form a covering of $\widetilde{Y}$ whose nerve is $\widehat{I}$ (i.e., a family of $\widetilde{Y}(s)$ has nonempty intersection if and only if the corresponding s are the vertices of a facet of $\widehat{I}$).

(ii) If a facet F of I has vertices $s_0, \ldots, s_p$, then

$\widetilde{Y}(F) = \widetilde{Y}(s_0) \cap \cdots \cap \widetilde{Y}(s_p)$.

(iii) If $F \subset S(A)$, then $\widetilde{Y}(o(A)) \cap \widetilde{Y}(F)$ is contractible; $\widetilde{Y}(o(A))$ is contractible.

(3.7.4.1) Let F_i $(i = 1, 2)$ be two facets of I and C_i $(i = 1, 2)$ two chambers of I such that $F_i \subset S(C_i)$ and $(\widetilde{Y}(b(C_1)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(qF_1)) \cap (\widetilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(qF_2)) \neq \varnothing$.

Then F_1 and F_2 span a facet F of I (unique by $(2.5)-(2.11)$), $F \subset S(C_i)$, and the preceding intersection coincides with

$\widetilde{Y}(b(C_1)) \cap \widetilde{Y}(b(C_2)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(F)$.

By hypothesis,

$Y(C_1, C_2) \cap \mathcal{R}^{-1}\operatorname{Et}(qF_1) \cap \mathcal{R}^{-1}\operatorname{Et}(qF_2) \neq \varnothing$;

F_1 and F_2 lie in $S(C_1) \cap S(C_2)$ and span a facet F there. Moreover, $\operatorname{Et}(qF_1) \cap \operatorname{Et}(qF_2) = \operatorname{Et}(qF)$, whence the stated formula.

From (3.7.4.1), one deduces that $\widetilde{Y}(F_1) \cap \widetilde{Y}(F_2)$ is empty if F_1 and F_2 do not span a common facet F, and otherwise equals

$\bigcap_{F \subset S(C)} \widetilde{Y}(bC) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(F) = \widetilde{Y}(F)$

(take $C_1 = C_2$). This proves (ii) and part of (i).

(3.7.4.2) If $o(A) \neq o(B)$, then $\widetilde{Y}(o(A)) \cap \widetilde{Y}(o(B)) = \varnothing$.

If $qA = qB$, one has $\widetilde{Y}(b(A)) \cap \widetilde{Y}(b(B)) = \varnothing$. If $qA \neq qB$, one has

$p\widetilde{Y}(o(A)) \cap p\widetilde{Y}(o(B)) = \varnothing$.

(3.7.4.3) If $F \not\subset S(A)$, then $\widetilde{Y}(o(A)) \cap \widetilde{Y}(F) = \varnothing$.

If $F \subset S(B)$, then $F \not\subset S(A) \cap S(B)$, and

$\operatorname{Et}(qF) \cap Y(A, B) = \operatorname{Et}(qF) \cap p(\widetilde{Y}(b(A)) \cap \widetilde{Y}(b(B))) = \varnothing$.

It is clear that:

(3.7.4.4) If $F \subset S(A)$, then p induces an isomorphism from $\widetilde{Y}(o(A)) \cap \widetilde{Y}(F)$ onto $Y(qA) \cap \mathcal{R}^{-1}\operatorname{Et}(qF)$.

Lemma (3.7.4) then follows from:

(3.7.5) Lemma. For every facet F of V and every chamber A,

$Y(A) \cap \mathcal{R}^{-1}\operatorname{Et}(F)$

is contractible.

Let v_0 be a vector such that $\mathcal{R}v_0 \in F$ and $\mathcal{I}v_0 \in A$. Then, for $0 \le t \le 1$, $\varphi_t : v \longmapsto v_0 + t(v - v_0)$ is a map of Y(A) and of $\mathcal{R}^{-1}\operatorname{Et}(F)$ into themselves. One has $\varphi_0(Y(A) \cap \mathcal{R}^{-1}\operatorname{Et}(F)) = \{v_0\}$ and $\varphi_1 = \operatorname{Id}$.

C. Study of $\widetilde{Y}(F)$ (F a facet of I).

By (3.5), (3.7.1), and (3.7.2), one has

(3.7.6) $\widetilde{Y}(F) = \bigcup_{F \subset B^-} \widetilde{Y}(b(B)) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(F)$.

Choose a "fundamental chamber" $A_{F,0}$ in $(V_F, \mathcal{M}_F)$, and denote the corresponding building by I_F. Also choose $\widetilde{A}^I_{F,0}$ in I above $\pi_F(A_{F,0})$. For every chamber $B = \widetilde{A}_0.g_F$ of I_F, denote by $\pi_F(B)$ the chamber $\widetilde{A}^I_{F,0}.\pi_F(g_F)$ of I. By (2.10), (3.7.6) may be rewritten

$$\widetilde{Y}(F) = \bigcup_{B \text{ in } I_F} \widetilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\operatorname{Et}(F).$$

Let

$$Y_F = (V_F)_C - \bigcup_{M \in \mathcal{M}_F} M_C,$$

and let $\widetilde{Y}_F$ be the covering of Y_F defined using I_F.

For any B and C in I_F, it follows from (1.30) that the chambers of S in $Y(\pi_F(B), \pi_F(C)) \cap \operatorname{Et}(F)$ are exactly the $\pi_F(D)$, for D in $Y_F(B, C)$.

The gluings defining $\widetilde{Y}(F)$ and $\widetilde{Y}_F$ are therefore compatible, and there is a unique Cartesian diagram (3.7.7)

$$
(3.7.7) \qquad
\begin{tikzcd}
\widetilde{Y}_F \arrow[d] & \widetilde{Y}(F) \arrow[l] \arrow[d] \\
Y_F & \mathcal{R}^{-1}\,Et(F) \subset Y \arrow[l, "pr_F"']
\end{tikzcd}
$$

which sends $\widetilde{Y}(b(\pi_F(B))) \cap p^{-1}\mathcal{R}^{-1}\,\mathrm{Et}(F)$ into $\widetilde{Y}_F(b(B))$ for B a chamber of I_F. The first vertical arrow therefore also the second are coverings.

It is easily verified that the map pr_F of (3.7.7) is a homotopy equivalence, whence

(3.7.8) Lemma. $\widetilde{Y}_F$ and $\widetilde{Y}(F)$ have the same homotopy type.

D. **End of the proof.**

We prove (3.7) by induction on $r = dim\ V$. Let $\mathcal{U}$ be the open covering of $\widetilde{Y}$ by the $\widetilde{Y}(s)$, for s a vertex of $\widehat{I}$. By (3.7.4)(i), $\widehat{I}$ is the nerve of $\mathcal{U}$. By (3.7.4), (3.7.8), and the induction hypothesis, the nonempty intersections of open sets belonging to $\mathcal{U}$ are contractible. This implies [6] that $\widetilde{Y}$ and $\widehat{I} = \mathrm{Nerf}(\mathcal{U})$ have the same homotopy type. The conclusion follows from (2.15).

4. Braid groups

(4.1) Let V be a finite-dimensional real vector space and $W \subset \mathrm{GL}(V)$ a finite group generated by reflections. Assume that $V^W = \{0\}$. Let Φ be any Euclidean structure on V invariant under W, and let $\mathcal{M}$ be the set of hyperplanes M such that the orthogonal reflection with respect to $\mathcal{M}$ belongs to W. One knows that

a) $(V, \mathcal{M})$ satisfies 1.6 ([1] *V* 3.9 prop. 7);

b) W permutes the chambers of $(V, \mathcal{M})$ in a strictly simply transitive manner ([1] *V* 3.2 Th. 1);

c) if $x \in V$ belongs to the closed chamber $\overline{A}$, its stabilizer is generated by the reflections in the walls of A containing x ([1] *V* 3.3 prop. 1);

d) (follows from c)) the group W acts freely on

$$Y_W = V_C - \bigcup_{M \in \mathcal{M}} M_C.$$

By b), if C_0 and C_1 are two chambers, there is a unique $w \in W$ such that $wC_0 = C_1$; these w induce a transitive system of bijections between the walls of the different chambers. Passing to the quotient, one obtains a set D with dim(V) elements and, for each chamber C, a bijec-
tion φ_C from D onto the set of walls of C. One has

(4.1.1) $\varphi_{wC}(i) = w\varphi_C(i)$ $(w \in W)$.

For the definition of the Coxeter matrix $(m_{ij})_{i,j \in D}$ and that of the Coxeter graph, we refer to [1] *V* 3.4.

(4.2) Choose a chamber A_0 of $(V, \mathcal{M})$. For $i \in D$, let w_i be the reflection in the wall $\varphi_{A_0}(i)$. [1] *V* 3.2 Th. 1 says that W is generated by the w_i, these generators being subject only to the relations
$(w_i w_j)^{m_{ij}} = e$.

(4.3) Let $X_W = Y_W/W$: by (4.1)d), Y_W is a covering of X_W, with group W. Let $y_0 \in A_0$ have image $x_0 \in X_W$. For each $i \in D$, let ℓ_i' be the homotopy class of paths in Y_W, from y_0 to $w_i(y_0)$, which, for $y \in A_0$, contains the broken line with successive vertices y_0, $y_0 + iy_0$, $w_i(y_0) + iy_0$, $w_i(y_0)$. The image ℓ_i of ℓ_i' in X_W is a loop based at x_0.

(4.4) Theorem. (i) The fundamental group $\pi_1(X_W, x_0)$ is generated by the ℓ_i; these are subject only to the relations

(4.4.1) $\mathrm{prod}(m_{ij};\, \ell_i,\, \ell_j) = \mathrm{prod}(m_{ji};\, \ell_j,\, \ell_i)$.

(ii) The universal covering of X_W is contractible: X_W is a $K(\pi;\, 1)$.

Assertion (i) is proved in Brieskorn [3]. Another proof will be sketched in (4.18) below. As explained in the introduction, (ii) follows from (3.7) and (4.1).

(4.5) In § 1, we have drawn heavily on Garside's study [4] of Artin's braid group; from the results obtained we can now derive information about all groups of the type $\pi_1(X_W,\, x_0)$ (Garside had already considered some of them). The translation rests on (4.6), (4.7) below.

(4.6) Let $G = (C_0,\, \dots,\, C_n)$ be a gallery and M_i the wall separating C_{i-1} from C_i $(1 \leq i \leq n)$. One has $\varphi^{-1}_{C_{i-1}}(M_i) = \varphi^{-1}_{C_i}(M_i)$. Let $\ell(G)$ be the element $\varphi^{-1}_{C_1}(M_1) \dots \varphi^{-1}_{C_n}(M_n)$ of $L^+(D)$ (0.1).

The map ℓ induces a bijection from the set of galleries G with a given source to $L^+(D)$. One has

(4.6.1) $\ell(w(G)) = \ell(G)$ (for $w \in W$);

(4.6.2) $\ell(G_0G_1) = \ell(G_0)\ell(G_1)$ (for composable G_0 and G_1).

Let $w : L(D) \longrightarrow W$ be the homomorphism extending the map $i \longmapsto w_i$ from D to W.

(4.7) Proposition. For every gallery G with source A_0, one has

$A_0.G = w(\ell(G))(A_0)$.

This is an avatar of the identity

$[(w_1 \dots w_{n-1})w_n(w_1 \dots w_{n-1})^{-1}]$

$\cdot[(w_1 \dots w_{n-2})w_{n-1}(w_1 \dots w_{n-2})^{-1}] \dots [w_1] = w_1 \dots w_n$.

(4.8) From this proposition one deduces that, for $w \in W$, the integer $d(A_0,\, w(A_0))$ is the least length of words in the w_i that equal w; more precisely, ℓ induces a bijection from the set of minimal galleries from A_0 to wA_0 to the set of words $m \in L(D)$, of minimal length, such that $w = w(m)$.

(4.9) Definition. (i) The braid monoid G^+ of type D is the monoid with unit generated by D, the generators $i \in D$ being subject only to the relations

(4.9.1) $\mathrm{prod}(m_{ij};\, \mathrm{i},\, j) = \mathrm{prod}(m_{ij};\, \mathrm{j},\, \mathrm{i})$.

(ii) The braid group G (or generalized braid group) of type D is the group generated by D, the generators being subject to (4.9.1).

(4.10) Notice that W also admits the presentation

(4.10.1) $\mathrm{prod}(m_{ij};\, w_i,\, w_j) = \mathrm{prod}(m_{ij};\, w_j,\, w_i)$,

(4.10.2) $w_i^2 = e$.

so that $i \longmapsto w_i$ extends to a homomorphism from G to W.

(4.11) One checks from the definitions that two galleries G_0 and G_1 with the same source are equivalent if and only if $\ell(G_0)$ and $\ell(G_1)$ have the same image in G^+. Taking account of (4.9), (1.12) therefore admits the following translation ([1] IV 1.6 prop. 5).

(4.12) Reminder. Let $w \in W$. The image in G^+ of the words $m \in L(D)$, of minimal length among those such that $w = w(m)$, depends only on w.

This image will be denoted $r(w)$; r is a section of $w : G^+ \longrightarrow W$. By (4.7), one has $r(w) = \ell(u(A_0,\, wA_0))$.

Similarly, [1] IV 1.4 lemma 2 corresponds to (1.11).

(4.13) The "quotient" category of $\mathrm{Gal}_+(V, \mathcal{M})$ by W is defined as follows:

a) Its set of objects is reduced to one element: it is
$\mathrm{Ob}(\mathrm{Gal}_+(V, \mathcal{M}))/W$.

b) The monoid of its arrows is $\mathrm{Fl}(\mathrm{Gal}_+(V, \mathcal{M}))/W$, and passage to the quotient
$\mathrm{Gal}_+(V, \mathcal{M}) \longrightarrow \mathrm{Gal}_+(V, \mathcal{M})/W$
is a functor.

This category, having only one object, is identified with the monoid of its arrows. By (4.6.1), (4.6.2), and (4.11), ℓ identifies this monoid with G^+.

Similarly one defines $\mathrm{Gal}(V, \mathcal{M})/W$. This category has only one object and, by its universal property, ℓ extends to an isomorphism from the group of its arrows to G :

$$\begin{tikzcd}
\mathrm{Gal}_+(V, \mathcal{M}) \arrow[r] \arrow[d] & \mathrm{Gal}_+(V, \mathcal{M})/W \arrow[r, "\sim", "\ell"'] \arrow[d] & G^+ \arrow[d] \\
\mathrm{Gal}(V, \mathcal{M}) \arrow[r] & \mathrm{Gal}(V, \mathcal{M})/W \arrow[r, "\sim", "\ell"'] & G
\end{tikzcd}$$

(4.14) Theorem. (i) In G^+, left and right translations are injective.

(ii) G^+ satisfies the Öre condition on the left and on the right. It follows from (i) that G^+ embeds in G.

(iii) For J a subset of D, let G_J^+ (resp. G_J) be the submonoid (resp. subgroup) of G^+ (resp. G) generated by the $i \in J$. Then the evident map identifies G_J^+ (resp. G_J) with the braid monoid (resp. braid group) of type J.

(iv) One has $G_J^+ = G_J \cap G^+$.

(v) If J and K are two subsets of D, one has $G_J \cap G_K = G_{J \cap K}$.

(vi) Let n(i) be the number of elements of G^+ of length i, and put $f = \sum n(i)t^i$. For $J \subset D$, let m(J) be the number of walls passing through the intersection of the walls of an (arbitrary) chamber A, indexed by J (= *the* number of reflections in the Weyl group W_J). One has

$$f = (\textstyle\sum_{J \subset D}(-1)^{|J|}t^{m(J)})^{-1}$$

These assertions translate respectively: (i): (1.1)(i) and (ii); (ii): (1.26)(ii) and its transform by passage to opposite galleries (cf. also (4.16) below); (iii): (1.30); (iv): (1.32); (v) : (1.31); (vi): (1.21) (the f_A of loc. cit. are all equal to f).

Let x, $y \in G^+$. We say that x begins with y if there exists z in G^+ with $x = yz$. Proposition (1.19)(iii) translates as follows

(4.15) Lemma. Let $x \in G^+$. There exists a chamber C such that, for every $w \in W$, x begins with $r(w)$ if and only if $wA_0 \in D(A_0, C)$.

(4.16) Let J be a subset of D and P the intersection of the walls $\varphi_{A_0}(i)$ $(i \in J)$. Let $\Delta(J)$ be the element $\ell(u(A_0, A_0.\Delta(P)))$ of G^+; put $\Delta = \Delta(D)$. Let $i \longmapsto \bar{\imath} = \varphi_{-C}^{-1}\varphi_C(i)$ be the opposition involution of D. We also denote by $w \longmapsto \overline{w}$ the automorphisms of $L^+(D)$, G^+, and G that send i to $\bar{\imath}$ (one has $m_{ij} = m_{\overline{ij}}$). Translating (1.26) with the aid of (4.4), one finds that for g in G^+ or G,

(4.16.1) $g\Delta = \Delta\overline{g}$.

In particular, Δ^2 is central.

For every $i \in D$, Δ "begins" with i: one has $\Delta = ix$, in G^+. Hence

(4.17) Proposition. G is obtained from G^+ by making the central element Δ^2 invertible.

(4.18) The description (2.11), (2.5) – (2.7) of the building I attached to $(V, \mathcal{M})$ can be translated as follows.

a) The set of vertices of I is

$$I^0 = \coprod_{i \in D} G/G_{D-i}$$

b) For a set of vertices to span a simplex, it is necessary and sufficient that it be contained in the set of classes gG_{D-i}, for some $g \in G$.

c) The fundamental sphere $S_0 = S(A_0)$ is the union of the fundamental simplex, spanned by the lateral classes of e in the G/G_{D-i}, and of its transforms by the $r(w)$ $(w \in W)$.

The preceding description makes evident a left action of G on I. It respects the set of spheres $S(A)$ and the correspondence $A \longmapsto S(A)$.

By transport of structure, G acts on the covering $\widetilde{Y}_W$ of Y_W (3.6), and one has an equivariant morphism $(G - \widetilde{Y}_W) \longrightarrow (W - Y_W)$.

It follows that G is the fundamental group of $X_W = Y_W/W$, and (4.4)(i).

(4.19) For completeness, let us show that the word problem, the conjugacy problem, and the question of computing the center of G can be solved as in [4]. If the Coxeter graph D of W is a disjoint union of graphs D_α, the braid group of type D is the product of the braid groups of types D_α. The essential case is therefore that of a connected graph D.

(4.20) The word problem: there is a decision procedure for determining whether two elements m and n of L(D) have the same image in G.

a) Whether m and n in $L^+(D)$ have the same image in G^+ is decidable, because the imposed relations do not change the length of the words, and there are only finitely many words of a given length. One may also use (1.22).

b) For any m_i $(i = 1, 2)$ in the free group L(D) generated by D, one can compute $m_i' \in L^+(D)$ and $k \geq 0$ such that m_i and $m_i'\Delta^{-k}$ have the same image in G (cf. (4.17)); m_1 and m_2 then have the same image in G if and only if m_1' and m_2' have the same image in G^+.

(4.21) Theorem. If the Coxeter graph D is connected (nonempty) and the opposition involution is trivial (resp. nontrivial), the center of G is infinite monogenic, generated by Δ (resp. Δ^2).

Consider the following property of $x \in G^+$.

(*) For every $i \in D$, there exists $i' \in D$ such that $ix = xi'$.

It is enough to prove that, if x satisfies (*) and $x \neq e$, then $x = \Delta y$ with $y \in G^+$. By induction on the length of x, it will follow that if x satisfies (*), x is of the form Δ^k. In particular, the center of G^+ is contained in the set of the Δ^k $(k \geq 0)$; by (4.14)(i) and (4.17), the center of G is contained in the set of the Δ^k $(k \in \mathbb{Z})$, and the assertion follows from (4.16.1).

Let $x \in G^+$ satisfy (*). Let C be as in (4.15), and let w be the image in W of $\ell u(A_0, C)$. We may write $x = r(w)y$, with $y \in G^+$. Let $i \in D$. By (*), $ix = r(w)yi'$ begins with $r(w)$; by (1.23), it follows that ir(w) begins with $r(w) : ir(w) = r(w)i''$. In particular, for every i, there exists i'' with $iw = wi''$; this means that every wall of A_0 is also a wall of wA_0. Among the $2^{|D|}$ open simplicial cones bounded by the walls of A_0, only A_0 and $-A_0$ are chambers: they are the only ones whose angles between the faces are all $\leq \pi/2$ (the connectedness of D is used here). If $x \neq e$, then $w \neq e$ and hence $wA_0 = -A_0$. Since then $\Delta = r(w)$, this completes the proof.

(4.22) The derived group. Let D' be the graph having D as its set of vertices, two vertices being joined by an edge if and only if m_{ij} is odd. Let D_0 be the set of connected components of D'. Define an epimorphism

Lg: $G \longrightarrow \mathbb{Z}^{D_0}$

by sending i to the basis vector indexed by the connected component of i in D'. One verifies at once that Lg identifies $\mathbb{Z}^{D_0}$ with the largest abelian quotient of G.

(4.23) The conjugacy problem. The problem is to give a decision procedure for determining whether two elements x and y of G (given explicitly as images of words in L(D)) are conjugate. Since Δ^2 is central, $\Delta^{-2k}a$ and $\Delta^{-2k}b$ are conjugate if and only if a and b are. By

(4.17), it therefore suffices to treat the case where $x, y \in G^+$. Similarly, if x and y are conjugate, there exists a in G^+ such that $xa = ay$. If x and y are conjugate, $\mathrm{Lg}(x) = \mathrm{Lg}(y)$. There are only finitely many $x \in G^+$ of given "length" $\mathrm{Lg}(x)$, and W is finite. The existence of a decision procedure then follows from (4.20) and the following lemma.

(4.24) Lemma. Let x, y be in G^+. If x and y are conjugate, there exists a sequence $x_0 = x$, x_1, $\ldots$, $x_n = y$ of elements of G^+ and a sequence of elements w_i of W such that

$x_{i+1}r(w_i) = r(w_i)x_i$.

Let $a \in G^+$ be such that $xa = ay$. Put $a = r(w_0)\ldots r(w_{n-1})$, where w_i is the element of W of maximum length such that $r(w_i)\ldots r(w_{n-1})$ can be written in the form $r(w_i)b_i$ with $b_i \in G^+$ (4.15). Let $a_i = r(w_0)\ldots r(w_i)$. We shall prove that $x_i = a_i^{-1}xa_i$ lies in G^+, so that the x_i and w_i answer the problem.

It follows from (1.23) that, for every $w \in W$, if xa begins with $r(w)$, then $xr(w_0)$ also begins with $r(w)$. In particular, since $xa = ay = r(w_0)b_0y$ begins with $r(w_0)$, $xr(w_0) = r(w_0)x_1$ with $x_1 \in G^+$. The proof is completed by induction on n.

Remark. Let σ be an automorphism of the Coxeter graph D, and again denote by σ the corresponding automorphism of G. One has $\Delta^\sigma = \Delta$. The preceding arguments still apply to the question of deciding, for x, $y \in G$, whether there exists a in G such that $xa = a^\sigma y$. We reduce to taking x, y, and a in G^+. In that case and, with the preceding notation, if $xa = a^\sigma y$, the $(a_i^{-1})^\sigma xa_i$ lie in G^+. Taking for σ the involution $i \longmapsto \bar{\imath}$, one obtains a criterion analogous to (4.24) for the conjugacy of Δx and Δy (x, $y \in G^+$).

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The reader will find a more complete bibliography on braid groups in [2].

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UNIRATIONAL NON-RATIONAL VARIETIES

[after M. ARTIN and D. MUMFORD]

by Pierre DELIGNE

Editorial note. This independent English edition follows the frozen French text and the IAS authority. Mathematical notation, cross-references, and typographical peculiarities of the printed source are retained rather than silently corrected. The seven source objects are reproduced from authority-grounded image derivatives. Running headers and printed page numbers are excluded from the body.

This exposé contains a description of the variety constructed by Artin and Mumford, and the proof that it is unirational and non-rational. It also contains the statements of the theorems proved by Clemens and Griffiths [4], and by Manin and Iskovskih [6], for constructing other examples. I attended talks by Artin and Mumford on the same subject and drew heavily on them.

In everything that follows, the expression “algebraic variety” will mean “a separated scheme of finite type over $\mathbb{C}$, reduced and irreducible”. We shall write $k(X)$ for the field of rational functions on an algebraic variety X.

1. Rationality and unirationality

Let K be an extension of finite type of $\mathbb{C}$. There then exists an algebraic variety X such that K is isomorphic to $k(X)$. X is called a *model* of K. By [5], K always admits a projective and smooth model.

Let K and L be two extensions of finite type of $\mathbb{C}$, with models X and Y. The homomorphisms $a^* : K \hookrightarrow L$ correspond bijectively to the dominant rational maps $a_* : Y \dashrightarrow X$, by the formula $a^*(f) = f \circ a_*$. By [5], if X and Y are projective and smooth, every rational map $a_* : Y \dashrightarrow X$ admits a decomposition

$$a = b\, c_n^{-1} \cdots c_1^{-1}$$

$$(1.1)\qquad \begin{tikzcd} Y_{n+1} \arrow[r, "c_n"] \arrow[d, "b"'] & \cdots \arrow[r] & Y_2 \arrow[r, "c_1"] & Y_1 = Y , \\ X \end{tikzcd}$$

where b and the c_i are morphisms (defined everywhere), and where Y_{i+1} is obtained from Y_i by blowing up a smooth subvariety Z_i (of codimension ≥ 2).

If $a^* : K \to L$ is an isomorphism, the preceding result can be applied to $((a^*)^{-1}, Y_{n+1}, X)$. By iteration, one obtains a commutative diagram of morphisms of schemes

$$(1.2)\qquad \begin{tikzcd} \cdots & Y^2 \arrow[r, "y^2"] \arrow[d, "a^2"] & Y^1 \arrow[r, equal] & Y^1 \arrow[r, "y^1"] \arrow[d, "a^1"] & Y \arrow[d, "a"] \\ \cdots & X^1 \arrow[r, equal] & X^1 \arrow[r, "x^1"] \arrow[u, "b^1"'] & X \arrow[r, equal] & X . \end{tikzcd}$$

In this diagram, the horizontal arrows x^i, y^i are composites of blow-ups with nonsingular centers, as in (1.1); the a^i and b^i are birational.

An invariant of a projective and smooth variety X is called *birational* if it depends only on $k(X)$. The birational character of an invariant is often verified with the aid of (1.2) (for a typical example, see 2.1). Invariants usable for a field K are generally defined as the value of a birational invariant of any projective and smooth model of K.

Definition 1.3.- Let X be an algebraic variety of dimension n. We say that X is unirational if the following conditions are satisfied:

(i) $k(X)$ is a subfield of a purely transcendental extension $\mathbb{C}(T_1, \dots, T_r)$ of $\mathbb{C}$;

(ii) a finite extension of $k(X)$ is isomorphic to $\mathbb{C}(T_1, \dots, T_n)$;

(iii) (for X projective and smooth), there exists a diagram (1.1) with $Y = \mathbb{P}^r(\mathbb{C})$;

(iv) (for X projective and smooth), there exists a diagram (1.1) with $Y = \mathbb{P}^n(\mathbb{C})$.

One may suppose X projective and smooth, and one has

$$\text{(ii)} \Rightarrow \text{(i)} \Leftrightarrow \text{(iii)} \Rightarrow \text{(iv)} \Leftrightarrow \text{(ii)}.$$

Definition 1.4.- We say that X is rational if, alternatively,

(i) $k(X)$ is a purely transcendental extension of $\mathbb{C}$;

(ii) (for X projective and smooth), there exists a diagram (1.2) with $Y = \mathbb{P}^n(\mathbb{C})$.

The most evident birational invariants do not permit one to distinguish unirational varieties from rational varieties.

Proposition 1.5.- Let X be a projective and smooth unirational algebraic variety of dimension $n > 0$.

(i) The plurigenera

$$P_k = \dim H^0\big(X, (\Omega_X^n)^{\otimes k}\big) \qquad (k > 0)$$

vanish. More generally, $H^0\big(X, (\Omega_X^1)^{\otimes k}\big) = 0$ for $k > 0$ (see [9], Remark, page 02).

(ii) X is simply connected ([10]).

For $n = 1$ or 2, property (i) above already characterizes rational varieties. A unirational variety of dimension ≤ 2 is therefore rational, and a subextension of transcendence degree ≤ 2 of $\mathbb{C}(T_1, \dots, T_k)$ is automatically purely transcendental. More precisely, one has the following:

a) For $n = 1$, if $H^0(X, \Omega_X^1) = 0$, then $X \simeq \mathbb{P}^1(\mathbb{C})$. By the Riemann–Roch theorem, for every $x \in X$, $\mathcal{O}(x)$ indeed defines an embedding of X into $\mathbb{P}^1(\mathbb{C})$.

b) For $n = 2$, the Castelnuovo–Enriques criterion asserts that X is rational if $P_a \simeq P_2 = 0$, i.e. if

$$H^0(X, \Omega_X^1) = H^0\big(X, (\Omega_X^2)^{\otimes 2}\big) = 0$$

(see [9]).

The symbols $P_a \simeq P_2$ and Ω_X^1 in this criterion reproduce the printed source.

For $n > 2$, it has been very difficult to define or calculate birational invariants capable of distinguishing between rational and unirational varieties.

2. Some birational invariants

A) Artin and Mumford

Proposition 2.1.- The torsion subgroup $H^3(X, \mathbb{Z})_{\text{tors}}$ of $H^3(X, \mathbb{Z})$ is a birational invariant of the projective and smooth variety X.

If a variety X_2 is obtained from a smooth variety X_1 by blowing up a smooth subvariety $Z \subset X_1$ of codimension $r \geq 2$, then (SGA 5 VII)

$$H^n(X_2, \mathbb{Z}) \simeq H^n(X_1, \mathbb{Z}) \oplus \bigoplus_{1}^{r-1} H^{n-2i}(Z, \mathbb{Z}).$$

The lower summation index 1, without an explicit $i =$, is retained from the printed formula. In particular,

$$H^3(X_2, \mathbb{Z}) \simeq H^3(X_1, \mathbb{Z}) \oplus H^1(Z, \mathbb{Z}).$$

Since $H^1(Z, \mathbb{Z})$ is torsion-free, one therefore has

$$H^3(X_1, \mathbb{Z})_{\text{tors}} \simeq H^3(X_2, \mathbb{Z})_{\text{tors}}.$$

Suppose that $k(X)$ is isomorphic to $k(Y)$, and let there be a diagram (1.2) From it one obtains a diagram

$$\begin{tikzcd}
H^3(X,\mathbb{Z})_{\mathrm{tors}} \arrow[r, "\simeq"] & H^3(X_1,\mathbb{Z})_{\mathrm{tors}} \arrow[r, equal] & H^3(X_1,\mathbb{Z})_{\mathrm{tors}} \arrow[r, "\simeq"] \arrow[d, "b^{1*}"] & H^3(X_2,\mathbb{Z})_{\mathrm{tors}} \\
H^3(Y,\mathbb{Z})_{\mathrm{tors}} \arrow[r, equal] & H^3(Y,\mathbb{Z})_{\mathrm{tors}} \arrow[r, "\simeq"] \arrow[u, "a^{1*}"] & H^3(Y_1,\mathbb{Z})_{\mathrm{tors}} \arrow[r, equal] & H^3(Y_1,\mathbb{Z})_{\mathrm{tors}} \arrow[u, "a^{2*}"] .
\end{tikzcd}$$

Since $a_2^* b_1^*$ is bijective, b_1^* is injective. Since $b_1^* a_1^*$ is bijective and b_1^* injective, a_1^* is bijective, and the assertion follows.
The example of Artin and Mumford is based on the

Construction 2.2.- We shall construct a unirational variety X, projective and smooth, of dimension 3, such that

$$H^3(X,\mathbb{Z})_{\mathrm{tors}} \neq 0 \qquad \big(\text{we shall have } H^3(X,\mathbb{Z}) \simeq \mathbb{Z}/(2)\big).$$

Multiplying this variety by $\mathbb{P}^r(\mathbb{C})$, one obtains an analogous example in every dimension ≥ 3.

B) Clemens and Griffiths

Let X be a projective and smooth variety of dimension 3, such that $H^0(X, \Omega^3_X) = 0$. The Hodge decomposition of $H^3(X,\mathbb{C})$ then reduces to

$$H^3(X,\mathbb{C}) = H^{2,1} \oplus H^{1,2}$$

and, according to Weil [11], the complex torus

$$J(X) = H^3(X,\mathbb{Z}) \backslash H^3(X,\mathbb{C}) / H^{2,1}$$

is an abelian variety: the *intermediate Jacobian* of X. Let

$$\psi : H^3(X,\mathbb{Z}) \otimes H^3(X,\mathbb{Z}) \longrightarrow H^6(X,\mathbb{Z}) \simeq \mathbb{Z}$$

be the cup product. By the Poincaré duality theorem, the alternating form on $H^3(X,\mathbb{Z})$/torsion deduced from ψ has discriminant one. If $H^1(X,\mathbb{Z}) = 0$, it follows from Weil [11] that this form defines a principal polarization on $J(X)$.

Proposition 2.3 ([4] 3.26).- If X is rational, the polarized abelian variety $J(X)$ is a product of Jacobians.

In the proof sketch that follows, the sign $\simeq$ denotes an isomorphism of *polarized* abelian varieties. One verifies successively:

a) $J(\mathbb{P}^3(\mathbb{C})) = 0$.

b) If Y'' is obtained from the nonsingular projective variety Y' by blowing up
a smooth curve Z with Jacobian $J(Z)$ (respectively, a point), one has ([4] 3.11)

$$J(Y'') \simeq J(Y') \times J(Z) \qquad \big(\text{respectively, } J(Y'') \simeq J(Y')\big).$$

c) If a morphism $b : Y' \to X$ is birational, $J(X)$ is a direct factor in $J(Y')$: there exists a principally polarized abelian variety of the principal series B such that

$$J(Y') \simeq J(X) \times B.$$

d) Let $(A_i)_{1 \leq i \leq n}$ be nonzero principally polarized abelian varieties. Suppose that no A_i admits a decomposition $A_i \simeq A_i' \times A_i''$ (for example, that A_i is a Jacobian). Then every decomposition of $A = \prod A_i$ is of the form

$$A \simeq \left(\prod_{i \in I} A_i\right) \times \left(\prod_{i \notin I} A_i\right) \quad \text{for } I \subset [1,n] \quad ([4]\ 3.23).$$

If X is rational, there exists a diagram 2.1 with $Y = \mathbb{P}^3(\mathbb{C})$ and b birational. By a) and b), $J(Y_{n+1})$ is a product of Jacobians. By c) and d), $J(X)$ is one as well. The reference "diagram 2.1" reproduces the printed source.

Take X to be a cubic hypersurface in $\mathbb{P}^4(\mathbb{C})$. One has

$$H^1(X,\mathbb{Z}) = H^3(X,\mathbb{Z})_{\mathrm{tors}} = H^0(X,\Omega_X^3) = 0, \qquad \dim J(X) = \dim H^1(X,\Omega_X^2) = 5,$$

and X is unirational by [7]. Let S be the surface parametrizing the lines contained in X, and let $\alpha : S \to \mathrm{Alb}(S)$ be the map of S into its Albanese variety. One of the essential results of [4] is the following

Theorem 2.4 ([4] 11.19 and 13.4).- There exists an isomorphism

$$J(X) \xrightarrow{\sim} \mathrm{Alb}(S),$$

which carries the divisor Θ of $J(X)$ to the image of $S \times S$ under the map

$$(x,y) \xrightarrow{\delta} \alpha(x) - \alpha(y).$$

Clemens and Griffiths then manage to use results of [1] to prove that $(\mathrm{Alb}(S), \delta(S \times S))$ cannot be the polarized Jacobian of a curve. Another proof of the irrationality of X is suggested in the appendix to [4]. It is a matter of proving

Theorem 2.5.- The singular locus of the divisor Θ of $J(X)$ has dimension 0, hence codimension > 4. It is even reduced to the image under δ of the diagonal of $S \times S$.

It is known that this is never the case for the divisor Θ of a Jacobian.

C) Manin and Iskovskih

The group of birational automorphisms of a variety X (equal to the group of $\mathbb{C}$-automorphisms of $k(X)$) is a birational invariant of X. It is this invariant that Manin and Iskovski use to show that a smooth quartic hypersurface in $\mathbb{P}^4(\mathbb{C})$ is never rational:

Theorem 2.6 ([6]).- Let X and Y be two smooth quartic hypersurfaces in $\mathbb{P}^4(\mathbb{C})$. Every birational isomorphism $a : X \longrightarrow Y$ is automatically biregular.

Since certain quartic hypersurfaces are unirational, [7] this theorem provides a new example of unirational non-rational varieties.

3. The example of Artin and Mumford

Consider in $\mathbb{P}^2(\mathbb{C})$ a configuration consisting of

a) two smooth cubic curves C_1 and C_2, with equations $f_1 = 0$ and $f_2 = 0$, meeting transversely;

b) a smooth conic Q, with equation $q = 0$, which meets each C_i in three distinct points of tangency.

We shall construct

a) a vector bundle V of rank 3 on $\mathbb{P}^2(\mathbb{C})$;

b) a quadratic form $\Phi : V \to \mathcal{L}$ ($\mathcal{L}$ an invertible sheaf) of discriminant $f_1 f_2$ (if Δ, a section of $\mathcal{L}^{\otimes 3} \otimes (\bigwedge^3 V)^{\otimes(-2)}$, is the discriminant, the subscheme of $\mathbb{P}^2(\mathbb{C})$ with equation $\Delta = 0$ is $C_1 \cup C_2$).

Let $X \subset \mathbb{P}(V^*)$ be the conic bundle over $\mathbb{P}^2(\mathbb{C})$, degenerating along $C_1 \cup C_2$, with homogeneous equation $\Phi = 0$. The following conditions will be satisfied.

c) Φ has rank ≥ 2 at every point: for $s \in C_1 \cup C_2$, the fibre $X_s = f^{-1}(s)$ is the union of two concurrent lines; the only singular points of X are the singular points of the fibres X_s for $s \in C_1 \cap C_2$.

d) Let S be the double cover of $\mathbb{P}^2(\mathbb{C})$, ramified along Q, and let $X_S = X \times_{\mathbb{P}^2(\mathbb{C})} S$. Then X_S/S has a section

$$\begin{tikzcd} & X \arrow[d] \\ S \arrow[ur, "\alpha"] \arrow[r, "\pi"] & \mathbb{P}^2(\mathbb{C}) . \end{tikzcd}$$

e) For $s \in (C_1 \cup C_2) - Q$, the two points $\alpha(t)$ for which $\pi(t) = s$ lie in the two irreducible components of X_s.

Locally (for the usual topology) on $C_1 \cup C_2$, the restriction of the bundle X to $C_1 \cup C_2$ is obtained by gluing two projective-line bundles along a section. However, when one traverses a loop in $C_1 \cup C_2$, these two bundles may be exchanged: there exists an unramified double cover $(C_1 \cup C_2)^*$ of $C_1 \cup C_2$ whose two local sections correspond, locally, to the two projective-line bundles of which $X|_{C_1 \cup C_2}$ is the union. Let $\pi_i : C_i^* \to C_i$

be the double cover of C_i induced by $(C_1 \cup C_2)^*$. We shall use from e) only the following corollary

Lemma 3.1.- C_i^* is irreducible.

By e), C_i^* is the normalization of the double cover of C_i induced by S. The lemma follows from the fact that the section q of $\mathcal{O}(2)|_{C_i}$ is not the square of a section of $\mathcal{O}(1)|_{C_i}$; if q were a square, the three intersection points of Q with C_i would indeed be collinear.

To the double cover C_i^* of C_i correspond a character of order 2 of the fundamental group of C_i and a local system A_i, locally isomorphic to $\mathbb{Z}$, defined by the exact sequence of sheaves on C_i

$$0 \longrightarrow A_i \longrightarrow \pi_{i*}\mathbb{Z} \xrightarrow{\mathrm{Tr}} \mathbb{Z} \longrightarrow 0,$$

where Tr is the "sum of the values on the two sheets of the covering". One has

$$H^0(C_i, A_i) = 0, \qquad H^1(C_i, A_i) = \mathbb{Z}/2, \qquad H^2(C_i, A_i) = \mathbb{Z}/2. \tag{3.2}$$

We shall use d) only to prove the following fact

Lemma 3.3.- X is unirational.

The surface S is a quadric in $\mathbb{P}^3(\mathbb{C})$, hence is rational. Since X_S/S has a section, there is a birational isomorphism $X_S \sim S \times \mathbb{P}^1(\mathbb{C})$, and X is the image of the rational variety X_S.

The nine singular points of X are ordinary quadratic points. Each can be resolved as follows (cf. [3])

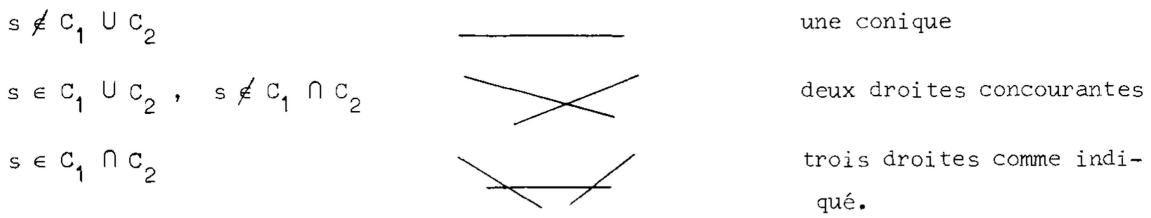
a) Blow it up; this resolves X and replaces the singular point by a quadric P, isomorphic to $\mathbb{P}^1(\mathbb{C}) \times \mathbb{P}^1(\mathbb{C})$.

b) Choose a projection $P \xrightarrow{\beta} \mathbb{P}^1(\mathbb{C})$, and contract in the blow-up $\widetilde{X}_1$ of X the fibres of this projection; this is possible because β is smooth, because its

fibres are isomorphic to $\mathbb{P}^1(\mathbb{C})$, and because the normal bundle of P in $\widetilde{X}_1$ induces on each fibre a bundle $\mathcal{O}(-1)$ ([8]).

Let $\widetilde{X}$ be the resulting variety. There is a map from $\widetilde{X}$ into X, and $\widetilde{X}$ is obtained from X by replacing each singular point by a projective line.

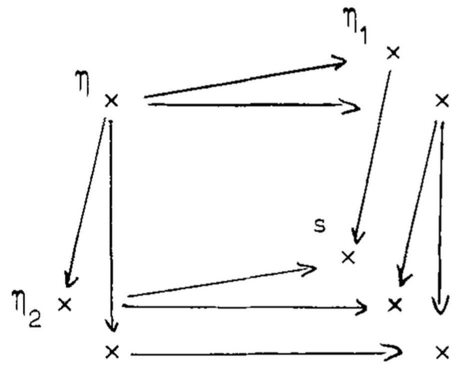
The fibres of the projection $p : \widetilde{X} \to \mathbb{P}^2(\mathbb{C})$ are of the following three types



English key: $s \notin C_1 \cup C_2$, a conic; $s \in C_1 \cup C_2$ and $s \notin C_1 \cap C_2$, two concurrent lines; $s \in C_1 \cap C_2$, three lines as shown.

When one passes from the generic point η of $\mathbb{P}^2(\mathbb{C})$ to the generic point η_i of C_i $(i = 1, 2)$, and to $s \in C_1 \cap C_2$, the irreducible components specialize as indicated in the following diagram

(3.4)



(the crosses represent irreducible components, all the multiplicities being one). The tedious verification of this local result is omitted. From it one obtains

Lemma 3.5.- One has

$$R^0p_*\mathbb{Z} = \mathbb{Z}, \qquad R^1p_*\mathbb{Z} = 0,$$

and an exact sequence of sheaves

$$0 \longrightarrow A_1 \oplus A_2 \longrightarrow R^2p_*\mathbb{Z} \xrightarrow{\mathrm{Tr}} \mathbb{Z} \longrightarrow 0.$$

We can now use the Leray spectral sequence of p to compute $H^*(\widetilde{X}, \mathbb{Z})$. The essential fact is the following (which implies the irrationality of X).

Lemma 3.6.- One has

$$H^3(\widetilde{X}, \mathbb{Z}) = \mathbb{Z}/(2).$$

Let us compute the E_2-terms of the Leray spectral sequence of p. One has

$$\begin{aligned} R^0p_*\mathbb{Z} &= \mathbb{Z} \quad \text{hence} \quad E_2^{p,0} = \mathbb{Z}, 0, \mathbb{Z}, 0, \mathbb{Z} \quad \text{for } p = 0 \text{ to } 4, \\ R^1p_*\mathbb{Z} &= 0 \quad \text{hence} \quad E_2^{p,1} = 0. \end{aligned}$$

To compute $E_2^{p,2} = H^p(R^2p_*\mathbb{Z})$, one uses the long exact sequence defined by the short exact sequence of sheaves 3.5. One finds

$$\begin{gathered} 0 \longrightarrow E^{0,2} \xrightarrow{a} \mathbb{Z} \xrightarrow{j} \mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow E^{1,2} \longrightarrow 0, \\ 0 \longrightarrow \mathbb{Z}/2 \oplus \mathbb{Z}/2 \longrightarrow E^{2,2} \longrightarrow \mathbb{Z} \longrightarrow 0, \\ E^{3,2} = 0 \qquad \text{and} \qquad E^{4,2} = \mathbb{Z}. \end{gathered}$$

One easily verifies that $a(E^{0,2}) = 2\mathbb{Z}$ (this fact is, moreover, not necessary for the proof that $H^3(\widetilde{X}, \mathbb{Z})_{\text{tors}} \neq 0$). The E_2^{pq} are therefore as follows

$$\begin{array}{c|ccccc} q & & & & & \\ & 2\mathbb{Z} & \mathbb{Z}/2 & E^{2,2} & 0 & \mathbb{Z} \\ & 0 & 0 & 0 & 0 & 0 \\ & \mathbb{Z} & 0 & \mathbb{Z} & 0 & \mathbb{Z} \\ \hline & & & & & p \end{array}$$

The differentials d_r can only be zero, so that $E_2^{pq} = E_\infty^{pq}$, and the lemma follows.

It is apparent that it was essential to have two curves $C_i \subset \mathbb{P}^2(\mathbb{C})$ along which X degenerates, for a group $\mathbb{Z}/2$ is killed in passing from

$$H^1\big(\mathrm{Ker}(R^2p_*\mathbb{Z} \to \mathbb{Z})\big) \quad \text{to} \quad H^1R^2p_*\mathbb{Z} = E^{1,2}.$$

It remains to carry out the promised constructions. Artin and Mumford know how to prove *a priori* the existence of a conic bundle of the required type. I shall confine myself to giving equations that define one.

1) The image of f_1f_2 in $H^0(Q,\mathcal{O}(6))$ is the square of $g_1 \in H^0(Q,\mathcal{O}(3))$, because Q is rational and the zeros of f_1f_2 on Q are double. This g_1 lifts to $g \in H^0(\mathbb{P}^2(\mathbb{C}),\mathcal{O}(3))$ because $H^1(\mathbb{P}^2(\mathbb{C}),\mathcal{O}(1)) = 0$. Since $f_1f_2 - g^2$ vanishes on Q, it is a multiple of q:

$$f_1f_2 = g^2 + qd.$$

2) Take

$$V = \mathcal{O}(-1) \oplus \mathcal{O}(-2) \oplus \mathcal{O},$$

and the form

$$\Phi : V \longrightarrow \mathcal{O} : (x, y, z) \longmapsto qx^2 + 2gxy - dy^2 - z^2.$$

One has $\Delta = g^2 + qd = f_1f_2$. At a point where Φ had rank ≤ 1, one would have $\Delta = q = d = 0$, whence $g = 0$, and $\Delta = g^2 + qd$ would have a double zero. This is impossible because $C_1 \cap C_2 \cap Q = \varnothing$. Finally, S is identified with the subscheme of X having equation $y = 0$, and d) and e) follow.

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MODULAR FORMS OF WEIGHT 1

BY PIERRE DELIGNE AND JEAN-PIERRE SERRE

To Henri Cartan,
on the occasion of his
70th birthday

Introduction

The Euler product decomposition and the functional equation of the Dirichlet series associated by Hecke with modular forms of weight 1 suggest that these forms correspond to Artin L-functions of degree 2 over $\mathbf{Q}$, in other words to representations of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in $\mathrm{GL}_2(\mathbf{C})$. It is such a correspondence, conjectured by Langlands, that we establish here.

The first three sections are preliminary. Section 4 contains the statement of the main theorem, together with several supplements. The proof occupies §§ 5 through 9. Its principle is as follows: for every prime number ℓ, one first constructs a representation of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in characteristic ℓ (cf. § 6); one then shows that the images of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ in these various representations are "small," which makes it possible to lift them to characteristic 0 and obtain the desired complex representation (§§ 7 and 8); the "smallness" in question itself follows from an average bound for the eigenvalues of the Hecke operators (Rankin, cf. § 5). Section 9 contains an estimate for the coefficients of modular forms of weight 1.

We point out that at one essential point (§ 6, Theorem 6.1) we have used results proved by one of us (P. Deligne), but for which no complete proof has yet been published; pending such a publication (as well as that of SGA 5, on which these results depend), we ask the reader to accept them.

§ 1. Review (analytic) of modular forms

1.1. Let N be an integer ≥ 1. With N we associate the subgroups

$$\Gamma(N) \subset \Gamma_1(N) \subset \Gamma_0(N).$$

of $\mathrm{SL}_2(\mathbf{Z})$ defined by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma(N) \iff a \equiv d \equiv 1 \pmod N \quad \text{and} \quad b \equiv c \equiv 0 \pmod N,$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_1(N) \iff a \equiv d \equiv 1 \pmod N \quad \text{and} \quad c \equiv 0 \pmod N,$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N) \iff c \equiv 0 \pmod N.$$

1.2. Let f be a function on the upper half-plane $H = \{z \mid \mathrm{Im}(z) > 0\}$. If k is an integer and $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is an element of $\mathrm{SL}_2(\mathbf{R})$, set

$$(f\mid_k \gamma)(z) = (cz+d)^{-k} f(\gamma z), \qquad \text{where} \qquad \gamma z = \frac{az+b}{cz+d}.$$

Let Γ be a subgroup of $\mathrm{SL}_2(\mathbf{Z})$ containing $\Gamma(N)$. We say that f is modular of weight k on Γ if:

(1.2.1) $f\mid_k \gamma = f$ for every $\gamma \in \Gamma$;

(1.2.2) f is holomorphic on H;

(1.2.3) f is "holomorphic at the cusps," i.e., for every $\sigma \in \mathrm{SL}_2(\mathbf{Z})$, the function $f\mid_k \sigma$ has a power-series expansion in $e^{2\pi i z/N}$ with exponents ≥ 0.

If, in (1.2.3), "exponents ≥ 0" is replaced by "exponents > 0," one obtains the notion of a cusp form.

1.3. Let f be a modular form of weight k on $\Gamma(N)$. For f to be a modular form on $\Gamma_1(N)$, it is necessary and sufficient that $f(z+1) = f(z)$, or equivalently that f have an expansion of the form

$$\sum_{n=0}^{\infty} a_n q^n, \qquad \text{where} \qquad q = e^{2\pi i z}.$$

In what follows, this expansion will be denoted by $f_\infty(q)$, or simply by f.

1.4. Let f be a modular form of weight k on $\Gamma_1(N)$. If $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ is an element of $\Gamma_0(N)$, the form $f\mid_k \gamma$ depends only on the image of d in $(\mathbf{Z}/N\mathbf{Z})^*$; it is denoted by $f \mid R_d$. We have $f \mid R_{-1} = (-1)^k f$.

1.5. Let ε be a Dirichlet character modulo N, that is, a homomorphism

$$\varepsilon : (\mathbf{Z}/N\mathbf{Z})^* \longrightarrow \mathbf{C}^*.$$

We say that ε is even (respectively odd) if $\varepsilon(-1) = 1$ (respectively if $\varepsilon(-1) = -1$).

Let k be an integer of the same parity as ε [i.e., $\varepsilon(-1) = (-1)^k$]. A modular form of type (k, ε) on $\Gamma_0(N)$ is a modular form f of weight k on $\Gamma_1(N)$ such that

$$f \mid R_d = \varepsilon(d) f$$

for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$, i.e.,

$$f\left(\frac{az+b}{cz+d}\right) = \varepsilon(d)(cz+d)^k f(z) \quad \text{for every} \quad \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma_0(N).$$

(Note that if ε and k did not have the same parity, this formula would imply $f = 0$.)

Every modular form of weight k on $\Gamma_1(N)$ is a linear combination of forms of types (k, ε_i) on $\Gamma_0(N)$, where the ε_i are the various characters of $(\mathbf{Z}/N\mathbf{Z})^*$ having the same parity as k.

This follows (cf. [16], p. IV-13) by observing that the map

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \longmapsto d \pmod N$$

induces an isomorphism from $\Gamma_0(N)/\Gamma_1(N)$ onto $(\mathbf{Z}/N\mathbf{Z})^*$.

1.6. HECKE OPERATORS. — Let $f = \sum a_n q^n$ be a modular form of type (k, ε) on $\Gamma_0(N)$, and let p be a prime number. Set

$$f \mid T_p = \sum a_{pn} q^n + \varepsilon(p) p^{k-1} \sum a_n q^{pn} \qquad \text{if } p \nmid N, \tag{1.6.1}$$

$$f \mid U_p = \sum a_{pn} q^n \qquad \text{if } p \mid N. \tag{1.6.2}$$

This gives another modular form of type (k, ε) on $\Gamma_0(N)$, which is a cusp form if f is.

1.7. PRIMITIVE FORMS. — We refer to [2] (when $\varepsilon = 1$) and to [6], [12], [13] (in the general case) for the definition of primitive cusp forms ("newforms") of type (k, ε) on $\Gamma_0(N)$.

If $f = \sum_{n=1}^{\infty} a_n q^n$ is such a form, then $a_1 = 1$, and f is an eigenform for the Hecke operators T_p and U_p, with corresponding eigenvalues a_p. It follows that the Dirichlet series

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s} \tag{1.7.1}$$

has the Euler product

$$\Phi_f(s) = \prod_{p \mid N} \frac{1}{1 - a_p p^{-s}} \prod_{p \nmid N} \frac{1}{1 - a_p p^{-s} + \varepsilon(p) p^{k-1-2s}}. \tag{1.7.2}$$

1.8. If f is as above and $p \mid N$, the absolute value of a_p is given by the following rule ([12], [14]):

$$a_p = 0 \quad \text{if } p^2 \mid N \text{ and if } \varepsilon \text{ can be defined modulo } N/p;$$

$$|a_p| = p^{(k-1)/2} \quad \text{if } \varepsilon \text{ cannot be defined modulo } N/p;$$

$$|a_p| = p^{k/2-1} \quad \text{if } p^2 \nmid N \text{ and if } \varepsilon \text{ can be defined modulo } N/p.$$

(It will follow from Theorem 4.6 below that the last case does not occur when $k = 1$.)

1.9. Every form f of type (k, ε) on $\Gamma_0(N)$ can be written

$$f(z) = E(z) + \sum \lambda_i f_i(d_i z),$$

where E is an Eisenstein series and f_i is a primitive cusp form of type (k, ε) on $\Gamma_0(N_i)$, with N_i a divisor of N such that ε can be defined modulo N_i, and d_i a divisor of N/N_i. Moreover, this decomposition is unique in the evident sense.

§ 2. Review (geometric) of modular forms

2.1. Let k and N be integers ≥ 1, and let μ_N be the group scheme of Nth roots of unity. From the geometric point of view, a modular form of weight k on $\Gamma_1(N)$ is a rule that assigns, to every elliptic curve E equipped with an embedding $\alpha : \mu_N \longrightarrow E$, a section of the kth tensor power $\omega_E^{\otimes k}$ of ω_E, where ω_E is the dual of the Lie algebra of E.

More precisely:

(a) An elliptic curve over a scheme S is a proper smooth morphism $E \longrightarrow S$, equipped with a section $e : S \longrightarrow E$, whose geometric fibers are elliptic curves. When S is the spectrum of a commutative ring A, we also say that E is an elliptic curve over A. We set $\omega_E = e^*\Omega^1_{E/S}$; when $S = \mathrm{Spec}(A)$, ω_E is identified with an invertible A-module.

(b) Let R be a commutative ring in which N is invertible. A modular form of weight k on $\Gamma_1(N)$, meromorphic at infinity and defined over R, is a rule that assigns, to every elliptic curve E over an R-algebra A, equipped with an embedding $\alpha : \mu_N \longrightarrow E$, an element $f(E, \alpha)$ of $\omega_E^{\otimes k}$. This rule is required to be compatible with isomorphisms and with extension of scalars.

(c) We say that f is holomorphic at infinity if it extends to a rule $\tilde{f}$ defined for pairs (E, α) in which E is a generalized elliptic curve ([7], II. 1.12) and α is an embedding of μ_N into E whose image meets every irreducible component of every geometric fiber ([7], IV. 4.14). If it exists, the rule $\tilde{f}$ is unique.

Suppose that R is a field. We say that f is a cusp form if it is holomorphic at infinity and if $\tilde{f}(E, \alpha) = 0$ whenever E is a degenerate (i.e., nonsmooth) elliptic curve over an algebraically closed extension of R.

These notions could also be defined in terms of Laurent expansions in q (cf. [7], VII, § 3).

2.2. Let f be as above. If $d \in (\mathbf{Z}/N\mathbf{Z})^*$, the modular form $f \mid R_d$ is defined by

$$(f \mid R_d)(E, \alpha) = f(E, d\alpha). \tag{2.2.1}$$

If ε is a homomorphism from $(\mathbf{Z}/N\mathbf{Z})^*$ to R^*, we say that f is of type (k, ε) on $\Gamma_0(N)$ if $f \mid R_d = \varepsilon(d) f$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$.

2.3. Let p be a prime number not dividing N. The Hecke operator T_p is then defined on the spaces of modular forms. If f is such a form and (E, α) is defined over an algebraically closed field of characteristic $\neq p$, then

$$(f \mid T_p)(E, \alpha) = \frac{1}{p} \sum_{\varphi} \varphi^*(f(\varphi E, \varphi \circ \alpha)),$$

where φ runs through the classes of isogenies of degree p with source E (two isogenies being in the same class if their kernels are equal).

The operators T_p commute with one another and with the R_d.

2.4. Take $R = \mathbf{C}$. The datum of $\alpha : \mu_N \longrightarrow E$ is then equivalent to that of the point

$$\alpha(\exp(2\pi i/N)),$$

which has order N. To a modular form f as above, we associate a function (also denoted f) on the upper half-plane H by the rule

$$f(z) = f(E_z, 1/N)/(2\pi i \, du)^{\otimes k}, \tag{2.4.1}$$

where E_z denotes the elliptic curve $\mathbf{C}/(\mathbf{Z} \oplus z\mathbf{Z})$.

Put $f(z) = f_\infty(e^{2\pi i z})$. Formula (2.4.1) can be rewritten as

$$f_\infty(q) = f(\mathbf{C}^*/q^{\mathbf{Z}}, \mathrm{Id})/(dt/t)^{\otimes k} \qquad (0 < |q| < 1), \tag{2.4.2}$$

where Id is induced by the inclusion of μ_N in $\mathbf{C}^*$.

This construction identifies the spaces of modular forms in the sense of 2.1 and 2.2 with the spaces having the same names in § 1; the same holds for the operators T_p and R_d.

2.5. For the definition of the Tate curve $\mathbf{G}_m/q^{\mathbf{Z}}$ over the ring $\mathbf{Z}((q)) = \mathbf{Z}[[q]](q^{-1})$, we refer to [7], VII, § 1. This curve is equipped with an invariant differential dt/t and a natural embedding $\mathrm{Id} : \mu_N \longrightarrow \mathbf{G}_m/q^{\mathbf{Z}}$. If f is a modular form of weight k on $\Gamma_1(N)$, meromorphic at infinity and defined over a ring R, set

$$f_\infty(q) = f(\mathbf{G}_m/q^{\mathbf{Z}}, \mathrm{Id})/(dt/t)^{\otimes k} \in \mathbf{Z}((q)) \otimes R \subset R((q)).$$

(In this formula, $\mathbf{G}_m/q^{\mathbf{Z}}$ denotes the curve over $\mathbf{Z}((q)) \otimes R$ obtained from the Tate curve by extension of scalars.) Put

$$f_\infty(q) = \sum a_n q^n \qquad \text{and} \qquad (f \mid R_d)_\infty(q) = \sum a_n(d) q^n, \quad d \in (\mathbf{Z}/N\mathbf{Z})^*;$$

if p is a prime number not dividing N, then

$$(f \mid T_p)_\infty(q) = \sum a_{pn} q^n + p^{k-1} \sum a_n(p) q^{np}. \tag{2.5.1}$$

In particular, if f is of type (k,ε) on $\Gamma_0(N)$, then

$$(f \mid T_p)_\infty(q) = \sum a_{pn}q^n + \varepsilon(p)p^{k-1}\sum a_n q^{pn}. \tag{2.5.2}$$

When $R = \mathbf{C}$, $f_\infty(q)$ is the series expansion in (2.4.2); this is proved in [7], VII, § 4 (at least for holomorphic f, the only case that concerns us). Formula (2.5.2) recovers (1.6.1).

2.6. If K is a field of characteristic 0, let S_K denote the vector space of cusp forms of weight k on $\Gamma_1(N)$ that are defined over K. We have

$$S_K = K \otimes_{\mathbf{Q}} S_{\mathbf{Q}}; \tag{2.6.1}$$

this follows by interpreting S_K as the space of sections of an invertible sheaf on the "algebraic stack" corresponding to $\Gamma_1(N)$ (cf. [7], VII, 3.2).

If K' is a subfield of K, a form $f \in S_K$ belongs to $S_{K'}$ if and only if the coefficients of the series $f_\infty(q)$ belong to K'. This follows by reducing to the case in which K is algebraically closed and observing that, for every K'-automorphism σ of K, the forms f and $\sigma(f)$ have the same series expansion and therefore coincide.

PROPOSITION 2.7. — Let L be the set of $f \in S_{\mathbf{C}}$ such that $(f \mid R_d)_\infty(q) \in \mathbf{Z}[[q]]$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$. Then:

(2.7.1) L is a free $\mathbf{Z}$-module of finite type, stable under the operators T_p and R_d.

(2.7.2) For every field K of characteristic 0, one has $S_K = K \otimes L$.

(2.7.3) The eigenvalues of the T_p on $S_{\mathbf{C}}$ are algebraic integers in a finite extension of $\mathbf{Q}$.

(2.7.4) If $f \in S_{\mathbf{C}} = \mathbf{C} \otimes L$ satisfies $f \mid T_p = a_p f$, then, for every automorphism σ of $\mathbf{C}$, the form $\sigma(f)$ satisfies $\sigma(f) \mid T_p = \sigma(a_p)\sigma(f)$. If f is of type (k,ε) on $\Gamma_0(N)$, then $\sigma(f)$ is of type $(k,\sigma(\varepsilon))$ on $\Gamma_0(N)$.

If $f \in S_{\mathbf{Q}}$, then $f \mid R_d \in S_{\mathbf{Q}}$ for every $d \in (\mathbf{Z}/N\mathbf{Z})^*$, and the series $(f \mid R_d)_\infty(q)$ belong to $\mathbf{Z}[[q]] \otimes \mathbf{Q}$ and hence have bounded denominators. It follows that a nonzero multiple of f belongs to L, whence $\mathbf{Q} \otimes L = S_{\mathbf{Q}}$ and, by (2.6.1), $K \otimes L = S_K$ for every field K of characteristic 0. The fact that L is of finite type follows because the linear functionals "nth coefficients of $(f \mid R_d)_\infty(q)$" separate the elements of $S_{\mathbf{Q}}$.

The stability of L under the R_d (respectively the T_p) is evident (respectively follows from (2.5.1)). Assertions (2.7.3) and (2.7.4) follow.

REMARK 2.8. — The fact that the series $f_\infty(q)$, for $f \in S_{\mathbf{Q}}$, has bounded denominators was deduced here from the fact that the Tate curve is defined over $\mathbf{Z}((q)) \otimes \mathbf{Q}$. One could also have used Shimura's Theorem 3.5.2 [24], valid when $k \geq 2$, and reduced weight 1 to weight 13 by multiplication by Δ.

§ 3. Review of Galois representations

3.1. Let $\overline{\mathbf{Q}}$ be an algebraic closure of $\mathbf{Q}$, and let $G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$. We shall have to consider linear representations of G, that is, continuous homomorphisms

$$\rho : G \longrightarrow \mathrm{GL}_n(k),$$

where k is of one of the following types:
(a) the field $\mathbf{C}$ (with the discrete topology);
(b) a finite field (with the discrete topology);
(c) a finite extension of an ℓ-adic field $\mathbf{Q}_\ell$ (with its natural topology).
In the first two cases, the image of ρ is finite.

If p is a prime number, we say that ρ is unramified at p if it is trivial on the inertia group of a place of $\overline{\mathbf{Q}}$ extending p. We then denote by $F_{p,\rho}$ the image under ρ of the Frobenius substitution [1] at p; it is an element of $\mathrm{GL}_n(k)$ defined up to conjugacy. Set

$$P_{p,\rho}(T) = \det(1 - F_{p,\rho}T) = 1 - \mathrm{Tr}(F_{p,\rho})T + \cdots + (-1)^n \det(F_{p,\rho})T^n. \tag{3.1.1}$$

Knowledge of the polynomials $P_{p,\rho}(T)$ almost determines ρ. More precisely:

LEMMA 3.2. — Let X be a set of prime numbers of density 1, and let ρ and ρ' be two semisimple linear representations of G. Suppose that, for every $p \in X$, ρ and ρ' are unramified and that $P_{p,\rho}(T) = P_{p,\rho'}(T)$ (respectively that $\mathrm{Tr}(F_{p,\rho}) = \mathrm{Tr}(F_{p,\rho'})$ when k has characteristic 0). Then ρ and ρ' are isomorphic.

This follows from the Chebotarev density theorem, combined with the fact that a semisimple linear representation of a group is determined up to isomorphism by the corresponding characteristic polynomials (respectively by the traces if the field has characteristic 0) ([3], § 30.16).

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3.3. In what follows, we shall apply Lemma 3.2 to the particular case in which X is the set of prime numbers not dividing a given integer N; we shall then say that ρ and ρ' are unramified outside N.

3.4. When $k = \mathbf{C}$, the semisimplicity condition is automatically satisfied, since $\rho(G)$ and $\rho'(G)$ are finite.

§ 4. Results

(a) STATEMENT OF THE MAIN THEOREM

THEOREM 4.1. — Let N be an integer ≥ 1, let ε be a Dirichlet character modulo N such that $\varepsilon(-1) = -1$, and let f be a nonzero modular form of type $(1, \varepsilon)$ on $\Gamma_0(N)$. Suppose that f is an eigenform for the T_p, $p \nmid N$, with eigenvalues a_p.

[1] We adopt here Artin's conventions [1]. Our "Frobenius substitution" is therefore the element denoted φ in [5]; its inverse is the "geometric Frobenius."

There then exists a linear representation

$$\rho : G \longrightarrow \mathrm{GL}_2(\mathbf{C}), \qquad \text{where} \qquad G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}),$$

which is unramified outside N and such that

$$\mathrm{Tr}(F_{p,\rho}) = a_p \qquad \text{and} \qquad \det(F_{p,\rho}) = \varepsilon(p) \quad \text{for every } p \nmid N. \tag{4.1.1}$$

This representation is irreducible if and only if f is a cusp form.

The proof will be given in § 8.

COROLLARY 4.2. — The eigenvalues a_p are sums of two roots of unity; in particular, $|a_p| \le 2$.

In other words, the "Ramanujan-Petersson conjecture" is true in weight 1; it is also known to be true in weights ≥ 2, cf. [5], 8.2.

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4.3. By Lemma 3.2, the representation ρ associated with f by 4.1 is unique up to isomorphism.

4.4. The formula $\det(F_{p,\rho}) = \varepsilon(p)$ shows that

$$\det(\rho) = \varepsilon,$$

where ε is identified with the character $G \longrightarrow \mathbf{C}^*$ corresponding to it under class field theory (it is simply the composite of ε with the homomorphism $G \longrightarrow (\mathbf{Z}/N\mathbf{Z})^*$ afforded by the action of G on the Nth roots of unity).

4.5. Let c denote the element of G corresponding to complex conjugation (for an embedding of $\overline{\mathbf{Q}}$ into $\mathbf{C}$). Since ε is odd, 4.4 shows that $\det(\rho(c)) = -1$; since c has order 2, this means that $\rho(c)$ is conjugate to the matrix

$$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

(b) ARTIN CONDUCTOR AND LOCAL FACTORS. — We retain the hypotheses and notation of Theorem 4.1.

THEOREM 4.6. — Suppose that f is a primitive cusp form, with coefficients a_n, $n \geq 1$. Let ρ be the corresponding representation of G. Then:

a. The Artin conductor of ρ is equal to N;

b. The Artin L-function $L(s, \rho)$ is equal to

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s}.$$

(For the definition of the L-series and the conductor of a representation, see [1].)

COROLLARY 4.7. — The representation ρ is ramified at every prime divisor of N.

This follows from (a).

COROLLARY 4.8. — The function $L(s, \rho)$ is entire.

[In other words, the "Artin conjecture" holds for ρ, cf. (c) below.]

Indeed, Hecke theory shows that $\Phi_f(s)$ is entire.

Proof of 4.6. — It uses the functional equations satisfied by $\Phi_f(s)$ and $L(s, \rho)$ (compare [9], pp. 172-177).

(i) Put $\tilde{f} = \sum \overline{a_n} q^n$. Since f is primitive, there exists a constant $\lambda \neq 0$ such that $f(-1/Nz) = \lambda z \tilde{f}(z)$, cf. [12] and [13]. Mellin transformation then gives

$$\Psi_f(1-s) = \mu \widetilde{\Psi}_f(s) \qquad \text{with} \qquad \mu = i\lambda/N^{1/2},$$

where

$$\Psi_f(s) = N^{s/2}(2\pi)^{-s}\Gamma(s)\Phi_f(s) \qquad \text{and} \qquad \widetilde{\Psi}_f(s) = \Psi_{\tilde{f}}(s).$$

(ii) By 4.5, the "factor at infinity" of $L(s, \rho)$ is $(2\pi)^{-s}\Gamma(s)$. If M is the conductor of ρ, and if we set

$$\xi(s, \rho) = M^{s/2}(2\pi)^{-s}\Gamma(s)L(s, \rho),$$

then

$$\xi(1-s, \rho) = \nu \cdot \xi(s, \overline{\rho}) \qquad \text{with} \qquad \nu \in \mathbf{C}^*.$$

(iii) Put

$$F(s) = (N/M)^{s/2}\Psi_f(s)/\xi(s, \rho) \qquad \text{and} \qquad \widetilde{F}(s) = (N/M)^{s/2}\widetilde{\Psi}_f(s)/\xi(s, \overline{\rho}).$$

The preceding formulas show that

$$F(1-s) = \omega \cdot \widetilde{F}(s) \qquad \text{with} \qquad \omega = \mu/\nu.$$

But if p is a prime number not dividing N, the p-factors of $\Psi_f(s)$ and $\xi(s, \rho)$ agree by 4.1. Hence

$$F(s) = A^s \prod_{p|N} F_p(s),$$

where

$$A = (N/M)^{1/2} \qquad \text{and} \qquad F_p(s) = \frac{1 - a_p p^{-s}}{(1 - b_p p^{-s})(1 - c_p p^{-s})},$$

where $1 - a_p p^{-s}$ is the p-factor of $\Psi_f(s)$ and $(1 - b_p p^{-s})(1 - c_p p^{-s})$ is that of $\xi(s, \rho)$ (note that b_p and c_p may be zero). It remains only to show that A and the F_p are equal to 1. For this we use the following elementary lemma:

LEMMA 4.9. — Let $G(s) = A^s \prod_p G_p(s)$ and $H(s) = A^s \prod_p H_p(s)$ be two finite Euler products. Suppose that

$$G(1-s) = \omega \cdot H(s) \qquad \text{with} \qquad \omega \in \mathbf{C}^*; \tag{4.9.1}$$

(4.9.2) Each of G_p and H_p is a finite product of terms of the form $(1-\alpha_p^{(i)}p^{-s})^{\pm 1}$, with $|\alpha_p^{(i)}| < p^{1/2}$.
Then $A = 1$ and $G_p = H_p = 1$ for every p.
If H_p is not equal to 1, the function H has infinitely many zeros (or poles) of the form

$$(\log(\alpha_p^{(i)}) + 2\pi i n)/\log p, \qquad n \in \mathbf{Z},$$

and it is easy to see that these cannot all be zeros (or poles) of $G(1-s)$; indeed, the hypothesis $|\alpha_p^{(i)}| < p^{1/2}$ ensures that none of the $\alpha_p^{(i)}$ can equal a $p/\alpha_p^{(j)}$.

(iv) It remains to verify that a_p, b_p, c_p, and their conjugates satisfy (4.9.2), i.e., have absolute value $< p^{1/2}$. This is clear for b_p and c_p, which are either 0 or roots of unity. For a_p, one may invoke 1.8, which shows that $|a_p| \leq 1$; alternatively, one may use Rankin's inequality:

$$|a_n| = O(n^{1/2-1/5}), \qquad cf.\ [18];$$

applying it to $n = p^m$ and observing that $a_n = (a_p)^m$, one obtains

$$|a_p| \leq p^{1/2-1/5} < p^{1/2}.$$

This completes the proof of 4.6.

(c) CHARACTERIZATION OF THE REPRESENTATIONS ATTACHED TO FORMS OF WEIGHT 1. — Retain the notation of (a), and suppose that the form f under consideration is a cusp form. The corresponding representation

$$\rho : G \longrightarrow \mathrm{GL}_2(\mathbf{C})$$

then has the following properties:
(i) ρ is irreducible (4.1);
(ii) $\det \rho$ is an odd character (4.4);
(iii) For every continuous character $\chi : G \longrightarrow \mathbf{C}^*$, the Artin L-function $L(s, \rho \otimes \chi)$ is entire [this follows from 4.8 applied to the cusp form

$$f_\chi = \sum \chi(n) a_n q^n].$$

Conversely:

THEOREM 4.10. (Weil-Langlands). — Let $\rho : G \longrightarrow \mathrm{GL}_2(\mathbf{C})$ be a continuous representation of $G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ satisfying conditions (i), (ii), and (iii) above. Put

$$L(s,\rho) = \sum a_n n^{-s}, \qquad f = \sum a_n q^n, \qquad \varepsilon = \det \rho, \qquad N = \text{conduct.}\, \rho.$$

Then f is a primitive cusp form of type $(1, \varepsilon)$ on $\Gamma_0(N)$, and ρ is the representation attached to f.

According to Langlands (cf. [27], pp. 152 and 160), the constants in the functional equations of the series $\sum a_n\chi(n)n^{-s}$ satisfy the identity required to apply the Hecke-Weil characterization of modular forms ([12], [26]). It follows that f is modular of type $(1,\varepsilon)$ on $\Gamma_0(N)$; it is clear that f is an eigenform for the T_p and the U_p, and that the representation associated with it is isomorphic to ρ. By 4.1, f is a cusp form. Let f' be the unique primitive cusp form (on some $\Gamma_0(N')$, where N' is a suitable divisor of N) such that $f' \mid T_p = a_p f'$ for $p \nmid N$. By 4.6, the Dirichlet series associated with f' is $L(s,\rho) = \sum a_n n^{-s}$. It follows that $f' = f$, which proves that f is primitive.

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4.11. A generalization of Theorem 4.10 to all global fields may be found in [27], p. 163.

4.12. Condition (iii) (the Artin conjecture for the $\rho\otimes\chi$) may be replaced by the weaker condition:

(iii′) There exists an integer $M \geq 1$ such that, for every character χ whose conductor is relatively prime to M, the function $L(s,\rho\otimes\chi)$ is entire.

This follows from [26] (see also [12]).

4.13. If the Artin conjecture is true, the theorems above provide a bijection between "classes of irreducible degree-2 representations of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q})$ with odd determinant" and "primitive cusp forms of weight 1."

4.14. Theorem 4.6 even gives a way to verify the Artin conjecture in particular cases. Given a representation ρ satisfying (i) and (ii), with conductor N and determinant ε, one can numerically determine the coefficients a_n of the series $L(s,\rho) = \sum a_n n^{-s}$ for n less than a given integer A, and one can seek to construct a primitive cusp form f of type $(1,\varepsilon)$ on $\Gamma_0(N)$ whose expansion begins with $\sum_{n\leq A} a_n q^n$. If A is sufficiently large,

$$\text{for example} \qquad A \geq (N/12)\prod_{p\mid N}(1+p^{-1}),$$

such a form is unique if it exists (if it does not exist, the Artin conjecture is false). Once f has been obtained, a representation ρ_f corresponds to it; if one can prove that ρ_f is isomorphic to ρ, it follows that ρ satisfies (iii).

EXAMPLES. — If ρ is as above, the image of ρ in the group

$$\mathrm{PGL}_2(\mathbf{C}) = \mathrm{GL}_2(\mathbf{C})/\mathbf{C}^*$$

is either a dihedral group or one of the groups $\mathfrak{A}_4$, $\mathfrak{S}_4$, or $\mathfrak{A}_5$ ([22], prop. 16). In the dihedral case, ρ is induced by a degree-1 representation of $\mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}(\sqrt{d}))$, where $\mathbf{Q}(\sqrt{d})$ is a quadratic extension of $\mathbf{Q}$. Condition (iii) is then satisfied, and ρ

corresponds indeed to a cusp form; the latter is a linear combination of theta series for binary forms of discriminant d, cf. [9], pp. 428-460. Non-dihedral examples have recently been constructed by Tate (for $N = 133, 229, 283, 331, \ldots$).

For one of these examples (the one with $N = 133$, corresponding to a group $\mathfrak{A}_4$), Tate, assisted by Atkin et al., was able to carry out the method outlined in 4.14 and prove the existence of a corresponding modular form—and hence also the Artin conjecture for the representation in question.

§ 5. Exploitation of a Result of Rankin

PROPOSITION 5.1. — Let f be a nonzero cusp form of type (k, ε) on $\Gamma_0(N)$. Suppose that f is an eigenform for the T_p, $p \nmid N$, with eigenvalues a_p. Then the series $\sum_{p\nmid N} |a_p|^2 p^{-s}$ converges for real $s > k$, and

$$\sum_{p\nmid N} |a_p|^2 p^{-s} \leq \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k. \tag{5.1.1}$$

Proof 5.2. — One immediately reduces to the case in which f is a primitive form $\sum_{n=1}^{\infty} a_n q^n$. For every $p \nmid N$, let $\varphi_p \in \mathrm{GL}_2(\mathbf{C})$ satisfy $\mathrm{Tr}(\varphi_p) = a_p$ and $\det(\varphi_p) = \varepsilon(p)p^{k-1}$. The Dirichlet series

$$\Phi_f(s) = \sum_{n=1}^{\infty} a_n n^{-s}$$

may then be written

$$\Phi_f(s) = \prod_{p\mid N} (1 - a_p p^{-s})^{-1} \prod_{p\nmid N} \det(1 - \varphi_p p^{-s})^{-1}, \qquad cf.\ (1.7.2).$$

Set

$$L(s) = \prod_{p\nmid N} \det(1 - \varphi_p \otimes \overline{\varphi}_p\, p^{-s})^{-1}.$$

This is an Euler product with four factors: if λ_p and μ_p denote the eigenvalues of φ_p, then

$$L(s) = \prod_{p\nmid N} [(1 - \lambda_p\overline{\lambda}_p p^{-s})(1 - \lambda_p\overline{\mu}_p p^{-s})(1 - \mu_p\overline{\lambda}_p p^{-s})(1 - \mu_p\overline{\mu}_p p^{-s})]^{-1}.$$

Using the formula $\lambda_p\overline{\lambda}_p\mu_p\overline{\mu}_p = |\varepsilon(p)p^{k-1}|^2 = p^{2k-2}$, one proves (cf., for example, [10], p. 33, or [12], [15]) that

$$L(s) = H(s)\zeta(2s - 2k + 2)\left(\sum_{n=1}^{\infty} |a_n|^2 n^{-s}\right),$$

where

$$H(s) = \prod_{p\mid N} (1 - p^{-2s+2k-2})(1 - |a_p|^2 p^{-s}).$$

According to [18] (cf. also [12], [13], and [15]), the series $\sum |a_n|^2 n^{-s}$ converges for $\mathrm{Re}(s) > k$, and its product with $\zeta(2s - 2k + 2)$ extends to a meromorphic function throughout the complex plane, with its only pole at $s = k$. Since $|a_p| < p^{k/2}$ when $p \mid N$ (cf. 1.8), the function $H(s)$ is holomorphic and nonzero in $\mathrm{Re}(s) \geq k$. It follows that $L(s)$ is meromorphic throughout the complex plane and holomorphic for $\mathrm{Re}(s) \geq k$, with the sole exception of $s = k$, where it has a simple pole; moreover, $L(s) \neq 0$ for real $s > k$, because this is true of $H(s)$, $\zeta(2s - 2k + 2)$, and $\sum |a_n|^2 n^{-s}$.

Set

$$g_m(s) = \sum_{p \nmid N} |\operatorname{Tr}(\varphi_p^m)|^2 p^{-ms}/m \qquad \text{and} \qquad g(s) = \sum_{m=1}^{\infty} g_m(s).$$

The series $g(s)$ is a Dirichlet series with nonnegative coefficients. For sufficiently large s, a direct calculation shows that it equals $\log L(s)$. Since $L(s)$ is holomorphic and nonzero for real $s > k$, a classical lemma of Landau ([21], p. 112) implies that $g(s)$ converges for $\mathrm{Re}(s) > k$. Since $L(s)$ has a simple pole at $s = k$, one has

$$g(s) = \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k.$$

But $g_1(s) = \sum |a_p|^2 p^{-s}$ is plainly $\leq g(s)$. Hence

$$\sum |a_p|^2 p^{-s} \leq \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k.$$

REMARKS 5.3. — Proposition 5.1 can be strengthened in various ways. First, once the Petersson conjecture is available, an easy bound shows that the series

$$\sum_{p \nmid N} \sum_{m \geq 2} |\operatorname{Tr}(\varphi_p^m)|^2 p^{-ms}/m = g_2(s) + g_3(s) + \cdots$$

converges for $\mathrm{Re}(s) \geq k$, and this permits the inequality (5.1.1) to be replaced by the equality

$$\sum |a_p|^2 p^{-s} = \log(1/(s-k)) + O(1) \qquad \text{as } s \longrightarrow k. \tag{5.3.1}$$

On the other hand, a Hadamard-de la Vallée Poussin argument shows that $L(s) \neq 0$ for every s such that $\mathrm{Re}(s) \geq k$ (including the critical line $\mathrm{Re}(s) = k$), and applying the Wiener-Ikehara theorem to $L'(s)/L(s)$ gives

$$\sum_{p \leq x} |a_p|^2 p^{-(k-1)} \sim x \qquad \text{as } x \longrightarrow \infty, \tag{5.3.2}$$

cf. Rankin [19].

5.4. APPLICATION TO FORMS OF WEIGHT 1. — Let P be the set of prime numbers and let X be a subset of P. Set

$$\operatorname{dens.sup} X = \limsup_{\substack{s \to 1 \\ s > 1}} \frac{\sum_{p \in X} p^{-s}}{\log(1/(s-1))}. \tag{5.4.1}$$

This is the upper density of X; it lies between 0 and 1.

PROPOSITION 5.5. — Retain the hypotheses of 5.1, and suppose in addition that the weight k of f is 1. Then, for every $\eta > 0$, there exists a set X_η of prime numbers and a finite subset Y_η of $\mathbf{C}$ such that

$$\text{dens. sup}\, X_\eta \leq \eta \qquad \text{and} \qquad a_p \in Y_\eta \quad \text{for every } p \notin X_\eta.$$

By 2.7, the a_p are algebraic integers in a finite extension K of $\mathbf{Q}$. If c is a constant ≥ 0, let $Y(c)$ be the set of integers a of K such that $|\sigma(a)|^2 \leq c$ for every embedding σ of K into $\mathbf{C}$; this is a finite set. Let $X(c)$ be the set of p such that $a_p \notin Y(c)$; it is enough to prove that dens. sup $X(c) \leq \eta$ when c is sufficiently large.

We know from 2.7 that the $\sigma(a_p)$ are likewise eigenvalues of the T_p in weight 1. By (5.1.1), therefore,

$$\sum_\sigma \sum_p |\sigma(a_p)|^2 p^{-s} \leq r \log(1/(s-1)) + O(1) \qquad \text{as } s \longrightarrow 1,$$

where $r = [K : \mathbf{Q}]$. Since $\sum_\sigma |\sigma(a_p)|^2 \geq c$ if $p \in X(c)$, it follows that

$$c \sum_{p \in X(c)} p^{-s} \leq r \log(1/(s-1)) + O(1) \qquad \text{as } s \longrightarrow 1,$$

and hence

$$\text{dens. sup}\, X(c) \leq r/c.$$

It is therefore enough to take $c \geq r/\eta$.

REMARK 5.6. — By using (5.3.2) instead of (5.1.1) in the preceding proof, one could have replaced the "analytic" density (5.4.1) by the "natural" density (cf. [21], VI, no. 4.5). In any event, 5.5 is only of provisional interest: once Theorem 4.1 has been proved, we shall know that the set of the a_p is finite.

§ 6. ℓ-adic Representation and Reduction mod ℓ

(a) ℓ-ADIC REPRESENTATIONS. — We shall use the following result:

THEOREM 6.1. — Let f be a nonzero modular form of type (k, ε) on $\Gamma_0(N)$. Suppose that $k \geq 2$ and that f is an eigenform for the T_p, $p \nmid N$, with eigenvalues a_p. Let K be a finite extension of $\mathbf{Q}$ containing the a_p and the $\varepsilon(p)$, cf. (2.7.3). Let λ be a finite place of K of residual characteristic ℓ, and let K_λ be the completion of K at λ. There then exists a continuous semisimple linear representation

$$\rho_\lambda : G \longrightarrow \mathrm{GL}_2(K_\lambda), \qquad \text{where} \qquad G = \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}),$$

which is unramified outside $N\ell$ and satisfies

$$\mathrm{Tr}(F_{\rho_\lambda, p}) = a_p \qquad \text{and} \qquad \det(F_{\rho_\lambda, p}) = \varepsilon(p) p^{k-1} \quad \text{if } p \nmid N\ell. \tag{6.1.1}$$

By 3.2, condition (6.1.1) determines ρ_λ uniquely up to isomorphism.

REMARK 6.2. — If f is an Eisenstein series, the assertion above follows immediately from Hecke's results ([9], p. 690) by taking ρ_λ to be the direct sum of two degree-1 representations. When f is a cusp form, 6.1 is proved in a special case in [4]. The general case is not much more difficult. It is treated, by another method (inspired by Ihara) and in another language, by Langlands [11] (where, however, a "trace formula" that seems accessible but that no one has proved is admitted without proof). In a future work by one of us, 6.2 will be reproved by a method due to Piateckii-Shapiro [17].

COROLLARY 6.3. — Let $(f, N, k, \varepsilon, (a_p))$ and $(f', N', k', \varepsilon', (a'_p))$ be as in Theorem 6.1. If the set of prime numbers p such that $a_p = a'_p$ has density 1, then $k = k'$, $\varepsilon = \varepsilon'$, and $a_p = a'_p$ for every $p \nmid NN'$.

Indeed, the representations attached to f and f' (for the same choice of K and λ) are isomorphic by 3.2.

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6.4. The image of G under ρ_λ is a compact subgroup of $\mathrm{GL}_2(K_\lambda)$, hence an ℓ-adic Lie group; it is not a finite group.

6.5. Once Theorem 4.1 has been proved, one sees easily [2] that 6.1 and 6.3 remain valid in weight 1; in that case, however, the image of the group G is finite.

(b) REDUCTION mod ℓ

6.6. Let $K \subset \mathbf{C}$ be a number field, let λ be a finite place of K, let O_λ be the corresponding valuation ring, $\mathfrak{m}_\lambda$ its maximal ideal, $k_\lambda = O_\lambda/\mathfrak{m}_\lambda$ its residue field, and ℓ the characteristic of k_λ. In what follows, we shall write "mod λ" for "mod $\mathfrak{m}_\lambda$."

Let f be a modular form of type (k, ε) on $\Gamma_0(N)$. We say that f is λ-integral (respectively, that $f \equiv 0 \pmod{\lambda}$) if the coefficients of the series $f_\infty(q)$ belong to O_λ (respectively, to $\mathfrak{m}_\lambda$). Suppose that f is λ-integral; we say that f is an eigenvector of T_p mod λ, with eigenvalue $a_p \in k_\lambda$, if

$$f \mid T_p - a_p f \equiv 0 \pmod{\lambda}. \tag{6.6.1}$$

THEOREM 6.7. — With the preceding notation, let f be a modular form of type (k, ε) on $\Gamma_0(N)$, with $k \geq 1$ and coefficients in K. Suppose that f is λ-integral, that $f \not\equiv 0 \pmod{\lambda}$, and that f is an eigenvector of the T_p mod λ, for $p \nmid N\ell$, with eigenvalues $a_p \in k_\lambda$. Let k_f be the subfield of k_λ generated by the a_p and the reductions mod λ of the $\varepsilon(p)$. There then exists a semisimple representation

$$\rho : G \longrightarrow \mathrm{GL}_2(k_f)$$

[2] This follows from the fact that the representation ρ of Theorem 4.1 is realizable over K: its image contains an element with distinct rational eigenvalues (the element $\rho(c)$ of 4.5), and this implies that its Schur index is 1 (cf. [20], IX a); it is therefore realizable over the field of values of its character.

which is unramified outside $N\ell$ and such that, for every $p \nmid N\ell$,

$$\mathrm{Tr}(F_{\rho,p}) = a_p \qquad \text{and} \qquad \det(F_{\rho,p}) \equiv \varepsilon(p)p^{k-1} \pmod{\lambda}. \tag{6.7.1}$$

Proof of Theorem 6.7

6.8. Let $(K', \lambda', f', k', \varepsilon', (a'_p))$ be as in Theorem 6.7, where K' contains K and λ' extends λ. If $a_p \equiv a'_p \pmod{\lambda'}$ and $\varepsilon(p)p^{k-1} \equiv \varepsilon'(p)p^{k'-1} \pmod{\lambda'}$ for every $p \nmid N\ell$, the theorem for f is equivalent to the theorem for f'. The second condition holds whenever $\varepsilon = \varepsilon'$ and $k \equiv k' \pmod{\ell - 1}$, and it implies the first provided that $f \equiv f' \pmod{\lambda'}$.

6.9. REDUCTION TO THE CASE $k \geq 2$. — For even $n > 2$, let E_n be the Eisenstein series of weight n on $\mathrm{SL}_2(\mathbf{Z})$, normalized so that its constant term is 1. If n is chosen divisible by $\ell - 1$, the series expansion of E_n is ℓ-integral, and $E_n \equiv 1 \pmod{\ell}$, cf. [25]. The product $f \cdot E_n$ is therefore congruent to f mod λ; its weight $k + n$ is congruent to k mod $(\ell - 1)$. By 6.8, the theorem for f is equivalent to the theorem for $f \cdot E_n$, which has weight > 2.

6.10. REDUCTION TO THE CASE IN WHICH f IS AN EIGENVECTOR OF THE T_p. — It is enough to verify that there exists an f' as in 6.8, with $(k', \varepsilon') = (k, \varepsilon)$, which is an eigenvector of the T_p. This follows from the next lemma, applied to the T_p acting on the $\mathcal{O}_\lambda$-module M of modular forms of type (k, ε) on $\Gamma_0(N)$ with coefficients in $\mathcal{O}_\lambda$:

LEMMA 6.11. — Let M be a free module of finite type over a discrete valuation ring $\mathcal{O}$; let $\mathfrak{m}$ denote the maximal ideal of $\mathcal{O}$, k its residue field, and K its field of fractions. Let $\mathcal{T}$ be a set of pairwise commuting endomorphisms of M. Let $f \in M/\mathfrak{m}M$ be a common nonzero eigenvector of the $T \in \mathcal{T}$, and let $a_T \in k$ be the corresponding eigenvalues. There then exists a discrete valuation ring $\mathcal{O}'$ containing $\mathcal{O}$, with maximal ideal $\mathfrak{m}'$ such that $\mathcal{O} \cap \mathfrak{m}' = \mathfrak{m}$, whose field of fractions K' is finite over K, and a nonzero element f' of

$$M' = \mathcal{O}' \otimes_{\mathcal{O}} M,$$

which is an eigenvector of the $T \in \mathcal{T}$, with eigenvalues a'_T such that $a'_T \equiv a_T \pmod{\mathfrak{m}'}$.

(Note that the assertion is not that the eigenvectors lift, but only that the eigenvalues do.)

Let $\mathcal{H}$ be the subalgebra of $\mathrm{End}(M)$ generated by $\mathcal{T}$. After making a finite extension of the scalars if necessary, we may suppose that $K \otimes \mathcal{H}$ is a product of Artinian rings with residue field K. Let $\chi : \mathcal{H} \longrightarrow k$ be the homomorphism such that $h \cdot f = \chi(h)f$ for every $h \in \mathcal{H}$. Since $\mathcal{H}$ is free over $\mathcal{O}$, there exists a prime ideal $\mathfrak{p}$ of $\mathcal{H}$ contained in the maximal ideal $\ker(\chi)$ and such that $\mathfrak{p} \cap \mathcal{O} = 0$; it is the kernel of a homomorphism $\chi' : \mathcal{H} \longrightarrow \mathcal{O}$ whose reduction mod $\mathfrak{m}$ is χ. The ideal of $K \otimes \mathcal{H}$ generated by $\mathfrak{p}$ belongs to the support of the module $K \otimes M$; it follows that there exists a nonzero element f'' of $K \otimes M$ annihilated by this ideal, i.e. such that $hf'' = \chi'(h)f''$ for every $h \in \mathcal{H}$. We then take for f' a nonzero multiple of f'' belonging to M.

Variant. — Reduce to the case in which M is $\mathcal{T}$-indecomposable and the eigenvalues of the $T \in \mathcal{T}$ belong to K. Show that there then exists a basis $(e_1, \ldots, e_n)$ of M relative to which the elements T of $\mathcal{T}$ have upper triangular matrices (T_{ij}); use the indecomposability of M to prove that $T_{ii} \equiv a_T \pmod{\mathfrak{m}}$ for every T and every i. The element $f' = e_1$ then answers the question.

6.12. End of the proof of 6.7. — By 6.9 and 6.10, we may suppose that $k \geq 2$ and that f is an eigenvector of the T_p, $p \nmid N\ell$; since T_ℓ commutes with the T_p, we may also suppose that f is an eigenvector of T_ℓ if $\ell \nmid N$. Let

$$\rho_\lambda : G \longrightarrow \mathrm{GL}_2(K_\lambda)$$

be the representation associated with f by Theorem 6.1. After replacing ρ_λ by an isomorphic representation if necessary, we may suppose that $\rho_\lambda(G)$ is contained in $\mathrm{GL}_2(\widehat{O}_\lambda)$, where $\widehat{O}_\lambda$ is the ring of integers of K_λ (i.e. the completion of O_λ). Reducing ρ_λ mod λ gives a representation

$$\tilde{\rho}_\lambda : G \longrightarrow \mathrm{GL}_2(k_\lambda).$$

Let φ be the semisimplification of $\tilde{\rho}_\lambda$; it is a semisimple representation, unramified outside $N\ell$, and satisfies (6.7.1). The group $\varphi(G)$ is finite; by the Čebotarev density theorem, every element of $\varphi(G)$ is of the form $F_{\varphi,p}$, with $p \nmid N\ell$. From the definition of k_f, therefore:

(6.12.1) For every $s \in \varphi(G)$, the coefficients of the polynomial $\det(1 - sT)$ belong to k_f.

The existence of the desired representation $\rho : G \longrightarrow \mathrm{GL}_2(k_f)$ now follows from the next lemma:

LEMMA 6.13. — Let $\varphi : \Phi \longrightarrow \mathrm{GL}_n(k')$ be a semisimple representation of a group Φ over a finite field k'. Let k be a subfield of k' containing the coefficients of the polynomials $\det(1 - \varphi(s)T)$, $s \in \Phi$. Then φ is realizable over k, i.e. it is isomorphic to a representation $\rho : \Phi \longrightarrow \mathrm{GL}_n(k)$.

For φ to be realizable over k, it is enough to verify that φ is isomorphic to $\sigma(\varphi)$ for every k-automorphism σ of k': this follows from the fact that the Brauer group of a finite field is trivial, so there is no "Schur index" to consider. But φ and $\sigma(\varphi)$ have the same characteristic polynomials and are semisimple; they are therefore isomorphic by [3], th. 30.16.

§ 7. Bounding the Orders of Certain Subgroups of $\mathrm{GL}_2(\mathbf{F}_\ell)$

If ℓ is a prime number, let $\mathbf{F}_\ell$ denote the field $\mathbf{Z}/\ell\mathbf{Z}$ with ℓ elements.

7.1. Let η and M be two positive numbers. We shall consider the following property of a subgroup G of $\mathrm{GL}_2(\mathbf{F}_\ell)$:

$C(\eta, M)$. — There exists a subset H of G such that $|H| \geq (1 - \eta)|G|$, and the set of polynomials $\det(1 - hT)$, $h \in H$, has at most M elements.

(If X is a finite set, $|X|$ denotes its cardinality.)

We shall say that G is semisimple if the identity representation

$$G \longrightarrow \mathrm{GL}_2(\mathbf{F}_\ell)$$

is semisimple.

PROPOSITION 7.2. — Let $\eta < 1/2$ and $M \geq 0$. There exists a constant $A = A(\eta, M)$ such that, for every prime number ℓ and every semisimple subgroup G of $\mathrm{GL}_2(\mathbf{F}_\ell)$ satisfying $C(\eta, M)$, one has $|G| \leq A$.

Proof. — Let G be a semisimple subgroup of $\mathrm{GL}_2(\mathbf{F}_\ell)$. Recall (cf. [22], § 2, props. 15 and 16) that one of the following conditions holds:

(a) G contains $\mathrm{SL}_2(\mathbf{F}_\ell)$;

(b) G is contained in a Cartan subgroup T;

(c) G is contained in the normalizer of a Cartan subgroup T, but is not contained in T;

(d) the image of G in $\mathrm{PGL}_2(\mathbf{F}_\ell) = \mathrm{GL}_2(\mathbf{F}_\ell)/\mathbf{F}_\ell^*$ is isomorphic to $\mathfrak{A}_4$, $\mathfrak{S}_4$, or $\mathfrak{A}_5$.

We shall bound the order of G in each case.

Case (a). — Put $r = (G : \mathrm{SL}_2(\mathbf{F}_\ell))$. Then $|G| = r\ell(\ell^2 - 1)$. On the other hand, the number of elements of $\mathrm{GL}_2(\mathbf{F}_\ell)$ with a given characteristic polynomial is $\ell^2 + \ell$, ℓ^2, or $\ell^2 - \ell$ according as the polynomial in question has 2, 1, or 0 roots in $\mathbf{F}_\ell$. If G satisfies $C(\eta, M)$, then

$$(1-\eta)r\ell(\ell^2 - 1) = (1-\eta)|G| \leq |H| \leq M(\ell^2 + \ell),$$

and hence

$$(1-\eta)r(\ell - 1) \leq M \qquad \text{and} \qquad \ell \leq 1 + \frac{M}{(1-\eta)r} \leq 1 + \frac{M}{1-\eta};$$

this gives a bound for ℓ, and therefore a fortiori a bound for $|G|$.

Case (b). — At most 2 elements of T have a given characteristic polynomial. The hypothesis $C(\eta, M)$ (with $\eta < 1$) therefore implies

$$(1-\eta)|G| \leq 2M,$$

and hence the bound

$$|G| \leq \frac{2M}{1-\eta}.$$

Case (c). — The group $G' = G \cap T$ has index 2 in G. If G satisfies $C(\eta, M)$, then G' satisfies $C(2\eta, M)$. Applying (b) to G', one obtains

$$|G| \leq \frac{4M}{1-2\eta}.$$

Case (d). — The image of G in $\mathrm{PGL}_2(\mathbf{F}_\ell)$ has order at most 60. The group

$$G \cap \mathrm{SL}_2(\mathbf{F}_\ell)$$

therefore has order at most 120, and G has at most 120 elements with a given determinant, and a fortiori with a given characteristic polynomial. If G satisfies $C(\eta, M)$, then

$$(1-\eta)|G| \leq 120M, \qquad \text{whence} \qquad |G| \leq \frac{120M}{1-\eta}.$$

§ 8. Proof of Theorem 4.1

We may suppose that the modular form f under consideration is either an Eisenstein series or a cusp form.

8.1. If f is an Eisenstein series, there exist characters χ_1 and χ_2 of $(\mathbf{Z}/N\mathbf{Z})^*$ such that $\chi_1\chi_2 = \varepsilon$ and $a_p = \chi_1(p) + \chi_2(p)$ for $p \nmid N$ (cf. [9], p. 690). We then take for ρ the reducible representation

$$\rho = \chi_1 \oplus \chi_2,$$

where the χ_i are identified with one-dimensional representations of G, cf. 4.4.

8.2. From now on, suppose that f is cuspidal. By 2.7, the a_p and the $\varepsilon(p)$ belong to the ring of integers O_K of a number field K, which we may suppose to be Galois over $\mathbf{Q}$. Let L be the set of prime numbers ℓ that split completely in K. For every $\ell \in L$, choose a place λ_ℓ of K above ℓ; the corresponding residue field is $\mathbf{F}_\ell$. By Theorem 6.7, there exists a continuous semisimple representation

$$\rho_\ell : \mathrm{Gal}(\overline{\mathbf{Q}}/\mathbf{Q}) \longrightarrow \mathrm{GL}_2(\mathbf{F}_\ell),$$

which is unramified outside $N\ell$ and satisfies

$$\det(1 - F_{\rho_\ell,p}T) \equiv 1 - a_pT + \varepsilon(p)T^2 \pmod{\lambda_\ell} \qquad \text{if } p \nmid N\ell.$$

Let G_ℓ be the subgroup of $\mathrm{GL}_2(\mathbf{F}_\ell)$ that is the image of ρ_ℓ.

LEMMA 8.3. — For every $\eta > 0$, there exists a constant M such that G_ℓ satisfies condition $C(\eta, M)$ of 7.1 for every $\ell \in L$.

By Proposition 5.5, there exists a subset X_η of the set P of prime numbers such that dens. sup $X_\eta \leq \eta$ and the a_p, for $p \notin X_\eta$, form a finite set. Let $\mathcal{M}$ denote the finite set of polynomials $1 - a_pT + \varepsilon(p)T^2$, for $p \notin X_\eta$, and put $M = |\mathcal{M}|$.

The group G_ℓ satisfies $C(\eta, M)$ for every $\ell \in L$. Indeed, let H_ℓ be the subset of G_ℓ formed by the Frobenius elements $F_{\rho_\ell,p}$, $p \notin X_\eta$, and their conjugates. By the Čebotarev density theorem, $|H_\ell| \geq (1-\eta)|G_\ell|$. On the other hand, if $h \in H_\ell$, the polynomial $\det(1 - hT)$ is the reduction modulo λ_ℓ of an element of $\mathcal{M}$, and therefore belongs to a set with at most M elements. Thus condition $C(\eta, M)$ is indeed satisfied.

LEMMA 8.4. — There exists a constant A such that $|G_\ell| \leq A$ for every $\ell \in L$.

This follows from the preceding lemma and Proposition 7.2.

8.5. Choose a constant A satisfying 8.4. After enlarging the field K if necessary (which decreases L), we may suppose that it contains all nth roots of unity for $n \leq A$. Let Y be the set of polynomials $(1 - \alpha T)(1 - \beta T)$, where α and β are roots of unity of order $\leq A$. If $p \nmid N$, then for every $\ell \in L$ with $\ell \neq p$ there exists $R(T) \in Y$ such that

$$1 - a_pT + \varepsilon(p)T^2 \equiv R(T) \pmod{\lambda_\ell}.$$

Since Y is finite, there is an R for which the preceding congruence holds for infinitely many ℓ, and hence

$$1 - a_p T + \varepsilon(p) T^2 = R(T),$$

that is, the polynomials $1 - a_p T + \varepsilon(p) T^2$ belong to Y.

8.6. Let L' be the set of $\ell \in L$ such that $\ell > A$ and such that, for $R, S \in Y$, $R \neq S$ implies $R \not\equiv S \pmod{\lambda_\ell}$; the set $L - L'$ is finite, and therefore L' is infinite. Let $\ell \in L'$. The order of the group G_ℓ is prime to ℓ. It follows, by a standard argument, that the identity representation $G_\ell \longrightarrow \mathrm{GL}_2(\mathbf{F}_\ell)$ is the reduction modulo λ_ℓ of a representation $G_\ell \longrightarrow \mathrm{GL}_2(O_{\lambda_\ell})$, where O_{λ_ℓ} is the valuation ring for λ_ℓ. Composing the latter with the canonical map $G \longrightarrow G_\ell$, we obtain a representation

$$\rho : G \longrightarrow \mathrm{GL}_2(O_{\lambda_\ell}).$$

By construction, ρ is unramified outside $N\ell$. If $p \nmid N\ell$, the eigenvalues of the Frobenius element $F_{\rho,p}$ are roots of unity of order $\leq A$ (since the image of ρ is isomorphic to G_ℓ, and hence has order $\leq A$); consequently $\det(1 - F_{\rho,p}T) \in Y$. On the other hand, since the reduction of ρ modulo λ_ℓ is ρ_ℓ, one has

$$\det(1 - F_{\rho,p}T) \equiv 1 - a_p T + \varepsilon(p) T^2 \pmod{\lambda_\ell}.$$

But the two polynomials $\det(1 - F_{\rho,p}T)$ and $1 - a_p T + \varepsilon(p) T^2$ belong to Y. Since they are congruent modulo λ_ℓ, they are equal, and

$$\det(1 - F_{\rho,p}T) = 1 - a_p T + \varepsilon(p) T^2 \qquad \text{for every } p \nmid N\ell.$$

Now replace ℓ by another prime number ℓ' in L'. We obtain a representation $\rho' : G \longrightarrow \mathrm{GL}_2(O_{\lambda_{\ell'}})$ having the same property as above, but for $p \nmid N\ell'$. In particular,

$$\det(1 - F_{\rho,p}T) = \det(1 - F_{\rho',p}T) \qquad \text{for } p \nmid N\ell\ell'.$$

By Lemma 3.2, this implies that ρ and ρ' are isomorphic as representations in $\mathrm{GL}_2(K)$, and a fortiori as complex representations. It follows that ρ is unramified outside N and that

$$\det(1 - F_{\rho,p}T) = 1 - a_p T + \varepsilon(p) T^2 \qquad \text{for every } p \nmid N.$$

8.7. It remains to show that ρ is irreducible. If it were not, it would be a sum of two one-dimensional representations; these would correspond to characters χ_1 and χ_2, unramified outside N, such that $\chi_1\chi_2 = \varepsilon$ and

$$a_p = \chi_1(p) + \chi_2(p) \qquad \text{for } p \nmid N.$$

Then

$$\sum |a_p|^2 p^{-s} = 2 \sum p^{-s} + \sum \chi_1(p)\overline{\chi}_2(p) p^{-s} + \sum \chi_2(p)\overline{\chi}_1(p) p^{-s}.$$

As s tends to 1, $\sum p^{-s} = \log(1/(s-1)) + O(1)$. On the other hand, the character $\chi_1\overline{\chi}_2$ is not 1 (otherwise one would have $\varepsilon = (\chi_1)^2$ and $\varepsilon(-1) = 1$); it follows (cf., for example, [21], VI.4.2) that

$$\sum \chi_1(p)\overline{\chi}_2(p) p^{-s} = O(1) \qquad \text{and} \qquad \sum \chi_2(p)\overline{\chi}_1(p) p^{-s} = O(1).$$

It follows that

$$\sum |a_p|^2 p^{-s} = 2\log(1/(s-1)) + O(1) \qquad \text{as } s \longrightarrow 1,$$

which contradicts Proposition 5.1 and completes the proof.

§ 9. Application to the Coefficients of Modular Forms of Weight 1

Let

$$f = \sum_{n=0}^{\infty} a_n e^{2\pi i n z/M}, \qquad M \geq 1,$$

be a modular form of weight 1 on a congruence subgroup of $\mathrm{SL}_2(\mathbf{Z})$.

(a) BOUND FOR $|a_n|$

THEOREM 9.1. — One has $|a_n| = O(d(n))$ as $n \longrightarrow \infty$.

(Recall that $d(n)$ denotes the number of divisors ≥ 1 of n.)

COROLLARY 9.2. — One has $|a_n| = O(n^{\delta})$ for every $\delta > 0$.

Indeed, $d(n)$ is known to have this property ([8], th. 315).

Proof of 9.1. — If n_0 is an integer ≥ 1, then $d(n_0 n)/d(n)$ lies between 1 and $d(n_0)$. It is therefore equivalent to prove estimate (9.1) for $f(z)$ or for $f(n_0 z)$, and this allows us to suppose that $M = 1$, i.e. that $f(z+1) = f(z)$. Using 1.5 and 1.9, we are reduced to the following two special cases:

(i) f is an Eisenstein series, in which case (9.1) follows from the formula for the a_n ([9], p. 475);

(ii) f is a primitive cusp form of type $(1, \varepsilon)$ on $\Gamma_0(N)$, for suitable N and ε. In this case one has the more precise result

$$|a_n| \leq d_N(n) \leq d(n), \tag{9.3}$$

where $d_N(n)$ is the number of positive divisors of n that are prime to N. Indeed, by the multiplicativity of a_n and $d_N(n)$, it is enough to verify (9.3) when n is a power p^m of a prime number p. We distinguish two cases:

(ii_1) $p \mid N$.

One has $a_n = (a_p)^m$, and Theorem 4.6 shows that a_p is either 0 or a root of unity. Hence

$$|a_n| \leq 1 = d_N(n).$$

(ii_2) $p \nmid N$.

If the polynomial $1 - a_p T + \varepsilon(p) T^2$ is written in the form $(1 - \lambda T)(1 - \mu T)$, then

$$a_n = \lambda^m + \lambda^{m-1}\mu + \cdots + \lambda\mu^{m-1} + \mu^m.$$

By Theorem 4.1, λ and μ are roots of unity. Thus

$$|a_n| \leq m + 1 = d_N(n).$$

REMARK 9.4. — If $f = \sum b_n e^{2\pi i n z/M}$, $M \geq 1$, is a cusp form of weight $k \geq 2$ on a congruence subgroup of $\mathrm{SL}_2(\mathbf{Z})$, the same argument as above (using [5], 8.2) shows that

$$|b_n| = O(n^{(k-1)/2} d(n)) \qquad \text{as } n \longrightarrow \infty.$$

(b) MAXIMAL ORDER OF MAGNITUDE OF $|a_n|$. — It is known ([8], th. 317) that the "maximal" order of magnitude of $d(n)$ is $2^{\log n/\log\log n}$, in the sense that

$$\limsup \frac{\log d(n) \log\log n}{\log n} = \log 2.$$

The same result holds for the $|a_n|$:

PROPOSITION 9.5. — If $f \neq 0$, then

$$\limsup \frac{\log |a_n| \log\log n}{\log n} = \log 2.$$

LEMMA 9.6. — Let N be an integer ≥ 1. There exist sets X_N and Y_N of prime numbers, of densities > 0, such that:

(x) For every $p \in X_N$, one has $p \equiv 1 \pmod N$ and $g \mid T_p = 2g$ for every modular form g of weight 1 on $\Gamma_1(N)$;

(y) For every $p \in Y_N$, one has $p \equiv -1 \pmod N$ and $g \mid T_p = 0$ for every modular form g of weight 1 on $\Gamma_1(N)$.

Let $\rho_1, \dots, \rho_h$ be the representations of G associated with the different systems of eigenvalues of the T_p acting on forms of type $(1, \varepsilon)$ on $\Gamma_0(N)$, as ε ranges over the odd characters of $(\mathbf{Z}/N\mathbf{Z})^*$. Let X_N be the set of $p \equiv 1 \pmod N$ such that

$$F_{\rho_i, p} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

for $i = 1, \dots, h$, and let Y_N be the set of $p \equiv -1 \pmod N$ such that $F_{\rho_i,p}$ is conjugate to $\rho_i(c)$, cf. 4.5. By the Čebotarev density theorem, X_N and Y_N have densities > 0. If $p \in X_N$, then 2 is the only eigenvalue of T_p; since T_p is semisimple, one indeed has $g \mid T_p = 2g$ for every g. The same argument shows that $g \mid T_p = 0$ if $p \in Y_N$, since the trace of the matrix $\rho_i(c)$ is 0.

PROOF OF 9.5. — As in (a), we reduce to the case in which f is a modular form of weight 1 on $\Gamma_1(N)$. Let X_N be as in Lemma 9.6, and choose an integer m such that $a_m \neq 0$. If x is an integer ≥ 1, let $p_1, \dots, p_{i(x)}$ be the distinct prime numbers $p \in X_N$ that are $\leq x$ and do not divide m. Put $n(x) = m p_1 p_2 \cdots p_{i(x)}$. Since the p_i belong to X_N, one has $f \mid T_{p_i} = 2f$ and $f \mid R_{p_i} = f$; by (2.5.1), this gives

$$a_{n(x)} = 2^{i(x)} a_m,$$

and hence

$$\log |a_{n(x)}| \sim i(x) \log 2 \qquad \text{as } x \longrightarrow \infty.$$

If c is the density of X_N, then $i(x) \sim cx/\log x$ and $\sum_{i\leq i(x)} \log p_i \sim cx$. It follows that

$$\log |a_{n(x)}| \sim cx \log 2/\log x,$$

$$\log n(x) \sim cx,$$

$$\log\log n(x) \sim \log x,$$

and hence

$$\limsup \frac{\log |a_n| \log\log n}{\log n} \geq \log 2.$$

The opposite inequality follows from $|a_n| = O(d(n))$.

(c) NORMAL ORDER OF MAGNITUDE OF $|a_n|$. — The "normal" (i.e. most frequent) order of magnitude of $d(n)$ is $2^{\log\log n}$ (cf. [8], th. 432). That of $|a_n|$ is smaller:

PROPOSITION 9.7. — The set of n such that $a_n = 0$ has density 1.

(A subset S of $\mathbf{N}$ is said to have density c if the number of elements of S that are $\leq x$ is $cx + o(x)$ as $x \longrightarrow \infty$.)

Here again, we may suppose that f is a modular form of weight 1 on $\Gamma_1(N)$. Let Y_N be as in Lemma 9.6. If $p \in Y_N$, then $f \mid T_p = 0$ and $f \mid R_p = -f$. By (2.5.1), it follows that if n is divisible by p but not by p^2, then $a_n = 0$. Now, if Y is a finite set of prime numbers, the set S_Y of integers n having the preceding property (for at least one $p \in Y$) has density

$$1 - \prod_{p\in Y}\left(1 - \frac{p-1}{p^2}\right).$$

Since Y_N has density > 0, the series $\sum_{p\in Y_N} 1/p$ diverges, and the product

$$\prod_{p\in Y_N}\left(1 - \frac{p-1}{p^2}\right)$$

has value 0. It follows that the union of the S_Y, $Y \subset Y_N$, has density 1, which proves the proposition.

REMARK 9.8. — For each x, let $M(x)$ denote the number of $n \leq x$ such that $a_n \neq 0$. Proposition 9.7 says that

$$M(x) = o(x) \qquad \text{as } x \longrightarrow \infty.$$

Using Theorem 2 of [23], one can prove the following more precise result: there exists $\alpha > 0$ such that

$$M(x) = O(x/\log^{\alpha} x) \qquad \text{as } x \longrightarrow \infty.$$

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GROUP THEORY

The Support of the Character of a Supercuspidal Representation

Pierre Deligne

Note (*) by Mr. Pierre Deligne, communicated by Mr. Jean-Pierre Serre.

The character of a supercuspidal representation of a semisimple group over a local field is supported on the union of the compact open subgroups.

1 Review of monogenic groups

Let V be a finite-dimensional vector space over a field k, and let $g \in \mathrm{GL}(V)$. If k is perfect, the Zariski closure $\langle g \rangle$ of the group generated by g is the product of a multiplicative group $\langle g \rangle_m$ and a unipotent group $\langle g \rangle_{\mathrm{un}}$. If $g = su = us$ is the Jordan decomposition of g into its semisimple and unipotent parts, then $\langle g \rangle_m = \langle s \rangle$ and $\langle g \rangle_{\mathrm{un}} = \langle u \rangle$. Without any hypothesis on k, define the subgroup $\langle g \rangle_m$ of $\langle g \rangle$ to be the Zariski closure of the union of the subgroups of $\langle g \rangle$ that are kernels of multiplication by n, for n relatively prime to the characteristic exponent of k. This definition is compatible with extension of scalars and is equivalent to the preceding one when k is perfect. Over an algebraic closure $\bar{k}$ of k, the character group of $\langle g \rangle_m$ is the subgroup X of $\bar{k}^*$ generated by the eigenvalues of g, and

$$s \in \langle g \rangle_m(\bar{k}) = \mathrm{Hom}(X, \bar{k}^*)$$

corresponds to the identity inclusion of X.

Suppose that k is complete with respect to a valuation and has uniformizer π. Let v be the valuation of $\bar{k}$, with values in $\mathbf{Q}$, normalized by $v(\pi) = 1$, and let $n > 0$ be an integer such that $nv(X) \subset \mathbf{Z}$. The homomorphism $nv(x) : X \to \mathbf{Z}$ transposes to $m_g : \mathbf{G}_m \to \langle g \rangle_m$.

For a linear algebraic group G over k and $g \in G(k)$, the preceding constructions may be repeated after choosing an embedding $G \subset \mathrm{GL}(V)$. The homomorphism $m_g : \mathbf{G}_m \to \langle g \rangle_m \subset G$ does not depend on the chosen embedding (for fixed n). For every linear representation W of G, it defines an action of $\mathbf{G}_m$ on W, that is, a grading of W (SGA 3 I 4.7.3). The degree-d part is the sum of the generalized eigenspaces of g corresponding to eigenvalues $\alpha \in \bar{k}$ such that $nv(\alpha) = d$.

Suppose that G is reductive. Every homomorphism m from $\mathbf{G}_m$ to G then defines two opposite parabolic subgroups of G, the parabolics contracted and dilated by m. We write P_g^+ for the parabolic contracted by m_g (also called contracted by g). It has the following equivalent characterizations:

(a) An element $x \in G(\bar{k})$ lies in P_g^+ (respectively, in its unipotent radical U_g^+) if and only if the morphism
$$\lambda \longmapsto m_g(\lambda) x m_g(\lambda)^{-1} : \mathbf{G}_m \longrightarrow G$$
extends to $\mathbf{G}_a$ [respectively, and $\lim_{\lambda \to 0} m_g(\lambda) x m_g(\lambda)^{-1} = e$] — equivalently, if and only if $\lim_{n \to \infty} g^n x g^{-n}$ exists (respectively, equals e).

(b) $\mathrm{Lie}\, P_g^+$ (respectively, $\mathrm{Lie}\, U_g^+$) is the sum, in $\mathrm{Lie}\, G$, of the generalized eigenspaces of $\mathrm{ad}\, g$ corresponding to eigenvalues of valuation ≥ 0 (respectively, > 0).

Set $P_g^- = P_{g^{-1}}^+$ [the parabolic dilated by g and contracted by $m_{g^{-1}} = (m_g)^{-1}$], and denote its unipotent radical by U_g^-. The parabolics P_g^+ and P_g^- are opposite, and we set $M_g = P_g^+ \cap P_g^-$.

2 Statement and proof of the theorem

Assume that k is locally compact and that G is reductive. Let τ denote the projection of G onto its adjoint group G^{ad}. An admissible $(\mathscr{H}, \rho)$ representation of $G(k)$ is *supercuspidal* if its matrix coefficients are supported in inverse images of compact subsets of $G^{\mathrm{ad}}(k)$.

If K is a compact subgroup of $G(k)$, let dk denote the normalized Haar measure on K, and let $[K]$ denote its image in the algebra of compactly supported distributions on $G(k)$. The endomorphism

$$\rho(K) := \rho([K])$$

is a projector from $\mathscr{H}$ onto the subspace $\mathscr{H}^K$ of K-invariants. Equivalently, supercuspidality of ρ means that, for every compact open subgroup K,

$$\rho(K)\rho(g)\rho(K) = 0$$

whenever $\tau(g) \in G^{\mathrm{ad}}$ lies outside a suitable compact subset (depending on K). Let G^{ad}_c denote the union of the compact subgroups (or, equivalently, the compact open subgroups) of $G^{\mathrm{ad}}(k)$.

Theorem — *If $(\mathscr{H}, \rho)$ is an admissible supercuspidal representation of $G(k)$, its character*(1) *is supported in $\tau^{-1}(G^{\mathrm{ad}}_c)$.*

Choose a system of local coordinates near e, that is, a lattice $L \subset \mathrm{Lie}(G)$ and an analytic isomorphism α from L onto a neighborhood of e, such that

(1) $\alpha(0) = e$, and $d\alpha$ at 0 is the identity of $\mathrm{Lie}(G)$.

Let $g \in G(k)$. Choose L so that

(2) $L = (L\cap \mathrm{Lie}\, U^+_g) \oplus (L\cap \mathrm{Lie}\, M_g) \oplus (L\cap \mathrm{Lie}\, U^-_g)$. Denote this decomposition by $L^+ \oplus L^0 \oplus L^-$.
(3) $\mathrm{ad}\, g(L^+) \subset L^+$, $\mathrm{ad}\, g(L^0) = L^0$, and $\mathrm{ad}\, g^{-1}(L^-) \subset L^-$.

Since P^+_g and P^-_g are stable under $\mathrm{ad}\, g$, condition (3) may be rewritten as

(3') $\mathrm{ad}\, g(L \cap \mathrm{Lie}\, P^+_g) \subset L$ and $\mathrm{ad}\, g^{-1}(L \cap \mathrm{Lie}\, P^-_g) \subset L$.

Let a subscript i denote intersection with $G_i = \alpha(\pi^i L)$. Under hypotheses (2) and (3), there exists i_0 such that

(4) for $i \geq i_0$, G_i is a subgroup of $G(k)$, $G_i = U^+_{gi} \cdot M_{gi} \cdot U^-_{gi}$, and
(5) $\mathrm{Int}\, g(U^+_{gi}) \subset U^+_{gi}$, $\mathrm{Int}\, g(M_{gi}) = M_{gi}$, and $\mathrm{Int}\, g^{-1}(U^-_{gi}) \subset U^-_{gi}$.

This follows from the decomposition

$$U^+_g \times M_g \times U^-_g \hookrightarrow G : (u^+, m, u^-) \longmapsto u^+ m u^-$$

of a neighborhood of e.

Moreover, for i_0 sufficiently large, there is a neighborhood V of g such that (2)–(5) remain valid when g is replaced by $g' \in V$. For (2) and (3'), one uses the continuous dependence of P^+_g and $\mathrm{ad}\, g$ on g; for (4) and (5), one uses the analytic dependence of m_g, and hence of $P^\pm_g$, on g. Thus, in the decomposition $x = u^+ m u^-$, the components u^+, m, and u^- are analytic functions of the pair (x, g).

Lemma — *If conditions (1)–(5) hold, $i \geq i_0$, and $\tau(g) \notin G^{\mathrm{ad}}_c$, then $\rho(G_i)\rho(g)\rho(G_i)$ is nilpotent.*

For the decomposition $U^+_g \times M_g \times U^-_g \to G$ of the big cell of G, Haar measures satisfy

$$dg = \lambda(m)\, du^+\, dm\, du^-,$$

where λ is a character of M_g. On every compact subgroup of M_g, $\lambda = 1$; hence

$$[G_i] = [U^+_{gi}] \star [M_{gi}] \star [U^-_{gi}].$$

We prove by induction on N that

$$([G_i]g[G_i])^N = [G_i]g^N[G_i]$$

in the algebra of compactly supported distributions on G (we write g for δ_g). Indeed,

$$\begin{aligned}([G_i]g^N[G_i])([G_i]g[G_i]) &= [G_i]g^N[G_i]g[G_i]\\ &= [G_i]g^N[U^+_{gi}][M_{gi}][U^-_{gi}]g[G_i]\\ &= [G_i][g^N U^+_{gi} g^{-N}][g^N M_{gi} g^{-N}]g^N g[g^{-1}U^-_{gi}g][G_i]\\ &= [G_i]g^{N+1}[G_i].\end{aligned}$$

If $\tau(g) \notin G^{\mathrm{ad}}_c$, the set of $\tau(g^N)$, $N \geq 1$, is not relatively compact. Since ρ is supercuspidal, there is therefore an N such that

$$(\rho(G_i)\rho(g)\rho(G_i))^N = \rho(G_i)\rho(g^N)\rho(G_i) = 0.$$

Proof of the theorem — For $g \in G(k)$ such that $\tau(g) \notin G^{\mathrm{ad}}_c$, choose L, α, i_0, and V as above, with $gG_{i_0} \subset V$. Let Θ denote the character of ρ. For every $g' \in gG_i$ and every $i \geq i_0$, we then have

$$\Theta(g'[G_i]) = \Theta([G_i]g'[G_i]) = \mathrm{Tr}(\rho(G_i)\rho(g')\rho(G_i)) = 0$$

by the lemma. Since the subgroups G_i form a fundamental system of neighborhoods of 0, Θ vanishes on gG_i, and its support does not contain g.

Remark 1 — The same theorem, with the same proof, holds for an analytic group that is a finite étale covering of an open subgroup of G. The hypothesis becomes "the image in G^{ad} of the support of the matrix coefficients is relatively compact"; the conclusion becomes "the image in G^{ad} of the support of Θ lies in G^{ad}_c".

Remark 2 — Let p be the residual characteristic of k. The proof makes essential use of the Haar measures $[K]$ only for K small, and in particular for a p-group. The theorem and its proof therefore hold for representations with coefficients in any field of characteristic different from p.

Source notes

(*) Session of 24 May 1976.

(1) Once a Haar measure dg on G has been chosen, the character Θ of ρ is the distribution on G whose value on a test function φ is

$$\mathrm{Tr}\left(\int \varphi(g)\rho(g)\,dg\right).$$

More intrinsically, it is the generalized function [the linear form on the space of measures $\varphi(g)\,dg$, where $\varphi(g)$ is locally constant with compact support and dg is a Haar measure] given by this same formula.

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Local Constants in the Functional Equation of the Artin L-Function of an Orthogonal Representation

Pierre Deligne (Bures-sur-Yvette)

To Jean-Pierre Serre, in token of admiration.

Contents

1. Statement of the theorem

(1.1) Let K be a global field (an A-field in Weil's sense), let K' be a finite Galois extension of K, and let V be a complex representation of the Galois group $\mathrm{Gal}(K'/K)$. We follow the conventions of [1] in defining the corresponding Artin L-function: the factors at infinity are included, and the local nonarchimedean factors are defined in terms of geometric Frobenius elements (the inverses of Frobenius substitutions). We further normalize the reciprocity isomorphism of local class field theory so that it sends geometric Frobenius elements to uniformizers (cf. [1] 3.6). These sign questions are in fact immaterial: our essential calculations will be made modulo 2. The function $L(V,s)$ satisfies a functional equation

$$L(V,s) = \varepsilon(V,s)L(V^*,1-s), \tag{1.1.1}$$

where $\varepsilon(V,s)$ is the product of a constant and an exponential factor, and V^* is the dual of V.

Let f be the norm of the Artin conductor of V, and let D be the absolute value of the discriminant of K (respectively q^{2g-2} for a function field); define

$$W(V) = \varepsilon\left(V, \tfrac{1}{2}\right). \tag{1.1.2}$$

Then

$$\varepsilon(V,s) = W(V)\left(f \cdot D^{\dim(V)}\right)^{\frac{1}{2}-s}, \tag{1.1.3}$$

$$W(V) \cdot W(V^*) = 1. \tag{1.1.4}$$

If V is isomorphic to V^*, i.e. if the character of V is real-valued, (1.1.4) shows that $W(V) = \pm 1$. Fröhlich and Queyrut [2] showed that if V is in fact orthogonal, i.e. realizable over $\mathbf{R}$, then $W(V) = +1$.

(1.2) Let K be a local field (= a nondiscrete locally compact field), let K' be a finite Galois extension of K contained in a separable closure $\bar{K}$ of K, let V be a complex representation of the Galois group $G = \mathrm{Gal}(K'/K)$, let $\psi : K \to \mathbf{C}^*$ be a nontrivial additive character, and let dx be a Haar measure on K. In fact, K' plays no role: all that is used is the representation of $\mathrm{Gal}(\bar{K}/K)$ obtained from V by inflation; it is again denoted by V. I refer to [1] §§ 4, 5 for the definition of $\varepsilon(V, \psi, dx, s)$. In [1], more generally, V is taken to be a representation of the Weil group $W(\bar{K}/K)$ (of which $\mathrm{Gal}(\bar{K}/K)$ is a completion in the nonarchimedean case and a quotient in the archimedean case); one defines $\varepsilon(V, \psi, dx)$, and sets $\varepsilon(V, \psi, dx, s) = \varepsilon(V \otimes \omega_s, \psi, dx)$, where ω_s is the quasicharacter $\|x\|^s$ of $K^* \simeq W(\bar{K}/K)^{\mathrm{ab}}$. As a function of s, $\varepsilon(V, \psi, dx, s)$ is the product of a constant and an exponential factor.

Starting from a global situation as in 1.1, let K_v run through the completions of K, let V_v be the restriction of V to a decomposition group at v, let the ψ_v be the local components of a character of A_K/K, and suppose that

$$\int_{A_K/K} \bigotimes_v dx_v = 1.$$

Then

$$\varepsilon(V, s) = \prod_v \varepsilon(V_v, \psi_v, dx_v, s). \tag{1.2.1}$$

Both locally and globally, ε is multiplicative in V: $\varepsilon(V \oplus V') = \varepsilon(V) \cdot \varepsilon(V')$. This makes it possible to define ε for virtual representations (elements of the Grothendieck group $R(G, \mathbf{C})$ of the category of representations of G).

Return to the local case. If V is a virtual representation of dimension 0, $\varepsilon(V, \psi, dx, s)$ is independent of dx, which is omitted from the notation ([1] 5.3). If V has dimension 0 and determinant $\chi : K^* \simeq W(\bar{K}/K)^{\mathrm{ab}} \to \mathbf{C}^*$, then ([1] 5.4)

$$\varepsilon(V, \psi(ax), s) = \chi(a)\varepsilon(V, \psi, s). \tag{1.2.2}$$

In particular, if $\chi = 1$, $\varepsilon(V, \psi, s)$ is independent of ψ; ψ is then omitted from the notation.

The Artin conductor f of V is by definition 1 when K is archimedean. When K is nonarchimedean, it is the number denoted $q^{a(V)}$ in [1] (q = the number of elements of the residue field). For V of dimension 0 and determinant 1, set

$$W(V) = \varepsilon\left(V, \tfrac{1}{2}\right). \tag{1.2.3}$$

Then

$$\varepsilon(V, s) = W(V) f^{\frac{1}{2} - s}, \tag{1.2.4}$$

$$W(V) \cdot W(V^*) = 1. \tag{1.2.5}$$

If V is isomorphic to its dual, i.e. if its character is real, (1.2.5) shows that $W(V) = \pm 1$.

The group $R(G, \mathbf{R})$ of real virtual representations of G, the Grothendieck group of the category of real representations of G, is a subgroup of the group of virtual representations of G having real character. The elements of this subgroup are said to be realizable over $\mathbf{R}$. We propose to calculate $W(V_v)$ when V_v has dimension 0, determinant 1, and is realizable over $\mathbf{R}$.

(1.3) Let G be a finite group. A real representation $\rho : G \to GL(V)$ has a Stiefel-Whitney class

$$w^*(V) = 1 + w^1(V) + \cdots \in H^*(G, \mathbf{Z}/2) = \prod_i H^i(G, \mathbf{Z}/2).$$

It may be defined as follows: the representation V defines a vector bundle (indeed, a local system) $\mathscr{V}$ on the classifying space BG of the group G, and $w^*(V)$ is the Stiefel-Whitney class of $\mathscr{V}$ in $H^*(BG, \mathbf{Z}/2) = H^*(G, \mathbf{Z}/2)$. The same construction applies to real virtual representations: w^* is the composite homomorphism

$$R(G, \mathbf{R}) \longrightarrow KO(BG) \xrightarrow{w^*} H^*(BG, \mathbf{Z}/2)^\times = H^*(G, \mathbf{Z}/2)^\times.$$

Via the isomorphism $H^1(G, \mathbf{Z}/2) = \mathrm{Hom}(G, \{\pm 1\})$, $w^1(V)$ identifies with the character $g \mapsto \det \rho(g)$. The group $H^2(G, \mathbf{Z}/2)$ classifies central extensions of G by the two-element group. If $w^1(V) = 0$, then $w^2(V)$ is the class of the extension obtained by pulling back under ρ the double cover $\mathrm{Spin}(V, Q)$ of $SO(V, Q)$, where Q is any positive-definite G-invariant quadratic form on V.

(1.4) Retain the notation of 1.2, and let $\mathrm{cl}(w^2(V))$ be the image of the class $w^2(V) \in H^2(G, \mathbf{Z}/2)$ under the composite map

$$H^2(G, \mathbf{Z}/2) \xrightarrow{\mathrm{infl}} H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2) \xrightarrow{(1)} H^2(\mathrm{Gal}(\bar{K}/K), \bar{K}^*) \xrightarrow{\mathrm{inv}} \mathbf{Q}/\mathbf{Z}, \tag{1.4.1}$$

where (1) is induced by $n \mapsto (-1)^n : \mathbf{Z}/2 \to \bar{K}^*$, and where inv is supplied by local class field theory. If K is complex or has characteristic 2, then $\mathrm{cl}\, w^2(V) = 0$. In the other cases, (1.4.1) identifies $H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2)$ with $\{0, \frac{1}{2}\} \subset \mathbf{Q}/\mathbf{Z}$.

Our principal result is the following:

(1.5) Theorem. If V is a real virtual representation of dimension 0 and determinant 1, then

$$W(V) = \exp\big(2\pi i\ \mathrm{cl}(w^2(V))\big).$$

Thus, in characteristic 2, $W(V) = 1$. In characteristic $\neq 2$, $W(V) = 1$ or -1 according as $w^2(V)$ is or is not trivial in $H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2)$.

(1.6) From this theorem one can deduce the theorem of Queyrut and Fröhlich:

(a) For V a real virtual representation of dimension 0 and determinant 1 over a global field K of characteristic $\neq 2$, the $\mathrm{cl}(w^2(V_v))$ are the local invariants of $w^2(V) \in H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2)$, and the reciprocity law ensures that their sum in $\mathbf{Q}/\mathbf{Z}$ is zero; hence the product of the $W(V_v)$ is one. In equal characteristic 2, all the $W(V_v)$ are 1.

(b) If V has dimension one and is defined by a character of order two (respectively, is trivial), the corresponding L-function is a quotient of two zeta functions (respectively, a zeta function), and one uses the fact that the global constant of a zeta function is $+1$.

(c) The conclusion follows by observing that every real virtual representation is a sum of representations of types (a) and (b).

(1.7) We shall also prove the variants of these results obtained by replacing Galois groups by Weil groups (§ 5).

2. Lemmas on real representations of finite groups

(2.1) Proposition. Let G be a finite group, H a subgroup, V a real virtual representation of H, and $\mathrm{Tr} : H^2(H, \mathbf{Z}/2) \to H^2(G, \mathbf{Z}/2)$ the trace morphism (corestriction). If the dimension of V is divisible by 4 and its determinant is 1 ($w^1(V) = 0$), then

$$w^1\left(\mathrm{Ind}_H^G(V)\right) = 0 \quad \text{and} \quad w^2\left(\mathrm{Ind}_H^G(V)\right) = \mathrm{Tr}\, w^2(V).$$

Let us translate this statement into topological terms. Let B_G be a classifying space for G, let E_G be the universal cover of B_G (the universal principal homogeneous G-space), and set $B_H = E_G/H$. This is a classifying space for H. Let f denote the projection $f : B_H \to B_G$. The diagrams

$$\begin{array}{ccc} R(H,\mathbf{R}) & \longrightarrow & KO(B_H) \\ \downarrow{\scriptstyle \mathrm{Ind}} & & \downarrow{\scriptstyle f_*} \\ R(G,\mathbf{R}) & \longrightarrow & KO(B_G) \end{array} \quad \text{and} \quad \begin{array}{ccc} H^*(B_H,\mathbf{Z}/2) & = & H^*(H,\mathbf{Z}/2) \\ \downarrow{\scriptstyle \mathrm{tr}} & & \downarrow{\scriptstyle f_*} \\ H^*(B_G,\mathbf{Z}/2) & = & H^*(G,\mathbf{Z}/2) \end{array}$$

commute (the definition of the maps f_* is recalled below), so that 2.1 follows from the following proposition.

(2.2) Proposition. Let $f : X \to Y$ be a finite covering of topological spaces. If $v \in KO(X)$ has dimension divisible by 4 and zero first Stiefel-Whitney class, then $w^1(f_* v) = 0$ and

$$w^2(f_* v) = f_*(w^2(v)).$$

Recall the definition of the maps f_*.

(a) In K-theory: f_* is induced by the operation that associates to a vector bundle $\mathscr{V}$ on X the vector bundle $f_*\mathscr{V}$ on Y whose fiber at $y \in Y$ is the direct sum

$$\bigoplus_{f(x)=y} \mathscr{V}_x.$$

(b) In cohomology. f_* is the composite of the isomorphism $H^*(X, \mathbf{Z}/2) = H^*(Y, f_*\mathbf{Z}/2)$ and the morphism induced functorially by the trace morphism:

$$\mathrm{Tr}_f : f_*\mathbf{Z}/2_X \longrightarrow \mathbf{Z}/2_Y : \text{ “sum of the values over the sheets.”}$$

We reduce to the case in which f has a constant number k of sheets and v is defined by an orthogonal vector bundle $\mathscr{V}$ of constant dimension n. Let P' be the corresponding $O(n)$-torsor and let $[P']$ be its class in $H^1(X, O(n))$. The descriptions in 1.3 of w^1 and w^2 have the following analogue.

(a) $w^1 \in H^1(X, \mathbf{Z}/2)$ is the image of $[P']$ under

$$\det : H^1(X, O(n)) \longrightarrow H^1(X, \{\pm 1\}) = H^1(X, \mathbf{Z}/2).$$

If $w^1 = 0$, P' lifts to an $SO(n)$-torsor P, of class $[P]$.

(b) The exact sequence

$$0 \longrightarrow \mathbf{Z}/2 \longrightarrow \mathrm{Spin}(n) \longrightarrow SO(n) \longrightarrow 0$$

induces $\partial : H^1(X, SO(n)) \to H^2(X, \mathbf{Z}/2)$, and, if $w^1 = 0$, $w^2 = \partial[P]$.

The construction $\mathscr{V} \mapsto f_*\mathscr{V}$ associates to an $O(n)$-torsor on X an $O(nk)$-torsor on Y. It factors as follows: the local isomorphisms from the constant orthogonal vector bundle $(\mathbf{R}^n)^k$ to $f_*\mathscr{V}$ which, at each point y, carry each of the k factors of $(\mathbf{R}^n)^k$ to one of the factors in the decomposition

$$(f_*\mathscr{V})_y = \bigoplus_{f(x)=y} \mathscr{V}_x,$$

identify with the local sections of a torsor under the semidirect product $O(n)^k \rtimes \mathfrak{S}_k$ of $O(n)^k$ by the symmetric group on k letters. The required $O(nk)$-torsor is obtained from this torsor and the morphism $O(n)^k \rtimes \mathfrak{S}_k \to O(nk)$.

If $w^1(\mathscr{V}) = 0$, the structure group of $f_*\mathscr{V}$ can be reduced to $SO(n)^k \rtimes \mathfrak{S}_k$. If n is even, this group is contained in $SO(nk)$, and $w^1(f_*\mathscr{V}) = 0$.

Suppose that the structure group of $\mathscr{V}$ has been reduced to $SO(n)$, giving an $SO(n)$-torsor P and an $SO(n)^k \rtimes \mathfrak{S}_k$-torsor Q on Y, of class $[Q]$. Consider the exact sequence

$$0 \longrightarrow (\mathbf{Z}/2)^k \longrightarrow \mathrm{Spin}(n)^k \rtimes \mathfrak{S}_k \longrightarrow SO(n)^k \rtimes \mathfrak{S}_k \longrightarrow 0.$$

The quotient $SO(n)^k \rtimes \mathfrak{S}_k$ acts on $(\mathbf{Z}/2)^k$ through $\mathfrak{S}_k$, and the local system $((\mathbf{Z}/2)^k)^Q$ on Y identifies with $f_*\mathbf{Z}/2$. Via the isomorphism $H^*(X, \mathbf{Z}/2) \simeq H^*(Y, f_*\mathbf{Z}/2)$, the class $w^2(\mathscr{V}) = \partial[P]$ identifies with $\partial[Q] \in H^2(Y, ((\mathbf{Z}/2)^k)^Q)$, as is shown by a cocycle computation.

The proposition now follows from the existence, when n is divisible by 4, of a middle arrow making the diagram

$$\begin{array}{ccccccccc} 0 & \to & (\mathbf{Z}/2)^k & \to & \mathrm{Spin}(n)^k \rtimes \mathfrak{S}_k & \to & SO(n)^k \rtimes \mathfrak{S}_k & \to & 0 \\ & & \downarrow{\scriptstyle\Sigma} & & \downarrow & & \downarrow & & \\ 0 & \to & \mathbf{Z}/2 & \to & \mathrm{Spin}(nk) & \to & SO(nk) & \to & 0 \end{array} \tag{2.2.1}$$

commute, where Σ is the sum-of-coordinates homomorphism.

To construct (2.2.1), it is enough to construct a lift

$$\begin{tikzcd} \mathfrak{S}_k \arrow[r] \arrow[dr, dashed] & SO(nk) \\ & \mathrm{Spin}(nk). \arrow[u] \end{tikzcd}$$

The obstruction to constructing it is defined for even n, and lies in $H^2(\mathfrak{S}_k, \mathbf{Z}/2)$. It is additive in n, because the diagram

$$\begin{array}{ccccccccc} 0 & \to & \mathbf{Z}/2 \times \mathbf{Z}/2 & \to & \mathrm{Spin}(n'k) \times \mathrm{Spin}(n''k) & \to & SO(n'k) \times SO(n''k) & \to & 0 \\ & & \downarrow{\scriptstyle a+b} & & \downarrow & & \downarrow & & \\ 0 & \to & \mathbf{Z}/2 & \to & \mathrm{Spin}((n'+n'')k) & \to & SO((n'+n'')k) & \to & 0 \end{array}$$

commutes. It is therefore zero when n is divisible by 4.

(2.3) Lemma. The real virtual representations of dimension 0 and determinant 1 of a finite group G form an ideal of $R(G, \mathbf{R})$. This ideal is stable under inflation, restriction, and induction.

That they form an ideal follows from the fact that $\dim : R(G, \mathbf{R}) \to \mathbf{Z}$ is a homomorphism, that $\det(V + W) = \det(V) \cdot \det(W)$, and that

$$\det(V \otimes W) = \det(V)^{\dim(W)} \cdot \det(W)^{\dim(V)}.$$

Stability under inflation and restriction is immediate, while stability under induction follows from 2.2 (the first assertion).

An analogous lemma holds for complex representations. It is proved in the same way ([1] 1.2).

(2.4) Let D_n be the dihedral group, the semidirect product of $\mathbf{Z}/2$ by $\mathbf{Z}/n$, and let $\chi : \mathbf{Z}/n \to \mathbf{C}^*$ be a character. The representation $\mathrm{Ind}(\chi)$ is realizable over $\mathbf{R}$: it is realized on the real vector space $\mathbf{C}$, on which $\mathbf{Z}/n$ acts by χ and $\mathbf{Z}/2$ acts by complex conjugation. Its determinant is independent of χ. The virtual representation

$$r(\chi) = \mathrm{Ind}_{\mathbf{Z}/n}^{D_n}([\chi] - [1])$$

is therefore realizable over $\mathbf{R}$, has dimension 0, and has trivial determinant.

(2.5) Proposition. Let G be a finite group. Every real virtual representation of G of dimension 0 and trivial determinant is a sum

(a) of the realification of a complex virtual representation of dimension 0, and

(b) of a $\mathbf{Z}$-linear combination of representations induced from representations $r(\chi)$ of dihedral quotients of subgroups of G.

Recall that the “realification” of a complex representation V of a group G is the representation of G on the same space V, regarded as a real vector space by restriction of scalars from $\mathbf{C}$ to $\mathbf{R}$.

(2.6) Lemma. Let G be a finite group having a nilpotent subgroup G' of index 1 or 2. Every real virtual representation of G is a sum

(a) of the realification of a complex virtual representation of G,

(b) of a $\mathbf{Z}$-linear combination of representations induced from representations $r(\chi)$,

(c) of a $\mathbf{Z}$-linear combination of one-dimensional real representations.

Let us deduce the proposition from the lemma.

Brauer’s induction theorem (as generalized by Witt, J. Crelle, 190 (1952); see, for example, Curtis-Reiner, Representation theory of finite groups and associative algebras, Intersc. Publ., N.Y. 1962) asserts that the representation 1 of G is a $\mathbf{Z}$-linear combination $\sum_i n_i \,\mathrm{Ind}_{H_i}^G(\rho_i)$ of representations induced from real representations of subgroups H_i of G, each H_i being a semidirect product of a p-group P by a cyclic group C, with P acting on C only by ± 1. Lemma 2.3 and the identity

$$V = V \otimes 1 = V \otimes \sum_i n_i \,\mathrm{Ind}_{H_i}^G(\rho_i) = \sum_i n_i \,\mathrm{Ind}_{H_i}^G(V \otimes \rho_i)$$

bring us back to the case where G itself is such a semidirect product, hence of the type considered in 2.6. For a virtual V of dimension 0 and determinant 1, consider the decomposition 2.6

$$V = A + B + C = (A - \dim A \cdot 1) + B + (C + \dim A \cdot 1).$$

The representation $(A - \dim A \cdot 1)$ is of type (a) in the proposition, and B is of type (b). Since V, $(A - \dim A \cdot 1)$, and B have dimension 0 and determinant 1, $(C + \dim A \cdot 1)$ has the same properties; this reduces us to proving the proposition for $(C + \dim A \cdot 1)$, hence for a $\mathbf{Z}$-linear combination of one-dimensional real representations. Thus let V be of the form

$$V = \sum_{i=1}^{k} [\varepsilon_i] - \sum_{i=1}^{l} [\eta_i]$$

with ε_i and η_i characters of order dividing 2, $k = l$ (dimension 0), and $\prod \varepsilon_i = \prod \eta_i$ (determinant 1).

We have

$$\begin{aligned}\sum_{i=1}^{k} [\varepsilon_i] &= k \cdot 1 + \sum_{i=1}^{k} ([\varepsilon_i] - 1) \\ &= k \cdot 1 - \sum_{i=1}^{k} ([\varepsilon_i] - 1) \otimes \left(\left[\prod_{j<i} \varepsilon_j \right] - 1 \right) + \sum_{i=1}^{k} ([\varepsilon_i] - 1) \otimes \left[\prod_{j<i} \varepsilon_j \right] \\ &= k \cdot 1 + \left(\left[\prod \varepsilon_i \right] - 1 \right) - \sum_{i=1}^{k} ([\varepsilon_i] - 1) \left(\left[\prod_{j<i} \varepsilon_j \right] - 1 \right).\end{aligned}$$

It follows that V has an expression of the form

$$V = \sum \pm \big([\alpha_i \beta_i] - [\alpha_i] - [\beta_i] + 1 \big),$$

where the α_i and β_i have order dividing 2. The terms in this sum are zero if $\alpha_i = 1$ or $\beta_i = 1$, are of type (a) if $\alpha_i = \beta_i$, and are of type (b) for a dihedral group $D_4 = (\mathbf{Z}/2)^2$ otherwise.

(2.7) Lemma. Lemma 2.6 holds for a representation of dimension 2.

Such a representation is in fact either

(a) the realification of a one-dimensional complex representation,

(b) dihedral, hence the sum of a virtual representation $r(\chi)$ and $[\varepsilon] + 1$, for a character ε of order two.

(2.8) Lemma. Lemma 2.6 holds for a representation of G induced from a one-dimensional real representation of a subgroup of G.

We proceed by induction on the index of the subgroup H.

Let $H' = G' \cap H$. If H has index ≤ 2, apply Lemma 2.7. Otherwise, there is a subgroup J of G', belonging to the ascending central series, such that $K = H'J/H'$ is nontrivial and abelian. Let l be a prime dividing the order of K, and let K_l be the kernel of multiplication by l. The group H/H' acts on K and K_l.

Since H/H' has order ≤ 2, all its irreducible representations (over $\mathbf{Q}$, or mod l) are one-dimensional, and the $\mathbf{F}_l$-vector space K_l contains a line stable under H/H'. Let I' be its inverse image in $H'J$. The group H normalizes I', so that $I = I'H$ is a subgroup. We have $I' = G' \cap I$, and $H' \subset I' \subset G'$, with I'/H' cyclic of order l.

Let $\varepsilon : H \to \{\pm 1\}$ be the character of the representation of H under consideration. The transitivity formula

$$\mathrm{Ind}_H^G(\varepsilon) = \mathrm{Ind}_I^G \mathrm{Ind}_H^I(\varepsilon)$$

and the induction hypothesis applied to I reduce us to showing that the representation $\mathrm{Ind}_H^I(\varepsilon)$ of I satisfies Lemma 2.6. If $l = 2$, this representation has dimension two, and Lemma 2.7 applies. If $l \neq 2$, the nilpotence of G' ensures that I' normalizes $\varepsilon|_{H'}$: otherwise, G' would have a subquotient that is a noncentral extension of a group of order l by a group of type $(2, \dots, 2)$, and such a group is not nilpotent.

Let us calculate the restriction of $\mathrm{Ind}_H^I(\varepsilon)$ to I'. It is $\mathrm{Ind}_{H'}^{I'}(\varepsilon)$, obtained by inflation from $\mathrm{Ind}_{H'/\ker(\varepsilon)}^{I'/\ker(\varepsilon)}(\varepsilon)$. The group $I'/\ker(\varepsilon)$ is the product of a cyclic group of order l by $H'/\ker(\varepsilon)$, of order ≤ 2. Thus, as a complex representation, the representation of I' is a sum of distinct complex characters, and I/I' acts on the set of these characters either as the identity or by $\chi \mapsto \chi^{-1}$ (according to its action on I'/H'). Grouping χ and χ^{-1} together, we find that the representation $\mathrm{Ind}_H^I(\varepsilon)$ of I is a sum of real representations of dimension ≤ 2, and Lemma 2.7 applies.

(2.9) Let us prove Lemma 2.6. We proceed by induction on the order of G.

We may, and do, assume that

(1) V is an irreducible representation (over $\mathbf{R}$);

(2) V is even absolutely irreducible (otherwise V is of type (a));

(3) V is faithful (otherwise the induction hypothesis is applied to a quotient of G);

(4) V is not induced from a real representation of a proper subgroup of G (otherwise the induction hypothesis is applied to that subgroup, together with Lemma 2.8);

(5) G has no abelian subgroup of index ≤ 2 (otherwise, by (2), $\dim(V) \leq 2$, and Lemma 2.7 applies).

Let B be an abelian normal subgroup of G, contained in G', and maximal with these properties. If the complex character χ of B occurs in the restriction of V to B, then V is known to be induced from a real representation of the subgroup H of G that stabilizes $\{\chi, \bar{\chi}\}$. By (1) and (4), $H = G$, and only χ and $\bar{\chi}$ occur in $V|_B$. Let G'' be the stabilizer of χ. We have $B \subset G' \cap G''$, and B is central in G''.

Case 1. $B \neq G' \cap G''$. By (3), B is central in the nilpotent group $G' \cap G''$. The group $G/(G' \cap G'')$ is of type $(\mathbf{Z}/2)^r$ (with $r = 0, 1$, or 2); all its irreducible representations (over $\mathbf{Q}$, or mod l) are one-dimensional. The argument at the beginning of the proof of Lemma 2.8 therefore applies again and shows that there is a subgroup B' of $G' \cap G''$, normal in G, with B'/B cyclic. This group B' is abelian, contradicting the maximality of B.

Case 2. $B = G' \cap G''$. Then B has index ≤ 2 in G'', which is therefore abelian. This contradicts (5).

3. Proof of 1.5

(3.1) Return to the notation of 1.2 and 1.4. For V of dimension 0 and determinant 1, the number $W(V) = \mathrm{sgn}(\varepsilon(V))$ is invariant under induction and multiplicative in V. By 2.1 and local class field theory (namely, the formula $\mathrm{inv}\,\mathrm{Tr} = \mathrm{inv}$), $\mathrm{cl}\, w^2(V)$ is likewise invariant under induction and additive in V. Proposition 2.5 therefore reduces the proof of the theorem to the following two cases.

Case 1. V is the realification of a complex virtual representation W of dimension zero.

Let $\chi : K^* \to \mathbf{C}^*$ be the determinant of W. For every ψ, $\varepsilon(V) = \varepsilon(W, \psi) \cdot \varepsilon(\bar{W}, \psi)$; and

$$\varepsilon(\bar{W}, \psi) = \chi(-1) \cdot \varepsilon(\bar{W}, \psi(-x)) = \chi(-1)\overline{\varepsilon(W, \psi)} \qquad (1.2.2),$$

whence $W(V) = \chi(-1)$.

In particular, $W(V) = 1$ when K has characteristic two, as it should. Now suppose that K has characteristic $\neq 2$. The class $w^2(V)$ is the reduction modulo 2 of the first Chern class $c^1(W)$. The class $c^1(W) \in H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z})$ is, up to sign, the image of $\chi \in H^1(\mathrm{Gal}(\bar{K}/K), \mathbf{C}^*)$ under the connecting homomorphism ∂ in the long exact sequence arising from the exponential exact sequence

$$0 \longrightarrow \mathbf{Z} \xrightarrow{2\pi i} \mathbf{C} \xrightarrow{\exp} \mathbf{C}^* \longrightarrow 0.$$

The class $w^2(V)$ is similarly obtained from χ and the sequence

$$0 \longrightarrow \mathbf{Z}/2 \longrightarrow \mathbf{C}^* \xrightarrow{z^2} \mathbf{C}^* \longrightarrow 0.$$

It is zero if and only if χ has a square root, i.e., since -1 is the unique element of order 2 in $\widehat{K^*} = \mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}}$, if and only if $\chi(-1) = 1$. This proves the theorem.

Case 2. V is obtained by inflation from a representation $r(\chi)$ of a dihedral quotient of $\mathrm{Gal}(\bar{K}/K)$.

For archimedean K, necessarily $V = 0$. We may therefore assume, and do assume, that K is nonarchimedean.

Let G be a finite group that is an extension of $\mathbf{Z}/2$ by $\mathbf{Z}/n$. If $u \in G$ does not belong to the subgroup $\mathbf{Z}/n$, its image under the transfer $G^{\mathrm{ab}} \to \mathbf{Z}/n$ is u^2. Thus G is dihedral if and only if the transfer is trivial.

Recall also that, for an extension L of K, the transfer $\mathrm{Gal}(\bar{K}/K)^{\mathrm{ab}} \to \mathrm{Gal}(\bar{K}/L)^{\mathrm{ab}}$ corresponds to the inclusion $K^* \to L^*$. The situation in Case 2 can therefore be described as follows: begin with a separable quadratic extension L/K and a character $\chi : L^* \to \mathbf{C}^*$ trivial on K^*. The representation $\mathrm{Ind}\,\chi$ of $\mathrm{Gal}(\bar{K}/K)$ is then realizable over $\mathbf{R}$, is dihedral, and $V = \mathrm{Ind}([\chi] - 1)$.

(3.2) Theorem (Fröhlich and Queyrut [2]). For V as above, and for a nonzero element Δ of L such that $\mathrm{Tr}_{L/K}(\Delta) = 0$ (Δ is well defined up to multiplication by an element of K^*), one has

$$W(V) = \chi(\Delta).$$

Let ψ be a nontrivial additive character of K, let $\psi_L = \psi \circ \mathrm{Tr}_{L/K}$, and let dx be a Haar measure on L. Then

$$\begin{aligned}\varepsilon(V, s) &= \varepsilon(\mathrm{Ind}([\chi] - 1), \psi, s) = \varepsilon([\chi] - 1, \psi_L, s) \\ &= \varepsilon(\chi, \psi_L, dx, s) \cdot \varepsilon(1, \psi_L, dx, s)^{-1}.\end{aligned}$$

Define the Fourier transform on the Schwartz-Bruhat functions on L by

$$\widehat{f}(y) = \int f(x)\psi_L(xy)\,dx.$$

The ε-factors are defined by Tate's functional equation

$$\frac{\int \widehat{f}(x)\|x\|^{1-s}\chi^{-1}(x)\,d^*x}{L(\chi^{-1}, 1-s)} = \varepsilon(\chi, \psi_L, dx, s)\frac{\int f(x)\|x\|^{s}\chi(x)\,d^*x}{L(\chi, s)} \tag{3.2.1}$$

(the same Haar measure d^*x on L^* is used on both sides). The two sides of this formula are generally defined by analytic continuation in s. However, if $f(0) = \widehat{f}(0) = 0$, the integrals reduce to finite sums. Let $x \mapsto \bar{x}$ be the nontrivial K-automorphism of L. We have $\chi^{-1}(x) = \chi(\bar{x})$, so that $L(\chi, s) = L(\chi^{-1}, s)$. Setting $s = \frac{1}{2}$ in (3.2.1), we find that, when $f(0) = \widehat{f}(0) = 0$,

$$\int \widehat{f}(x)\|x\|^{1/2}\chi^{-1}(x)\,d^*x = \varepsilon\Big(\chi, \psi_L, dx, \tfrac{1}{2}\Big)\int f(x)\|x\|^{1/2}\chi(x)\,d^*x.$$

We have

$$\int f(x)\|x\|^{1/2}\chi(x)\,d^*x = \int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)\int_{K^*} f(ax)\,da,$$

where da is a Haar measure on K, and

$$d^*a = \frac{da}{\|a\|_K}$$

(note that $\|a\|_L = \|a\|_K^2$). Similarly,

$$\begin{aligned}\int \widehat{f}(x)\|x\|^{1/2}\chi^{-1}(x)\,d^*x &= \int_{L^*/K^*}\|x\|^{1/2}\chi^{-1}(x)(d^*x/d^*a)\int_K \widehat{f}(ax)\,da\\ &= \int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)\int_K \widehat{f}(a\bar{x})\,da.\end{aligned}$$

For a constant c depending on ψ_L, dx, da, and Δ, one has

$$\int_K \widehat{f}(a\bar{x})\,da = c\int_K f(ax\Delta)\,da,$$

so, putting $c_1 = c\cdot\|\Delta\|^{-1/2}$,

$$\begin{aligned}\int \widehat{f}(x)\|x\|^{1/2}\chi^{-1}(x)\,d^*x &= \int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)c\cdot\int_K f(ax\Delta)\,da\\ &= c\cdot\|\Delta\|^{-1/2}\chi(\Delta)\int_{L^*/K^*}\|x\|^{1/2}\chi(x)(d^*x/d^*a)\int_K f(ax)\,da\\ &= c_1\chi(\Delta)\int f(x)\|x\|^{1/2}\chi(x)\,d^*x.\end{aligned}$$

For suitable choices of f, the integrals can be made nonzero. Hence

$$\varepsilon\Big(\chi, \psi_L, dx, \tfrac{1}{2}\Big) = c_1\chi(\Delta)$$

and

$$W(V) = \varepsilon\Big(V, \tfrac{1}{2}\Big) = \chi(\Delta).$$

(3.3) Return to Case 2 in the proof of the theorem. If K has characteristic 2, then $\Delta \in K^*$, and Theorem 3.2 shows that $W(V) = 1$, as it should. Now suppose that K has characteristic $\neq 2$.

The group $O(2)$ is the semidirect product of $\{e, s\}$ (with s a reflection) by $SO(2)$. Let $\widetilde{O}(2)$ be the semidirect product of $\{e, s\}$ by the double cover of $SO(2)$. For a representation $\rho : G \to O(2)$ of the finite group G, the obstruction to lifting ρ to $\widetilde{O}(2)$ (the class of the central extension of G by $\mathbf{Z}/2$ obtained by pulling back $\widetilde{O}(2)$ under ρ) is of the form $\alpha w_1^2 + w_2$, with $\alpha = 0$ or 1. In fact, $\alpha = 0$, as is seen by testing on $G = \{e, s\}$.

Put $w^1 = w^1(\operatorname{Ind}\chi) = w^1(\operatorname{Ind} 1)$ and $w^2 = w^2(\operatorname{Ind}\chi)$. We have

$$\begin{aligned} w^* \operatorname{Ind}([\chi] - 1) &= (w^* \operatorname{Ind}(\chi))(w^* \operatorname{Ind} 1)^{-1} \\ &= (1 + w^1 + w^2 + \cdots)(1 - w^1 + (w^1)^2 - \cdots) = 1 + w^2 + \cdots . \end{aligned}$$

The class $w^2 \operatorname{Ind}([\chi] - 1) \in H^2(\operatorname{Gal}(\bar{K}/K), \mathbf{Z}/2)$ is therefore zero if and only if the representation

$$\operatorname{Ind}\chi : \operatorname{Gal}(\bar{K}/K) \longrightarrow O(2)$$

lifts to $\widetilde{O}(2)$. This means that $\chi : L^* \to \mathbf{C}^*$ has a square root trivial on K^*, or equivalently that $\chi(\Delta) = 1$, where Δ is the unique element of order 2 in L^*/K^*. This completes the proof of the theorem.

4. Weil Groups and the Hasse Reciprocity Law

(4.1) Let K be a local field and $\bar{K}$ a separable closure of K. For the definition of the (absolute) Weil group $W(\bar{K}/K)$, see [1] 2.2. If K is nonarchimedean, it is a subgroup of $\operatorname{Gal}(\bar{K}/K)$, an extension of $\mathbf{Z}$ by inertia. Its cohomology with values in a discrete module M is defined by

$$H^i(W(\bar{K}/K), M) = \varinjlim_U H^i(W(\bar{K}/K)/U, M^U)$$

(U an open subgroup of the inertia group, normal in $W(\bar{K}/K)$). If K is archimedean, $W(\bar{K}/K)$ is a Lie group, an extension of $\operatorname{Gal}(\bar{K}/K)$ by $\bar{K}^*$; its cohomology is that of the classifying space $BW(\bar{K}/K)$.

Let n be an integer prime to the characteristic exponent of K, and let μ_n be the group of n-th roots of unity in $\bar{K}$. It is a $W(\bar{K}/K)$-module.

(4.2) Construction. One constructs an isomorphism

$$\operatorname{inv} : H^2(W(\bar{K}/K), \mu_n) \xrightarrow{\sim} (\mathbf{Q}/\mathbf{Z})_n.$$

(4.2.1) Lemma. If K is nonarchimedean and M is a discrete torsion $\operatorname{Gal}(\bar{K}/K)$-module, then

$$H^i(\operatorname{Gal}(\bar{K}/K), M) \xrightarrow{\sim} H^i(W(\bar{K}/K), M).$$

Let I be the inertia group. The map in 4.2.1 is the abutment of a morphism of spectral sequences

$$E_2^{pq} = H^p(\widehat{\mathbf{Z}}, H^q(I, M)) \Rightarrow H^{p+q}(\operatorname{Gal}(\bar{K}/K), M),$$

$$E_2^{pq} = H^p(\mathbf{Z}, H^q(I, M)) \Rightarrow H^{p+q}(W(\bar{K}/K), M),$$

and one uses the fact that $\mathbf{Z}$ and $\widehat{\mathbf{Z}}$ have the same cohomology with values in a discrete torsion module (on which $\widehat{\mathbf{Z}}$ acts continuously).

For nonarchimedean K, therefore,

$$H^2(\operatorname{Gal}(\bar{K}/K), \mu_n) \xrightarrow{\sim} H^2(W(\bar{K}/K), \mu_n).$$

One then uses the Kummer exact sequence

$$0 \longrightarrow \mu_n \longrightarrow \bar{K}^* \xrightarrow{n} \bar{K}^* \longrightarrow 0$$

and Theorem 90 to identify this H^2 with the kernel of multiplication by n in

$$H^2(\mathrm{Gal}(\bar{K}/K), \bar{K}^*) = \mathrm{Br}(K) = \mathbf{Q}/\mathbf{Z}.$$

Given the state of the literature, this defines 4.2 up to sign. The sign will be discussed in 4.7. In the following section, in any case, we shall use only the case $n = 2$.

For archimedean K, let $\mathbf{Z}(1) = 2\pi i\mathbf{Z} \subset \bar{K}$. We shall prove that $H^3(W(\bar{K}/K), \mathbf{Z}(1)) = 0$ and construct an isomorphism

$$\mathrm{inv} : H^2(W(\bar{K}/K), \mathbf{Z}(1)) \xrightarrow{\sim} \mathbf{Z}. \tag{4.2.2}$$

The long exact cohomology sequence associated with the short exact sequence

$$0 \longrightarrow \mathbf{Z}(1) \xrightarrow{n} \mathbf{Z}(1) \xrightarrow{\exp(x/n)} \mu_n \longrightarrow 0$$

will give the required isomorphism

$$\mathrm{inv} : H^2(W(\bar{K}/K), \mu_n) \xrightarrow{\sim} \mathbf{Z}/n \xrightarrow[x\mapsto x/n]{\sim} (\mathbf{Q}/\mathbf{Z})_n.$$

If K is complex, then $\bar{K} = K$, and $H^2(W(\bar{K}/K), \mathbf{Z}(1))$ classifies extensions of K^* by $\mathbf{Z}(1)$, i.e. homomorphisms from $\pi_1(K^*)$ to $\mathbf{Z}(1)$. This is the group $\mathbf{Z}$, generated by the class of the extension

$$0 \longrightarrow \mathbf{Z}(1) \longrightarrow \bar{K} \xrightarrow{\exp} \bar{K}^* \longrightarrow 0. \tag{4.2.3}$$

The cohomology of $W(\bar{K}/K)$ is that of infinite-dimensional complex projective space: the successive H^i, with values in $\mathbf{Z}(1)$, are

$$\mathbf{Z}(1), 0, \mathbf{Z}, 0, \mathbf{Z}(1), \dots .$$

For real K, use the spectral sequence

$$E_2^{pq} = H^p(\mathbf{Z}/2, H^q(\bar{K}^*, \mathbf{Z}(1))) \Rightarrow H^{p+q}(W(\bar{K}/K), \mathbf{Z}(1)).$$

$H^q(\bar{K}^*, \mathbf{Z}(1))$	H^0H^q	H^1H^q	H^2H^q	H^3H^q	H^4H^q	$\cdots$
0	0	0	0	0	0	$\cdots$
$\mathbf{Z}$	$\mathbf{Z}$	0	$\mathbf{Z}/2$	0	$\mathbf{Z}/2$	$\cdots$
0	0	0	0	0	0	$\cdots$
$\mathbf{Z}(1)$	0	$\mathbf{Z}/2$	0	$\mathbf{Z}/2$	0	$\cdots$

We have $E_2 = E_3$, since $E_2^{pq} = 0$ for odd q.

(4.2.4) Lemma. The homomorphism

$$H^2(W(\bar{K}/K), \mathbf{Z}(1)) \longrightarrow H^2(\bar{K}^*, \mathbf{Z}(1))$$

is not surjective.

It is enough to see that extension 4.2.3 is not the restriction to $\bar{K}^*$ of an extension E of $W(\bar{K}/K)$ by $\mathbf{Z}(1)$. The group $W(\bar{K}/K)$ is generated by $\bar{K}^*$ and F, with relations $F^2 = -1$ and $FzF^{-1} = \bar{z}$. If E existed, one could lift F to $\widetilde{F}$ in E. We would have $\widetilde{F}z\widetilde{F}^{-1} = \bar{z}$ ($z \in \bar{K}$) and $\widetilde{F}^2 \equiv \pi i \pmod{\mathbf{Z}(1)}$. It would follow that $\widetilde{F}^2$ did not commute with $\widetilde{F}$, which is absurd.

By the lemma, $d_3 : E_3^{0,2} \to E_3^{3,0}$ is nonzero, and the first H^i are

$$H^0 = 0.$$

$$H^1 = \mathbf{Z}/2 \quad (\text{equal to } H^1(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}(1))).$$

$H^2 = \mathbf{Z}$, and $H^2(W(\bar{K}/K), \mathbf{Z}(1)) \hookrightarrow H^2(\bar{K}^*, \mathbf{Z}(1))$ as a subgroup of index two, the inclusion being identified with $n \mapsto 2n$.

$H^3 = 0$.

(4.3) Remark. One can show that the cohomology is periodic with period 4. We shall not use this fact.

(4.4) Remark. Let K' be a finite extension of K in $\bar{K}$, and let $\alpha \in H^2(W(\bar{K}/K'), \mu_n)$. Then $\operatorname{inv} \operatorname{Tr} \alpha = \operatorname{inv} \alpha$. This follows from the analogous statement for $\operatorname{Br}(K)$ when K is nonarchimedean, and from the same formula for $\alpha \in H^2(W(\bar{K}/K'), \mathbf{Z}(1))$ when K is archimedean. The latter follows from the formulas $\operatorname{Tr} \operatorname{res} \beta = 2\beta$ and $\operatorname{inv} \operatorname{res} \beta = 2 \operatorname{inv} \beta$ for $\beta \in H^2(W(\bar{K}/K), \mathbf{Z}(1))$, with K real and K' complex.

(4.5) Example. Let $\chi : K^* \to \mathbf{C}^*$ be a quasicharacter. The class $\partial\chi$ of the extension of K^* by $\mathbf{Z}$ obtained by pulling back along χ the extension

$$0 \longrightarrow \mathbf{Z} \xrightarrow{2\pi i} \mathbf{C} \xrightarrow{\exp} \mathbf{C}^* \longrightarrow 0 \tag{4.5.1}$$

is an element of $H^2(K^*, \mathbf{Z})$. If α is a root of unity in K whose order divides n, its image under $\mathbf{Z} \to \mu_n : m \mapsto \alpha^m$ gives

$$(\chi, \alpha) \in H^2(W(\bar{K}/K), \mu_n).$$

(4.6) Proposition. We have

$$\exp\bigl(2\pi i \operatorname{inv}(\chi, \alpha)\bigr) = \chi(\alpha).$$

The archimedean case is verified by a direct calculation. When K is nonarchimedean, neither side changes if χ is multiplied by an unramified character. We may therefore suppose that χ has finite order. It then extends to

$$\chi : \operatorname{Gal}(\bar{K}/K)^{\mathrm{ab}} \longrightarrow \mathbf{C}^*,$$

which defines $[\chi] \in H^1(\operatorname{Gal}(\bar{K}/K), \mathbf{C}^*)$. Its image $\partial\chi \in H^2(\operatorname{Gal}(\bar{K}/K), \mathbf{Z})$, under the connecting homomorphism ∂ associated with (4.5.1), is the class of the extension of $\operatorname{Gal}(\bar{K}/K)$ by $\mathbf{Z}$ obtained by pulling back (4.5.1) along χ. Its image under $\mathbf{Z} \to \mu_n$, $m \mapsto \alpha^m$, is the cup product $\partial\chi \cup \alpha \in H^2(\operatorname{Gal}(\bar{K}/K), \mu_n)$. Its restriction to $W(\bar{K}/K)$ is (χ, α), and $\operatorname{inv}(\chi, \alpha)$ is the invariant of the cup product $\partial\chi \cup \alpha \in H^2(\operatorname{Gal}(\bar{K}/K), \bar{K}^*)$. In this form, the formula

$$\exp\bigl(2\pi i \operatorname{inv}(\partial\chi \cup \alpha)\bigr) = \chi(\alpha)$$

holds for every $\alpha \in K^*$: this is the definition of the reciprocity isomorphism.

(4.7) Signs. Since we have already specified our normalization (= choice of sign) of the reciprocity isomorphism, the validity of 4.6 fixes our normalization of inv for nonarchimedean K.

(4.8) Let K be a global field. For a finite Galois extension L of K, Weil defined the Weil group $W_{L/K}$, an extension of $\operatorname{Gal}(L/K)$ by the idèle class group $C(L)$ of L. It is an inverse limit of Lie groups; its cohomology is by definition the direct limit of the cohomologies of their classifying spaces. Let $\bar{K}$ be a separable closure of K, and let M be a discrete $\operatorname{Gal}(\bar{K}/K)$-module. We set

$$H^*(W(\bar{K}/K), M) = \varinjlim_{L \subset \bar{K}} H^*(W_{L/K}, M^{\operatorname{Gal}(\bar{K}/L)}).$$

Let n be prime to the characteristic exponent of K, and let $\alpha \in H^2(W(\bar{K}/K), \mu_n)$. The restriction of α to $H^2(W(\bar{K}_v/K_v), \mu_n)$ is zero for almost all places v of K (because it factors through an $H^2(\mathbf{Z}, \mu_n) = 0$).

The maps inv of 4.2 therefore give a map

$$\mathrm{inv} : H^2(W(\bar{K}/K), \mu_n) \longrightarrow \bigoplus_v (\mathbf{Q}/\mathbf{Z})_n.$$

(4.9) Theorem. The sequence

$$0 \longrightarrow H^2(W(\bar{K}/K), \mu_n) \xrightarrow{\mathrm{inv}} \bigoplus_v (\mathbf{Q}/\mathbf{Z})_n \xrightarrow{\Sigma} (\mathbf{Q}/\mathbf{Z})_n \longrightarrow 0$$

is exact.

Let $\chi : C(K) \to \mathbf{C}^*$ be a quasicharacter. The class $\partial\chi$ of the extension of $C(K)$ by $\mathbf{Z}$ obtained by pulling back along χ the extension

$$0 \longrightarrow \mathbf{Z} \xrightarrow{2\pi i} \mathbf{C} \xrightarrow{\exp} \mathbf{C}^* \longrightarrow 0$$

is an element of $H^2(C(K), \mathbf{Z})$. If α is a root of unity in K whose order divides n, its image under $\mathbf{Z} \to \mu_n : m \mapsto \alpha^m$ gives

$$(\chi, \alpha) \in H^2(C(K), \mu_n).$$

(4.9.1) Lemma. We have

$$\sum_v \mathrm{inv}_v(\chi, \alpha) = 0.$$

By 4.6, this is a reformulation of $\prod_v \chi_v(\alpha_v) = 1$.

(4.9.2) Lemma. There exists $c \in H^2(W(\bar{K}/K), \mu_n)$ having prescribed local invariants at the infinite places (or, indeed, at any finite set of places) and satisfying

$$\sum_v \mathrm{inv}_v(c) = 0.$$

If $\mu_n \subset K$, one may take c of the form (χ, α). Otherwise, take a finite extension K'/K such that K' contains μ_n, and take c of the form $\mathrm{Tr}_{K'/K}(\chi, \alpha)$; the reciprocity law follows from 4.4 and 4.9.1.

(4.9.3) Lemma. Let $c \in H^2(W(\bar{K}/K), \mu_n)$. If $\mathrm{inv}_v(c) = 0$ for complex v, and has order dividing two for real v, then c comes from $\bar{c} \in H^2(\mathrm{Gal}(\bar{K}/K), \mu_n)$.

When K has nonzero characteristic, one passes from a Weil group to a Galois group by replacing a quotient $\mathbf{Z}$ by $\widehat{\mathbf{Z}}$, and it is enough to argue as in 4.2.1.

Now suppose that K is a number field. There exists a finite totally imaginary Galois extension L of K, containing μ_n if desired, such that c is defined by an extension of $W_{L/K}$ by μ_n. There is even a modulus $\mathfrak{m}$ such that c comes from an extension E', by μ_n, of the quotient ${}_{\mathfrak{m}}W_{L/K}$ of $W_{L/K}$, which is an extension of $\mathrm{Gal}(L/K)$ by $C_{\mathfrak{m}}(L)$. The identity component of $C_{\mathfrak{m}}(L)$ is the quotient $L^*_{\mathbf{R}}/U_{\mathfrak{m}}$ of $(L \otimes \mathbf{R})^*$ by the group of units congruent to 1 modulo $\mathfrak{m}$. The hypothesis ensures that the extension induced by E' of $L^*_{\mathbf{R}}$ by μ_n is trivial. The extension of $L^*_{\mathbf{R}}/U_{\mathfrak{m}}$ by μ_n therefore comes from $\phi : U_{\mathfrak{m}} \to \mu_n$. By a theorem of Chevalley (deux théorèmes d'arithmétique, J. Math. Soc. Jap. 3 (1951), pp. 36–44), $\mathrm{Ker}(\phi)$ is a congruence subgroup: for a suitable modulus $\mathfrak{m}' \geq \mathfrak{m}$, c is defined by an extension E of ${}_{\mathfrak{m}'}W_{L/K}$ by μ_n, trivial on the identity component of ${}_{\mathfrak{m}'}W_{L/K}$. The extension E therefore comes from the Galois group $\pi_0({}_{\mathfrak{m}'}W_{L/K})$; this proves 4.9.3.

Set

$$(\mathbf{Q}/\mathbf{Z})_{n,v} = \begin{cases} \text{the subgroup of } \mathbf{Q}/\mathbf{Z} \text{ of order } n, & v \text{ nonarchimedean,} \\ \text{the subgroup of order } (n,2), & v \text{ real,} \\ \text{the subgroup of order } 1, & v \text{ complex,} \end{cases}$$

and

$$A = \left(\bigoplus_v (\mathbf{Q}/\mathbf{Z})_n \right) \Big/ \left(\bigoplus_v (\mathbf{Q}/\mathbf{Z})_{n,v} \right).$$

The proof of 4.9 is completed by considering the following diagram, in which exactness of the last row is the Hasse reciprocity law:

$$\begin{tikzcd}
 & & 0 & & \\
 & A \arrow[r, equal] & A \arrow[u] & & \\
 & H^2(W(\bar{K}/K), \mu_n) \arrow[u] \arrow[r, "\mathrm{inv}"] & \bigoplus_v (\mathbf{Q}/\mathbf{Z})_n \arrow[u] \arrow[r, "\Sigma"] & (\mathbf{Q}/\mathbf{Z})_n \arrow[r] & 0 \\
0 \arrow[r] & H^2(\mathrm{Gal}(\bar{K}/K), \mu_n) \arrow[u] \arrow[r, "\mathrm{inv}"] & \bigoplus_v (\mathbf{Q}/\mathbf{Z})_{n,v} \arrow[u] \arrow[r, "\Sigma"] & (\mathbf{Q}/\mathbf{Z})_n \arrow[u, equal] \arrow[r] & 0 \\
 & & 0 \arrow[u] & &
\end{tikzcd}$$

5. Orthogonal representations of Weil groups

(5.1) Rather than considering real representations of Galois groups, it is more general and more natural to consider complex orthogonal representations of Weil groups.

Let K be a local field, let $\bar{K}$ be a separable closure of K, let V be a complex vector space equipped with a nondegenerate quadratic form Q, and let $\rho : W(\bar{K}/K) \to O(V,Q)$ be a continuous orthogonal representation of $W(\bar{K}/K)$. The Stiefel–Whitney classes $w^i \in H^i(W(\bar{K}/K), \mathbf{Z}/2)$ are still defined. Since the space of nondegenerate quadratic forms on V invariant under $W(\bar{K}/K)$ is connected, they depend only on the linear representation underlying ρ. The total Stiefel–Whitney class has the following additivity properties:

(5.1.1) If V is the orthogonal direct sum of subrepresentations V' and V'', then

$$w^*(V) = w^*(V') \cdot w^*(V'').$$

(5.1.2) If W is an invariant totally isotropic subspace of V, then

$$w^*(V) = w^*(W \oplus V/W^\perp) \cdot w^*(W^\perp/W).$$

The quadratic form Q on V induces a duality between W and $V/W^\perp$—hence a quadratic form on $W \oplus V/W^\perp$—whereas $Q|W^\perp$ has kernel W—hence a quadratic form on $W^\perp/W$. This gives meaning to 5.1.2. To prove the formula, it is enough to prove the analogous statement for a complex orthogonal vector bundle $\mathscr{V}$ on a paracompact topological space and for $\mathscr{W} \subset \mathscr{V}$

totally isotropic. Here $\mathscr{V}$ is isomorphic to $(\mathscr{W} \oplus \mathscr{V}/\mathscr{W}^{\perp}) \oplus (\mathscr{W}^{\perp}/\mathscr{W})$, and one applies the analogue of 5.1.1.

Let $RO(W(\bar{K}/K), \mathbf{C})$ be the Grothendieck group of semisimple complex orthogonal representations of $W(\bar{K}/K)$, with respect to orthogonal direct sum. It is a subgroup of $R(W(\bar{K}/K), \mathbf{C})$, the Grothendieck group of the category of semisimple complex representations of $W(\bar{K}/K)$. Property (5.1.1) makes it possible to define the total Stiefel–Whitney class of an element of $RO(W(\bar{K}/K), \mathbf{C})$ (a virtual orthogonal representation).

If V is an orthogonal representation of $W(\bar{K}/K)$, its semisimplification V' is again orthogonal, and (5.1.2) gives $w^*(V) = w^*(V')$. This justifies the restriction above to semisimple representations.

For $x \in H^2(W(\bar{K}/K), \mathbf{Z}/2)$, the class $\mathrm{cl}(x)$ is 0 when K has characteristic 2, and otherwise is the invariant of its image in $H^2(W(\bar{K}/K), \mu_2)$ (with $\mathbf{Z}/2$ and μ_2 canonically isomorphic). When K is not complex, one has

$$H^2(\mathrm{Gal}(\bar{K}/K), \mathbf{Z}/2) \xrightarrow{\sim} H^2(W(\bar{K}/K), \mathbf{Z}/2),$$

and this definition is compatible with 1.4. For K of characteristic different from 2, archimedean or not, cl is an isomorphism

$$\mathrm{cl} : H^2(W(\bar{K}/K), \mathbf{Z}/2) \xrightarrow{\sim} (\mathbf{Q}/\mathbf{Z})_2.$$

For a complex virtual representation V of $W(\bar{K}/K)$, of dimension 0 and determinant 1 and isomorphic to its dual, define, as in 1.2, a number $W(V) = \pm 1$. Again one has:

(5.2) Proposition. Let V be a virtual orthogonal representation of dimension 0 and determinant 1. Then

$$W(V) = \exp\big(2\pi i\ \mathrm{cl}(w^2(V))\big).$$

For V of the form $W \oplus W^*$ (where W is a virtual representation of dimension 0 and determinant χ), one proceeds as in 3.1, Case 1, using the formulas

$$w^2(V) = c^1(W) \pmod 2$$

and

$$\varepsilon(W, \psi) \cdot \varepsilon(W^*(1), \psi(-x)) = 1.$$

This suffices to prove 5.2 when K is complex. When K is real, it remains to treat the dihedral case of a representation $r(\chi)$, where χ is a character of $\bar{K}^*$ trivial on K^*. Theorem 3.2 remains valid, with the same proof [this time, for f and $\hat{f}$, smooth and rapidly decreasing and having a zero of high multiplicity at 0, the integrals converge in a wide strip], and one repeats the proof of 3.3.

When K is nonarchimedean, one verifies using ([1] 4.10) that every semisimple orthogonal representation V of $W(\bar{K}/K)$ is the sum of a representation of the form $W \oplus W^*$ and a representation factoring through a finite quotient of $\mathrm{Gal}(\bar{K}/K)$. For virtual V of dimension 0, one obtains a similar decomposition with W of dimension 0, and the proposition therefore follows from Theorem 1.5.

(5.3) Corollary. Let K be a global field, let L be a finite extension of K, and let V be an orthogonal representation of the Weil group $W_{L/K}$. The constant $W(V)$ in the functional equation of the corresponding L-function, defined as in (1.1.2), is +1.

The proof proceeds as in 1.6, now using 5.2 and the reciprocity law 4.9 (for $n = 2$).

(5.4) ℓ-adic variant. Let K be a nonarchimedean local field, and let V be a complex virtual representation of $W(\bar{K}/K)$, of dimension 0, with trivial determinant, and isomorphic to its dual. The definition of $\varepsilon(V) = \varepsilon(V, 0)$ is purely algebraic, i.e. independent of the topology of $\mathbf{C}$. If the conductor of V is not a square, the same is not true of $W(V) = \varepsilon(V) \cdot f^{-1/2}$. However, if V is orthogonal, Serre proved [3] that the conductor exponent is even; in this case $W(V)$ therefore has a purely algebraic meaning. The same holds for $w^2(V)$. By transport of structure, it follows that the formula (to which a meaning has just been given)

$$W(V) = (-1)^{2\,\mathrm{cl}(w^2(V))} \tag{5.4.1}$$

holds for a virtual orthogonal representation V of $W(\bar{K}/K)$, of dimension 0 and trivial determinant, over any algebraically closed field k of characteristic 0. The hypothesis "algebraically closed" is unnecessary, as is seen by replacing k with its algebraic closure.

(5.5) Let (V, ρ) be a continuous ℓ-adic representation of $W(\bar{K}/K)$, and fix ψ and dx. Following [1] 8.4.2, one associates with it a "representation" (ρ', N) of $'W(\bar{K}/K)$, consisting of

(a) a representation ρ' of $W(\bar{K}/K)$ on V, continuous for the discrete topology on V;

(b) a nilpotent endomorphism N of V, such that

$$\rho'(w) N \rho'(w)^{-1} = q^{v(w)} N.$$

If ρ preserves a quadratic form Q, then so does ρ', and $N \in \mathrm{Lie}(O(Q))$.

The constant $\varepsilon(V, \psi, dx, t)$ (where t is morally q^{-s}), defined by [1] 8.12, is the product of $\varepsilon(\rho', \psi, dx, t)$ and

$$\det\bigl(-Ft,\ V^{\rho'(I)} / \operatorname{Ker}(N)^{\rho'(I)}\bigr).$$

This monomial depends only on $V^{\rho'(I)}$, on which F and N act (with $FNF^{-1} = q^{-1}N$). The lemma below shows that, if V is orthogonal, it has even degree and takes the value 1 at $t = q^{-1/2}$.

For a virtual ℓ-adic orthogonal representation V of degree 0 and determinant one, the conductor of V therefore still has even exponent, and $W(V)$ may be defined as before. Moreover, $W(V) = W(V^{\mathrm{ss}})$, where V^{ss} is the semisimplification of V. This result may be used to calculate the right-hand side of (1.2.1) for an ℓ-adic orthogonal representation V of the Weil group of a global field; the method of [2] § 9 calculates the left-hand side, and one finds that (1.2.1) remains valid in this case.

(5.6) Lemma. Let k be a field, let $q \in k^*$, let V be a vector space over k, let Q be a nondegenerate quadratic form on V, and let $F \in O(V, Q)$ and $N \in \mathrm{Lie}(O(Q))$. Suppose that $FNF^{-1} = q^{-1}N$. Then $\dim(V / \operatorname{Ker} N)$ is an even integer $2m$, and

$$\det(-Ft, V / \operatorname{Ker} N) = q^m t^{2m}.$$

Let B be the bilinear form associated with Q. By hypothesis, the form $\langle x, y\rangle = B(Nx, y)$ is alternating, and $\langle Fx, Fy\rangle = q\langle x, y\rangle$. The kernel of $\langle x, y\rangle$ is $\operatorname{Ker} N$, and F induces a symplectic similitude of multiplier q on $V / \operatorname{Ker} N$. The lemma follows.

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Representations of reductive groups over finite fields

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Introduction

Let us consider a connected, reductive algebraic group G, defined over a finite field $\mathbb{F}_q$, with Frobenius map F. We shall be concerned with the representation theory of the finite group G^F, over fields of characteristic 0.

In 1968, Macdonald conjectured, on the basis of the character tables known at the time (GL_n, Sp_4), that there should be a well defined correspondence which, to any F-stable maximal torus T of G and a character θ of T^F in general position, associates an irreducible representation of G^F; moreover, if T modulo the centre of G is anisotropic over $\mathbb{F}_q$, the corresponding representation of G^F should be cuspidal (see Seminar on algebraic groups and related finite groups, by A. Borel et al., Lecture Notes in Mathematics, 131, pp. 117 and 101). In this paper we prove Macdonald's conjecture. More precisely, for T as above and θ an arbitrary character of T^F we construct virtual representations R_T^θ which have all the required properties.

These are defined in Chapter 1 as the alternating sum of the cohomology with compact support of the variety of Borel subgroups of G which are in a fixed relative position with their F-transform, with coefficients in certain G^F-equivariant locally constant l-adic sheaves of rank one. This generalizes a construction made by Drinfeld for SL_2 (see Ch. 2).

In Chapter 4 we prove a character formula for R_T^θ, based on the fixed point formula of Chapter 3. This character formula is in terms of certain "Green functions" on the unipotent elements; in Chapters 6, 7, 8 we prove that these Green functions satisfy all the axioms predicted by Springer, Kilmoyer and Macdonald ([9], [12]).

By 6.8, $\pm R_T^\theta$ is irreducible if θ is in general position and the vanishing theorem (9.9) gives an explicit model for it provided that q is not too small (if G is a classical group or G_2 any q will do; in the general case $q \geq 30$ is sufficient).

In Chapter 10 we study the irreducible components of the Gelfand-Graev representation of G^F, assuming that the centre of G is connected. The proof uses the results of Chapter 5 together with the disjointness theorem (6.2).

Finally, in Chapter 11 we discuss the case of the Suzuki and Ree groups.

It would be very desirable to find formulas for the Green functions more explicit than (4.1.2). Such formulas are known for GL_n (Green [4]), Sp_4 (Srinivasan [13]) and G_2 (Chang, Ree [2]). Very recently, Kazhdan has proved, using results of Springer, that the Green functions can be expressed as exponential sums on the Lie algebra (see [12]) provided that the characteristic is good and q is not too small.

Some of the results in this paper were announced in [7], [8].

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TABLE OF CONTENTS

Conventions and standard notations

0.1. In this paper, k is always an algebraically closed field and p is its characteristic exponent. The following assumptions and definitions will be in force in Chapters 4 to 8, in Chapter 10, and in parts of Chapters 1 and 9.

(0.1.1) p is a prime, and k is an algebraic closure of the prime field $\mathbb{F}_p$ of characteristic p. For q a power of p, $\mathbb{F}_q$ is the subfield with q elements of k. G is a reductive algebraic group over k, obtained by extension of scalars from G_0 over $\mathbb{F}_q$. We denote by F the corresponding Frobenius endomorphism $F : G \to G$ (as well as the Frobenius endomorphism of any k-scheme defined over $\mathbb{F}_q$).

0.2. l is a prime different from p, and $\overline{\mathbb{Q}}_l$ is an algebraic closure of the l-adic field. When there is no ambiguity on l, we will write simply $H^i_c(X)$ (resp. $H^i(X)$) for $H^i_c(X, \overline{\mathbb{Q}}_l)$ (resp. $H^i(X, \overline{\mathbb{Q}}_l)$) ($l$-adic cohomology of X, a scheme over k). The groups $\mathbb{Z}/l^n(1)$ are the groups $\mu_{l^n}(k)$ of l^n-roots of unity in k; they form a projective system, with transition maps $x \mapsto x^{l^{n-m}}$; the projective limit is $\mathbb{Z}_l(1)$. One defines $\mathbb{Z}_l(n) = \mathbb{Z}_l(1)^{\otimes n}$, and $\mathbb{Z}_l(-n)$ the dual of $\mathbb{Z}_l(n)$. The symbol (n) is for a Tate twist: tensoring with $\mathbb{Z}_l(n)$. We will make an accidental use of the similarly defined group $\widehat{\mathbb{Z}}_{p'}(1) = \varprojlim_{(n,p)=1} \mu_n(k)$.

0.3. For H a finite group, $\mathscr{R}(H)$ is the Grothendieck group of the finite dimensional representations of H over $\overline{\mathbb{Q}}_l$. If f and f' are class functions on H, with values in a cyclotomic field (often $\subset \overline{\mathbb{Q}}_l$), the inner product $\langle f, f' \rangle_H$ (or simply $\langle f, f' \rangle$) is

$$\frac{1}{|H|} \sum_{x \in H} f(x)\, \overline{f'(x)},$$

where the bar is the automorphism inducing $\zeta \mapsto \zeta^{-1}$ on the roots of unity. The same notation $\langle\ ,\ \rangle$ will be used for the inner product of elements of $\mathscr{R}(H) \otimes \mathbb{Q}$, identified with their characters.

0.4. The length function on a Coxeter group W (with canonical generators $s_1, \ldots, s_n$) will be denoted by $l(\ )$. A reduced expression for $w \in W$ is a decomposition $w = s_{i_1} \cdots s_{i_k}$ with $l(w) = k$.

0.5. Miscellaneous:

pr_i: ith projection (as in $\mathrm{pr}_1 : X \times X \to X$);

X^T: the fixed point subscheme of the endomorphism T of the scheme X (for instance: $G^F = G_0(\mathbb{F}_q)$);

p' (in index): away from p (as in $\widehat{\mathbb{Z}}_{p'}$, completion away from p) or the subgroup of elements of order prime to p (as in $(\mathbb{Q}/\mathbb{Z})_{p'}$);

$\mathrm{ad}\, g$: the inner automorphism $x \mapsto gxg^{-1}$, or maps deduced from it;

1, or e: the identity element of a group;

$\mathrm{Tr}(f, V^*)$, for f an endomorphism of a graded vector space, is

$$\sum (-1)^i \,\mathrm{Tr}(f, V^i);$$

reductive: reductive groups are meant to be connected and smooth.

The Jordan decomposition of an element $g \in G$ (G as in (0.1.1)) is $g = su$ where s is a semisimple element and u is a unipotent element commuting with s.

0.6. We will often identify a scheme over k with the set of its k-rational points. This should cause no confusion.

1. Some basic definitions

1.1. Suppose that in some category we are given a family $(X_i)_{i \in I}$ of objects and a compatible system of isomorphisms $\varphi_{ji} : X_i \xrightarrow{\sim} X_j$. This is as good as giving a single object X, the "common value" or "projective limit" of the family. This projective limit is provided with isomorphisms $\sigma_i : X \xrightarrow{\sim} X_i$ such that $\varphi_{ji}\sigma_i = \sigma_j$. We will use such a construction to define the maximal torus $\mathbf{T}$ and the Weyl group $\mathbf{W}$ of a connected reductive algebraic group G over k.

As index set I, we take the set of pairs (T, B) consisting of a maximal torus T and a Borel subgroup B containing T. For $i \in I$, $i = (T, B)$, we take

$T_i = T, \quad W_i = N(T)/T.$

The isomorphism φ_{ji} is the isomorphism induced by ad g where g is any element of G conjugating i into j; these elements g form a single right T_i-coset, so that φ_{ji} is independent of the choice of g.

One similarly defines the root system of $\mathbf{T}$, its set of simple roots, the action of $\mathbf{W}$ on $\mathbf{T}$ and the fundamental reflections in $\mathbf{W}$.

Let $F : G \to G$ be an isogeny. For any $i \in I$, $i = (T, B)$, $F(i) = (F(T), F(B))$ is again in I; F induces an isogeny $F : T_i \to T_{F(i)}$ and an isomorphism $F : W_i \to W_{F(i)}$. The corresponding endomorphism (resp. automorphism) $\sigma_{F(i)}^{-1} F \sigma_i$ of the torus $\mathbf{T}$ (resp. of the Weyl groups $\mathbf{W}$) is independent of the choice of i; we say that it is induced by F.

1.2. Let X (or X_G) be the set of all Borel subgroups of G. The group G acts on X by conjugation, and X is a smooth projective homogeneous space of G. For each Borel subgroup B of G, B is the stabilizer of the corresponding point of X, hence there is a natural isomorphism $G/B \to X : g \mapsto gBg^{-1}$.

The set of orbits of G in $X \times X$ can be identified with the Weyl group $\mathbf{W}$ of G as follows: for any $i = (T, B)$ as in (1.1), use the composite bijection

$$\mathbf{W} \xrightarrow[\sigma_i]{\sim} N(T)/T \xrightarrow{\sim} B\backslash G/B \xrightarrow[(e,g)]{\sim} G\backslash(G/B \times G/B) \xrightarrow{\sim} G\backslash(X \times X);$$

this is independent of the choice of (T, B). We denote by $O(w)$ the orbit corresponding to $w \in \mathbf{W}$; this is the orbit of $(B, \dot{w}B\dot{w}^{-1})$, where $\dot{w} \in N(T)$ represents w. We shall say that two Borel subgroups B', B'' of G are in relative position w, $w \in \mathbf{W}$, if and only if $(B', B'') \in O(w)$. In diagrams, we will picture this by

$$B' \xrightarrow{w} B''.$$

The basic properties of Bruhat decomposition can be expressed as follows:

(a) If $w = w_1 w_2$, with $l(w) = l(w_1) + l(w_2)$ then

(a$_1$) $(B', B'') \in O(w_1)$ and $(B'', B''') \in O(w_2) \Rightarrow (B', B''') \in O(w)$;

(a$_2$) if $(B', B''') \in O(w)$, there is one and only one B'' such that $(B', B'') \in O(w_1)$ and $(B'', B''') \in O(w_2)$.

On the scheme level:

$$O(w_1) \times_X O(w_2) \xrightarrow{\sim} O(w).$$

(b) Let s be an elementary reflection and let B be a Borel subgroup.

(b$_1$) $P = B \cup BsB$ is a (minimal parabolic) subgroup, i.e., if $(B', B'') \in O(s)$ and $(B'', B''') \in O(s)$, then either $(B', B''') \in O(s)$ or $B' = B'''$.

(b$_2$) The quotient $\overline{L}$ of $L = P/U_P$ (U_P = unipotent radical of P) by its centre is isomorphic to $PGL(2)$. The space $X_{\overline{L}}$ is hence a projective line. The inverse image map $X_{\overline{L}} \to X$ has as image the set of Borel subgroups B' in relative position e or s with B.

(b$_3$) Via either projection, $\overline{O(s)} = O(s) \cup O(e) \subset X \times X$ is hence a fibre space over X, with fibre P_1. It is provided with the section $O(e)$; this reduces its structural group from the projective to the affine group. The complement $O(s)$ of that section is a fibre space with fibre the affine line.

1.3. The assumptions (0.1.1) are in force in the rest of this chapter. The scheme X is hence defined over $\mathbb{F}_q$ and provided with a Frobenius map $F : X \to X$.

DEFINITION 1.4. For w in the Weyl group $\mathbf{W}$ of G, $X(w) \subset X$ is the locally closed subscheme of X consisting of all Borel subgroups B of G such that B and $F(B)$ are in relative position w.

One can also regard $X(w)$ as the intersection, in $X \times X$, of $O(w)$ with the graph of Frobenius. It is easily checked that this intersection is transverse. The orbit $O(w)$ being smooth of dimension $\dim(X)+l(w)$, it follows that $X(w)$ is smooth and purely of dimension $l(w)$. The subscheme $X(w)$ of X is G^F-stable. Hence, for each prime number $l \neq p$, G^F acts on the l-adic cohomology with compact supports of $X(w)$.

DEFINITION 1.5. $R^1(w)$ is the virtual representation

$$\sum(-1)^i H_c^i(X(w), \overline{\mathbb{Q}}_l)$$

of G^F (an element of the Grothendieck group of representations of G^F over $\overline{\mathbb{Q}}_l$).

For $w = e$, $X(w)$ is of dimension 0; it is the set of rational Borel subgroups, and $R^1(w)$ is induced by the unit representation of B^F, where B is an F-stable Borel subgroup.

It will follow from (3.3) that the character of $R^1(w)$ has integral values, independent of l. This will justify omitting l from the notation; $R^1(w)$ could be also regarded as a complex virtual representation of G^F.

An element of the Weyl group $\mathbf{W}$ is said to be F-conjugate to $w \in \mathbf{W}$, if it is of the form $w_1 w F(w_1)^{-1}$, for some $w_1 \in \mathbf{W}$. We denote by $W_F^{\sim}$ the set of F-conjugacy classes in $\mathbf{W}$. We shall recall in (1.14), that the G^F-conjugacy classes of F-stable (i.e., $\mathbb{F}_q$-rational) maximal tori in G are parametrized by $W_F^{\sim}$.

THEOREM 1.6. $R^1(w)$ depends only on the F-conjugacy class of w.

Let w and w' be F-conjugate.

Case 1. $w = w_1 w_2$, $w' = w_2 F(w_1)$ and $l(w) = l(w_1) + l(w_2) = l(w_2) + l(F(w_1)) = l(w')$ (0.4). For $B \in X(w)$, there is a unique Borel subgroup σB such that $(B, \sigma B) \in O(w_1)$ and $(\sigma B, FB) \in O(w_2)$; we have the diagram of relative positions:

$$\begin{array}{ccc} B & \xrightarrow{w_1 w_2} & F(B) \\ w_1 \downarrow & \nearrow_{w_2} & \downarrow F(w_1) \\ \sigma B & \xrightarrow{w_2 F(w_1)} & F(\sigma B) \end{array}$$

(the bottom one because $l(w_2F(w_1)) = l(w_2) + l(F(w_1)))$, and $\sigma B \in X(w')$. By the same argument applied to $w_2F(w_1)$ and $F(w_1)F(w_2) = F(w)$, we get a map $\tau : X(w') \to X(Fw)$. The diagram

$$\begin{array}{ccc} X(w) & \xrightarrow{\sigma} & X(w') \\ F \downarrow & \swarrow_{\tau} & \downarrow F \\ X(Fw) & \xrightarrow{\sigma^{(q)}} & X(Fw') \end{array}$$

is commutative. The vertical maps induce equivalences of étale sites, hence so do τ and σ [SGA 1, IX, 4.10]. The resulting isomorphism

$$H_c^*(X(w')) \xrightarrow{\sim} H_c^*(X(w))$$

is G^F-equivariant, whence (1.6) in this case.

Case 2. For some fundamental reflection s, we have $w' = swF(s)$ and $l(w') = l(w) + 2$. For $B \in X(w')$, we can find two Borel subgroups γB, δB such that $(B, \gamma B) \in O(s)$, $(\gamma B, \delta B) \in O(w)$, $(\delta B, FB) \in O(F(s))$; moreover, γB and δB are uniquely determined by these requirements. We define a partition $X(w') = X_1 \cup X_2$ by

$$X_1 = \{B \in X(w') \mid \delta B = F(\gamma B)\}, \qquad X_2 = \{B \in X(w') \mid \delta B \neq F(\gamma B)\}.$$

Note that X_1 is a closed subscheme of $X(w')$, while X_2 is an open one. We have $\gamma : X_1 \to X(w)$ and, for $B' \in X(w)$, $\gamma^{-1}(B')$ is the set of all Borel subgroups B such that $(B, B') \in O(s)$; hence $\gamma^{-1}(B')$ is an affine line over k, and X_1 is (via γ) an affine line bundle over $X(w)$. It follows that γ induces an isomorphism

$$H_c^i(X_1) \xrightarrow{\sim} H_c^{i-2}(X(w))(-1), \qquad i \geq 0. \tag{1.6.1}$$

If $B \in X_2$, we have

$$(\delta B, FB) \in O(F(s)), \quad (FB, F(\gamma B)) \in O(F(s)), \quad \delta B \neq F(\gamma B),$$

hence $(\delta B, F(\gamma B)) \in O(F(s))$. Since

$$(F(\gamma B), F(\delta B)) \in O(F(w)), \qquad l(F(sw)) = l(F(s)) + l(F(w)),$$

it follows that $(\delta B, F(\delta B)) \in O(F(sw))$. Thus we have $\delta : X_2 \to X(F(sw))$. Let

$$X_2' = \{(B, \widetilde{B}) \in X_2 \times X(sw) \mid F(\widetilde{B}) = \delta B\}$$

and let $\delta' : X_2' \to X(sw)$ be defined by $\delta'(B, \widetilde{B}) = \widetilde{B}$; we define also $\varphi : X_2' \to X_2$ by $\varphi(B, \widetilde{B}) = B$. We have a cartesian diagram

$$\begin{array}{ccc} X_2' & \xrightarrow{\delta'} & X(sw) \\ {\scriptstyle\varphi}\downarrow & & \downarrow{\scriptstyle F} \\ X_2 & \xrightarrow{\delta} & X(F(sw)). \end{array} \tag{1.6.2}$$

For any $\widetilde{B} \in X(sw)$ we denote by $s\widetilde{B}$ the unique Borel subgroup such that $(\widetilde{B}, s\widetilde{B}) \in O(s)$, $(s\widetilde{B}, F\widetilde{B}) \in O(w)$. It is easy to see that $\delta'^{-1}(\widetilde{B})$ can be identified with the set of Borel subgroups B such that $(\widetilde{B}, B) \in O(s)$, $B \neq s\widetilde{B}$; it follows that $\delta'^{-1}(\widetilde{B})$ is an affine line with a point removed. Thus X_2' is (via δ') a line bundle over $X(sw)$ with the zero-section removed. Since the vertical arrows in (1.6.2) induce equivalences of étale sites, we have a canonical exact sequence (see [5]):

$$\cdots \xrightarrow{\partial} H_c^{i-1}(X(sw)) \longrightarrow H_c^i(X_2) \longrightarrow H_c^{i-2}(X(sw))(-1) \xrightarrow{\partial} H_c^i(X(sw)) \longrightarrow \cdots. \tag{1.6.3}$$

It can be proved that the maps ∂ in (1.6.3) are zero; this fact will not be used here.

Note that (1.6.1) and (1.6.3) are G^F-equivariant and that G^F acts trivially on $\overline{\mathbb{Q}}_l(-1)$. It follows that for any $g \in G^F$:

$$\operatorname{tr}(g^*, H_c^*(X_2)) = 0,$$

$$\operatorname{tr}(g^*, H_c^*(X_1)) = \operatorname{tr}(g^*, H_c^*(X(w))).$$

If one uses the exact sequence

$$\cdots \longrightarrow H_c^{i-1}(X_1) \longrightarrow H_c^i(X_2) \longrightarrow H_c^i(X(w')) \longrightarrow H_c^i(X_1) \longrightarrow \cdots,$$

it follows that

$$\begin{aligned} \operatorname{tr}(g^*, H_c^*(X(w'))) &= \operatorname{tr}(g^*, H_c^*(X_1)) + \operatorname{tr}(g^*, H_c^*(X_2)) \\ &= \operatorname{tr}(g^*, H_c^*(X(w))) \end{aligned}$$

and 1.6 is proved in this case.

The general case. It suffices to treat the case where $w' = swF(s)$ for a fundamental reflection s. By permuting, if necessary, w and w' ($w = sw'F(s)$), we may even assume that $l(w') \geq l(w)$. If $l(w') > l(w)$ we are in case 2. If $l(w') = l(w)$, the following lemma shows that either we are in case 1 (with w and w' possibly interchanged) or that $w = w'$.

LEMMA 1.6.4. Let s, t be two fundamental reflections in W and let $w \in W$ be such that $l(w) = l(swt)$. Then either $w = swt$, or

$$l(sw) = l(w) - 1, \qquad \text{or} \qquad l(wt) = l(w) - 1.$$

Let $w = s_1 s_2 \cdots s_k$ be a reduced expression for w (0.4). Assume that $l(wt) = l(w) + 1$; then $wt = s_1 s_2 \cdots s_k t$ is also a reduced expression. We have $l(swt) = l(wt) - 1$. It follows (cf. Bourbaki, [1, Ch. IV, §1, Lemme 3]) that either there exists j, $1 \leq j \leq k$, with

$$ss_1 \cdots s_{j-1} = s_1 \cdots s_{j-1} s_j,$$

or we have

$$ss_1 \cdots s_k = s_1 \cdots s_k t.$$

In the first case, we have $w = ss_1 \cdots s_{j-1} s_{j+1} \cdots s_k$ and $l(sw) = l(w) - 1$; in the second case we have $w = swt$ and the lemma is proved.

1.7. Let us choose a maximal torus T^* in G and a Borel subgroup $B^* \subset G$ containing T^*, with unipotent radical U^*. The quotient $E = G/U^*$ is a T^*-torsor (= right principal homogeneous space of T^*) over $X = G/B^*$. For $x \in X$, the fibre $E(x)$ of the projection $E \to X$ is

$$E(x) = \{g \in G \mid ge^* = x\}/U^*,$$

where e^* is the point of X corresponding to B^*.

Let $\dot{w} \in N(T^*)$ define the element w in the Weyl group $\mathbf{W}$ via the isomorphism $\sigma(T^*, B^*) : \mathbf{W} \xrightarrow{\sim} N(T^*)/T^*$. If $x, y \in X$ are in relative position w, the g's in G such that $ge^* = x$ and $g\dot{w}e^* = y$ form a torsor $A(x, y)$ under

$$B^* \cap \dot{w}B^*\dot{w}^{-1} = T^* \cdot (U^* \cap \dot{w}U^*\dot{w}^{-1}).$$

For $g \in A(x, y)$, the class of $g\dot{w}$ in $E(y)$ depends only on the class of g in $E(x)$. This defines a map $E(x) \to E(y)$, which we denote as right multiplication by $\dot{w}$. We have the formulas

$$(ut)\dot{w} = (u\dot{w}) \operatorname{ad} \dot{w}^{-1}(t), \tag{1.7.1}$$

$$u(\dot{w}t) = (u\dot{w})t. \tag{1.7.2}$$

We will express (1.7.1) by saying that $\cdot\dot{w}$ is a w-map of T^*-torsors. It is induced by a w-map of T^*-torsors over $O(w)$

$$\cdot\dot{w} : \operatorname{pr}_1^* E \longrightarrow \operatorname{pr}_2^* E.$$

Assume that $w = w_1 w_2$, $\dot{w} = \dot{w}_1 \dot{w}_2$ and that $l(w) = l(w_1) + l(w_2)$; then for $x, y, z \in X$ with $(x, y) \in O(w_1)$ and $(y, z) \in O(w_2)$, we have $(x, z) \in O(w)$ and

$$u\dot{w} = (u\dot{w}_1)\dot{w}_2. \tag{1.7.3}$$

1.8. We now assume that T^* and B^* are F-stable. The identification $\sigma(T^*, B^*)$ of T^* and $N(T^*)/T^*$ with the torus $\mathbf{T}$ and the Weyl group $\mathbf{W}$ is then compatible with F. For w in the Weyl group, we denote by $\mathbf{T}(w)$ the torus $\mathbf{T}$, provided with the rational structure for which the Frobenius is $\operatorname{ad}(\dot{w})F$. We have

$$\mathbf{T}(w)^F \simeq \{t \in T^* \mid \operatorname{ad}(\dot{w})F(t) = t\}.$$

For $x \in X$, the Frobenius map induces a map $F : E(x) \to E(F(x))$, with $F(ut) = F(u)F(t)$. For $x \in X(w)$, we put

$$E(x, \dot{w}) = \{u \in E(x) \mid F(u) = u\dot{w}\}.$$

This is a $\mathbf{T}(w)^F$-torsor. The $E(x, \dot{w})$ are the fibres of a map

$$\pi : \widetilde{X}(\dot{w}) \longrightarrow X(w),$$

with $\widetilde{X}(\dot{w}) \subset E \mid X(w)$ a $\mathbf{T}(w)^F$-torsor over $X(w)$. The action of G on E restricts to an action of G^F on $\widetilde{X}(\dot{w})$.

Up to isomorphism, the G^F-equivariant $\mathbf{T}(w)^F$-torsor $\widetilde{X}(\dot{w})$ over $X(w)$ is independent of the lifting $\dot{w}$ of w in $N(T^*)$: for $\dot{w}' = \dot{w}t$, there exists t_1 with $\operatorname{ad}\dot{w}^{-1}(t_1)F(t_1)^{-1} = t$ and the map $u \mapsto ut_1$ induces an isomorphism

$$\widetilde{X}(\dot{w}) \xrightarrow{\sim} \widetilde{X}(\dot{w}').$$

The groups G^F and $\mathbf{T}(w)^F$ act on $H_c^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)$ by transport of structure. For any $\theta \in \operatorname{Hom}(\mathbf{T}(w)^F, \overline{\mathbb{Q}}_l^*)$ we denote by $H_c^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta$ the subspace of $H_c^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)$ on which $\mathbf{T}(w)^F$ acts by θ.

DEFINITION 1.9. $R^\theta(w)$ is the virtual representation

$$\sum (-1)^i H_c^i(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta$$

of G^F (an element of the Grothendieck group of representations of G^F over $\overline{\mathbb{Q}}_l$).

The character θ can be used to transform the $\mathbf{T}(w)^F$-torsor $\widetilde{X}(\dot{w})$ into a local system of $\overline{\mathbb{Q}}_l$-vector spaces of rank one $\mathscr{F}_\theta$ over $X(w)$, provided with

$$\theta : \widetilde{X}(\dot{w}) \longrightarrow \mathscr{F}_\theta, \qquad \theta(xt) = \theta(x)\theta(t).$$

The morphism $\pi : \widetilde{X}(\dot{w}) \to X(w)$ is finite and

$$\pi_* \overline{\mathbb{Q}}_l = \bigoplus_\theta \mathscr{F}_\theta.$$

The sheaf $\mathscr{F}_\theta$ is the subsheaf of $\pi_* \overline{\mathbb{Q}}_l$ on which $\mathbf{T}^F$ acts by θ, hence

$$H_c^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta = H_c^*(X(w), \mathscr{F}_\theta).$$

In particular, for $\theta = 1$,

$$R^1(w) = \sum (-1)^i H_c^i(X(w), \overline{\mathbb{Q}}_l)$$

so that Definition 1.9 is compatible with (1.5).

Example 1.10. For $w = \dot{w} = e$, $\pi : \widetilde{X}(\dot{w}) \to X(w)$ becomes the projection

$$\pi : G^F/U^{*F} \longrightarrow G^F/B^{*F},$$

and $R^\theta(w)$ is the representation of G^F on the space of functions on G^F satisfying

$$f(gtu) = \theta(t)^{-1} f(g),$$

G^F acting by $(g * f)(x) = f(g^{-1}x)$ (induced representation).

1.11. The Borel subgroup $\operatorname{ad} gB^*$ is in $X(w)$ if and only if $\operatorname{ad} gB^*$ and $\operatorname{ad} FgB^*$ are in relative position w, i.e., if and only if $g^{-1}Fg \in B^*\dot{w}B^*$ (where $\dot{w} \in N(T^*)$ represents w):

$$X(w) = \{g \in G \mid g^{-1}Fg \in B^*\dot{w}B^*\}/B^*. \tag{1.11.1}$$

If a Borel subgroup B is in $X(w)$, one can find $g \in G$ such that $\operatorname{ad} gB^*$ is

B and $\operatorname{ad} g \operatorname{ad} \dot{w} B^* = FB$:

$$X(w) = \{g \in G \mid g^{-1}Fg \in \dot{w}B^*\}/\big(B^* \cap \operatorname{ad} \dot{w}B^*\big), \tag{1.11.2}$$

where

$$B^* \cap \operatorname{ad} \dot{w}B^* = T^* \cdot (U^* \cap \operatorname{ad} \dot{w}U^*).$$

Changing g to gt, we can normalize g so that we have also $g^{-1}Fg \in \dot{w}U^*$:

$$X(w) = \{g \in G \mid g^{-1}Fg \in \dot{w}U^*\}/\big(\mathbf{T}(w)^F \cdot (U^* \cap \operatorname{ad} \dot{w}U^*)\big). \tag{1.11.3}$$

A point in $\widetilde{X}(\dot{w})$ is defined by a Borel subgroup B, plus $g \in G$ such that $\operatorname{ad} gB^* = B$, $\operatorname{ad} g \operatorname{ad} \dot{w}B^* = FB$ and $g\dot{w} = Fg \bmod U^*$:

$$\widetilde{X}(\dot{w}) = \{g \in G \mid g^{-1}Fg \in \dot{w}U^*\}/\big(U^* \cap \operatorname{ad} \dot{w}U^*\big). \tag{1.11.4}$$

COROLLARY 1.12. The following assertions are equivalent:

(i) $X(w)$ is affine;

(ii) $\widetilde{X}(\dot{w})$ is affine;

(iii) Let ρ be the action of $U^* \cap \operatorname{ad} \dot{w}U^*$ on U^* defined by

$$\rho(u)v = \operatorname{ad} \dot{w}^{-1}(u)vF(u^{-1});$$

then $U^*/\rho(U^* \cap \operatorname{ad} \dot{w}U^*)$ is affine.

Put $S = \{g \in G \mid g^{-1}Fg \in \dot{w}U^*\}$. The map $f : S \to U^* : g \mapsto \dot{w}^{-1}g^{-1}Fg$ induces an isomorphism $G^F\backslash S \to U^*$ and is such that, for $u \in U^* \cap \operatorname{ad} \dot{w}U^*$,

$$f(gu) = \rho(u)^{-1}f(g).$$

Hence,

$$G^F\backslash\widetilde{X}(\dot{w}) \simeq U^*/(U^* \cap \operatorname{ad} \dot{w}U^*).$$

As $\widetilde{X}(\dot{w})/\mathbf{T}(w)^F = X(w)$, it only remains to use the fact that a space and a quotient of it by a finite group are simultaneously affine or not.

We will have to use another description of $\pi : \widetilde{X}(\dot{w}) \to X(w)$. First, an easy lemma:

LEMMA 1.13. Let J be the set of pairs (T, B), T an F-stable maximal torus and B a Borel subgroup containing T. The map h which to (T, B) associates the relative position of B and FB induces a bijection

$$G^F\backslash J \xrightarrow{\sim} \mathbf{W}.$$

The proof will be given in (1.15).

If we use $(T, B) \in J$ to identify $\mathbf{W}$ with $N(T)/T$, we have

$$h(T, \operatorname{ad} \dot{w}B) = w^{-1}h(T, B)F(w), \tag{1.13.1}$$

where $\dot{w} \in N(T)$ represents w, hence

COROLLARY 1.14. The map h induces a bijection

$$\{G^F\text{-conjugacy classes of } F\text{-stable maximal tori}\} \xrightarrow{\sim} \mathbf{W}_{\widetilde{F}}.$$

Here is another description of h: for $(T, B) \in J$, if $\sigma : \mathbf{T} \to T$ and $\sigma : \mathbf{W} \to W$

($W = N(T)/T$) are defined by (T, B), then

$$\sigma F \sigma^{-1} = \mathrm{ad}\, h(T, B) \circ F \,:\, \mathbf{T} \longrightarrow \mathbf{T}.$$

To give $h(T, B)$ is the same as to give $\mathrm{ad}\, h(T, B) \circ F \in \mathbf{W} \circ F \subset \mathrm{End}(\mathbf{T})$, and to give the F-conjugacy class of $h(T, B)$ is the same as to give $\mathrm{ad}\, h(T, B) \circ F$ up to $\mathbf{W}$-conjugacy.

1.15. The space of maximal tori of G can be identified with the homogeneous space $G/N(T^*)$, and the space of maximal tori marked by a containing Borel subgroup can be identified with G/T^*. The group T^* being the connected component of $N(T^*)$, (1.13) is a special case of the general result described below.

Let G_0 be a connected algebraic group over $\mathbb{F}_q$ and let $x \,:\, \widetilde{E}_0 \to E_0$ be a morphism of G_0-homogeneous spaces. We denote by $G, \widetilde{E}, E$ the corresponding objects over k, and we assume that the stabilizer $S(\tilde{e})$ of $\tilde{e} \in \widetilde{E}$ is the connected component of the stabilizer $S(\pi(\tilde{e}))$ of $\pi(\tilde{e}) \in E$. Since any G_0-homogeneous space has a rational point, the existence of $\widetilde{E}_0$ imposes no condition on E_0.

The groups $S(e)/S(e)^0$ form a local system on E, which becomes constant on $\widetilde{E}$; we denote by W its constant value on $\widetilde{E}$. For $\tilde{e} \in \widetilde{E}$, we have an isomorphism

$$\alpha(\tilde{e}) \,:\, W \xrightarrow{\sim} S(\pi(\tilde{e}))/S(\pi(e)),$$

and, for $\widetilde{f} = g\tilde{e}$, we have $\alpha(\widetilde{f}) = \mathrm{ad}\, g\, \alpha(\tilde{e})$. We let the group W act on $\widetilde{E}$ on the right, by

$$\tilde{e}\, w = \alpha(\tilde{e})(w)\tilde{e};$$

in this way, $\widetilde{E}$ becomes a W-torsor (= principal homogeneous space) over E.

The group W is acted on by F, with $F(\tilde{e}\, w) = F(\tilde{e})F(w)$. The set $W_F^{\sim}$ of F-conjugacy classes in W is the set of orbits of the action of W on itself by

$$w \longmapsto w_1 w F(w_1)^{-1}.$$

PROPOSITION 1.16. For $e \in E^F$ and $\tilde{e} \in E$ above it, define $h(e, \tilde{e}) \in W$ by

$$F(\tilde{e}) = \tilde{e} \cdot h(e, \tilde{e}).$$

(i) The map h induces a bijection from the set of G^F-orbits in

$$\{(e, \tilde{e}) \mid e \in E^F,\ \pi(\tilde{e}) = e\}$$

to W.

(ii) We have

$$h(e, \tilde{e}w) = w^{-1} h(e, \tilde{e}) F(w).$$

Hence the map h induces a bijection from $G^F \backslash E^F$ to the set of F-conjugacy classes in W.

We will only prove (i). Let Y be the set of $\tilde{e} \in E$ such that $\pi(\tilde{e}) \in E^F$. If $\tilde{e}_0 \in E^F$, the map $g \mapsto g\tilde{e}_0$ identifies X with

$$\{g \in G \mid g^{-1} F g \in S(\pi(\tilde{e}_0))\}/S(\tilde{e}_0).$$

The Lang isogeny $g \mapsto g^{-1}Fg$ is an isomorphism $G^F \backslash G \to G$, hence the map $ge_0 \mapsto g^{-1}Fg$ induces a bijection

$$G^F \backslash X \xrightarrow{\sim} S(\pi(\tilde{e}_0))/\big(S(\tilde{e}_0) \text{ acting by } x \mapsto s^{-1} x F(s)\big).$$

The orbits of this action of $S(\tilde{e}_0)$ are just the usual cosets, and the resulting bijection $G^F\backslash X \xrightarrow{\sim} W$ is the h above.

DEFINITION 1.17. Let T be an F-stable maximal torus and let B be a Borel subgroup containing T, with unipotent radical U; let w be the relative position of B and FB.

(i) $X_{T\subset B}$ is $X(w)$. The map $g \mapsto \operatorname{ad} gB$ induces isomorphisms

$$\begin{aligned} X_{T\subset B} &= \{g \in G \mid g^{-1}Fg \in B\cdot F(B)\}/B \\ &= \{g \in G \mid g^{-1}Fg \in FB\}/(B\cap FB) \\ &= \{g \in G \mid g^{-1}Fg \in FU\}/\big(T^F\cdot(U\cap FU)\big). \end{aligned}$$

(ii) $\widetilde{X}_{T\subset B}$ is

$$\{g \in G \mid g^{-1}Fg \in FU\}/(U\cap FU).$$

We have a projection map $\pi : \widetilde{X}_{T\subset B} \to X_{T\subset B}$, for which $\widetilde{X}_{T\subset B}$ is a G^F-equivariant T^F-torsor over $X_{T\subset B}$; G^F acts by left multiplication and T^F by right multiplication (it normalizes $U\cap FU$).

1.18. Let $\dot{w} \in N(T^*)$ be a representative of w. If $x' \in G$ is such that $\operatorname{ad} x'(T^*, B^*) = (T, B)$ then B and $FB = \operatorname{ad} Fx'B^*$ are in relative position w, and FB contains T, hence $FB = \operatorname{ad} x' \operatorname{ad} \dot{w}B^*$ and

$$x'^{-1}F(x') \in \dot{w}B^* \cap N(T^*) = \dot{w}T^*.$$

Replacing x' by $x = x't$ ($t \in T^*$) one can achieve $x^{-1}F(x) = \dot{w}$. The x such that $\operatorname{ad} x(T^*, B^*) = (T, B)$ and $x^{-1}F(x) = \dot{w}$ form a T^{*F}-torsor. For such an x, $\operatorname{ad} x$ induces an isomorphism $\mathbf{T}(w) \to T$ (hence $\mathbf{T}(w)^F \to T^F$); this isomorphism is independent of x.

PROPOSITION 1.19. Let T, B, U, w be as in (1.17) and $x, \dot{w}$ as in (1.18). The map $g \mapsto gx^{-1}$ induces an isomorphism from the G^F-equivariant $\mathbf{T}(w)^F$-torsor $\widetilde{X}(\dot{w})$ over $X(w)$ (or rather its model (1.11)) to the G^F-equivariant T^F-torsor $\widetilde{X}_{T\subset B}$ over $X_{T\subset B}$ (or rather its model (1.17)).

This is a straightforward computation.

1.20. The cohomology of $\widetilde{X}_{T\subset B}$ is acted on by G^F and T^F. For any character $\theta : T^F \to \overline{\mathbb{Q}}_l^*$ we put as in (1.8):

$$R^\theta_{T\subset B} = \sum_i (-1)^i H^i_c(\widetilde{X}_{T\subset B}, \overline{\mathbb{Q}}_l)_\theta$$

(an element in the Grothendieck group of representations of G^F over $\overline{\mathbb{Q}}_l$).

By (1.19), for x as in (1.18), we have

$$R^\theta_{T\subset B} = R^{\theta\circ\operatorname{ad} x}(w).$$

We shall see in Chapter 4 that $R^\theta_{T\subset B}$ is independent of B. For $g \in G^F$ such that $\operatorname{ad} g$ carries T, B, θ to T', B', θ', we have clearly

$$R^\theta_{T\subset B} = R^{\theta'}_{T'\subset B'},$$

hence $R^\theta_{T\subset B}$ will eventually depend only on the G^F-conjugacy class of T and on the orbit of θ under $(N(T)/T)^F$.

The end of this chapter will be used in the proof of 7.10 only.

1.21. Isogenies. Let T be an F-stable maximal torus of G, and let Z be the centre of G. We denote $G \times^Z T$ the quotient of $G \times T$ by the subgroup

$$\{(z, z^{-1}) \mid z \in Z\}.$$

Let B be a Borel subgroup containing T. The action $x \mapsto gxt$ of $G^F \times T^F$ on $\widetilde{X}_{T\subset B}$ is induced by an action of $(G \times^Z T)^F$, given by the same formula.

COROLLARY 1.22. On $H^*(\widetilde{X}_{T\subset B}, \overline{\mathbb{Q}}_l)_\theta$, Z^F acts by the character $\theta \mid Z^F$. In particular, on any irreducible representation occurring in $R^\theta_{T\subset B}$, Z^F acts by $\theta \mid Z^F$.

Let $\pi : \widetilde{G} \to G$ be the simply connected covering of the derived group of G, $\widetilde{T} = \pi^{-1}(T)$, $\widetilde{B} = \pi^{-1}(B)$ and let $\widetilde{Z}$ be the centre of $\widetilde{G}$.

PROPOSITION 1.23. One has

$$T^F/\pi(\widetilde{T}^F) \xrightarrow{\sim} G^F/\pi(\widetilde{G}^F).$$

Injectivity is clear; we have to check the surjectivity of the map

$$(T \times \widetilde{G})^F \longrightarrow G^F$$

induced by

$$\varphi : T \times \widetilde{G} \longrightarrow G : \quad t, \tilde{g} \longmapsto t\pi(\tilde{g}).$$

Via φ, $T \times \widetilde{G}$ is a $\widetilde{T}$-torsor over G (with

$$(t, \tilde{g}) * \tilde{t} = (t\tilde{t}, \tilde{t}^{-1}\tilde{g})),$$

and one applies Lang's theorem to the connected group $\widetilde{T}$.

1.24. Let $h : A \to B$ be a homomorphism of finite groups and let X be a space on which A acts. The induced space $\mathrm{Ind}_A^B(X)$ (unique up to unique isomorphism) is any B-space I, provided with an A-equivariant map $\varphi : X \to I$, such that for any B-space Y,

$$\mathrm{Hom}_B(I, Y) \xrightarrow[\varphi]{\sim} \mathrm{Hom}_A(X, Y).$$

One has

$$\mathrm{Ind}_A^B(X) = \coprod_{b\in B/A} b\varphi(X), \qquad \varphi(X) \simeq \ker(h)\backslash X.$$

PROPOSITION 1.25. The $(G \times^Z T)^F$-space $\widetilde{X}_{T\subset B}$ is induced by the $(\widetilde{G} \times^{\widetilde{Z}} \widetilde{T})^F$-space $\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}$.

LEMMA 1.26. In the diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \ker & \longrightarrow & \widetilde{T}^F & \longrightarrow & T^F & \longrightarrow & \mathrm{coker} & \longrightarrow & 0 \\
 & & {\scriptstyle (1)}\downarrow & & {\scriptstyle (t,1)}\downarrow & & {\scriptstyle (t,1)}\downarrow & & {\scriptstyle (2)}\downarrow & & \\
0 & \longrightarrow & \ker & \longrightarrow & (\widetilde{T} \times^{\widetilde{Z}} \widetilde{G})^F & \longrightarrow & (T \times^Z G)^F & \longrightarrow & \mathrm{coker} & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & & & & & \\
 & & (\widetilde{G}/\widetilde{Z})^F & = & (G/Z)^F & & & & & &
\end{array}$$

the maps (1) and (2) are isomorphisms.

Indeed, $\widetilde{T} \times^{\widetilde{Z}} \widetilde{G}$ (resp. $T \times^Z G$) is a $\widetilde{T}$ (resp. T)-torsor over $\widetilde{G}/\widetilde{Z} = G/Z$, hence, since T and $\widetilde{T}$ are connected, $(\widetilde{T} \times^{\widetilde{Z}} \widetilde{G})^F$ is a $\widetilde{T}^F$-torsor over $(\widetilde{G}/\widetilde{Z})^F$, and $(T \times^Z G)^F$ is the induced T^F-torsor.

Proof of 1.25. By 1.26, we are reduced to prove that $\widetilde{X}_{T\subset B}$, as a T^F-space, is induced by the $\widetilde{T}^F$-space $\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}$. The spaces $\widetilde{X}_{T\subset B}$ and $\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}$ are indeed respectively T^F and $\widetilde{T}^F$-torsor over $X_{T\subset B} = X_{\widetilde{T}\subset\widetilde{B}}$, and $X_{\widetilde{T}\subset\widetilde{B}}$ is hence the T^F-torsor induced by the $\widetilde{T}^F$-torsor $X_{T\subset B}$.

COROLLARY 1.27. Let θ be a character of $G^F/\pi(\widetilde{G}^F)$. We denote again by θ its restriction to T^F. One has

$$R^{\theta\eta}_{T\subset B} = \theta \otimes R^{\eta}_{T\subset B}.$$

It follows from 1.25 that

$$H^*(\widetilde{X}_{T\subset B}, \overline{\mathbb{Q}}_l) = \mathrm{Ind}^{(G\times^Z T)^F}_{(\widetilde{G}\times^{\widetilde{Z}}\widetilde{T})^F}(\widetilde{X}_{\widetilde{T}\subset\widetilde{B}}, \overline{\mathbb{Q}}_l).$$

The character $\theta(gt)$ of $(G\times^{\widetilde{Z}} T)^F$ is trivial on the image of $\widetilde{G}\times^{\widetilde{Z}}\widetilde{T}$. The induced representation we consider is hence isomorphic to its tensor product with $\theta(gt)$, and 1.27 is a formal consequence of this.

2. Examples in the classical groups

2.1. Let V be an n-dimenssional vector space over k and put $G = \mathrm{GL}(V)$. If $b = (b_1, \dots, b_n)$ is a basis of V, we may take for T^* the group of diagonal matrices and for B^* the group of upper triangular matrices. The Weyl group lifts into the subgroup of $N(T^*)$ consisting of the $\dot{w}$'s inducing a permutation of basis vectors. In this case, $\mathbf{T}, \mathbf{W}$ (1.1), X (1.2), E, $\cdot\dot{w}$ (1.7), have the following alternative description.

(a) $\mathbf{T} = \mathbf{G}_m^n$, $\mathbf{W} = \mathfrak{S}_n$, the fundamental reflections are the transpositions $(i, i+1)$ and the action of $\mathbf{W}$ on $\mathbf{T}$ is by permutation.

(b) X is the space of complete flags $D_1 \subset \cdots \subset D_{n-1}$ in V: a flag D is an increasing filtration of V with $\dim D_i = i$ for $1 \le i \le n-1$.

(c) E is the space of complete flags marked by non-zero vectors

$$e_i \in D_i/D_{i-1} = \mathrm{Gr}^D_i(V) \qquad (1 \le i \le n),$$

where we use the convention $D_0 = 0$, $D_n = V$; $\mathbf{T}$ acts on E by

$$(D, (e_i))(\lambda_i) = (D, (\lambda_i e_i)).$$

This is a G-equivariant $\mathbf{T}$-torsor over X.

(d) If D' and D'' are two flags, their relative position is labelled by the permutation w such that

$$\mathrm{Gr}^{D'}_{w(i)}\,\mathrm{Gr}^{D''}_i(V) \neq 0.$$

The isomorphisms

$$\mathrm{Gr}^{D'}_{w(i)}(V) \simeq \mathrm{Gr}^{D''}_i\,\mathrm{Gr}^{D'}_{w(i)}(V) \simeq \mathrm{Gr}^{D'}_{w(i)}\,\mathrm{Gr}^{D''}_i(V) \simeq \mathrm{Gr}^{D''}_i(V)$$

induce a w-isomorphism between the $\mathbf{T}$-torsor $E(D')$ of markings of D' and $E(D'')$:

$$e \longmapsto e\cdot\dot{w}.$$

When w is the n-cycle $(1, \dots, n)$, D' and D'' are in relative position w if and only if

$$D''_i + D'_i = D'_{i+1} \quad (1 \le i < n-1), \qquad D''_{n-1} + D'_1 = V.$$

2.2. We now take k and $\mathbb{F}_q$ as in (0.1.1) and assume that V is provided with an $\mathbb{F}_q$-structure. Frobenius maps are then defined. For $w = (1, \dots, n)$, the condition for a flag D to be in relative position w with its image FD by Frobenius is that D be the flag

$$D_1 \subset D_1 + FD_1 \subset D_1 + FD_1 + F^2D_1 \subset \cdots \qquad \text{and that} \qquad V = \bigoplus_0^{n-1} F^i D_1.$$

If we denote by $\mathbf{P}(V)$ the set of homogeneous lines in V, the map $D \mapsto D_1$ is an isomorphism from $X(w)$ to the set of all $x \in \mathbf{P}(V)$ which do not lie on any $\mathbb{F}_q$-rational hyperplane. A marking e of F is such that $F(e) = e \cdot \dot{w}$ if and only if

$$\begin{aligned} &e_2 \equiv F(e_1) \pmod{e_1}, \\ &e_3 \equiv F^2(e_1) \pmod{e_1, F(e_1)}, \quad \ldots, \\ &e_n \equiv F^{n-1}(e_1) \pmod{e_1, F(e_1), \ldots, F^{n-2}(e_1)} \end{aligned}$$

and

$$e_1 \equiv F^n(e_1) \pmod{F(e_1), \ldots, F^{n-1}(e_1)};$$

e is defined by $e_1 \in D_1$ subject to the condition that

$$e_1 \wedge F(e_1) \wedge \cdots \wedge F^{n-1}(e_1) = F^n(e_1) \wedge F(e_1) \wedge \cdots \wedge F^{n-1}(e_1),$$

i.e.:

$$F(e_1 \wedge \cdots \wedge F^{n-1}(e_1)) = (-1)^{n-1}(e_1 \wedge \cdots \wedge F^{n-1}(e_1)). \tag{2.2.1}$$

If (x_i) are the coordinates of e_1 with respect to some rational basis, the condition (2.2.1) can be rewritten

$$(-1)^{n-1} \left(\det(x_i^{q^{j-1}})_{1 \le i,j \le n}\right)^{q-1} = 1. \tag{2.2.2}$$

The form on the left is invariant under $\mathrm{GL}(n, \mathbb{F}_q)$. Up to a scalar factor, it is the product of all non-zero $\mathbb{F}_q$-rational linear forms. The map $(D, e) \mapsto e_1$ induces an isomorphism of $\widetilde{X}(\dot{w})$ with the affine hypersurface (2.2.2). This hypersurface is stable under $x \mapsto \lambda x$ for $\lambda \in \mathbb{F}_{q^n}^*$, and this is the action of $\mathbf{T}(w)^F$.

Our work has been inspired by results of Drinfeld, who proved that the discrete series representations of $\mathrm{SL}(2, \mathbb{F}_q)$ occur in the cohomology of the affine curve

$$xy^q - x^q y = 1$$

(the form

$$xy^q - x^q y = \begin{vmatrix} x & x^q \\ y & y^q \end{vmatrix}$$

is $\mathrm{SL}_2(\mathbb{F}_q)$-invariant).

Let $\varphi(q, r, n)$ be the number of $\mathbb{F}_{q^r}$-rational points of $X(w)$, $r \ge 1$. We have the following

PROPOSITION 2.3.

$$\varphi(q, r, n) = \prod_{1 \le i \le n-1} (q^r - q^i).$$

Proof. We define a partition

$$\mathbf{P}(V) = X_0 \cup X_1 \cup \cdots \cup X_{n-1}$$

as follows: X_i is the set of points $x \in \mathbf{P}(V)$ such that $x, F(x), F^2(x), \ldots$ span a linear subspace of dimension i of $\mathbf{P}(V)$. This partition is invariant under Frobenius, and clearly X_i has precisely

$$\varphi(q, r, i+1) \frac{(1 - q^{n-i}) \cdots (1 - q^n)}{(1 - q) \cdots (1 - q^{i+1})}$$

$\mathbb{F}_q$-rational points. It follows that

$$\frac{q^{nr}-1}{q-1} = \sum_{0\leq i\leq n-1} \varphi(q,r,i+1)\frac{(1-q^{n-i})\cdots(1-q^n)}{(1-q)\cdots(1-q^{i+1})}.$$

This shows that, for fixed n, $\varphi(q,r,n)$ is a polynomial of degree $(n-1)$ in $Q=q^r$, with coefficient polynomials in q, and leading term Q^{n-1}. Since $\varphi(q,r,n)=0$ for $1\leq r\leq n-1$, this polynomial must be divisible by $(Q-q)$, $(Q-q^2)$, ..., $(Q-q^{n-1})$. It follows that

$$\varphi(q,r,n)=(Q-q)(Q-q^2)\cdots(Q-q^{n-1})$$

and the proposition is proved.

2.4. Let V be a vector space as in (2.2), with a fixed non-degenerate symplectic form $\langle\ ,\ \rangle$ defined over $\mathbb{F}_q$. We must have $n=2m$. The symplectic group $\mathrm{Sp}(V)$ is then a group as in (0.1.1). Let Y be the set of all complete isotropic flags $D_1\subset D_2\subset\cdots\subset D_m$ in V ($\dim D_i=i$) such that

$$D_1\neq FD_1\subset D_2,\quad D_2\neq FD_2\subset D_3,\quad \ldots,\quad D_{m-1}\neq FD_{m-1}\subset D_m,\quad D_m\neq FD_m.$$

Then Y can be identified with $X(w)$, where w is a Coxeter element in the Weyl group $\mathbf{W}$ of $\mathrm{Sp}(V)$; if we identify $\mathbf{W}$ with the group of all permutations σ of $-m,\ldots,-2,-1,1,2,\ldots,m$ such that $\sigma(-i)=-\sigma(i)$ for all i, then w is the permutation

$$\begin{aligned} -i &\longmapsto -i+1 \quad (2\leq i\leq m), &\qquad -1&\longmapsto m,\\ i&\longmapsto i-1 \quad (2\leq i\leq m), & 1&\longmapsto -m.\end{aligned}$$

On the other hand, Y (hence also $X(w)$) can be identified with the set of $x\in\mathbf{P}(V)$ such that

$$\langle x,F(x)\rangle=\langle x,F^2(x)\rangle=\cdots=\langle x,F^{m-1}(x)\rangle=0,\qquad \langle x,F^m(x)\rangle\neq 0.$$

For example, if $\dim V=4$, this is just the set of all $x\in\mathbf{P}(V)$ such that $\langle x,F(x)\rangle=0$, $\langle x,F^2(x)\rangle\neq 0$. Note that the equation $\langle x,F(x)\rangle=0$ defines a non-singular surface S in $\mathbf{P}(V)$ and that $\langle x,F^2(x)\rangle\neq 0$ means that we remove from S a union of rational curves, one for each isotropic plane in V, defined over $\mathbb{F}_q$. One can prove that in this case ($n=4$), the number of $\mathbb{F}_{q^r}$-rational points of $X(w)$ is given by:

$$q^{2r}-\frac{q}{2}(1+q)^2q^r+\frac{q}{2}(1-q)^2(-q)^r+q^4=\begin{cases}(q^r-q^2)^2, & r\text{ even},\\ (q^r-q)(q^r-q^3), & r\text{ odd}.\end{cases}$$

3. A fixed point formula

3.1. Let X be a scheme, separated and of finite type over k and let $\sigma: X\to X$ be an automorphism of finite order of X. We decompose σ as $\sigma=s\cdot u$ where s and u are powers of σ respectively of order prime to p and a power of p. The main result of this chapter is the following:

THEOREM 3.2. With the above notations (see also 0.5),

$$\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \mathrm{Tr}(u^*, H_c^*(X^s, \overline{\mathbb{Q}}_l)).$$

The first step is to prove that the left hand side is an integer independent of l. One might conjecture that for any endomorphism f of X and each i, $\mathrm{Tr}(f^*, H_c^i(X, \overline{\mathbb{Q}}_l))$ is integral and independent of l, but such a more general and precise result is known only for X proper and smooth (Katz and Messing, Inv. Math. 23 (1974), 73-77).

PROPOSITION 3.3. With the notations of 3.1, $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is an integer independent of l ($l \neq p$).

A standard specialization argument allows us to assume that $p > 1$ and that k is an algebraic closure of the prime field $\mathbb{F}_p$. This is anyway the only case we will need in the rest of the paper. The scheme X and σ can then be defined over some finite extension $\mathbb{F}_q \subset k$ of $\mathbb{F}_p$; we denote by $F : X \to X$ the corresponding Frobenius endomorphism.

Let us first assume that X is quasi-projective. Then, for $n \geq 1$, the composite $F^n \circ \sigma$ is the Frobenius map relative to some new way of lowering the field of definition of X from k to $\mathbb{F}_{q^n}$ and the Lefschetz fixed point formula for Frobenius ([5] and [11]) shows that $\mathrm{Tr}((F^n\sigma)^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is the number of fixed points of $F^n\sigma$ ($n \geq 1$). The automorphisms F^* and σ^* of the cohomology commute; as a function of n, $\mathrm{Tr}((F^n\sigma)^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is hence of the form $\sum_\lambda \alpha_\lambda \lambda^n$ where λ runs through the multiplicative group $\overline{\mathbb{Q}}_l^*$ of $\overline{\mathbb{Q}}_l$ and where α_λ is zero for all λ except for a finite number of them.

The functions $\mathbb{N}^+ \to \overline{\mathbb{Q}}_l^* : n \mapsto \lambda^n$ (for $\lambda \in \overline{\mathbb{Q}}_l^*$) are linearly independent. This can be checked either by using Vandermonde determinants or by appealing to Dedekind's theorem on the linear independence of characters (Bourbaki, Algèbre V, § 7, 5). For $n \geq 1$ the numbers $\sum_\lambda \alpha_\lambda \lambda^n = |X^{F^n\sigma}|$ are rational; hence for any automorphism τ of $\overline{\mathbb{Q}}_l$ one has

$$\sum_\lambda \tau(\alpha_\lambda)\tau(\lambda)^n = \sum_\lambda \tau(\alpha_{\tau^{-1}(\lambda)})\lambda^n = \sum_\lambda \alpha_\lambda \lambda^n \qquad (n \geq 1),$$

and, by the linear independence of the functions λ^n,

$$\alpha_{\tau(\lambda)} = \tau(\alpha_\lambda).$$

In particular, $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \sum_\lambda \alpha_\lambda$ is invariant by any automorphism of $\overline{\mathbb{Q}}_l$, hence rational. Similarly, as $\mathrm{Tr}((F^n\sigma)^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = |X^{F^n\sigma}|$ is independent of l, for any isomorphism $\tau : \overline{\mathbb{Q}}_l \to \overline{\mathbb{Q}}_{l'}$ one has

$$\tau\,\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_{l'})),$$

hence the rational number $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is independent of l.

Since the automorphism σ has finite order, $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l))$ is a sum of roots of unity, hence an algebraic integer. Being rational it is an ordinary integer.

If we do not assume X to be quasi-projective, we can either repeat the previous argument by working with algebraic spaces or reduce to the quasi-projective case: if (X_i) is a finite partition of X into locally closed quasi-projective subschemes stable under σ, then (cf. [5])

$$\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \sum_i \mathrm{Tr}(\sigma^*, H_c^*(X_i, \overline{\mathbb{Q}}_l)). \tag{3.3.1}$$

3.4. Proof of 3.2. Let (X_i) be a partition of X into locally closed subschemes stable under σ and such that, on each X_i, σ defines a free action of a cyclic group. By applying (3.3.1) to the decompositions $X = \bigcup X_i$ and $X^s = \bigcup X_i^s$, one reduces to the case where σ generates a free action of a cyclic group H. In this case, either

(a) σ is of order a power of p, hence $s = \mathrm{Id}$, $u = \sigma$ and (3.2) is trivial, or

(b) the order of σ is divisible by a prime number $l' \neq p$ and we must prove that $\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = 0$. As

$$\mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_l)) = \mathrm{Tr}(\sigma^*, H_c^*(X, \overline{\mathbb{Q}}_{l'})) \tag{3.3}$$

we may as well assume that $l = l'$. Let us now grant the

PROPOSITION 3.5. Let H be a finite group acting freely on X. Then

$$\chi(h) = \mathrm{Tr}(h^*, H_c^*(X, \overline{\mathbb{Q}}_l))$$

is the character of a virtual projective $\mathbb{Z}_l[H]$-module.

If H is abelian, and if H' is its largest subgroup of order prime to l, a representation of H over $\mathbb{Z}_l$ gives rise to a projective $\mathbb{Z}_l[H]$-module if and only if it is induced from a representation of H'. Its character vanishes then on $H - H'$ and we get the vanishing (b) by applying (3.5) to the cyclic group generated by σ.

3.6. Let A be a torsion ring with unit and let $\mathscr{F}$ be a sheaf of left A-modules on a scheme Y, with Y separated and of finite type over k. The A-modules $H_c^i(Y, \mathscr{F})$ are then the cohomology modules of a finer object $R\Gamma_c(Y, \mathscr{F})$ in the derived category $D^b(A)$ of the category of A-modules. The following is the key to the proof of 3.5.

PROPOSITION 3.7. Assume A to be right and left noetherian. If $\mathscr{F}$ is a constructible sheaf of projective A-modules, then $R\Gamma_c(Y, \mathscr{F})$ can be represented by a finite complex of projective A-modules of finite type.

We repeat the proof in [10, XVII (5.2.10)].

(a) The $H^i_c(Y, \mathscr{F})$ are A-modules of finite type, and vanish for $i > 2\dim(Y)$ ([10, XVII (5.2.8.1) and (5.3.6)]).

(b) For any right A-module of finite type N, one has

$$R\Gamma_c(Y, N \otimes_A \mathscr{F}) \simeq N \otimes^L_A R\Gamma_c(Y, \mathscr{F})$$

([10, XVII (5.2.9)]); the proof rests on replacing N by a free resolution of N, to reduce to the trivial case where N is free of finite type; we are allowed to use such left infinite resolutions because $R\Gamma_c$ is of finite cohomological dimension. The assumption that N is of finite type is in fact unnecessary.

(c) By (a) we can represent $R\Gamma_c(Y, \mathscr{F})$ by a complex of A-modules $K^\bullet$, with $K^i = 0$ for $i \notin [0, 2\dim Y]$ and K^i free of finite type for $i > 0$. For any right A-module of finite type N and any $i > 0$,

$$\mathrm{Tor}^A_i(N, K^0) = H^{-i}(N \otimes^L R\Gamma_c(Y, \mathscr{F})) =_{(b)} H^{-i}R\Gamma_c(Y, N \otimes \mathscr{F}) = 0.$$

The A-module K^0 is hence flat; it is of finite type because $H^0(K^\bullet)$ is, hence it is projective.

3.8. Proof of 3.5. We will assume that X is quasi-projective (the general case can be handled as in 3.3). Put $Y = X/H$ and denote by π the projection $\pi : X \to Y$. The group H acts on $\pi_*\mathbb{Z}/l^n$ and this action turns $\pi_*\mathbb{Z}/l^n$ into a locally constant sheaf of free $\mathbb{Z}/l^n[H]$-modules of rank one. By 3.7, $R\Gamma_c(Y, \pi_*\mathbb{Z}/l^n)$ can be represented by a complex $K^\bullet_n$ of projective $\mathbb{Z}/l^n[H]$-modules of finite type. One has

$$R\Gamma_c(Y, \pi_*\mathbb{Z}/l^n) \simeq R\Gamma_c(Y, \pi_*\mathbb{Z}/l^{n+1}) \otimes^L_{\mathbb{Z}/l^{n+1}} \mathbb{Z}/l^n$$

(3.7 (b)), and one checks easily ([11, XV, 3.3, Lemme 1]) that once $K^\bullet_n$ is chosen, one can choose $K^\bullet_{n+1}$ such that $K^\bullet_n$ is the reduction mod l^n of $K^\bullet_{n+1}$. Taking a projective limit, we get a complex $K^\bullet_\infty$ of projective $\mathbb{Z}_l[H]$-modules and an H-equivariant isomorphism

$$H^*_c(X, \mathbb{Z}_l) = \varprojlim H^*_c(X, \mathbb{Z}/l^n) = \varprojlim H^*_c(Y, \pi_*\mathbb{Z}/l^n) = H^*(K^\bullet_\infty)$$

(the middle equality because π is finite). We then have

$$\chi(h) = \sum_i (-1)^i \,\mathrm{Tr}(h, K^i_\infty).$$

3.9. Let X/k be as in (3.1), T a finite abelian group of order prime to p, G a finite group and ρ an action of $T \times G$ on X. We assume that the action of T on X is free. We let $T \times G$ act on $H^*_c(X)$ by transport of structure. For any character θ of T with values in $\overline{\mathbb{Q}}_l^*$, we denote by H^*_θ the subspace of $H^*_c(X)$ on which T acts by θ; on H^*_θ, t^* is multiplication by $\theta(t)^{-1}$.

Let us assume for simplicity that X is quasi-projective. We denote by π the projection $\pi : X \to Y = X/T$. The group G acts on Y. For each $g \in G$, we denote by $I(g)$ the set of connected components of the fixed point set Y^g

and by Y_i^g the component corresponding to $i \in I(g)$.

For $y \in Y$, $\pi^{-1}(y)$ is a principal homogeneous set for the action of the abelian group T. If $y \in Y^g$, g acts on $\pi^{-1}(y)$ and commutes with T, it hence acts like some $t(g, y) \in T$:

$$gx = t(g,x)x, \qquad x \in \pi^{-1}(y);$$

$t(g, y)$ is constant for y in each connected component Y_i^g of Y^g; we denote it by $t(g, i)$.

The Theorem 3.2 will be used in the following form.

COROLLARY 3.10. With the above notations, let $g = su$ be the decomposition of $g \in G$ as the product of commuting elements respectively of order prime to p and a power of p. Then,

$$\operatorname{Tr}(g^*, H_\theta^*) = \sum_{i\in I(s)} \theta(t(s,i)^{-1}) \operatorname{Tr}(u^*, H_c^*(Y_i^s)).$$

For $t \in T$, the decomposition 3.1 of $\sigma = t \cdot g$ is $tg = (st) \cdot u$. We have

$$X^{st} = \coprod_{i\in I(s)} \pi^{-1}(Y_i^s)^{st} = \coprod_{\substack{i\in I(s)\\ t=t(s,i)^{-1}}} \pi^{-1}(Y_i^s),$$

hence

$$\begin{aligned}\operatorname{Tr}(g^*, H_\theta^*) &= \frac{1}{|T|}\sum_t \theta(t)^{-1} \operatorname{Tr}(g^* t^{*-1}, H_c^*(X))\\ &= \frac{1}{|T|}\sum_t \theta(t) \operatorname{Tr}(g^* t^*, H_c^*(X))\\ &= \frac{1}{|T|}\sum_t \theta(t) \sum_{\substack{i\in I(s)\\ t=t(s,i)^{-1}}} \operatorname{Tr}(u^*, H_c^*(\pi^{-1}(Y_i^s)))\\ &= \frac{1}{|T|}\sum_{i\in I(s)} \theta(t(s,i)^{-1}) \operatorname{Tr}(u^*, H_c^*(\pi^{-1}(Y_i^s))).\end{aligned}$$

The decomposition (3.1) of the automorphism tu of $\pi^{-1}(Y_i^s)$ is $t \cdot u$. For $t \neq e$, t has no fixed point, hence

$$\operatorname{Tr}((tu)^*, H_c^*(\pi^{-1}(Y_i^s))) = 0 \qquad (t \neq e).$$

The endomorphism

$$\frac{1}{|T|}\sum_t t^* \quad \text{of} \quad H_c^*(\pi^{-1}(Y_i^s))$$

is a retraction onto the subspace $\pi^* H_c^*(Y_i^s)$, hence

$$\frac{1}{|T|} \operatorname{Tr}(u^*, H_c^*(\pi^{-1}(Y_i^s))) = \operatorname{Tr}(u^*, H_c^*(Y_i^s))$$

and, substituting into the expression for $\operatorname{Tr}(g^*, H_\theta^*)$, we get (3.10).

3.11. The end of this chapter will not be used in this paper. Let G be a finite group acting on X/k as in (3.1). For simplicity we assume X to be quasi-projective. Assume further that G acts freely on X, and put $Y = X/G$. The covering X of Y is said to be tame if Y can be imbedded as a Zariski open dense subset of a proper scheme $\overline{Y}$ in such a way that the p-Sylow-subgroups of G act freely on the normalisation $\overline{X}$ of $\overline{Y}$ in X:

$$\begin{array}{ccc} X & \stackrel{j}{\hookrightarrow} & \overline{X} \\ \downarrow & & \downarrow \\ Y & \hookrightarrow & \overline{Y}. \end{array}$$

PROPOSITION 3.12. With the above notations, if X/Y is tame, then the virtual representation $\sum_i (-1)^i H^i_c(X)$ of G is a multiple of the regular representation.

It suffices to prove that $\mathrm{Tr}(g^*, H^*_c(X)) = 0$ for $g \neq e$. If g is not of order a power of p, this follows from (3.2). If g is of order a power of p, g acts without fixed points on $\overline{X}$ and

$$\mathrm{Tr}(g^*, H^*_c(X)) = \mathrm{Tr}(g^*, H^*(\overline{X}, j_! \overline{\mathbb{Q}}_l)) = 0$$

by the Lefschetz fixed point theorem applied to $\overline{X}$ and the sheaf $j_! \overline{\mathbb{Q}}_l$.

3.13. Historical remark. The method we have followed, using (3.5), was first used by Zarelua (On finite groups of transformations, Proc. Int. Symp. on Topology and its Applications, 1968, pp. 334-339) and independently by J. L. Verdier to prove that if a finite group G acts freely on a topological space X of finite cohomological dimension, and if the $H^i(X, \mathbb{Z})$ are of finite type, then $\sum_i (-1)^i H^i(X, \mathbb{Z}) \otimes \mathbb{Q}$ is a multiple of the regular representation of G.

4. The character formula

The assumptions (0.1.1) are in force in this chapter and the next four.

DEFINITION 4.1. Let T be an F-stable maximal torus of G. The Green function $Q_{T,G}(u)$ is the restriction to the unipotent elements of the character of the virtual representation $R^1_{T \subset B}$, where B is any Borel subgroup containing T.

This Green function does not depend on B (by (1.6) and (1.17 (i))); it only depends on the G^F-conjugacy classes of T and u. The natural map $x \mapsto \overline{x}$ from G to its adjoint group induces a bijection on the unipotent sets, and

$$Q_{T,G}(u) = Q_{\overline{T}, G^{\mathrm{ad}}}(\overline{u}). \tag{4.1.1}$$

The Green function is integer-valued (3.3) and is a restriction of a character of G^F, hence $Q_{T,G}(u) = Q_{T,G}(u^n)$ if $(n, p) = 1$.

Let α be the smallest integer ≥ 1 such that F^α is the identity on $\mathbf{W}$. From the proof of (3.3) we see that $\sum_{n\geq 1} X_{T\subset B}^{F^{\alpha n}u} t^n$ is a rational function of t and that

$$Q_{T,G}(u) = -\left\{\sum_{n\geq 1} X_{T\subset B}^{F^{\alpha n}u} t^n\right\}_{t=\infty}. \tag{4.1.2}$$

In this chapter, we will express the character of $R_{T\subset B}^\theta$ in terms of θ and of Green functions.

THEOREM 4.2. Let $x = su$ be the Jordan decomposition of $x \in G^F$. Then

$$\mathrm{Tr}(x, R_{T\subset B}^\theta) = \frac{1}{|Z^0(s)^F|} \sum_{\substack{g\in G^F \\ \mathrm{ad}\, gT\subset Z^0(s)}} Q_{\mathrm{ad}\, gT, Z^0(s)}(u)\, \mathrm{ad}\, g(\theta)(s).$$

Let ε_x be the function on G^F whose value is 1 at $x \in G^F$ and 0 elsewhere. One can rewrite 4.2 as giving the following formula for the character of $R_{T\subset B}^\theta$:

$$\mathrm{Tr}(\ , R_{T\subset B}^\theta) = \sum_{t\in T^F} \frac{1}{|Z^0(t)^F|} \theta(t) \sum_{\substack{u\in Z^0(t)^F \\ g\in G^F}} Q_{T,Z^0(t)}(u) \varepsilon_{\mathrm{ad}\, g(tu)} \tag{4.2.1}$$

where we put $Q_{T,Z^0(t)}(u) = 0$ whenever u is not unipotent.

COROLLARY 4.3. $R_{T\subset B}^\theta$ is independent of the choice of B ($B \supset T$).

From now on we shall write R_T^θ for $R_{T\subset B}^\theta$. We will deduce 4.2 from 3.7 and the following geometrical facts.

PROPOSITION 4.4. (i) If a Borel subgroup B_1 of G contains s, then $B_1 \cap Z^0(s)$ is a Borel subgroup of $Z^0(s)$.

(ii) Pick $T \subset B$ in G. Any Borel subgroup B_1 such that $s \in B_1$ is of the form $\mathrm{ad}\, gB$ with $\mathrm{ad}\, gT \subset Z^0(s)$ (i.e., $g^{-1}sg \in T$). The left coset

$$\tau_{T,B}(B_1) \overset{\text{def.}}{=} Z^0(s) \cdot g$$

depends only on T, B, and B_1.

(iii) The map $B_1' \mapsto B_1' \cap Z^0(s)$ is an isomorphism from the space of Borel subgroups containing s, such that $\tau_{T,B}(B_1') = \tau_{T,B}(B_1)$ and the space of Borel subgroups of $Z^0(s)$.

Proof. (i) is well known. Put $B_1 = \mathrm{ad}\, g_1 B$. We have $g_1^{-1}sg_1 \in B$ hence there exists u in the unipotent radical of B such that $u^{-1}g_1^{-1}sg_1u \in T$. We take $g = g_1u$. Put $T' = \mathrm{ad}\, gT$. If $g' = hg$ is also such that $\mathrm{ad}\, g'B = B_1$ (resp. $\mathrm{ad}\, g'T \subset Z^0(s)$) then $h \in B_1$ (resp. $h \in Z^0(s)N(T')$, by the conjugacy of maximal tori in $Z^0(s)$). If $W = N(T')/T'$ is the Weyl group of G and $W^0 = (N(T')\cap Z^0(s))/T'$ that of $Z^0(s)$, the Bruhat decomposition for $Z^0(s)$ reads

$$Z^0(s) = (B_1 \cap Z^0(s))W^0(B_1 \cap Z^0(s)),$$

and

$$(B_1Z^0(s)) \cap N(T') = B_1W^0(B_1 \cap Z^0(s)) \cap N(T') \subset B_1W^0B_1 \cap N(T') = W^0(T') \subset Z^0(s).$$

Hence

$$B_1 \cap (Z^0(s) \cdot N(T')) \subset Z^0(s)$$

and

$$B_1 \cap (Z^0(s) \cdot N(T')) = B_1 \cap Z^0(s).$$

The element g such that $\mathrm{ad}\, gB = B_1$ and $\mathrm{ad}\, gT \subset Z^0(s)$ is unique modulo $B_1 \cap Z^0(s)$, hence (ii) and (iii).

A $Z^0(s)$-left coset $\tau = Z^0(s)g$, with $\mathrm{ad}\, gT \subset Z^0(s)$, defines an isomorphism $\overline{\tau}$ from the maximal torus of G to that of $Z^0(s)$; $\mathrm{ad}\, g$ maps T (contained in the Borel subgroup B) to $\mathrm{ad}\, gT \subset Z^0(s)$ (contained in the Borel subgroup $\mathrm{ad}\, gB \cap Z^0(s)$). This construction will allow us to compare the relative position of Borel subgroups in G and $Z^0(s)$.

PROPOSITION 4.5. Let B_1' and B_1'' be Borel subgroups containing s,

$$\tau' = \tau_{T,B}(B_1'), \qquad \tau'' = \tau_{T,B}(B_1'');$$

let w be the relative position of B_1' and B_1'' (in G) and w_1 that of $B_1' \cap Z^0(s)$ and $B_1'' \cap Z^0(s)$ (in $Z^0(s)$). Then

$$w_1 = \overline{\tau}' w \overline{\tau}''^{-1}.$$

Replacing (T, B) by a conjugate, we may assume that $B = B_1'$ and that $T \subset B_1' \cap Z^0(s) \cap B_1''$. We will identify the maximal torus of G and that of $Z^0(s)$ with T, using $T \subset B$ and $T \subset (B \cap Z^0(s))$. We then have $\overline{\tau}' = \mathrm{Id}$. Replacing B_1'' by an $N(T) \cap Z^0(s)$-conjugate, we may further assume that $w_1 = 1$, i.e., $B_1' \cap Z^0(s) = B_1'' \cap Z^0(s)$. We must prove that $w = \overline{\tau}''$, which is clear.

LEMMA 4.6. If B and B', containing T, are in the same relative position as B_1 and B_1' containing s, then $\tau_{T,B}(B_1) = \tau_{T,B'}(B_1')$.

Replacing (B, T, B') by a conjugate, we may again assume that $B = B_1$ and that $T \subset B_1 \cap Z^0(s) \cap B_1'$. In this case $B' = B_1'$.

To prove (4.2) we first compute the space $X^s_{T\subset B}$ of Borel subgroups B_1 containing s and such that B_1 and FB_1 are in the same relative position as B and FB.

PROPOSITION 4.7. Let $X^s_{T\subset B}(g)$ be the subspace of $X^s_{T\subset B}$ consisting of those B_1 with $\tau_{T,B}(B_1) = Z^0(s)g$. Then

$$X^s_{T\subset B} = \coprod_{\substack{g \in Z^0(s)^F \backslash G^F \\ \mathrm{ad}\, gT \subset Z^0(s)}} X^s_{T\subset B}(g) \tag{4.7.1}$$

and, for $g \in G^F$, the map $B_1 \mapsto B_1 \cap Z^0(s)$ induces an isomorphism

$$X^s_{T\subset B}(g) \xrightarrow{\sim} X_{\mathrm{ad}\, gT\subset \mathrm{ad}\, gB\cap Z^0(s)}. \tag{4.7.2}$$

By 4.6, we have $F\tau_{T,B}(B_1) = \tau_{T,FB}(FB_1) = \tau_{T,B}(B_1)$. By Lang's Theorem, the $Z^0(s)$-principal homogeneous space $\tau_{T,B}(B_1)$, being F-stable, has a rational point, hence (4.7.1). By (4.4 (iii)) and (4.5), the map (4.7.2) is an isomorphism from $X^s_{T\subset B}(g)$ to some $X(w)$ (relative to $Z^0(s)$); to check which w appears, we observe that $\mathrm{ad}\, gB \in X^s_{T\subset B}(g)$.

Proof of 4.2. The theorem is now an immediate application of (4.7) and (3.10) applied to the action (1.17) of $G^F \times T^F$ on $\widetilde{X}_{T\subset B}$.

5. Characters of tori

We will assume chosen isomorphisms

$$k^* \xrightarrow{\sim} (\mathbb{Q}/\mathbb{Z})_{p'}, \tag{5.0.1}$$

and

$$(\text{roots of unity of order prime to } p \text{ in } \overline{\mathbb{Q}}_l^*) \xrightarrow{\sim} (\mathbb{Q}/\mathbb{Z})_{p'}. \tag{5.0.2}$$

5.1. Let T be a torus over k. Besides the character group $X(T) = \mathrm{Hom}(T, \mathbb{G}_m)$, we will consider its dual $Y(T) = \mathrm{Hom}(\mathbb{G}_m, T)$. The duality is given by

$$\langle \alpha, h\rangle = \alpha \circ h \in \mathrm{Hom}(\mathbb{G}_m, \mathbb{G}_m) = \mathbb{Z}$$

($\alpha \in X(T)$ and $h \in Y(T)$). The maps $(h, x) \mapsto h(x)$ and $t \mapsto (\alpha \mapsto \alpha(t))$ induce isomorphisms

$$Y(T) \otimes k^* = T(k) = \mathrm{Hom}(X(T), k^*). \tag{5.1.1}$$

If T is a maximal torus in G, the roots of T in G are elements of $X(T)$, while the coroots belong to $Y(T)$. If α is a root, the corresponding coroot H_α is characterised as follows: there is a homomorphism $u : \mathrm{SL}(2) \to G$, which maps the subgroup $\begin{pmatrix} 1 & * \\ 0 & 1 \end{pmatrix}$ onto the root subgroup U_α, and whose restriction to the group of diagonal matrices (identified with $\mathbb{G}_m$ by $x \mapsto \begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix}$) is H_α.

5.2. Using the chosen isomorphism (5.0.1), we can rewrite (5.1.1) as

$$T(k) = Y(T) \otimes (\mathbb{Q}/\mathbb{Z})_{p'} \simeq (Y(T) \otimes \mathbb{Q}/Y(T))_{p'}. \tag{5.2.1}$$

If T is obtained by extension of scalars from a torus $T_0/\mathbb{F}_q$, the Frobenius map F induces a map $F : Y(T) \to Y(T)$ (Y is a covariant functor) and T^F is the subgroup $\mathrm{Ker}(F-1)$ of $T(k) = Y(T) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$; since F is divisible by p, it is also the kernel of $F-1$ in $Y(T) \otimes \mathbb{Q}/\mathbb{Z}$:

$$0 \longrightarrow T^F \longrightarrow Y(T)\otimes \mathbb{Q}/\mathbb{Z} \xrightarrow{F-1} Y(T)\otimes \mathbb{Q}/\mathbb{Z} \longrightarrow 0. \tag{5.2.2}$$

By applying the Snake Lemma to

$$\begin{array}{ccccccccc}
0 & \longrightarrow & Y(T) & \longrightarrow & Y(T)\otimes\mathbb{Q} & \longrightarrow & Y(T)\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0 \\
 & & {\scriptstyle F-1}\downarrow & & {\scriptstyle F-1}\downarrow & & {\scriptstyle F-1}\downarrow & & \\
0 & \longrightarrow & Y(T) & \longrightarrow & Y(T)\otimes\mathbb{Q} & \longrightarrow & Y(T)\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0,
\end{array}$$

we get another exact sequence

$$0 \longrightarrow Y(T) \xrightarrow{F-1} Y(T) \longrightarrow T^F \longrightarrow 0. \tag{5.2.3}$$

The maps in (5.2.2), (5.2.3) depend on the choice (5.0.1). An intrinsic expression for these sequences would be as follows: T^F is the image of the middle map in the exact sequence

$$0 \longrightarrow Y(T)_{\widehat{p'}}(1) \xrightarrow{F-1} Y(T)_{\widehat{p'}}(1) \xrightarrow{(F-1)^{-1}} Y(T)_{\widehat{p'}}(1)\otimes\mathbb{Q}/\mathbb{Z} \xrightarrow{F-1} Y(T)_{\widehat{p'}}(1)\otimes\mathbb{Q}/\mathbb{Z} \longrightarrow 0.$$

Let us map the sequences (5.2.2), (5.2.3) into $\mathbb{Q}/\mathbb{Z}$. The isomorphism (5.0.2) provides an isomorphism

$$(T^F)^{\sim} \overset{\text{def}}{=} \mathrm{Hom}(T^F, \overline{\mathbb{Q}}_l^*) = \mathrm{Hom}(T^F, \mathbb{Q}/\mathbb{Z}),$$

and we get exact sequences

$${}^*0 \longrightarrow X(T) \xrightarrow{F-1} X(T) \longrightarrow (T^F)^{\sim} \longrightarrow 0, \tag{5.2.2}$$

$${}^*0 \longrightarrow (T^F)^{\sim} \longrightarrow X(T)\otimes\mathbb{Q}/\mathbb{Z} \xrightarrow{F-1} X(T)\otimes\mathbb{Q}/\mathbb{Z} \longrightarrow 0. \tag{5.2.3}$$

The dual torus T^* of T is defined by the rule $X(T^*) = Y(T)$, (hence $Y(T^*) = X(T)$); its $\mathbb{F}_q$-structure is defined by (F on $Y(T^*)$) = (${}^tF$ on $X(T)$). Via some isomorphism

$$(T^F)^{\sim} = T^{*F}, \tag{5.2.4}$$

the sequences (5.2.3), (5.2.2) for T^* are identical with (5.2.2)*, (5.2.3)*.

5.3. If we go from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$, F is replaced by F^n. Composition with the norm map

$$N = \frac{F^n-1}{F-1} = \sum_0^{n-1} F^i \,:\, T^{F^n} \longrightarrow T^F$$

is an injection ${}^tN \,:\, (T^F)^{\sim} \to (T^{F^n})^{\sim}$. We have commutative diagrams

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & Y & \xrightarrow{F^n-1} & Y & \longrightarrow & T^{F^n} & \longrightarrow & Y\otimes\mathbb{Q}/\mathbb{Z} & \xrightarrow{F^n-1} & Y\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0\\
 & & {\scriptstyle N}\downarrow & & \| & & {\scriptstyle N}\downarrow & & {\scriptstyle N=\sum_0^{n-1}F^i}\downarrow & & \| & & \\
0 & \longrightarrow & Y & \longrightarrow & Y & \longrightarrow & T^F & \longrightarrow & Y\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & Y\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0,
\end{array}$$

$$\begin{array}{ccccccccccc}
0 & \longrightarrow & X & \longrightarrow & X & \longrightarrow & (T^F)^{\sim} & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0\\
 & & \| & & {\scriptstyle {}^tN=\sum_0^{n-1}{}^tF^i}\downarrow & & {\scriptstyle {}^tN}\downarrow & & \| & & {\scriptstyle {}^tN}\downarrow & & \\
0 & \longrightarrow & X & \longrightarrow & X & \longrightarrow & (T^{F^n})^{\sim} & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & X\otimes\mathbb{Q}/\mathbb{Z} & \longrightarrow & 0.
\end{array}$$

where X and Y stand for $X(T)$ and $Y(T)$. In particular, for θ a character of T^F, θ and $\theta\circ N$ have the same image in $X(T)\otimes\mathbb{Q}/\mathbb{Z}$, and (5.2.4) gives rise to a commutative diagram

$$\begin{array}{ccc}
(T^F)^{\sim} & \simeq & (T^*)^F\\
{\scriptstyle {}^tN}\downarrow & & {\scriptstyle N}\downarrow\\
(T^{F^n})^{\sim} & \simeq & (T^*)^{F^n}.
\end{array}$$

PROPOSITION 5.4. Let T and T' be two F-stable maximal tori of G, and let θ,θ' be characters of $T^F,(T')^F$. We identify them with characters of $Y(T)$ and $Y(T')$, by (5.2.3). The following conditions are equivalent:

(i) For some $x\in G$, with $\operatorname{ad}x(T)=T'$, the map induced by $\operatorname{ad}x: Y(T)\to Y(T')$ carries θ to θ';

(ii) For some n, the pairs $(T,\theta\circ N)$, $(T',\theta'\circ N)$, where N is the norm from T^{F^n} to T^F (resp. $(T')^{F^n}$ to $(T')^F$) are G^{F^n}-conjugate.

By 5.3, the condition (i) is invariant under the replacement of F by F^n and of θ,θ' by $\theta\circ N,\theta'\circ N$. It hence suffices to check (5.4) in the trivial case where T and T' are split.

DEFINITION 5.5. The pairs (T,θ), (T',θ') are said to be geometrically conjugate when the equivalent conditions of (5.4) hold.

5.6. Let (T,θ) be as above. The choice of a Borel subgroup B containing T defines an isomorphism of T with the torus $\mathbf{T}$; the corresponding isomorphism $Y(T)\to Y(\mathbf{T})$ carries θ to a character of $Y(\mathbf{T})$ with values in the roots of unity of order prime to p in $\overline{\mathbb{Q}}_l$. The isomorphism (5.0.2) identifies it with an element of $X(\mathbf{T})\otimes(\mathbb{Q}/\mathbb{Z})_{p'}$. Its class $[\theta]$ in $(X(\mathbf{T})\otimes(\mathbb{Q}/\mathbb{Z})_{p'})/\mathbf{W}$ is independent of the choice of B. It is invariant under F. Let us put

$$\mathcal{S}\overset{\text{def}}{=}[(X(\mathbf{T})\otimes(\mathbb{Q}/\mathbb{Z})_{p'})/\mathbf{W}]^F\xrightarrow{\sim}[(X(\mathbf{T})\otimes\mathbb{Q}/\mathbb{Z})/\mathbf{W}]^F.$$

PROPOSITION 5.7. (i) The map $\theta\mapsto[\theta]$ induces a bijection from the set of geometric conjugacy classes of pairs (T,θ) to $\mathcal{S}$.

(ii) (Compare Steinberg [15, p. 93]). The number of geometric conjugacy classes of pairs (T, θ) is $|Z^{0F}|q^l$ where Z^0 is the identity component of the centre of G and l is the semisimple rank of G.

(i) The injectivity is clear on 5.4(i). Let us prove surjectivity. If the class mod $\mathbf{W}$ of $x \in X(\mathbf{T}) \otimes (\mathbb{Q}/\mathbb{Z})_{p'}$ is invariant by F, one indeed has ${}^t(wF)x = x$ for some $w \in \mathbf{W}$, and x corresponds to a character of $\mathbf{T}(w)^F$, hence of T^F, for some F-stable maximal torus T in G (1.14).

(ii) As in (5.2), we may now identify geometric conjugacy classes of pairs (T, θ) with $\mathbb{F}_q$-rational points of the quotient $\mathbf{T}^*/\mathbf{W}$. Our task is to compute

$$|(\mathbf{T}^*/\mathbf{W})^F| = \sum_i (-1)^i \operatorname{Tr}(F^*, H_c^i(\mathbf{T}^*/\mathbf{W})) = \sum_i (-1)^i \operatorname{Tr}(F^*, H_c^i(\mathbf{T}^*)^{\mathbf{W}}).$$

Let G' be the derived group of G; the torus $\mathbf{T}^*$ is isogenous to (hence has the same cohomology as) the product of Z^{0*} with $\mathbf{T'}^*$ ($\mathbf{T'}$ being the torus of G'). Since any torus has as many rational points as its dual, we get by Künneth's Theorem

$$|(\mathbf{T}^*/\mathbf{W})^F| = |Z^{0F}| \sum_i (-1)^i \operatorname{Tr}(F^*, H_c^i(\mathbf{T'}^*)^{\mathbf{W}}).$$

We have

$$H^*(\mathbf{T'}^*, \overline{\mathbb{Q}}_l) = \Lambda^* H^1(\mathbf{T'}^*, \overline{\mathbb{Q}}_l) = \Lambda^*(Y(\mathbf{T'}) \otimes \overline{\mathbb{Q}}_l(-1));$$

hence by Poincaré duality,

$$H_c^{2l-i}(\mathbf{T'}^*, \overline{\mathbb{Q}}_l) = \operatorname{Hom}(\Lambda^i Y(\mathbf{T'}), \overline{\mathbb{Q}}_l(i-l)),$$

and

$$\operatorname{Tr}(F^*, H_c^{2l-i}(\mathbf{T'}^*)^{\mathbf{W}}) = q^{l-i} \operatorname{Tr}(F, \Lambda^i Y(\mathbf{T'})^{\mathbf{W}}).$$

The next lemma shows that this trace is zero for $i \neq 0$, hence the q^l factor.

LEMMA 5.8. Let $W \subset \operatorname{Aut}(V)$ be a finite group generated by reflections of a real vector space V of dimension d. We assume that $V^W = 0$. Then $(\Lambda^i V)^W = 0$ for $i \neq 0$.

For a proof, see Bourbaki [1, Ch. V, Ex. 3 of § 2].

5.9. Let T be an F-stable maximal torus in G and let α be a root of T. For θ a character of T^F, we will say that θ is orthogonal to H_α: $\langle H_\alpha, \theta \rangle = 0$ if θ, viewed as a character of $Y(T)$ (5.2.3), is identically 1 on H_α. The character θ of $Y(T)$ is then invariant by the corresponding reflection s_α.

5.10. Let $\pi : \widetilde{G} \to G$ be the simply connected covering of the derived group of G, and let $\widetilde{T}$ be the inverse image of T in $\widetilde{G}$. The group $\widetilde{T}$ is connected: it is a maximal torus in $\widetilde{G}$. It acts on $T \times \widetilde{G}$ by

$$\tilde{t} * (t, g) = (t\pi(\tilde{t})^{-1}, \tilde{t}g),$$

and the map $(t, g) \mapsto tg$ induces an isomorphism $\widetilde{T}\backslash(T \times \widetilde{G}) \xrightarrow{\sim} G$, hence (Lang's Theorem) $\widetilde{T}^F\backslash(T^F \times \widetilde{G}^F) \xrightarrow{\sim} G^F$. We have

$$T^F/\pi(\widetilde{T}^F) \xrightarrow{\sim} G^F/\pi^F(\widetilde{G}). \tag{5.10.1}$$

PROPOSITION 5.11. (i) A character of T^F is the restriction to T^F of a character of $G^F/\pi(\widetilde{G}^F)$ if and only if it is orthogonal to all coroots.

(ii) Let θ be a character of $G^F/\pi(\widetilde{G}^F)$, T and T' two F-stable maximal tori and let $x \in G$ be such that $\operatorname{ad} x(T) = T'$. Then $\operatorname{ad} x : Y(T) \to Y(T')$ carries the restriction of θ to T^F (viewed as a character of $Y(T)$) onto the restriction of θ to $(T')^F$: the restrictions of θ to the F-stable maximal tori are all geometrically conjugate.

The assertion (i) follows from (5.10.1), and the fact that $Y(\widetilde{T}) \subset Y(T)$ is spanned by the coroots H_α. To prove (ii), we will use the criterion 5.4 (ii). Let n be such that $x \in G^{F^n}$. We will prove that if $y \in T^{F^n}$, $N(y)$ and $N(\operatorname{ad} x(y))$ have the same image in $G^F/\pi(\widetilde{G}^F)$, hence that $\operatorname{ad} x$ transforms $(\theta \mid T^F) \circ N$ into $(\theta \mid (T')^F) \circ N$.

The map

$$t \longmapsto N(t)^{-1}N(\operatorname{ad} x(t)) : T \longrightarrow G$$

lifts uniquely into a map $\varphi : T \to \widetilde{G}$ mapping e to e. Indeed,

(a) it factors through T/Z;

(b) the map $t \mapsto N(t)^{-1}N(\operatorname{ad} x(t)) : \widetilde{T} \to \widetilde{G}$ factors through $\widetilde{T}/\widetilde{Z} = T/Z$ and the desired lifting is the composite $T \to T/Z = \widetilde{T}/\widetilde{Z} \to \widetilde{G}$.

Similarly, the map

$$(a, t) \longmapsto a^{-1}N(t)^{-1}N(\operatorname{ad} x(t)) \operatorname{ad} x(a) : T \times T \longrightarrow G$$

lifts into $\psi : T \times T \to \widetilde{G}$, with $\psi(e, t) = \varphi(t)$. The identity

$$\begin{aligned} F(N(t)^{-1}N(\operatorname{ad} x(t))) &= N(Ft)^{-1}N(F \operatorname{ad} x(t)) \\ &= (t^{-1}F^n t)^{-1}N(t)^{-1}N(\operatorname{ad} x(t)) \operatorname{ad} x(t^{-1})F^n \operatorname{ad} x(t) \\ &= (t^{-1}F^n t)^{-1}N(t)^{-1}N(\operatorname{ad} x(t)) \operatorname{ad} x(t^{-1}F^n t) \end{aligned}$$

lifts into

$$F\varphi(t) = \psi(t^{-1}F^n t, \varphi(t)).$$

Putting $t = y$, we have $y^{-1}F^n y = e$, hence $F\varphi(y) = \varphi(y)$ and $\varphi(y) \in \widetilde{G}^F$. We have $N(y)^{-1}N(\operatorname{ad} x(y)) \in \pi(\widetilde{G}^F)$, as required.

Remark 5.12. The argument above can be used to show the existence of a norm map

$$N : G^{F^n}/\pi(\widetilde{G}^{F^n}) \longrightarrow G^F/\pi(\widetilde{G}^F)$$

which for each F-stable maximal torus T gives rise to a commutative diagram

$$\begin{array}{ccc} T^{F^n}/\pi(\widetilde{T}^{F^n}) & \xrightarrow{\sim} & G^{F^n}/\pi(\widetilde{G}^{F^n}) \\ {\scriptstyle N}\downarrow & & {\scriptstyle N}\downarrow \\ T^F/\pi(\widetilde{T}^F) & \xrightarrow{\sim} & G^F/\pi(\widetilde{G}^F). \end{array}$$

THEOREM 5.13. Let (T,θ) be as above. We use (5.2.3) to identify θ with a character of $Y(T)$. If the centre of G is connected, then the stabilizer of θ in the Weyl group $W = N(T)/T$ is generated by the reflections s_α, for H_α a coroot orthogonal to θ.

The key point in the proof is the same as in Steinberg's Theorem that if the derived group G' is simply connected, then the centralizer of any semisimple element is connected.

Let us identify θ with the corresponding element in $X(T)\otimes(\mathbb{Q}/\mathbb{Z})_{p'}$ (invariant under F). Let $\theta_1 \in X(T)\otimes\mathbb{Q}$ be a representative for it. This θ_1 has no p in the denominator. We have the dictionary:

(a) $\langle H_\alpha, \theta\rangle = 0 \iff \langle H_\alpha, \theta_1\rangle \in \mathbb{Z}$;

(b) θ is fixed by $w \in W \iff$ for some $x \in X(T)$, $w\theta_1 + x = \theta_1$.

Let X^{ad} be the subgroup of X generated by the roots. It is the character group of the image T^{ad} of T in the adjoint group. The character group of the centre $Z = \mathrm{Ker}(T \to T^{\mathrm{ad}})$ of G is X/X^{ad}; the character group of Z/Z^0_{red} is the torsion subgroup of X/X^{ad}, hence

(c) Z is connected and smooth (resp. connected) if and only if X/X^{ad} is torsion free (resp. has no torsion prime to p).

For each root α, $s_\alpha(\theta_1) = \theta_1 - \langle H_\alpha, \theta_1\rangle\alpha$. The number $\langle H_\alpha, \theta_1\rangle$ has no p in the denominator; hence $\langle H_\alpha, \theta_1\rangle\alpha \in X(T)$ if and only if $\langle H_\alpha, \theta_1\rangle \in \mathbb{Z}$; $s_\alpha(\theta) = \theta$ if and only if $\langle H_\alpha, \theta\rangle = 0$. It remains to check that the stabilizer of θ is generated by the reflections it contains.

Fix N such that $(1/p^N)X^{\mathrm{ad}} \supset X \cap (X^{\mathrm{ad}} \otimes \mathbb{Q})$ and such that $p^N \equiv 1$ (mod the order of θ). For any $w \in W$, $w\theta_1 - \theta_1$ is in $X^{\mathrm{ad}} \otimes \mathbb{Q}$, hence if $w\theta_1 - \theta_1$ is in X, $w(p^N\theta_1) - (p^N\theta_1) \in X^{\mathrm{ad}}$. Replacing θ_1 by $p^N\theta_1$, which is also a representative of θ, we may assume that θ is fixed by $w \in W$ if and only if $w\theta_1 + x = \theta_1$ for some $x \in X^{\mathrm{ad}}$. The stabilizer of θ in W is the image in W of the stabilizer of θ_1 in the affine Weyl group. It remains to apply Bourbaki [1, Ch. VI, Ex. 1 of § 2].

Remark 5.14. The proof shows that, if we use a suitable Borel subgroup $B \supset T$ to identify W with $\mathbf{W}$, the stabilizer of θ becomes the subgroup of $\mathbf{W}$ generated by some of the reflections corresponding either to a simple root or to the negative of the highest coroot.

DEFINITION 5.15. (i) The character θ of T^F is nonsingular if it is not orthogonal to any coroot.

(ii) θ is in general position if it is not kept fixed by any non-trivial element of $(N(T)/T)^F$.

PROPOSITION 5.16. If the centre of G is connected, a character is non-singular if and only if it is in general position.

The stabilizer of θ in $N(T)/T$ is a group generated by reflections, and is stable by Frobenius. We have to prove that the subgroup fixed by Frobenius is non-trivial. This follows from the next lemma.

LEMMA 5.17. Let $V \neq \{0\}$ be a euclidian vector space, and let W be a finite group generated by orthogonal reflections. We assume that $V^W = \{0\}$. If a belongs to the normalizer A of W in $O(V)$, the centralizer W^a of a in W is non-trivial.

(a) Reduction to the irreducible case. Let $V = \bigoplus V_i$ be the decomposition of V as a direct sum of irreducible root systems; we have $W = \prod W_i$. The automorphism a permutes the V_i; by taking a direct factor, we may assume that it permutes them cyclically: $V = \bigoplus_{i \in \mathbb{Z}/n} V_i$ and $aV_i = V_{i+1}$. An element $w = (w_i) \in W = \prod W_i$ is in W^a if and only if $w_i = \operatorname{ad} a^i(w_0)$ and w_0 is fixed by a^n; we are reduced to proving that $W_0^{a^n}$ is non-trivial.

(b) In the irreducible case, one always has either $A = W \cup -W$ or $-1 \in W$; both cases are clear.

COROLLARY 5.18. For any G, if θ is in general position, then θ is non-singular.

Let us embed G in a group G_1 with connected centre and the same derived group. For instance, one can take $G_1 = G \times T/\{(z, z^{-1}) \mid z \in Z(G)\}$. The torus T is then contained in an F-stable maximal torus T_1 of G_1, and θ is the restriction to T^F of some character θ_1 of T_1^F. If θ is in general position, θ_1 is so a fortiori, hence non-singular (5.16). A character θ_1 of T_1^F is non-singular if and only if its restriction to T^F is, hence (5.18).

When all roots of G have the same length, geometric conjugacy can be given the following convenient description.

DEFINITION 5.19. (Assuming all roots to be of the same length) Let T be an F-stable maximal torus and let θ be a character of T^F. The connected centralizer $S(T, \theta)$ of (T, θ) in G is the F-stable reductive subgroup of G with maximal torus T and with root subgroups relative to T the U_α for which $\langle H_\alpha, \theta \rangle = 0$.

If $\langle H_\alpha, \theta \rangle = \langle H_\beta, \theta \rangle = 0$ and $\alpha + \beta$ is a root, then $H_{\alpha+\beta} = H_\alpha + H_\beta$, hence $\langle H_{\alpha+\beta}, \theta\rangle = 0$ and the definition makes sense. Let $\pi : \widetilde{S(T,\theta)} \to S(T,\theta)$ be the simply connected covering of the derived subgroup of $S(T,\theta)$. By 5.11 (i), θ extends to a character θ_S of $S(T,\theta)^F/\pi(\widetilde{S(T,\theta)}^F)$.

PROPOSITION 5.20. If the centre of G is connected and all roots are of the same length, then a pair (T',θ') is geometrically conjugate to (T,θ) if and only if $(S(T,\theta),\theta_S)$ and $(S(T',\theta'),\theta'_{S'})$ are G^F-conjugate.

The "if" part follows from 5.11. Conversely, let (T',θ') be geometrically conjugate to (T,θ): for some $x \in G$, $\operatorname{ad} x(T) = T'$ and $\operatorname{ad} x : Y(T) \to Y(T')$ maps θ to θ' (via (5.2.3)). If $F' : T' \to T'$ is the Frobenius map of T', $\operatorname{ad} x^{-1}(F') = \operatorname{ad} wF$ for some w in the Weyl group of T. We have ${}^tF\theta = \theta$, and ${}^t(\operatorname{ad} wF)\theta = \theta$, hence θ is fixed by w, which by part (i) belongs to the Weyl group of T in $S(T,\theta)$. Let $T'' = \operatorname{ad} y(T)$, $y \in S(T,\theta)$, be an F-stable torus in $S(T,\theta)$, with Frobenius F'', such that $(\operatorname{ad} y)^{-1}(F'') = (\operatorname{ad} x)^{-1}(F')$. Replacing (T,θ) by $(T'', \operatorname{ad} y(\theta))$ (by applying 5.11) and replacing x by xy^{-1}, we are reduced to the case where $\operatorname{ad} xF = F' \operatorname{ad} x$. In this case, (T,θ) and (T',θ') are G^F-conjugate; the identity $\operatorname{ad} x(Ft) = F' \operatorname{ad} x(t)$ for $t \in T$ amounts to $x^{-1}Fx \in T$; and replacing x by xt^{-1} with $t \in T$, $t^{-1}Ft = x^{-1}Fx$, makes x rational.

When roots are not all of the same length, one has to work in the dual group G^* of G.

DEFINITION 5.21. A group G^* dual to G is a reductive group G^* defined over $\mathbb{F}_q$, whose maximal torus $\mathbf{T}^*$ is provided with an isomorphism with the dual of the maximal torus $\mathbf{T}$ of G, this isomorphism carrying simple roots to simple coroots.

Here are some properties of this duality. Let G and G^* be dual.

(5.21.1) G and G^* have the same Weyl group $\mathbf{W}$. The action of $\mathbf{W}$ on $Y(\mathbf{T}^*)$ is the contragredient of its action on $Y(\mathbf{T})$.

(5.21.2) However, Frobenius does not act in the same way on $\mathbf{W}$ in G and G^*: it acts in inverse ways (the origin of this is that F on $Y(\mathbf{T}^*)$ is the transpose of F on $Y(\mathbf{T})$).

(5.21.3) The conjugacy classes of pairs (T,B) (T an F-stable maximal torus, B a Borel subgroup containing it) in G and G^* correspond. With the notations of (1.13), (1.14), to the class of (T,B) we associate the class of (T^*,B^*) with

$$ {}^t(\operatorname{ad} h(T,B)^{-1} \circ F) = \operatorname{ad} h(T^*,B^*)^{-1} \circ {}^tF; $$

the tori T and T^* are in duality.

(5.21.4) This induces a bijection between the rational conjugacy classes of maximal tori in G and G^*. If T and T' are in corresponding classes, we have a natural class (mod $(N(T)/T)^F$) of isomorphisms between T^* and T'.

(5.21.5) Let T be an F-stable torus on G, and let θ be a character of T^F. Fix T', a corresponding torus in G^*. The character θ defines an element of T^{*F}, then a $(N(T')/T')^F$-conjugacy class of elements θ' of T'. In that way, we get a bijection between G^F-conjugacy classes of pairs (T,θ) as above, and G^{*F}-conjugacy classes of pairs (T',θ'), T' an F-stable maximal torus of G^* and θ' an element of T'^F. This correspondence is compatible with extension of scalars from $\mathbb{F}_q$ to $\mathbb{F}_{q^n}$ (cf. 5.3.).

(5.21.6) By forgetting T', we see that each G^F-conjugacy class of pairs (T,θ) defines an element $\theta' \in G^{*F}$, well-defined up to G^{*F}-conjugacy.

PROPOSITION 5.22. Two pairs (T_1,θ_1) and (T_2,θ_2) are geometrically conjugate if and only if θ_1' and θ_2' are geometrically conjugate.

An extension of scalars (cf. 5.4, 5.5) reduces us to the case where T_1 and T_2 are split. Conjugating, we may assume further that $T_1 = T_2$. In this case, geometric conjugacy is W-conjugacy, and the proposition is clear.

PROPOSITION 5.23. Let G and G^* be dual. Then, the centre of G is connected and smooth (resp. connected) if and only if the derived group of G^* is simply connected (resp. if its simply connected covering is unseparable).

Put $Y = Y(T^*)$, T^* the maximal torus of G^*; let Y_1 be the subgroup generated by the coroots; and put $Y_2 = Y \cap Y_1 \otimes \mathbb{Q}$. Then, Y_2 is the Y-group of the maximal torus of the derived group, and Y_1 that of the maximal torus of its universal covering. The group of this covering is the dual of the Pontrjagin dual of Y_2/Y_1, and (5.23) follows by comparison with point (c) in the proof of (5.13).

COROLLARY 5.24. If the centre of G is connected, two pairs (T_1,θ_1), (T_2,θ_2) as in (5.22) are geometrically conjugate if and only if θ_1' and θ_2' are G^{*F}-conjugate.

Indeed, by a theorem of Steinberg, geometric conjugacy amounts in this case (5.23) to conjugacy, for semi-simple elements of G^* (their centralizers are connected).

DEFINITION 5.25. Let (T',θ') correspond to (T,θ) as in (5.21.5). The pair (T,θ) is maximally split if the $\mathbb{F}_q$-rank of T is equal to that of the centralizer of θ', i.e., if T' is maximally split in that centralizer.

Any geometric conjugacy class of pairs (T,θ) contains maximally split pairs.

PROPOSITION 5.26. If the centre of G is connected, two geometrically conjugate maximally split pairs (T_1,θ_1), (T_2,θ_2) are G^F-conjugate.

Via (5.21.5) and (5.24), this amounts to the known fact that two maximally split maximal tori in the centralizer $Z(\theta')$ of a semi-simple element θ' of G^{*F} are conjugate by an element of $Z(\theta)^F$.

PROPOSITION 5.27. If (T, θ) is maximally split, and the image of T in the adjoint group is anisotropic, then θ is non-singular.

Fix (T', θ') as in (5.21.5). By assumption, $Z^0(\theta')$ contains no non-central F-stable split torus, hence $Z^0(\theta') = T'$. The roots of T' in $Z^0(\theta')$ correspond to the coroots of T orthogonal to θ, hence the proposition.

COROLLARY 5.28. If (T, θ) is maximally split, T is containd in an F-stable Levi subgroup L of an F-stable parabolic subgroup P, and, in L, (T, θ) is non-singular.

One chooses P and L so that the image of T in the adjoint group of L is anisotropic. Being maximally split in G, (T, θ) is a fortiori maximally split in L, and one applies (5.27).

The following result will be used in the next chapter.

PROPOSITION 5.29. Let T' be a subtorus of a torus T, with T defined over $\mathbb{F}_q$ (no assumption on T'). Let θ be a character of T^F. It is trivial on $T' \cap T^F$ if and only if, when viewed as a character of $Y(T)$, it is trivial on

$$((F-1)Y(T') \otimes \mathbb{Q} \cap Y(T)).$$

The condition is that $\theta(x) = 0$ for

$$(F-1)^{-1}(x) \in (Y(T') \otimes \mathbb{Q} + Y(T)),$$

i.e., for

$$x \in ((F-1)Y(T') \otimes \mathbb{Q} \cap Y(T)) + ((F-1)Y(T)).$$

The character θ being trivial on $(F-1)Y(T)$, this means $\theta(x)$ is trivial for $x \in ((F-1)Y(T') \otimes \mathbb{Q} \cap Y(T))$.

6. Intertwining numbers

6.1. Let T, T' be two F-stable maximal tori in G and let θ, θ' be characters of T^F and T'^F. We put $N_G(T, T') = \{g \in G \mid Tg = gT'\}$ and

$$W_G(T, T') = T \backslash N_G(T, T') = N_G(T, T')/T'.$$

We will drop the index G if there is no ambiguity. F acts on $W(T, T')$, and

$$W(T, T')^F = T^F \backslash N(T, T')^F = N(T, T')^F / T'^F.$$

THEOREM 6.2. If θ^{-1} is not geometrically conjugate to θ', then

$$\left[H_c^*(\widetilde{X}_{T\subset B})_\theta \otimes H_c^*(\widetilde{X}_{T'\subset B'})_{\theta'}\right]^{G^F} = 0,$$

i.e., if an irreducible representation of G^F occurs in $H_c^*(\widetilde{X}_{T\subset B})_\theta$, its dual does not occur in $H_c^*(\widetilde{X}_{T'\subset B'})_{\theta'}$.

The virtual representation $R_T^{\theta^{-1}}$ is the dual of R_T^θ (as can be seen on the character formula 4.2), hence we have as a corollary:

COROLLARY 6.3. If θ and θ' are not geometrically conjugate, no irreducible representation of G^F can occur in both virtual representations R_T^θ and $R_{T'}^{\theta'}$.

The proof of 6.2 will make use of the following homotopy argument:

PROPOSITION 6.4. Let H be a connected algebraic group acting on a scheme Y, separated and of finite type over k. For any $h \in H$, the action of h on $H_c^*(Y, \mathbb{Z}/n)$ is trivial.

Let π be the projection of $H \times Y$ onto H and let $f : H \times Y \to H \times Y$ be defined by $f(h, y) = (h, hy)$:

$$\begin{array}{cccc} & H \times Y & \xrightarrow{f} & H \times Y \\ & \searrow_\pi & & \swarrow_\pi \\ [-1ex] & & H & \end{array}$$

By the change-of-basis theorem in cohomology with compact support applied to

$$\begin{array}{ccc} H \times Y & \longrightarrow & Y \\ \downarrow & & \downarrow \\ H & \longrightarrow & \operatorname{Spec} k, \end{array}$$

$R^i\pi_!\mathbb{Z}/n$ is the constant sheaf $H_c^i(Y, \mathbb{Z}/n)$ on H. The automorphism f acts on it and, at $h \in H$, it acts the way h acts on $H_c^i(Y, \mathbb{Z}/n)$. At the identity element of H, the action of f is trivial. An endomorphism of a constant sheaf over a connected base is constant, hence f acts trivially everywhere, and the proposition is proved.

COROLLARY 6.5. The conclusion of (6.4) holds in l-adic cohomology.

The proof is a passage to limit.

6.6. Proof of 6.2. On $\widetilde{X}_{T\subset B} \times \widetilde{X}_{T'\subset B'}$ we have commuting actions of G^F, T^F, and T'^F (G^F acts diagonally). By Künneth's Theorem, we have to prove that

$$\left[H_c^*(\widetilde{X}_{T\subset B} \times \widetilde{X}_{T'\subset B'})_{\theta,\theta'}\right]^{G^F} = 0,$$

where the symbol $M_{\theta,\theta'}$, applied to any $T^F \times T'^F$-module M, means the subspace of M, where T^F and T'^F act by θ and θ'.

$$\begin{cases}\text{Let } U \text{ (resp. } U'\text{) be the unipotent radical of } B \text{ (resp. } B'\text{)}.\\ \text{Let } S_{T\subset B} = \{g \in G \mid g^{-1}Fg \in FU\}, \quad S_{T'\subset B'} = \{g' \in G \mid g'^{-1}Fg' \in FU'\}.\end{cases}$$

The unipotent group $(U \cap FU) \times (U' \cap FU')$ acts freely on $S_{T\subset B} \times S_{T'\subset B'}$ and the orbit space is $\widetilde{X}_{T\subset B} \times \widetilde{X}_{T'\subset B'}$ (see (1.17)). It follows that

$$H_c^i(\widetilde{X}_{T\subset B} \times \widetilde{X}_{T'\subset B'})(-d) \simeq H_c^{i+2d}(S_{T\subset B} \times S_{T'\subset B'}),$$

where d is the dimension of $(U \cap FU) \times (U' \cap FU')$. This isomorphism is compatible with the action of $G^F \times T^F \times T'^F$; moreover this group acts trivially on $\overline{\mathbb{Q}}_l(-d)$. Hence it is sufficient to prove that

$$H_c^*(S_{T\subset B} \times S_{T'\subset B'})^{G^F}_{\theta,\theta'} = H_c^*((S_{T\subset B} \times S_{T'\subset B'})/G^F)_{\theta,\theta'} = 0.$$

The map

$$(g, g') \longmapsto (x, x', y), \qquad x = g^{-1}Fg, \quad x' = g'^{-1}Fg', \quad y = g^{-1}g',$$

defines an isomorphism of $(S_{T\subset B} \times S_{T'\subset B'})/G^F$ with

$$\bar{S} = \{(x, x', y) \in FU \times FU' \times G \mid xF(y) = yx'\}.$$

Under this isomorphism, $T^F \times T'^F$ acts on $\bar{S}$ by the formula

$$(x, x', y) \longmapsto (t^{-1}xt, t'^{-1}x't', t^{-1}yt'), \qquad (t, t') \in T^F \times T'^F. \tag{6.6.1}$$

Let U'^- be opposed to U' with respect to T'. Any $g \in G$ can be written uniquely in the form $g = u_g n_g u'_g$, with with $u_g \in U \cap n_g U'^- n_g^{-1}$, $n_g \in N(T, T')$ (see (6.1)), $u'_g \in U'$; this follows easily from Bruhat's Lemma. For any $w \in W(T, T')$, let G_w be the set of all $g \in G$ such that n_g represents w. Then $(G_w)_{w\in W(T,T')}$ is a finite partition of G into locally closed subschemes. It has the property (6.6.2) below, expressing the fact that the closure of a Bruhat cell is a union of Bruhat cells.

(6.6.2.) For a suitable ordering of $W(T, T')$, the unions

$$\Sigma_w = \bigcup_{w'<w} G_{w'}$$

are closed.

Let $\bar{S}_w = \{(x, x', y) \in \bar{S} \mid y \in G_w\}$. Then $(\bar{S}_w)_{w\in W(T,T')}$ is a finite partition of $\bar{S}$ into locally closed subschemes, stable under $T^F \times T'^F$. It inherits a property analogous to (6.6.2), that is, the unions $\bigcup_{w'<w} \bar{S}_{w'}$ are closed for any w.

The spectral sequence associated to the filtration of $\bar{S}$ by these unions shows that, in order to prove that $H_c^*(\bar{S})_{\theta,\theta'} = 0$, it is sufficient to prove that $H_c^*(\bar{S}_w)_{\theta,\theta'} = 0$ for any $w \in W(T, T')$.

Let

$$H_w = \{(t, t') \in T \times T' \mid t'F(t')^{-1} = F(\dot{w})^{-1}tF(t)^{-1}F(\dot{w})\},$$

where $\dot{w} \in N(T, T')$ represents w. Then H_w is a closed subgroup of $T \times T'$, containing $T^F \times T'^F$. We define an action of H_w on $\bar{S}_w$ as follows. Let $(t, t') \in H_w$; for $(x, x', y) \in \bar{S}_w$ define

$$f_{t,t'}(x,x',y) = (\tilde{x},\tilde{x}',\tilde{y}),$$

where

$$\begin{aligned}\tilde{x} &= t^{-1}xF(u_y)tF(t^{-1}u_y^{-1}t),\\ \tilde{x}' &= t'^{-1}x'F(u'_y)^{-1}t'F(t'^{-1}u'_yt'),\\ \tilde{y} &= t^{-1}yt'.\end{aligned}$$

It is easy to check that $(\tilde{x},\tilde{x}',\tilde{y}) \in \bar{S}_w$, hence $f_{t,t'} : \bar{S}_w \longrightarrow \bar{S}_w$. It is also easy to check that

$$f_{t_1,t'_1} \circ f_{t_2,t'_2} = f_{t_1t_2,t'_1t'_2}$$

for $(t_i,t'_i) \in H_w$, $i = 1, 2$, hence $f_{t,t'}$ defines an action of H_w on $\bar{S}_w$. It is clear that this action extends the action (6.6.1) of $T^F \times T'^F$ on $\bar{S}_w$. The theorem now follows from 6.5 and the

LEMMA 6.7. If the character $\theta\theta'$ of $T^F \times T'^F$ is trivial on $H_w^0 \cap (T^F \times T'^F)$, then $\theta'^{-1} = \theta \circ \mathrm{ad}(F(w))$ (as characters of $Y(T')$).

The subgroup H_w of $T \times T'$ is the kernel of the composite map

$$T \times T' \xrightarrow{x^{-1}Fx} T \times T' \xrightarrow{t^{-1}\,\mathrm{ad}\,F(w)(t')} T.$$

Applying the functor Y, we get that $Y(H_w^0)$ is the kernel K of

$$Y(T) \times Y(T') \xrightarrow{Fx-x} Y(T) \times Y(T') \xrightarrow{\mathrm{ad}\,F(w)(x')-x} Y(T).$$

By (5.29), the triviality of $\theta\theta'$ on $H_w^0 \cap (T^F \times T'^F)$ means that, when we view $\theta\theta'$ as a character of $Y(T) \times Y(T')$, it is trivial on

$$(F-1)(K \otimes \mathbb{Q}) \cap (Y(T) \times Y(T')).$$

Since the map $F - 1$ is injective, this intersection is

$$\mathrm{Ker}(\mathrm{ad}\,F(w)(x') - x \,:\, Y(T) \times Y(T') \longrightarrow Y(T))$$

and the assumption becomes the triviality of $\theta\theta'$ on $\mathrm{Ker}(\mathrm{ad}\,F(w)(x') - x)$, i.e., the identity

$$\theta \circ \mathrm{ad}\,F(w)(\theta') = 1.$$

We will now conjugate (6.2) with the character formula (4.2) to get quantitative results.

THEOREM 6.8. Let $\theta \in (T^F)^\vee$, $\theta' \in (T'^F)^\vee$. Then

$$\langle R_T^\theta, R_{T'}^{\theta'}\rangle_{G^F} = \#\{w \in W(T,T')^F \mid \mathrm{ad}\,w(\theta') = \theta\}.$$

THEOREM 6.9.

$$\frac{1}{|G^F|}\sum_{\substack{u\in G^F\\ \text{unipotent}}} Q_{T,G}(u)Q_{T',G}(u) = \frac{|N(T,T')^F|}{|T^F|\,|T'^F|}. \tag{6.9.1}$$

We will prove (6.8) and (6.9) simultaneously, by induction on the dimension of G.

LEMMA 6.10. If (6.9) holds, for G replaced by $Z^0(s)$, where $s \in G^F$ is any semisimple element not contained in the centre Z of G, then

$$\langle R_T^\theta, R_{T'}^{\theta'}\rangle_{G^F} = \#\{w \in W(T,T')^F \mid \mathrm{ad}\,w(\theta') = \theta\} + \alpha \tag{6.10.1}$$

where

$$\alpha = \sum_{s\in T^F\cap Z} \theta(s)\theta'(s^{-1})\left(\frac{1}{|G^F|}\sum_{\substack{u\in G^F\\ \text{unipotent}}} Q_{T,G}(u)Q_{T',G}(u) - \frac{|N(T,T')^F|}{|T^F|\,|T'^F|}\right).$$

In particular, (6.9) implies (6.8).

According to (4.2) we have:

$$\begin{aligned}\langle R_T^\theta, R_{T'}^{\theta'}\rangle &= \frac{1}{|G^F|}\sum_{g_1\in G^F}\operatorname{Tr}(g_1,R_T^\theta)\operatorname{Tr}(g_1^{-1},R_{T'}^{\theta'})\\ &= \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|^2}\sum_{\substack{g,g'\in G^F\\ g^{-1}sg\in T^F\\ g'^{-1}sg'\in T'^F}}\theta(g^{-1}sg)\theta'(g'^{-1}sg')^{-1}\\ &\quad\times\sum_{\substack{u\in Z^0(s)^F\\ \text{unipotent}}} Q_{gTg^{-1},Z^0(s)}(u)Q_{g'T'g'^{-1},Z^0(s)}(u).\end{aligned}$$

With our assumption, this can be written as

$$\alpha + \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|}\sum_{\substack{g,g'\in G^F\\ g^{-1}sg\in T^F\\ g'^{-1}sg'\in T'^F}}\theta(g^{-1}sg)\theta'(g'^{-1}sg')^{-1}\times\frac{|N_{Z^0(s)}(gTg^{-1},g'T'g'^{-1})^F|}{|T^F|\,|T'^F|}.$$

The formula $(g,g',n_1)\longmapsto(g,n,n_1)$, $n=g'^{-1}n_1g$, establishes a one-to-one correspondence between the sets

$$\{(g,g',n_1)\in G^F\times G^F\times G^F \mid g^{-1}sg\in T^F,\ g'^{-1}sg'\in T'^F,\ n_1\in N_{Z^0(s)}(gTg^{-1},g'T'g'^{-1})^F\},$$

$$\{(g,n,n_1)\in G^F\times N_G(T,T')^F\times Z^0(s)^F \mid g^{-1}sg\in T^F\}.$$

It follows that

$$\begin{aligned}\langle R_T^\theta, R_{T'}^{\theta'}\rangle &= \alpha + \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|}\sum_{\substack{g\in G^F\\ n\in N_G(T,T')^F\\ g^{-1}sg\in T^F}}\theta(g^{-1}sg)\theta'(ng^{-1}sgn^{-1})^{-1}\frac{|Z^0(s)^F|}{|T^F|\,|T'^F|}\\ &= \alpha + \frac{1}{|G^F|}\sum_{\substack{t\in T^F\\ n\in N_G(T,T')^F}}\theta(t)\theta'(ntn^{-1})^{-1}\frac{|G^F|}{|T^F|\,|T'^F|}\\ &= \alpha + \frac{1}{|T'^F|}\sum_{n\in N_G(T,T')^F}\frac{1}{|T^F|}\sum_{t\in T^F}\theta(t)\theta'(ntn^{-1})^{-1}\\ &= \alpha + \#\{w\in W(T,T')^F \mid \operatorname{ad} w(\theta')=\theta\},\end{aligned}$$

and (6.10.1) is proved.

Proof of 6.9. Let $\bar G = G/Z$; the centre of $\bar G$ consists of the identity element. Both sides of (6.9.1) remain unchanged when G is replaced by $\bar G$; for the left hand side this follows from (4.1.1), while for the right hand side this is easy to check. Hence in order to prove (6.9), we may assume that the centre of G has a single element; moreover we may assume by induction that the conclusion of (6.9) is true for G replaced by $Z^0(s)$, where $s \in G^F$ is an arbitrary non-central semisimple element, and for any two F-stable maximal tori in $Z^0(s)$. (To start the induction we may take G to be the group with only one element in which case (6.9) is clear.) It follows from (6.3) that for any non-trivial character θ of T^F one has

$$\langle R_T^\theta, R_{T'}^1 \rangle = 0.$$

Substituting this information into (6.10), we see that if T^F has any non-trivial character, then $\alpha = 0$ which proves (6.9). A similar argument applies when $|T'^F| \neq 1$. It remains to check the special case where $|T^F| = |T'^F| = 1$. In this case, we must have $q = 2$, and T, T' must be $\mathbb{F}_q$-split tori. In particular, T and T' are conjugate under G^F, hence

$$\#\{w \in W(T, T')^F \mid \operatorname{ad} w(\theta') = \theta\} = |W(T)^F| = |W(T)|.$$

Moreover R_T^1 and $R_{T'}^1$ are both equal to the representation of G^F induced by the unit representation of B^F, where B is an F-stable Borel subgroup of G. It follows that

$$\langle R_T^1, R_{T'}^1 \rangle = |B^F \backslash G^F / B^F| = |W(T)|. \tag{1.10}$$

Substituting in (6.10) we get (6.9) in this case. This completes the proof of (6.9) (and of (6.8) by (6.10)).

7. Computations on semisimple elements

The following result describes the Euler characteristic of the scheme $X_{T\subset B}$. Let $\sigma(G)$ (resp. $\sigma(T)$) be the $\mathbb{F}_q$-rank of G (resp. T).

THEOREM 7.1.

$$\chi(X_{T\subset B}) = Q_{T,G}(e) = (-1)^{\sigma(G)-\sigma(T)} \frac{|G^F|}{\mathrm{St}_G(e)|T^F|}.$$

(Here St_G is the Steinberg representation of G^F, identified with its character.)

We may assume that the centre of G is reduced to a single element and that the statement is true when G is replaced by $Z^0(s)$, where $s \in G^F$ is any non-central semisimple element.

We first assume that T^F has a non-trivial character θ. The Steinberg representation occurs in

$$R_{T^*}^1 = \mathrm{Ind}_{B^{*F}}^{G^F}(1)$$

($T^* \subset B^*$ as in 1.8), hence, by (6.3), it does not occur in R_T^θ: $\langle R_T^\theta, \mathrm{St}_G \rangle = 0$. It is known that

$$\mathrm{St}_G(g) = \begin{cases} (-1)^{\sigma(G)-\sigma(Z^0(g))} \mathrm{St}_{Z^0(g)}(e), & g \in G^F \text{ semisimple}, \\ 0, & g \in G^F \text{ non-semisimple}. \end{cases}$$

If we use 4.2, it follows:

$$\sum_{s\in G^F}(-1)^{\sigma(G)-\sigma(Z^0(s))}\frac{\mathrm{St}_{Z^0(s)}(e)}{|Z^0(s)^F|}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T}}Q_{gTg^{-1},Z^0(s)}(e)\theta(g^{-1}sg)=0$$

(sum over the semisimple elements s of G^F).

By our assumption, we may substitute

$$Q_{gTg^{-1},Z^0(s)}(e)=(-1)^{\sigma(Z^0(s))-\sigma(T)}\frac{|Z^0(s)^F|}{\mathrm{St}_{Z^0(s)}(e)|T^F|},$$

for all $s\neq e$:

$$\frac{(-1)^{\sigma(G)-\sigma(T)}}{|T^F|}\sum_{\substack{s\in G^F\\ s\neq e}}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}}\theta(g^{-1}sg)+\mathrm{St}_G(e)Q_{T,G}(e)=0,$$

$$(-1)^{\sigma(G)-\sigma(T)}\frac{|G^F|}{|T^F|}\sum_{\substack{t\in T^F\\ t\neq e}}\theta(t)+\mathrm{St}_G(e)Q_{T,G}(e)=0.$$

Since the character θ in non-trivial, $\sum_{t\in T^F}\theta(t)=0$ and the desired formula for $Q_{T,G}(e)$ follows.

In the case where $|T^F|=1$, T must be an $\mathbb{F}_q$-split torus and $q=2$. In this case, $Q_{T,G}(e)=|G^F|/|B^{*F}|$, where B^* is any F-stable Borel subgroup (1.8). This agrees with the statement of the theorem and ends the proof.

COROLLARY 7.2. For any semisimple element $s\in G^F$,

$$\mathrm{Tr}(s,R_T^\theta)=(-1)^{\sigma(Z^0(s))-\sigma(T)}\frac{1}{\mathrm{St}_{Z^0(s)}(e)|T^F|}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}}\theta(g^{-1}sg).$$

This follows from (7.1) and 4.2. An equivalent statement is:

PROPOSITION 7.3.

$$(-1)^{\sigma(G)-\sigma(T)}R_T^\theta\otimes\mathrm{St}_G=\mathrm{Ind}_{T^F}^{G^F}(\theta).$$

PROPOSITION 7.4. If θ is a non-singular character of T^F (5.19) then $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ can be represented by an actual G^F-module. If θ is in general position (5.15), $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ is irreducible.

By embedding G in a group G_1 with connected centre and the same derived group, as in (5.18), we reduce to the case where θ is in general position. It remains to use (6.8) and (7.1).

PROPOSITION 7.5. Let $s\in G^F$ be a semisimple element. The character of

$$\frac{1}{\mathrm{St}_G(s)}\sum_{\substack{T\\ s\in T}}\sum_{\theta\in(T^F)^\vee}\theta(s)^{-1}(-1)^{\sigma(G)-\sigma(T)}R_T^\theta \tag{7.5.1}$$

equals $|Z(s)^F|$ on elements conjugate to s in G^F and zero on all other elements of G^F.

Let μ be the character of (7.5.1) and let μ' be the class function on G^F defined by

$$\mu'(g) = \begin{cases} |Z(s)^F|, & g \text{ conjugate to } s \text{ in } G^F, \\ 0, & \text{otherwise.} \end{cases}$$

In order to prove that $\mu = \mu'$ it is sufficient to prove that

$$\langle \mu, \mu \rangle = \langle \mu, \mu' \rangle = \langle \mu', \mu' \rangle \qquad \text{(see (0.3))}.$$

We have $\langle \mu', \mu' \rangle = |Z(s)^F|$; from (6.8) we see that

$$\begin{aligned}
\langle \mu, \mu \rangle &= \frac{1}{\mathrm{St}_G(s)^2} \sum_{\substack{T \ni s \\ T' \ni s}} \sum_{\substack{\theta \in (T^F)^\vee \\ \theta' \in (T'^F)^\vee}} \theta(s^{-1})\theta'(s)\#\{n \in N(T, T')^F \mid \mathrm{ad}\, n(\theta') = \theta\}|T^F|^{-1} \\
&= \frac{1}{\mathrm{St}_G(s)^2} \sum_{\substack{T \ni s \\ \theta \in (T^F)^\vee \\ n \in G^F \\ nsn^{-1} \in T}} \theta(s^{-1})\theta(nsn^{-1})|T^F|^{-1} \\
&= \frac{1}{\mathrm{St}_G(s)^2} \#\{T \ni s,\ n \in G^F \mid sn = ns\} \\
&= \frac{|Z(s)^F|}{\mathrm{St}_G(s)^2} \#\{F\text{-stable maximal tori in } Z^0(s)\} = |Z(s)^F|
\end{aligned}$$

(by [15, Cor. 14.16]).

By (7.2), we have

$$\begin{aligned}
\langle \mu, \mu' \rangle = \mu(s) &= \frac{1}{\mathrm{St}_G(s)} \sum_{\substack{T \\ s \in T}} \sum_{\theta \in (T^F)^\vee} \theta(s^{-1}) \frac{(-1)^{\sigma(G) - \sigma(Z^0(s))}}{\mathrm{St}_{Z^0(s)}(e)|T^F|} \sum_{\substack{g \in G^F \\ g^{-1}sg \in T^F}} \theta(g^{-1}sg) \\
&= \frac{1}{\mathrm{St}_G(s)^2} \sum_{\substack{T \\ s \in T}} \sum_{\substack{g \in G^F \\ g^{-1}sg \in T^F}} |T^F|^{-1}\theta(s^{-1})\theta(g^{-1}sg) \\
&= \frac{1}{\mathrm{St}_G(s)^2} \#\{T \ni s,\ g \in G^F \mid sg = gs\} \\
&= \frac{|Z(s)^F|}{\mathrm{St}_G(s)^2} \#\{F\text{-stable maximal tori in } Z^0(s)\} = |Z(s)^F|,
\end{aligned}$$

and the proposition is proved.

COROLLARY 7.6. For any $\rho \in \mathscr{R}(G^F)$ and any semisimple element $s \in G^F$,

$$\mathrm{Tr}(s, \rho) = \frac{1}{\mathrm{St}_G(s)} \sum_{\substack{T \\ s \in T}} \sum_{\theta \in (T^F)^\vee} \theta(s)(-1)^{\sigma(G) - \sigma(T)} \langle \rho, R_T^\theta \rangle. \tag{7.6.1}$$

In particular, if $s \in G^F$ and is regular semisimple, contained in a unique torus T,

$$\mathrm{Tr}(s, \rho) = \sum_{\theta \in (T^F)^\vee} \theta(s) \langle \rho, R_T^\theta \rangle, \tag{7.6.2}$$

$$\dim \rho = \frac{1}{\mathrm{St}_G(e)} \sum_{T} \sum_{\theta \in (T^F)^\vee} (-1)^{\sigma(G) - \sigma(T)} \langle \rho, R_T^\theta \rangle. \tag{7.6.3}$$

COROLLARY 7.7. For any irreducible representation ρ of G^F there exists an F-stable maximal torus T and a character θ of T^F such that $\langle \rho, R_T^\theta \rangle \neq 0$.

This follows from (7.6.3).

DEFINITION 7.8. An irreducible representation ρ of G^F is said to be unipotent if $\langle \rho, R_T^1 \rangle \neq 0$ for some F-stable maximal torus T.

By (6.3) and (7.7), ρ is unipotent if and only if $\langle \rho, R_T^\theta \rangle = 0$ for any F-stable maximal torus T and any $\theta \in (T^F)^\vee$, $\theta \neq 1$. For example, any irreducible component of $\mathrm{Ind}_{B^{*F}}^{G^F}(1)$ (B^{*F} an F-stable Borel subgroup) is a unipotent representation. For unipotent representations, (7.6) becomes:

PROPOSITION 7.9. Let ρ be a unipotent representation of G^F and let $s \in G^F$ be a semisimple element. Then

$$\mathrm{Tr}(s, \rho) = \frac{1}{\mathrm{St}_G(s)} \sum_{\substack{T \\ s \in T}} (-1)^{\sigma(G)-\sigma(T)} \langle \rho, R_T^1 \rangle.$$

In particular, if $s \in G^F$ is regular semisimple contained in a unique maximal torus T, $\mathrm{Tr}(s, \rho) = \langle \rho, R_T^1 \rangle$ and

$$\dim \rho = \frac{1}{\mathrm{St}_G(e)} \sum_T (-1)^{\sigma(G)-\sigma(T)} \langle \rho, R_T^1 \rangle.$$

It follows that if B is an F-stable Borel subgroup of G and T is an F-stable maximal torus in B, we have $\mathrm{Tr}(s, \rho) = \langle \rho, \mathrm{Ind}_{B^F}^{G^F}(1) \rangle$; in the case where $\langle \rho, \mathrm{Ind}_{B^F}^{G^F}(1) \rangle \neq 0$, this is a result of Curtis, Kilmoyer and Seitz (see C. W. Curtis, On the Values of Certain Irreducible Characters of Finite Chevalley Groups, Ist. Naz. di Alta Mat. Symp. Mat. XIII, 1974, 343–355).

PROPOSITION 7.10. Let G^{ad} be the adjoint group of G. Then, the restriction to G^F of a unipotent representation of $(G^{\mathrm{ad}})^F$ is irreducible; non-isomorphic unipotent representations have non-isomorphic restrictions; and every unipotent representation of G is such a restriction.

It suffices to check that, if σ, τ are unipotent representations of $(G^{\mathrm{ad}})^F$, then $\langle \sigma, \tau \rangle_{G^{\mathrm{ad}}F} = \langle \sigma, \tau \rangle_{G^F}$. Let G_1 be the image of G^F in $(G^{\mathrm{ad}})^F$. One has

$$\begin{aligned} \langle \sigma, \tau \rangle_{G^F} &= \langle \sigma, \tau \rangle_{G_1} = \left\langle \mathrm{Ind}_{G_1}^{G^{\mathrm{ad}}F}(\mathrm{Res}\,\sigma), \tau \right\rangle_{G^{\mathrm{ad}}F} \\ &= \sum_{\theta \in (G^{\mathrm{ad}}F/G_1^F)^\vee} \langle \sigma \otimes \theta, \tau \rangle_{G^{\mathrm{ad}}F}. \end{aligned}$$

By the exclusion theorem 6.3 and (1.23), (1.27), $\langle \sigma \otimes \theta, \tau \rangle_{G^{\mathrm{ad}}F} = 0$ for $\theta \neq 1$, hence the proposition.

PROPOSITION 7.11. Let ρ be a virtual representation of G^F such that $\mathrm{Tr}(su, \rho) = \mathrm{Tr}(s, \rho)$ for any $s \in G^F$ semisimple, $u \in G^F$ unipotent, $su = us$. Let T be an F-stable maximal torus and let θ be a character of T. Then

$$\langle \rho, R_T^\theta \rangle_{G^F} = (-1)^{\sigma(G)-\sigma(T)} \langle \rho \otimes \mathrm{St}_G, R_T^\theta \rangle_{G^F} = \langle \rho, \theta \rangle_{T^F}.$$

The Steinberg character is integral valued, hence by (7.3),

$$(-1)^{\sigma(G)-\sigma(T)}\langle\rho\otimes\mathrm{St}_G, R_T^\theta\rangle_{G^F} = \langle\rho,(-1)^{\sigma(G)-\sigma(T)}\,\mathrm{St}_G\otimes R_T^\theta\rangle_{G^F} = \langle\rho,\mathrm{Ind}_{T^F}^{G^F}(\theta)\rangle_{G^F} = \langle\rho,\theta\rangle_{T^F}.$$

We now prove the identity

$$\langle\rho,R_T^\theta\rangle_{G^F} = \langle\rho,\theta\rangle_{T^F}. \tag{7.11.1}$$

The special case of this identity

$$\langle 1,R_T^1\rangle_{G^F} = 1 \tag{7.11.2}$$

follows from the fact that the Euler characteristic of $X_{T\subset B}/G^F$ (which is the same as the Euler characteristic of

$$\{g\in G\mid g^{-1}Fg\in FU\}/G^F\times T^F = FU/T^F$$

by (1.17)) equals 1. If we use (4.2), (7.11.2) can be written in the form

$$\frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}}\sum_{\substack{u\in Z^0(s)^F\\ \text{unipotent}}} Q_{gTg^{-1},Z^0(s)}(u) = 1. \tag{7.11.3}$$

This implies, by induction on the dimension of G, that

$$\frac{1}{|G^F|}\sum_{\substack{u\in G^F\\ \text{unipotent}}} Q_{T,G}(u) = \frac{1}{|T^F|}. \tag{7.11.4}$$

Indeed, by substituting (7.11.4) in (7.11.3) we find a true identity:

$$\frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|T^F|}\#\{g\in G^F\mid g^{-1}sg\in T^F\} = 1.$$

We apply (4.2) again and use (7.11.4) to compute:

$$\begin{aligned}\langle\rho,R_T^\theta\rangle_{G^F} &= \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\frac{1}{|Z^0(s)^F|}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}}\sum_{\substack{u\in Z^0(s)^F\\ \text{unipotent}}} Q_{gTg^{-1},Z^0(s)}(u)\theta(g^{-1}sg)^{-1}\,\mathrm{Tr}(s,\rho)\\ &= \frac{1}{|G^F|}\sum_{\substack{s\in G^F\\ \text{semis.}}}\sum_{\substack{g\in G^F\\ g^{-1}sg\in T^F}}\frac{1}{|T^F|}\theta(g^{-1}sg)^{-1}\,\mathrm{Tr}(s,\rho)\\ &= \frac{1}{|G^F|}\sum_{t\in T^F}\theta(t)^{-1}\,\mathrm{Tr}(t,\rho) = \langle\rho,\theta\rangle_{T^F}.\end{aligned}$$

PROPOSITION 7.12. Let ρ be as in (7.11). Then

$$\rho = \sum_{(T)}\frac{1}{|W(T)^F|}\sum_{\theta\in(T^F)^\vee}\langle\rho,\theta\rangle_{T^F}R_T^\theta, \tag{7.12.1}$$

$$\rho\otimes\mathrm{St}_G = \sum_{(T)}\frac{(-1)^{\sigma(G)-\sigma(T)}}{|W(T)^F|}\sum_{\theta\in(T^F)^\vee}\langle\rho,\theta\rangle_{T^F}R_T^\theta, \tag{7.12.2}$$

where $\sum_{(T)}$ means sum over all G^F-conjugacy classes of F-stable maximal tori T.

The character of $\rho\otimes\mathrm{St}_G$ is a linear combination of the form $\sum_{(T,\theta)}c_{T,\theta}R_T^\theta$ (sum over all G^F-conjugacy classes of pairs (T,θ)). The coefficients $c_{T,\theta}$ are determined by (7.11) and (7.12.2) follows.

Now let ρ' (resp. ρ'') be the right hand side of (7.12.1) (resp. (7.12.2)). We have clearly

$$\langle \rho', \rho' \rangle = \langle \rho'', \rho'' \rangle \tag{7.12.3}$$

and by (7.11),

$$\langle \rho, \rho' \rangle = \langle \rho \otimes \mathrm{St}_G, \rho'' \rangle. \tag{7.12.4}$$

We note also that

$$\langle \rho, \rho \rangle = \langle \rho \otimes \mathrm{St}_G, \rho \otimes \mathrm{St}_G \rangle. \tag{7.12.5}$$

This follows from the fact that the number of unipotents in $Z^0(s)^F$ equals $\mathrm{St}_G(s)^2$ for any semisimple element $s \in G^F$ (see [15, Thm. 15.1]). By (7.12.2) we have

$$\langle \rho \otimes \mathrm{St}_G - \rho'', \rho \otimes \mathrm{St}_G - \rho'' \rangle = 0.$$

This, together with (7.12.3), (7.12.4) and (7.12.5) implies:

$$\langle \rho - \rho', \rho - \rho' \rangle = 0,$$

hence

$$\rho = \rho'.$$

Remark 7.13. It is known that

$$\langle \rho, \rho \rangle_{G^F} = \sum_{(T)} \frac{1}{|W(T)^F|} \langle \rho, \rho \rangle_{T^F}$$

(N. Kawanaka, A theorem on finite Chevalley groups, Osaka J. Math. 10 (1973), 1–13); this could be also deduced from (7.12.1) and (6.8).

COROLLARY 7.14.

$$1 = \sum_{(T)} \frac{1}{|W(T)^F|} R_T^1. \tag{7.14.1}$$

$$\mathrm{St}_G = \sum_{(T)} \frac{(-1)^{\sigma(G)-\sigma(T)}}{|W(T)^F|} R_T^1. \tag{7.14.2}$$

In particular,

$$\langle \mathrm{St}_G, R_T^1 \rangle = (-1)^{\sigma(G)-\sigma(T)}.$$

Here is an alternative proof of (7.14.1). In the language of (1.4), (7.14.1) asserts that

$$\sum_{w \in W} R^1(w) = |W|\, 1.$$

Since $(X(w))_{w \in W}$ is a partition of the flag manifold into locally closed subschemes, we have

$$\sum_{w\in W} R^1(w) = \sum_i (-1)^i H^i_c(X_G)$$

as virtual representations of G^F. The action of G^F on X_G is the restriction of the action of the connected group G; by (6.5), G^F acts trivially on $H^*_c(X_G)$ and it remains to use the fact that the Euler characteristic of X_G equals $|W|$.

COROLLARY 7.15.

$$\mathrm{St}_G = \sum_{(T)} \frac{(-1)^{\sigma(G)-\sigma(T)}}{|W(T)^F|} \operatorname{Ind}_{T^F}^{G^F}(1), \tag{7.15.1}$$

$$\mathrm{St}_G \otimes \mathrm{St}_G = \sum_{(T)} \frac{1}{|W(T)^F|} \operatorname{Ind}_{T^F}^{G^F}(1). \tag{7.15.2}$$

This follows from (7.1) by tensoring with St_G, and using (7.3). The formula (7.15.1) is due to B. Srinivasan (On the Steinberg character of a finite simple group of Lie type, J. Australian Math. Soc. 12 (1971), 1–14).

8. Induced and cuspidal representations

8.1. Let P be an F-stable parabolic subgroup of G and let $T \subset P$ be an F-stable maximal torus. We denote by U_P the unipotent radical of P. The quotient group P/U_P is a connected reductive algebraic group acted on by F. Let $\pi : P \to P/U_P$ be the canonical projection. π induces an isomorphism $T \xrightarrow{\sim} \pi(T)$ hence also an isomorphism $T^F \xrightarrow{\sim} \pi(T)^F$. Let θ be a character of T^F and let $\bar\theta$ be the corresponding character of $\pi(T)^F$. We denote by $R^\theta_{T,P}$ the image of the virtual representation $R^{\bar\theta}_{\pi(T)}$ of $(P/U_P)^F$ under the canonical embedding $\mathscr{R}((P/U_P)^F) \subset \mathscr{R}(P^F)$. With these notations, we have the following generalisation of 1.10:

PROPOSITION 8.2.

$$R^\theta_T = \operatorname{Ind}_{P^F}^{G^F}(R^\theta_{T,P}).$$

We choose a Borel subgroup B in G such that $T \subset B \subset P$. Let $\mathscr{P}$ be the set of all parabolic subgroups P' in G such that P' and P are conjugate under G^F; this is a finite set. We have

$$\widetilde{X}_{T\subset B} = \coprod_{P'\in\mathscr{P}} \widetilde{X}(P')$$

where

$$\widetilde{X}(P') = \{g \in G \mid g^{-1}Fg \in FU,\ gPg^{-1} = P'\}/U \cap FU,$$

with U the unipotent radical of B.

(Note that $g^{-1}F(g) \in FU$ implies $F(gPg^{-1}) = gPg^{-1}$.) Let $P' \in \mathscr{P}$ and let $g_1 \in G^F$ be such that $g_1Pg_1^{-1} = P'$. If $g \in G$, $g^{-1}F(g) \in FU$, $gPg^{-1} = P'$, we have

$$g_1^{-1}g \in P, \qquad (g_1^{-1}g)^{-1}F(g_1^{-1}g) \in FU;$$

by sending g to $\pi(g_1^{-1}g)$ we get an isomorphism

$$\widetilde{X}(P') \xrightarrow{\sim} \widetilde{X}_{\pi(T)\subset\pi(B)}.$$

It follows easily that the G^F-module $H_c^i(\widetilde{X}_{T\subset B})$ is canonically isomorphic to the G^F-module induced by $H_c^i(\widetilde{X}_{\pi(T)\subset\pi(B)})$ regarded as a P^F-module; moreover this isomorphism is compatible with the action of T^F. The proposition follows.

THEOREM 8.3. Let T be an F-stable maximal torus in G such that T is not contained in any F-stable, proper parabolic subgroup of G. Let θ be a non-singular character of T^F (cf. 5.19). Then $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ can be represented by a cuspidal G^F-module.

Applying (7.5) for s equal to the identity element of G^F, we see that the regular representation of G^F is equal to:

$$\mathrm{Ind}_e^{G^F}(1) = \frac{1}{\mathrm{St}_G(e)} \sum_T \sum_{\theta\in(T^F)^\vee} (-1)^{\sigma(G)-\sigma(T)} R_T^\theta.$$

We apply this formula with G replaced by P/U_P (P as in (8.1)); we then regard the regular representation of $(P/U_P)^F$ as a representation of P^F on which U_P^F acts trivially (i.e., as $\mathrm{Ind}_{U_P^F}^{P^F}(1)$) and we induce it to G^F:

$$\begin{aligned}\mathrm{Ind}_{U_P^F}^{G^F}(1) &= \mathrm{Ind}_{P^F}^{G^F}\big(\mathrm{Ind}_{U_P^F}^{P^F}(1)\big)\\ &= \frac{1}{\mathrm{St}_{P/U_P}(e)} \sum_{T'\subset P/U_P} \sum_{\theta'\in(T'^F)^\vee} (-1)^{\sigma(P/U_P)-\sigma(T')}\,\mathrm{Ind}_{P^F}^{G^F}(R_{T'}^{\theta'}),\end{aligned}$$

where $R_{T'}^{\theta'}$ is regarded as an element in $\mathscr{R}(P^F)$ in the natural way. We now use (8.2) and the fact that any F-stable maximal torus T' in P/U_P can be lifted to an F-stable maximal torus T in P in precisely $|U_P^F|$ different ways:

$$\mathrm{Ind}_{U_P^F}^{G^F}(1) = \frac{1}{\mathrm{St}_G(e)} \sum_{T\subset P} \sum_{\theta\in(T^F)^\vee} (-1)^{\sigma(G)-\sigma(T)} R_T^\theta. \tag{8.3.1}$$

If $P \neq G$ and T is as in the statement of the theorem, we see from (8.3.1) and (6.8) that

$$\Big\langle(-1)^{\sigma(G)-\sigma(T)}R_T^\theta, \mathrm{Ind}_{U_P^F}^{G^F}(1)\Big\rangle = 0 \qquad \text{for any } \theta\in(T^F)^\vee. \tag{8.3.2}$$

If θ is non-singular, $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ can be represented by an actual G^F-module (7.4). This G^F-module is cuspidal by (8.3.2).

9. A Vanishing Theorem

9.1. Let G be a reductive algebraic group over k. Let $w = s_1\cdots s_n$ be a minimal expression for an element of the Weyl group W of G (0.4). The following desingularization of the closure $\overline{O(w)}$ of $O(w)\subset X\times X$ (1.2) has been considered by H. C. Hansen [6] and M. Demazure [3].

DEFINITION 9.2. $\overline{O}(s_1,\dots,s_n)$ is the space of sequences $(B_0,\dots,B_n)$ of

Borel subgroups, with B_{i-1} and B_i in relative position e or s_i.

The maps

$$\overline{O}(s_1,\dots,s_n)\longrightarrow\overline{O}(s_1,\dots,s_{n-1})\longrightarrow\cdots\longrightarrow\overline{O}(\,)=X$$

express $\overline{O}=\overline{O}(s_1,\dots,s_n)$ as an iterated fibre space over X, with fibre $\mathbb{P}^1$ (1.2). The subspace $D_i\subset\overline{O}$ where $B_{i-1}=B_i$ is the inverse image of a section of $\overline{O}(s_1,\dots,s_i)\to\overline{O}(s_1,\dots,s_{i-1})$; it is a smooth divisor, and $D=\bigcup D_i$ is a divisor with normal crossings. The map $(B_0,\dots,B_n)\mapsto(B_0,B_n)$ induces an isomorphism $\overline{O}-D\xrightarrow{\sim}O(w)$, and hence maps $\overline{O}$ to $\overline{O(w)}$: $\overline{O}$ is a resolution of singularities for $\overline{O(w)}$.

9.3. Let T^*, B^* and U^* be as in (1.7). For any character $\lambda\,:\,T^*\to\mathbb{G}_m$, the T^*-torsor E over X defined in (1.7) gives rise to a line bundle E_λ, provided with $\lambda\,:\,E\to E_\lambda$ such that $\lambda(et)=\lambda(e)\lambda(t)$. For $\dot w\in N(T^*)$ with image w in $W=N(T^*)/T^*$, the w-map of T^*-torsors over $O(w)\subset X\times X$ constructed in (1.7) induces an isomorphism

$$\Psi(\dot w)\,:\,\mathrm{pr}_1^*\,E_\lambda\xrightarrow{\sim}\mathrm{pr}_2^*\,E_{\lambda\circ\mathrm{ad}\,\dot w}=\mathrm{pr}_2^*\,E_{w^{-1}(\lambda)}$$

which makes the following diagram commute:

$$\begin{array}{ccc}\mathrm{pr}_1^*\,E & \xrightarrow{\cdot\dot w} & \mathrm{pr}_2^*\,E\\ \lambda\downarrow & & \downarrow w^{-1}(\lambda)\\ \mathrm{pr}_1^*\,E_\lambda & \xrightarrow{\Psi(\dot w)} & \mathrm{pr}_2^*\,E_{w^{-1}(\lambda)}.\end{array}$$

We will investigate its behaviour at infinity.

9.4. Let $X(T^*)$ be the character group of T^* and let $C\subset X(T^*)\otimes\mathbb{R}$ be the fundamental chamber (corresponding to B^*). If C_1 and C_2 are two chambers, we write $D(C_1,C_2)$ for the intersection of the (closed) radicial half spaces containing both C_1 and C_2, and $D^0(C_1,C_2)$ for its interior.

PROPOSITION 9.5. Let $\overline{O(w)}'$ be the normalization of the closure $\overline{O(w)}$ of $O(w)$ in $X\times X$. The map $\Psi(\dot w)\,:\,\mathrm{pr}_1^*\,E_\lambda\to\mathrm{pr}_2^*\,E_{w^{-1}(\lambda)}$ extends over $\overline{O(w)}'$ if and only if $\lambda\in D(C,-wC)$. It vanishes outside of $O(w)$ if and only if $\lambda\in D^0(C,-wC)$.

Let us consider $\Psi(\dot w)$ as a rational section of $\mathrm{pr}_1^*\,E_{\lambda^{-1}}\otimes\mathrm{pr}_2^*\,E_{w^{-1}(\lambda)}$; it suffices to prove that, with the notations of (9.2), for each i, the order v_i of $\Psi(\dot w)$ along the divisor $D_i\subset\overline{O}(s_1,\dots,s_n)$ is ≥ 0 for $\lambda\in D(C,-wC)$, >0 for $\lambda\in D^0(C,-wC)$. The convex region $D(C,-wC)$ is the intersection of the radicial half spaces containing C but not wC. Put $w_i=s_1\cdots s_i$. The gallery $(C,w_1C,w_2C,\dots,wC)$ connects C to wC; its walls are all the radicial hyperplanes that separate C and wC. The wall between $w_{i-1}C$ and w_iC is defined by the root $w_{i-1}(\alpha_i)$, where α_i is the simple root corresponding to s_i; C and $w_{i-1}C$ are on the same side of it. The convex region $D(C, -wC)$ is hence defined by the inequalities

$$\langle \mu, H_{w_{i-1}(\alpha_i)} \rangle \geq 0.$$

(Recall that H_α denotes the coroot corresponding to a root α, cf. (5.1).) We will prove that

$$v_i = \langle \lambda, H_{w_{i-1}(\alpha_i)} \rangle. \tag{9.5.1}$$

Let us lift the decomposition $w = (s_1 \cdots s_{i-1})s_i(s_{i+1} \cdots s_n)$ into a decomposition $\dot{w} = \dot{w}_{i-1}\dot{s}_i\dot{w}_i'$ in $N(T^*)$. We have

$$\Psi(\dot{w}) = \Psi(\dot{w}_i')\Psi(\dot{s}_i)\Psi(\dot{w}_{i-1}).$$

When we approach a general point of D_i, the isomorphisms $\Psi(\dot{w}_i')$ and $\Psi(\dot{w}_{i-1})$ extend, and we are left to prove that the order along D_i of

$$\Psi(\dot{s}_i) : E_{w_{i-1}^{-1}(\lambda)} \longrightarrow E_{w_i^{-1}(\lambda)}$$

is

$$\langle \lambda, H_{w_{i-1}(\alpha_i)} \rangle = \langle w_{i-1}^{-1}(\lambda), H_{\alpha_i} \rangle;$$

we are reduced to the case where w is a fundamental reflection: all computations occur within a minimal parabolic subgroup P, or P/U_P, or the derived group of P/U_P, or its universal covering SL(2). Let us make a direct check for $G = \mathrm{SL}(2)$, $w \neq e$ and λ the fundamental weight.

(a) We take SL(2) in its obvious representation k^2, T^* = diagonal matrices, B^* = upper triangular matrices. The weight λ is

$$\begin{pmatrix} a & * \\ 0 & a^{-1} \end{pmatrix} \longmapsto a, \qquad \text{and} \qquad \langle \lambda, H_\alpha \rangle = 1.$$

(b) $X = \mathbb{P}^1$, the space of homogeneous lines of k^2 and the fibre of E_λ at x is the corresponding line.

(c) Up to a scalar, $\Psi(\dot{w})$ associates to $u \in (E_\lambda)_x$ the linear form $u \wedge v$ on $(E_\lambda)_y$, for $x \neq y$. It vanishes simply for $y \to x$.

9.6. The assumptions (0.1.1) are in force from now on. Under the assumptions of (1.8), we have

$$F^* E_\lambda = E_{F^*\lambda} \qquad (\text{where } F^*\lambda = \lambda \circ F).$$

On the inverse image $\overline{X(w)}'$ of the graph of the Frobenius map $F : X \to X$ in $\overline{O(w)}'$, the line bundle $\mathrm{pr}_1^* E_{\lambda^{-1}} \otimes \mathrm{pr}_2^* E_{w^{-1}(\lambda)}$ is hence isomorphic to

$$\mathrm{pr}_1^*(E_{F^*w^{-1}\lambda - \lambda}).$$

It will be ample if $E_{F^*w^{-1}\lambda-\lambda}$ is ample on X, i.e., if $F^*w^{-1}\lambda - \lambda \in -C^0$. On the other hand, if $\lambda \in D^0(C, -wC)$, the section $\Psi(\dot{w})$ of it has as zero set the complement of $X(w)$ in the projective variety $\overline{X(w)}'$. If both conditions can be simultaneously fulfilled, then $X(w)$ is affine. In terms of $\mu = -w^{-1}(\lambda)$, the conditions read

$$\mu \in D^0(C, -w^{-1}C), \tag{9.6.1}$$

$$F^*\mu - w\mu \in C^0. \tag{9.6.2}$$

THEOREM 9.7. If there is $\mu \in X(T^*) \otimes \mathbb{R}$ satisfying (9.6.1) and (9.6.2) then $X(w)$ is affine. As a consequence, $X(w)$ is affine as soon as q is larger than the Coxeter number h of G.

It remains to prove that if $q \geq h$, then some μ satisfies (9.6.1) and (9.6.2). The first condition will be fulfilled if $\mu \in C^0$, that is, if $\langle \mu, H_\alpha \rangle > 0$ for each simple root α. We take μ so that $\langle \mu, H_\alpha \rangle = 1$ for each simple root α. We then have $\langle F\mu, H_\alpha \rangle = q$, and $\langle w\mu, H_\alpha \rangle = \langle \mu, w^{-1}H_\alpha \rangle$. If $H = \sum n_\alpha H_\alpha$ is the highest coroot, $\sum n_\alpha = h - 1$ and

$$\langle F\mu - w\mu, H_\alpha \rangle \geq q - \langle \mu, H \rangle = q - h + 1 > 0,$$

hence (9.6.2).

THEOREM 9.8. If the character $\theta : T(w)^F \to \overline{\mathbb{Q}}_l^*$ is non-singular, then

$$H_c^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta \xrightarrow{\sim} H^*(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta.$$

COROLLARY 9.9. If $X(w)$ is affine and θ is non-singular, then

$$H_c^i(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta = 0 \qquad \text{for } i \neq l(w).$$

Let us deduce (9.9) from (9.8). With the notations of (1.9), (9.8) means that

$$H_c^*(X(w), \mathscr{F}_\theta) \xrightarrow{\sim} H^*(X(w), \mathscr{F}_\theta). \tag{9.9.1}$$

If $X(w)$ is affine, then $H^i(X(w), \mathscr{F}_\theta) = 0$ for $i > \dim X(w) = l(w)$ ([10, XIV, 3.2]). The sheaf $\mathscr{F}_\theta$ is locally constant, its dual is $\mathscr{F}_{\theta^{-1}}$ and $X(w)$ is smooth and purely of dimension $l(w)$. By Poincaré duality, $H_c^i(X(w), \mathscr{F}_\theta)$ is dual (up to a twist) to $H^{2l(w)-i}(X(w), \mathscr{F}_{\theta^{-1}})$, hence vanishes for $2l(w) - i > l(w)$, i.e., for $i < l(w)$. This proves the corollary.

9.10. We first construct a nice compactification of $X(w)$. Let $w = s_1 \cdots s_n$ be a minimal expression for w. We define $\overline{X}(s_1, \dots, s_n)$ to be the space of sequences $(B_0, \dots, B_n)$ of Borel subgroups of G, with $B_n = FB_0$ and with B_{i-1} and B_i in relative position s_i or e. It is the inverse image of the graph of Frobenius by the map

$$\alpha : (B_0, \dots, B_n) \longmapsto (B_0, B_n) : \overline{O}(s_1, \dots, s_n) \longrightarrow X \times X.$$

In other words, if $\Gamma \xrightarrow{i} X \times X$ is the inclusion in $X \times X$ of the graph of Frobenius, it is the fibre product $\overline{O} \times_{X \times X} \Gamma$:

$$\begin{array}{ccc} \overline{X}(s_1, \dots, s_n) & \longrightarrow & \Gamma \\ \downarrow & & \downarrow i \\ \overline{O}(s_1, \dots, s_n) & \longrightarrow & X \times X. \end{array}$$

LEMMA 9.11. Γ is transverse to $\overline{O}(s, \dots, s)$, as well as to any intersection of the divisors D_i. The fibre product $\overline{X}(s, \dots, s_n)$ is hence a smooth compactification of $X(w)$, with a divisor with normal crossings (sum of the traces $\overline{D}_i$ of the D_i) at infinity.

We will show that Γ is transverse to any smooth G-equivariant scheme $\pi : Y \to X \times X$ over $X \times X$ (with G acting diagonally on $X \times X$); that is, if $\pi(y) \in \Gamma$, the sum of the tangent space to Γ at $\pi(y)$ and of the image of $d\pi$ is the whole tangent space of $X \times X$ at $\pi(y)$. Indeed

(a) the tangent space of Γ at $\pi(y)$ is $T_x \times \{0\}$;

(b) the image of $d\pi$ contains the image of $\mathrm{Lie}(G)$ by the derivative at $e \in G$ of $g \mapsto g\pi y$ (by the equivariance of Y). Since the space X is homogeneous with reduced stabilizers this image projects onto T_x by the second projection.

9.12. We now investigate how the covering $\widetilde{X}(\dot{w})$ (with structural group $T(w)^F$) ramifies along the divisor $\overline{D}_i$. The structural group being of order prime to p, the ramification is tame. On each connected component of $\overline{D}_i$, it gives rise to a homomorphism $\tau_i : \hat{\mathbb{Z}}(1) \to T(w)^F$.

Let $x(t) \in \overline{X}(s_1, \dots, s_n)$ be a one parameter family with $x(0)$ a point of $\overline{D}_i$ (not on any $\overline{D}_j$, $j \neq i$), and with the tangent vector $\dot{x}(0)$ transverse to $\overline{D}_i$. In technical terms: for S = spectrum of the Henselization of $k[t]$ at (t), x is a morphism $x : S \to \overline{X}(s_1, \dots, s_n)$ such that the closed point $t = 0$ is mapped to a point of $\overline{D}_i$ not on any $\overline{D}_j$, $j \neq i$, and that the inverse image of $\overline{D}_i$ is the reduced scheme $t = 0$. Put $x(t) = (B_0(t), \dots, B_n(t))$ and let E_i be the pull back by $t \mapsto B_i(t)$ of the T-torsor E on X (1.7). If we factor $\dot{w}$ as in 9.5, the isomorphism $\Psi(\dot{w}) : E_0 \to E_n$ (over the generic point of S) factors as

$$\Psi(\dot{w}) = \Psi(\dot{w}_i')\Psi(\dot{s}_i)\Psi(\dot{w}_{i-1})$$

and $\Psi(\dot{w}_{i-1})$ and $\Psi(\dot{w}_i')$ extend over S; $\Psi(\dot{s}_i)$ doesn't; rather, the composite $x \mapsto \Psi(\dot{s}_i)(xH_{-\alpha}(t))$ does (9.5), hence $\Psi(\dot{w})$ is of the form

$$\Psi(\dot{w})(x) = \Psi_0\big(xH_{w_{i-1}(\alpha_i)}(t)\big)$$

where $\Psi_0 : E_0 \to E_n$ extends over S. The pull back under x of the $T(w)^F$-torsor $\widetilde{X}(\dot{w})$, to the generic point of S, is given by the equation

$$F(u) = \Psi_0\big(uH_{w_{i-1}(\alpha_i)}(t)\big). \tag{9.12.1}$$

Let u_0 be a solution, over S, of the equation

$$F(u_0) = \Psi_0(u_0)$$

(this exists, because S is strictly Henselian and F and Ψ_0 extend over S); if we put $u = u_0 y$, (9.12.1) becomes

$$y^{-1}\,\mathrm{ad}\,\dot{w}F(y) = H_{w_{i-1}(\alpha_i)}(t). \tag{9.12.2}$$

In other words:
LEMMA 9.13. The $T(w)^F$-torsor $\widetilde{X}(\dot{w})$ ramifies along $\overline{D}_i$ in the same way as the pull back under $H_{w_{i-1}(\alpha_i)} : \mathbb{G}_m \to T$ of the Lang covering of $T(w)$ ramifies at 0.

The tame fundamental group of $\mathbb{G}_m$ is $\widehat{\mathbb{Z}}_{p'}(1)$, that of T is $Y^\wedge_{p'}(1)$ (for Y the dual of the character group of T) and $H_{w_{i-1}(\alpha_i)}$ induces $x \mapsto xH_{w_{i-1}(\alpha_i)}$:

$$\widehat{\mathbb{Z}}_{p'}(1) \longrightarrow Y^\wedge_{p'}(1).$$

The Lang covering of $T(w)^F$ gives rise to the map $Y^\wedge_{p'}(1) \to T(w)^F$ described in (5.2). Hence the pull back (9.13) corresponds to the composite

$$\widehat{\mathbb{Z}}_{p'}(1) \xrightarrow{H_{w_{i-1}(\alpha_i)}} Y^\wedge_{p'}(1) \longrightarrow T(w)^F.$$

For θ a character of $T(w)^F$, $\mathscr{F}_\theta$ will ramify along $\overline{D}_i$ if and only if θ is not trivial on $H_{w_{i-1}(\alpha_i)}\widehat{\mathbb{Z}}_{p'}(1)$. In that case, all higher direct images $R^i j_* \mathscr{F}_\theta$ of $\mathscr{F}_\theta$ ($i \geq 0$) under $j : X(w) \to \overline{X}(s_1, \dots, s_n)$ will vanish on $\overline{D}_i$. In particular, we have
LEMMA 9.14. If θ is non-singular, $j_*\mathscr{F}_\theta = j_!\mathscr{F}_\theta$ (extension by zero) and $R^i j_* \mathscr{F}_\theta = 0$ for $i > 0$.

Proof of 9.8 (in the guise 9.9.1). By (9.14), the Leray spectral sequence for j reads

$$H^*(\overline{X}(s_1, \dots, s_n), j_!\mathscr{F}_\theta) \xrightarrow{\sim} H^*(X(w), \mathscr{F}_\theta).$$

The space $\overline{X}(s_1, \dots, s_n)$ being a compactification of $X(w)$, the left hand side is by definition $H^*_c(X(w), \mathscr{F}_\theta)$.

Remark 9.15.1. By using arguments parallel to those of (1.6), one can show that the conclusion of (9.9): “for θ non-singular, $H^i_c(\widetilde{X}(\dot{w}), \overline{\mathbb{Q}}_l)_\theta = 0$ for $i \neq l(w)$” holds true as soon as there is w' in the F-conjugacy class of w such that $X(w')$ is affine. The criterion (9.7) can be used to check that this is always the case for the classical groups and for G_2 except the case ($G_2, q = 2, w =$ Coxeter) for which a different method applies.

Remark 9.15.2. Let (T', θ') be maximally split (5.17); under an isomorphism $T(w) \xrightarrow{\sim} T'$ (inducing $T(w)^F \xrightarrow{\sim} T'^F$), as in (1.18), θ' becomes a character θ of $T(w)^F$. Proposition 5.24 and the proof of the Induction Theorem

8.2 show that the conclusion of (9.8) still holds for θ. The same applies to (9.9).

THEOREM 9.16. If $u \in G^F$ is a regular unipotent element, then, for any F-stable maximal torus T, we have $Q_{T,G}(u) = 1$.

We must prove that for any $w \in \mathbf{W}$, $\mathrm{Tr}(u, H_c^*(X(w))) = 1$. We know that $\mathrm{Tr}(u, H_c^*(X(w)))$ is an integer, and that

$$\sum_{w \in \mathbf{W}} \mathrm{Tr}(u, H_c^*(X(w))) = |\mathbf{W}|$$

(see (7.14) and its proof). Hence it suffices to prove the

LEMMA 9.17. Under the assumptions of (9.16), $\mathrm{Tr}(u, H_c^*(X(w))) > 0$, for any $w \in \mathbf{W}$.

Take $w = s_1 \cdots s_n$ as in (9.10) and let $\overline{X}(s_1, \ldots, s_n)$ be the corresponding compactification of $X(w)$. For $P \subset [1, n]$ we write $\overline{D}_P$ for the intersection, in $\overline{X}(s_1, \ldots, s_n)$, of the divisors $\overline{D}_i$ $(i \in P)$ (9.11). For $P = \emptyset$, $\overline{D}_P$ is $\overline{X}(s_1, \ldots, s_n)$. The additivity property of cohomology with compact support shows that

$$\mathrm{Tr}(u, H_c^*(X(w))) = \sum_{P \subset [1,n]} (-1)^{|P|}\, \mathrm{Tr}(u, H^*(\overline{D}_P)). \tag{9.17.1}$$

Let b be the unique fixed point of u in X. Then $a = (b, \ldots, b)$ is the unique fixed point of u in $\overline{X}(s_1, \ldots, s_n)$; it is contained in each $\overline{D}_P$. The variety $\overline{D}_P$ being smooth and compact, $\mathrm{Tr}(u, H^*(\overline{D}_P))$ is just the multiplicity of the unique fixed point a of u acting on $\overline{D}_P$; to compute it, we may replace $\overline{D}_P$ by its completion $\widehat{D}_P$ at a. Let (x_i) be a formal coordinate system for $\overline{X}(s_1, \ldots, s_n)$ at a, such that $x_i = 0$ is an equation for $\widehat{D}_i$. In terms of these coordinates, u maps the point with coordinates $x_1, \ldots, x_n$ into the point with coordinates $P_1(x_1, \ldots, x_n), \ldots, P_n(x_1, \ldots, x_n)$, for some formal power series P_i. Since the divisor $\widehat{D}_i$ is preserved by u, P_i is divisible by x_i. The jacobian matrix of u at $(0, \ldots, 0)$ is hence diagonal; u being of order a power of p, it is the identity, and we have

$$P_i = x_i(1 + Q_i).$$

with $Q_i(0, \ldots, 0) = 0$. The fixed point scheme of u is given by the equations $x_i Q_i = 0$.

Let E_i be the divisor $Q_i = 0$. It is non-empty. The formula (9.17.1) can be rewritten

$$\begin{aligned}
\mathrm{Tr}(u, H_c^*(X(w))) &= \sum_{P \subset [1,n]} (-1)^{|P|} \prod_{i \in P} \widehat{D}_i \prod_{j \notin P} (\widehat{D}_j + E_j) \\
&= \sum_{P \subset Q \subset [1,n]} (-1)^{|P|} \prod_{i \in Q} \widehat{D}_i \prod_{j \notin Q} E_j \\
&= \sum_{Q \subset [1,n]} \left(\sum_{P \subset Q} (-1)^{|P|} \right) \prod_{i \in Q} \widehat{D}_i \prod_{j \notin Q} E_j \\
&= \prod_j E_j > 0,
\end{aligned}$$

(the product of n divisors stands for the multiplicity of their intersection).

PROPOSITION 9.18. Let $s \in G^F$ be a semisimple element. Let ν be the class function on G^F defined by:

$$\nu(g) = \begin{cases} |Z^0 Z^0(s)^F|\, q^{l(Z^0(s))} \dfrac{|Z(s)^F|}{|Z^0(s)^F|}, & \text{if } g \in G^F \text{ is regular with semisimple} \\ & \text{part conjugate in } G^F \text{ to } s, \\ 0, & \text{otherwise.} \end{cases}$$

Then ν is the character of

$$\frac{1}{|Z^0(s)^F|} \sum_{\substack{T \\ s \in T}} |T^F| \sum_{\theta \in (T^F)^\sim} \theta(s)^{-1} R_T^\theta.$$

(Here $Z^0 Z^0(s)$ is the connected centre of the connected centralizer of s, and $l(Z^0(s))$ is the semisimple rank of $Z^0(s)$.) The proof is similar to that of (7.5); it uses (9.16) and (4.2) instead of (7.2).

For any $\rho \in \mathscr{R}(G^F)$ we have

$$\langle \rho, \nu \rangle = \frac{|Z^0 Z^0(s)^F|}{|Z^0(s)^F|} q^{l(Z^0(s))} \sum_{u \in A(s)} \rho(su) \tag{9.18.1}$$

where $A(s)$ is the set of regular unipotent elements in $Z^0(s)^F$.

COROLLARY 9.19. For any $\rho \in \mathscr{R}(G^F)$, the average value of the character of ρ on the regular elements in G^F with semisimple part s, equals

$$\frac{1}{|A(s)|} \sum_{u \in A(s)} \mathrm{Tr}(su, \rho) = \frac{1}{|Z^0(s)^F|} \sum_{\substack{T \\ s \in T}} |T^F| \sum_{\theta \in (T^F)^\sim} \theta(s) \langle \rho, R_T^\theta \rangle.$$

This follows from (9.18), (9.18.1) and the following lemma.

LEMMA 9.20. The number of regular unipotent elements in G^F equals

$$|G^F| / |Z^{0F}|\, q^l.$$

(Here Z^0 is the connected centre of G and l is the semisimple rank of G.)

To count regular unipotents in G^F, we observe that any such element is contained in a unique Borel subgroup and then count the regular unipotents in G^F contained in a fixed F-stable Borel subgroup.

10. A decomposition of the Gelfand-Graev representation

10.1. Let $(G^F)^\sim$ be the set of isomorphism classes of irreducible representations of G^F (over $\overline{\mathbb{Q}}_l$). For any $\rho \in (G^F)^\sim$, there exists an F-stable maximal torus T and $\theta \in (T^F)^\sim$ such that $\langle \rho, R_T^\theta \rangle \neq 0$ (7.7); moreover the geometric conjugacy class $[\theta]$ of (T, θ) (see 5.5), is uniquely determined by ρ (6.3). We thus get a well-defined surjective map

$$(G^F)^\sim \longrightarrow \mathfrak{S} \tag{10.1.1}$$

where $\mathfrak{S}$ is the set of all geometric conjugacy classes of pairs (T,θ).

10.2. In the rest of this chapter we shall assume that the centre Z of G is connected. Let T^*, B^* be as in (1.8). Let U^* be the unipotent radical of B^* and let $U^*_\bullet$ be the subgroup of U^* generated by the root subgroups corresponding to non-simple roots. The quotient $U^*/U^*_\bullet$ is commutative and is a direct product over the simple roots α: $\prod_\alpha U^*_\alpha$, with U^*_α one-dimensional. Let I be the set of orbits of F on the simple roots. For any $i \in I$, let $U^*_i = \prod_{\alpha\in i} U^*_\alpha$; then $U^*/U^*_\bullet = \prod_{i\in I} U^*_i$. This decomposition is F-stable, hence we have also $U^{*F}/U^{*F}_\bullet = \prod_{i\in I} U^{*F}_i$.

We consider the Gelfand-Graev representation $\Gamma_G = \mathrm{Ind}_{U^{*F}}^{G^F}(\chi)$ where χ is any character of U^{*F} which is trivial on $U^{*F}_\bullet$ and defines a non-trivial character of U^{*F}_i for all $i \in I$. All such χ are conjugate under T^{*F}, since Z is connected, hence the G^F-module Γ_G is well-defined up to isomorphism.

Let Δ_G be the class function on G^F which equals $|Z^F|q^l$ on any regular unipotent element in G^F and vanishes on all other elements. (l is the semisimple rank of G.) The following result shows that this is the character of a virtual representation of G^F (which will be also denoted by Δ_G):

PROPOSITION 10.3. For any subset $J \subset I$, let $P(J) \supset B^*$ be the parabolic subgroup generated by B^* and by the root subgroups corresponding to minus the simple roots in F-orbits in J. Let $L(J)$ be $P(J)$ modulo its unipotent radical. Then

$$\Delta_G = \sum_{J\subset I}(-1)^{|J|}\,\mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)}). \tag{10.3.1}$$

(Note that the centre of $L(J)$ is connected, since that of G is, hence $\Gamma_{L(J)}$ is well-defined.)

Remark 10.4. The following identity is a formal consequence of (10.3.1):

$$\Gamma_G = \sum_{J\subset I}(-1)^{|J|}\,\mathrm{Ind}_{P(J)^F}^{G^F}(\Delta_{L(J)}).$$

10.5. Proof of 10.3. For any subset $J \subset I$, let $\mathscr{C}(J)$ be the set of characters χ of U^{*F} with $\chi|U^{*F}_\bullet = 1$ and such that χ induces a non-trivial character of U^*_i ($i \in I$) if and only if $i \in J$. It is easy to see that, for any $\chi \in \mathscr{C}(J)$, we have

$$\mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)}) = \mathrm{Ind}_{U^{*F}}^{G^F}(\chi).$$

Let $u_0 \in U^{*F}$ be a fixed regular unipotent element; in other words, the image of u_0 in U^{*F}_i is non-zero for any $i \in I$. It follows that

$$\sum_{\chi\in\mathscr{C}(J)} \chi(u_0) = (-1)^{|J|}.$$

Hence (10.3.1) is equivalent to:

$$\Delta_G = \sum_{\chi \in \mathscr{C}} \chi(u_0) \operatorname{Ind}_{U^{*F}}^{G^F}(\chi) \tag{10.5.1}$$

where $\mathscr{C} = \bigcup_{J \subset I} \mathscr{C}(J)$. The character of the right hand side of (10.5.1) vanishes at non-unipotent elements. Its value at $u \in U^{*F}$ is:

$$\begin{aligned} &|U^{*F}|^{-1} \sum_{\substack{g \in G^F \\ g^{-1}ug \in U^{*F}}} \sum_{\chi \in \mathscr{C}} \chi(g^{-1}ugu_0) \\ &= q^l |U^{*F}|^{-1} \#\{g \in G^F \mid g^{-1}ug \in U^{*F},\ g^{-1}ugu_0 \in U_{\bullet}^{*F}\}. \end{aligned} \tag{10.5.2}$$

This is zero unless $u \in U^{*F}$ is regular unipotent; we now assume that $u \in U^{*F}$ is regular unipotent. For any $g \in G^F$ with $g^{-1}ug \in U^{*F}$, we have $u \in B^* \cap gB^*g^{-1}$; but B^* is the only Borel subgroup containing u, hence $g \in B^{*F}$. If $g = tu'$, $t \in T^{*F}$, $u' \in U^{*F}$, the relation $g^{-1}ugu_0 \in U_{\bullet}^{*F}$ determines t uniquely modulo Z^F and leaves u' arbitrary. Thus the expression (10.5.2) becomes

$$q^l |U^{*F}|^{-1} |Z^F| |U^{*F}| = |Z^F| q^l,$$

and (10.3) is proved.

We shall now prove the following

LEMMA 10.6. Let M be a virtual representation of G^F such that

$$\langle M, M \rangle = |Z^F| q^l \qquad \text{and} \qquad \langle M, R_T^\theta \rangle = \varepsilon_T^\theta = \pm 1$$

for any F-stable maximal torus $T \subset G$ and any $\theta \in T^{\vee F}$. Then

$$M = \sum_{x \in \mathfrak{S}} \sigma_x M_x$$

where $\sigma_x = \pm 1$ and M_x are distinct irreducible representations of G^F such that

$$M_x = \sigma_x \sum_{\substack{(T,\theta) \bmod G^F \\ [\theta] = x}} \frac{\varepsilon_T^\theta}{\langle R_T^\theta, R_T^\theta \rangle} R_T^\theta.$$

Let $\{M\}$ be the set of all $\rho \in (G^F)^\sim$ such that $\langle \rho, M \rangle \neq 0$; $\{M\}$ has at most $|Z^F| q^l$ elements. By our assumption and by 5.7(i), the map $\{M\} \to \mathfrak{S}$ (the restriction of (10.1.1)) is surjective. Since $|\mathfrak{S}| = |Z^F| q^l$ (5.7), it is bijective and $\{M\}$ has exactly $|Z^F| q^l$ elements. Let $M_x \in \{M\}$ be the element corresponding to $x \in \mathfrak{S}$ under this bijection; we must have $M = \sum_{x \in \mathfrak{S}} \sigma_x M_x$, $\sigma_x = \pm 1$, since $\langle M, M \rangle = |\{M\}|$. If (T, θ) is such that $[\theta] = x$, we have

$$\langle M, R_T^\theta \rangle = \sigma_x \langle M_x, R_T^\theta \rangle = \varepsilon_T^\theta;$$

from the orthogonality of the R_T^θ's (6.8) we have

$$M_x = \sigma_x \sum_{\substack{(T,\theta) \bmod G^F \\ [\theta] = x}} \frac{\varepsilon_T^\theta}{\langle R_T^\theta, R_T^\theta \rangle} R_T^\theta + M'_x$$

where M'_x is orthogonal to all the R_T^θ's. It remains to prove that $M'_x = 0$. Since M_x is irreducible, this would follow if we prove that $\langle M_x - M'_x, M_x - M'_x \rangle = 1$; we certainly have $\langle M_x - M'_x, M_x - M'_x \rangle \leq 1$, hence it is sufficient to prove that

$$\sum_{x\in\mathfrak{S}}\langle M_x - M'_x, M_x - M'_x\rangle = |Z^F|q^l$$

or, in other words, that

$$\sum_{(T,\theta)\bmod G^F}\frac{1}{\langle R_T^\theta, R_T^\theta\rangle} = |Z^F|q^l.$$

By (6.8), this is equivalent to the identity

$$\sum_{T\bmod G^F}\frac{1}{|W(T)^F|}|T^F| = |Z^F|q^l,$$

which is the same as

$$\frac{1}{|\mathbf{W}|}\sum_{w\in\mathbf{W}}\det(q - w\tau) = q^l$$

where w is regarded as an automorphism of the lattice $X_0 = X(T/Z)$ and τ is defined by $F^* = q\cdot\tau^{-1}$. The last identity (compare Steinberg [15, p. 91]) can be written as

$$\sum_{0\le i\le l}(-1)^i q^{l-i}\frac{1}{|\mathbf{W}|}\sum_{w\in\mathbf{W}}\mathrm{Tr}(w\tau, \Lambda^i(X_0)) = q^l$$

and follows from the fact that $\Lambda^i(X_0)^{\mathbf{W}} = 0$ for $i > 0$ (5.8).

THEOREM 10.7. We recall that Z is connected. Let $x \in \mathfrak{S}$, and x' be a representative of the corresponding semisimple conjugacy class in the dual group (5.24). We define δ_x to be the $\mathbf{F}_q$-rank of the centralizer of x'.

(i) The formulae

$$\rho_x = \sum_{\substack{(T,\theta)\bmod G^F\\ [\theta]=x}}\frac{(-1)^{\sigma(G)-\sigma(T)}}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta \tag{10.7.1}$$

and

$$\rho'_x = (-1)^{\sigma(G)-\delta_x}\sum_{\substack{(T,\theta)\bmod G^F\\ [\theta]=x}}\frac{1}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta \tag{10.7.2}$$

define irreducible representations of G^F. The $|Z^F|q^l$ elements $\rho_x \in (G^F)^\sim$, $x \in \mathfrak{S}$ (resp. $\rho'_x \in (G^F)^\sim$, $x \in \mathfrak{S}$) are distinct. The maps $x \mapsto \rho_x$, $x \mapsto \rho'_x$ are two cross-sections of the map (10.1.1).

(ii) One has

$$\Gamma_G = \sum_{x\in\mathfrak{S}}\rho_x = \sum_{(T,\theta)\bmod G^F}\frac{(-1)^{\sigma(G)-\sigma(T)}}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta, \tag{10.7.3}$$

$$\Delta_G = \sum_{x\in\mathfrak{S}}(-1)^{\sigma(G)-\delta_x}\rho'_x = \sum_{(T,\theta)\bmod G^F}\frac{1}{\langle R_T^\theta, R_T^\theta\rangle}R_T^\theta. \tag{10.7.4}$$

(iii) Assume that all the roots have the same length. Fix (T,θ) in the geometric conjugacy class x, let S be the connected centralizer of (T,θ) (5.19) and let θ_S be the corresponding character of S^F. We have

$$\rho_x \otimes \mathrm{St}_G = \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S \otimes \mathrm{St}_S) \tag{10.7.5}$$

$$\rho'_x \otimes \mathrm{St}_G = \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S). \tag{10.7.6}$$

(iv) Let S^* be the centralizer of x'. We still have

$$\dim \rho_x = \frac{|G^F|/\,\mathrm{St}_G(e)}{|S^{*F}|/\,\mathrm{St}_{S^*}(e)} \cdot \mathrm{St}_{S^*}(e), \tag{10.7.7}$$

$$\dim \rho'_x = \frac{|G^F|/\,\mathrm{St}_G(e)}{|S^{*F}|/\,\mathrm{St}_{S^*}(e)}. \tag{10.7.8}$$

If we tensor the right hand side of (10.7.1) by St_G we find (by 7.3):

$$\sum_{\substack{(T,\theta) \bmod G^F \\ [\theta]=x}} \frac{1}{\langle R_T^\theta, R_T^\theta\rangle} \mathrm{Ind}_{T^F}^{G^F}(\theta). \tag{10.7.9}$$

Let us first assume that all roots have the same length. By 5.14(ii), the G^F-conjugacy classes of pairs (T,θ), $T \subset G$, with $[\theta] = x$ are in one-to-one correspondence with S^F-conjugacy classes of pairs $(T', \theta_S|_{T'})$, with T' an F-stable maximal torus in S. Moreover, by 5.14(i) and (6.8), we have $\langle R_T^\theta, R_T^\theta\rangle = |W_S(T')^F|$. Thus the expression (10.7.9) equals

$$\begin{aligned}
&\sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{1}{|W_S(T')^F|} \mathrm{Ind}_{T'^F}^{G^F}(\theta_S|_{T'}) \\
&= \sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{1}{|W_S(T')^F|} \mathrm{Ind}_{S^F}^{G^F}\left(\mathrm{Ind}_{T'^F}^{S^F}(1) \otimes \theta_S\right) \\
&= \mathrm{Ind}_{S^F}^{G^F}\left(\sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{1}{|W_S(T')^F|} \mathrm{Ind}_{T'^F}^{S^F}(1) \otimes \theta_S\right) \\
&= \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S \otimes \mathrm{St}_S), \qquad \text{by 7.14.}
\end{aligned}$$

Similarly, by tensoring the right hand side of (10.7.2) by St_G, we find

$$\mathrm{Ind}_{S^F}^{G^F}\left(\sum_{\substack{T'\subset S \\ \bmod S^F}} \frac{(-1)^{\sigma(S)-\sigma(T')}}{|W_S(T')^F|} \mathrm{Ind}_{T'^F}^{S^F}(1) \otimes \theta_S\right) = \mathrm{Ind}_{S^F}^{G^F}(\theta_S \otimes \mathrm{St}_S), \qquad \text{by 7.14.}$$

In general, the G^F-conjugacy classes of pairs (T,θ), with $[\theta] = x$, are in one-to-one correspondence with the G^{*F}-conjugacy classes of pairs (T',θ'), with θ' conjugate to x' (5.24), i.e., with the S^{*F}-conjugacy classes of pairs (T', x'), $(T' \subset S^*)$. The dimension of (10.7.9) is that of

$$\sum_{T'\subset S^*} \frac{1}{|W_{S^*}(T')^F|} \mathrm{Ind}_{T'^F}^{G^{*F}}(1)$$

(because $|G^{*F}| = |G^F|$), while that of (10.7.2) $\otimes\, \mathrm{St}_G$ is

$$\sum_{T'\subset S^*} \frac{(-1)^{\sigma(S)-\sigma(T')}}{|W_{S^*}(T')^F|} \mathrm{Ind}_{T'^F}^{G^F}(1)$$

and the same proof as above gives (10.7.7), (10.7.8).

We now show that (10.6) can be applied with $M = \Gamma_G$ or Δ_G. We have

$$\begin{aligned}\langle \Delta_G, \Delta_G\rangle &= \frac{1}{|G^F|}(|Z^F|q^l)^2 \#\{\text{regular unipotents in } G^F\}\\ &= \frac{1}{|G^F|}(|Z^F|q^l)^2 \frac{|G^F|}{|Z^F|q^l} = |Z^F|q^l \qquad \text{(by 9.20).}\end{aligned}$$

The analogous identity $\langle \Gamma_G, \Gamma_G\rangle = |Z^F|q^l$ follows from R. Steinberg ([14]). Now, for any (T, θ), $\langle \Delta_G, R_T^\theta\rangle$ equals the average value of the character of R_T^θ on the regular unipotents in G^F hence equals 1, by (9.16).

We now prove that $\langle \Gamma_G, R_T^\theta\rangle = (-1)^{\sigma(G)-\sigma(T)}$ for any (T, θ). Using a result of Rodier (cf. T. A. Springer, Caractères de groupes de Chevalley finis, Sém. Bourbaki 429, Fév. 1973) and the induction formula 8.2, we are reduced to the case where T is not contained in any proper F-stable parabolic subgroup of G. Using (10.3.1) and the already known fact that $\langle \Delta_G, R_T^\theta\rangle = 1$, we see that it is enough to prove

$$\left\langle \mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)}), R_T^\theta \right\rangle = 0, \qquad J \neq I. \tag{10.7.8}$$

(Note that in this case $(-1)^{\sigma(G)-\sigma(T)} = (-1)^{|I|}$.) Assuming that the theorem is already proved for groups of dimension strictly smaller than that of G and, in particular, for $L(J)$, we see from (10.7.5) and (8.2) that $\mathrm{Ind}_{P(J)^F}^{G^F}(\Gamma_{L(J)})$ is a linear combination of terms of the form $R_{T'}^{\theta'}$, with $T' \subset P(J)$, $FT' = T'$, so that (10.7.8) follows from the orthogonality formula (6.8).

Now (10.7.1), (10.7.2), (10.7.5), (10.7.6) follow directly from (10.6) applied to $M = \Gamma_G$ or Δ_G, except for an indeterminacy of sign. To remove this indeterminacy of sign, we only have to check that the right hand sides of (10.7.1) and (10.7.2) have strictly positive dimension; but they are both of the form $|G^F|/|S^{*F}|$ up to a power of q, by the first part of the proof. The theorem is proved.

COROLLARY 10.8. For any $\rho \in (G^F)^\sim$, the average value of the character of ρ on the regular unipotents in G^F equals 0, 1, or -1. This value is ± 1 if and only if $\rho = \rho'_x$ for some $x \in \mathfrak{S}$.

Indeed, this average value is just $\langle \rho, \Delta_G\rangle$.

Remark 10.9. If we assume that p is a good prime for G, then all regular unipotents in G^F fall in a single G^F-conjugacy class so that the character of ρ at any regular unipotent in G^F equals 0, 1, or -1. This was first proved in the article by J. A. Green, G. I. Lehrer, and G. Lusztig (On the degrees of certain group characters, to appear in The Quarterly Journal of Mathematics).

PROPOSITION 10.10. Let $x \in \mathfrak{S}$. Let (T, θ) be a maximally split pair in x

(see (5.17); (T,θ) is uniquely defined up to G^F-conjugacy by (5.18)). Then the virtual representation $(-1)^{\sigma(G)-\sigma(T)}R_T^\theta$ of G^F can be represented by an actual G^F-module, in which ρ_x and ρ'_x occur with multiplicity 1.

By (5.23), there exists an F-stable parabolic subgroup $P \subset G$ such that T is contained in some F-stable Levi subgroup L of P and (T,θ) is non-singular in L. L has connected centre, hence, by 5.20, (T,θ) is non-degenerate in L. Let $R_{T,L}^\theta$ be the virtual representation corresponding to (T,θ) with respect to L (see 1.20). By 7.4,

$$(-1)^{\sigma(L)-\sigma(T)}R_{T,L}^\theta = (-1)^{\sigma(G)-\sigma(T)}R_{T,L}^\theta$$

is irreducible. By 8.2,

$$(-1)^{\sigma(G)-\sigma(T)}R_T^\theta = \mathrm{Ind}_{P^F}^{G^F}\big((-1)^{\sigma(G)-\sigma(T)}R_{T,L}^\theta\big),$$

hence it can be realized as an actual G^F-module.

It remains to observe that for any (T,θ) in x (not necessarily maximally split) we have (by (10.7.1), (10.7.2), and (6.8)):

$$\langle \rho_x, R_T^\theta\rangle = (-1)^{\sigma(G)-\sigma(T)},$$
$$\langle \rho'_x, R_T^\theta\rangle = (-1)^{\sigma(G)-\delta_x}.$$

11. Suzuki and Ree groups

Our methods apply also to the Suzuki and Ree groups ${}^2B_2(q)$, ${}^2G_2(q)$, ${}^2F_4(q)$ (with q an odd power of $\sqrt{2}$, resp. $\sqrt{3}$, $\sqrt{2}$).

These groups are the fixed point set of an exceptional isogeny $F' : G \to G$, with G respectively of type B_2, G_2 and F_4, and F'^2 being the Frobenius endomorphism relative to a rational structure over the field with q^2 elements (R. Steinberg [15]). For the sake of induction, one should more generally consider groups of the form $G^{F'}$, with G reductive and F'^2 the Frobenius endomorphism relative to a rational structure over $\mathbf{F}_{q^2}$ (q an odd power of $\sqrt{2}$ or $\sqrt{3}$). Besides the Suzuki and Ree groups, this includes for instance groups of the form $G = G_1 \times G_1$, with $F'(x,y) = (Fy,x)$; for such a group, $G^{F'} = G_1^F$.

All definitions and results in Chapter 1 can be given with F replaced by F' throughout; in particular, for any F'-stable maximal torus T in G and any Borel subgroup B containing T, the subscheme $X_{T\subset B}$ of X_G and the $G^{F'}$-equivariant $T^{F'}$-torsor $\widetilde{X}_{T\subset B}$ over $X_{T\subset B}$ are well-defined (1.17). Thus for any $\theta \in (T^{F'})\check{}$, the virtual representation $R_{T\subset B}^\theta$ of $G^{F'}$ is well-defined (1.20). For $G = G_1 \times G_1$, $F'(x,y) = (Fy,x)$, this coincides with the similar representation of $G_1^F = G^{F'}$.

All the results in Chapters 4, 5, 9 (except (9.18), (9.19)), the disjointness

Theorem 6.2 (with its Corollary 6.3), the results (7.13) and (8.2) continue to be true. On the other hand, the proofs of the orthogonality relations (6.8), (6.9), and of the dimension formula (7.1) require the assumption $|T^{F'}| > 1$. If $q > 2$ this is automatically satisfied: $|T^{F'}|$ is of the form $\prod(q - \rho_i)$, where $|\rho_i| = 1$, hence $|T^{F'}| > \prod(q - 1)$. Thus, if we exclude the case where q equals $\sqrt{2}$ or $\sqrt{3}$, all the results in Chapters 6, 7, 8, 10 hold, as well as (9.18), (9.19). They can all be deduced from the orthogonality relations (6.8), (6.9). Even in the case $q = \sqrt{2}$ or $\sqrt{3}$, we can prove them for ${}^2B_2$ and ${}^2G_2$, but we cannot handle ${}^2F_4(\sqrt{2})$.

ADDED IN PROOF (2nd Dec. 1975). We can now handle ${}^2F_4(\sqrt{2})$, too; this will be considered by one of us, in a future article.

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CIRCLE DIFFEOMORPHISMS
[after M. R. HERMAN]
by Pierre DELIGNE

Let f be an orientation-preserving diffeomorphism of the circle $\mathbf{R}/\mathbf{Z}$.

If the rotation number $\alpha \in \mathbf{R}/\mathbf{Z}$ of f is irrational (i.e. if f has no periodic point) and the derivative Df of f has bounded variation (i.e. if the second derivative $D^2 f$—taken in the sense of distributions—is a measure), Denjoy proved that f is topologically conjugate to the rotation

$$R_\alpha : x \longmapsto x + \alpha :$$

there exists an orientation-preserving homeomorphism $h : \mathbf{R}/\mathbf{Z} \to \mathbf{R}/\mathbf{Z}$ such that $hfh^{-1} = R_\alpha$. This homeomorphism is unique up to composition on the left by a rotation, because the centralizer of an irrational rotation consists only of rotations.

It amounts almost to the same thing to prove a regularity property for h, or the same property uniformly for all iterates f^n of f. To pass from h to the f^n, one writes

$$f^n = (h^{-1} R_\alpha h)^n = h^{-1} R_{n\alpha} h.$$

Conversely, it is useful to change notation and pass to the universal cover $\mathbf{R}$ of the circle $\mathbf{R}/\mathbf{Z}$: f becomes an increasing diffeomorphism of $\mathbf{R}$, and h an increasing homeomorphism of $\mathbf{R}$, defined up to addition of a constant; both are of the form $x + \varphi(x)$, with φ periodic of period 1, and

$$hfh^{-1} = R_\alpha : x \longmapsto x + \alpha,$$

with $\alpha \in \mathbf{R}$ (the rotation number). Set

$$h_n = \frac{1}{n} \sum_{i=0}^{n-1} (f^i - i\alpha).$$

We shall see that h, suitably normalized, is the limit of the h_n (3.1). Suppose now that the f^n are uniformly $C^{r+\beta}$, i.e. that f is C^r and that the r-th derivatives $D^r f^n$ satisfy, uniformly in n, a Hölder condition of exponent β ($0 < \beta \leq 1$):

$$|D^r f^n(x) - D^r f^n(y)| \leq A|x - y|^\beta.$$

The $f^i - R_{i\alpha}$ satisfy the same condition and are, moreover, bounded (by one). The same is true of the $h_n - \mathrm{Id}$, which are their barycenters. Thus the $h_n - \mathrm{Id}$ form a relatively compact set of functions on **R**/**Z** in the C^r topology, and one can extract a convergent subsequence from the sequence h_n. The conjugacy h is therefore C^r, and passage to the limit shows that $D^r h$ satisfies the same Hölder condition as the $D^r f^n$.

Herman proved a theorem slightly stronger than the following.

THEOREM.— If the continued-fraction expansion of α has bounded coefficients and f is C^∞, then h is C^∞.

First (§§ 1 to 6), one proves that, as q ranges over the denominators of the convergents p_n/q_n of the continued-fraction expansion of α, $|Df^q|$ tends rapidly to 0 (at rate $1/q^\varepsilon$, with $\varepsilon > 0$). Such a result is plausible: the rotation $R_{q\alpha}$ is close to the identity, and $f^q = h^{-1} R_{q\alpha} h$.

It follows that h is $C^{1+\varepsilon'}$, with $\varepsilon' > 0$; then, from the fact that f is $C^{1+\varepsilon'}$-conjugate to R_α, one deduces that the conjugates $f_n = h_n f h_n^{-1}$ of f tend to R_α in the $C^{2+\varepsilon''}$ topology ($\varepsilon'' > 0$) (§§ 7 and 8).

An implicit-function theorem of Arnold-Moser ([1], [3]), improved by Herman following Rüssmann [4], finally shows that if f is sufficiently close to R_α in the $C^{2+\varepsilon}$ topology ($\varepsilon > 0$), then f is C^∞-conjugate to a rotation.

Terminology and notation

Write $\|\psi\|_\infty$ and $\|\psi\|_1$ for the L^∞ and L^1 norms of a function ψ on **R**/**Z**: $\sup|\psi(x)|$ and $\int_{\mathbf{R}/\mathbf{Z}} |\psi(x)|\,dx$. For $0 \le \beta \le 1$, write $|\psi|_\beta$ for its Hölder norm of exponent β:

$$|\psi|_\beta = \sup_{x \neq y} \frac{|\psi(x) - \psi(y)|}{|x-y|^\beta}.$$

Write Var(ψ) for the total variation of ψ. If ψ is C^1, this is $\|D\psi\|_1$.
R_s = the "rotation" $x \longmapsto x+s$.
$f : \mathbf{R} \to \mathbf{R}$ is an increasing diffeomorphism of the form $x+\varphi(x)$, with φ periodic of period 1. Its iterates are denoted by f^n ($n \in \mathbf{Z}$). By hypothesis, f and f^{-1} are C^1; in addition, Df (and hence log Df) is assumed to have bounded variation.

$$V = \mathrm{Var}(\log Df).$$

α = the rotation number of f. We have $\|f^n - R_{n\alpha}\|_\infty < 1$. We assume that α is irrational.
μ = the probability measure on $\mathbf{R}/\mathbf{Z}$ invariant under f. If h conjugates f to R_α, it carries μ to dx: $\mu = dh$ (the distributional derivative, or Stieltjes integral).
h = the increasing homeomorphism of $\mathbf{R}/\mathbf{Z}$ conjugating f to R_α, normalized by the condition

$$\int (h(x)-x)\,\mu = 0.$$

$$h_n = \frac{1}{n}\sum_{i=0}^{n-1}(f^i - i\alpha).$$

This is a barycenter of increasing diffeomorphisms, hence an increasing diffeomorphism. We have $h = \lim h_n$ (3.1).

$$f_n = h_n f h_n^{-1}.$$

A rational approximation to $s \in \mathbf{R}$ is an irreducible fraction p/q such that $|s-p/q| < 1/q^2$, i.e. $|qs-p| < 1/q$. Every irrational number has infinitely many rational approximations (Dirichlet).
q denotes the denominator of a rational approximation to α (and p its numerator).
§ 1. First bound for $|Df^q|$; the Denjoy-Koksma inequality
THEOREM 1.1 (Denjoy-Koksma).— Let $s \in \mathbf{R}$, let m be the denominator of a rational approximation to s, and let ψ be a function of bounded variation on $\mathbf{R}/\mathbf{Z}$. Then

$$\left|\sum_{i=0}^{m-1}\psi(x+is) - m\int_{\mathbf{R}/\mathbf{Z}}\psi(t)\,dt\right| \le \mathrm{Var}(\psi).$$

After the change of variable $t \longmapsto \pm(t-x)$, it suffices to assume that $x=0$ and $n/m \leq s \leq n/m + 1/m^2$ (with $(n,m)=1$). For $0 \leq i \leq m-1$, we have $in/m \leq is < in/m + 1/m$, and the points in/m run through $\frac{1}{m}\mathbf{Z}/\mathbf{Z}$. Hence (the sums run from $i=0$ to $i=m-1$)

$$\sum_{i=0}^{m-1} \psi(is) - m \int_{\mathbf{R}/\mathbf{Z}} \psi(x)\,dx = \sum_{i=0}^{m-1} \left(\psi(is) - m \int_{in/m}^{in/m+1/m} \psi(x)\,dx \right),$$

and

$$\left| \psi(is) - \int_{in/m}^{in/m+1/m} \psi(x)\,(m\,dx) \right|$$

is bounded by the variation of ψ between in/m and $in/m+1/m$, proving the theorem.
Recall that the variation $\int_{\mathbf{R}/\mathbf{Z}} |d\psi|$ is invariant under change of variables. Take $s=\alpha$, and make the change of variable that conjugates the rotation R_α to f; it carries dx to the measure μ, and one obtains:
COROLLARY 1.2.— For a function ψ of bounded variation on $\mathbf{R}/\mathbf{Z}$,

$$\left| \sum_{i=0}^{q-1} \psi \circ f^i - q \int_{\mathbf{R}/\mathbf{Z}} \psi(x)\,\mu \right| \leq \mathrm{Var}(\psi).$$

1.3. We have

$$Df^n = Df \circ f^{n-1} \cdot Df^{n-1} = Df \circ f^{n-1} \cdot Df \circ f^{n-2} \cdots Df,$$

whence

$$\log Df^n = \sum_{i=0}^{n-1} \log Df \circ f^i. \tag{1.3.1}$$

Applying the bound 1.2 with $\psi = \log Df$ gives

$$\left| \log Df^q - q \int_{\mathbf{R}/\mathbf{Z}} \log Df \cdot \mu \right| \leq V.$$

If the integral were nonzero, $\log Df^q$ would tend uniformly to $\pm\infty$ as $q \to \infty$, and Df^q would tend uniformly to 0 or ∞. Since

$$\int_{\mathbf{R}/\mathbf{Z}} Df^q(x)\,dx = 1,$$

this is impossible, and therefore:
COROLLARY 1.4

$$\int_{\mathbf{R}/\mathbf{Z}} \log Df \cdot \mu = 0.$$

COROLLARY 1.5

$$e^{-V} \leq Df^q \leq e^V \qquad \left(\text{when } |\alpha - p/q| < 1/q^2\right).$$

Remark 1.6.— The proof method of Theorem 1.1 proves 1.2 directly for every increasing homeomorphism f (and for μ an invariant measure). If $V = \int |d \log Df| < \infty$, one obtains 1.5. In his note [2], Denjoy starts from 1.5 to prove that f is topologically conjugate to a rotation.
1.7. Let s be irrational, let q_i ($i \geq 0$) be an increasing sequence of rational approximations to s, and let a_i be the integer part of q_i/q_{i-1} ($i \geq 1$). Assume that $q_0 = 1$. Every integer $N < q_n$ can be written

$$N = \sum_{i<n} b_i q_i \qquad \text{with } b_i \leq a_{i+1} :$$

if $n = 0$, this is clear; if $n > 0$, write $N = b_{n-1}q_{n-1} + N'$ with $N' < q_{n-1}$, and conclude by induction. The sum $\sum_{i<N} \psi(x + is)$ then decomposes into $(\sum b_i)$ partial sums, each having one of the q_i as its length, and (1.1) gives

$$\left| \sum_{i=0}^{N-1} \psi(x+is) - N \int_{\mathbf{R}/\mathbf{Z}} \psi(t)\, dt \right| \leq \sum_i b_i \operatorname{Var}(\psi) \leq \sum_{q_i \leq N} a_{i+1} \operatorname{Var}(\psi), \tag{1.7.1}$$

and, for $s = \alpha$, (1.2) and (1.3) give

$$\left\| \sum_{i=0}^{N-1} \psi \circ f^i - N \int \psi\, \mu \right\|_\infty \leq \sum_{q_i \leq N} a_{i+1} \operatorname{Var}(\psi), \tag{1.7.2}$$

$$\| \log Df^N \|_\infty \leq \sum_{q_i \leq N} a_{i+1} V. \tag{1.7.3}$$

§ 2. Bound for $|D^2 f^q|$
2.1. Differentiate formula (1.3.1) twice:

$$\log Df^n = \sum_{i<n} \log Df \circ f^i :$$

$$D \log Df^n = \sum_{i<n} D \log Df \circ f^i \cdot Df^i, \tag{2.1.1}$$

that is,

$$D^2 f^n = Df^n \cdot \sum_{i<n} D \log Df \circ f^i \cdot Df^i. \tag{2.1.2}$$

$$D^2 \log Df^n = \sum_{i<n} \left(D^2 \log Df \circ f^i \cdot (Df^i)^2 + D \log Df \circ f^i \cdot D^2 f^i\right)$$
$$= \sum_{i<n} D^2 \log Df \circ f^i \cdot (Df^i)^2 + \sum_{j<i<n} D \log Df \circ f^i \cdot Df^i \cdot D \log Df \circ f^j \cdot Df^j. \tag{2.1.3}$$

In the last term one recognizes (Oh! miracle) the cross terms in the expansion of the square of the right-hand side of (2.1.1). Completing the square gives

$$D^2 \log Df^n = \frac{1}{2}(D \log Df^n)^2 - \frac{1}{2}\sum_{i<n}(D \log Df \circ f^i \cdot Df^i)^2 + \sum_{i<n} D^2 \log Df \circ f^i \cdot (Df^i)^2. \tag{2.1.4}$$

If in (2.1.4) one bounds the $|Df^i|$ by $\sup_{i<n} \|Df^i\|_\infty$, one obtains

$$\|D^2 \log Df^n\|_\infty \leq \frac{1}{2}\|D \log Df^n\|_\infty^2 + \frac{1}{2} An \sup_{i<n} \|Df^i\|_\infty^2, \tag{2.1.5}$$

where

$$A = \|D \log Df\|_\infty^2 + 2\|D^2 \log Df\|_\infty. \tag{2.1.6}$$

2.2. By Hadamard, if ψ is a C^2 function on **R**/**Z**, then (cf. 7.1.4)

$$\|D\psi\|_\infty \leq \left(2\|\psi\|_\infty \|D^2\psi\|_\infty\right)^{1/2}. \tag{2.2.1}$$

Assume that f is C^3, and apply this interpolation inequality to $\psi = \log Df^q$, taking account of (1.5) and (2.1.5):

$$\|D \log Df^q\|_\infty \leq \left(V\|D \log Df^q\|_\infty^2 + qAV \sup_{i<q} \|Df^i\|_\infty^2\right)^{1/2}. \tag{2.2.2}$$

If $V < 1$, this gives an estimate for $\|D \log Df^q\|_\infty$ (which occurs on both sides of the formula): squaring, one finds

$$\|D \log Df^q\|_\infty^2 \leq \frac{AV}{1-V} \cdot q \cdot \sup_{i<q} \|Df^i\|_\infty^2.$$

Writing $D \log Df^q = D^2 f^q / Df^q$, and applying 1.5 once again, one finally obtains:
THEOREM 2.3.— Assume that f is C^3 and that $V < 1$. Let

$$B = \left(\frac{AV}{1-V}\right)^{1/2} \cdot e^V,$$

where A is defined by (2.1.6). If q is the denominator of a rational approxi-

mation to α, then
$$\|D^2 f^q\|_\infty \leq B\, q^{1/2} \sup_{i<q} \|Df^i\|_\infty.$$
The essential gain in this estimate is that a factor $\sqrt{q}$, rather than q, has been obtained.
§ 3. Var$(\log Df_n)$ tends to 0
In order to apply 2.3, it is necessary to reduce to the case $V < 1$. Later, in order to control the $\|Df^i\|_\infty$, ($i < q$), we shall even need to reduce to the case in which V is small. The main result 3.3 of this section will make this possible.
PROPOSITION 3.1.— The h_n converge uniformly to h.
Put $\psi(x) = h(x) - x$. By definition, h is normalized so that
$$\int_{\mathbf{R}/\mathbf{Z}} \psi(x)\, \mu = 0.$$
The formula $R_\alpha h = hf$ may be written $h = h \circ f - \alpha = f - \alpha + \psi \circ f$. Likewise, since h conjugates f^i to $R_{i\alpha}$, we have
$$h = f^i - i\alpha + \psi \circ f^i.$$
Take the mean of these identities for $0 \leq i < n$:
$$h = \frac{1}{n}\sum_{i<n}(f^i - i\alpha) + \frac{1}{n}\sum_{i<n} \psi \circ f^i = h_n + \frac{1}{n}\sum_{i<n} \psi \circ f^i.$$
It remains to show that $\frac{1}{n}\sum_{i<n} \psi \circ f^i$ converges uniformly to $\int \psi\, \mu = 0$. This can be deduced from 1.7 (ψ has variation ≤ 2), or, by a change of variable (using h), this convergence can be reduced to a theorem of H. Weyl:
Reminder 3.2 (H. Weyl).— Let φ be a continuous function on $\mathbf{R}/\mathbf{Z}$, and let α be irrational. The sums
$$S_n(\varphi) = \frac{1}{n}\sum_{i<n} \varphi(x + i\alpha)$$
converge uniformly to
$$\int_{\mathbf{R}/\mathbf{Z}} \varphi(x)\, dx.$$
This is seen by approximating φ by trigonometric polynomials P. One checks directly that, for a trigonometric polynomial, $S_n(P)$ tends to $\int_{\mathbf{R}/\mathbf{Z}} P(x)\, dx$. Since $|S_n(\varphi)| \leq \|\varphi\|_\infty$, it follows that

$$\limsup \left\| S_n(\varphi) - \int_{\mathbf{R}/\mathbf{Z}} \varphi(x)\,dx \right\|_\infty = \limsup \left\| S_n(\varphi - P) - \int_{\mathbf{R}/\mathbf{Z}} (\varphi(x) - P(x))\,dx \right\|_\infty$$
$$\leq 2\|\varphi - P\|_\infty,$$

which is arbitrarily small.
THEOREM 3.3.— If f is C^2, the variation

$$V_n = \|D \log Df_n\|_1$$

tends to 0 as $n \to \infty$.
Let us compute Df_n. We have

$$h_n \circ f = \frac{1}{n}\sum_{i=0}^{n-1}(f^{i+1} - i\alpha) = \frac{1}{n}\sum_{i=1}^{n}(f^i - i\alpha) + \alpha$$
$$= h_n + \alpha + \frac{1}{n}(f^n - x - n\alpha),$$

whence

$$f_n = R_\alpha + \frac{1}{n}(f^n - x - n\alpha) \circ h_n^{-1}$$

and

$$Df_n = 1 + \left[\frac{1}{n}(Df^n - 1)\cdot(Dh_n)^{-1}\right] \circ h_n^{-1}.$$

Thus

$$Df_n - 1 = \frac{Df^n - 1}{\sum_{i<n} Df^i} \circ h_n^{-1}.$$

Furthermore, since $Df^n = Df \circ f^{n-1} \cdot Df^{n-1}$ and

$$(Df^{n-1})^{-1} \circ f^{1-n} = D(f^{1-n}),$$

we have

$$\frac{Df^n}{\sum_{i<n} Df^i} \circ f^{1-n} = \frac{Df}{\sum_{i<n} Df^i \circ f^{1-n} \cdot D(f^{1-n})} = \frac{Df}{\sum_{i<n} Df^{-i}},$$

that is,

$$Df_n - 1 = \left(\frac{Df}{\sum_{i<n} Df^{-i}} \circ f^{n-1} - \frac{1}{\sum_{i<n} Df^i}\right) \circ h_n^{-1}. \tag{3.3.1}$$

Lemma 3.4.— The sum $\sum_{i<n} Df^i$ tends uniformly to ∞.
Indeed, all the Df^i are positive, and infinitely many of them are $\geq e^{-V}$ (1.5). Applying this lemma to f and f^{-1}, we deduce from 3.4 that:
Lemma 3.5.— The Df_n converge uniformly to 1; it is therefore equivalent to prove that the V_n tend to 0, or that the variation of Df_n tends to 0.

Since variation is invariant under change of variable, we have

$$\begin{aligned}\mathrm{Var}(Df_n) = \mathrm{Var}(Df_n - 1) &\le \mathrm{Var}\left(\frac{Df}{\sum_{i<n} Df^{-i}}\right) + \mathrm{Var}\left(\frac{1}{\sum_{i<n} Df^{i}}\right)\\ &= \left\| D\frac{Df}{\sum_{i<n} Df^{-i}}\right\|_1 + \left\| D\frac{1}{\sum_{i<n} Df^{i}}\right\|_1\\ &\le \left\| \frac{D^2 f}{\sum_{i<n} Df^{-i}}\right\|_1 + \|Df\|_\infty \left\| D\frac{1}{\sum_{i<n} Df^{-i}}\right\|_1 + \left\| D\frac{1}{\sum_{i<n} Df^{i}}\right\|_1 .\end{aligned}$$

By 3.4, the function in the first term tends uniformly to 0. Theorem 3.1 therefore follows from the following result, applied to f and f^{-1}.

THEOREM 3.6.— If f is C^2,

$$D\left(\frac{1}{\sum_{i<n} Df^{i}}\right)$$

is uniformly bounded and tends almost everywhere to 0. Consequently, this function tends to 0 in the L^1 norm.

Let G be the automorphism

$$\varphi \longmapsto \varphi\circ f\cdot Df = f^*(\varphi\, dx)/dx$$

of $L^1(\mathbf{R}/\mathbf{Z})$. It preserves the L^1 norm and maps positive functions to positive functions. The Chacon-Ornstein ergodic theorem can therefore be applied to it (see, for example, Garsia, Topics in almost everywhere convergence): for every φ in L^1, the quantities

$$\psi_n(\varphi) = \frac{\sum_{i=0}^{n-1} G^i(\varphi)}{\sum_{i=0}^{n-1} G^i(1)} = \frac{\sum_{i=0}^{n-1} \varphi\circ f^i\cdot Df^i}{\sum_{i=0}^{n-1} Df^i}$$

converge almost everywhere to a limit $\psi(\varphi)$.

We shall apply this theorem to $\varphi = D\log Df$. Put

$$\psi_n = \frac{\sum_{i<n} D\log Df\circ f^i\cdot Df^i}{\sum_{i<n} Df^i},$$

and denote by ψ the almost-sure limit of the ψ_n. The $|\psi_n|$, and hence $|\psi|$, are bounded above by $\|D\log Df\|_\infty$.

Lemma 3.7.—

$$D\left(\frac{1}{\sum_{i<n} Df^i}\right)$$

is uniformly bounded and tends almost everywhere (and hence also in the L^1 norm) to $-\frac{1}{2}\psi$.
With the summation indices running from 0 to $n-1$, we have

$$-D\left(\frac{1}{\sum Df^i}\right) = \frac{\sum D^2 f^i}{(\sum Df^i)^2} = \frac{\sum_{i>j} Df^i \cdot D\log Df \circ f^j \cdot Df^j}{(\sum Df^i)^2}.$$

The quantities

$$\lambda_{ij} = \frac{Df^i Df^j}{\left(\sum_{i<n} Df^i\right)^2} \qquad (j<i<n)$$

are positive and have sum ≤ 1; hence one already sees that

$$-D\left(\frac{1}{\sum Df^i}\right) = \sum \lambda_{ij} D\log Df \circ f^j$$

is bounded in absolute value by $\|D\log Df\|_\infty$.
With the summation indices still running from 0 to $n-1$, we have

$$\begin{aligned}
\sum_{i>j} Df^i \cdot D\log Df \circ f^j \cdot Df^j &= \sum_i Df^i \cdot \psi_i \cdot \sum_{j<i} Df^j \\
&= \psi \cdot \sum_{j<i} Df^i Df^j + \sum_i (\psi - \psi_i) Df^i \sum_{j<i} Df^j \\
&= \frac{1}{2}\psi \cdot \left(\sum Df^i\right)^2 - \frac{1}{2}\psi \sum (Df^i)^2 + \sum_i (\psi-\psi_i) Df^i \sum_{j<i} Df^j,
\end{aligned}$$

and therefore

$$D\left(\frac{1}{\sum Df^i}\right) + \frac{1}{2}\psi = \frac{1}{2}\psi \cdot \frac{\sum (Df^i)^2}{(\sum Df^i)^2} + \frac{\sum_i (\psi-\psi_i) Df^i \sum_{j<i} Df^j}{(\sum Df^i)^2}.$$

Bound $\sum (Df^i)^2 = \sum Df^i \cdot Df^i$ by $\sup Df^i \cdot \sum Df^i$, and $\sum_{j<i} Df^j$ by $\sum Df^j$. This gives

$$\left| D\left(\frac{1}{\sum Df^i}\right) + \frac{1}{2}\psi \right| \leq \frac{1}{2}|\psi| \frac{\sup Df^i}{\sum Df^i} + \frac{\sum |\psi - \psi_i| Df^i}{\sum Df^i} = A_n + B_n.$$

For every j between 0 and $n-1$,

$$\frac{Df^j}{\sum Df^i} = \left(\sum Df^i \cdot (Df^j)^{-1}\right)^{-1} = \left(\sum Df^{i-j} \circ f^j\right)^{-1} \leq \inf\left(\frac{1}{\sum_{i<n-j} Df^i} \circ f^j, \frac{1}{\sum_{i\leq j} Df^{-i}} \circ f^j\right).$$

By 3.4, A_n therefore tends uniformly to 0. As for B_n, the following lemma, whose verification is left to the reader, shows that B_n tends to 0 at every point at which ψ_n tends to ψ.

Lemma 3.8.— Let a_n be a sequence tending to 0, and let (b_n) be the terms of a divergent positive series. Then

$$\frac{\sum_{i<n} a_i b_i}{\sum_{i<n} b_i}$$

tends to 0.

Finally, let us deduce 3.6 from 3.7. Since $1/\sum_{i<n} Df^i$ tends to 0,

$$D\left(1/\sum_{i<n} Df^i\right)$$

tends to 0 in the sense of distributions. By 3.7, this sequence tends in L^1 to $-\frac{1}{2}\psi$. Hence $\psi = 0$, and 3.7 applies.

Supplement 3.9.— Herman can deduce from 3.7 that an increasing C^2 diffeomorphism of **R**/**Z** with irrational rotation number is dx-ergodic: every Borel subset (indeed, every dx-measurable subset) of **R**/**Z** that is invariant under f has Lebesgue measure 0 or 1.

Remark 3.10.— The Chacon-Ornstein theorem is not effective: it gives no estimate for the rate at which, for each ε, the ψ_n converge uniformly to ψ outside a set of measure ε. Consequently, I do not know how to deduce an effective bound for the V_n from the proof above.

§ 4. Approximation of a number by rationals

Except in 4.5, the general conventions concerning α, p, q are suspended in this section. We begin with some reminders about continued fractions, for which Hardy and Wright may be consulted.

4.1. Let $[a_0, \ldots, a_n, \ldots] = a_0 + 1/(a_1 + 1/\cdots$ ($a_i \in$ **Z**, $a_i > 0$ for

$i > 0$) be the continued-fraction expansion of an irrational number α. Let $p_n/q_n = [a_0, \dots, a_n]$ be the nth convergent. We have $p_n/q_n < \alpha$ for even n, and $p_n/q_n > \alpha$ for odd n. If, for $i = -1, -2$, (p_i, q_i) is defined by

$$\begin{pmatrix} p_{-2} & p_{-1} \\ q_{-2} & q_{-1} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

then for every $n \geq 0$,

$$\begin{pmatrix} p_{n-1} & p_n \\ q_{n-1} & q_n \end{pmatrix} = \begin{pmatrix} p_{n-2} & p_{n-1} \\ q_{n-2} & q_{n-1} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & a_n \end{pmatrix}, \tag{4.1.1}$$

that is,

$$\begin{cases} p_n = a_n p_{n-1} + p_{n-2}, \\ q_n = a_n q_{n-1} + q_{n-2}. \end{cases} \tag{4.1.2}$$

It follows from (4.1.2) that

$$a_{n+1} q_n \leq q_{n+1} \leq (a_{n+1} + 1) q_n, \tag{4.1.3}$$

$$2q_n \leq (a_{n+1} a_{n+2} + 1) q_n \leq q_{n+2}, \tag{4.1.4}$$

$$q_n \geq \sqrt{2}^{\,n} \quad \text{for } n \geq 2 \quad (\text{since } q_0 = 1,\ q_3 \geq 3 > 2^{3/2}). \tag{4.1.5}$$

By induction, (4.1.1) gives $p_{n+1} q_n - p_n q_{n+1} = \pm 1$, i.e.

$$\left| \frac{p_n}{q_n} - \frac{p_{n+1}}{q_{n+1}} \right| = \frac{1}{q_n q_{n+1}}. \tag{4.1.6}$$

Since α lies between p_n/q_n and p_{n+1}/q_{n+1}, it follows from (4.1.6) that the p_n/q_n are rational approximations to α, from below for even n and from above for odd n.

4.2. The following relations hold between convergents and rational approximations:

(4.2.1) Every rational approximation to s is either a convergent or is of the form $[a_0, \dots, a_{n-1}, a_n \pm 1]$. If the rational approximations are arranged in order of increasing denominators, then among any three successive ones at least one is a convergent.

(4.2.2) Of two successive convergents, at least one satisfies

$$\left|\alpha - \frac{p}{q}\right| < \frac{1}{2q^2}.$$

A rational approximation such that $|s - p/q| < 1/(2q^2)$ is a convergent.

4.3. Set

$$\|x\| = \inf_{n\in\mathbf{Z}} |x-n|.$$

The denominators q_n are the numbers q such that $\|q\alpha\| < \|q'\alpha\|$ for every $q' < q$. On $\mathbf{R}/\mathbf{Z}$, three successive points $q_n\alpha$ are arranged as follows (the drawing is for $a_{n+2} = 2$):

$$q_{n+1}\alpha \quad \Big| \quad 0 \quad \Big| \quad q_{n+2}\alpha \quad \underbrace{\hspace{6em}}_{a_{n+2}(q_{n+1}\alpha)} \quad q_n\alpha. \tag{4.3.1}$$

If the case $n=0$, $a_1 = q_1 = 1$, is excluded, then $\|q_n\alpha\|$ is the distance from 0 to $q_n\alpha$, as it appears in this figure, and

$$a_{n+2} \text{ is the integer part of } \frac{\|q_n\alpha\|}{\|q_{n+1}\alpha\|} \quad (\text{for } n \geq 1), \tag{4.3.2}$$

$$\|q_{n+2}\alpha\| \leq \frac{1}{a_{n+2}+1}\|q_n\alpha\| \leq \frac{1}{2}\|q_n\alpha\|. \tag{4.3.3}$$

Set $p'_n/q'_n = [a_0, \ldots, a_n + 1]$. The arrangement is

$$\frac{p_{n+1}}{q_{n+1}} \quad \Big| \quad \alpha \quad \Big| \quad \frac{p'_{n+1}}{q'_{n+1}} \quad \Big| \quad \frac{p_n}{q_n}. \tag{4.3.4}$$

Hence, by (4.1.6) and (4.1.3),

$$\frac{1}{q_n q'_{n+1}} \leq \left|\alpha - \frac{p_n}{q_n}\right| \leq \frac{1}{q_n q_{n+1}}, \tag{4.3.5}$$

that is,

$$\frac{1}{(a_{n+1}+2)q_n} \leq \|q_n\alpha\| \leq \frac{1}{a_{n+1}q_n} \quad (\text{except for } n=0,\ q_0 = q_1 = 1). \tag{4.3.6}$$

Comparing (4.1.3) and (4.3.6), and taking (4.2.2) into account, one sees that the following requirements are equivalent:

(a) $\|q\alpha\|$ is never too small;

(b) a_n is never too large;

(c) the denominators of the rational approximations to α do not grow too quickly.

DEFINITION 4.4.— An irrational number α has bounded density if its density

$$d = \sup_{n>0} \frac{1}{n} \sum_{i=1}^{n} a_i$$

is finite.
Apply 1.7 with $s = \alpha$ of density $d < \infty$, taking the q_i to be the denominators of the successive convergents. For $n \geq 2^{N/2}$, one has $N \leq q_n$ by (4.1.5), and

$$\sum_{q_i \leq N} a_{i+1} \leq nd.$$

Formula 1.7.3 gives the following result.
PROPOSITION 4.5.— There are universal constants C, D such that, when α has bounded density d,

$$\|D^N f\|_\infty \leq C^{dV} \cdot N^{DdV}.$$

4.6. Let $T : [0,1[\to [0,1[$ send x to the fractional part of x^{-1}. The probability measure

$$\nu = \frac{1}{\log 2} \frac{dx}{1+x}$$

on $[0,1[$ satisfies $T_*\nu = \nu$. Khintchine proved that T is ν-ergodic. The n-th coefficient in the continued-fraction expansion of an irrational $x \in [0,1]$ is

$$a_n(x) = [(T^{n-1}(x))^{-1}] \qquad (n \geq 1).$$

For almost every x (with respect to ν—or equivalently Lebesgue measure), an integer k therefore occurs in the sequence $a_n(x)$, $n \geq 1$, with frequency

$$\lim_{N\to\infty} \frac{1}{N} \big(\#\{n \leq N \mid a_n(x) = k\}\big) = \int_{[x^{-1}]=k} \nu \sim \frac{1}{k^2}.$$

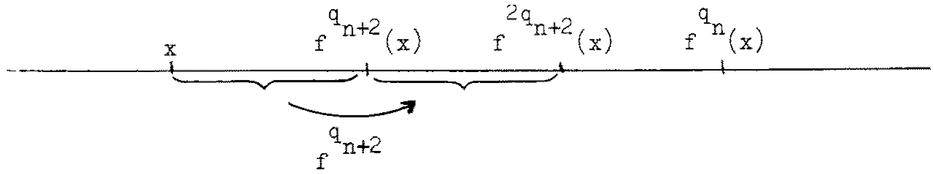
Since $\sum_k (1/k^2)\, k$ diverges, the set of numbers of bounded density has Lebesgue measure zero.
Proposition 4.5 is the only step in Herman's proof at which one must impose on α a hypothesis that is not satisfied by almost every number.

§ 5. Estimate of $|f^{q_n} - x - p_n|$

Write p_n/q_n for the n-th convergent of α. By 4.3.3, the following arrangement holds on the circle **R**/**Z**:

Authority-derived positional diagram P15-A01.

By conjugacy, the same arrangement holds for the iterates of f:



Authority-derived positional diagram P15-A02.

On **R**, the same arrangement holds if f^{q_i} is replaced by $f^{q_i} - p_i$. The axis must be oriented from left to right for odd n, and from right to left for even n.

Since $Df^{q_i} \geq e^{-V}$, it follows that

$$(1 + e^{-V})\,|f^{q_{n+2}}(x) - p_{n+2} - x| \leq |f^{q_n}(x) - p_n - x|\,,$$

and hence:

PROPOSITION 5.1.— The quantities $\|f^{q_n} - x - p_n\|_\infty$ decrease at least as fast as a geometric progression.

A slight refinement of this argument proves the following theorem.

THEOREM 5.2.— Suppose that α satisfies

$$(*) \qquad \lim_{A\to\infty} \limsup_{N\to\infty} \frac{\displaystyle\sum_{\substack{a_i > A \\ 1\leq i\leq N}} \log(a_i + 1)}{\displaystyle\sum_{1\leq i\leq N} \log(a_i+1)} = 0,$$

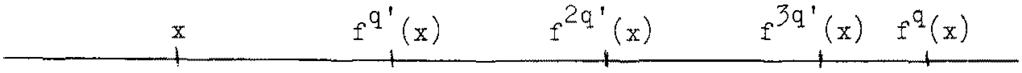
or, equivalently,

$$(*)' \qquad \text{For every } \varepsilon \text{ there is } A \text{ such that } \prod_{\substack{a_i > A \\ 1\leq i\leq N}} (a_i + 1) \leq O(q_N^{\varepsilon}).$$

There is then a function $V \longmapsto \varepsilon(V)$, tending to 0 with V, such that

$$|f^{q_n}(x) - p_n - x| \leq O\Big(q_n^{-1+\varepsilon(V)}\Big)\,.$$

Sketch of proof.
a) Let p/q and p'/q' be two rational approximations to α from above, and put $a = q\alpha - p$, $b = q'\alpha - p'$, and $c = a/b$. If $0 \leq rb \leq a$, where r is an integer ≥ 1, then on **R**/**Z** one has the following arrangement (drawn for $r = 3$):



Authority-derived positional diagram P16-A01.

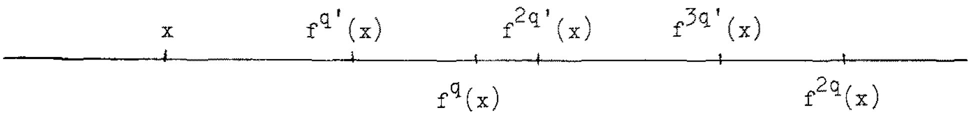
Since $Df^{q'} \geq e^{-V}$,

$$\left(\sum_{i=0}^{r-1} e^{-iV}\right)\left|f^{q'}(x) - p' - x\right| \leq |f^q(x) - p - x|\,.$$

b) More generally, if $rb \leq sa$, i.e. if $r/s \leq c$, then

$$\left(\sum_{i=0}^{r-1} e^{-iV}\right)\left|f^{q'}(x) - p' - x\right| \leq \left(\sum_{i=0}^{s-1} e^{iV}\right)|f^q(x) - p - x|\,. \tag{5.2.1}$$

For $r = 3$, $s = 2$, the drawing is:



Authority-derived positional diagram P16-A02.

c) Given ε and A, there is B such that

$$\forall x\ (2 \leq x \leq A) \quad \exists r, s \quad (r \leq B,\ s \leq B,\ x^{1-\varepsilon} < r/s < x).$$

There is then V_0 such that, if $V < V_0$, for all $r, s < B$ with $r \geq 2s$,

$$\frac{\displaystyle\sum_{i=0}^{s-1} e^{iV}}{\displaystyle\sum_{i=0}^{r-1} e^{-iV}} \leq \left(\frac{s}{r}\right)^{1-\varepsilon}.$$

d) If $V < V_0$ and $2 \leq c \leq A$, (5.2.1) gives

$$\left|f^{q'}(x) - p' - x\right| \leq c^{-1+\varepsilon'}\,|f^q(x) - p - x|\,, \tag{5.2.2}$$

where $(1 - \varepsilon') = (1 - \varepsilon)^2$.

e) Take the sequence of rational approximations p_{2n+1}/q_{2n+1}, and set

$$b_n = |q_{2n+1}\alpha - p_{2n+1}| = \|\|\|q_{2n+1}\alpha\|\|\|.$$

Then

$$2 \le \frac{b_n}{b_{n+1}} \le (a_{2n+2}+1)(a_{2n+3}+1).$$

If $V < V_0$, it follows that

$$\begin{cases} |f^{q_{2n+3}}(x) - p_{2n+3} - x| \le \left(\dfrac{b_{n+1}}{b_n}\right)^{1-\varepsilon'} |f^{q_{2n+1}}(x) - p_{2n+1} - x| & (1) \\ \qquad\qquad \text{if } (a_{2n+2}+1)(a_{2n+3}+1) \le A, & \\ |f^{q_{2n+3}}(x) - p_{2n+3} - x| \le |f^{q_{2n+1}}(x) - p_{2n+1} - x| & (2) \\ \qquad\qquad \text{in every case.} & \end{cases}$$

Chaining these inequalities gives

$$|f^{q_{2n+1}}(x) - p_{2n+1} - x| \le O\left(\left[\prod (a_{2i}+1)(a_{2i+1}+1)\right]^{1-\varepsilon'} \cdot b_n^{1-\varepsilon'}\right),$$

where the product is over $i \le n$ such that $(a_{2i}+1)(a_{2i+1}+1) \ge A$.
It is bounded by the product $B_n = \prod (a_j+1)^2$, taken over $j \le 2n+1$ such that $a_j + 1 \ge A^{1/2}$; also, b_n is bounded by $1/q_{2n+1}$. Thus

$$|f^{q_{2n+1}}(x) - p_{2n+1} - x| \le O\left(B_n q_{2n+1}^{-1+\varepsilon'}\right).$$

The hypothesis on α allows B_n to be absorbed into ε', which proves the result.
Remark 5.3.— Condition (*) holds for almost every number and for numbers of bounded density.
Remark 5.4.— Unless the a_i are bounded, this method unfortunately does not provide a lower bound for $|f^{q_n}(x) - p_n - x|$.

§ 6. Estimate of $|Df^{q_n} - 1|$

Let $\varphi_n = f^{q_n} - p_n - x$. We shall bound $|Df^{q_n} - 1| = |D\varphi_n|$ by Hadamard's interpolation formula (2.2.1). For α of bounded density and $f \in C^3$,

$$\|\varphi_n\|_\infty \le O\Big(q_n^{-1+\varepsilon(V)}\Big) \quad \text{with } \varepsilon(V) \to 0 \text{ as } V \to 0 \qquad (5.2),$$

$$\|D^2\varphi_n\|_\infty \le O\left(q_n^{1/2} \cdot \sup_{i<q_n} \|Df^i\|_\infty\right) \quad \text{for } V < 1 \qquad (2.3),$$

$$\sup_{i<q_n} \|Df^i\|_\infty \le O\Big(q_n^{\varepsilon'(V)}\Big) \quad \text{with } \varepsilon'(V) \to 0 \text{ as } V \to 0 \qquad (4.5).$$

Consequently,

$$\|D\varphi_n\|_\infty \le O\Big(q_n^{-1/4+\varepsilon''(V)}\Big), \qquad \varepsilon''(V) = \tfrac{1}{2}(\varepsilon(V) + \varepsilon'(V)).$$

For every $\varepsilon > 0$, there is m such that

$$\varepsilon''(\text{Var} \log Df_m) < \varepsilon \qquad (3.3),$$

and hence

$$\|Df_m^{q_n} - 1\|_\infty \le O\Big(q_n^{-1/4+\varepsilon}\Big).$$

By conjugacy ($f^{q_n} = h_m f_m^{q_n} h_m^{-1}$, $f_m^{q_n}$ is close to $\mathbf{R}/\mathbf{Z}$ by 5.2, and h_m is C^3), we finally obtain:

THEOREM 6.1.— If f is C^3 and α has bounded density, then for every $\varepsilon > 0$,

$$\|Df^{q_n} - 1\|_\infty \le O\Big(q_n^{-1/4+\varepsilon}\Big).$$

§ 7. h is $C^{1+\beta}$ for $\beta < 1/3$

7.1. We shall use the following properties of the Hölder seminorm $|\varphi|_\beta$.

(7.1.1) If φ has period 1 and $g : \mathbf{R}/\mathbf{Z} \to \mathbf{R}/\mathbf{Z}$, then

$$|\varphi \circ g|_\beta \le |\varphi|_\beta \|Dg\|_\infty^\beta.$$

(7.1.2) $|\varphi|_\beta$ is an increasing log-convex function of β.

Indeed, log $|\varphi|_\beta$ is the supremum of the increasing linear functions

$$\log \frac{|\varphi(x)-\varphi(y)|}{|x-y|^\beta}.$$

$$|\varphi|_0 \leq 2\|\varphi\|_\infty \qquad \text{and} \qquad |\varphi|_1 = \|D\varphi\|_\infty. \tag{7.1.3}$$

$$\|D\varphi\|_\infty \leq \left(\frac{1+\beta}{2\beta}\right)^{\beta/(\beta+1)} \cdot \left(|\varphi|_0^\beta |D\varphi|_\beta\right)^{1/(1+\beta)}. \tag{7.1.4}$$

Suppose that the maximum of $|D\varphi|$ is attained at x_0, and, to fix ideas, that $D\varphi(x_0) \geq 0$. Then

$$D\varphi(x) \geq \|D\varphi\|_\infty - |D\varphi|_\beta (x-x_0)^\beta$$

and

$$\varphi(x)-\varphi(x_0) \geq \|D\varphi\|_\infty (x-x_0) - |D\varphi|_\beta \frac{(x-x_0)^{1+\beta}}{1+\beta}, \qquad x \geq x_0.$$

Evaluate this lower bound at x such that $\|D\varphi\|_\infty - |D\varphi|_\beta (x-x_0)^\beta = 0$. Combining it with the symmetric result for $x < x_0$, one obtains

$$\frac{1}{2}|\varphi|_0 \geq \|D\varphi\|_\infty X - |D\varphi|_\beta \frac{X^{1+\beta}}{1+\beta} = X\left(\|D\varphi\|_\infty - \frac{|D\varphi|_\beta X^\beta}{1+\beta}\right)$$

for X such that $\|D\varphi\|_\infty = |D\varphi|_\beta X^\beta$. Formula (7.1.4), of which Hadamard's formula (2.2.1) is the special case $\beta = 1$, follows.

(7.1.5) $\|\varphi\|_\beta = \|\varphi\|_\infty + |\varphi|_\beta$ is a Banach-algebra norm on the algebra of C^β functions. The formula $\|\varphi_1\varphi_2\|_\beta \leq \|\varphi_1\|_\beta \|\varphi_2\|_\beta$ follows from the printed identity

$$\varphi_1(x)\varphi_2(x) - \varphi_1(y)\varphi_2(y) = \varphi_1(x)(\varphi_2(x)-\varphi_2(y)) + (\varphi_1(x)-\varphi_2(x))\varphi_2(y).$$

7.2. Assume that α has bounded density and that $f \in C^3$. Let $\varepsilon < 1/4$. By 6.1,

$$|\log Df^{q_n}| \leq A \cdot q_n^{-\varepsilon}. \tag{7.2.1}$$

Let N be an integer and write $N = \sum b_i q_i$ as in 1.7. Breaking the sum

$$\log Df^N = \sum_{i<N} \log Df \circ f^i$$

into partial sums as in 1.7 gives

$$\log Df^N = \sum_i \left(\sum \text{ of } b_i \text{ terms of the form } \log Df^{q_i} \circ f^j\right). \tag{7.2.2}$$

Applying (7.2.1):

$$\|\log Df^N\|_\infty \le A \sum_{i=0}^{\infty} a_{i+1} q_i^{-\varepsilon}. \tag{7.2.3}$$

This series converges if α has bounded density (and for almost every number), hence:

PROPOSITION 7.3.— $\|\log Df^n\|_\infty$ is uniformly bounded.

7.4. Suppose $V < 1$, and interpolate by 7.1.2 between the estimate in 6.1 for $\|Df^{q_n} - 1\|_\infty$ and the estimate in 2.3 for $\|D(Df^{q_n} - 1)\|_\infty$. For $\beta < 1/3$, this gives

$$|Df^{q_n}|_\beta \le O(q_n^{-\varepsilon}) \qquad \text{with } \varepsilon > 0. \tag{7.4.1}$$

Take $|\;|_\beta$ in (7.2.2). The compositions by f^j are harmless by (7.1.1) and 7.3, so

$$|Df^n|_\beta \le O\Big(\sum a_{i+1} q_i^{-\varepsilon}\Big) = O(1) :$$

the f^n are uniformly $C^{1+\beta}$, as are the h_n, and therefore h is $C^{1+\beta}$, as explained in the introduction. Finally, 3.3 removes the hypothesis $V < 1$.

PROPOSITION 7.5.— For every $\beta < 1/3$, h is $C^{1+\beta}$.

§ 8. $C^{2+\beta}$ convergence of the f_n

THEOREM 8.1.— Let $0 < \beta' < \beta \le 1$. Suppose that f is $C^{2+\beta}$ and h is $C^{1+\beta}$. Then the f_n converge to R_α in the $C^{2+\beta'}$ topology.

The convergence is already known to be C^1 (3.5). Here this is also immediate from (3.3.1), since the Df^i are bounded below:

$$Df^i > A > 0.$$

The interpolation formulas (7.1.2) and (7.1.4) therefore reduce the problem to showing that the f_n remain bounded in the $C^{2+\beta}$ topology, i.e. that the $|D^2 f_n|_\beta$ remain bounded.

The f^n are uniformly $C^{1+\beta}$. Thus

$$\left|\frac{1}{n}\sum_{i<n} Df^i\right|_\beta$$

is bounded. Since the Df^i are bounded below, and

$$|\varphi^{-1}|_\beta \le \frac{|\varphi|_\beta}{(\inf|\varphi|)^2},$$

we also have boundedness of

$$\left|\frac{n}{\sum_{i<n} Df^i}\right|_\beta .$$

By the same method as in 3.6, using the fact that the h_n and f^n are uniformly C^1 (which justifies the changes of variables), it remains to bound uniformly in the norm $\| \ \|_\beta$ of (7.1.5) the functions

$$-D\left(\frac{1}{\sum_{i<n} Df^i}\right) = \sum_{j<i<n} \lambda_{ij}\, D \log Df \circ f^j$$

where

$$\lambda_{ij} = \frac{Df^i Df^j}{\left(\sum_{i<n} Df^i\right)^2} = \frac{1}{n^2}\left[Df^i \cdot Df^j \cdot \left(\frac{n}{\sum_{i<n} Df^i}\right)^2\right] \qquad \text{(proof of } (3.7)).$$

The expression in brackets and the functions $D \log Df \circ f^j$ are uniformly bounded in the norm $\| \ \|_\beta$, so this is immediate.

§ 9. Application of an implicit-function theorem

Herman's final result is the following.

THEOREM 9.1.— If α has bounded density and f is C^r, $r \geq 3$ ($r = n + \beta$, with integer n and $0 \leq \beta < 1$), then h is $C^{r-1-\varepsilon}$ for every $\varepsilon > 0$.

He obtains it by combining 7.5 and 8.1 with an implicit-function theorem:

THEOREM 9.2.— Let α be irrational and let $1 \leq \theta < \theta'$. Suppose there is C such that

$$|q\alpha - p| \geq C \cdot q^{-\theta}$$

for all integers p, q. There is then a neighborhood V of R_α in the $C^{2\theta'}$ topology such that, for f in this neighborhood and of class C^r (respectively C^∞, C^ω), the equation

$$f = R_\lambda \circ g \circ R_\alpha \circ g^{-1}$$

with $\lambda \in \mathbf{R}$ and g an increasing homeomorphism has a $C^{r-\theta'}$ solution (respectively a C^∞, C^ω solution).

To obtain 9.1, take $1 < \theta < \theta' = 1 + \varepsilon$, take α to be the rotation number of f, reduce by 7.5 and 8.1 to the case where $f \in V$, and observe that the rotation number of

$$R_\lambda \circ g \circ R_\alpha \circ g^{-1}$$

can equal α only when $\lambda = 0$.
As already stated, the proof of 9.2 is inspired by [1], [3], and [4].

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GROUP THEORY. — Non-residually finite central extensions of arithmetic groups. Note (*) by Mr. Pierre Deligne, Member of the Academy.

⟦English translation of the French abstract⟧

A variant of known methods for treating the congruence subgroup problem gives results such as the following: for n $\geq$ 2, the group $\widetilde{Sp}$(2n, $\mathbb{Z}$), the inverse image of Sp(2n, $\mathbb{Z}$) in the universal covering of Sp(2n, $\mathbb{R}$) (a central extension of Sp(2n, $\mathbb{Z}$) by π_1 Sp(2n, $\mathbb{R}$) = $\mathbb{Z}$), is not Hausdorff for the topology defined by finite-index subgroups. More precisely, every finite-index subgroup of $\widetilde{Sp}$(2n, $\mathbb{Z}$) contains $2\pi_1$ Sp(2n, $\mathbb{R}$).

⟦English abstract printed in the 1978 source, retained as source matter⟧

A variant of known methods to handle the congruence subgroup problem yields results like the following. Let $\widetilde{Sp}$(2n, $\mathbb{Z}$) be the inverse image of Sp(2n, $\mathbb{Z}$) in the universal covering of Sp(2n, $\mathbb{R}$); it is a central extension of Sp(2n, $\mathbb{Z}$) by π_1 Sp(2n, $\mathbb{R}$) $\simeq \mathbb{Z}$. If n $\geq$ 2, the group $\widetilde{Sp}$(2n, $\mathbb{Z}$) is not residually finite : any subgroup of finite index contains $2\pi_1$ Sp(2n, $\mathbb{R}$).

1. S-ARITHMETIC GROUPS. — Let G be an almost simple, simply connected group isotropic over a global field F. To simplify the statements, we shall also suppose that G(F) is perfect (i.e. equal to its commutator subgroup); this is the case except, perhaps, when G is a trialitarian form of D_4. Let S also be a finite set of places of F containing all the infinite places, and suppose that

(i) $\sum_{v\in S}$ rank(G/F_v) $\geq$ 2.

For each place v of F, put G_v = G(F_v). For each finite set T of places, likewise put $G_T = \prod_{v\in T} G_v$ and $G_{T'} = \prod_{v\notin T} G_v$ (restricted product): for $\mathbb{A}$, the ring of adèles of F, one has G($\mathbb{A}$) = $G_T \times G_{T'}$.

A subgroup of G(F) is called an S-congruence subgroup if it is the inverse image of a compact open subgroup of $G_{S'}$. It is called S-arithmetic if it is commensurable with an S-congruence subgroup, i.e. if it contains, with finite index, a subgroup of finite index in an S-congruence group. The S-congruence subgroups (respectively, S-arithmetic subgroups) form a neighborhood basis of the identity for a topology (of a topological group) $\mathcal{T}$(c) [respectively, $\mathcal{T}$(a)] on G(F). For a fixed S-congruence subgroup Γ, one still obtains a neighborhood basis by considering only the S-congruence subgroups normal in Γ (respectively, only the normal finite-index subgroups of Γ). This ensures the existence of the completion $\widehat{G(F)}$(c) [respectively, $\widehat{G(F)}$(a)] of G(F) for $\mathcal{T}$(c) [respectively, $\mathcal{T}$(a)]. The topology $\mathcal{T}$(c) is the topology induced from that of $G_{S'}$. Since G(F) is dense in $G_{S'}$ (the strong approximation theorem), one has $\widehat{G(F)}$(c) = $G_{S'}$. The completion $\widehat{G(F)}$(a) of G(F) is an extension of $\widehat{G(F)}$(c) = $G_{S'}$ by the profinite group C(S, G) = lim proj Γ/Δ (the projective limit over pairs Δ ⊂ Γ, with Δ normal of finite index in Γ and Γ an S-congruence group).

For G of rank $\geq$ 2, Raghunathan (1) proved that:

(ii) $\widehat{G(F)}$(a) is a central extension of $\widehat{G(F)}$(c) = $G_{S'}$,

and he completes the verification that the same conclusion holds under hypothesis (i) alone.

Let $\widetilde{G}_S$ be a topological central extension of G_S by a discrete group A, and let $\widetilde{G(F)}$ be the inverse image of G(F) in $\widetilde{G}_S$. A subgroup of $\widetilde{G(F)}$ will be called an S-congruence subgroup if it is the inverse image of an S-congruence subgroup of G(F), and S-arithmetic if it is commensurable with an S-congruence subgroup. We again denote the corresponding topologies by $\mathcal{T}$(c) and $\mathcal{T}$(a), and the separated completions by $\widehat{\widetilde{G(F)}}$(c), $\widehat{\widetilde{G(F)}}$(a), for these

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topologies. One again has $\widehat{\widetilde{G(F)}}(c) = G_{S'}$, while $\widehat{\widetilde{G(F)}}(a)$ is an extension of $G_{S'}$ by a profinite group $\widetilde{C}$(S, G), itself an extension of C(S, G) by

$\widehat{A}$ = lim proj A/(A ∩ Γ) (Γ S-arithmetic).

PROPOSITION 1. — If condition (ii) holds, $\widehat{\widetilde{G(F)}}(a)$ is a central extension of $\widehat{\widetilde{G(F)}}(c) = G_{S'}$.

Recall that if a group $\widetilde{H}$ is a central extension of a group H, the commutator map $\widetilde{H} \times \widetilde{H} \to \widetilde{H}$ factors through a map, again denoted (,) : H × H → $\widetilde{H}$. In particular, since $\widehat{\widetilde{G(F)}}(a)$ is a central extension of $\widehat{G(F)}(a)$ by $\widehat{A}$, the commutator map $\widetilde{C}$(S, G) × $\widehat{\widetilde{G(F)}}(a) \to \widehat{\widetilde{G(F)}}(a)$ factors through C(S, G) × $\widehat{G(F)}(a)$. By (ii), it factors through a bimultiplicative map C(S, G) × $\widehat{G(F)}(a) \to \widehat{A}$;

since G(F) is perfect and dense in $\widehat{G(F)}(a)$, this map is trivial.

More concretely, the proposition means that for every S-arithmetic subgroup Δ one has:

(1 a) For every conjugate Δ^g of Δ (g ∈ $\widetilde{G(F)}$), there exists an S-congruence group Γ such that Γ ∩ Δ = Γ ∩ Δ^g.

(1 b) $\widetilde{G(F)}$ acts trivially on the finite group C_Δ = lim proj Γ/(Γ ∩ Δ).

These conditions are also equivalent to:

(1 c) For every g ∈ G(F), there exists an S-congruence group Γ such that (g, Γ) ⊂ Δ.

One has $\widetilde{C}$(S, G) = lim proj C_Δ.

2. CENTRAL EXTENSIONS OF ADELIC GROUPS. — From now on we shall suppose that:

(iii) the group $\widetilde{G}_S$ is perfect.

The most interesting case would be that in which $\widetilde{G}_S$ is the universal central extension of G_S by a discrete group (the "universal covering" of G_S), but we do not wish to assume the existence of such a universal extension.

A topological central extension E of G(𝔸) by a discrete group X is determined by the extensions E_S and $E_{S'}$ of G_S and $G_{S'}$ by X, obtained as the inverse images of G_S and $G_{S'}$ in E: because G_S is perfect, one has

E ←~ $(E_S \times E_{S'})$/{(x, −x) | x ∈ X}.

If the extension E_S is dominated by $\widetilde{G}_S$, i.e. is obtained from $\widetilde{G}_S$ by pushout along f : A → X, then f is unique, and the data of E amount to the data of the extension $E_{S'}$ together with f (an equivalence of categories); one has

E ←~ $\widetilde{G}_S \times E_{S'}$/{(a, −f(a)) | a ∈ A}.

Let $\mathcal{L}$ be the category of topological central extensions of G(𝔸) by a finite group, split over G(F), and with E_S dominated by $\widetilde{G}_S$:

1 → X → E → G(𝔸) → 1
↖s ↑
G(F)

It would amount to the same thing to require "splittable over G(F)": since G(F) is perfect, the splitting, if it exists, is unique. The category $\mathcal{L}$ is filtered on the left. For (E, s) in $\mathcal{L}$, the closure in E of (image of $\widetilde{G}_S$) · sG(F) maps onto G(𝔸): its image is closed, because X is finite, and it contains G_S · G(F), which is dense (strong approximation). The subcategory L of $\mathcal{L}$ consisting of the pairs (E, s) for which (image of $\widetilde{G}_S$) · sG(F) is dense in E is therefore cofinal; since there is at most one arrow between any two objects of L, it reduces to a filtered ordered set.

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PROPOSITION 2. — Under hypotheses (ii) and (iii), we construct a natural isomorphism between $\widetilde{C}$(S, G) and the projective limit over $\mathcal{L}$, or L, lim proj X. Under this isomorphism, the natural map from A to $\widetilde{C}$(S, G) is the limit of the maps −f : A → X, where f is such that E_S is obtained from $\widetilde{G}_S$ by f.

Let E be a topological central extension of G($\mathbb{A}$) by a discrete group X. Giving a splitting s of E over G(F) amounts to giving an isomorphism between the extension of G(F) by X induced from E_S and the opposite of the extension induced from $E_{S'}$. When E_S is dominated by $\widetilde{G}_S$ and f is as above, this amounts to giving a commutative diagram

1 → X → $E_{S'}$ → $G_{S'}$ → 1

↑ −f ↑t ↑

1 → A → $\widetilde{G(F)}$ → G(F) → 1.

Hence there is an equivalence between the category $\mathcal{L}$ and the category $\mathcal{L}'$ of commutative diagrams

(1) 1 → X → $E_{S'}$ —q→ $G_{S'}$ → 1

↑t ↑

$\widetilde{G(F)}$ → G(F)

(where $E_{S'}$ is a topological central extension of $G_{S'}$ by a finite group): E is reconstructed as $\widetilde{G}_S \times E_{S'}/\{(a, ta) \mid a \in A\}$, and the splitting over G(F) is given by s(g) = ($\tilde{g}$, t$\tilde{g}$) modulo the elements (a, ta), for $\tilde{g} \in \widetilde{G(F)} \subset \widetilde{G}_S$ mapping to g in G(F). The subcategory L of $\mathcal{L}$ corresponds to the subcategory L′ of diagrams (1) in which t has dense image.

Let ($E_{S'}$, t) be in L′, and let $\mathcal{T}$ be the topology on $\widetilde{G(F)}$ induced by that of $E_{S'}$. If K is a compact open subgroup of $E_{S'}$ disjoint from X, the intersections of $\Delta = t^{-1}$K with the S-congruence subgroups form a neighborhood basis for $\mathcal{T}$. For Γ an S-congruence subgroup defined by a compact open subgroup $K_1 \subset$ qK, one furthermore has

(2) X →~ XK/K ←~ $q^{-1}K_1/(K \cap q^{-1}K_1)$ ←~ Γ/(Δ ∩ Γ) ←~ C_Δ.

Since X is finite, Δ is S-arithmetic. Conversely, Proposition 1 ensures that a neighborhood basis of S-arithmetic groups is obtained in this way: the filtered ordered set L′ identifies with the set of germs (for the S-congruence topology) of S-arithmetic groups, and the isomorphism of Proposition 2 is defined as the projective limit of the isomorphisms (2).

3. QUASI-SPLIT GROUPS. — Let μ be the group of roots of unity in F and; for every non-complex place v of F, let μ_v be the group of roots of unity in F_v. For complex v, put $\mu_v = \{1\}$. Deodhar proved that, for G quasi-split, there exists a universal topological central extension $\widetilde{G}_v$ of G_v by a discrete group (the "universal covering"), and that for v non-real its kernel $\pi_1(G_v)$ is canonically a quotient of μ_v. I completed this result by proving that in fact $\pi_1(G_v) = \mu_v$. For real v, one has $\pi_1(G_v) = \mathbb{Z}$ or $\mathbb{Z}/2$. In every case, μ_v occurs as a quotient of $\pi_1(G_v)$.

For almost every v, the extension $\widetilde{G}_v$ splits over the natural maximal compact subgroup of G_v. This makes it possible to define the restricted product of the $\widetilde{G}_v$, an extension of G($\mathbb{A}$) by the direct sum of the $\pi_1(G_v)$. Push it out by

R : $\oplus_v \pi_1(G_v) \to \oplus_v \mu_v$ —σ → μ : $\sigma((x_v)) = \prod_v x_v^{[\mu_v:\mu]}$.

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This gives a central extension of G($\mathbb{A}$) by μ, and it follows from Deodhar (2) and C. Moore (3) that this is the universal topological central extension of G($\mathbb{A}$) by a discrete group, split over G(F):

(3) $1 \to \mu \to \widetilde{G(\mathbb{A})} \to$ G($\mathbb{A}$) $\to 1$

$\nwarrow \uparrow$

G(F)

Take $\widetilde{G}_S$ to be the universal covering of G_S, and let R_S be the restriction of R to $\pi_1 G_S = \prod_{v \in S} \pi_1(G_v)$. Proposition 2 identifies the completion $\widehat{\widetilde{G(F)}}$(a) of $\widetilde{G(F)}$ with the inverse image $\widetilde{G(\mathbb{A})}_{S'}$ of $G_{S'} = \widehat{\widetilde{G(F)}}$(c) in (3), and gives an exact commutative diagram

(4) $1 \to \mu \to \widetilde{G(\mathbb{A})}_{S'} \to G_{S'} \to 1$

$\uparrow -R_S \uparrow$i $\uparrow$

$\prod_{v \in S} \pi_1(G_v) \to \widehat{\widetilde{G(F)}}$(a) $\to \widehat{G(F)}$(a) $\to 1$

In particular:

THEOREM. — Let G/F be quasi-split, and let $\widetilde{G(F)}$ be the inverse image of G(F) in the universal covering of G_S. If $\sum_{v \in S}$ rank$(G_v) \geq 2$, the intersection of the S-arithmetic subgroups of $\widetilde{G(F)}$ is the kernel of $R_S : \prod_{v \in S} \pi_1(G_v) \to \mu$.

For Sp(2n, $\mathbb{Q}$) and S $= \infty$, this is the result announced in the introduction. For SL(n, K), with K real quadratic and S $=$ ∞, the resulting statement generalizes that of J. Millson (4).

PROBLEM. — Does every discrete finite-covolume subgroup of the universal covering of Sp(2n, $\mathbb{R}$) (n $\geq$ 2) contain $2\pi_1$ Sp(2n, $\mathbb{R}$)? (5).

4. THE SYSTEM OF FACTORS OF THE θ FUNCTIONS. — When condition (i) is not satisfied, Proposition 1 is no longer available; it nevertheless remains possible to study S-arithmetic subgroups satisfying a condition of type (1 a).

Let S therefore be a nonempty finite set of places containing all the infinite places, and let T $\supset$ S be such that (G, T) satisfies condition (ii). The interesting case is that in which S consists of a single place at which G has rank 1. Let $\widetilde{G}_S$ again be a topological central extension of G_S by a discrete group, and let $\widetilde{G(F)}$ be the inverse image of G(F) in $\widetilde{G}_S$. For an S-arithmetic subgroup Δ of $\widetilde{G(F)}$, consider the condition:

($\star$) there exists a T-arithmetic subgroup U of $\widetilde{G(F)}$ such that, for every conjugate Δ^g of Δ (g $\in$ U), there exists an S-congruence group Γ for which $\Gamma \cap \Delta = \Gamma \cap \Delta^g$ (U-invariance of the germ of Δ).

PROPOSITION 3. — The completion $\widehat{\widetilde{G(F)}}$(a*) of $\widetilde{G(F)}$ for the topology of S-arithmetic subgroups satisfying ($\star$) is a central extension of $\widehat{\widetilde{G(F)}}$(c) $= G_{S'}$.

More concretely, we must verify that if (U, Δ) satisfies ($\star$), then Δ satisfies (1 c). To prove this, we may first shrink U and Δ. In particular, we may suppose that $\Delta \subset$ U (replacing Δ by $\Delta \cap$ U). The group U then acts by conjugation on the

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finite group C^U_Δ = lim proj (Γ ∩ U)/(Γ ∩ Δ) (Γ S-congruence); replacing U by the kernel of this action and Δ by some Δ ∩ Γ, we may further suppose that U acts trivially on C^U_Δ.

Since U is T-arithmetic, its completion U(c) ⊂ $G_{S'}$ has finite index in a subgroup G_{T-S} × L, with L ⊂ $G_{T'}$ compact open; since G_{T-S} has no open subgroup of finite index, U(c) itself has the form G_{T-S} × L. Condition (⋆) ensures that the subgroups Δ ∩ Γ (Γ S-congruence) form a neighborhood basis of the identity for a group topology on U. The corresponding completion U(Δ) is an extension of U(c) by C^U_Δ—and hence a central extension of G_{T-S} × L. Replace L by an open subgroup L_1 over which this extension is trivial, and replace U and Δ by the inverse images of L_1. Since G_{T-S} is perfect, a central extension of G_{T-S} × L is determined by its restrictions to the two factors; this reduces us to the case in which the central extension U(Δ) of G_{T-S} × L is the inverse image of a central extension $\widetilde{G}_{T-S}$ of G_{T-S} by C^U_Δ. By construction, U lifts to $\widetilde{G}_{T-S}$, and Δ is the inverse image of a compact open subgroup K of $\widetilde{G}_{T-S}$ × L:

(5) K ↪ $\widetilde{G}_{T-S}$ × L

↑ ↑

Δ ——————————————————— → U

Let $\widetilde{G(F)}_T$ be the central extension of G(F) obtained as the inverse image of G(F) in $\widetilde{G}_T = \widetilde{G}_S \times \widetilde{G}_{T-S}$, and let $\widetilde{U}$ be the lifting (5) of U in $\widetilde{G(F)}_T$. For g ∈ G(F), Proposition 1 gives (g, Γ_T) ⊂ $\widetilde{U}$ for a T-congruence group Γ_T defined by a sufficiently small L′ ⊂ L. If the S-congruence subgroup Γ defined by K′ ⊂ G_{T-S} × L′, whose inverse image in $\widetilde{G}_{T-S}$ × L′ is K″, is small enough that (g, K′) ⊂ K, then (g, Γ) ⊂ Δ, which verifies (1 c).

The kernel $\widetilde{C}^*$(S, G) = Ker($\widehat{\widetilde{G(F)}}$(a*) → $\widehat{G(F)}$(c)) is still described by Proposition 2 (with the same proof), and the calculation can be carried through in the quasi-split case (cf. §3). Here is an application, with G = SL(2, ℚ), S = {∞}, and T = {∞, p}. Let $\widetilde{SL(2,\mathbb{R})}$ be the double covering of SL(2, ℝ). The factor systems of the θ functions, restricted to arbitrarily small congruence subgroups of SL(2, ℤ), may be interpreted as defining a germ τ (for the congruence-subgroup topology) of liftings of SL(2, ℤ) into $\widetilde{SL(2,\mathbb{R})}$. This germ is invariant under GL(2, ℚ). If a modular form of half-integral weight k/2 is defined to be a function on the Poincaré half-plane which, under the action of a congruence subgroup of SL(2, ℤ), transforms like the θ function of a quadratic form in k variables, this amounts to saying that if

(a b)

(c d) ∈ GL(2, ℚ)

has positive determinant and f is modular of weight k/2, then f($-z^{-1}$) and $(cz+d)^{k/2}$ f[(az + b)/(cz + d)] are likewise modular. Conversely:

PROPOSITION 4. — For every prime p, the germ τ is the unique germ of liftings of SL(2, ℤ) into $\widetilde{SL(2,\mathbb{R})}$ that is invariant under a finite-index subgroup of SL(2, ℤ[1/p]).

The arguments leading to the theorem of §3 show that every arithmetic subgroup of SL(2, ℤ) whose germ is invariant under a finite-index subgroup of SL(2, ℤ[1/p]) is a congruence subgroup.

Let Γ ⊂ SL(2, ℤ) be a congruence subgroup, and let $\tau_1 = \tau_2$ be two liftings of Γ into $\widetilde{SL(2,\mathbb{R})}$.

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Let Δ be the subgroup of Γ on which $\tau_1 = \tau_2$. If the germs of τ_1 and τ_2 are distinct and invariant under a {p}-arithmetic group, then Δ satisfies ($\star$) and is not a congruence subgroup.

(*) Session of July 3, 1978.

(1) M. S. Raghunathan, Publ. Math. I.H.E.S., 46, 1976, pp. 107-162.

(2) V. V. Deodhar, On central extensions of rational points of algebraic groups (forthcoming).

(3) C. C. Moore, Publ. Math. I.H.E.S., 35, 1968, pp. 5-70.

(4) J. Millson, Real Vector Bundles with Discrete Structure Groups (forthcoming).

(5) I learn from Mostow that such a subgroup Γ contains, in any case, a finite-index subgroup of π_1 Sp(2n, $\mathbb{R}$): one verifies that the projection of Γ into Sp(2n, $\mathbb{R}$) is discrete, since otherwise the Lie algebra of its closure would be nontrivial and central.

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THE FUNDAMENTAL GROUP OF THE COMPLEMENT OF A PLANE CURVE HAVING ONLY ORDINARY DOUBLE POINTS IS ABELIAN

[after W. FULTON]

by Pierre DELIGNE

Introduction

THEOREM 1.- Let C $\subset \mathbb{P}^2(\mathbb{C})$ be a plane curve whose only singularities are double points with distinct tangents. Then the fundamental group of $\mathbb{P}^2(\mathbb{C})$ - C is abelian.

This theorem was stated by Zariski in 1929 ([8]). See also [9], Ch. VIII. His proof used Severi's theorem, according to which the space of irreducible plane curves of a given degree having a given number of double points is irreducible (Vorlesungen über algebraische Geometrie, 1921, Anhang F). It is still unknown whether Severi's assertion is true or false.

The π_1 considered in the theorem is the genuine fundamental group, the topologists' one. If $\mathbb{C}$ is replaced by an arbitrary algebraically closed field k of characteristic 0, the same statement holds for the algebraic fundamental group, the one classifying étale coverings: the Lefschetz principle reduces us to assuming k = $\mathbb{C}$, in which case the algebraic fundamental group is the profinite completion of the genuine fundamental group. In arbitrary characteristic one has the variant:

THEOREM 2 (Fulton [3]).- Let k be an algebraically closed field and, over k, let C $\subset \mathbb{P}^2$ be a plane curve whose only singularities are double points with distinct tangents. Then the tame fundamental group of $\mathbb{P}^2$ - C is abelian.

In other words, coverings (in the sense of 0.2) of $\mathbb{P}^2$, ramified only along C and tamely ramified, are abelian. Recall that tameness of the ramification is tested after base change from $\mathbb{P}^2$ to its localizations at the generic points of the irreducible components of C. These localizations are traits (spectra of discrete valuation rings).

Suppose k has characteristic p > 0. If C is as in Theorem 2, there always exists in relative projective space $\mathbb{P}^2$ over Spec(W(k)) a relative normal-crossings divisor $\widetilde{C}$ whose special fiber is C (see, for example, Mumford, Appendix to Ch. VIII of [9]). This makes it possible to deduce Theorem 2 from Theorem 1 by a specialization argument.

Once one knows that $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C) is abelian, homological arguments immediately give its structure: if C is the union of irreducible curves of degrees d_1, ..., d_r, then $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C) is the quotient of $\mathbb{Z}^r$ by the subgroup generated by the element (d_1, ..., d_r).

In his proof of Theorem 2, Fulton uses, on the one hand, methods introduced by Abhyankar [1] to prove the special case of Theorem 2 in which the irreducible components of C are assumed smooth - methods very clearly presented here by Serre ([7]) - and, on the other hand, Theorem 3 below. An analysis of the proof then enabled the expositor to give a variant yielding Theorem 1.

Abhyankar and Fulton both use instances of the principle that, in algebraic geometry, what is very mobile is connected. Let C $\subset \mathbb{P}^2$ be as in Theorem 2, let D be an irreducible component of C, and let f : X $\to \mathbb{P}^2$ be a covering ramified only along C and tamely ramified. Since D $\subset \mathbb{P}^2$ is very mobile, f^{-1}(D) is connected. More precisely, the divisor D $\subset \mathbb{P}^2$ is ample, hence the divisor f^{-1}(D) $\subset$ X is also ample because f is finite, and one applies Bertini. By a local study of f, Abhyankar shows that if D is smooth then (f^{-1}(D))red is smooth as well, and concludes that f^{-1}(D) is irreducible.

In the general case, let g : $\widetilde{D} \to$ D be the normalization of D. The same local study shows that the reduced fiber product (X $\times_{\mathbb{P}^2} \widetilde{D}$)red is smooth. It is therefore the normalization of f^{-1}(D). If it can be shown to be connected, it follows that f^{-1}(D) is irreducible, and one can repeat the end of Abhyankar's proof as presented in [7].

Recall the definition of the fiber product: if h : X $\times \widetilde{D} \to \mathbb{P}^2 \times \mathbb{P}^2$ is the product of the maps f : X $\to \mathbb{P}^2$ and g : $\widetilde{D} \to$ D $\subset \mathbb{P}^2$, then the fiber product X $\times_{\mathbb{P}^2} \widetilde{D}$ is the inverse image $h^{-1}(\Delta)$ of the diagonal. Since $\mathbb{P}^2$ has many automorphisms, the diagonal $\Delta \subset \mathbb{P}^2 \times \mathbb{P}^2$ is very mobile, and $h^{-1}(\Delta)$ is connected. This completes the proof. More precisely, one applies to h the special case r = 2, m = 2 of the following theorem:

THEOREM 3 (W. Fulton and J. Hansen [4]).- Let P = $(\mathbb{P}^m)^r$ be the product of r copies of projective space $\mathbb{P}^m$, let δ be the diagonal map from $\mathbb{P}^m$ to P, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be a complete irreducible variety, let h : Z $\to$ P, let n = dim h(Z), and let c = (r - 1)m be the codimension of Δ in P. If n - c $\geq$ 1, then $h^{-1}(\Delta)$ is connected.

This theorem has many other applications, for which I refer to [4] and to the article of T. Gaffney and R. Lazarsfeld [5].

At the end of this introduction, we explain by an example the method used to transpose Fulton's proof of Theorem 2 into a proof of Theorem 1. The example is this: how to transpose Abhyankar's arguments to prove Theorem 1 in the special case where C is smooth (hence irreducible).

Let U be a tubular neighborhood of C in $\mathbb{P}^2(\mathbb{C})$, let U* = U - C, and let u $\in$ U*. For every covering f : X $\to$ $\mathbb{P}^2(\mathbb{C})$ ramified only along C, one has

(f^{-1}(C) connected) $\Leftrightarrow$ (f^{-1}(U) connected) $\Leftrightarrow$ (f^{-1}(U*) connected).

The key result - that for every covering X the curve f^{-1}(C) is connected - is therefore equivalent to the following: for every finite quotient H of $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u), the composite map

π_1(U*,u) —j*→ $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u) $\to$ H

is surjective. A Morse-theoretic argument proves more: the map j* : π_1(U*,u) $\to$ $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u) is surjective.

After choosing a Riemannian metric on $\mathbb{P}^2(\mathbb{C})$, for $\varepsilon > 0$ sufficiently small one may take U to be the image under the exponential map exp of the subbundle of balls of radius ε in the normal bundle of C in $\mathbb{P}^2(\mathbb{C})$. For ε sufficiently small, exp identifies U* with an oriented punctured-disk bundle over C, giving an exact sequence

$\mathbb{Z} \to \pi_1$(U*,u) $\to \pi_1$(C,$\bar{u}$) $\to 1$

($\bar{u}$ is the image of u). The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop γ going once around C in the positive direction.

Since j* is surjective, γ also defines a central element of $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u). The quotient of $\pi_1(\mathbb{P}^2(\mathbb{C})$ - C,u) by the normal subgroup generated by γ (in this case, the cyclic subgroup generated by γ) is $\pi_1(\mathbb{P}^2(\mathbb{C})$,u) = {e}, and the special case of the theorem under consideration follows.

0. Terminology

0.1. Connected means connected and nonempty. The same convention applies to irreducible.

0.2. A covering of a scheme S, assumed normal and connected, is a morphism f : T $\to$ S whose source T is normal and connected and whose restriction to a dense open subset of T is étale.

0.3. Throughout the rest of this exposition we work over $\mathbb{C}$. We shall write simply $\mathbb{P}^n$ for $\mathbb{P}^n(\mathbb{C})$. All schemes considered are assumed separated.

1. Theorems of Lefschetz and Bertini type

The special case of 1.1 below in which $Z \subset \mathbb{P}^n$ is the complement of a hypersurface is proved by Hamm and Lê [6], using Morse theory. Our aim in this section is to deduce from it the variant 1.6 of Theorem 3.

Assertion 1.1.- Let $Z \subset \mathbb{P}^n$ be a nonempty Zariski-open subset. If H is a general hyperplane, the natural map $\pi_i(Z \cap H) \to \pi_i(Z)$ is bijective for $i < n - 1$ and surjective for $i = n - 1$.

Assertion 1.1 is stated in [2], but Cheniot's proof covers only the case in which the complement of Z has codimension ≥ 2. Under this hypothesis, Z and $Z \cap H$ are simply connected, and it is harmless to work, as he does, with the H_i rather than the π_i. The case i = 1 can also be deduced from [6] or [2 bis].

Recall that the adjective "general," qualifying an element H of an irreducible algebraic variety S (here the dual projective space $\mathbb{P}^n\vee$), is a quantifier. It means that there exists a nonempty (hence dense) Zariski-open subset S' of S such that for every H in S', ...

1.2.- Let $Y \subset \mathbb{P}^n \times \mathbb{P}^n\vee$ be the space of pairs (z,H) such that $z \in Z$ and $z \in H$. The isotopy theorem, applied to the second projection $\mathrm{pr}_2 : Y \to \mathbb{P}^n\vee$, gives a nonempty Zariski-open subset U of $\mathbb{P}^n\vee$ over which Y is a topologically locally trivial fiber space. It is enough to take U to be the set of hyperplanes transverse to the strata of an algebraic Whithney stratification of the complement of Z in $\mathbb{P}^n$. Locally over U, after choosing a local trivialization of this fiber space, the inclusion $Z \cap H = \mathrm{pr}_2^{-1}(\{H\})$ into Z appears as a continuously H-dependent map from a fixed space into Z. Thus the conclusion of 1.1 can hold for one H in U only if it holds for all of them, and 1.1 is equivalent to saying that it holds for every H in U.

Suppose that Z depends on parameters: take an irreducible scheme S and an open subset Z of $S \times \mathbb{P}^n$; set $\mathrm{pr}_2\ \mathrm{pr}_1^{-1}(\{s\}) = Z(s)$, and assume every Z(s) is nonempty. Let $Y \subset S \times \mathbb{P}^n \times \mathbb{P}^n\vee$ be the space of triples (s,z,H) such that $z \in Z(s)$ and $z \in H$. The isotopy theorem, applied to the projection of Y onto $S \times \mathbb{P}^n\vee$ (respectively, of Z onto S), gives a nonempty Zariski-open subset U of $S \times \mathbb{P}^n\vee$ (respectively V of S) over which Y (respectively Z) is a topologically locally trivial fiber space.

As above, if $s \in V$ and $(s,H) \in U$, the conclusion of 1.1 holds for Z(s) and H. This parameterized variant of 1.1 will be used implicitly in the proof of 1.4.

I conjecture more generally:

Conjecture 1.3.- Let Z be smooth and connected of dimension n, let $f : Z \to \mathbb{P}^N$ be a morphism with finite fibers, let c be an integer, and let L be a linear subspace of $\mathbb{P}^N$ of codimension c, with n* = n - c. Choose a Riemannian metric on $\mathbb{P}^N$, and let V(ε) be the set of points p of $\mathbb{P}^N$ such that d(p,L) < ε. Then, for ε sufficiently small, the natural map

$\pi_i(f^{-1}(\mathrm{V}(\varepsilon))) \to \pi_i(\mathrm{Z})$

is bijective for i < n* and surjective for i = n*.

For general L, the inclusion of f^{-1}(L) into f^{-1}(V(ε)) is, for ε sufficiently small, a homotopy equivalence: 1.3 is indeed a generalization of 1.1.

If f is no longer assumed quasi-finite, and if the set of points of $\mathbb{P}^N$ where the fiber of f has dimension k has dimension φ(k), then n* should be redefined as

n* = 2 dim Z - sup(2k + φ(k) + inf(φ(k), c - 1)) - 1.

Conjecture 1.3 should moreover follow from a more precise statement saying that Z is obtained from f^{-1}(V(ε)) by attaching handles that kill spheres of dimension ≥ n*.

We shall be content with weaker results than 1.3. We deduce them from the special case of 1.1 in which Z is the complement of a hypersurface.

Lemma 1.4.- Let Z be normal and connected, let f : Z → $\mathbb{P}^N$ be a morphism, let n be the dimension of f(Z), and let c be an integer. Assume n - c ≥ 1. If L is a general linear subspace of $\mathbb{P}^N$ of codimension c, then f^{-1}(L) is connected and, for z ∈ f^{-1}(L), the natural map from $\pi_1(f^{-1}$(L),z) to π_1(Z,z) is surjective.

One may replace Z by a nonempty Zariski-open subset Z': for general L, f^{-1}(L) is still normal, and Z' (respectively Z' ∩ f^{-1}(L)) is obtained from Z (respectively f^{-1}(L)) by removing an algebraic subset with empty interior. By normality, this does not change π_0, can only enlarge π_1, and one considers the commutative diagram

π_1(Z' ∩ f^{-1}(L),z) ↠ $\pi_1(f^{-1}$(L),z)
↓ ↓
π_1(Z',z) ↠ π_1(Z,z).

In particular, we may and do assume that Z is smooth. A reader interested only in the proof of Theorem 1 could in fact have assumed this from the outset; throughout the remainder of the section, that reader may replace "normal" by "smooth."

After shrinking Z, we may and do assume that f(Z) is locally closed and that Z is a topologically locally trivial fiber space over f(Z).

If c = 1 and f is dominant, then f(Z) is a Zariski-open subset of $\mathbb{P}^N$ - after shrinking Z, we may assume it is the complement of a hypersurface - and it remains only to apply 1.1 to f(Z) and compare the long exact homotopy sequences for Z → f(Z) and f^{-1}(L) → L ∩ f(Z).

Continue to assume c = 1. If f is not dominant, let A be a linear subspace of dimension N - n - 1, let B be the projective space of linear subspaces of dimension N - n containing A, and let q : $\mathbb{P}^N$ - A $\to$ B send x to the subspace containing it. For A disjoint from $\overline{f(Z)}$, the restriction of q to $\overline{f(Z)}$ is a finite morphism, hence dominant because dim $\overline{f(Z)}$ = dim B, and qf : Z $\to$ B is dominant. Applying the lemma to qf with c = 1 gives the conclusion of the lemma for a general hyperplane L among those containing A. Since A is only required to lie in a dense open subset of a suitable Grassmannian, the conclusion of the lemma also holds for general L (cf. 1.2).

The proof is completed by induction on c: a general linear subspace of codimension c - 1 in a general hyperplane is a general linear subspace of codimension c (cf. 1.2).

Lemma 1.5.- Let Z, f, n, and c be as in 1.4. Every linear subspace L of $\mathbb{P}^N$ of codimension c has a fundamental system of open neighborhoods V such that f^{-1}(V) is connected and, for z $\in f^{-1}$(V), the natural map from $\pi_1(f^{-1}$(V),z) to π_1(Z,z) is surjective.

Let Gr be the Grassmannian of linear subspaces of $\mathbb{P}^N$ of codimension c, let Y $\subset$ Z $\times$ Gr be the set of pairs (z,L) such that f(z) $\in$ L, and let G be a nonempty Zariski-open subset of Gr such that over G, Y is a topologically locally trivial fiber space. The conclusion of 1.4 then holds for L $\in$ G (cf. 1.2).

For every open subset U of Gr, write U* for the open subset of $\mathbb{P}^N$ that is the union of the L $\in$ U. We shall prove that if U is connected, the conclusion of 1.5 holds for V = U*. The lemma will follow.

Let us prove that if U is connected, f^{-1}(U*) is connected as well. The intersection U' = U $\cap$ G is connected, since it is obtained from U by removing a triangulable subset of real codimension $\geq$ 2. By 1.4, the union f^{-1}(U'*) of the f^{-1}(L) for L $\in$ U' is therefore connected. We shall show that it is dense in f^{-1}(U*).

The image f(Z) of Z is not contained in the complement F of G*: since n - c $\geq$ 0, the L $\in$ G meet f(Z). If z $\in$ Z, the image of any neighborhood of z is not contained in F; otherwise the image of Z would be contained in F by analytic continuation. If z $\in f^{-1}$(U*), there are therefore points z' arbitrarily near z such that f(z') $\in$ G*. In the set of L $\in$ Gr containing z', the trace of G is dense, since it is nonempty. If z $\in f^{-1}$(L), with L $\in$ U, this makes it possible to find z' as above lying in some f^{-1}(L'), with L' in G arbitrarily close to L. In particular one may take L' $\in$ U', in which case z' $\in f^{-1}$(U'*). This proves the density of f^{-1}(U'*) in f^{-1}(U*) and the connectedness of f^{-1}(U*).

Finally let L $\in$ U'. We have f^{-1}(L) $\subset f^{-1}$(U*) $\subset$ Z. By 1.4, $\pi_1(f^{-1}$(L)) maps onto π_1(Z). A fortiori the same is true for $\pi_1(f^{-1}$(U*)).

We shall deduce from this a variant of Theorem 3.

THEOREM 1.6.- Let P = $(\mathbb{P}^m)^r$ be the product of r copies of projective space $\mathbb{P}^m$, let δ be the diagonal map from $\mathbb{P}^m$ to P, let $\Delta = \delta(\mathbb{P}^m)$ be the diagonal, let Z be normal and connected, let h : Z $\to$ P, let n = dim h(Z), and let c = (r - 1)m be the codimension of Δ in P. If n - c $\geq$ 1, then Δ has a fundamental system of open neighborhoods V such that h^{-1}(V) is connected. If h^{-1}(V) is connected and z $\in h^{-1}$(V), the natural map from $\pi_1(h^{-1}$(V),z) to π_1(V,z) is surjective.

As in 1.4, one may shrink Z, and in particular assume that Z is smooth. We do so. Let $(H_\alpha)_0 \leq \alpha \leq$m be a family of hyperplanes in $\mathbb{P}^m$ such that

a) the intersection of the H_α is empty: in suitable projective coordinates (X_α), they are given by $X_\alpha = 0$;

b) h(Z) is not contained in any pr_$i^{-1}(H_\alpha)$. After shrinking Z, we may and do assume that $\overline{h(Z)}$ is disjoint from the pr _$i^{-1}(H_\alpha)$;

c) $\overline{h(Z)} \cap \Delta$ is not contained in the union of the $\delta(H_\alpha)$: since Δ belongs to a connected family of cycles filling all of P, one has $\overline{h(Z)} \cap \Delta \neq \emptyset$, and it is possible to choose H_α satisfying c).

Let $A_\alpha = \mathbb{P}^m$ - H_α, let $B_\alpha = A_\alpha^r \subset$ P, and let $\Delta_\alpha = \delta(A_\alpha)$ be the diagonal of B_α. The spaces A_α and B_α are affine spaces and Δ_α is an affine subspace of B_α. The projective completion of A_α is $\mathbb{P}^m$. Denote by $\bar{B}$_α the projective completion of B_α, and by Δ_$\alpha \subset \bar{B}$_α the completion of Δ_α.

Fix α, and write simply H, A, B, Δ_A, $\bar{B}$, Δ_A for H_α, A_α, B_α, Δ_α, $\bar{B}$_α, and Δ_α.

Lemma 1.7.- For every neighborhood U of $\Delta \subset$ P, there exists a neighborhood V of Δ_A in $\bar{B}$ such that V $\cap$ B $\subset$ U $\cap$ B.

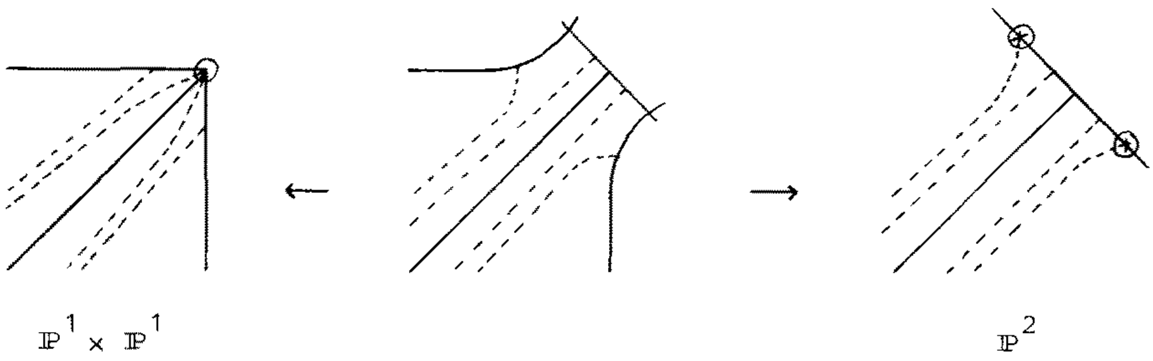
Choose on A the structure of a Hermitian vector space compatible with its affine structure. A fundamental system of neighborhoods U of Δ (respectively V of Δ_A) then has traces on B of the following form:

U $\cap$ B = $\{(x_i)$ | If $|x_i| \leq$ R for some i, then $\forall$j $\forall$k $|x_j$ - $x_k| < \varepsilon$; if the $|x_i|$ are $\geq$ R, then for all j,k the angle between the complex lines generated by x_j and x_k is $< \varepsilon\}$

V $\cap$ B = $\{(x_i)$ | If $|x_i| \leq$ R for some i, then $\forall$j $\forall$k $|x_j$ - $x_k| < \varepsilon$; if the $|x_i|$ are $\geq$ R, then $\forall$j $\forall$k $|x_j$ - $x_k| \leq \varepsilon$ inf$\{|x_i|\}\}$

The lemma follows.

When r = m = 1, the two compactifications $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ of $\mathbb{A}^2$ are topped by a third as follows: the drawing represents the locus at infinity and the diagonal; the circled points are those that must be blown up to obtain the third compactification; U and V are indicated by the dotted lines:



A similar construction can be made for arbitrary r and m.

Return to the proof of 1.6. Apply 1.5 to h : Z $\to \bar{B}\_\alpha$ and to Δ_α. It follows that Δ_α has an arbitrarily small open neighborhood V_α such that $h^{-1}(V_\alpha) = h^{-1}(V_\alpha \cap B_\alpha)$ is connected and its π_1 maps onto that of Z. By 1.7, whatever the neighborhood U of Δ in P, V_α can be chosen so that $V_\alpha \cap B_\alpha \subset$ U.

The union V of the open subsets $V_\alpha \cap B_\alpha$ of P is an arbitrarily small neighborhood of Δ. Hypothesis b) on the H_α ensures that the intersection of the $V_\alpha \cap B_\alpha$ meets h(Z). Hence h^{-1}(V), a union of connected open subsets $h^{-1}(V_\alpha)$ with nonempty common intersection, is connected. Since the π_1 of one of the $h^{-1}(V_\alpha)$ maps onto that of Z, a fortiori the same is true for h^{-1}(V), and this completes the proof.

2. Proof of Theorem 1

2.1.- Let C $\subset \mathbb{P}^2$ be a reduced plane curve, let D be an irreducible component of C, let C' = $\overline{(C-D)}$ be the union of the other components, let Sing(C) be the set of singular points of C, and let D* = D - Sing(C). Choose a base point O $\in \mathbb{P}^2$ - C, and let γ be a loop in $\mathbb{P}^2$ - C of the following type: γ goes from O along a path ch to a point p' near a point p of D*, goes once around D* in the positive direction, and returns to O along ch^{-1}. Since D* is connected, all loops of this type are conjugate. The quotient of $\pi_1(\mathbb{P}^2$ - C,O) by the normal subgroup they generate is $\pi_1(\mathbb{P}^2$ - C').

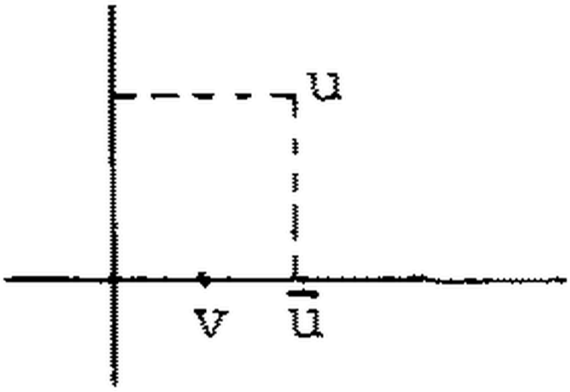
PROPOSITION 2.2.- Suppose that the singular points of C lying on D are double points with distinct tangents. Then γ is central in $\pi_1(\mathbb{P}^2$ - C,O).

Let j be the inclusion of $\mathbb{P}^2$ - C into $\mathbb{P}^2$, let g : $\widetilde{D} \to$ D be the normalization of D, and apply 1.6 to

h = j $\times$ g : ($\mathbb{P}^2$ - C) $\times\ \widetilde{D} \to \mathbb{P}^2 \times \mathbb{P}^2$.

To describe conveniently the inverse images of certain neighborhoods of the diagonal, choose on $\mathbb{P}^2$ a Riemannian structure such that, near each point $P \in D \cap \mathrm{Sing}(C)$, $\mathbb{P}^2$ is Euclidean and the two branches of C through P are orthogonal real 2-planes.

If $(u,v) \in (\mathbb{P}^2 - C) \times \widetilde{D}$ lies in the inverse image of a sufficiently small neighborhood of the diagonal, that is, if u and v are sufficiently close, then u is near D and one can drop a short perpendicular from u to D. When u is near Sing(D), there are even two such perpendiculars, one for each branch. Choose the one whose foot $\bar{u}$ is close to v on $\widetilde{D}$:



The point $\bar{u}$ lies in D^*.

Conversely, let $\varepsilon > 0$ be sufficiently small, and let $q : U^* \to D^*$ be the punctured-disk subbundle of the normal bundle of D^* in $\mathbb{P}^2$ consisting of nonzero normal vectors of length $< \varepsilon$. Let E be the set of pairs (x,v) with $x \in U^*$, $v \in \widetilde{D}$, and $d(q(x),v) < \varepsilon$ (distance measured on $\widetilde{D}$). The map $\mathrm{pr}_1 : E \to U^*$ is a homotopy equivalence and has the space of pairs (x,q(x)) as a section. The map $\exp : (x,v) \mapsto (\exp(x),v)$ from E to $(\mathbb{P}^2 - C) \times \widetilde{D}$ identifies E with the inverse image of a neighborhood of Δ, and as $\varepsilon \to 0$ one obtains inverse images of a fundamental system of neighborhoods. Applying 1.6, one finds that for every $O \in U^*$, exp and q induce a surjection

$(\exp,q)^* : \pi_1(U^*,O) \twoheadrightarrow \pi_1(\mathbb{P}^2 - C,\exp(O)) \times \pi_1(\widetilde{D},q(O)).$

There is an exact homotopy sequence

$\mathbb{Z} \to \pi_1(U^*,O) \to \pi_1(D^*,q(O)) \to 1.$

The image of $\mathbb{Z}$ is central; the image of $1 \in \mathbb{Z}$ is represented by a small loop going once around D^* in the positive direction. Since $(\exp,q)^*$, and hence $\exp^* : \pi_1(U^*,O) \to \pi_1(\mathbb{P}^2 - C,\exp(O))$, is surjective, this loop also defines a central element of $\pi_1(\mathbb{P}^2 - C,\exp(O))$. It is γ.

2.3 Proof of Theorem 1: For every irreducible component D of C, let $\gamma(D)$ be the corresponding central element of $\pi_1(\mathbb{P}^2 - C,O)$. The quotient of $\pi_1(\mathbb{P}^2 - C,O)$ by the - central - subgroup generated by the $\gamma(D)$ is $\pi_1(\mathbb{P}^2,O) = \{e\}$. The theorem follows.

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I have just received (1/10/80) a very fine account surveying the methods studied here, their variants, and their many applications: W. FULTON and R. LAZARSFELD - Connectivity and its applications in algebraic geometry.

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